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M.S. CHOUHAN

ADVANCED PROBLEMS IN
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Advanced Problems in
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by:

M.S. Chouhan

Director

Vibrant Academy, Kota



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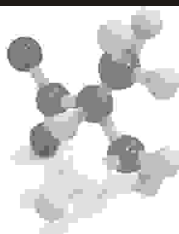
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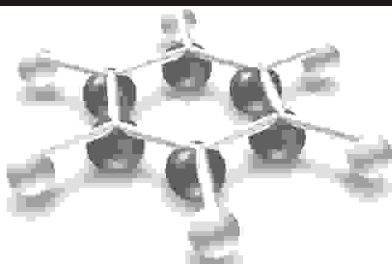
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1

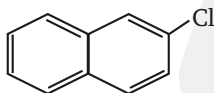


GENERAL ORGANIC CHEMISTRY



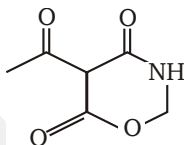
Level - 1

1. How many 2° Hydrogen atoms are present in the given following compound ?



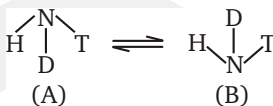
- (a) 2 (b) 5 (c) 7 (d) 8

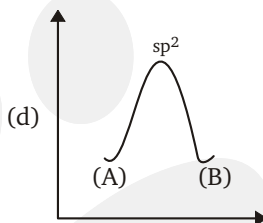
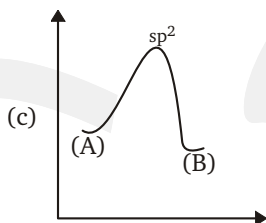
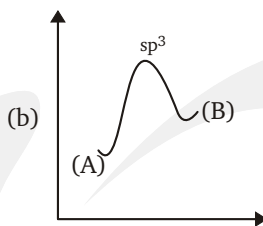
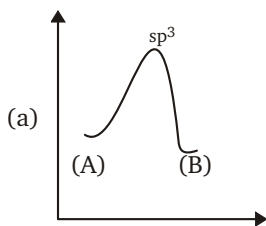
2. Identify which functional group is **Not** present in the given following compound?



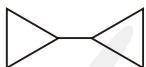
- (a) Ketone (b) Ester (c) Amide (d) Ether

3. Correct energy profile for amine inversion and hybridization of nitrogen in transition state is:





4.



(i)



(ii)

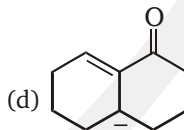
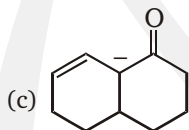
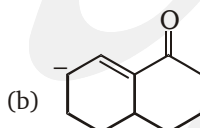
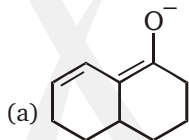


(iii)

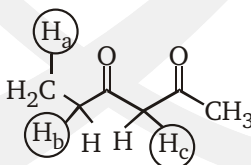
Correct order of the heats of combustion of above compounds is:

- (a) (i) > (ii) > (iii) (b) (i) > (iii) > (ii) (c) (ii) > (i) > (iii) (d) (ii) > (iii) > (i)

5. Which of the following is not a resonance structure of the others ?



6. Rank the hydrogen atoms (H_a , H_b , H_c) present in the following molecule in decreasing order of their acidic strength.

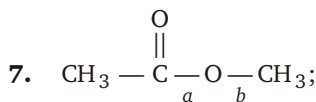


(a) $a > b > c$

(b) $b > a > c$

(c) $b > c > a$

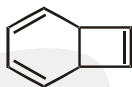
(d) $c > b > a$



The correct relation between the bond lengths a and b is:

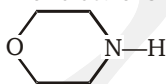
- (a) $a = b$ (b) $b > a$
(c) $b < a$ (d) Impossible to predict

8. The number of sp^2 sp^2 sigma bonds in the compound given below is :



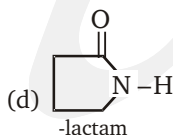
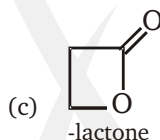
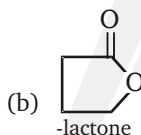
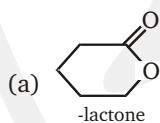
- (a) 1 (b) 3 (c) 4 (d) 5

9. The total number of lone pair of electrons in the given molecule is :

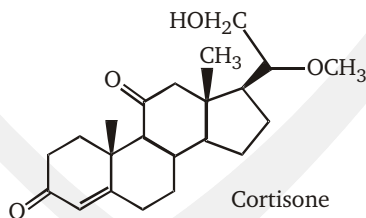


- (a) 2 (b) 3 (c) 4 (d) 5

10. Which of the following rings is highly strained ?

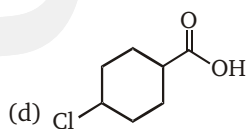
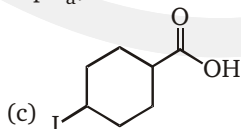
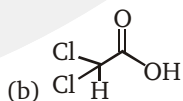
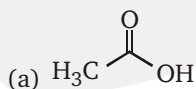


11. The functional groups present in Cortisone are :

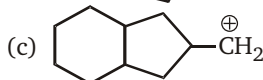
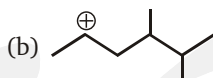


- (a) ether, alkene, alcohol (b) alcohol, ketone, alkene, ether
(c) alcohol, ketone, amine (d) ether, amine, ketone

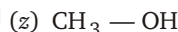
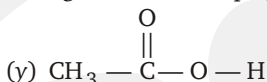
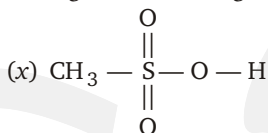
12. Select the acid with the highest K_a (i.e., lowest pK_a).



13. Most stable carbocation among the following is :



14. Arrange the following in increasing order of their $\text{p}K_a$ values.



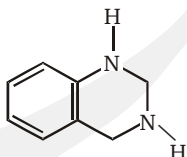
(a) $y < x < z$

(b) $x < y < z$

(c) $y < z < x$

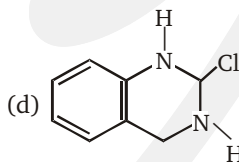
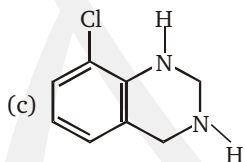
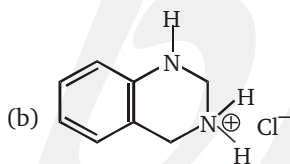
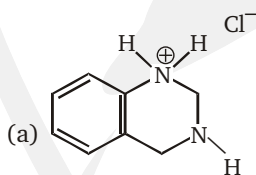
(d) $x < z < y$

15. Which is the major product of the following reaction ?

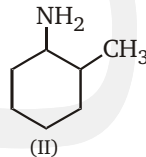
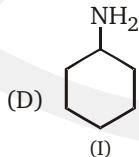
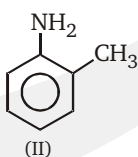
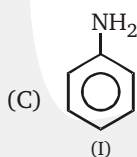
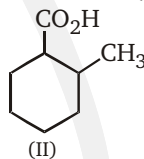
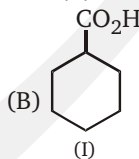
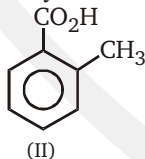
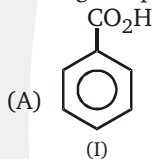


HCl
1 mole

Product



16. In the given pair identify most acidic compound in (A) and (B). Most basic in (C) and (D).



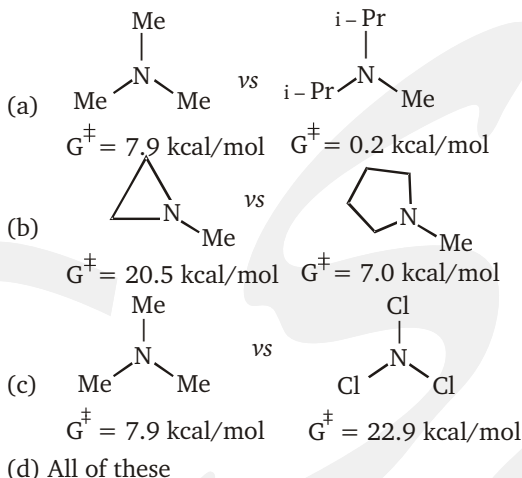
(a) A - I, B - II, C - I, D - II

(c) A - II, B - II, C - II, D - II

(b) A - II, B - I, C - I, D - II

(d) A - I, B - II, C - I, D - I

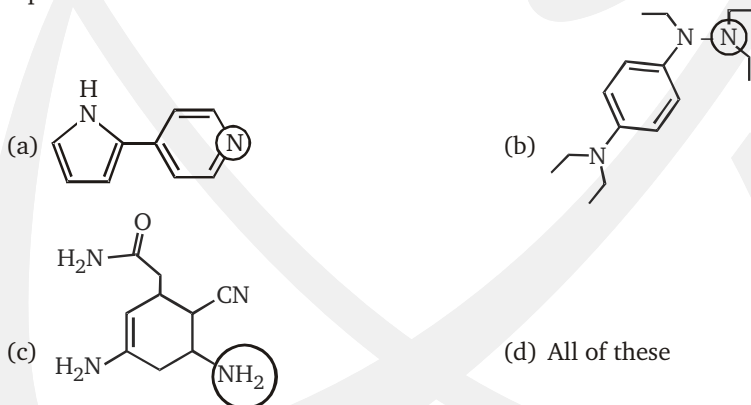
17. Several factors (steric, electronic, orbital interactions etc.) can affect the inversion barrier of an amine. In the given pair which data is correctly placed ?



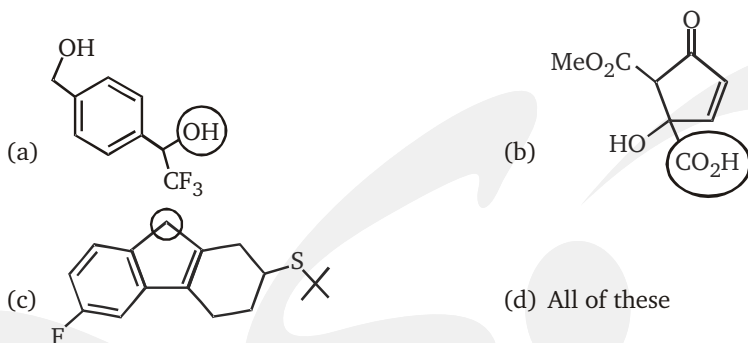
18. Select the response that correctly identifies the number of carbon atoms of each type of hybridization in the compound given below

	H_2C	C	$\text{CH} - \text{CH}$	O			
	sp^3	sp^2	sp		sp^3	sp^2	sp
(a)	2	2	0	(b)	1	3	0
(c)	0	3	1	(d)	1	2	1

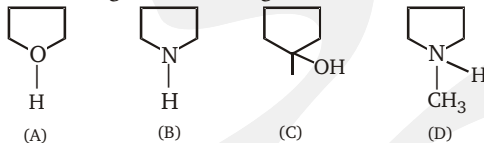
19. Circle represents most basic atoms in these molecule. Which of the following is correct representation ?



20. Circle represent most acidic hydrogens in these molecules. Which of the following is correct representation ?

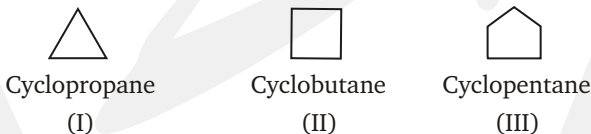


21. Arrange the following in decreasing order of their acidic strengths.



- (a) $A > C > B > D$ (b) $A > D > B > C$ (c) $A > D > C > B$ (d) $D > A > C > B$

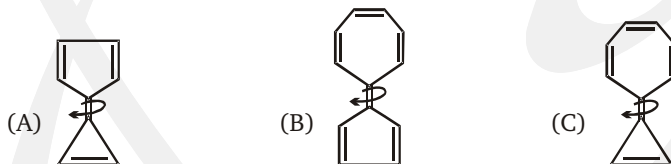
22.



The correct order of heats of combustion of above compounds is :

- (a) $I > II > III$ (b) $II > I > III$ (c) $III > II > I$ (d) $III > I > II$

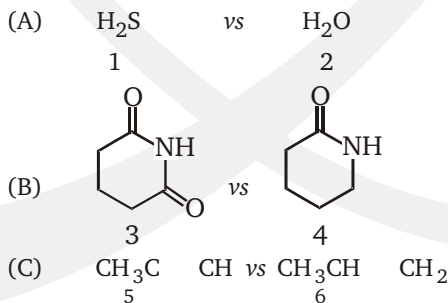
23.



Compare carbon-carbon bond rotation across A, B, and C

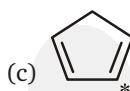
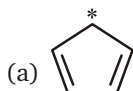
- (a) $A > B > C$ (b) $A > C > B$ (c) $B > A > C$ (d) $B > C > A$

24. Which of the following acids would have a STRONGER CONJUGATE BASE ?



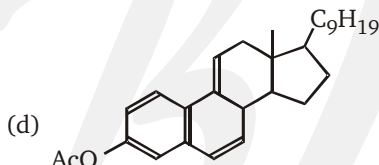
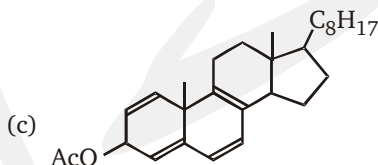
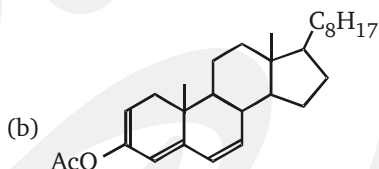
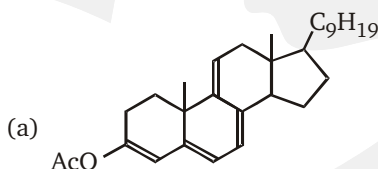
- (a) 2, 4, 6 (b) 1, 3, 5 (c) 2, 3, 5 (d) 1, 3, 6

25.  $\xrightarrow[\text{H}_2\text{O}]{\text{KH}}$ A ; Major products of the reaction is (are):
 (C C^{14}) Major Product

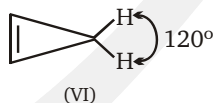
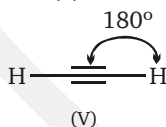
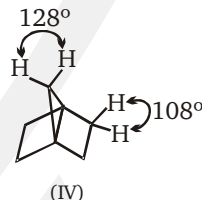
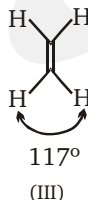
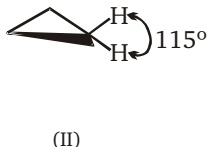
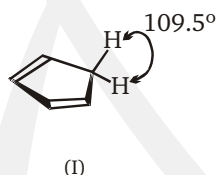


(d) both (b) & (c)

26. Which of the following compound is most stable ?

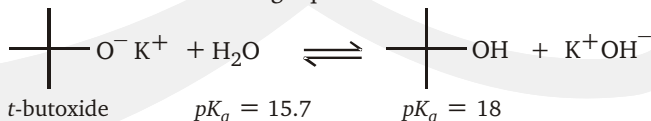


27. Selected bond angles for six hydrocarbons are shown below. Arrange these hydrocarbons according to their pK_a values, from the lowest to the highest.



- (a) $\text{V} < \text{I} < \text{VI} < \text{II} < \text{III} < \text{IV}$
 (c) $\text{II} < \text{IV} < \text{I} < \text{VI} < \text{V} < \text{III}$
 (b) $\text{IV} < \text{I} < \text{II} < \text{III} < \text{V} < \text{VI}$
 (d) $\text{I} < \text{V} < \text{IV} < \text{III} < \text{II} < \text{VI}$

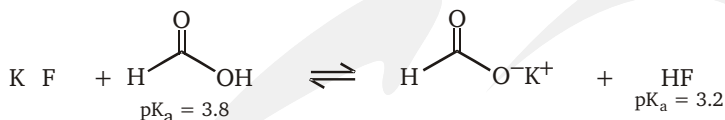
28. Which statement about the following equilibrium is true ?



- (a) The equilibrium favours the products
 (b) *t*-Butoxide is the dominant anionic species in the equilibrium

- (c) Water is the weaker acid
 (d) *t*-Butoxide is stabilized by resonance

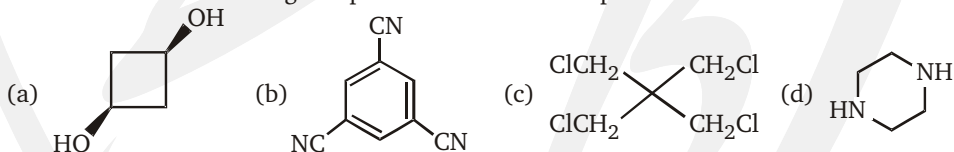
29. Consider the following reaction involving two acids shown below : formic acid and HF.



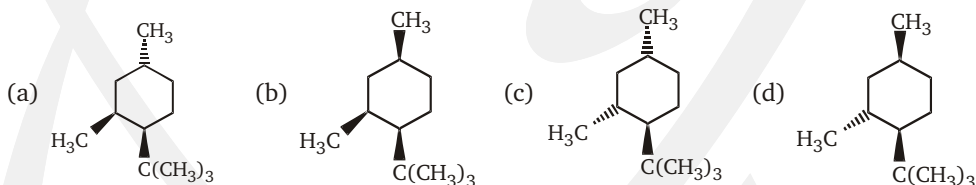
Which of the following statements about this reaction are true ?

- (A) Formic acid is the strongest Bronsted acid in the reaction
 (B) HF is the strongest Bronsted acid in the reaction
 (C) KF is the strongest Bronsted base in the reaction
 (D) KO_2CH is the strongest Bronsted base in the reaction
 (E) The equilibrium favours the reactants
 (F) The equilibrium favours the products
 (G) Formic acid has a weaker conjugate base
 (H) HF has a weaker conjugate base
- (a) A, D and F (b) B, D, and H (c) A, C, and H (d) B, D, E and H

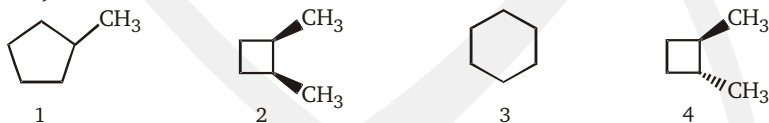
30. Which one of the following compounds has non zero dipole moment?



31. Which one of the following has the smallest heat of combustion ?



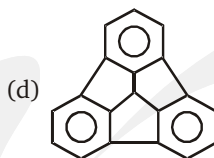
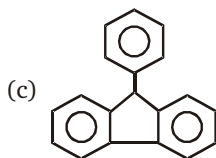
32. Rank the following substances in order of decreasing heat of combustion (maximum minimum).



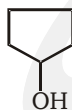
- (a) $1 > 2 > 4 > 3$ (b) $3 > 4 > 2 > 1$
 (c) $2 > 4 > 1 > 3$ (d) $1 > 3 > 2 > 4$

33. Which of the following has lowest pK_a value ?





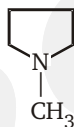
34. Arrange the following (w, x, y, z) in decreasing order of their boiling points:



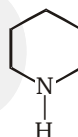
(w)



(x)



(y)



(z)

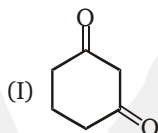
(a) $w > x > z > y$

(b) $w > x > y > z$

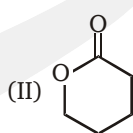
(c) $w > z > y > x$

(d) $w > z > x > y$

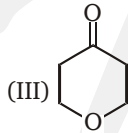
35. Arrange the following in increasing order of their acidic strength.



(a) $\text{III} < \text{I} < \text{IV} < \text{II}$

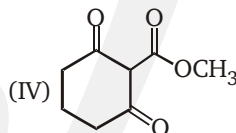


(c) $\text{I} < \text{III} < \text{IV} < \text{II}$

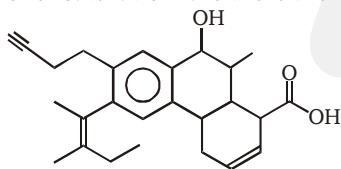


(b) $\text{II} < \text{I} < \text{IV} < \text{III}$

(d) $\text{II} < \text{III} < \text{I} < \text{IV}$



36. How many degrees of unsaturation are there the following compound ?



(a) 6

(b) 7

(c) 10

(d) 11

37. The heat of hydrogenation for 3-methylbutene and 2-pentene are -30 kcal/mol and -28 kcal/mol respectively. The heats of combustion of 2-methylbutane and pentane are -784 kcal/mol and -782 kcal/mol respectively. All the values are given under standard conditions. Taking into account that combustion of both alkanes give the same products, what is H (in kcal/mol) for the following reaction under same conditions ?



(a) 0

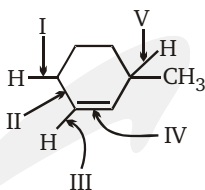
(b) -4

(c) -2

(d) 2

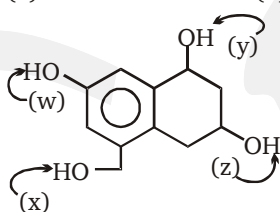


38. Which of the following σ -bonds participate in hyperconjugation ?



- (a) I and II (b) I and V (c) II and V (d) III and IV

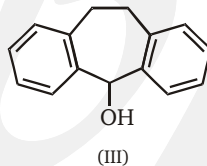
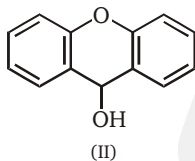
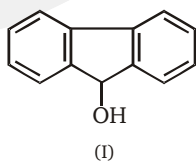
39.



Decreasing order of acidic strength of different ($-\text{OH}$) groups is :

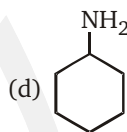
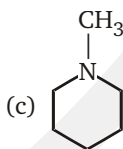
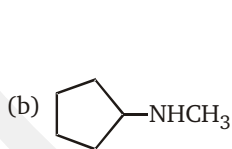
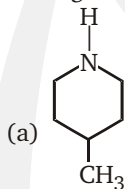
- (a) $w > x > y > z$ (b) $w > z > x > y$
(c) $z > w > x > y$ (d) $z > x > w > y$

40. Arrange the following alcohols in decreasing order of the ease of ionization under acidic conditions.

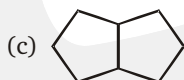
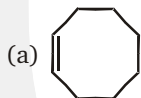


- (a) $\text{I} > \text{III} > \text{II}$ (b) $\text{I} > \text{II} > \text{III}$ (c) $\text{II} > \text{III} > \text{I}$ (d) $\text{II} > \text{I} > \text{III}$

41. Among the isomeric amines select the one with the lowest boiling point.



42. Which one of the compounds shown below, is not an isomer of the others ?



43. Arrange the anions (p) CH_3^- , (q) NH_2^- , (r) OH^- , (s) F^- , in decreasing order of their basic strength.

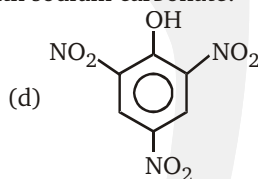
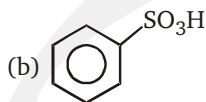
(a) $p > q > r > s$

(b) $q > p > r > s$

(c) $r > q > p > s$

(d) $r > p > q > s$

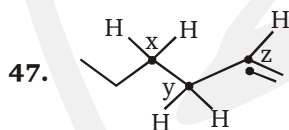
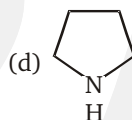
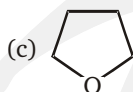
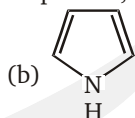
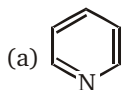
44. One among the following compounds will not give effervescence with sodium carbonate:



45. The carboxylic acid which has maximum solubility in water is:



46. Among the following compounds, the most basic compound is :



Arrange the (C - H) bonds x, y and z in decreasing order of their bond dissociation energies in homolysis.

(a) $y > x > z$

(b) $z > x > y$

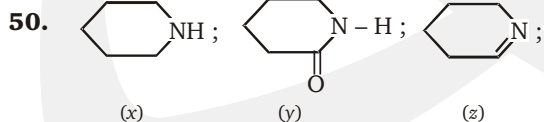
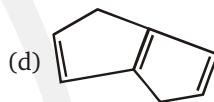
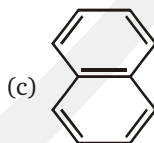
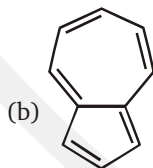
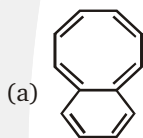
(c) $z > y > x$

(d) $y > z > x$

48. 23 g of sodium will react with methyl alcohol to give :



49. Which of the following is most polar?



The correct order of decreasing basic strengths of x, y and z is :

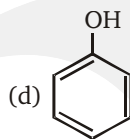
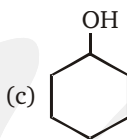
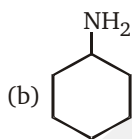
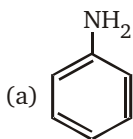
(a) $x > y > z$

(b) $x > z > y$

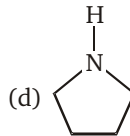
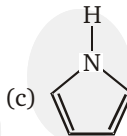
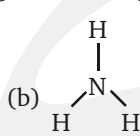
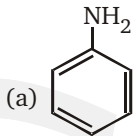
(c) $y > x > z$

(d) $y > z > x$

51. Which of the following is the strongest Bronsted acid ?



52. Which of the following is the strongest Bronsted base ?



53. Which of the following is polar aprotic solvent ?

(a) DMSO

(b) Crown ether

(c) DMG

(d) All of these

54. Some pairs of acids are given below. Select the pair in which second acid is stronger than first

(a) $\text{CH}_3\text{CO}_2\text{H}$ and $\text{CH}_2\text{FCO}_2\text{H}$

(b) $\text{CH}_2\text{FCO}_2\text{H}$ and $\text{CH}_2\text{ClCO}_2\text{H}$

(c) $\text{CH}_2\text{ClCO}_2\text{H}$ and $\text{CH}_2\text{BrCO}_2\text{H}$

(d) $\text{CH}_3\text{CH}_2\text{CHFCO}_2\text{H}$ and $\text{CH}_3\text{CHFCH}_2\text{CO}_2\text{H}$

55. $\text{H}-\text{C} \quad \text{C} \xrightarrow{a} \text{C} \quad \text{C} \xrightarrow{b} \text{CH}_3$;

Compare the bond lengths a and b :

(a) $a = b$

(b) $a > b$

(c) $b > a$

(d) $a \gg b$

56. Which (isomeric) amine has lowest boiling point ?

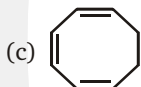
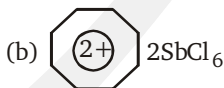
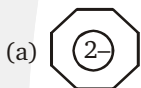
(a) 1° amine

(b) 2° amine

(c) 3° amine

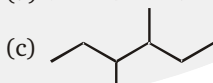
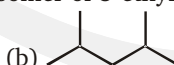
(d) cannot predict

57. 2SbCl_5 P ; P will be :



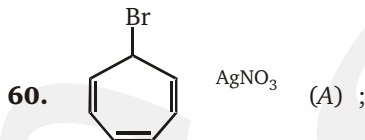
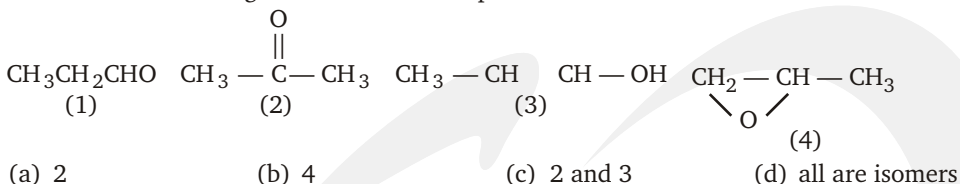
(d) mixture of (a) and (b)

58. Which of the following substances is not an isomer of 3-ethyl 2-methyl pentane ?



(d) All are isomers

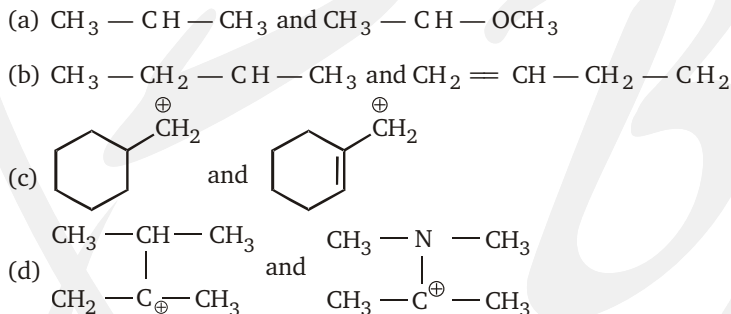
59. Which of the following is an isomer of compound 1 ?



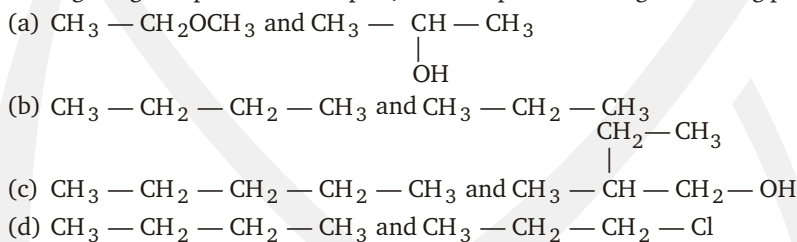
Which statement is incorrect in respect of the above reaction ?

- (a) Product is aromatic (b) Product has high dipole moment
(c) Product has less resonance energy (d) Product is soluble in polar solvent

61. Some pairs of ions are given below. In which pair, first ion is more stable than second?



62. Among the given pairs in which pair, first compound has higher boiling point than second ?



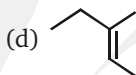
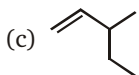
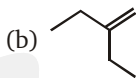
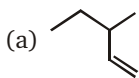
63. Which of the following alcohols is the least soluble in water ?

- (a) Ethanol (b) 1-Propanol
(c) 1-Butanol (d) 1-Pentanol

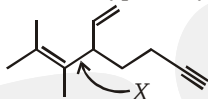
64. Which of the following alcohols is expected to have a lowest pK_a value ?

- (a) Ethanol (b) 1-propanol
(c) 2, 2, 2-trifluorethanol (d) 2-chloroethanol

65. Which of the following alkenes is the most stable ?



66. Bond X is made by the overlap of which type of hybridized orbitals ?



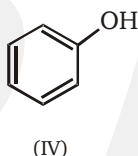
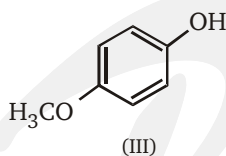
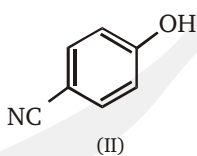
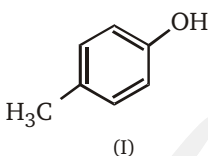
(a) sp and sp^3

(b) sp and sp^2

(c) sp^2 and sp^3

(d) none of these

67. Increasing order of acidic strength of given compounds is :



(a) $\text{III} < \text{I} < \text{IV} < \text{II}$

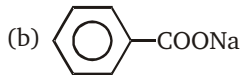
(b) $\text{II} < \text{I} < \text{IV} < \text{III}$

(c) $\text{I} < \text{III} < \text{IV} < \text{II}$

(d) $\text{I} < \text{III} < \text{II} < \text{IV}$

68. $\text{CO}_2 + \text{C}_6\text{H}_5\text{COONa}$ is with the product :

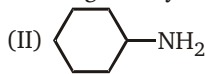
(a) CO_2



(c) both

(d) none of these

69. Rank in the order of increasing acidity.



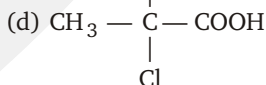
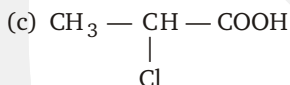
(a) $\text{III} < \text{I} < \text{II}$

(b) $\text{I} < \text{III} < \text{II}$

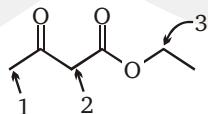
(c) $\text{III} < \text{II} < \text{I}$

(d) $\text{II} < \text{I} < \text{III}$

70. Which compound has the highest value of pK_a ?



71. Consider the hydrogen atoms attached to three different carbon atoms (labeled 1, 2 & 3). Rank the attached hydrogen atoms in order from most acidic to least acidic.



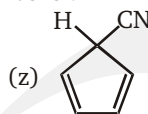
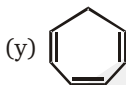
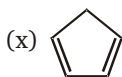
(a) $2 > 1 > 3$

(b) $1 > 2 > 3$

(c) $2 > 3 > 1$

(d) $3 > 2 > 1$

72. Decreasing order of acidic strengths of following compounds is :



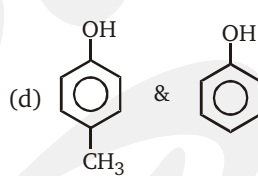
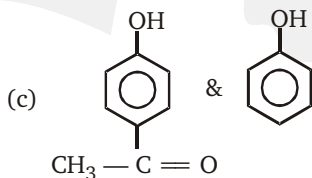
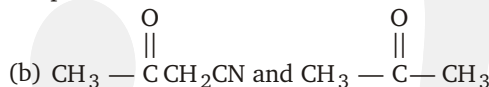
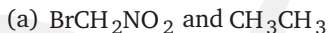
(a) $x > y > z$

(b) $y > x > z$

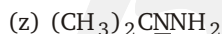
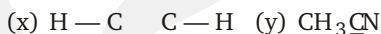
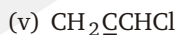
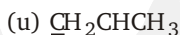
(c) $z > y > x$

(d) $z > x > y$

73. Among the given pairs, in which pair second compound is more acidic than first ?



74. Which of the underlined atoms in the molecules shown below have sp -hybridization ?



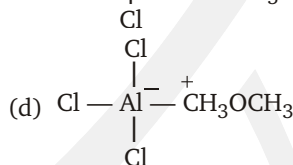
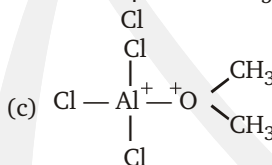
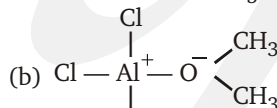
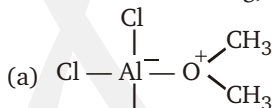
(a) x and z

(b) x, y, and z

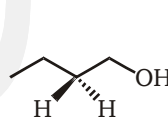
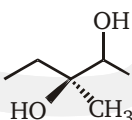
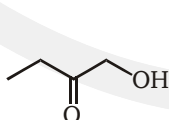
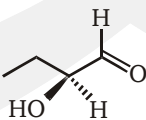
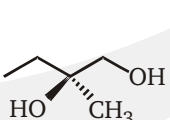
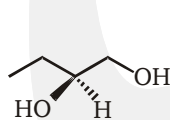
(c) u, w and x

(d) v, x and y

75. Which of the following, is the product of the reaction between AlCl_3 and CH_3OCH_3 ?



76. Which of the following compounds contain at least one secondary alcohol ?



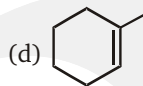
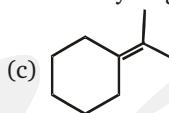
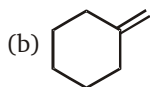
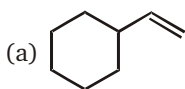
(a) I, II, IV, VI

(b) I, III

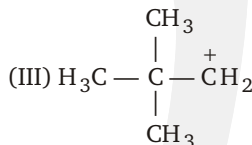
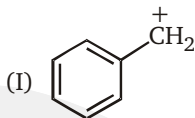
(c) I, II, III, V

(d) I, III, V

77. Which of the following has the most negative heat of hydrogenation ?



78. Which of the following options is the correct order of relative stabilities of cations I, II and III as written below (most stable first) ?



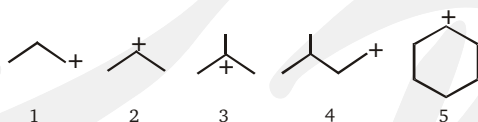
(a) $\text{I} > \text{II} > \text{III}$

(b) $\text{II} > \text{III} > \text{I}$

(c) $\text{III} > \text{I} > \text{II}$

(d) $\text{I} > \text{III} > \text{II}$

79. What is the decreasing order of stability (most stable to least stable) of the following carbocations ?



(a) $3 > 2 > 1 > 4 > 5$

(b) $3 > 2 > 5 > 4 > 1$

(c) $1 > 4 > 2 > 5 > 3$

(d) $3 > 1 > 4 > 2 > 5$

80.



the hydrogen indicated by arrow will be easily removed as :

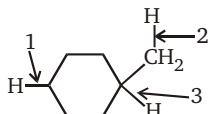
(a) H

(b) H

(c) H

(d) H^2

81. Rank the bond dissociation energies of the bonds indicated with the arrows. (from smallest to largest).



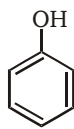
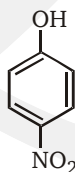
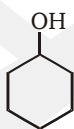
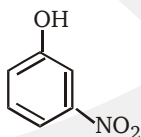
(a) $1 < 2 < 3$

(b) $3 < 2 < 1$

(c) $2 < 3 < 1$

(d) $3 < 1 < 2$

82. Rank the following compounds in order of decreasing acid strength (most acidic to least acidic) .



1

2

3

4

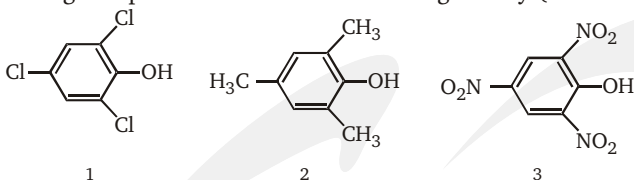
(a) $2 > 4 > 1 > 3$

(b) $1 > 3 > 4 > 2$

(c) $3 > 1 > 2 > 4$

(d) $3 > 1 > 4 > 2$

83. Rank the following compounds in order of increasing acidity (weakest acid first).

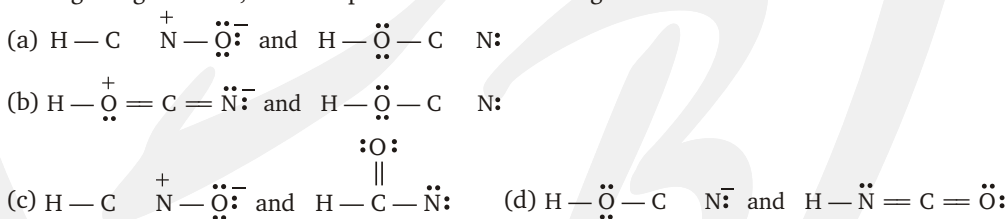


- (a) $2 < 3 < 1$ (b) $3 < 1 < 2$ (c) $1 < 2 < 3$ (d) $2 < 1 < 3$

84. Which of the following phenols has the largest pK_a value (i.e., is least acidic) ?



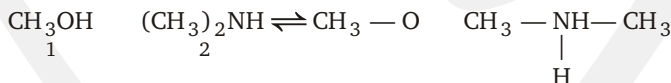
85. Among the given sets, which represents the resonating structures ?



86. Identify each species in the following equilibrium according to the code :

SA = stronger acid ; SB = stronger base ; WA = weaker acid ; WB = weaker base.

The pK_a of $(\text{CH}_3)_2\text{NH}$ is 36 ; the pK_a of CH_3OH is 15.2.



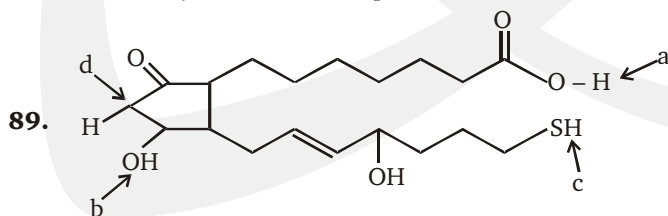
- (a) WA WB (b) WB WA (c) SA SB (d) SB SA
- (e) WA WA

87. The hydrogen bonding is strongest in which one of the following set ?

- (a) $\text{F}-\text{H} \cdots \text{F}$ (b) $\text{O}-\text{H} \cdots \text{S}$ (c) $\text{S}-\text{H} \cdots \text{F}$ (d) $\text{F}-\text{H} \cdots \text{O}$

88. Intermolecular hydrogen bonding is strongest in :

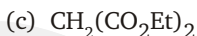
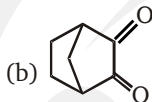
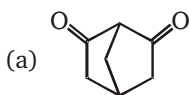
- (a) methylamine (b) phenol (c) formaldehyde (d) methanol



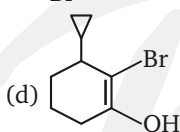
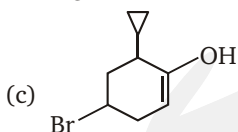
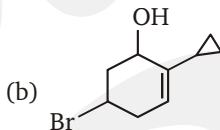
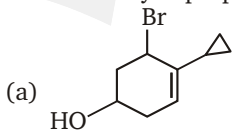
Identify most acidic hydrogen in given compound.

- (a) *a* (b) *b* (c) *c* (d) *d*

90. Which of the following compounds would you expect to be strongest carbon acid ?



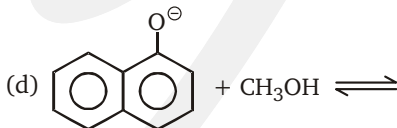
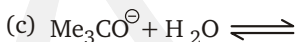
91. 5-Bromo-2-cyclopropyl cyclohex-2-enol have correct structure is:



92. Rearrange the following in the increasing order of acidic strength.

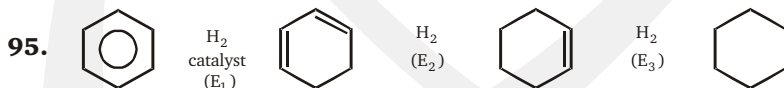
- (i) benzoic acid (ii) *p*-methoxybenzoic acid (iii) *o*-methoxybenzoic acid
 (a) $i < ii < iii$ (b) $iii < i < ii$ (c) $ii < i < iii$ (d) $iii < ii < i$

93. In the following acid-base reaction, in which can backward reaction if favoured?



94. Which compound posses highest dipole moment ?

- (a) naphthalene (b) phenanthrene
 (c) anthracene (d) azulene



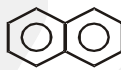
(E = activation energy)

Relation between activation energies of above reactions is :

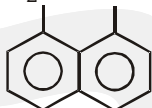
- (a) $E_2 > E_1 > E_3$ (b) $E_3 > E_1 > E_2$
 (c) $E_3 > E_2 > E_1$ (d) $E_1 > E_2 > E_3$

96. Rank the following solvents in decreasing order of ability to dissolve given compound.

Solvent
Et ₂ O
H ₂ O
EtOH



Solute

(a) Et₂O H₂O EtOH(c) H₂O Et₂O EtOH(b) H₂O EtOH Et₂O(d) Et₂O EtOH H₂OMe₂N: :NMe₂

1, 8-Bis (dimethylamino) naphthalene is after referred so as (Proton sponge)

97.

Its basic strength is 10¹⁰ more than 1-dimethyl amino naphthalene.

Reason for high basic strength is :

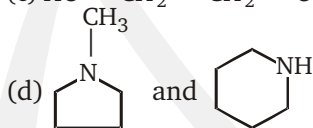
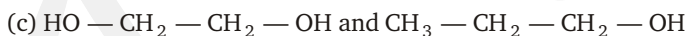
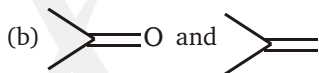
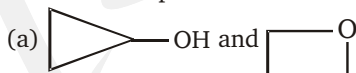
(a) resonance

(b) steric inhibition of resonance

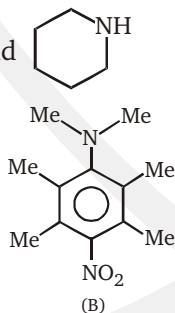
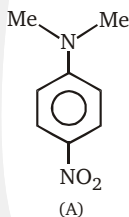
(c) ortho effect

(d) hyperconjugation

98. In the given pair of compounds, in which pair second compound has higher boiling point than first compound ?



99.



; Dipole moments of given compound will be :

(a) (A) = 6.87D, (B) = 4.11D

(b) (A) = 4.11 D , (B) = 6.87 D

(c) (A) = 4.11 D , (B) = 4.11 D

(d) (A) = 6.87 D, (B) = 6.87 D

100. Order of decreasing basic strengths of halides is :

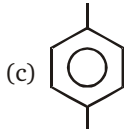
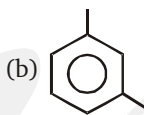
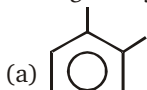
(a) F Cl I Br

(b) F Cl Br I

(c) I Br Cl F

(d) I Cl Br F

101. Among the xylenes, which is thermodynamically most stable ?

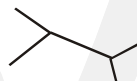
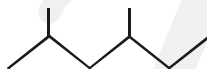
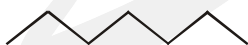


(d) All are equally stable

102. Heat of combustion of two isomer x and y are 17 kJ/mol and 12 kJ/mol respectively. From this information it may be concluded that :

- (a) isomer x is 5 kJ/mol more stable
 (b) isomer y is 5 kJ/mol less stable
 (c) isomer y has 5 kJ/mol more potential energy
 (d) isomer x is 5 kJ/mol less stable

103. Rank the following substances in decreasing order of heat of combustion (most exothermic least exothermic)



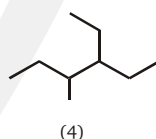
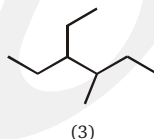
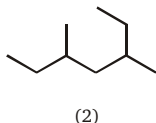
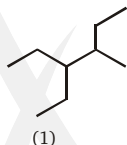
(a) $B > A > C$

(b) $A > B > C$

(c) $C > A > B$

(d) $C > B > A$

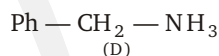
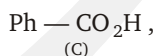
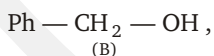
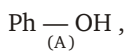
104.



Choose the statement that best describes given compounds.

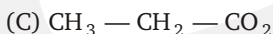
- (a) 1, 3, 4 represent same compound (b) 1 and 3 are isomer of 2 and 4
 (c) 1, 4 are isomer of 2 and 3
 (d) All the structure represent the same compound

105. Decreasing order of acid strengths is :



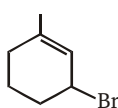
- (a) $B > A > C > D$ (b) $C > A > B > D$
 (c) $C > A > D > B$ (d) $C > B > A > D$

106. Rank the following in decreasing order of basic strength is :

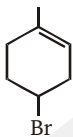


- (a) $B > A > D > C$ (b) $D > A > B > C$ (c) $A > D > B > C$ (d) $A > D > C > B$

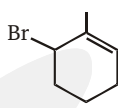
107. Among the given compound choose the two that yield same carbocation on ionization.



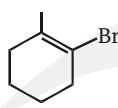
(A)



(B)



(C)



(D)

(a) A, C

(b) B, D

(c) A, B

(d) B, C

108. Oxalic acid

 pK_1

Malonic acid

 pK_2

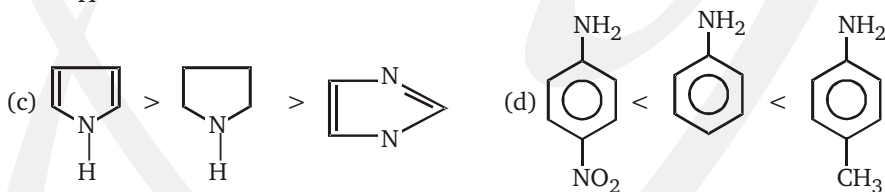
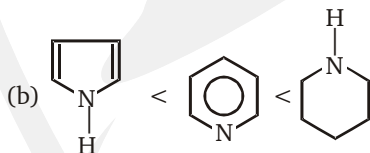
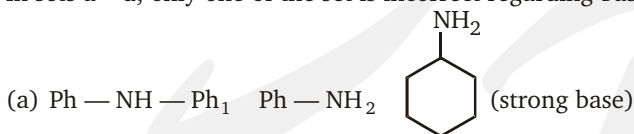
Heptanedioic acid

 pK_3

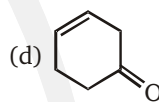
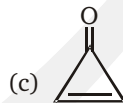
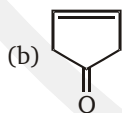
where pK_1, pK_2, pK_3 are first ionization constants. Correct order is :

(a) $pK_1 \ pK_2 \ pK_3$ (b) $pK_1 \ pK_2 \ pK_3$ (c) $pK_3 \ pK_2 \ pK_1$ (d) $pK_3 \ pK_1 \ pK_2$

109. In sets a – d, only one of the set is incorrect regarding basic strength. Select it :



110. Dipole moment of which ketone is maximum ?



111. Correct order of basic strengths of given amines is :

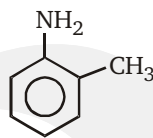
(a) $\text{Me}_2\text{NH} \quad \text{MeNH}_2 \quad \text{Me}_3\text{N} \quad \text{NH}_3$ (Protic solvent)

(b) $\text{Et}_2\text{NH} \quad \text{Et}_3\text{H} \quad \text{EtNH}_2 \quad \text{NH}_3$ (Protic solvent)

(c) $\text{Me}_3\text{N} \quad \text{Me}_2\text{NH} \quad \text{Me} \quad \text{NH}_2 \quad \text{NH}_3$ (Gas phase)

(d) All are correct

112. Order of basic strength $\text{Ph}-\text{NH}_2$, $\text{Ph}-\text{NH}-\text{Me}$, $\text{Ph}-\text{N}(\text{Me})_2$
- (A) (B) (C) (D)

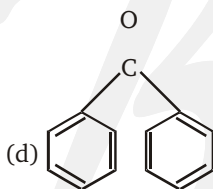
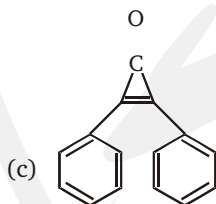
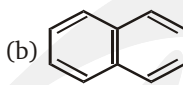
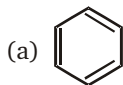


- (a) $A > B > C > D$ (b) $B > A > C > D$ (c) $C > B > A > D$ (d) $C > B > D > A$

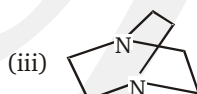
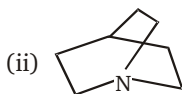
113. Carbon-carbon double bond length will be maximum in which of the following compounds ?

- (a) $\text{CH}_3-\text{CH}=\text{CH}_2$ (b) $\text{CH}_3-\text{CH}=\text{CH}-\text{CH}_3$
 (c) $\text{CH}_3-\text{C}(\text{CH}_3)=\text{C}(\text{CH}_3)-\text{CH}_3$ (d) $\text{CH}_2=\text{CH}_2$

114. Which has maximum dipole moment ?



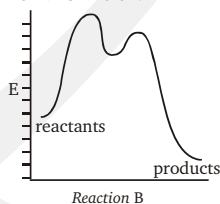
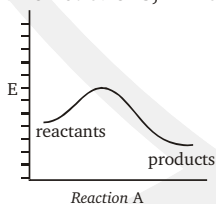
115. (i) Et_3N



Compare the basic strengths of compounds given:

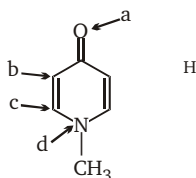
- (a) (i) > (ii) > (iii) (b) (ii) > (i) > (iii)
 (c) (ii) > (iii) > (i) (d) (iii) > (ii) > (i)

116. For the following two reactions, which statement is true ?

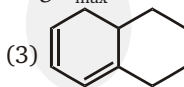
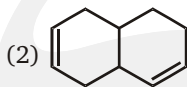
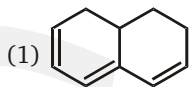


- (a) Reaction A is faster and less exergonic than B
 (b) Reaction B is faster and more exergonic than A
 (c) Reaction A is faster and less endergonic than B
 (d) Reaction B is faster and more endergonic than A

117.

Identify the site, where attack of H⁺ is most favourable.

- (a) a (b) b (c) c (d) d

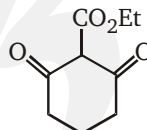
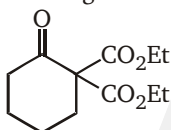
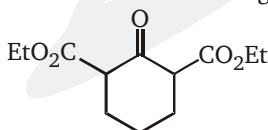
118. Rank the following alkenes on order of increasing λ_{max} .

- (a) $1 < 2 < 3$ (b) $1 < 3 < 2$ (c) $2 < 1 < 3$ (d) $2 < 3 < 1$

119. Which of the following cyclic amine has lowest $G^\ddagger$ for inversion?

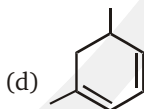
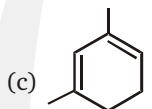
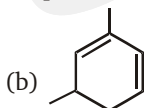
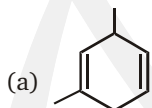
- (a) (b) (c) (d)

120. Rank in the order of increasing acidic strength:



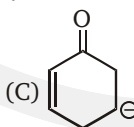
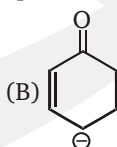
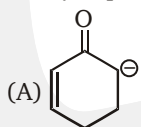
- (a) $A < B < C$ (b) $A < C < B$ (c) $B < A < C$ (d) $B < C < A$

121. Which one of the following dienes would you expect to be the most stable?

122. Which metal catalyzed reaction would release the maximum amount of heat per CH_2 unit?

- (a) cyclopropane H_2 propane (b) cyclobutane H_2 butane
(c) cyclopentane H_2 pentane (d) cyclohexane H_2 hexane

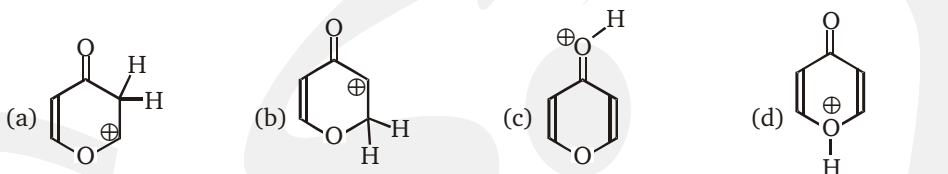
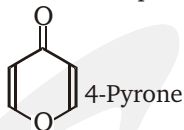
123.



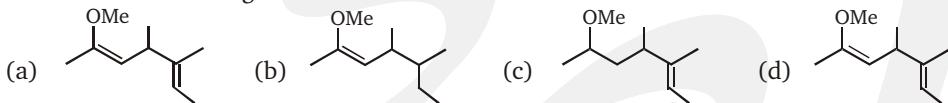
Compare basic strengths of the above compounds:

- (a) $A > B > C$ (b) $B > A > C$ (c) $C > A > B$ (d) $C > B > A$

- 124.** On reaction with acid, 4-pyrone gives a very stable cationic product. Which of the following structures shows the protonation site in that product ?



- 125.** Which of the following is the most stabilized carbocation ?



- 126.** Which carbocation is the most stable ?



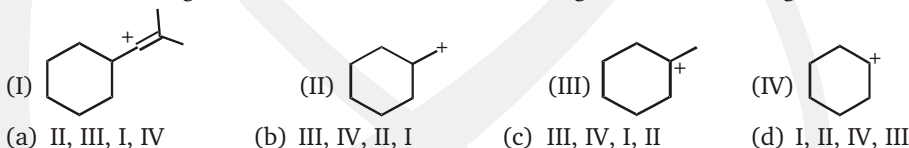
- 127.** Consider a positively charged C_2H_3 species in which the positively charged carbon is sp -hybridized, the uncharged carbon is sp^2 -hybridized and an empty p -orbital is perpendicular to the system. What is the best description of this cation ?

(a) vinyl (b) allenyl (c) alkyl (d) allyl

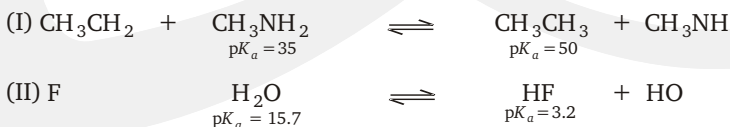
- 128.** Which of the following reactions is not exothermic ?



- 129.** List the following carbocations in order of decreasing stabilization energies.

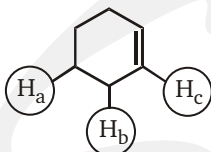


- 130.** For the following two acid-base reactions, which statement is true ?



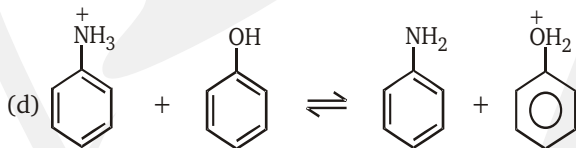
- (a) I is favoured to the right, II is favoured to the left
 (b) I is favoured to the left, II is favoured to the right
 (c) I is favoured to the right, II is favoured to the right
 (d) I is favoured to the left, II is favoured to the left

131. Rank the hydrogen atoms (H_a , H_b , H_c) in the following molecules according to their acidic strengths:

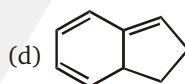
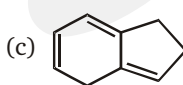
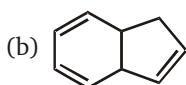
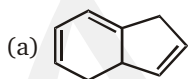


- (a) $a > b > c$ (b) $b > a > c$ (c) $b > c > a$ (d) $a > c > b$

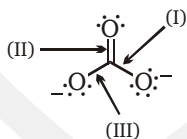
132. In which of the following reactions, backward reaction is favoured ?



133. Which carbocation is the most stabilized ?

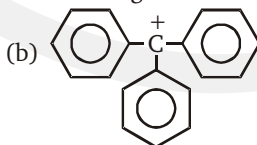
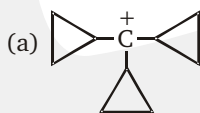


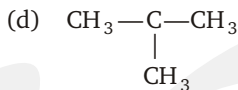
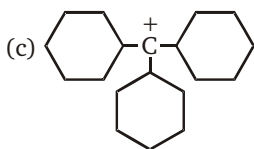
134. Taking into account of hybridization and resonance effects, rank the following bonds in order of decreasing bond length.



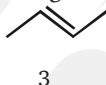
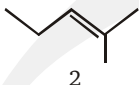
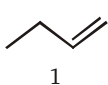
- (a) $I > II = III$ (b) $II > III > I$ (c) $I > III > II$ (d) $II = III = I$

135. Which one among the following carbocations has the longest half-life ?





136. Rank the following alkenes in order of decreasing heats of hydrogenation (largest first)



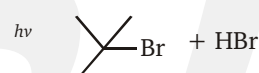
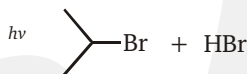
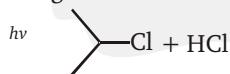
(a) $2 > 3 > 4 > 1$

(b) $2 > 4 > 3 > 1$

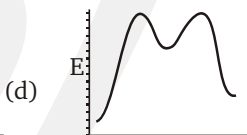
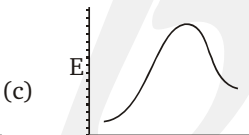
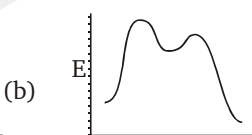
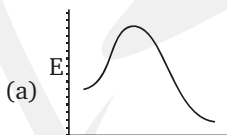
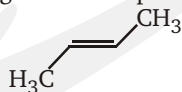
(c) $1 > 3 > 4 > 2$

(d) $1 > 4 > 3 > 2$

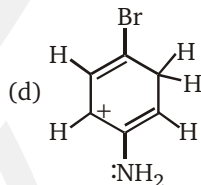
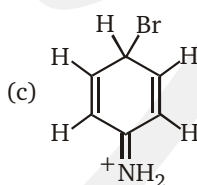
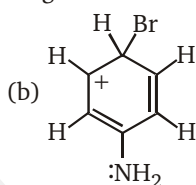
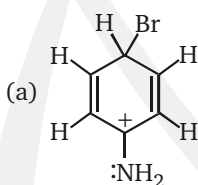
137. Which of the following reactions is most exothermic ?



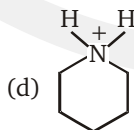
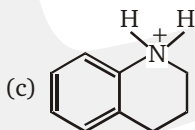
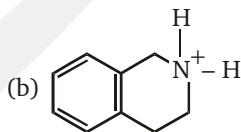
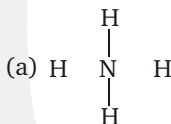
138. Which energy diagram best represents the given reaction ?



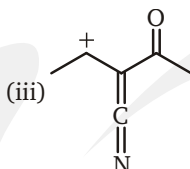
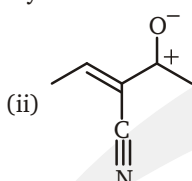
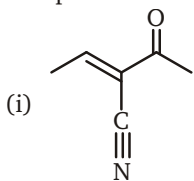
139. Which one of the following is most stable ?



140. Which of the following is strongest acid ?

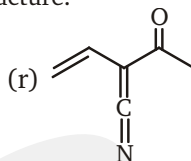
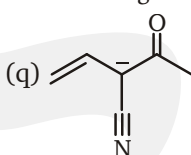
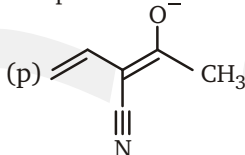


141. Compare relative stability of the following resonating structure.



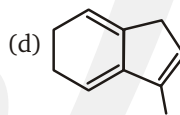
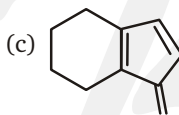
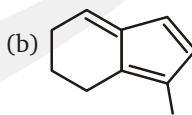
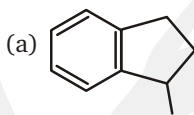
- (a) (i) > (ii) > (iii) (b) (ii) > (i) > (iii) (c) (i) > (iii) > (ii) (d) (ii) > (iii) > (i)

142. Compare relative stability of the following resonating structure.

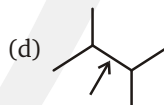
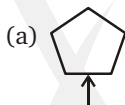


- (a) $p > q > r$ (b) $q > p > r$ (c) $q > r > p$ (d) $p > r > q$

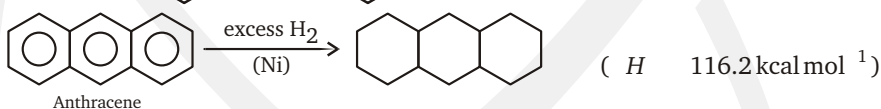
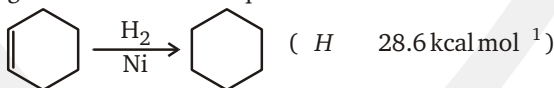
143. Which of the following isomeric hydrocarbons is most acidic ?



144. Which of the following has the lowest barrier to rotation about the indicated bond ?



145. Use the following data to answer the question below.



Anthracene

Calculate the resonance energy of anthracene:

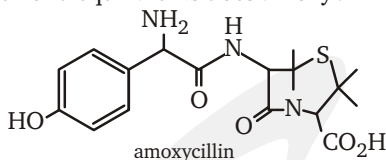
- (a) 84 kcal/mol (b) 100 kcal/mol (c) 110 kcal/mol (d) 116 kcal/mol

146. How many double bond equivalents does a compound of molecular formula $C_6H_{12}O_6$ possess?

- (a) 0 (b) 1
(c) 2 (d) 3

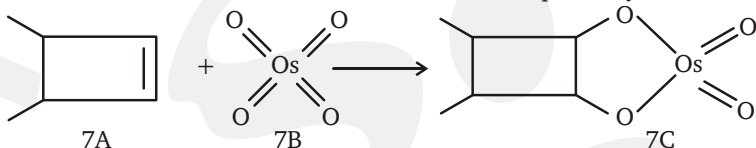


147. How many double bond equivalents does amoxicillin (shown below) possess ?



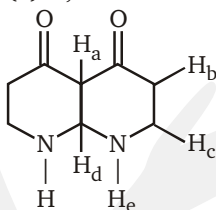
- (a) 5 (b) 6 (c) 7 (d) 9

148. What is the oxidation state of osmium in 7B and 7C, respectively ?



- (a) 6, 8 (b) 8, 6 (c) 6, 6 (d) 8, 8

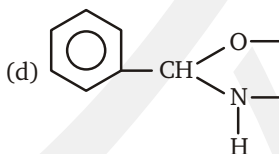
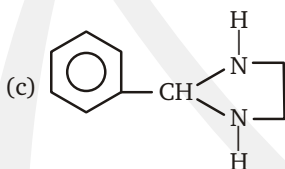
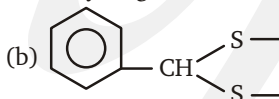
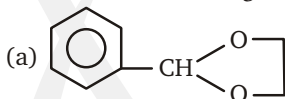
149.



Identify most acidic hydrogen present in the above compound:

- (a) a (b) b (c) c (d) d

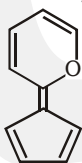
150. Which of the following compounds has most acidic hydrogen ?



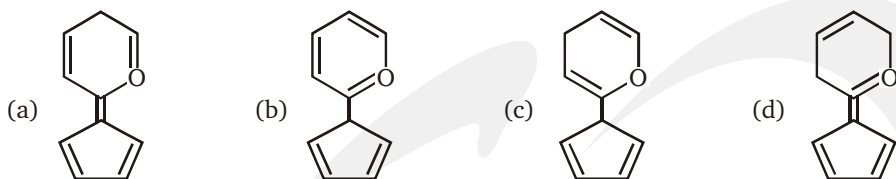
151. Acetic acid, (CH_3COOH), has a $\text{p}K_a$ of 4.8. Ethanol, ($\text{CH}_3\text{CH}_2\text{OH}$), has a $\text{p}K_a$ of 16.0. What are the major species present, when acetic acid and ethanol are added to water and the pH is adjusted to 7.0 ?

- (a) $\text{CH}_3\text{CO}_2\text{H}$ and $\text{CH}_3\text{CO}_2\text{H}$ (b) $\text{CH}_3\text{CH}_2\text{O}^-$ and $\text{CH}_3\text{CH}_2\text{OH}$
 (c) $\text{CH}_3\text{CO}_2\text{H}$ and $\text{CH}_3\text{CH}_2\text{O}^-$ (d) CH_3CO_2^- and $\text{CH}_3\text{CH}_2\text{OH}$

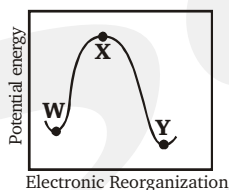
152.



The most stable canonical structure of given molecule is:



153. In the potential energy diagram to the right, the point X represents :

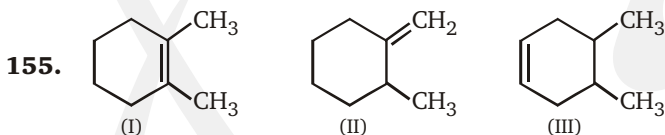


- (a) a transition state (b) a reaction intermediate
(c) a resonance hybrid (d) a reactant



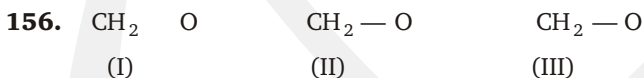
Which of the following orders is correct for heat of hydrogenation of these compounds ?

- (a) I > III > II (b) III > II > I (c) III > I > II (d) II > I > III



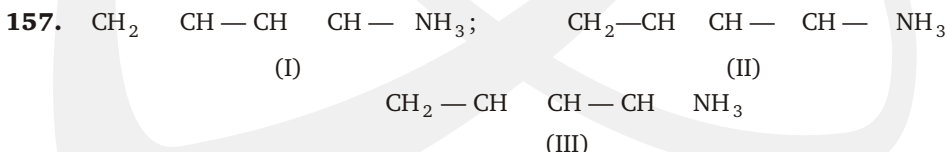
Which of the following orders is correct for heat of hydrogenation of these compounds ?

- (a) I > II > III (b) III > II > I (c) II > III > I (d) III > I > II



Which of these structures is practically not a valid canonical structure for formaldehyde ?

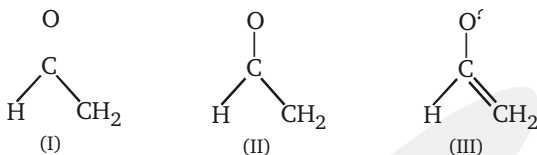
- (a) I (b) II (c) III (d) None of these



Which of these structures is not a valid canonical structure ?

- (a) I (b) II (c) III (d) none of these

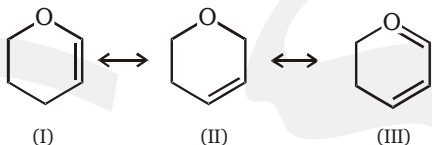
158.



The correct order of stability for the given canonical structures is :

- (a) I > III > II (b) III > I > II (c) II > III > I (d) II > I > III

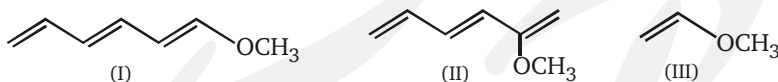
159.



The most stable canonical structure among the given structure is :

- (a) I (b) II (c) III (d) all are equally stable

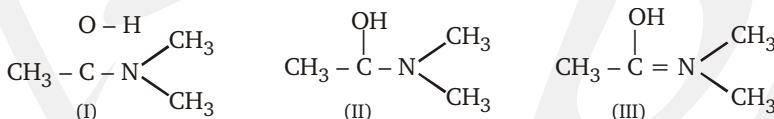
160.



For the given compounds the correct order of resonance energy is :

- (a) III > I > II (b) II > I > III (c) I > II > III (d) III > II > I

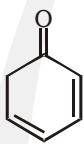
161.



The correct stability order of the given canonical structures is :

- (a) I > II > III (b) III > I > II (c) I > III > II (d) II > III > I

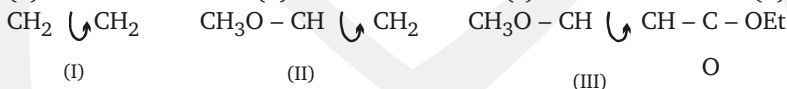
162.



In the above compound, how many sites are available for the attack of CH_3O^- ?

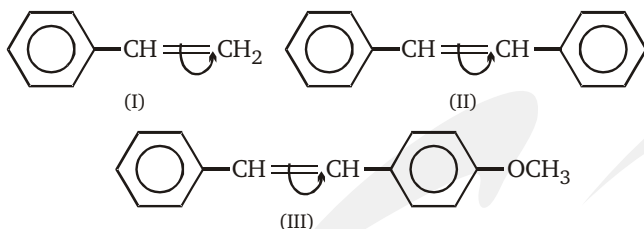
- (a) 1 (b) 2 (c) 3 (d) 4

163.



Which of the following orders of rotation barrier about the $\text{C} = \text{C}$ bond, as indicated, is correct ?

- (a) I > II > III (b) III > II > I (c) III > I > II (d) II > I > III



(a) $I > II > III$ (b) $III > II > I$ (c) $III > I > II$ (d) $II > I > III$

(a)  (b)  (c)  (d) 

(a) General formula (b) Empirical formula
(c) Structural formula (d) Molecular formula

(a)  (b)  (c)  (d) 

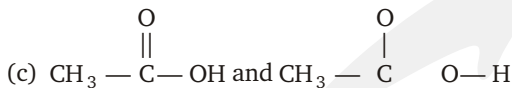
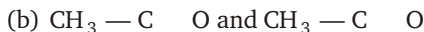
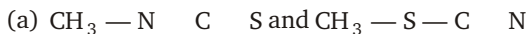
(a) $-\text{NO}_2$ (b) $-\text{CN}$ (c) $-\text{CH}_3$ (d) $-\text{CHO}$

(a) $\text{CH}_2 - \text{CH} = \text{CH} - \text{CH}_2$ (b) $\text{CH}_2 - \text{CH} - \text{C}(\text{CH}_3)_2$
(c) $\text{CH}_2 - \text{CH}(\text{CH}=\text{CH}_2) - \text{CH}_2$ (d) $\text{CH}_2 - \text{CH} - \text{C}(\text{CH}_2\text{CH}_3)_2$

$$\begin{array}{ccccccc} \text{O} & & \text{O}^- & & \text{O}^- & & \text{O}^+ \\ \parallel & & | & & | & & | \\ \text{H}-\text{C}-\text{OH} & \longleftrightarrow & \text{H}-\text{C}=\text{O}^+ & \longleftrightarrow & \text{H}-\text{C}^+-\text{OH} & \longleftrightarrow & \text{H}-\text{C}^--\text{OH} \\ \text{I} & & \text{II} & & \text{III} & & \text{IV} \end{array}$$

(a) II I III IV (b) I II III IV (c) IV III I II (d) IV III I II

171. Which of the following is not resonating structure of each other?



172. In the molecule $\text{CH}_3\text{C} \equiv \text{CCH} = \text{CH}_2$, the maximum number of carbon atoms arranged linearly is:

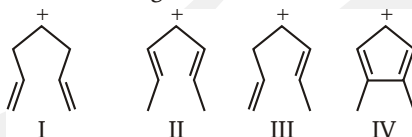
(a) 2

(b) 3

(c) 4

(d) 5

173. The stability order of the following carbocations is:



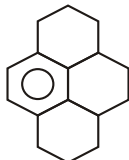
(a) II IV III I

(b) IV II III I

(c) II III I IV

(d) I III II IV

174. Total number of α -hydrogen in given compound is:



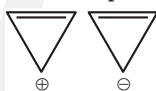
(a) 4

(b) 5

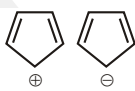
(c) 6

(d) 7

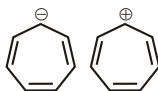
175. In which pair second ion is more stable than first?



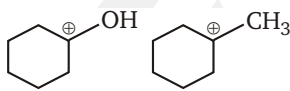
(i)



(ii)



(iii)



(iv)

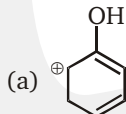
(a) (i) and (ii)

(b) (ii) and (iii)

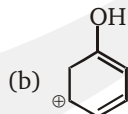
(c) (ii) and (iv)

(d) (iii) and (iv)

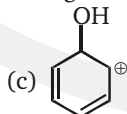
176. Which one is the most stable cation in the following ?



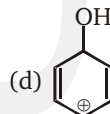
(a)



(b)

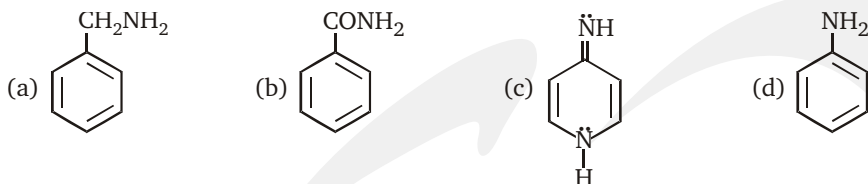


(c)

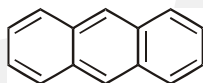


(d)

177. The most reactive amine towards dilute hydrochloric acid is _____.



178. How many resonance structures are there for anthracene ?

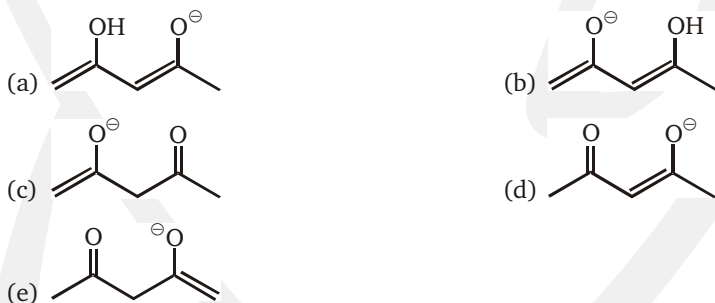
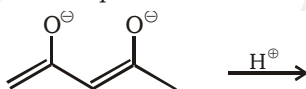


- (a) 6 (b) 5 (c) 4 (d) 2

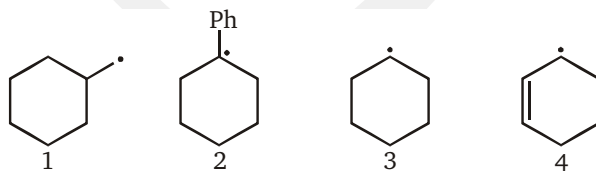
179. Which base is strong enough to convert $(\text{CH}_3)_3\text{COH}$ into $(\text{CH}_3)_3\text{CONa}$ in a reaction that goes to completion ?

- (a) NaNH_2 (b) $\text{CH}_3\text{CH}_2\text{Na}$ (c) NaOH (d) $\text{CH}_3\text{CO}_2\text{Na}$
(e) More than one of the above

180. Based upon an understanding of product stability, predict the product formed when the following dianion reacts with one equivalent of acid



181. Rank the following alkyl radicals in order of increasing stability (least < < < most).



- (a) 4 2 1 3 (b) 3 1 2 4 (c) 1 3 4 2 (d) 2 4 3 1

182. Among the given cations, the most stable carbonium ion is ?

- (a) sec-butyl (b) tert-butyl (c) n-butyl (d) None of these

183. Cyclohexadiene contains ____ vinylic and ____ allylic hydrogen atoms ?



- (a) 2 and 2 respectively (b) 4 and 4 respectively
(c) 2 and 4 respectively (d) 4 and 2 respectively

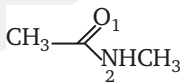
184. The dipole moments of halo compounds are in the order

- (a) CHCl_3 CCl_4 CHCl_2 cis CHCl CHCl
(b) cis CHCl CHCl CHCl_3 CH_2Cl CCl_4
(c) cis CHCl CHCl CH_2Cl_2 CHCl_3 CCl
(d) CHCl_3 CHCl_2 cis CHCl CHCl CCl_4

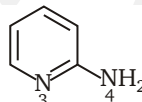
185. The pK_a value in H_2O of picric acid, acetic acid and phenol are in the order :

- (a) Picric acid 0.4, acetic acid 4.75, phenol 10.0
(b) Acetic acid 0.4, picric acid 4.75, phenol 10.0
(c) Picric acid 0.4 phenol 4.75, acetic acid 10.0
(d) Phenol 0.4, acetic acid 4.75 picric acid 10.0

186. The preferred sites of protonation in the following compounds are:



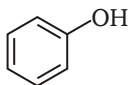
(i)



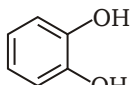
(ii)

- (a) 1 and 3 (b) 2 and 4 (c) 1 and 4 (d) 2 and 3

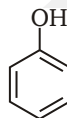
187. Among i-iii



(i)



(ii)

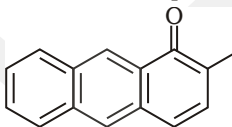


(iii)

the boiling point follows the order

- (a) (ii) < (i) < (iii) (b) (iii) < (ii) < (i) (c) (i) < (ii) < (iii) (d) (ii) < (iii) < (i)

188. The number of C — C sigma bonds in the compound



- (a) 16 (b) 14 (b) 18 (d) 11

189. The correct order of dipole moment for the following molecules is



(I)



(II)



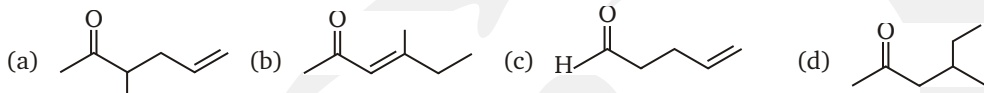
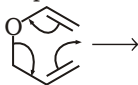
(III)



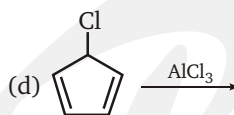
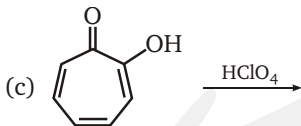
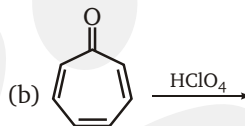
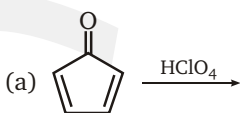
(IV)

- (a) IV > I > III > II (b) I > IV > III > II (c) III > I > II > IV (d) II > III > IV > I

- 190.** Curved arrows are used in Organic Chemistry to show the movement of electrons in the mechanism of a reaction. The correct product of the following reaction is



- 191.** Which of the following will form carbocation most readily ?



- 192.** Observed heat of hydrogenation for cyclohexa-1,4-diene and cyclohexa-1,3-diene is x & y kcal/mol respectively, calculate the resonance energy of cyclohexa-1,3-diene :

(a) $\frac{3x}{2}$ y
(c) $\frac{3y}{2}$ x

(b) $\frac{2x}{2}$ y
(d) $\frac{2y}{2}$ x

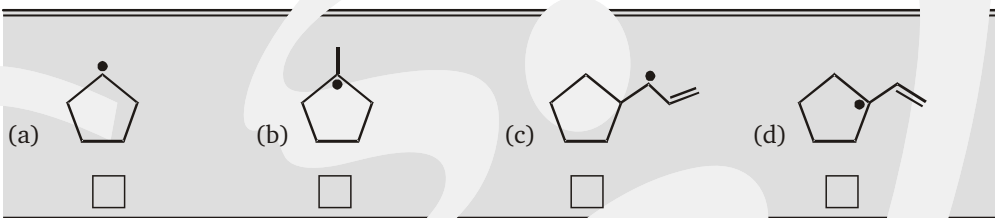


ANSWERS – LEVEL 1

1.	(c)	2.	(d)	3.	(d)	4.	(a)	5.	(d)	6.	(d)	7.	(b)	8.	(c)
9.	(b)	10.	(c)	11.	(b)	12.	(b)	13.	(a)	14.	(b)	15.	(b)	16.	(b)
17.	(d)	18.	(c)	19.	(d)	20.	(d)	21.	(c)	22.	(c)	23.	(c)	24.	(a)
25.	(d)	26.	(d)	27.	(d)	28.	(a)	29.	(d)	30.	(a)	31.	(c)	32.	(c)
33.	(d)	34.	(d)	35.	(d)	36.	(d)	37.	(b)	38.	(b)	39.	(a)	40.	(c)
41.	(c)	42.	(d)	43.	(a)	44.	(c)	45.	(c)	46.	(d)	47.	(b)	48.	(d)
49.	(b)	50.	(b)	51.	(d)	52.	(d)	53.	(d)	54.	(a)	55.	(c)	56.	(c)
57.	(b)	58.	(b)	59.	(d)	60.	(c)	61.	(b)	62.	(b)	63.	(d)	64.	(c)
65.	(d)	66.	(c)	67.	(a)	68.	(a)	69.	(d)	70.	(b)	71.	(a)	72.	(d)
73.	(d)	74.	(d)	75.	(a)	76.	(d)	77.	(a)	78.	(a)	79.	(b)	80.	(a)
81.	(d)	82.	(d)	83.	(d)	84.	(c)	85.	(b)	86.	(a)	87.	(a)	88.	(b)
89.	(a)	90.	(d)	91.	(b)	92.	(c)	93.	(d)	94.	(d)	95.	(d)	96.	(d)
97.	(b)	98.	(d)	99.	(a)	100.	(b)	101.	(b)	102.	(d)	103.	(a)	104.	(a)
105.	(c)	106.	(c)	107.	(c)	108.	(b)	109.	(c)	110.	(c)	111.	(d)	112.	(c)
113.	(c)	114.	(c)	115.	(c)	116.	(a)	117.	(a)	118.	(d)	119.	(c)	120.	(c)
121.	(c)	122.	(a)	123.	(c)	124.	(c)	125.	(d)	126.	(b)	127.	(a)	128.	(d)
129.	(b)	130.	(a)	131.	(c)	132.	(d)	133.	(c)	134.	(d)	135.	(a)	136.	(d)
137.	(c)	138.	(d)	139.	(c)	140.	(c)	141.	(a)	142.	(d)	143.	(b)	144.	(c)
145.	(a)	146.	(b)	147.	(d)	148.	(b)	149.	(a)	150.	(b)	151.	(d)	152.	(b)
153.	(a)	154.	(a)	155.	(c)	156.	(c)	157.	(c)	158.	(b)	159.	(c)	160.	(c)
161.	(b)	162.	(c)	163.	(a)	164.	(a)	165.	(c)	166.	(a)	167.	(d)	168.	(c)
169.	(d)	170.	(b)	171.	(a)	172.	(c)	173.	(c)	174.	(c)	175.	(b)	176.	(b)
177.	(c)	178.	(c)	179.	(e)	180.	(d)	181.	(c)	182.	(b)	183.	(b)	184.	(c)
185.	(a)	186.	(a)	187.	(a)	188.	(b)	189.	(b)	190.	(c)	191.	(c)	192.	(b)

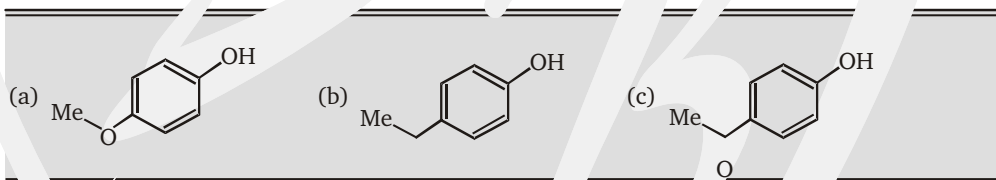
Level - 2

1. Rank in order of radical stability (1 = most stable).



2. Predict the acidity order for the three phenols shown below :

Acidity order : 1 (most) to 3 (least)

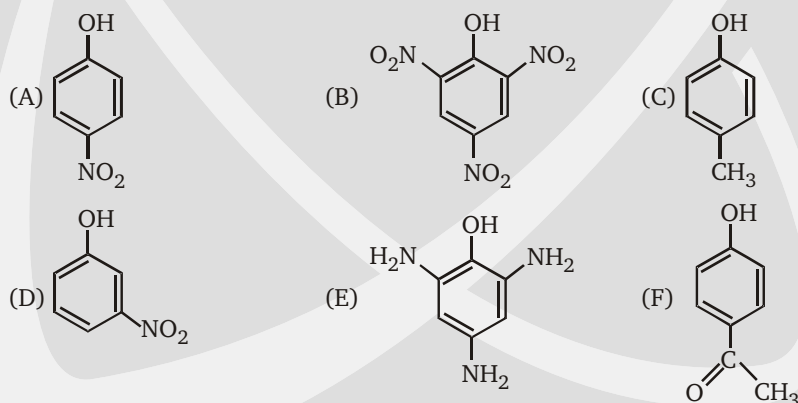


Acidity order :

.....

.....

3. Comprehension



A. Which of the phenol derivatives above is the strongest acid ?

- ☐ Compound A ☐ Compound B ☐ Compound C
☐ Compound D ☐ Compound E ☐ Compound F

B. Which of the phenol derivatives above is the weakest acid ?

- ☐ Compound A ☐ Compound B ☐ Compound C
☐ Compound D ☐ Compound E ☐ Compound F

C. Which of the mono-nitrophenol derivatives above is the strongest acid ?

- ☐ Compound A ☐ Compound D

D. Which of the carbon-substituted phenol derivatives above is the strongest acid ?

- ☐ Compound C ☐ Compound F

4. The following questions refer to the twelve compounds given below. You may enter as many as six choices in each answer box.

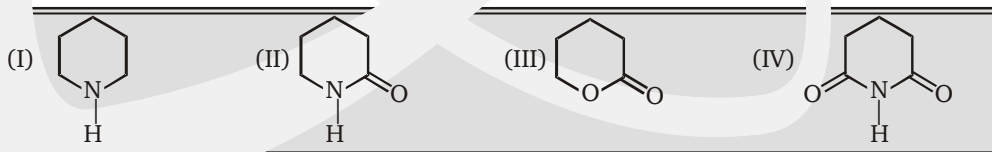
(a)		(b)		(c)	$\text{H}_3\text{C}-\text{C}(=\text{O})-\text{O}-\text{C}_2\text{H}_5$
(d)	$\text{H}-\text{F}$	(e)		(f)	
(g)		(h)		(i)	
(j)		(k)		(l)	

A. Which compound may serve only as H-bond donors ?

B. Which may serve both as H-bond donors and acceptors?

C. Which compounds will not participate in H-bonding ?

5. Consider the following compounds and answer A and B.



A. Which of the compounds is the strongest Bronsted acid ?

(a) I

(b) II

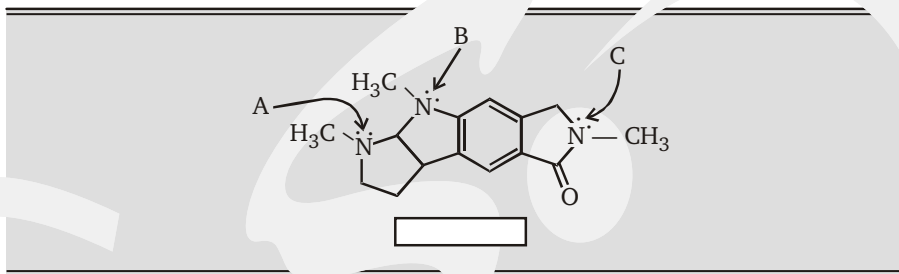
(c) III

(d) IV

B. Which of the compounds is the strongest Lewis base ?

- (a) I (b) II (c) III (d) IV

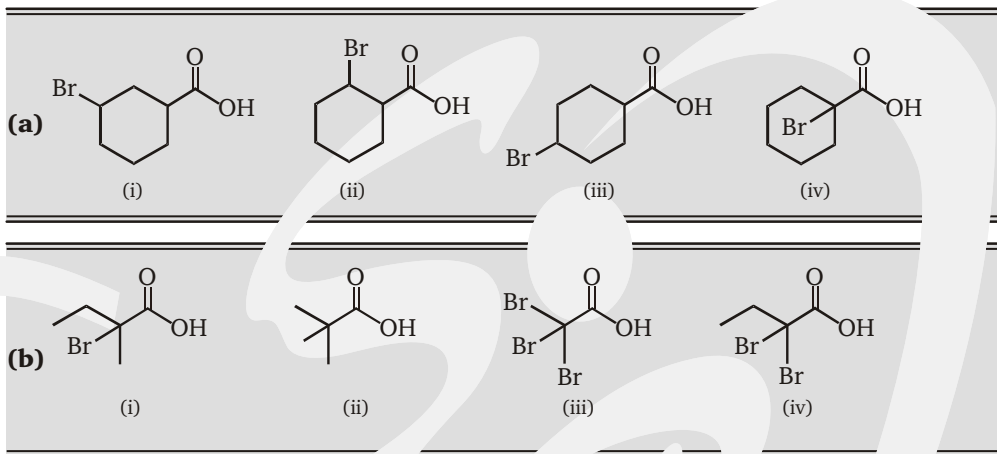
6. Rank the non-bonding electrons indicated by the arrows in order of increasing energy.



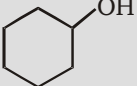
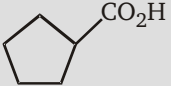
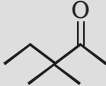
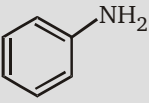
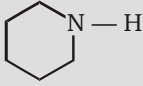
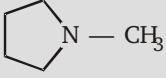
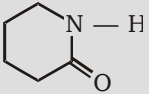
7. In each of the following sections four nitrogen containing compounds are listed. In the box under each formula write a number (1 to 4) indicating the order of base strength.

(a)				
(b)				
(c)				
(d)				

8. For the two sets of acids shown below, rank their acidity most acidic to least acidic.

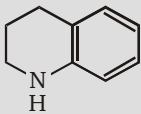
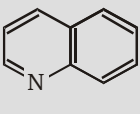
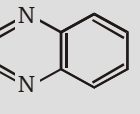
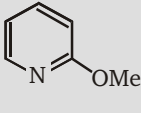
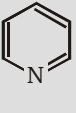
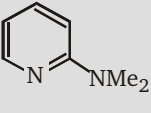
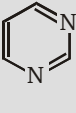


9. In each of the following sections four compounds are listed. In the box under each formula enter a number (1 to 4) indicating the order of acid strength (1 is strongest & 4 is weakest).

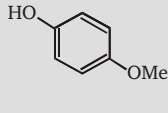
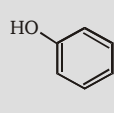
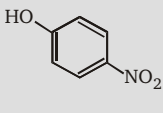
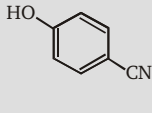
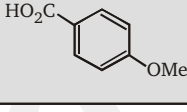
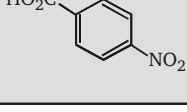
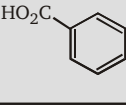
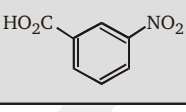
(a)	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CO}_2\text{H}$	$\text{CH}_3\text{CH}_2\text{CHBrCO}_2\text{H}$	$\text{ClCH}_2\text{CH}_2\text{CH}_2\text{CO}_2\text{H}$	$\text{CH}_3\text{CCl}_2\text{CO}_2\text{H}$
(b)	$\text{C}_6\text{H}_5\text{CH}_2\text{OH}$	$\text{C}_6\text{H}_5\text{CO}_2\text{H}$	$\text{C}_6\text{H}_5\text{OCH}_3$	$\text{C}_6\text{H}_5\text{OH}$
(c)				
(d)				

10. In the two questions below, you are asked to rank the relative strengths of illustrated acids and bases. Use your knowledge of resonance and inductive to answer this.

A. For the series of bases shown below, rank the set from strongest to weakest.

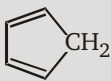
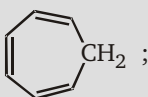
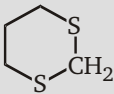
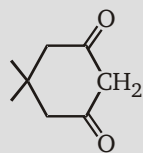
(i)					Strongest
	(a)	(b)	(c)	(d)	Weakest
(ii)					Strongest
	(a)	(b)	(c)	(d)	Weakest

B. For the series of acids shown below, rank the set from strongest to weakest.

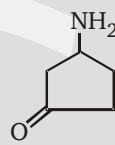
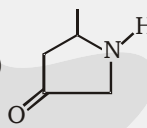
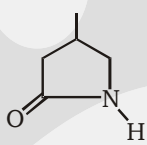
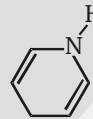
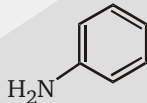
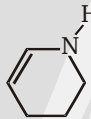
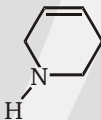
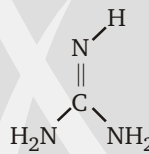
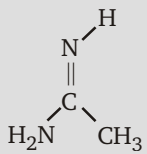
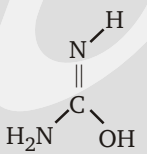
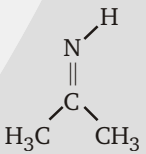
(i)					Strongest
	(a)	(b)	(c)	(d)	Weakest
(ii)					Strongest
	(a)	(b)	(c)	(d)	Weakest

11. In each of the following sections four compounds are listed. (Decreasing order of acidic strength, 1 is strongest & 4 is weakest).

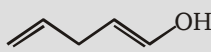
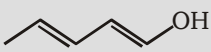
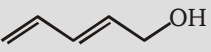
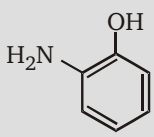
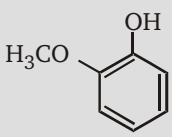
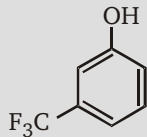
(a)	$\text{CH}_2(\text{CO}_2\text{C}_2\text{H}_5)_2$	$\text{CH}_3\text{COCH}_2\text{CO}_2\text{C}_2\text{H}_5$	$(\text{CH}_3\text{CO})_2\text{CH}_2$	$\text{RC}-\text{CH}$
(b)	RCH_2NO_2	RSO_2CH_3	$(\text{C}_6\text{H}_5)_3\text{CH}$	RCOCH_3
(c)	$\text{CH}_2(\text{C}=\text{N})_2$	$\text{CH}_2(\text{NO}_2)_2$	$\text{HC}=\text{N}$	$\text{RCH}_2\text{CO}_2\text{C}_2\text{H}_5$

(d)				

12. Rank in the order of increasing basic strength.

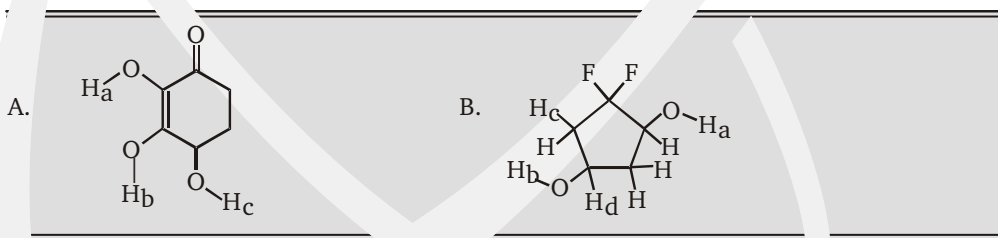
A.	(a)		(b)		(c)			
B.	(a)		(b)		(c)		(d)	
C.	(a)		(b)		(c)		(d)	

13. Compare acidic strength of the following (Write your answer in box).

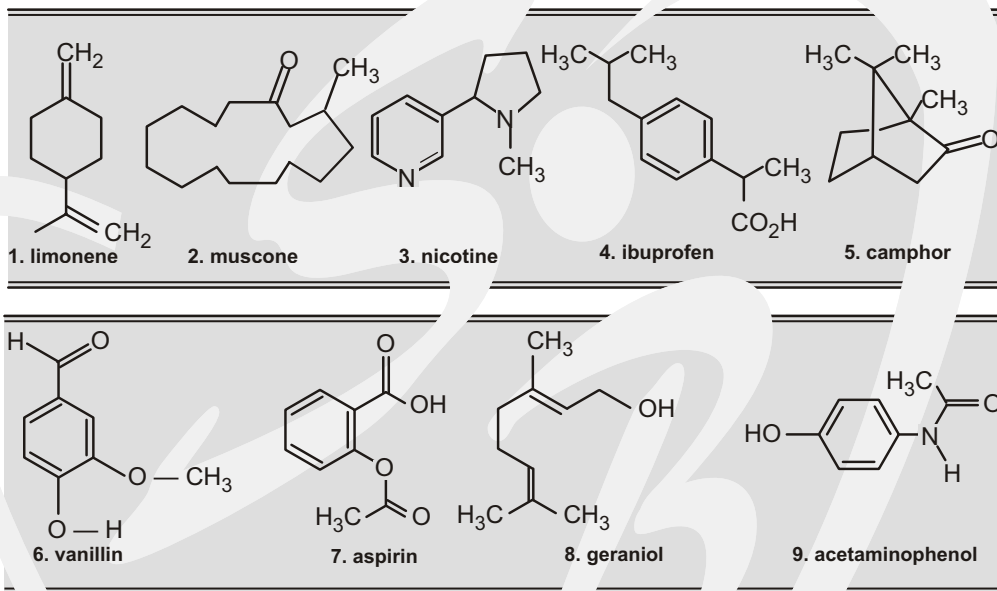
A.					
	(a)	(b)	(c)		
B.					
	(a)	(b)	(c)		

C.						
	(a)	(b)	(c)			
D.						
	(a)	(b)	(c)	(d)		
E.						
	(a)	(b)	(c)			
F.						
	(a)	(b)	(c)			

14. Arrange the hydrogens in increasing order of their acidic strengths.

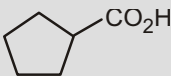
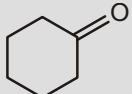
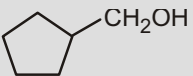
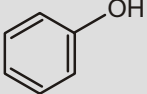
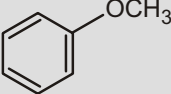
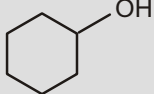
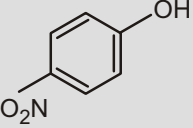
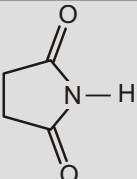
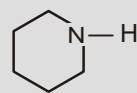
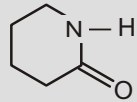
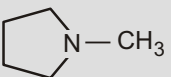
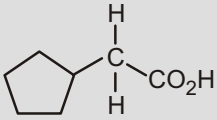
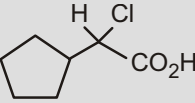
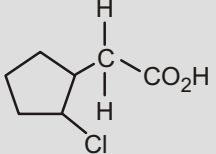
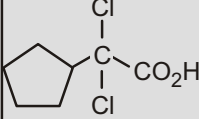


15. The compounds whose structures are shown below, incorporate a variety of functional groups. The question on the right ask you to identify which compounds have a specific functional group. For each compound that has the designed group, enter the appropriate number. The aromatic rings should not be counted as double bonds.



A.	Which have carbon-carbon double bonds ?	
B.	Which have a ketone carbonyl group ?	
C.	Which have an aldehyde carbonyl group ?	
D.	Which have aromatic rings ?	
E.	Which have a hydroxy group ?	
F.	Which have ether groups ?	
G.	Which have an ester group ?	
H.	Which have an amide group ?	
I.	Which have a carboxylic acid group ?	

16.

Problem	A	B	C	D
1				
2				
3				
4				

A. Which is the strongest acid in 1?

- (a) A (b) B (c) C (d) D

B. Which is weakest acid in 1?

- (a) A (b) B (c) C (d) D

C. Which is the strongest acid in 2?

- (a) A (b) B (c) C (d) D

D. Which is weakest acid in 2?

- (a) A (b) B (c) C (d) D

E. Which is the strongest acid in 3?

- (a) A (b) B (c) C (d) D

F. Which is weakest acid in 3?

- (a) A (b) B (c) C (d) D

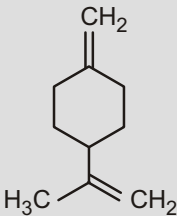
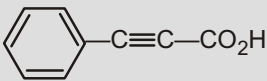
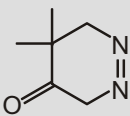
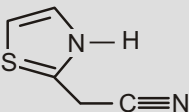
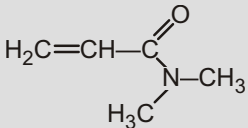
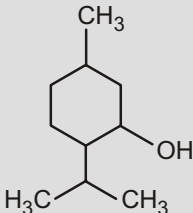
G. Which is the strongest acid in 4?

- (a) A (b) B (c) C (d) D

H. Which is weakest acid in 4?

- (a) A (b) B (c) C (d) D

17. For each of the six structural formulae (A through F), shown below, five questions are posed. The answer to each is a number that should be entered in the appropriate answer box.

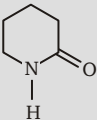
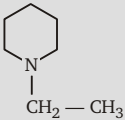
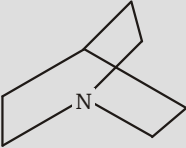
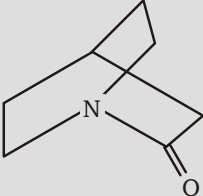
		
A	B	C
		
D	E	F

- | | | |
|--|--|--|
| <p>A. (i) Number of sp^3 carbons :</p> <p>(ii) Number of sp^2 carbons :</p> <p>(iii) Number of sp carbons :</p> <p>(iv) Number of carbon-carbon -bonds :</p> <p>(v) Number of -bonds to carbon :</p> | <p>B. Number of sp^3 carbons :</p> <p>Number of sp^2 carbons :</p> <p>Number of sp carbons :</p> <p>Number of carbon-carbon -bonds :</p> <p>Number of -bonds to carbon :</p> | <p>C. Number of sp^3 carbons :</p> <p>Number of sp^2 carbons :</p> <p>Number of sp carbons :</p> <p>Number of carbon-carbon -bonds :</p> <p>Number of -bonds to carbon :</p> |
| <p>D. (i) Number of sp^3 carbons :</p> <p>(ii) Number of sp^2 carbons :</p> <p>(iii) Number of sp carbons :</p> <p>(iv) Number of carbon-carbon -bonds :</p> <p>(v) Number of -bonds to carbon :</p> | <p>E. Number of sp^3 carbons :</p> <p>Number of sp^2 carbons :</p> <p>Number of sp carbons :</p> <p>Number of carbon-carbon -bonds :</p> <p>Number of -bonds to carbon :</p> | <p>F. Number of sp^3 carbons :</p> <p>Number of sp^2 carbons :</p> <p>Number of sp carbons :</p> <p>Number of carbon-carbon -bonds :</p> <p>Number of -bonds to carbon :</p> |

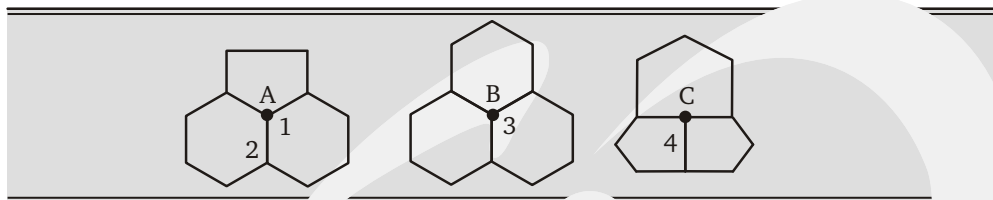
18. Match the column (I) and (II). (Matrix)

Column (I)		Column (II)	
	Molecule		Property
(a)		(p)	cis-compound
(b)		(q)	trans-compound
(c)		(r)	Highest heat of combustion
(d)		(s)	lowest heat of combustion

19. Match the column (I) and (II).

Column (I)		Column (II)	
	Molecule		pK_a of Conjugate acid
(a)		(p)	0.8
(b)		(q)	5.33
(c)		(r)	10.65
(d)		(s)	10.95

20. The junctures centred on atoms A, B and C on the given structure.



- A. Which juncture has the greatest deviation from planarity ?
 (a) A (b) B (c) C (d) Cannot be predicted
- B. Of the carbon-carbon bonds, (shown above) numbered from 1 to 4, which represent the most favourable site for H_2 addition ?
 (a) 1 (b) 2 (c) 3 (d) 4

21. Select the most stable structure in each of the following

Part (A)			
(a)	(b)	(c)	(d)
Part (B)			
(a)	(b)	(c)	(d)
Part (C)			
(a) $H_2C \quad CH-CH \quad CH-CH_3$		(b) $H_2C \quad C \quad CH-CH_2-CH_3$	
(c) $H_3C-CH \quad C \quad CH-CH_3$		(d) $H_2C \quad CH-CH_2-CH \quad CH_2$	

22. Match the column I and II. (Matrix)

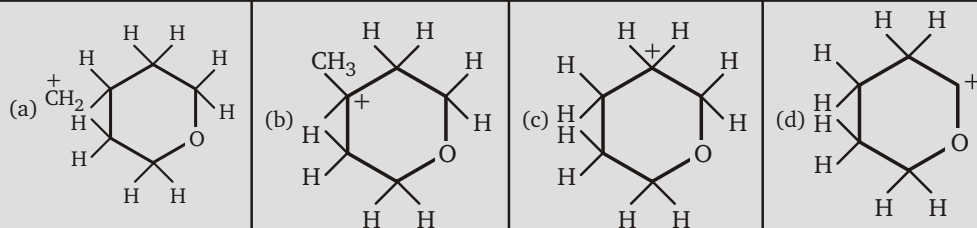
Column (I)		Column (II)	
(a)	$-NO_2$	(p)	m effect
(b)	$-O^-$	(q)	m effect
(c)	$-O-CH_3$	(r)	+ I effect
(d)	$-C \quad N$	(s)	I effect

23. Match the column I and II. (Matrix)

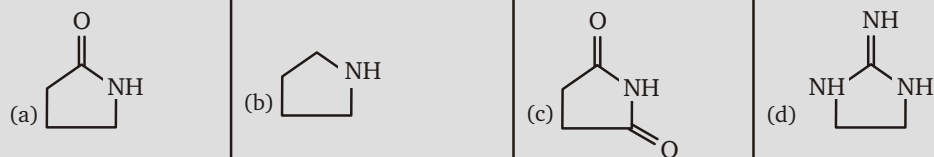
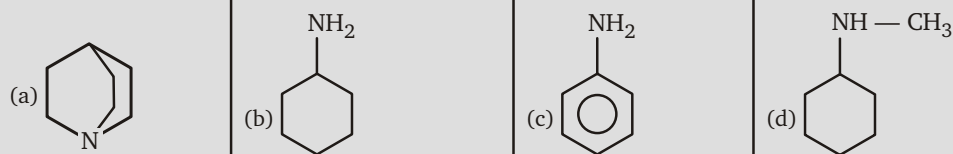
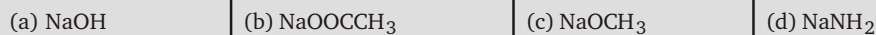
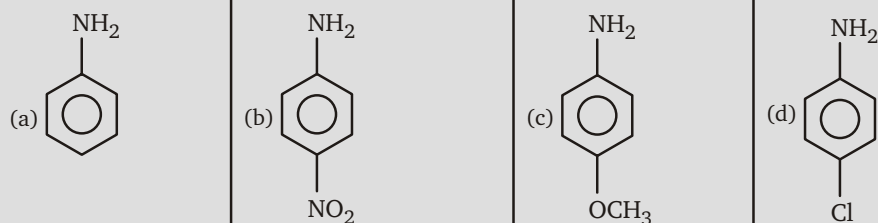
Column (I)		Column (II)	
(a)	$H_3C-CH \quad CH-CH_3$	(p)	Dipole (<i>cis</i> > <i>trans</i>)
(b)	$H_3C-CH \quad CH-CN$	(q)	Dipole (<i>trans</i> > <i>cis</i>)
(c)	$H_3C-CH \quad CH-Cl$	(r)	Melting point ((<i>trans</i> > <i>cis</i>))
(d)	$Cl-CH \quad CH-Cl$	(s)	Boiling point (<i>cis</i> > <i>trans</i>)

24. Identify the most stable structure in each of the following :

Part (A)			
(a)	(b)	(c)	(d)
Part (B)			
(a)	(b)		
(c)	(d)		
Part (C)			
(a)	(b)	(c)	(d)
Part (D)			
(a)	(b)	(c)	(d)
Part (E)			
(a)	(b)		
(c)	(d)		
Part (F)			
(a)	(b)	(c)	(d)

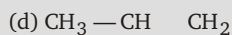
Part (G)

25. Identify the most basic compound in the following.

Part (A)**Part (B)****Part (C)****Part (D)**

26. Identify the most acidic hydrogen containing compound from the following.

Part (A)			
(a)	(b)	(c)	(d)
Part (B)			
(a)	(b)	(c)	(d)
Part (C)			
(a)	(b)	(c)	(d)
Part (D)			
(a)	(b)	(c)	(d)
Part (E)			
(a)	(b)	(c)	(d)



(d)

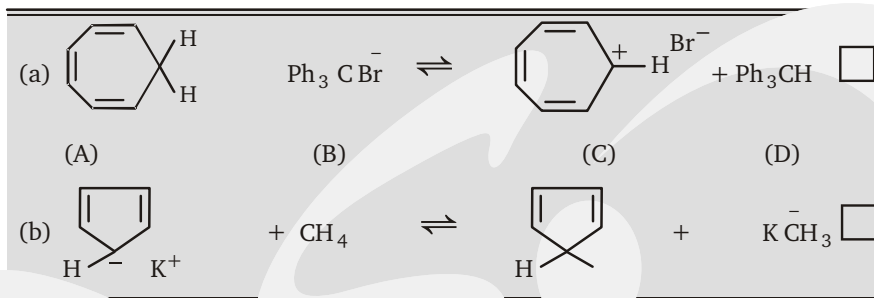
(d)

(d)

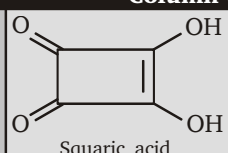
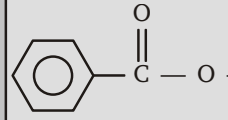
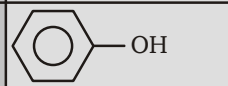
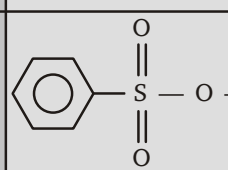
$$\text{N} \equiv \text{C}^1 - \text{C}^2 = \text{O}^1 - \text{CH}_2 - \text{C}^3 - \text{H}$$

(vi) C₃ —.....

28. Predict the direction of the following equilibrium. Write your answer in the box given below.



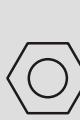
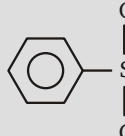
29. Match the column I and II. (Matrix)

Column (I)		Column (II)	
(a)	NaHCO ₃ will react with	(p)	 Squaric acid
(b)	Na will react with	(q)	
(c)	NaOH will react with	(r)	
(d)	NaNH ₂ will react with	(s)	

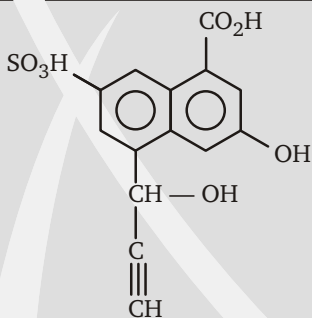
30. Match the column I and II.

Column (I)		Column (II)	
Acid		pK _a	
(a)	CH ₃ — CO ₂ H	(p)	5.69
(b)	(CH ₃) ₃ N CH ₂ CO ₂ H	(q)	4.27
(c)	(CH ₃) ₃ N(CH ₂) ₄ CO ₂ H	(r)	1.83
(d)	$\bar{O}_2C - CH_2 - CO_2H$	(s)	4.80

31. Match the column I and II.

Column (I)		Column (II)	
(a)	 $\text{C} = \text{O}$ $\text{C} - \text{O} - \text{H} + \text{NaHCO}_3$	(p)	NH_3
(b)	 $\text{C} = \text{O}$ $\text{C} - \text{O} - \text{H} + \text{NaHCO}_3$	(q)	$^{14}\text{CO}_2$
(c)	 $\text{C} = \text{O}$ $\text{C} - \text{O} - \text{H} + \text{Na}$	(r)	CO_2
(d)	 $\text{S} = \text{O}$ $\text{S} - \text{O} - \text{H} + \text{NaNH}_2 \longrightarrow$	(s)	H_2

32.

(A)
gas

(B)

(C)
gas

(D)

Sum of molecular mass of gas (A + C) is :

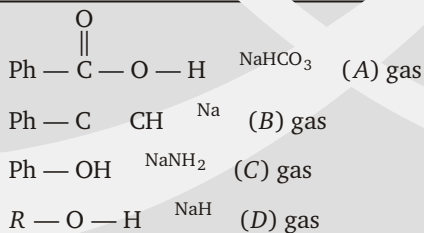
(a) 88

(b) 90

(c) 92

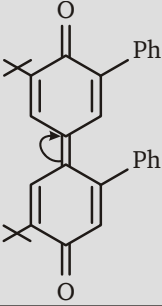
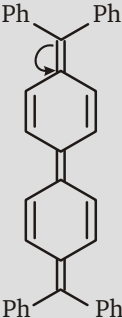
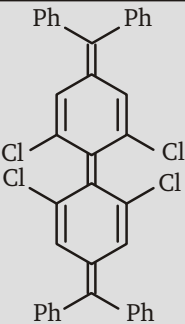
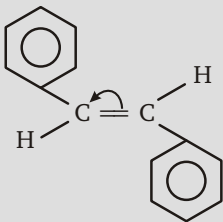
(d) 40

33.

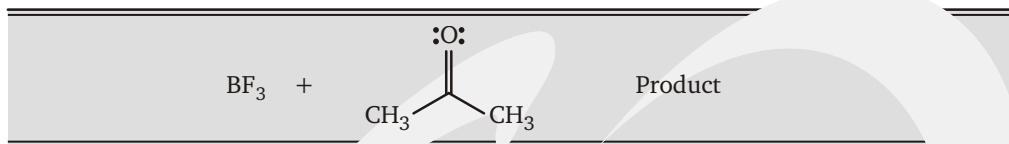


Sum of molecular mass of gas A B C D is :

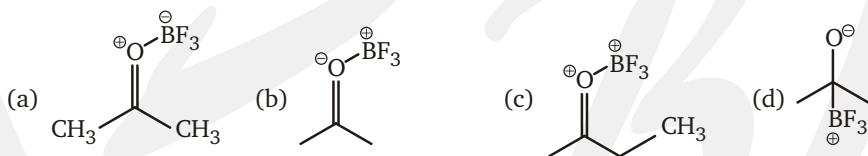
34. Match the column I and II.

Column (I)		Column (II)	
	Molecule		Rotational free energy barrier
(a)		(p)	180 kJ/mol
(b)		(q)	88.3 kJ/mol
(c)		(r)	21 kJ/mol
(d)		(s)	Negative barrier

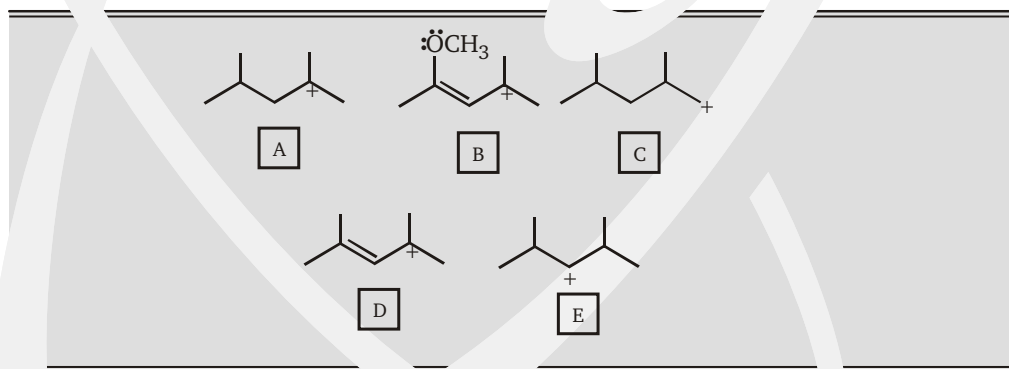
35. Consider the following reaction of boron trifluoride (BF_3) and acetone :



- A.** What is the critical HOMO (nucleophile) of this reaction ?
- (a) non-bonding orbital on boron (b) π -orbital of acetone
 (c) π -orbital of acetone (d) non-bonding electron pair orbital on oxygen
- B.** What is the critical LUMO (electrophile) of the reaction ?
- (a) p -orbital of BF_3 (b) π -orbital of BF_3
 (c) π^* orbital of acetone (d) non-bonding electron pair orbital on oxygen
- C.** Which of the following is the correct product of this reaction ?
 (Lone electron pairs are not shown explicitly).

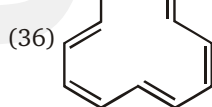
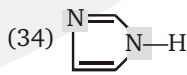
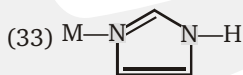
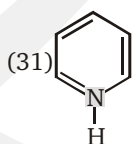
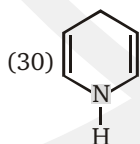
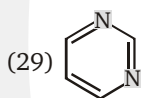
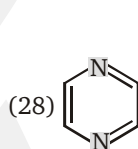
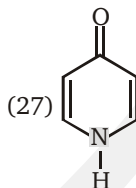
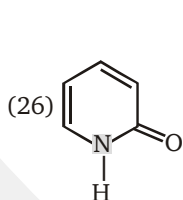
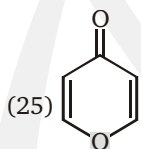
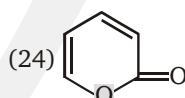
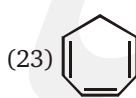
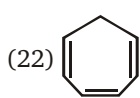
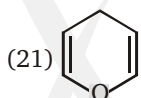
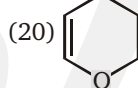
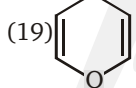
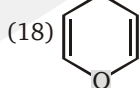
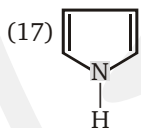
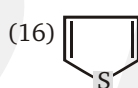
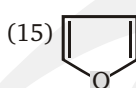
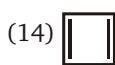
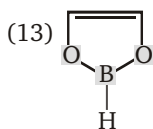
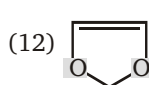
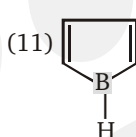
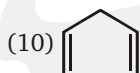
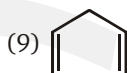
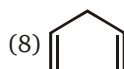
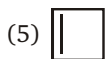
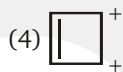


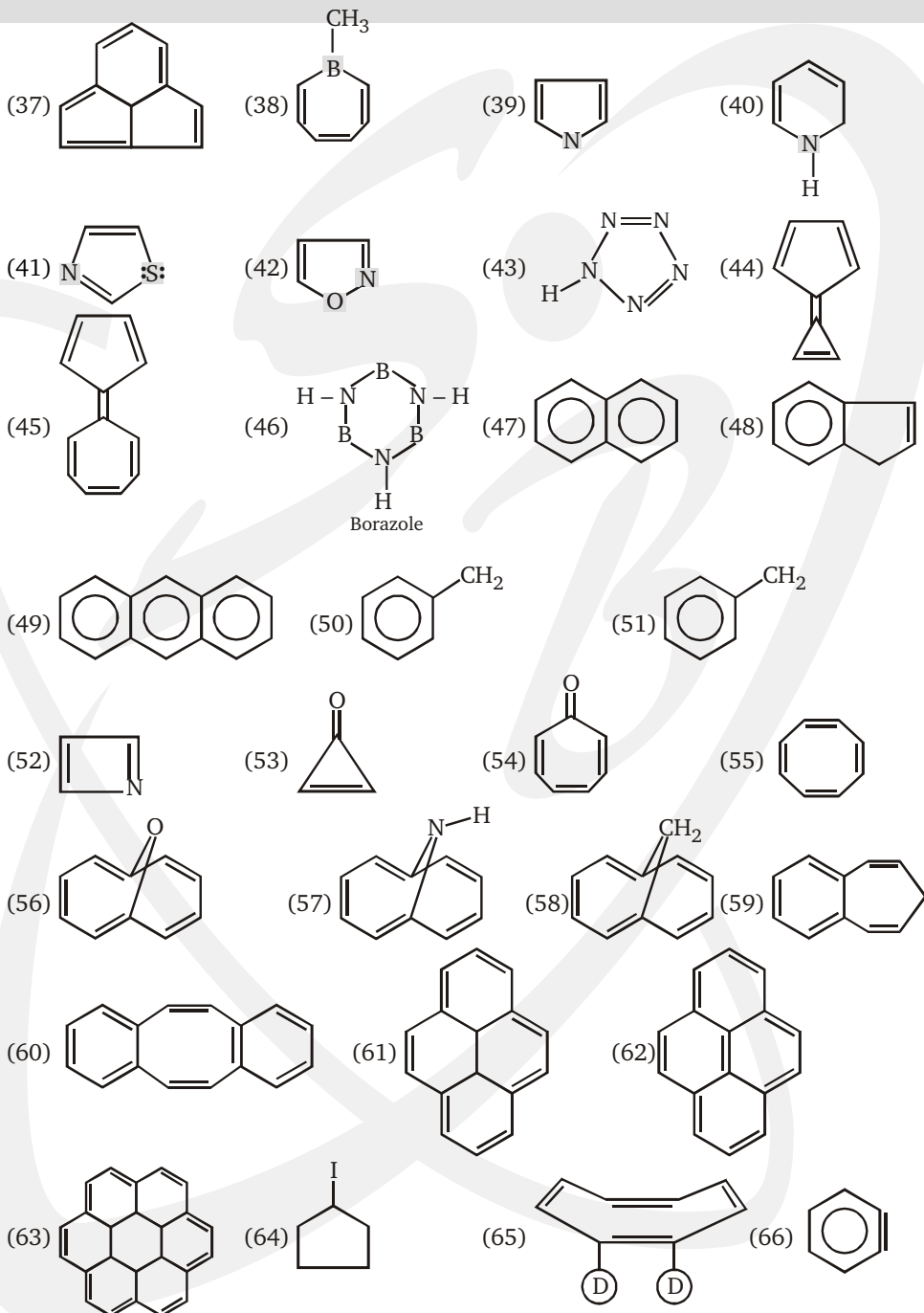
36. Rank the following carbocations according to stability (1 = most stable, 5 = least stable).



Put the answer in the boxes.

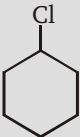
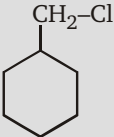
37. Among the given molecules, identify aromatic, anti-aromatic and non-aromatic molecules.



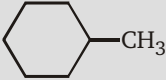
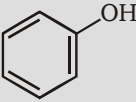
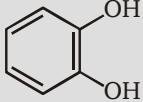
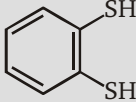


38. Among the given pairs, which is more reactive towards AgNO_3 (or) toward hydrolysis.

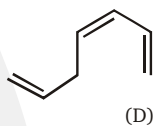
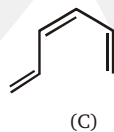
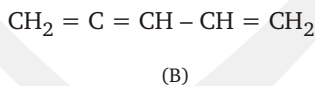
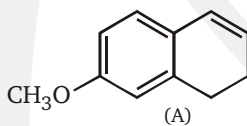
	Compound (A)	Compound (B)	Put the Answer here
1.			
2.			
3.			
4.			
5.			
6.	$\text{CH}_3 \quad \text{O} \quad \text{CH}_2 \quad \text{Cl}$	$\text{CH}_3 \quad \text{CH}_2 \quad \text{CH}_2 \quad \text{Cl}$	
7.			
8.			
9.			

10.			
-----	---	---	--

39. Put the answer in boxes given as directed.

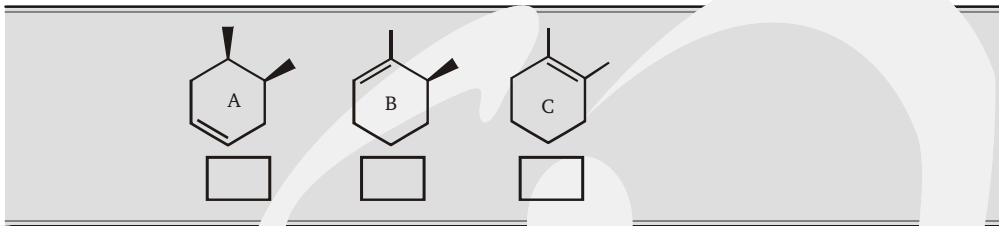
S.No.	Property	Molecules	Correct Answer	Name of force responsible for the property
A.	highest boiling point	NCl_3 ClNH_2 NH_4Cl NH_3		
B.	highest boiling point	  		
C.	most soluble in water	  		
D.	highest solubility in benzene	 		

40. Circle any conjugated portions of these molecules.



41. Arrange in the order as directed -

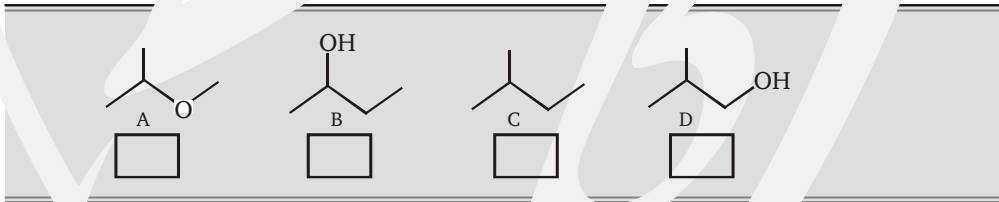
A. The given alkenes in the order of their stability (1- most stable, 3-least stable).



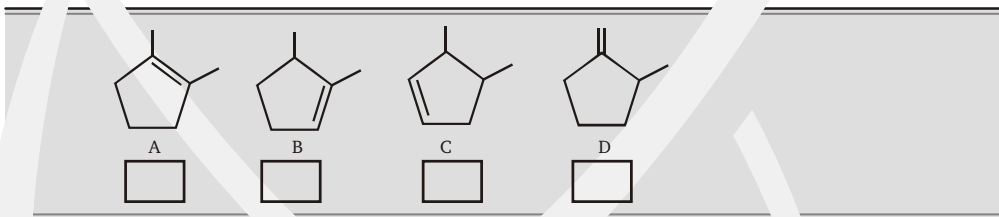
B. Arrange the following in the order of their acidic strength (1-most acidic, 4-least acidic)



C. Arrange the following molecules in order of expected boiling point. (1=highest bpt ; 4=lowest bpt.)



D. Arrange the following alkenes in order of their stability. (1 = most stable ; 5 = least stable).



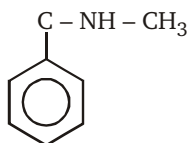
42. Match the column. (Matrix)

Column (I)		Column (II)	
Compounds		Number of Benzylic hydrogen	
(a)		(p)	2
(b)		(q)	3
(c)		(r)	4
(d)		(s)	5

43. Identify (+M) mesomeric & (-M) group of following.

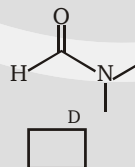
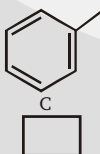
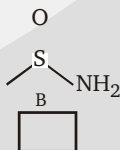
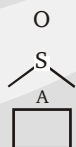
	+M	-M	-I	+I

O

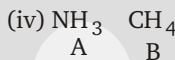
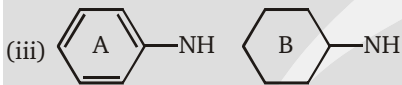
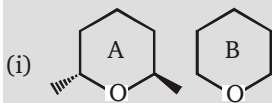


$\begin{array}{c} \text{O} \\ \\ \text{C} - \text{NH} - \text{CH}_3 \\ \\ \text{C}_6\text{H}_5 \end{array}$				
$\begin{array}{c} \text{O} \\ \\ \text{C} - \text{CH}_3 \\ \\ \text{C}_6\text{H}_5 \end{array}$				
$\begin{array}{c} \text{NO}_2 \\ \\ \text{C}_6\text{H}_5 \end{array}$				
$\begin{array}{c} \text{OCH}_3 \\ \\ \text{C}_6\text{H}_5 \end{array}$				
$\begin{array}{c} \text{CH} = \text{O} \\ \\ \text{C}_6\text{H}_5 \end{array}$				
$\begin{array}{c} \text{O} = \text{C} - \text{OH} \\ \\ \text{C}_6\text{H}_5 \end{array}$				
$\begin{array}{c} \text{NH} - \text{CH}_3 \\ \\ \text{C}_6\text{H}_5 \end{array}$				

44. Identify the following solvents as polar protic (PP), polar aprotic (PA), non-polar protic (NPP) or non-polar aprotic (NPA).



45. Identify the stronger nucleophile in each pair.



46. Encircle the molecule as directed :

(a) Which has higher boiling point : HBr or HCl

(b) Which has a higher boiling point : $\text{CH}_3 - \text{CH}_2 - \text{OH}$ or $\text{CH}_3 - \text{CH} = \text{O}$

(c) Which is more miscible with methanol (CH_3OH) : $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{CH}_3$

or $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{CH}_3$

(d) Which has a higher melting point : CH_4 or $\text{CH}_3 - \text{CH}_2 - \text{CH}_3$

(e) Which has a higher boiling point : $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{CH}_3$ or

47. Encircle the molecule as directed :

(a) Which is more stable : BH_3 or BF_3

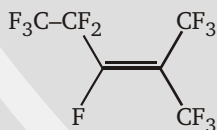
(b) Which is a stronger base : HO^- or H_2O

(c) Which is a stronger base : HO^- or HS^-

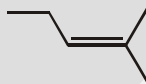
(d) Which is a stronger acid : HCl or HI

(e) Which is a stronger acid : HOCl or HCl

48. Explain why **A** has lower boiling point than **B** ?

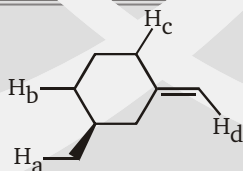


(A)
b.p. = 58°C

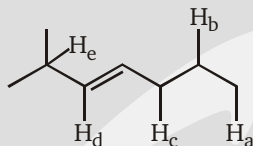


(B)
b.p. = 67°C

49. Arrange the protons shown in the decreasing order of their approximate bond energies.



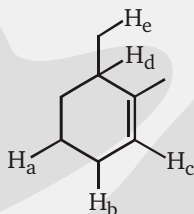
50. Consider the H-atoms in the molecule given below and answer the following.



(A) Identify the type (1°, 2° or 3° alkyl, vinyl, allyl etc.) of these H-atoms.

(B) Arrange them in the decreasing order of their ease of abstraction (easiest first)

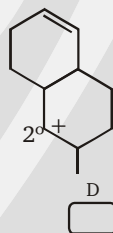
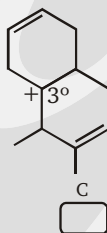
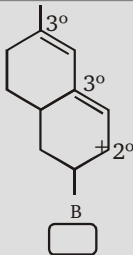
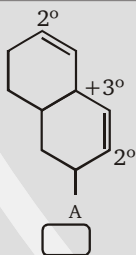
51. Consider the molecule shown below and answer with respect to H_a H_e



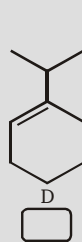
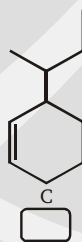
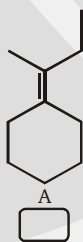
(A) Identify the type of H-atom (1°, 2°, 3° alkyl, vinyl or allyl)

(B) Arrange them in decreasing order of their bond energy.

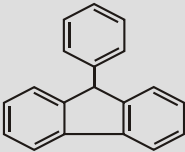
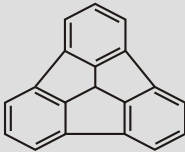
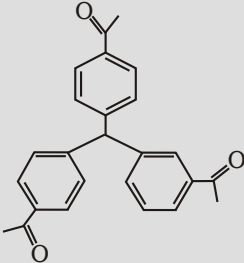
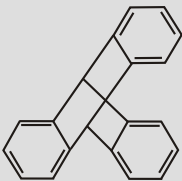
52. Rank the following carbocations in order of stability (1 = most stable).



53. Rank the following alkenes according to energy (1 = lowest energy).

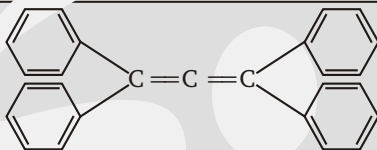


54. Match the column:

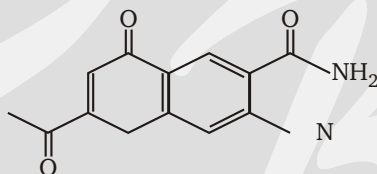
Column (I)		Column (II)	
(Compounds)		(Double bond equivalent value)	
(a)		(p)	11
(b)		(q)	12
(c)		(r)	13
(d)		(s)	14
		(t)	15

Subjective Problems

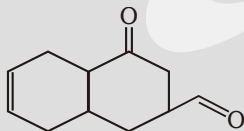
1. How many 2° carbon in the following ?



2. Find out the double bond equivalent (DBE) value of the given following compound:



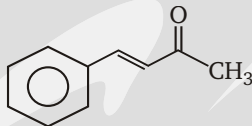
3. Total number of functional groups present in the given following compound :



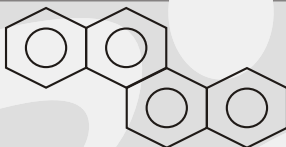
4. Total number of β -hydrogen in the given following compound is:



5. How many carbon atom present in the parent chain in the given following compound?

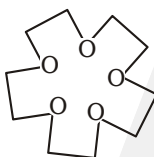


6. Total number of DBE value in :



7. How many isomers of $C_4H_{10}O$ reacts with Na metal to evolve H_2 gas ? (excluding stereoisomer)

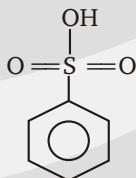
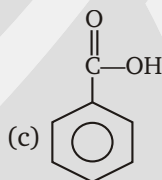
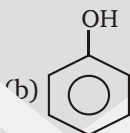
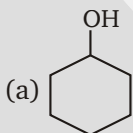
8.



[x]-crown-[y]-ether.

value of $\frac{x}{3} \times \frac{y}{3}$?

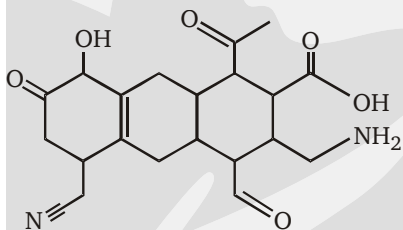
9. Which of the given following compound will react with $NaHCO_3$ or soluble in $NaHCO_3$?



10. How many compound are stable after deprotonation ?

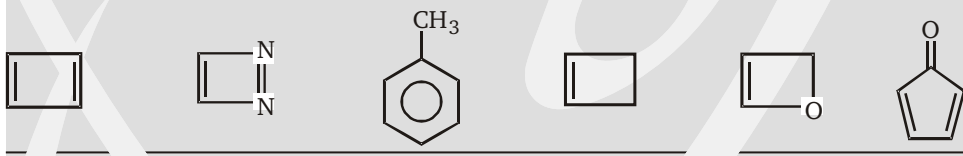


11.



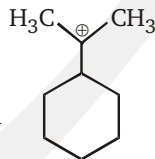
Sum of types of functional group and DBE value for given compound is X so the value of $X - 10$ is

12. P Number of anti-aromatic compound, so the value of x is :



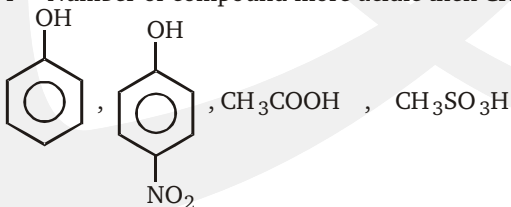
Q Total number of resonating structures of carbonate ion $[\text{CO}_3^{2-}]$

R Number of β -hydrogen in given carbocation



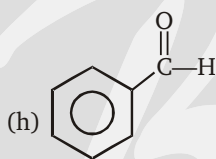
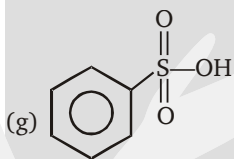
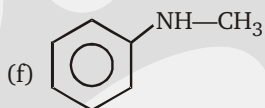
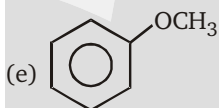
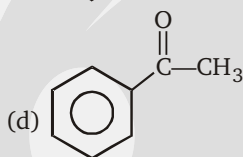
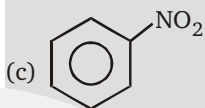
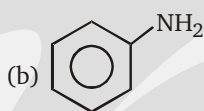
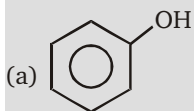
S Total number of geometrical isomers of $\text{CH}_3 - \text{CH} = \text{CH} - \text{CH} = \text{CH}_2$

T Number of compound more acidic than $\text{CH}_3\text{CH}_2\text{OH}$



Sum of (P Q R S T) 15 is :

13. x number of (M) group attached with phenyl ring, so the value of x is.



ANSWERS — LEVEL 2

1. a - 4; b - 3; c - 2; d - 1
2. a - 3; b - 2; c - 1
3. A - B; B - E; C - A; D - F
4. A - a, c, f, g, k, l; B - b, d, h, j; C - e, i
5. A - d; B - a
6. $C < B < A$
7. a - 2, 1, 4, 3; b - 1, 2, 3, 4; c - 3, 4, 1, 2; d - 3, 2, 4, 1
8. a - iv > ii > i > iii; b - iii > iv > i > ii
9. a - 4, 2, 3, 1; b - 3, 1, 4, 2; c - 2, 1, 3, 4; d - 2, 3, 4, 1
10. A (i) - d > b > c > a; (ii) - c > a > b > d
B (i) - c > d > b > a; (ii) - b > d > c > a
11. (a) 3 2 1 4; (b) 2 1 4 3; (c) 3 1 2 4; (d) 3 4 1 2
12. A - c < a < b; B - b < a < c < d; C - d < b < c < a
13. A - c < a < b; B - a < b < c; C - c < b < a; D - d < c < a < b; E - c < a < b; F - a > b > c
14. A - H_c H_a H_b; B - H_d H_c H_b H_a
15. A - 1, 3, 4, 6, 7, 8, 9; B - 2, 5; C - 6; D - 3, 4, 6, 7, 9; E - 6, 8, 9; F - 6; G - 7; H - 9; I - 4, 7
16. A - b; B - a; C - d; D - b; E - a; F - d; G - d; H - a
17.

	(i)	(ii)	(iii)	(iv)	(v)
A	6	4	0	10	2
B	0	7	2	9	6
C	5	1	0	5	1
D	1	3	1	3	4
E	2	3	0	2	2
F	10	0	0	10	0
18. a - q; b - p, r; c - p, s; d - q
19. a - p; b - r; c - s; d - q
20. A - c; B - d
21. A - b; B - b; C - a
22. a - p, s; b - q, r; c - q, s; d - p, s
23. a - p, r, s; b - q, r; c - q, r; d - p, r, s

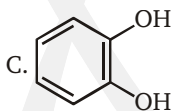
24. A – b; B – c; C – a; D – c; E – a; F – b; G – d
 25. A – d; B – a; C – d; D – c
 26. A – c; B – b; C – b; D – b; E – b; F – a; G – b; H – c; I – b
 27. i. – sp ; ii. – sp ; iii. – sp^2 ; iv. – sp^2 ; v. – sp^3 ; vi. – sp^3
 28. a – forward b – backward
 29. a – p, q, s; b – p, q, r, s; c – p, q, r, s; d – p, q, r, s
 30. a – s; b – r; c – q; d – p
 31. a – r; b – q; c – s; d – p
 32. b
 33. 65
 34. a – q; b – r; c – s; d – p
 35. A – d; B – a; C – a
 36. A 3; B 1; C 5; D 2; E 4
 37. Aromatic— 3, 4, 5, 9, 12, 13, 15, 16, 17, 19, 22, 24, 25, 26, 27, 28, 29, 31, 32, 33, 34, 35, 37, 38, 39, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 53, 54, 56, 57, 58, 61, 62, 63, 66
 Non-aromatic— 1, 6, 7, 8, 18, 20, 23, 30, 40, 55, 64, 65

Anti-aromatic— 2, 10, 11, 14, 21, 36, 52, 59, 60

38. 1 B; 2 A; 3 B; 4 A; 5 A; 6 A; 7 B; 8 B; 9 A; 10 A

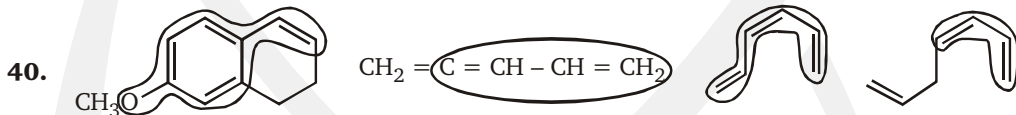
39. A. NH_4Cl , cation-anion interaction

B.  CH_3 , van der Waals' forces



, H-bonding (Also dipole-dipole)

D.  NH , Aromatic stacking



41. A. A 3, B 2, C 1,

B. A 1, B 3, C 2, D 4

C. A 3, B 1, C 4, D 2

D. A 1, B 2, C 3, D 4

42. a s; b r; c q; d p

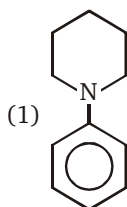
43.

+M

-M

-I

+I

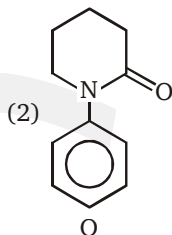


✓

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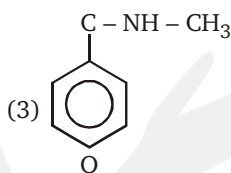


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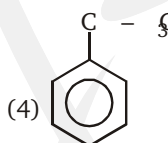


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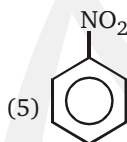


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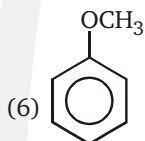


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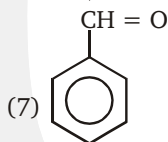


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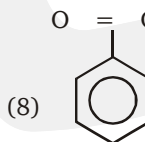


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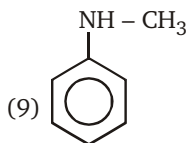


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44. A PA; B PP; C NPA; D PA

45. (i) B, (ii) B, (iii) B, (iv) A

46. (a) HBr ; (b) $\text{CH}_3 - \text{CH}_2 - \text{OH}$; (c) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{CH}_3$;(d) $\text{CH}_3 - \text{CH}_2 - \text{CH}_3$; (e) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{CH}_3$ 47. (a) BF_3 ; (b) HO^- ; (c) HO^- ; (d) HI ; (e) HOCl

48. In A, highly electronegative F-atoms are present at the periphery. In liquid term these F-atoms will repel each other due to partial negative charge and thus A will have lower b.pt.

49. H_d H_a H_b H_c 50. A- H_a 1° alkyl; H_b 2° alkyl; H_c 2° allyl; H_d vinyl; H_e 3° allyl
B- Easiest to abstract: H_e H_c H_b H_a H_d Hardest to abstract51. A- H_a 2° alkyl; H_b 2° allyl; H_c vinyl; H_d 3° allyl; H_e 1° alkyl
B- H_c H_e H_a H_b H_d

52. A-2; B-1; C-3; D-4

53. A-1; B-4; C-3; D-2

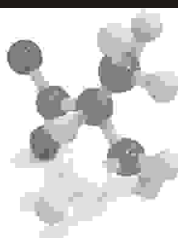
54. a-r; b-t; c-t; d-s

Subjective Problems

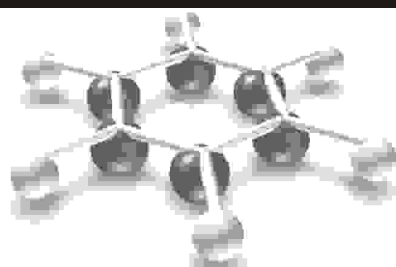
1. 21 2. 11 3. 3 4. 6 5. 4 6. 13 7. 4 8. 7

9. 2 (c, d) 10. 3 (a, c, f) 11. 7

12. P 3, Q 3, R 7, S 2, T 4 19 15 4 13. 4



2

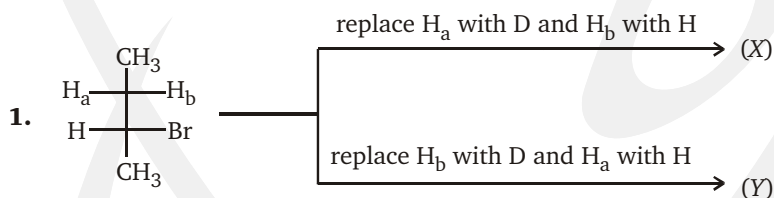


ISOMERISM

(Structural & Stereoisomerism)



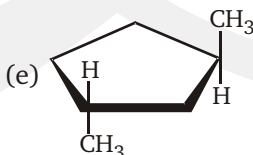
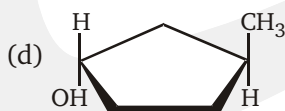
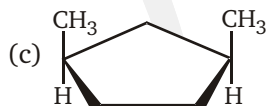
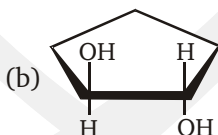
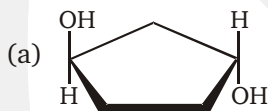
Level - 1



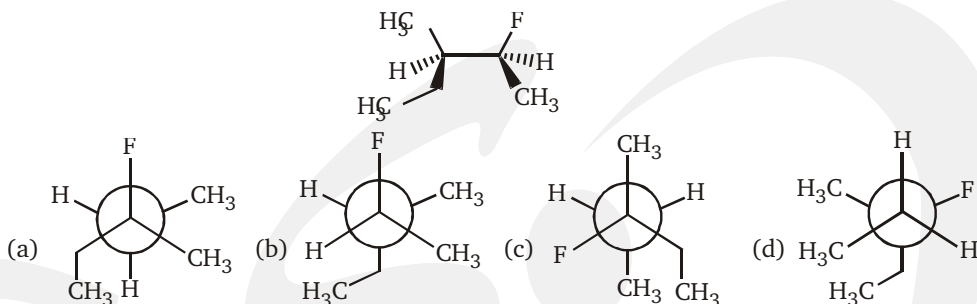
Relation between (X) and (Y) is :

- (a) enantiomers (b) diastereomers
(c) *E* and *Z* isomer (d) constitutional isomer

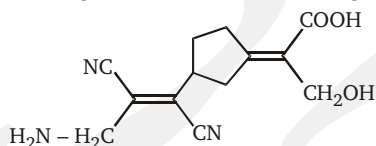
2. Which of the following cyclopentane derivative is optically **inactive** ?



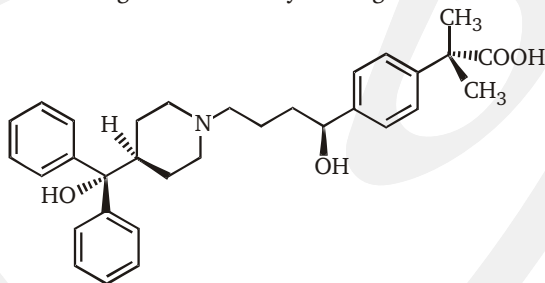
3. Which is the most stable conformer along the 2, 3 C – C bond axis of the compound ?



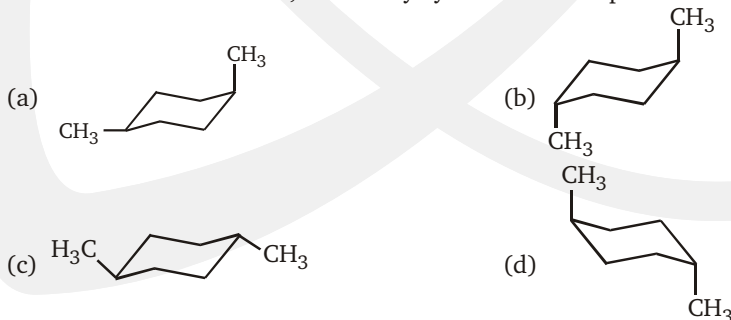
4. Assign double bond configurations to the following :



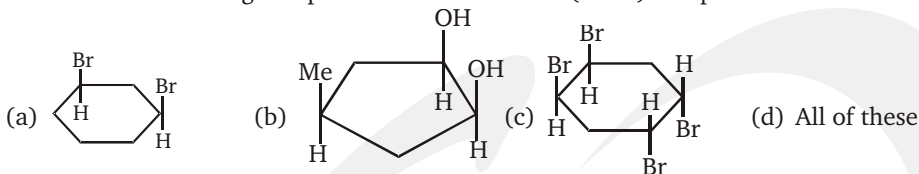
- (a) E (b) Z (c) E, E (d) Z, Z
5. Allegra, a common prescription drug with the structure shown below, is given for the treatment of seasonal allergies. How many stereogenic carbon does Allegra possess ?



- (a) 1 (b) 2 (c) 3 (d) 4
6. How many meso isomers of $C_4H_8Cl_2$ will be ?
- (a) 0 (b) 1 (c) 2 (d) 3
7. The stable form of *trans*-1, 4-dimethylcyclohexane is represented as:



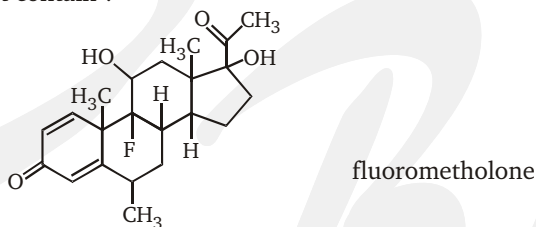
8. Which of the following compound is non-resolvable (meso) compounds ?



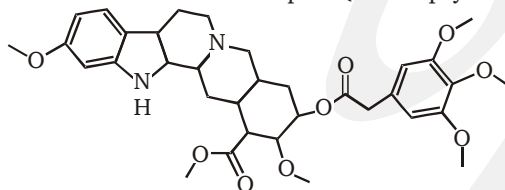
9. $\text{HO}-\underset{(2)}{\text{CH}_2}-\underset{(3)}{\text{CH}_2}-\text{F}$

Which conformer of above compound is most stable across C_2-C_3 ?

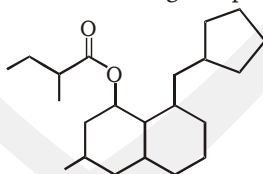
- (a) staggered (b) eclipsed (partially)
(c) gauche (d) fully eclipsed
10. The following molecule is fluorometholone, a steroidal anti-inflammatory agent. How many stereogenic centers does it contain ?



- (a) 5 (b) 6 (c) 7 (d) 8
11. How many chiral carbons are there in Reserpine (an antipsychotic drug) ?

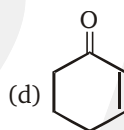
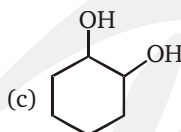
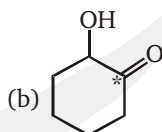
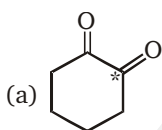
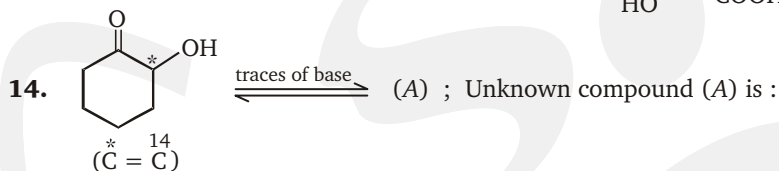
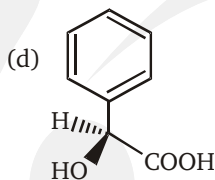
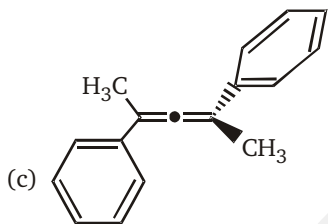


- (a) 9 (b) 8 (c) 7 (d) 6
12. How many chiral centers are in the following compound ?

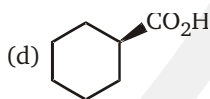
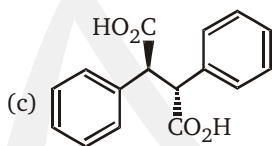
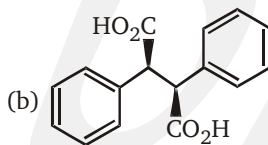
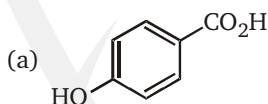


- (a) 4 (b) 5 (c) 6 (d) 7
13. Among the following, the optically inactive compound is:

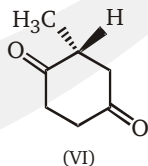
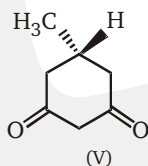
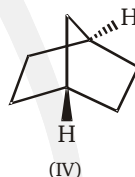
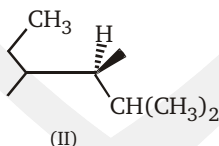
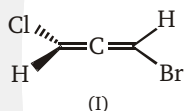




15. Which of the following compounds might be useful to the chemist trying to increase the optical purity of the (d) sample ?



16. Which of the following molecules is (are) chiral ?



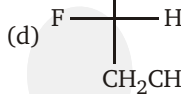
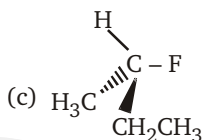
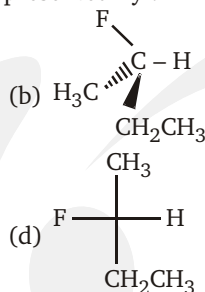
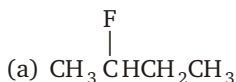
(a) I and II

(b) III and IV

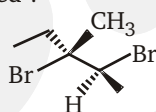
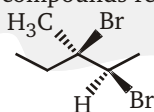
(c) II, IV and VI

(d) I, II, III and VI

17. The structure of (S)-2-fluorobutane is best represented by :

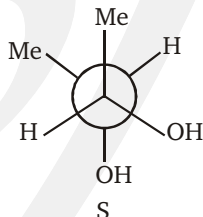
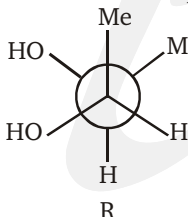
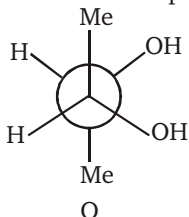
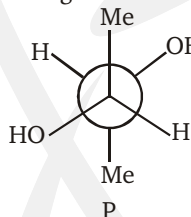


18. How are the following compounds related ?



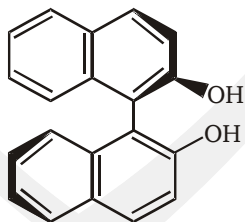
- (a) Diastereomers
(b) Enantiomers
(c) Meso compounds
(d) Identical
19. Which one of the following is chiral ?
(a) 1, 1-Dibromo-1-chloropropane
(b) 1, 3-Dibromo-1-chloropropane
(c) 1, 1-Dibromo-3-chloropropane
(d) 1, 3-Dibromo-2-chloropropane

20. Among the following, the Newmann projections of meso-2, 3-butanediol are :



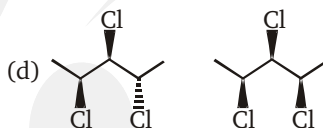
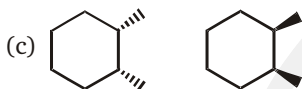
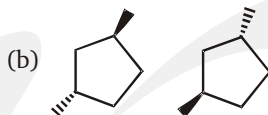
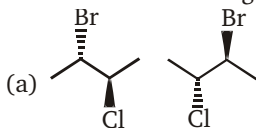
- (a) P, Q
(b) P, R
(c) R, S
(d) Q, S

21. The binaphthol (**Bnp**) is:

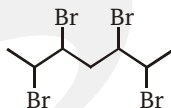


- (a) an optically active compound having chiral centre
(b) an optically inactive compound
(c) a meso compound
(d) an optically active compound without having chiral centre

22. Which of the following pairs of compounds is a pair of enantiomers ?



23. The maximum number of stereoisomers that could exist for the compound below ?



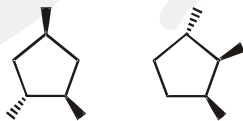
(a) 6

(b) 8

(c) 10

(d) 16

24. The following pair of compounds is best described as :



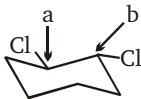
(a) identical

(b) diastereomers

(c) enantiomers

(d) none of the above

25. Determine the absolute configurations of the labeled carbons (a and b):



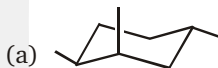
(a) $a = R ; b = R$

(b) $a = R ; b = S$

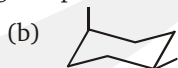
(c) $a = S ; b = R$

(d) $a = S ; b = S$

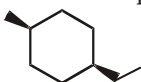
26. Which of the structures (a – d) will be produced if a “ring flip” occurs in the following compound in chair form ?

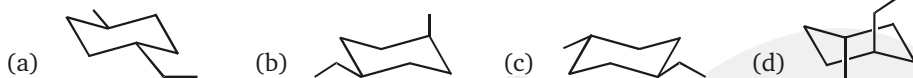


27. Which of the following compounds is most stable ?

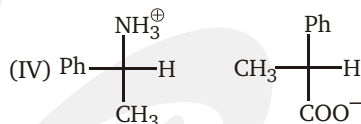
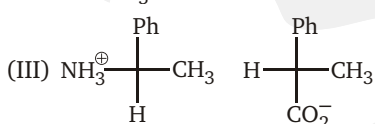
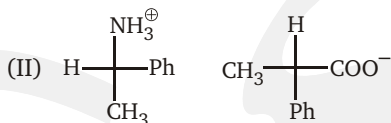
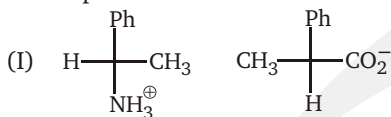


28. Which is the most stable chair form of this compound ?



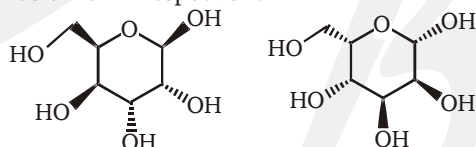


29. Which pairs of the salts would have identical solubilities in methanol ?



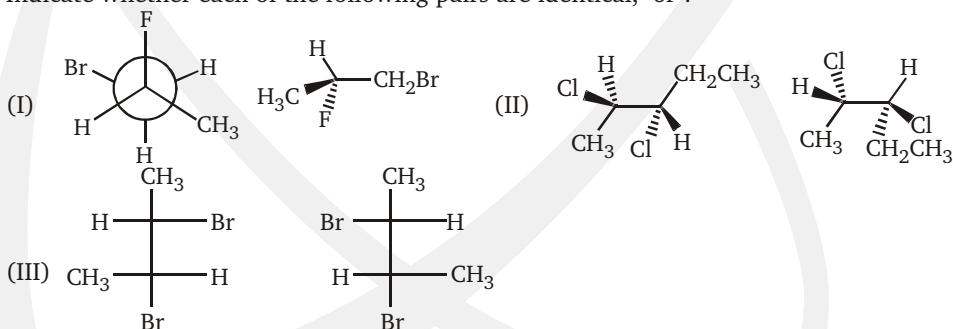
- (a) I & IV (b) I & III (c) I & II (d) II & IV

30. The following compounds differ in respect of :



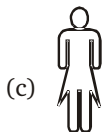
- (a) their chemical and physical properties
(b) nothing
(c) the direction in which they rotate plane of polarized light
(d) their interactions with molecules

31. Indicate whether each of the following pairs are identical, or ?



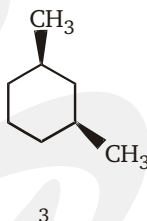
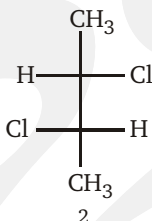
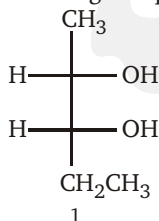
- I (a) enantiomers (b) identical (c) enantiomers (d) enantiomers
- II (a) diastereomers (b) enantiomers (c) diastereomers (d) identical
- III (a) enantiomers (b) enantiomers (c) identical (d) identical

32. Which of the following is achiral ?



(d) a molecule of 3-methylheptane

33. Which of the following compounds are meso forms ?



- (a) 1 only
(c) 1 and 2

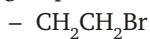
- (b) 3 only
(d) 2 and 3

34. The separation of a racemic mixture into pure enantiomers is termed as :

- (a) Racemization
(c) Resolution

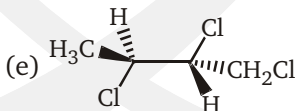
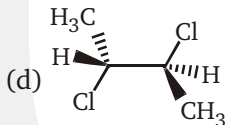
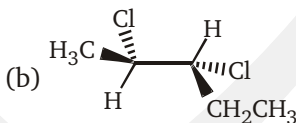
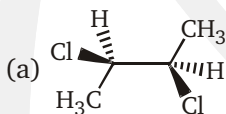
- (b) Isomerization
(d) Equilibration

35. Rank of the following groups in order of R, S precedence (IV is highest) :

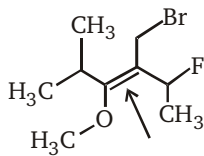


1	2	3	4	I	II	III	IV
I	II	III	IV	I	II	III	IV
(a) 3	2	4	1	(b) 1	4	2	3
(c) 3	4	1	2	(d) 3	4	2	1

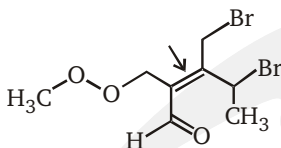
36. Which of the following is a meso compound ?



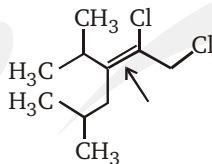
37. Among the following structures, select E isomers (arrows indicate the bonds to be considered) ?



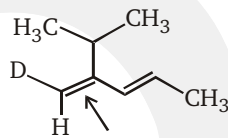
(1)



(2)



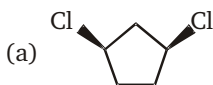
(3)



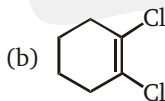
(4)

- (a) 1 and 2 (b) 1 and 3 (c) 1 and 4 (d) 2 and 3

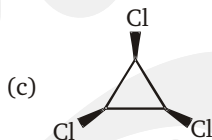
38. Which of the following compounds has a zero dipole moment ?



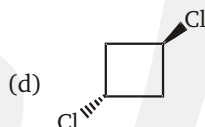
(a)



(b)



(c)

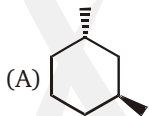


(d)

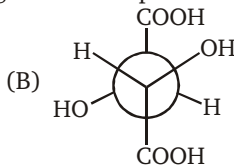
39. On Pluto, where everything is frozen, astronauts discovered two forms of butane gauche and anti. Assuming that there are no rotations around single bonds, which statement about the two forms is correct ?

- (a) They are enantiomers
(b) They are diastereoisomers
(c) They are meso compounds
(d) The gauche form has two stereogenic centers, and the anti has only one

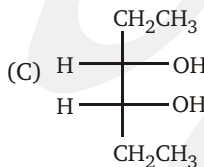
40. Which of the following will show optical activity ?



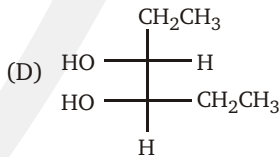
(A)



(B)



(C)

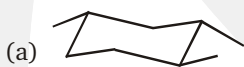


(D)

(E) 50/50 mixture of C and D

- (a) A, D and E (b) A and E only (c) B, C and D (d) All except C

41. Among the structure shown below, which has lowest potential energy ?



(a)



(b)

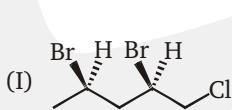


(c)

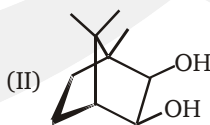


(d)

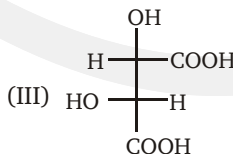
42. Which of the following molecules is/are chiral ?



(I)



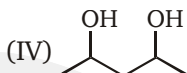
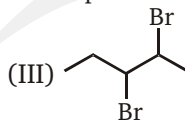
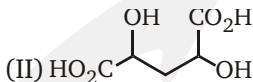
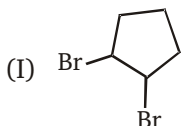
(II)



(III)

- (a) I (b) II (c) III (d) I, II

43. A compound was synthesized by a student, but its structure was not identified. However, his wonderfully helpful instructor told him that it was a meso compound with 5 carbons and 2 stereogenic centers. Which of the following structures should the student consider as possibilities for his compound?



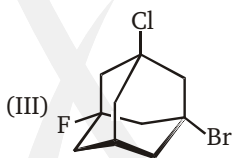
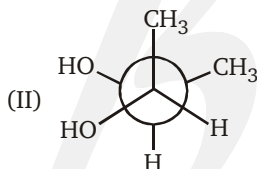
- (a) I, II, IV (b) II, IV (c) I, III, V (d) II, IV, V

44. How many isomers are possible for the following molecule?



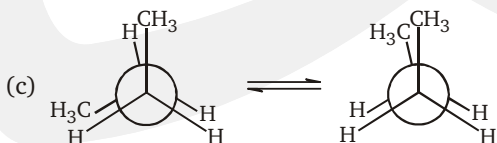
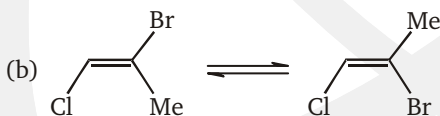
- (a) 1 (b) 2 (c) 3 (d) 4

45. Which of the following molecules are chiral?

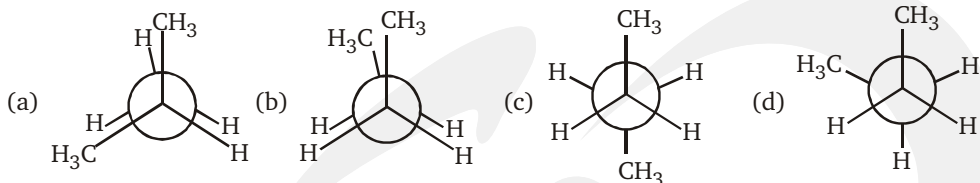


- (a) I, II, III and IV (b) II, III and IV (c) II and IV (d) I and II

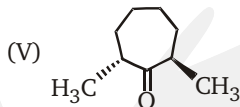
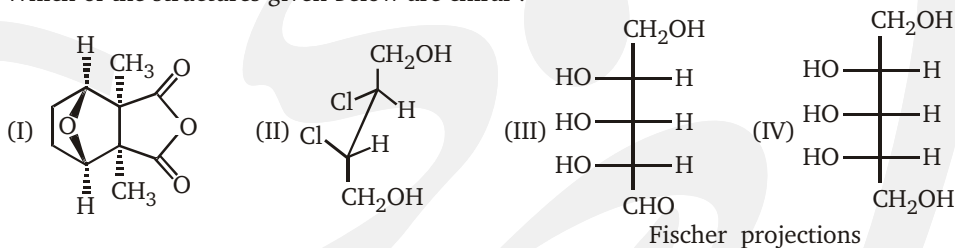
46. Which equilibrium is not rapid at room temperature?



47. Which is the lowest energy conformation of butane ?

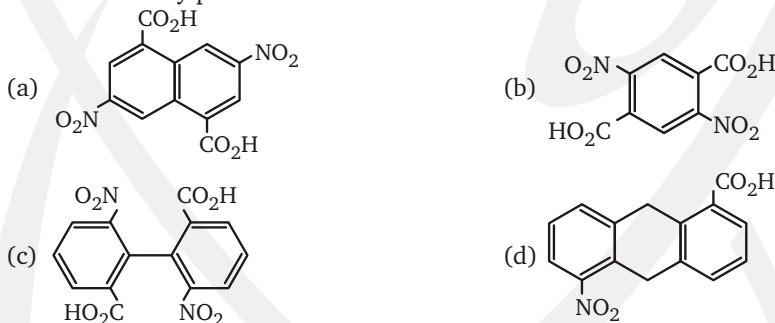


48. Which of the structures given below are chiral ?

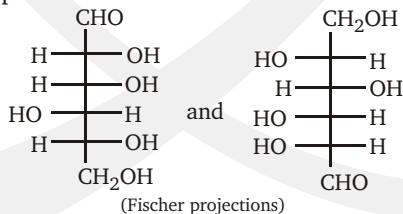


- (a) I, II, III (b) II, III, V (c) II, III (d) I, II

49. Which of the following carboxylic acids could be resolved by reaction with an enantiomerically pure chiral amine ?

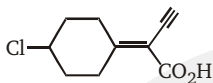


50. What is the relationship between the molecules in the following pairs ?



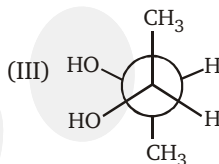
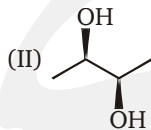
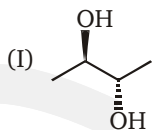
- (a) enantiomers (b) diastereomers (c) identical (d) structural isomers

51. What are the correct designations for the structure below ?



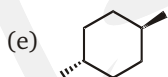
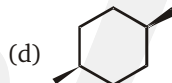
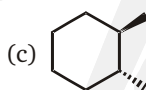
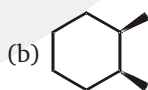
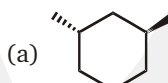
- (a) E, E (b) Z, E
(c) E, Z (d) No geometrical isomers are possible

52. Which of the following molecules are chiral ?

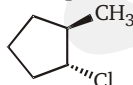
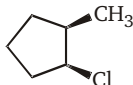


- (a) I and III (b) I and V (c) II and III (d) II, III, IV

53. Which one of the following isomeric structures has the lowest energy ?

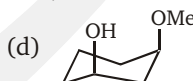
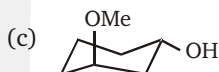
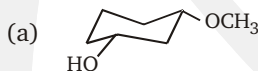


54. The following compounds are identical with respect to :

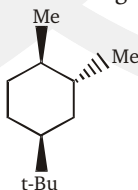


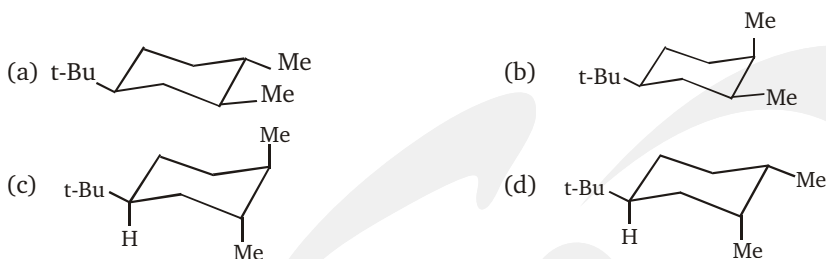
- (a) molecular composition (b) boiling point
(c) melting point (d) IUPAC name

55. Among the following, the most stable isomer is :



56. The most stable conformation of the following compound is :



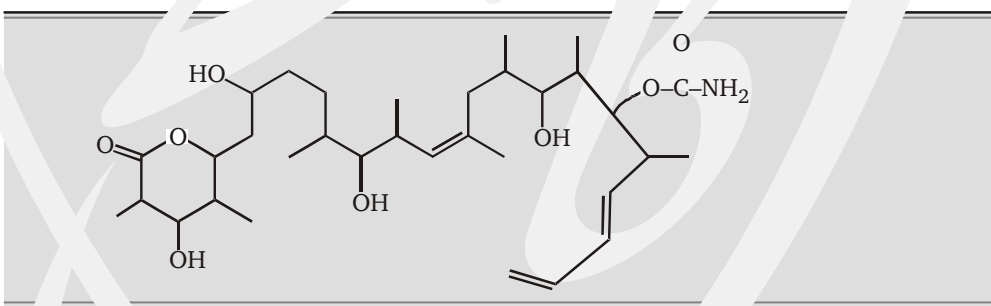


57. Which of the following molecules have non-zero dipole moments ?

- (I) gauche conformation of 1, 2-dibromoethane
 (II) anti conformation of 1, 2-dibromoethane
 (III) *trans*-1, 4-dibromocyclohexane
 (IV) *cis*-1, 4-dibromocyclohexane
 (V) tetrabromomethane
 (VI) 1, 1-dibromocyclohexane

- (a) I and II
 (b) I and IV
 (c) II and V
 (d) I, IV and VI

58. What is the maximum number of stereoisomers possible for discodermolide ?



- (a) 2^{14} (b) 2^{15} (c) 2^{16} (d) 2^{17}

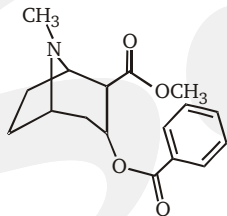
59. An aqueous solution containing compounds *A* and *B* shows optical activity. *A* and *B* are stereoisomers. Which of the following possibilities cannot be correct ?

- (a) *A* has two chiral centers, but *B* does not have any because it has a symmetry plane
 (b) *A* and *B* are enantiomers
 (c) *A* and *B* are diastereomers
 (d) *A* and *B* are not present in equal amounts

60. Which of the following structures represents the lowest-energy form of (1*S*, 2*S*, 4*R*)-trimethyl-cyclohexane ?

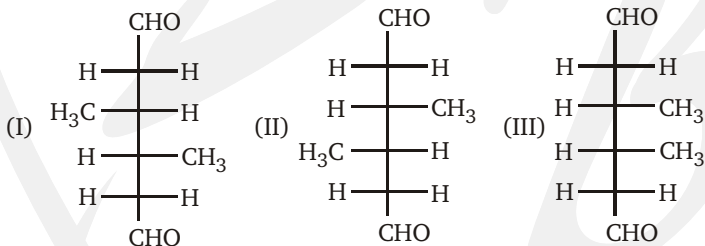


61. Which one of the following is a diastereomer of (R)-4-bromo-cis-2-hexene ?
- (S)-4-bromo-cis-2-hexene
 - (S)-5-bromo-trans-2-hexene
 - (R)-4-bromo-trans-2-hexene
 - (R)-5-bromo-trans-2-hexene
62. The structural formula of cocaine is shown below. How many stereogenic carbon atoms are there in this molecule ?

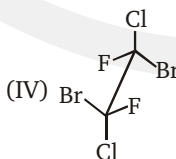
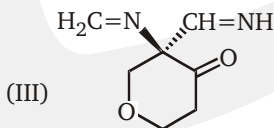
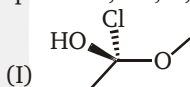


- 1
- 2
- 3
- 4

63. Which of the following statements best describes the stereochemical relationships of compound I, II and III shown below ?

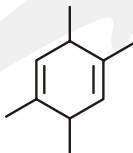


- All compounds are chiral
 - None of the compounds is chiral
 - I and II are meso compounds
 - I and II are diastereomers, and III is a meso compound
 - I and II are chiral
64. What is the absolute configuration of the following molecules ? (NS = the molecule has no center) Note : For the purpose of this question only, the order of stereocenters is not specified ; i.e., R, S = S, R.



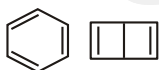
- | | I | II | III | IV | | I | II | III | IV |
|-----|---|------|-----|----|-----|---|------|-----|------|
| (a) | R | R, S | R | NS | (b) | R | R, R | S | R, R |
| (c) | R | R, S | NS | NS | (d) | R | R, S | R | R, S |

65. The number of all the possible stereoisomers formed by the given compound is :

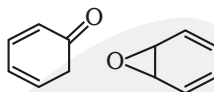


- (a) 2 (b) 3 (c) 32 (d) 64

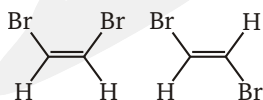
66. The relationship among the following pairs of isomers is:



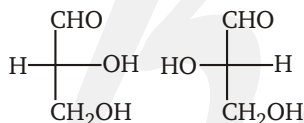
(I)



(II)



(III)



(IV)

I	A: Constitutional
II	B: Configurational
III	C: Conformational
IV	D: Optical

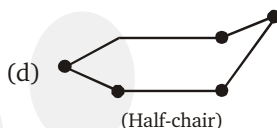
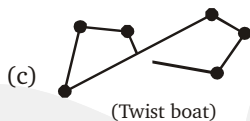
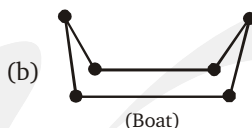
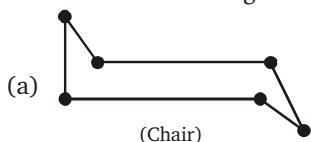
- (a) I A,II B,III B,IV D (b) I A,II A,III B,IV D
 (c) I B,II A,III B,IV D (d) I B,II B,III A,IV B

67. The structural formula of sativene is shown below. How many stereogenic centers are there in this molecule ?

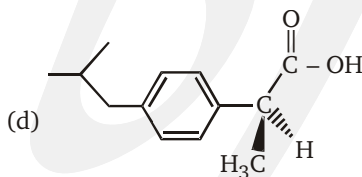
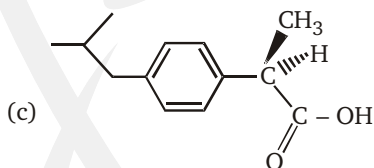
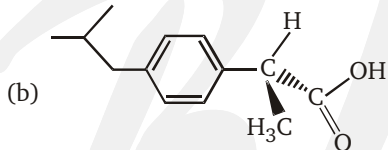
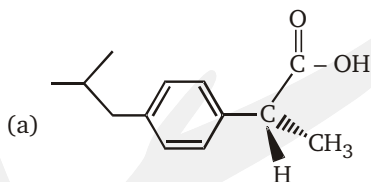


- (a) 2 (b) 3
 (c) 4 (d) 5

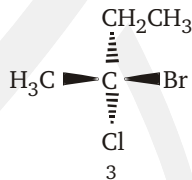
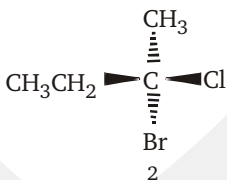
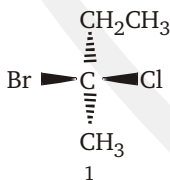
68. Which of the following is the least stable conformer of cyclohexane ?



69. The S- enantiomer of ibuprofen is responsible for its pain-relieving properties. Which one of the following structures shown below is (S)-ibuprofen ?



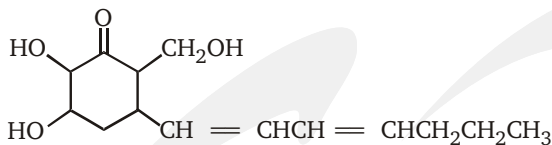
70. Which of the following depict the same ?



- (a) 1 and 2
(c) 2 and 3

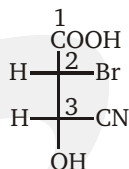
- (b) 1 and 3
(d) 1, 2, and 3

71. A naturally occurring substance has the constitution shown below. How many may have this constitution ?



- (a) 2 (b) 8 (c) 16 (d) 64

72. The absolute configurations of the two centers in the following molecule are :

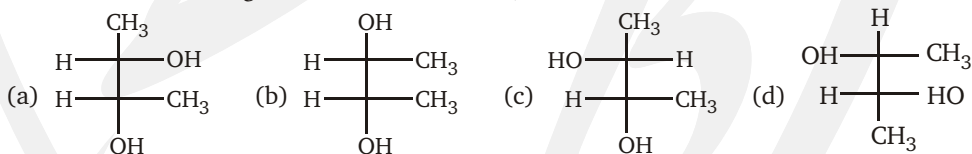


- (a) 2(R), 3(S) (b) 2(R), 3(R) (c) 2(S), 3(S) (d) 2(S), 3(R)

73. The total number of stereoisomer possible for 2, 3-dichloro butane :

- (a) 2 (b) 3 (c) 4 (d) 5

74. Which of the following structure is not meso-2,3-butanediol ?

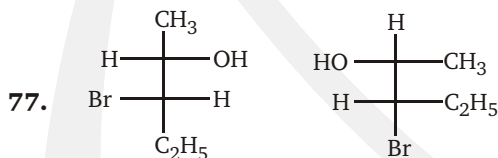


75. A solution of optically active 1-phenylethanol racemizes in acidified aqueous medium. It is due to :

- (a) enolization (b) carbonium ion formation
(c) carbanion formation (d) reversible oxidation-reduction

76. The most stable conformation of ethylene glycol is :

- (a) Anti (b) Gauche (c) Partially eclipsed (d) Fully eclipsed



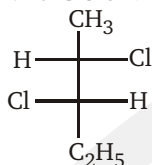
77. The molecules represented by the above two structures are :

- (a) identical (b) enantiomers
(c) diastereomers (d) epimers

78. The correct order of priority of groups SCH_3 (I), —NO_2 (II), —CH (III) and $\text{—CH}_2\text{C}_6\text{H}_5$ (IV), on the basis of CIP classification, is (increasing order) :

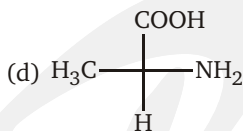
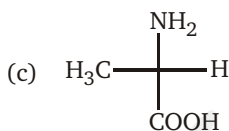
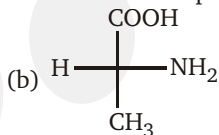
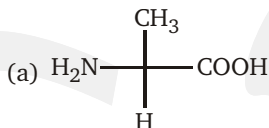
- (a) I, III, II, IV (b) IV, III, II, I
(c) II, IV, I, III (d) III, IV, II, I

79. The configuration at C-2 and C-3 of the compound given :

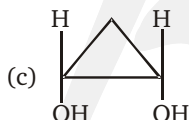
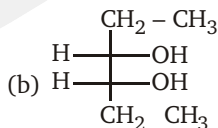
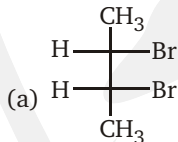


- (a) 2R, 3S (b) 2S, 3R (c) 2S, 3S (d) 2R, 3R

80. Amongst the following amino acids, the (R) - enantiomer is represented by :



81. Which of the following is a meso compound ?

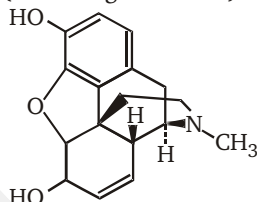


(d) All of these

82. Predict stereochemistry of product when *d* and *l*-amine reacts with *l*-acid:

- (a) Diastereomers (b) Meso (c) Racemic (d) Pure Enantiomer

83. How many chiral center (excluding N centres) are there in morphine?

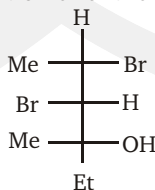


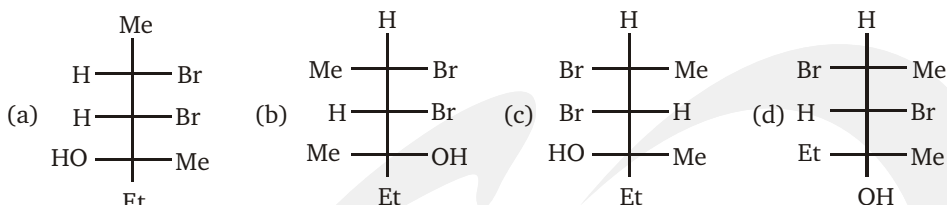
- (a) 4 (b) 5 (c) 6 (d) More than 6

84. Which dimethylcyclobutane is optically active ?

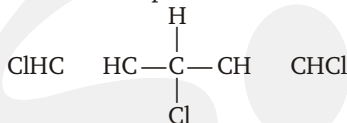
- (a) *trans*-1, 2 (b) *cis*-1, 2 (c) *trans*-1, 3 (d) *cis*-1, 3

85. Which of the following is the enantiomer of the compound shown below ?





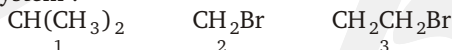
86. How many different stereoisomers are possible for the following compound ?



- (a) 1 (b) 2 (c) 3 (d) 4
87. The following compounds are best described as :
(R)-PhCH(OH)CH₃ and (S)-PhCH(OH)CH₃

- (a) enantiomers
(b) diastereomers
(c) not stereoisomers
(d) conformational isomers (differing by single bond rotation)

88. Rank the following substituent groups in order of decreasing priority according to the Cahn-Ingold-Prelog system :



- (a) 2 > 3 > 1 (b) 1 > 3 > 2 (c) 3 > 1 > 2 (d) 2 > 1 > 3

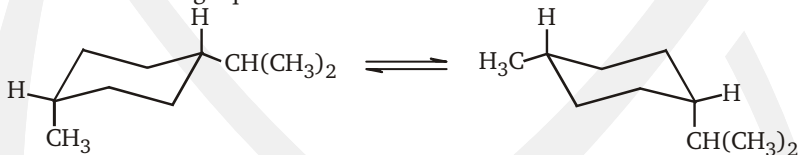
89. Compare the stabilities of the following two compounds :

A : *cis*-1-Ethyl-3-methylcyclohexane

B : *trans*-1-Ethyl-3-methylcyclohexane

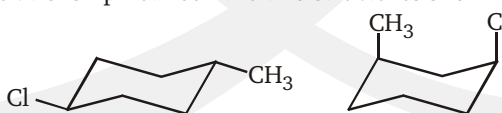
- (a) A is more stable (b) A and B are of equal stability
(c) B is more stable (d) No comparison can be made

90. What, if anything, can be said about the magnitude of the equilibrium constant K for the following equilibrium ?



- (a) $K = 1$ (b) $K < 1$
(c) $K > 1$ (d) No estimate of K can be made

91. What is the relationship between the two structures shown ?



- (a) Constitutional isomers
(b) Stereoisomers
(c) Different drawing of the same conformation of the same compound
(d) Different conformation of the same compound



92. Which of the following statements is true ?

- (a) van der Waals' strain in *cis*-1, 2-dimethylcyclopropane is the principal reason for its decreased stability relative to the *trans* isomer
 (b) Cyclohexane gives off more heat per CH_2 group on being burned in air than any other cycloalkane
 (c) The principal source of strain in the boat conformation of cyclohexane is angle strain
 (d) The principal source of strain in the gauche conformation of butane is torsional strain

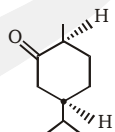
93. $\text{Ph}-\text{CH}(\text{NO}_2)\text{H}$ isomerises in 3 days to (x), Isomer (x) is :

- (a) $\text{Ph}-\text{NO}-\text{CH}_2\text{OH}$ (b) $\text{Ph}-\text{CH}_2-\text{NO}_2$
 (c) $\text{Ph}-\text{NH}-\text{CO}_2\text{H}$ (d) None

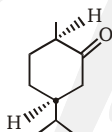
94. Which of the following will not show geometrical isomerism ?

- (a) $\text{CH}_3-\text{C}(\text{CH}_3)(\text{CH}_3)-\text{CH}(\text{CH}_3)-\text{CH}_2-\text{CH}_3$ (b) $\text{CH}_3-\text{CH}(\text{CH}_3)-\text{CH}(\text{CH}_3)-\text{CH}_2-\text{CH}_3$
 (c) $\text{CH}_3-\text{CH}(\text{CH}_3)-\text{CH}(\text{CH}_3)-\text{CH}_3$ (d) $\text{CH}_3-\text{CH}_2-\text{CH}(\text{CH}_3)-\text{CH}_2-\text{CH}_3$

95. The two compounds shown below are :

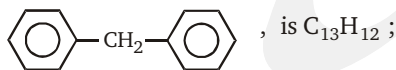


and



- (a) diastereomers (b) enantiomers (c) epimers (d) regiomers

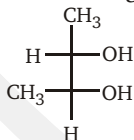
96. The molecular formula of diphenylmethane,



How many structural isomers are possible when one of the hydrogen is replaced by a chlorine atom ?

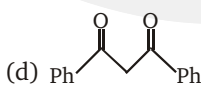
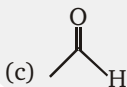
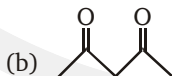
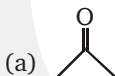
- (a) 6 (b) 4 (c) 8 (d) 7

97. Correct configuration of the following molecule is :



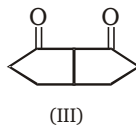
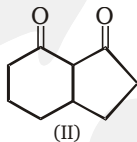
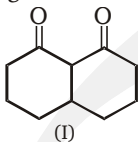
- (a) 2S, 3S (b) 2S, 3R (c) 2R, 3S (d) 2R, 3R

98. Maximum enol content is in :

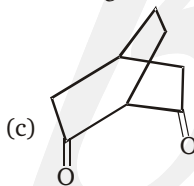
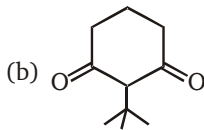
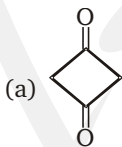


99. Which of the following will have one of the stereoisomer meso ?
 (a) 2-chlorobutane (b) 2, 3-dichlorobutane
 (c) 2,3-dichloropentane (d) 2-hydroxypropanoic acid

100. The correct decreasing order in the enol content of following molecules is :

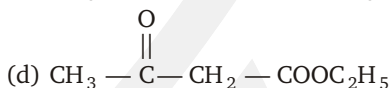
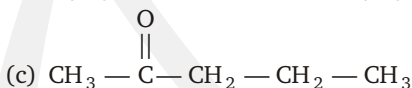
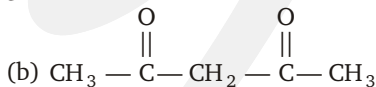
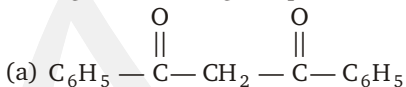


- (a) I > II > III (b) II > I > III (c) III > II > I (d) II > III > I
101. Total number of stereoisomers of the compound 1-bromo-3-chlorocyclobutane is:
 (a) 0 (b) 1 (c) 2 (d) 3
102. Total number of stereoisomers of the 1,3-dichlorocyclohexane is:
 (a) 0 (b) 1 (c) 3 (d) 4
103. Total number of stereoisomers of the compound 1, 4-dichlorocyclohexane is :
 (a) 0 (b) 1 (c) 2 (d) 4
104. Total number of stereoisomers of the compound 2-4-dichloroheptane is:
 (a) 0 (b) 2 (c) 3 (d) 4
105. In which of the following keto form is more dominating than enol form:

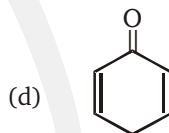
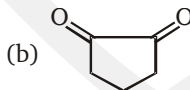
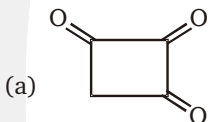


(d) all of these

106. Among the following compounds, which will give maximum enol content in solution :



107. Which of the following has unstable enol form ?

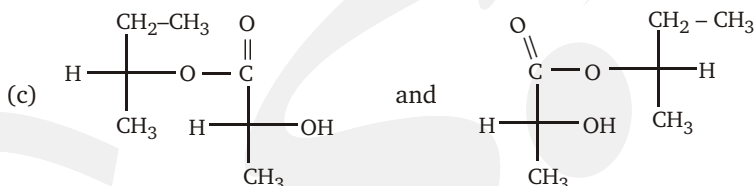
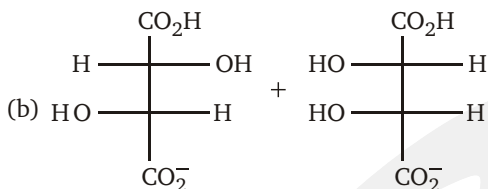


108. Calculate enantiomeric excess of mixture containing 6g of (+) 2-butanol and 4g of (–) -2-butanol.

(a) 10% (b) 20% (c) 40% (d) 33%

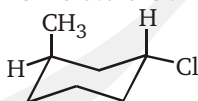
109. Which of the following pair represent pair of diastereomers ?

(a) Meso tartaric acid and (l) tartaric acid



(d) All of these

110. The stereochemistry of this molecule is :



(a) 1R, 3R

(b) 1R, 3S

(c) 1S, 3S

(d) 1S, 3R

111. Pure (S)-2-butanol has a specific rotation of +13.52 degrees. A sample of 2-butanol prepared in the lab and purified by distillation has a calculated specific rotation of +6.76 degrees. What can you conclude about the composition ?

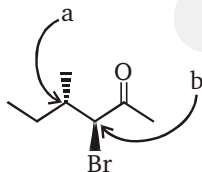
(a) 50% (S), 50% impurity

(b) 50% (S), 50% (R)

(c) 50% (S), 50% racemic

(d) some other mixture

112. Determine the absolute configurations of the chiral centres in the following compound.



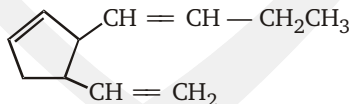
(a) a R; b S

(b) a R; b R

(c) a S; b S

(d) a S; b R

113. Total number of stereoisomers possible for following compound is :



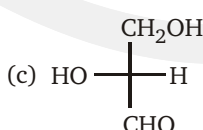
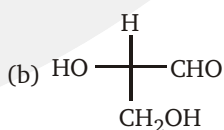
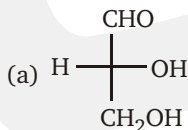
(a) 8

(b) 16

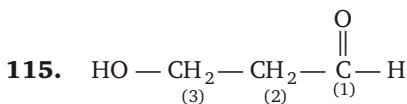
(c) 32

(d) 64

114. Which is the correct structure of D-glyceraldehyde ?

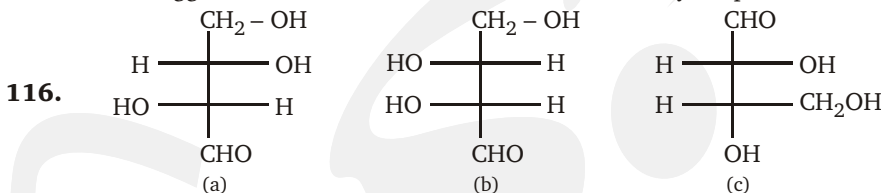


(d) All of these



Which conformer of above compound is most stable (consider conformer across C_2-C_3)

- (a) Staggered (b) Gauche (c) Fully eclipsed (d) Partially eclipsed



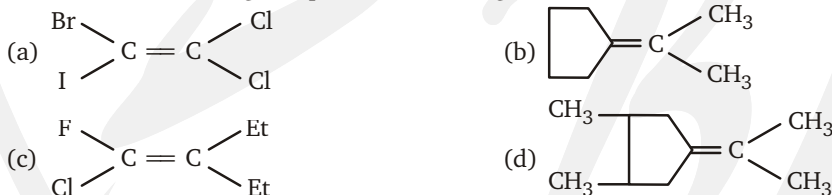
(D) & (L) Configuration of above carbohydrate is :

- (a) L, L, D (b) L, D, L (c) L, L, L (d) L, D, D

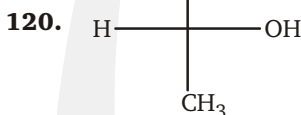
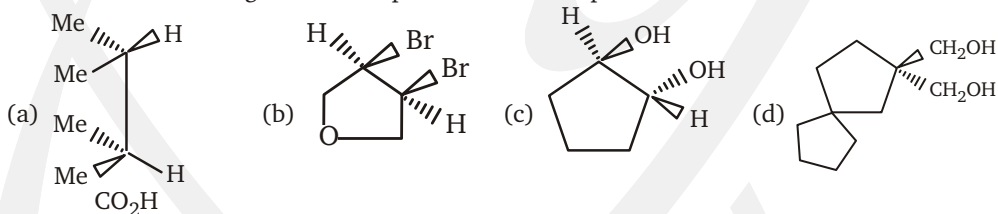
117. How many isomers have the name bromomethylcyclopentane ? (ignoring chirality)

- (a) 4 (b) 5 (c) 6 (d) 7

118. Which of the following compound can show geometrical isomerism ?



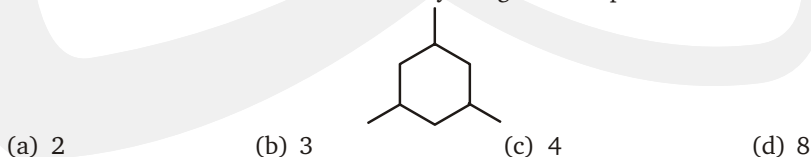
119. Which of the following structure represent meso-compound ?



How many representations of lactic acid are possible in Fischer projection (d & l) ?

- (a) 8 (b) 12 (c) 24 (d) 36

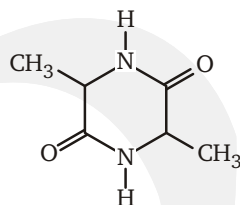
121. Total number of stereoisomer formed by the given compound is :



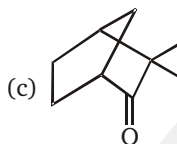
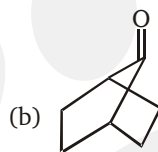
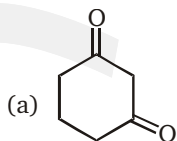
122. The number of stereoisomers formed by the given compound is :

- (a) 2
(c) 4

- (b) 3
(d) 5



123. Which of the following compound does not undergo base - catalyzed exchange in D_2O even though it has an α -hydrogen?

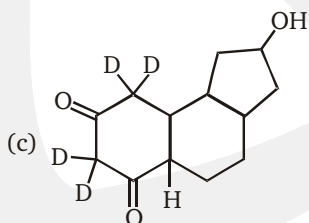
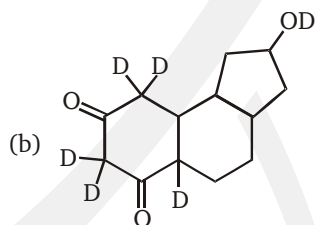
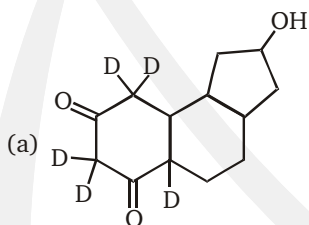


- (d) both (b) & (c)



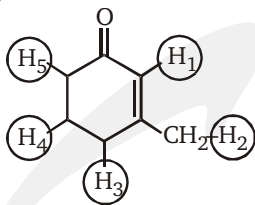
124. D_2O/OH
long time Product

Identify the product formed in the above reaction:

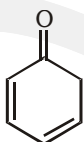


- (d) None of these

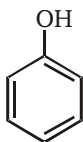
125. In 3-methyl-2-cyclohexenone which hydrogen cannot undergo deuterium exchange when it reacts with $\text{CH}_3\text{O}^- / \text{CH}_3\text{OD}$?

(a) H_1, H_4 (b) H_4 (c) H_3, H_2 (d) H_5, H_3

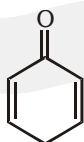
126.



(I)



(II)



(III)

The tautomer of II is :

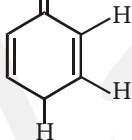
(a) I

(b) III

(c) both I and III

(d) none of these

127.



In the enolization of the given molecule, the H-atom involved is :

(a) -H

(b) -H

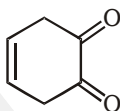
(c) -H

(d) cannot be enolized

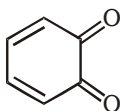
128.



(I)



(II)



(III)

Among the given structure which can exhibit tautomerism ?

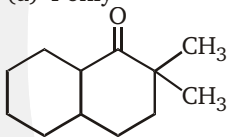
(a) I only

(b) II only

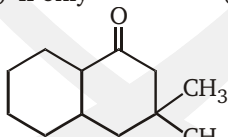
(c) III only

(d) none of these

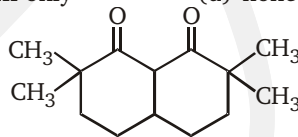
129.



(I)



(II)



(III)

Identify the which can exhibit tautomerism ?

(a) I only

(b) II only

(c) III only

(d) all of these

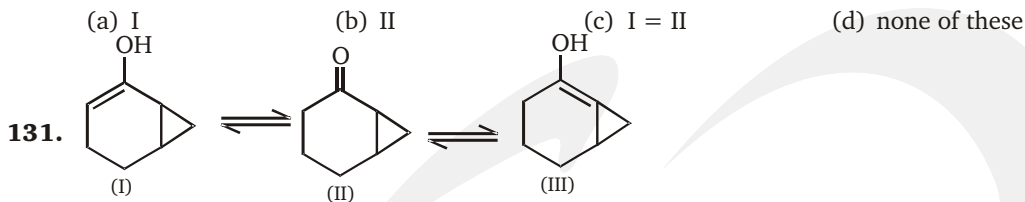
130.



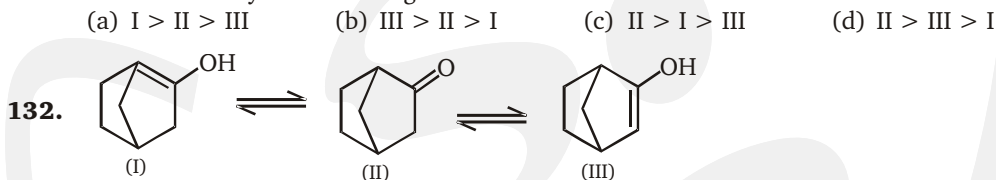
(I)

(II)

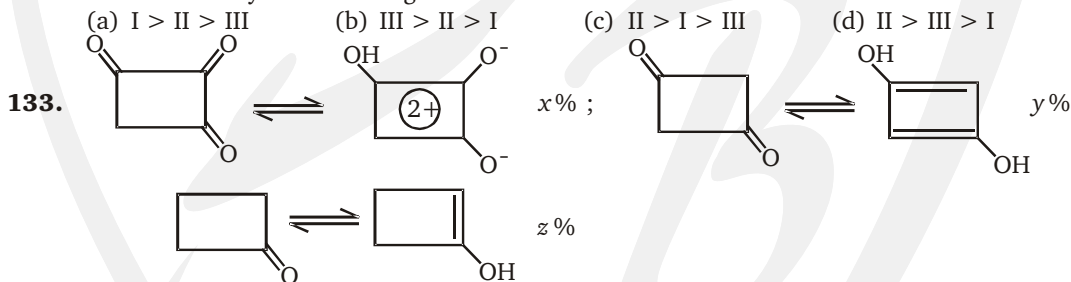
Between the two tautomers which is more stable ?



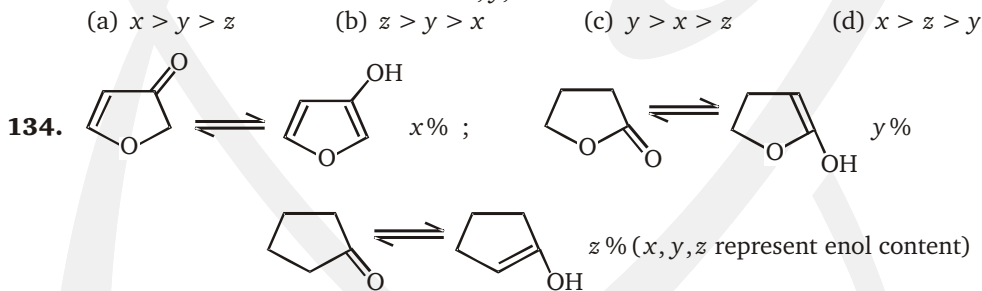
Correct stability order of the given tautomers is :



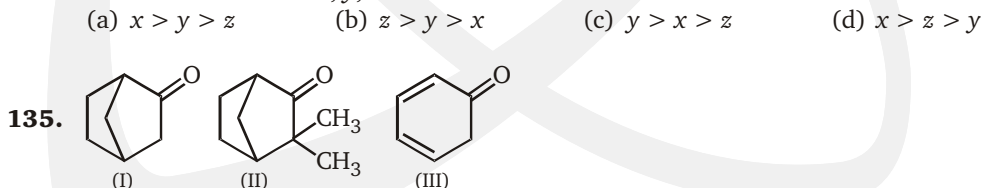
Correct stability order of the given tautomers is :



The correct order of enol contents x, y, z is :

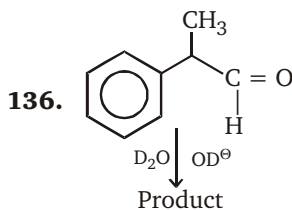


The correct order of x, y, z is :

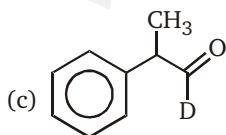
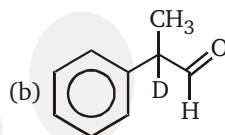
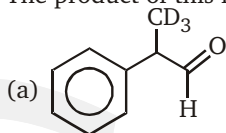


Among the given ketones, the one which does not enolize is :

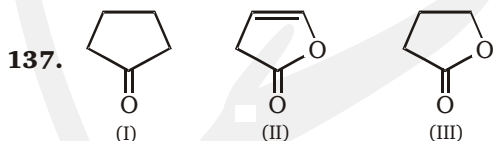
(a) I (b) II (c) III (d) none of these



The product of this reaction should be :



(d) All of these



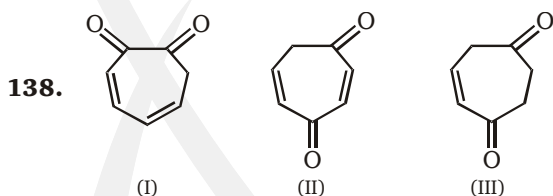
Among the given compounds, the correct order of enol content is :

(a) I > II > III

(b) III > II > I

(c) II > I > III

(d) II > III > I



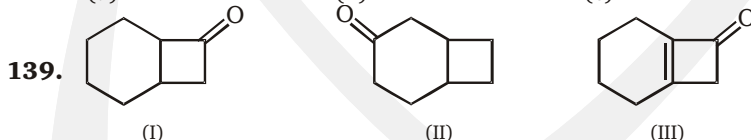
Among the given compounds, the correct order of enol content is :

(a) I > II > III

(b) III > II > I

(c) II > I > III

(d) II > III > I



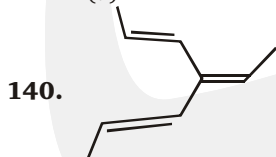
Among the given compounds, the correct order of enol content is :

(a) I > II > III

(b) III > II > I

(c) III > I > II

(d) II > I > III



How many geometrical isomers are possible for the above compound ?

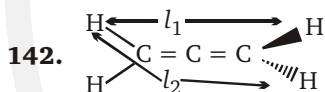
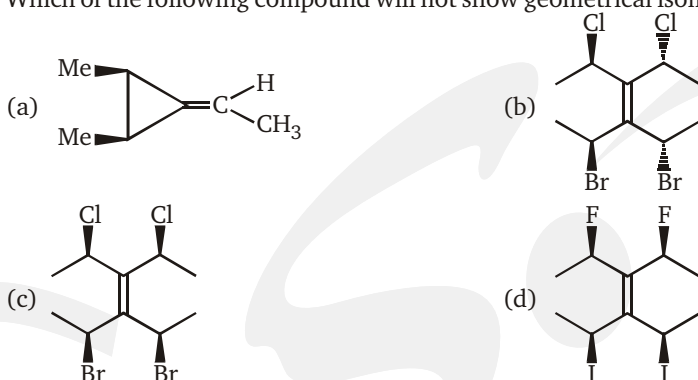
(a) 3

(b) 4

(c) 6

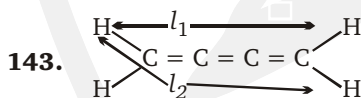
(d) 8

141. Which of the following compound will not show geometrical isomerism across the $\text{C}=\text{C}$ bond ?



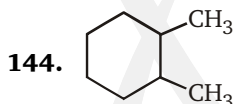
Choose the correct relation between l_1 and l_2 ?

- (a) $l_1 = l_2$ (b) $l_1 < l_2$ (c) $l_1 > l_2$ (d) $l_2 = 2l_1$



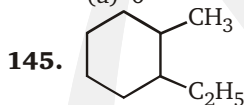
Choose the correct relation between l_1 and l_2 ?

- (a) $l_1 = l_2$ (b) $l_1 < l_2$ (c) $l_1 > l_2$ (d) $l_2 = 2l_1$



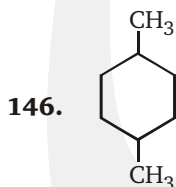
How many geometrical isomers are possible for the above compound ?

- (a) 0 (b) 2 (c) 3 (d) 4



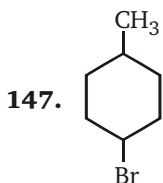
How many geometrical isomers are possible for the above compound ?

- (a) 0 (b) 2 (c) 3 (d) 4



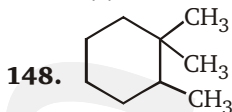
How many geometrical isomers are possible for the above compound ?

- (a) 0 (b) 2 (c) 3 (d) 4



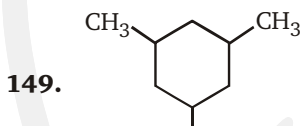
How many geometrical isomers are possible for the above compound ?

- (a) 0 (b) 2 (c) 3 (d) 4



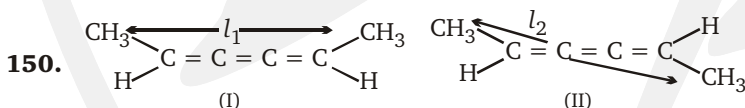
How many geometrical isomers are possible for the above compound ?

- (a) 0 (b) 2 (c) 3 (d) 4



How many geometrical isomers are possible for the above compound ?

- (a) 0 (b) 2 (c) 3 (d) 4



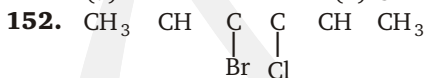
I and II are geometrical isomers of each other because

- (a) l_1 l_2 (b) l_1 l_2
(c) l_2 l_1 (d) l_1 and l_2 cannot be compared.



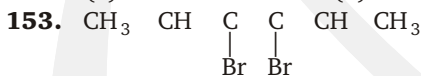
How many geometrical isomers are possible for this compound ?

- (a) 2 (b) 3 (c) 4 (d) 8



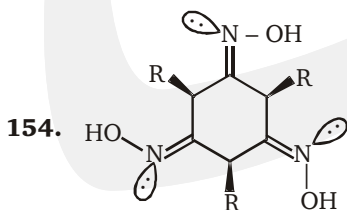
How many geometrical isomers are possible for this compound?

- (a) 2 (b) 3 (c) 4 (d) 6



How many geometrical isomers of this compound are possible ?

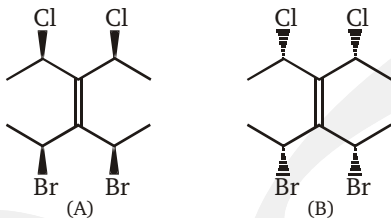
- (a) 2 (b) 3 (c) 4 (d) 6



- (a) chiral
(c) Optically active

- (b) C_3 axis of symmetry
(d) All of these

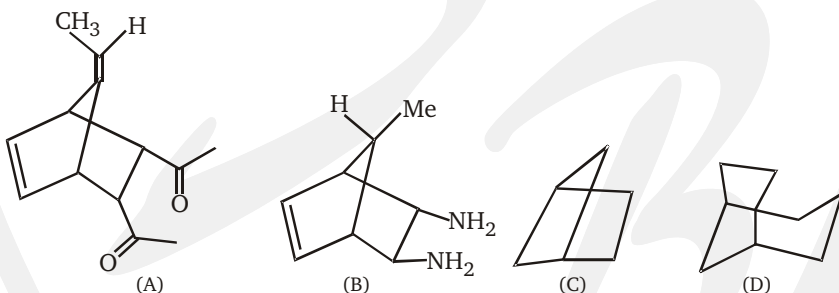
155.



Relationship between above pair (A) & (B) is :

- (a) Enantiomer (b) Diastereomers
(c) Identical (d) Structural isomer

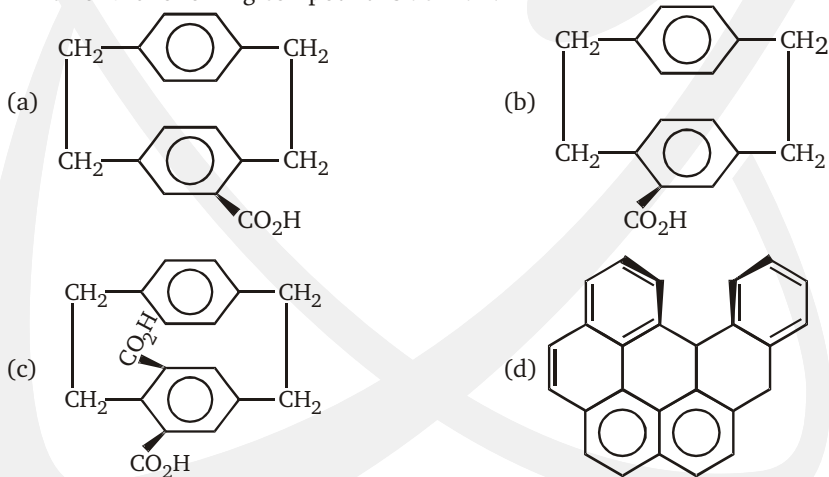
156.



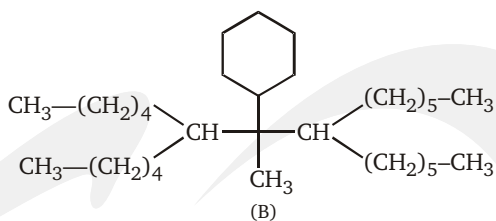
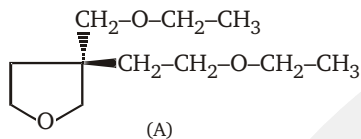
From the above compound (A), (B), (C) & (D) chiral compound is :

- (a) A (b) B (c) C (d) D

157. Which of the following compound is achiral ?



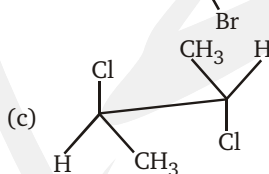
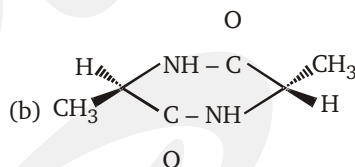
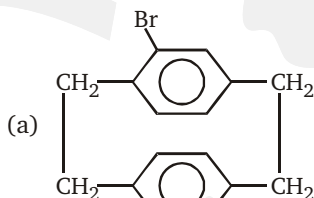
158.



R and S configuration of compound (A) & (B) will be :

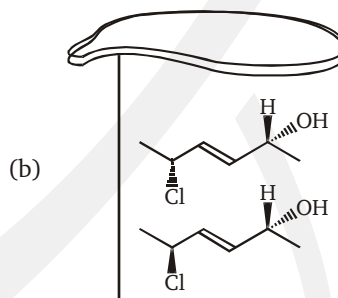
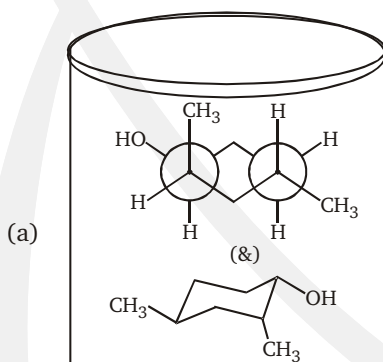
- (a) R, R (b) R, S (c) S, R (d) S, S

159. Which of following compound has center of symmetry?

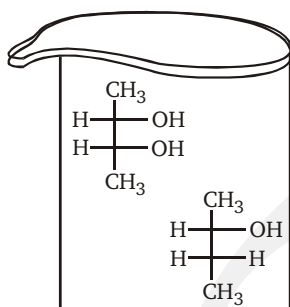


(d) All of these

160. Which mixture of structure in each beaker would rotate plane polarized light ?



(c)



(d) All of these

161. Which of following compound will rotate the plane polarized light at room temperature?

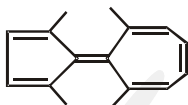
(a)



(b)



(c)

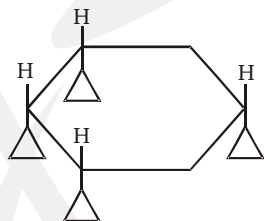


(d)

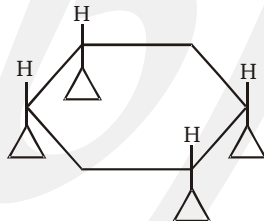


162. Which of the following having plane of symmetry ?

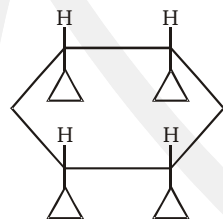
(a)



(b)



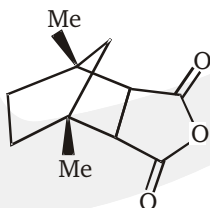
(c)



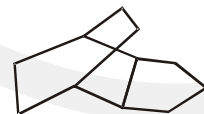
(d) All of these

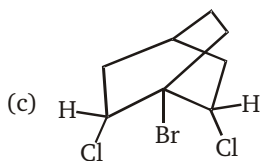
163. Which of following compound is achiral ?

(a)



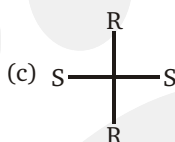
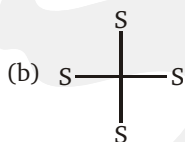
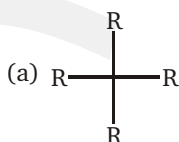
(b)





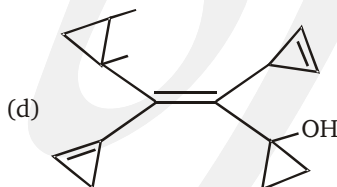
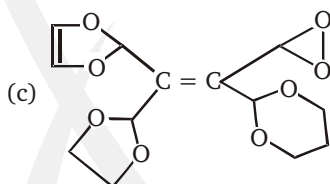
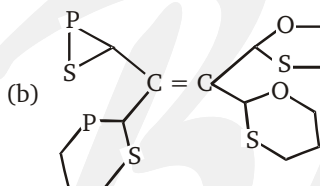
(d) All of these

164. Which of the following compound has plane of symmetry ?

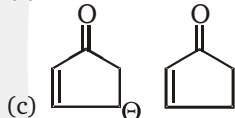
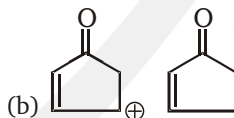
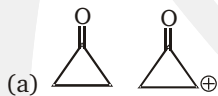


(d) None of these

165. Which of following is E isomer ?

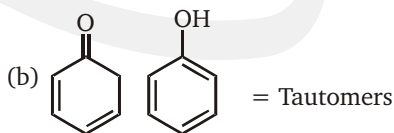
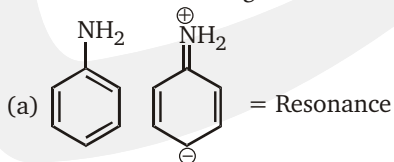


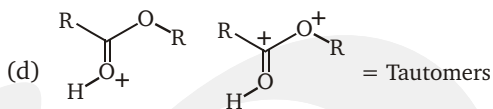
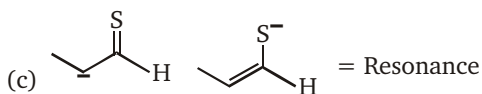
166. Among the given pairs, in which pair second compound has less enol content than first compound?



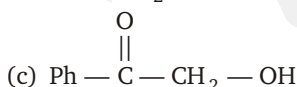
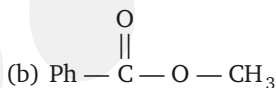
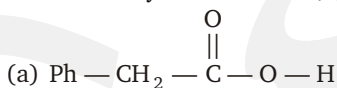
(d) none of these

167. Which of the following is incorrect relation between given pairs ?



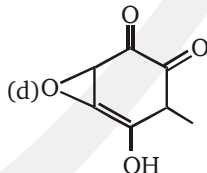
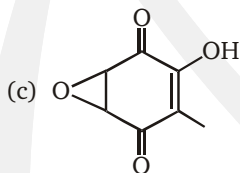
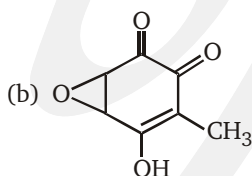
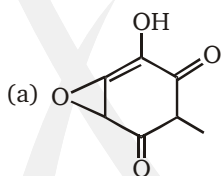
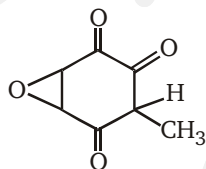


- 168.** $\text{Ph}-\underset{\text{OH}}{\underset{|}{\text{CH}}}-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$ $\xrightarrow[\text{H}_2\text{O}]{\text{HO}}$ (B) ; (A) and (B) are isomer and isomerization effectively carried out by trace of base (B). Identify (B).

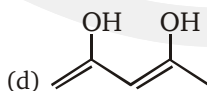
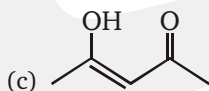
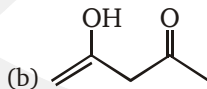
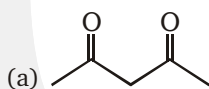


- 169.** $\text{CH}_3-\text{CH}=\text{CH}-\text{CH}=\text{CH}-\text{CH}_3$; total number of geometrical isomer is :
 (a) 2 (b) 3 (c) 4 (d) 6

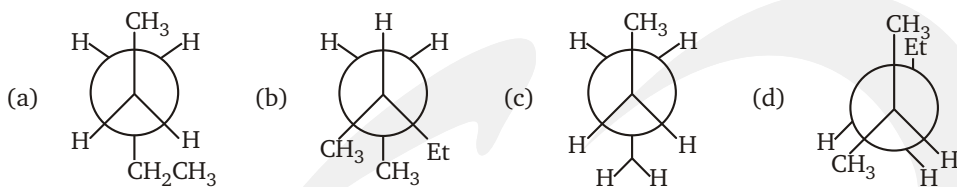
- 170.** Identify most stable enol form of teric acid:



- 171.** Which structure is most stable ?



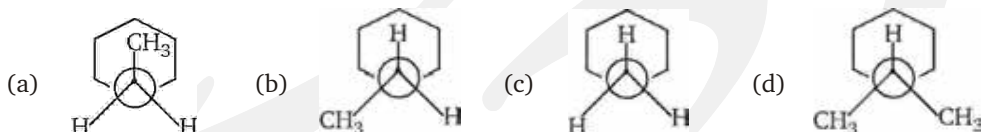
172. Identify conformer of 2-methyl pentane :



173. The lowest energy conformer of



is:



174.



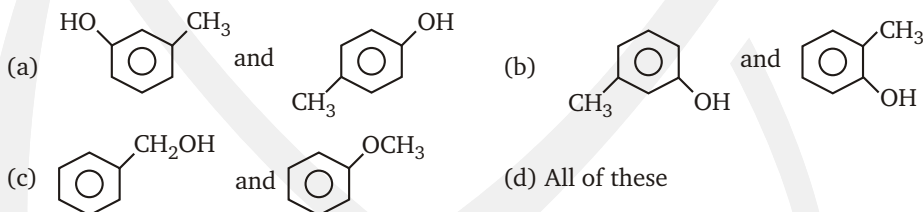
How many atoms will be bisect during plane of symmetry ?

- (a) 2 (b) 4 (c) 6 (d) 8

175. The number of all types of isomers of chlorobutane is :

- (a) 2 (b) 4 (c) 6 (d) 5

176. Which of the following pairs of compounds are not positional isomers ?



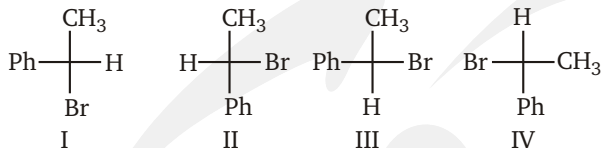
177. Which of the following pairs of compounds are functional isomers ?



178. The isomeric alcohol which has a chiral carbon atom is:

- (a) *n*-butyl alcohol (b) *iso*-butyl alcohol (c) *sec*-butyl alcohol (d) *tert*-butyl alcohol

179. The pair of enantiomers among the following compound is:

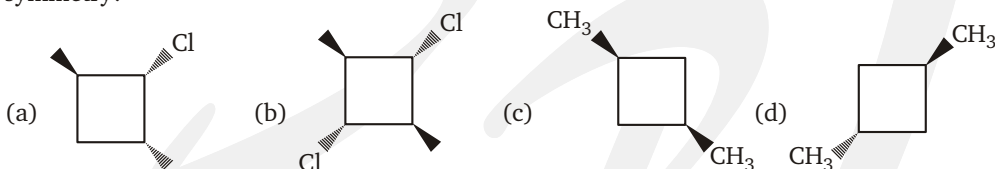


- (a) I and IV (b) II and IV (c) II and III (d) I and II

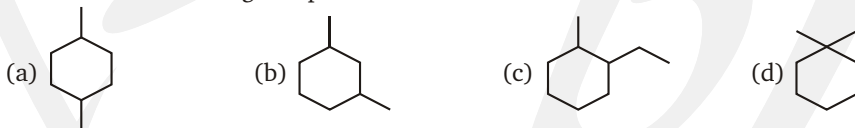
180. Which of the following is chiral?

- (a) Cell phone (b) Spiral staircase (c) Scissor (d) All of these

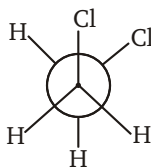
181. In which of the following compound, possess plane of symmetry as well as centre of symmetry?



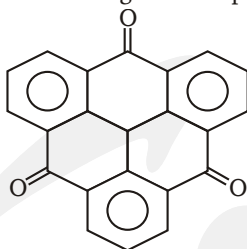
182. Which of the following compound has one of the stereoisomers as a meso compound?



183. For the following Newman projection

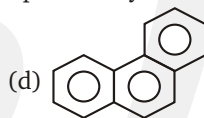
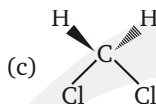
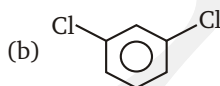
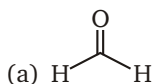


184. Which of the following is correct for the given compound?



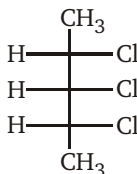
- (a) It possess centre of symmetry (b) It possess C_4 axis of symmetry
(c) It possess plane of symmetry (d) Compound is chiral

185. Which of the following molecules has axis of symmetry and a coaxial plane of symmetry?



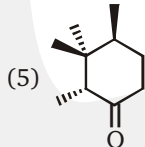
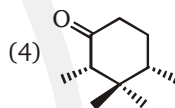
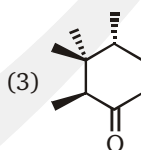
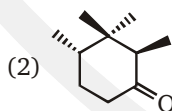
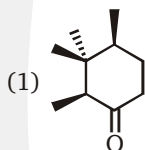
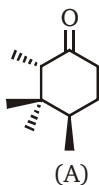
- (e) All of these

186. Number of diastereomer of given compound :



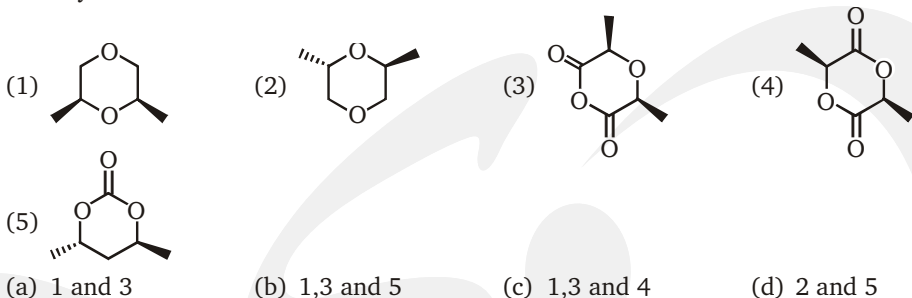
- (a) 2 (b) 3 (c) 4 (d) 6

187. Which of the structures is/are diastereomer of **A** ?



- (a) 3 (b) 1 and 4 (c) 2 and 3 (d) 5

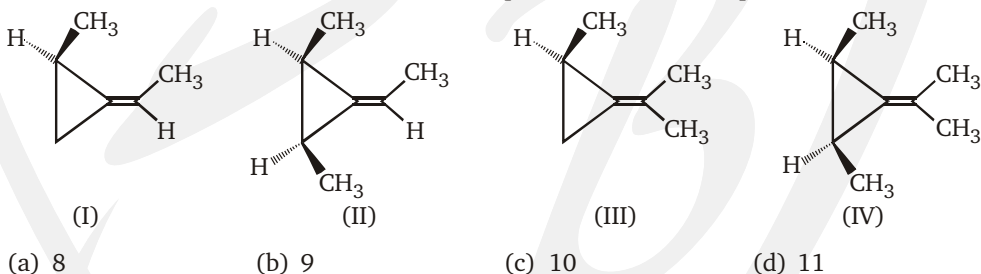
188. Identify which of the structures below are meso structures ?



189. How many enol form is possible for $\text{CH}_3 - \text{CH}_2 - \text{C}(=\text{O}) - \text{CH}_2 - \text{CH}_2 - \text{CH}_3$ (including stereoisomers) will be ?

- (a) 2 (b) 3 (c) 4 (d) 5

190. Find the sum of all the stereocenters that are present in below compounds :



191. A pair of stereoisomers might be classified in various ways. Which of the following statement are true with respect to pairs of stereoisomers ?

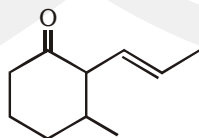
- (a) They might be configurational isomers (b) They might be diastereomers
(c) They might be constitutional isomers (d) They might be tautomers
(e) They might be conformational isomers (f) They might be enantiomers
(g) They might be positional isomers
- (a) a, b, c, e (b) b, d, e, f, g (c) a, b, f (d) a, b, c, f

192. Ignoring specific markings, which of the following objects are chiral ?

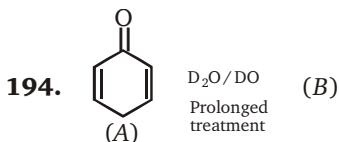
- (I) a shoe (II) a book (III) a pencil
(IV) a pair of shoes (consider the pair as one object)
(V) a pair of scissors

- (a) I only (b) I & V (c) I, IV, V (d) III, IV, V

193. Calculate the total number of stereoisomers when alkene having trans configuration :

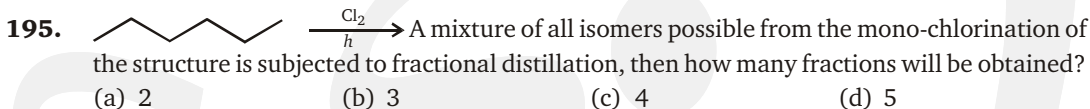


- (a) 2 (b) 3 (c) 4 (d) 8

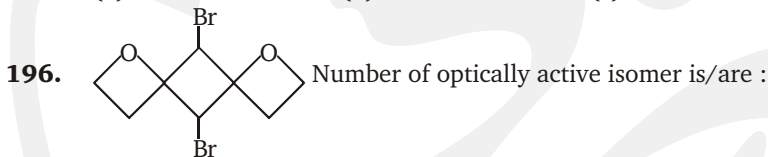


After prolonged treatment of (A) by $\text{D}_2\text{O}/\text{DO}$, the difference in molecular weights of compounds (A) and (B) is :

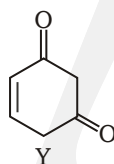
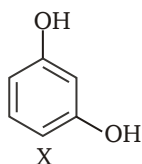
- (a) 2 (b) 3 (c) 4 (d) 8



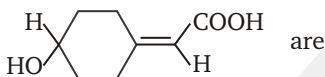
- (a) 2 (b) 3 (c) 4 (d) 5



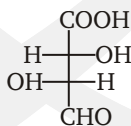
- (a) 0 (b) 1 (c) 2 (d) 3



- (a) resonance structures (b) tautomers
(c) functional isomers (d) positional isomers



- (a) enantiomers (b) diastereomers (c) conformers (d) rotamers



- (a) R, R (b) S, S (c) R, S (d) S, R

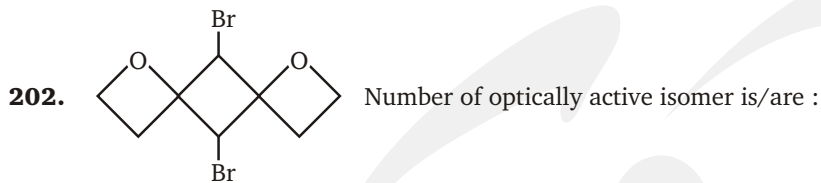


- (a) 3-methyl-3-hexene (b) 4-chloro-1-methylcyclohexane
(c) 2-phenylpentane (d) 1,3-diisopropylbenzene



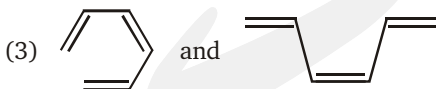
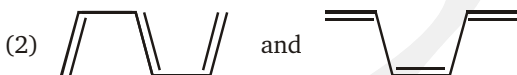
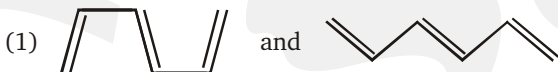
201. Number of optically active tartaric acid is/are possible :

- (a) 1 (b) 2 (c) 3 (d) 4



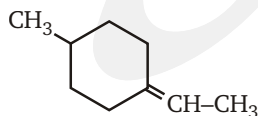
- (a) 0 (b) 1 (c) 2 (d) 3

203. Correct relationship b/w pair of compounds.

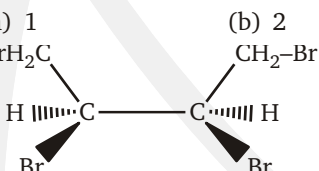


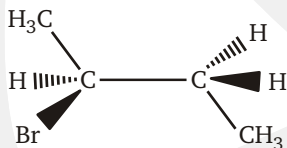
- (a) 1-Conformer ; 2-Conformer ; 3-Conformer
 (b) 1-Conformer ; 2-Stereoisomers (GI) ; 3-Stereoisomers (GI)
 (c) 1-Conformer ; 2-Stereoisomers (GI) ; 3-Conformer
 (d) 1-Stereoisomerism (GI) ; 2-Stereoisomerism (GI) ; 3-Conformer

204. Number of diastereomers possible for



- (a) 1 (b) 2 (c) 4 (d) 0

205.  No. of enantiomer = X



No. of diastereomers = Y

Sum X Y ?

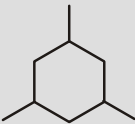
- (a) 1 (b) 3 (c) 4 (d) 0

ANSWERS – LEVEL 1

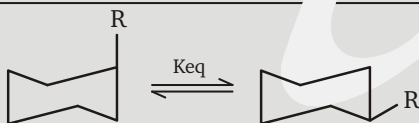
1.	(b)	2.	(c)	3.	(b)	4.	(c)	5.	(a)	6.	(b)	7.	(c)	8.	(d)
9.	(c)	10.	(d)	11.	(b)	12.	(c)	13.	(a)	14.	(b)	15.	(b)	16.	(d)
17.	(c)	18.	(a)	19.	(b)	20.	(b)	21.	(d)	22.	(b)	23.	(c)	24.	(d)
25.	(a)	26.	(b)	27.	(d)	28.	(b)	29.	(a)	30.	(c)	31.	(c)	32.	(a)
33.	(b)	34.	(c)	35.	(c)	36.	(d)	37.	(c)	38.	(d)	39.	(b)	40.	(a)
41.	(a)	42.	(d)	43.	(a)	44.	(d)	45.	(a)	46.	(b)	47.	(c)	48.	(b)
49.	(c)	50.	(c)	51.	(d)	52.	(d)	53.	(e)	54.	(a)	55.	(d)	56.	(c)
57.	(d)	58.	(b)	59.	(a)	60.	(a)	61.	(c)	62.	(d)	63.	(e)	64.	(d)
65.	(b)	66.	(b)	67.	(d)	68.	(d)	69.	(d)	70.	(d)	71.	(d)	72.	(a)
73.	(b)	74.	(a)	75.	(b)	76.	(b)	77.	(a)	78.	(b)	79.	(c)	80.	(b)
81.	(d)	82.	(a)	83.	(b)	84.	(a)	85.	(a)	86.	(d)	87.	(a)	88.	(d)
89.	(a)	90.	(b)	91.	(a)	92.	(a)	93.	(b)	94.	(a)	95.	(b)	96.	(b)
97.	(a)	98.	(d)	99.	(b)	100.	(a)	101.	(c)	102.	(c)	103.	(c)	104.	(d)
105.	(d)	106.	(a)	107.	(c)	108.	(b)	109.	(d)	110.	(a)	111.	(c)	112.	(c)
113.	(a)	114.	(d)	115.	(b)	116.	(b)	117.	(c)	118.	(d)	119.	(b)	120.	(c)
121.	(a)	122.	(b)	123.	(d)	124.	(b)	125.	(b)	126.	(c)	127.	(c)	128.	(b)
129.	(d)	130.	(a)	131.	(c)	132.	(d)	133.	(d)	134.	(d)	135.	(b)	136.	(b)
137.	(c)	138.	(a)	139.	(d)	140.	(b)	141.	(b)	142.	(a)	143.	(c)	144.	(b)
145.	(b)	146.	(b)	147.	(b)	148.	(a)	149.	(b)	150.	(c)	151.	(a)	152.	(c)
153.	(b)	154.	(d)	155.	(c)	156.	(a)	157.	(c)	158.	(d)	159.	(d)	160.	(d)
161.	(b)	162.	(d)	163.	(d)	164.	(d)	165.	(d)	166.	(c)	167.	(d)	168.	(c)
169.	(b)	170.	(c)	171.	(c)	172.	(d)	173.	(b)	174.	(c)	175.	(d)	176.	(c)
177.	(b)	178.	(c)	179.	(c)	180.	(d)	181.	(d)	182.	(b)	183.	(b)	184.	(c)
185.	(e)	186.	(b)	187.	(b)	188.	(a)	189.	(c)	190.	(c)	191.	(c)	192.	(b)
193.	(c)	194.	(c)	195.	(b)	196.	(a)	197.	(b)	198.	(a)	199.	(a)	200.	(c)
201.	(b)	202.	(a)	203.	(c)	204.	(d)	205.	(d)						

Level - 2

1. Match the Column (I) and (II).

Column (I)				Column (II)	
Reaction				Stereoisomers	
(a)	$\text{CH}_3 - \text{CH}$	$\text{CH} - \text{CH}$	$\text{N} - \text{OH}$	(p)	2
(b)				(q)	4
(c)	$\text{CH}_3 - \text{CH}$	$\text{CH} - \text{CH}$	$\text{CH} - \text{CH}$	(r)	6
(d)	$\text{CH}_3 - \text{CH}$	$\text{CH} - \text{CH}$	$\text{CH} - \text{CH}$	(s)	8
			$\text{CH} - \text{Ph}$		

2. Match the Column (I) and (II).

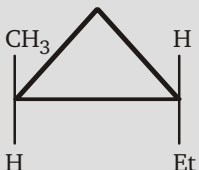
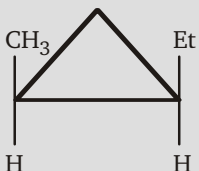
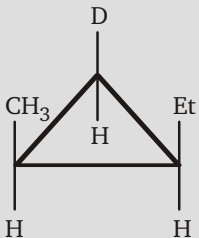
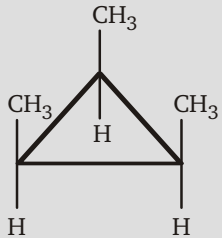


Column (I)			Column (II)	
Group			Equilibrium Constant	
(a)	$\text{R} - \text{H}$		(p)	38
(b)	$\text{R} - \text{CH}_3$		(q)	23
(c)	$\text{R} - \text{Et}$		(r)	18
(d)	$\text{R} - \text{CH} - \text{CH}_3$ $\quad \quad \quad \text{CH}_3$		(s)	1

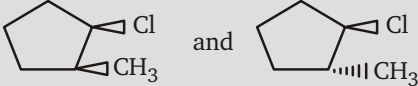
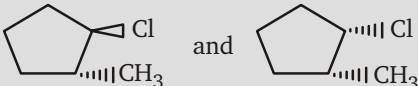
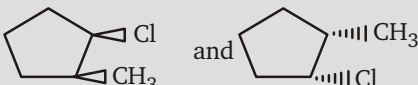
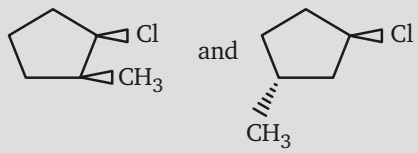
3. Match the Column (I) and (II). (Matrix)

Column (I)		Column (II)	
Molecule		Nature	
(a)		(p)	Chiral
(b)		(q)	Achiral
(c)		(r)	Meso
(d)		(s)	Compound containing even number of chiral centers

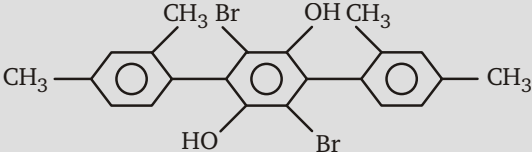
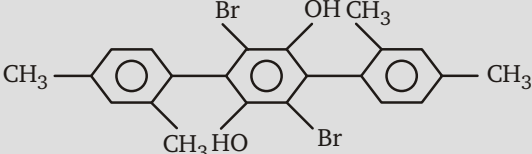
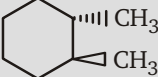
4. Match the Column (I) and (II). (Matrix)

Column (I)		Column (II)	
Compound		Isomerism	
(a)		(p)	Geometrical isomerism
(b)		(q)	Optical isomerism
(c)		(r)	Compound containing plane of symmetry
(d)		(s)	Compound containing center of symmetry

5. Match the Column (I) and (II).

Column (I)		Column (II)	
Molecules		Relationship	
(a)		(p)	Identical
(b)		(q)	Enantiomer
(c)		(r)	Diastereomer
(d)		(s)	Structural Isomerism

6. Match the Column (I) and (II). (Matrix)

Column (I)		Column (II)	
Compound		Nature	
(a)		(p)	cis-compound
(b)		(q)	trans-compound
(c)		(r)	Optically active
(d)		(s)	Optically inactive

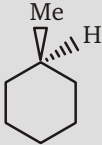
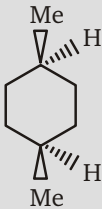
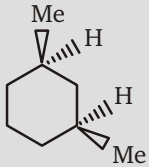
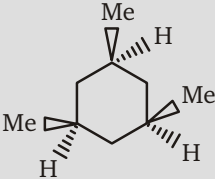
7. Match the Column (I) and (II). (Matrix)

Column (I)		Column (II)	
Molecule		Property	
(a)		(p)	Chiral centers containing compound
(b)		(q)	Presence of stereocenter
(c)		(r)	Optically active compound
(d)		(s)	Compound containing plane of symmetry

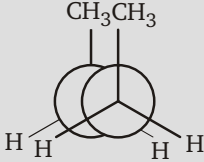
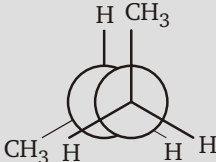
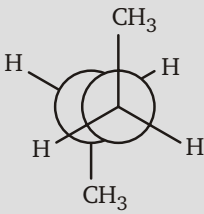
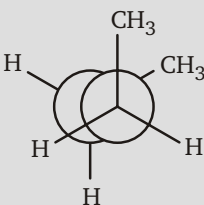
8. Match the Column (I) and (II). (Matrix)

Column (I)		Column (II)	
Molecule		Property	
(a)		(p)	Polar molecule
(b)		(q)	Optically active
(c)		(r)	Optically inactive
(d)		(s)	Plane of symmetry

9. Match the Column (I) and (II). (Matrix)

Column (I)		Column (II)	
Molecule		Property	
(a)		(p)	Meso compound
(b)		(q)	Achiral
(c)		(r)	Chiral compound
(d)		(s)	Compound will show geometrical isomerism

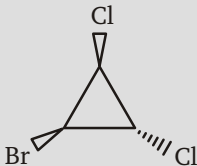
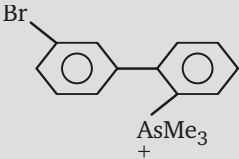
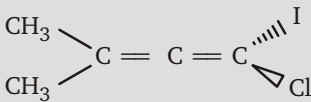
10. Match the Column (I) and (II).

Column (I)		Column (II)	
Modified Newmann Projection		Conformers	
(a)		(p)	Fully eclipsed
(b)		(q)	Partially eclipsed
(c)		(r)	Gauche
(d)		(s)	Staggered

11. Match the Column (I) and (II).

Column (I)		Column (II)	
Newmann Projection		Name of the Compound	
(a)		(p)	3-methyl pentane
(b)		(q)	<i>n</i> -butane
(c)		(r)	Methyl-cyclopentane
(d)		(s)	1,2,4-trimethyl cyclohexane

12. Match the Column (I) and (II). (Matrix)

Column (I)		Column (II)	
Molecule		Property	
(a)		(p)	Rotates plane polarized light
(b)		(q)	Cannot rotate plane polarized light
(c)		(r)	Plane of symmetry
(d)		(s)	Centre of symmetry

13. Match the Column (I) and (II).

Column (I)		Column (II)	
Molecule		Stereocenters	
(a)	$\begin{array}{c} \text{CH}_3 - \text{CH} \quad \text{CH} - \text{CH} - \text{CH}_3 \\ \\ \text{Br} \end{array}$	(p)	1
(b)	$\begin{array}{c} \text{H} - \text{C} \quad \text{C} - \text{CH} \quad \text{CH} - \text{CH} - \text{CH} - \text{CH}_3 \\ \quad \quad \\ \text{Br} \quad \quad \text{Br} \end{array}$	(q)	2
(c)	$\begin{array}{c} \text{O} \\ \\ \text{Ph} - \text{S} - \text{CH} \quad \text{CH} - \text{CH} - \text{CH}_3 \\ \\ \text{CH}_3 \end{array}$	(r)	3
(d)	$\begin{array}{c} \text{Ph} - \text{CH} - \text{Et} \\ \\ \text{Cl} \end{array}$	(s)	4

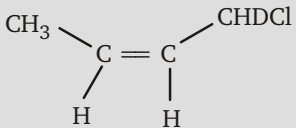
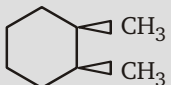
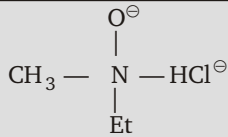
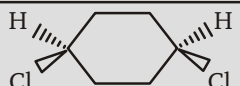
14. Match the Column (I) and (II).

Column (I)		Column (II)	
Molecule		Stereoisomers	
(a)		(p)	2
(b)		(q)	0
(c)		(r)	4
(d)		(s)	8

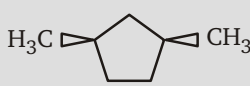
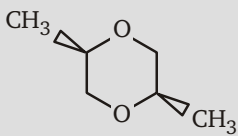
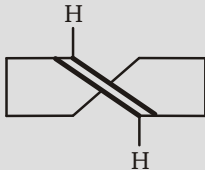
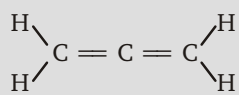
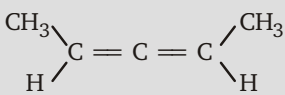
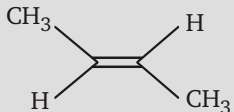
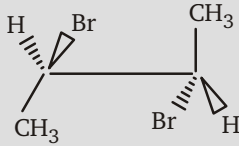
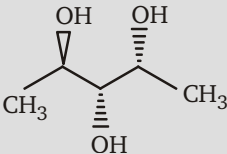
15. Match the Column (I) and (II). (Matrix)

Column (I)		Column (II)	
Molecule		Property	
(a)		(p)	Meso Compound
(b)		(q)	Compound having even no. of chiral centres
(c)		(r)	Optically active compound
(d)		(s)	Compound having odd no. of chiral centres.

16. Match the Column (I), (II) and (III). (Matrix)

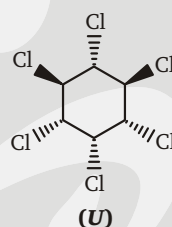
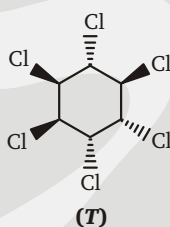
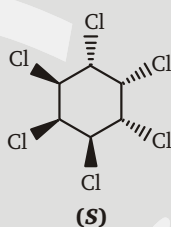
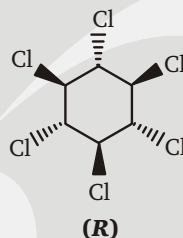
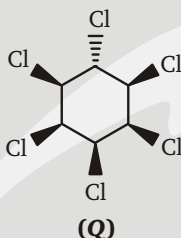
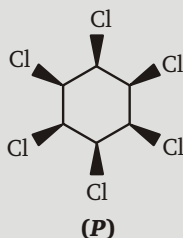
Column (I)		Column (II)		Column (III)	
Property		Molecule		No. of Chiral Center	
(a)		(p)	Optically active	(w)	0
(b)		(q)	Optically inactive	(x)	1
(c)		(r)	Plane of symmetry	(y)	2
(d)		(s)	Centre of symmetry	(z)	3

17.

(a)		(b)		(c)	
(d)		(e)		(f)	
(g)		(h)			

From the above compounds select :

(A)	two of which are chiral and contain chiral centre :	
(B)	two of which are achiral and contains chiral centre :	
(C)	two of which are chiral and does not contain chiral centre :	
(D)	two of which are achiral and does not contain chiral centre :	

18. Comprehension

Consider the given structures and answer A, B & C.

A. Which of the compound is optically active ?

- (a) P (b) R (c) S (d) T

B. Which of the isomer is most stable ?

- (a) R (b) S (c) T (d) U

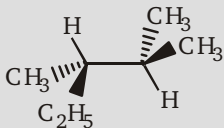
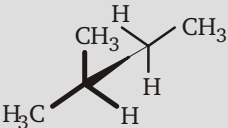
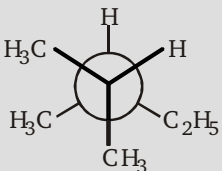
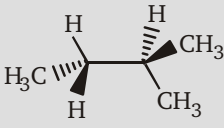
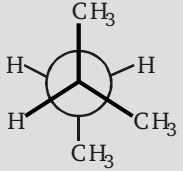
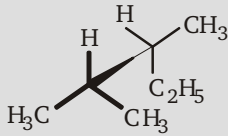
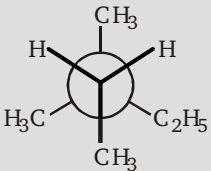
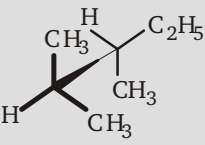
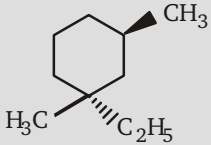
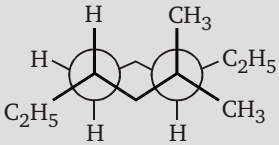
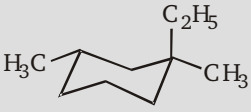
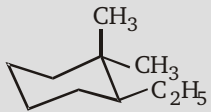
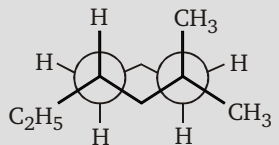
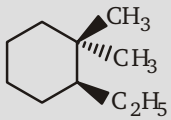
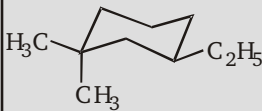
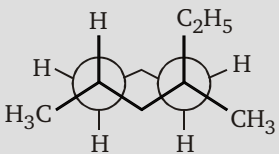
19. Identify relationship between following pairs :

(a)		(b)	
(c)		(d)	

If they are enantiomer answer will be 1, if they are diastereomers answer will be 2, if they are constitutional isomers answer will be 3 and if they are identical present 4 as the answer. Sum of answer of each part $a + b + c + d$ is :

20. In each of the following three questions a hydrocarbon is named. For each select from among the sixteen conformational structures (a through p) all structures that represent possible conformers of that compound. Write letters (a through p), corresponding to your selections, in each answer box.

A.	2-methylbutane	
B.	2,3-dimethylpentane	
C.	1-ethyl-1, 3-dimethyl cyclohexane	

(a)		(b)		(c)	
(d)		(e)		(f)	
(g)		(h)		(i)	
(j)		(k)		(l)	
(m)		(n)		(o)	
(p)					

21. Examine structures a through j, shown below, with respect to their symmetry or lack of it. Assume that the five-membered rings and the ring in compound g are planar. The wedge-hatched bonds in b, c, d & e designate specific configurations. Also, for the acyclic compounds assume stable anti conformations. Answer each of the following questions by writing letters (a through j), corresponding to your selections, in each answer box. If there is no structure that fits the description enter an x in the answer box.

A.	Which structures are chiral ?	
B.	Which structures have a plane of symmetry ?	
C.	Which structures have a center of symmetry ?	

(a)		(b)		(c)	
(d)		(e)		(f)	
(g)		(h)	$C_2H_5CHCl_2$	(i)	$C_2H_5CHClC_2H_5$
(j)	$C_2H_5CHClCH_3$				

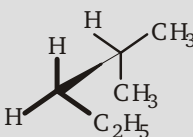
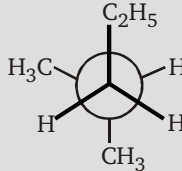
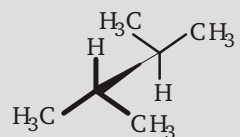
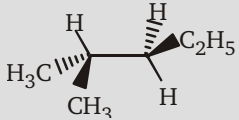
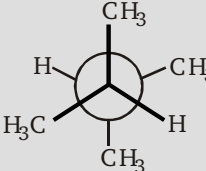
22. (i) 1,2-dichlorocyclopropane *w*
(ii) 1,3-dimethyl-cyclobutane *x*
(iii) 2-bromo-3-chlorobutane *y*
(iv) 1,3-dimethyl cyclohexane *z*

Calculate total number of stereoisomer of the above compounds.

Sum of *w* *x* *y* *z*

23. Examine the following formulas and select those pairs that satisfy the following conditions :

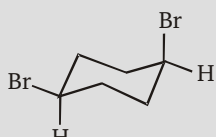
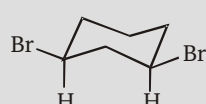
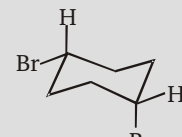
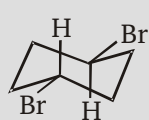
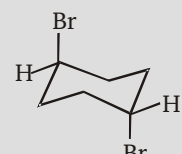
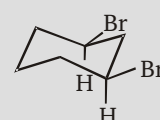
Be sure to write two letters (and only two) in each answer box, unless you select f. In the second and third parts more than one answer is possible.

(a)		(b)		(c)	
(d)		(e)		(f)	No formulas meet the designated condition

A.	Which are identical in all respects?	
B.	Which are conformational isomers?	
C.	Which are constitutional isomers?	

24. Examine the following formulas and select those pairs that satisfy the following conditions :

Be sure to write two letters (and only two) in each answer box. In the second and fourth parts more than one answer is possible.

(a)		(b)		(c)	
(d)		(e)		(f)	

A.	Which are identical in all respects?	
B.	Which are configuration isomers?	
C.	Which are conformational isomers?	

25. Consider the following statements regarding the given projection (True or False).

(W)	$\begin{array}{c} \text{CH}_2\text{Cl} \\ \\ \text{H} - \text{C} - \text{Br} \\ \\ \text{Cl} - \text{C} - \text{H} \\ \\ \text{CH}_3 \end{array}$	(X)	$\begin{array}{c} \text{CH}_2\text{Cl} \\ \\ \text{Cl} - \text{C} - \text{H} \\ \\ \text{H} - \text{C} - \text{Cl} \\ \\ \text{CH}_3 \end{array}$
(Y)	$\begin{array}{c} \text{CH}_3 \quad \text{H} \\ \quad \\ \text{Cl} - \text{C} - \text{C} - \text{Br} \\ \quad \\ \text{H} \quad \text{CH}_2\text{Cl} \end{array}$	(Z)	$\begin{array}{c} \text{CH}_2\text{Cl} \\ \\ \text{H} - \text{C} - \text{Cl} \\ \\ \text{H} - \text{C} - \text{Cl} \\ \\ \text{CH}_3 \end{array}$

(a)	W and Y are diastereomers	
(b)	Z is the projection of X	
(c)	W, X, Y and Z are optically active	
(d)	Y and Z are isomer	

26. Examine the following structural formulas and select those that are chiral.

(a)		(b)		(c)	
(d)		(e)	$\begin{array}{c} \text{CO}_2\text{H} \\ \\ \text{H} - \text{C} - \text{OH} \\ \\ \text{H} - \text{C} - \text{OH} \\ \\ \text{CH}_2 - \text{CO}_2\text{H} \end{array}$	(f)	$\begin{array}{c} \text{CN} \\ \\ \text{H} - \text{C} - \text{OH} \\ \\ \text{HO} - \text{C} - \text{CH}_3 \\ \\ \text{CH}_2 - \text{CO}_2\text{H} \end{array}$
(g)		(h)		(i)	
(j)	$\begin{array}{c} \text{CH}_3 \\ \\ \text{H} - \text{C} - \text{Cl} \\ \\ \text{H} - \text{C} - \text{Cl} \\ \\ \text{CH}_3 \end{array}$				
Write your choice here					

27. The configuration of eight compounds, a through h are shown below, using various kind of stereo representations. To answer the question given below, write (a through h) indicating your choice.

(a)		(b)		(c)	
(d)		(e)		(f)	
(g)		(h)			

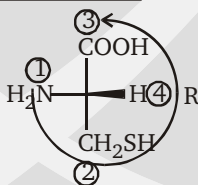
A.	Which of these configuration are achiral?	
B.	Which configuration has no stereogenic center?	
C.	Which configuration has more than one stereogenic center?	
D.	Which of these configuration are meso compound?	

28. The structural formula of ten compounds, (I) through (X) are drawn below, you may select any one of these structure.

Answer the following question about that compound.

I		II		III	
IV		V		VI	
VII		VIII		IX	
X					

- A.** How many chiral centre are present in this compound ?
 (a) 0 (b) 1 (c) 2 (d) 3
 (e) 4 (f) 5
- B.** Is this compound chiral or achiral ?
 (a) Chiral (b) Achiral
- C.** What symmetry element are present in this compound ?
 (a) None (b) Plane of symmetry (c) Center of symmetry
- 29.** The structure of one of the enantiomers of the amino acid cysteine is shown below.



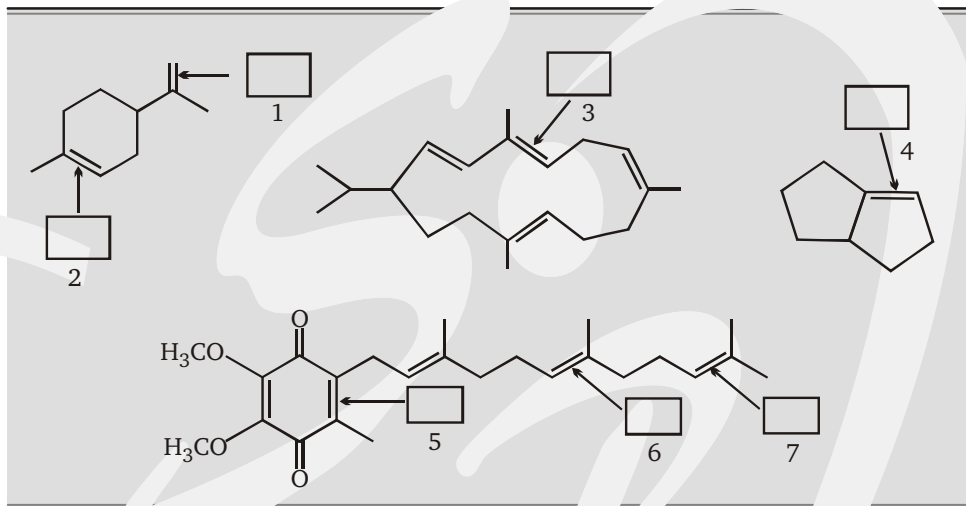
Classify this structure as :

(a) R or S

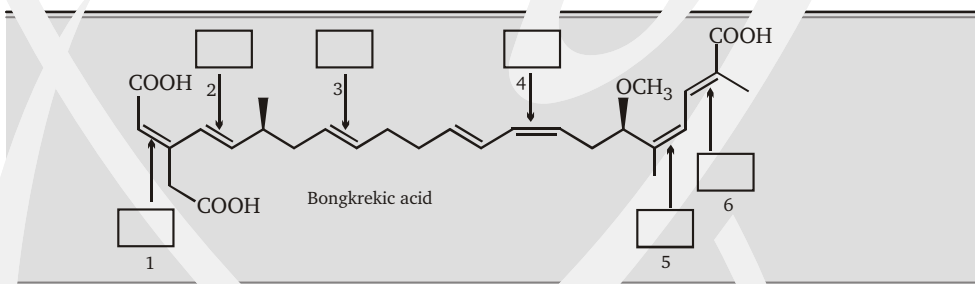
(b) D or L

30. Identify the following double bonds either E, Z or None (N) in the compounds given below either.

A.

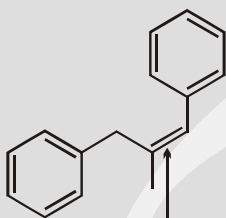


- B. (a) Bongkreic acid is a toxic compound produced by *Pseudomonas cocovenenans*, and isolated from a mold that grows on bongkreik, a fermented Indonesian coconut dish. (a) Label each double bond as E, Z or neither (N).

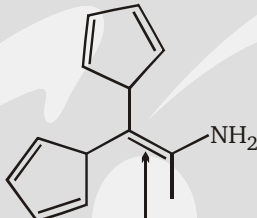


- (b) How many total stereoisomers (including all types) are possible for bongkreic acid ?
.....
- (c) How many sites of unsaturation are present in bongkreic acid ?

31. Designate the following double bonds as E, Z or none (N) configuration in the boxes provided below.



1

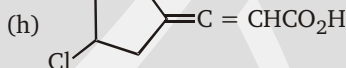
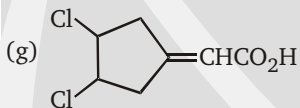
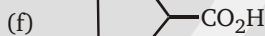
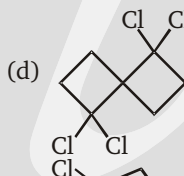
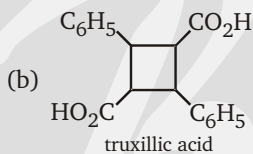
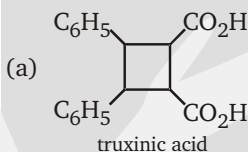


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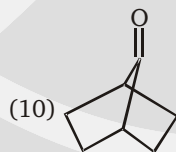
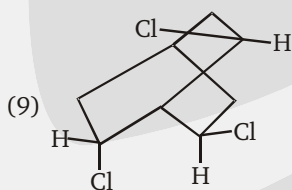
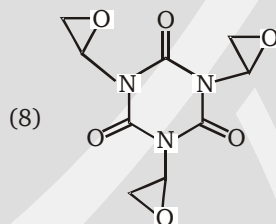
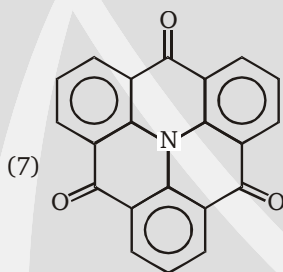
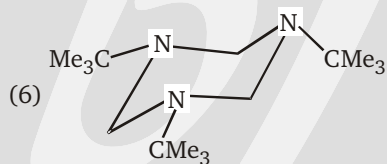
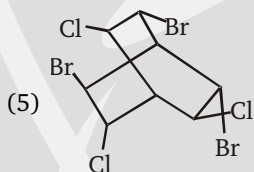
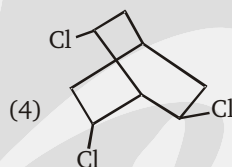
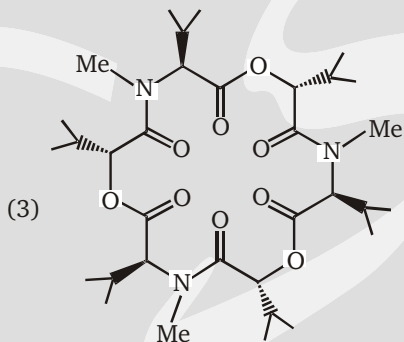
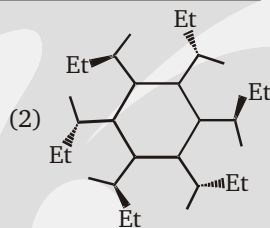
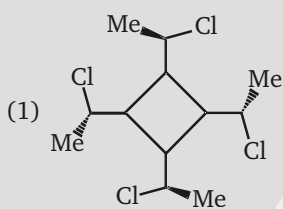
32. The following compounds may exist as two or more stereoisomers. These may be classified as enantiomer pairs or meso compounds.

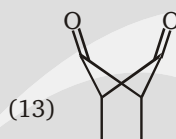
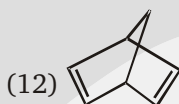
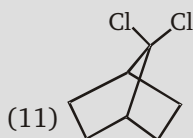


Answer the following question about the above structure.

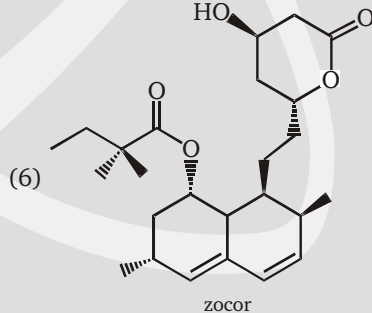
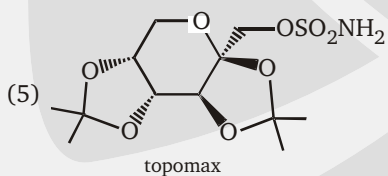
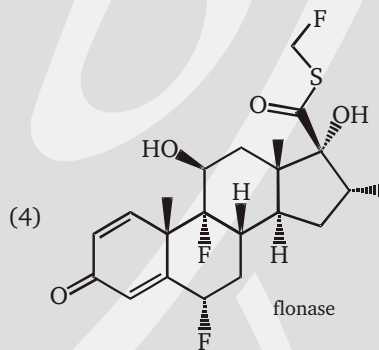
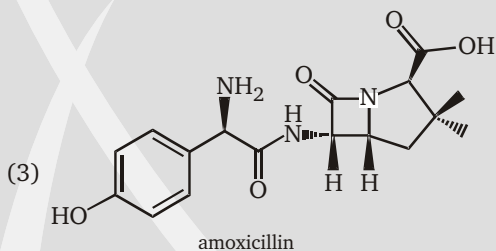
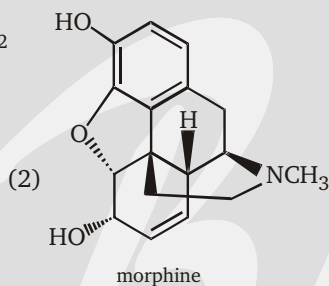
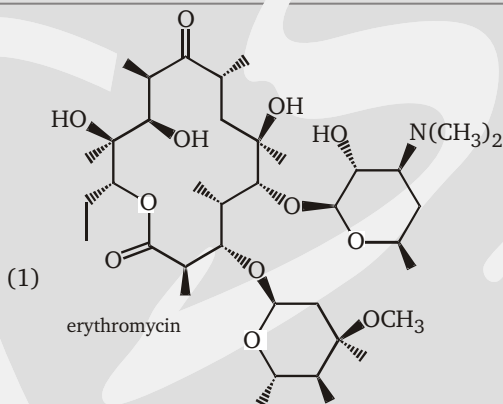
- (A) Total number of stereoisomers :
 (B) Number of enantiomeric pairs :
 (C) Number of meso compounds :

33. Identify axis of symmetry in the given compound.





34. For each of the following pharmaceutical compounds, identify all stereogenic (i.e., all asymmetric carbon atoms) and label the configuration of each as being either (R) or (S).



35. Find relationship between given pair :

		Identical	Enantiomer	Diastereomer	Constitutional Isomer
1.					
2.					
3.					
4.					
5.					
6.					
7.					

8.					
9.					
10.					

36. Comprehension

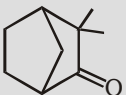
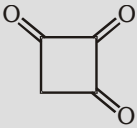
Structural formula of compound (A) is following:



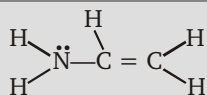
- A.** The correct statement(s) about the compound (A) is/are:
- The total number of stereoisomers possible for (A) is 3
 - The total number of mesoisomer possible for (A) is 1
 - The total number of pair of enantiomer possible for (A) is 1
 - All of these
- B.** Number of plane of symmetry in *cis*-form of compound (A) is:
- 0
 - 1
 - 2
 - 3
- 37.** Match the column. **(Matrix)**

Column (I)		Column (II)	
No. of Carbon		No. of structural isomer	
(a)	C_4H_{10}	(p)	
(b)	C_5H_{12}	(q)	3
(c)	C_6H_{14}	(r)	5
(d)	C_7H_{16}	(s)	9

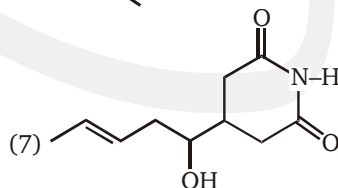
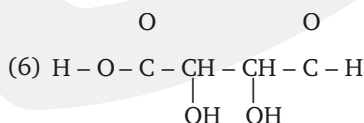
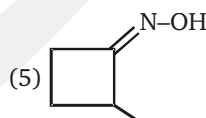
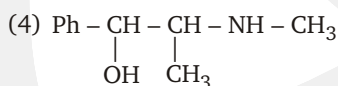
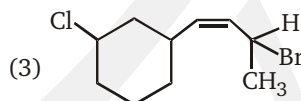
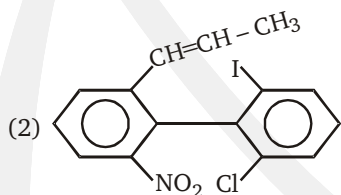
38. Match the column. (Matrix)

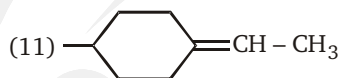
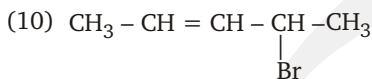
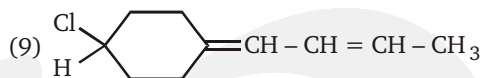
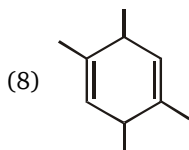
Column (I)		Column (II)	
Compound		% of enol content	
(a)		(p)	100 %
(b)		(q)	76 %
(c)	$\text{CH}_3 - \overset{\text{O}}{\parallel}{\text{C}} - \text{CH}_2 - \overset{\text{O}}{\parallel}{\text{C}} - \text{CH}_3$	(r)	8 %
(d)	$\text{CH}_3 - \overset{\text{O}}{\parallel}{\text{C}} - \text{CH}_2 - \overset{\text{O}}{\parallel}{\text{C}} - \text{O} - \text{Et}$	(s)	Keto-Enol is not possible

39. Draw a most stable conformation (N - C) bond in the following compound.



40. Find total number of stereoisomers for each compound given below :





41. Find the total number of stereoisomer for each compound :



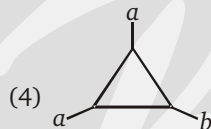
a



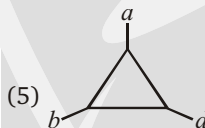
b



c



d



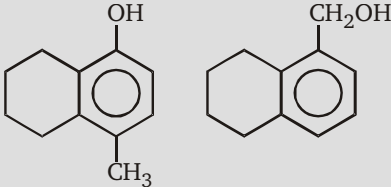
e



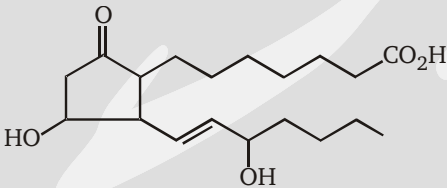
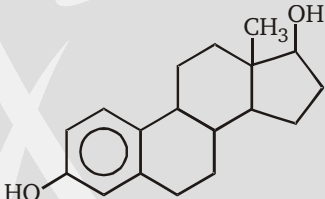
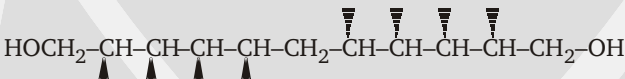
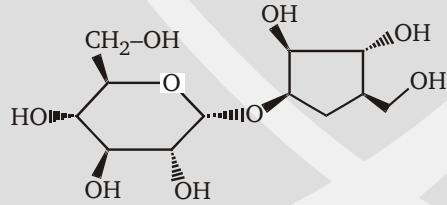
f

42. Match the column :

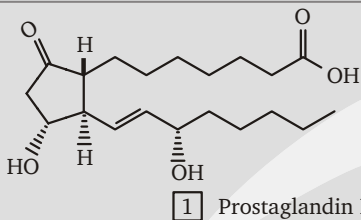
Column (I)		Column (II)	
Pair		Isomeric Relationship	
(a)		(p)	Chain
(b)		(q)	Positional

(c)		(r)	Functional
(d)		(s)	Metamers

43. Find sum of stereoisomer of following compound.

	Number of stereoisomers
(a) 	(a)
(b) 	(b)
(c) 	(c)
(d) 	(d)

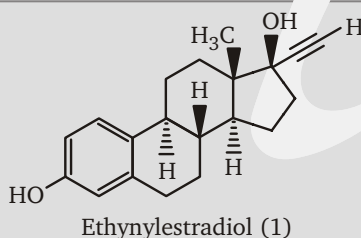
44.



Prostaglandin E₁ 1 is a compound produced by the body to regulate a variety of processes including blood clotting, fever, pain and inflammation.

- A.** Which of the following functional groups is not contained in 1 ?
 (a) A ketone (b) An alcohol (c) A carboxylic acid (d) An alkene
 (e) A nitrile
- B.** How many asymmetric (stereogenic) centres are present in compound 1 ?
 (a) 3 (b) 4 (c) 5 (d) 6
- C.** How many sp^2 hybridised carbon atoms are present in compound 1 ?
 (a) 1 (b) 2 (c) 3 (d) 4
- D.** What is the geometric configuration about the double bond in compound 1 ?
 (a) *E* (b) *Z*

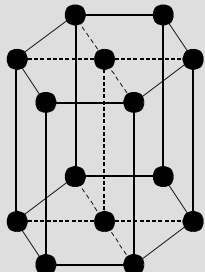
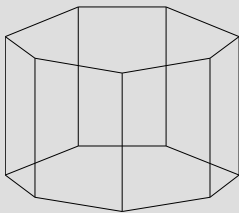
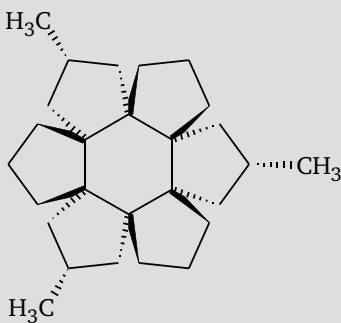
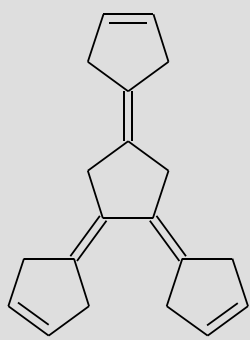
45.



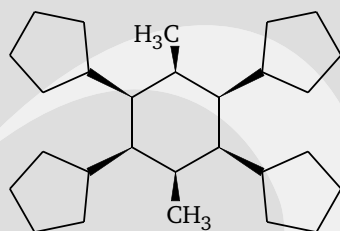
The synthetic steroid ethynylestradiol (1) is a compound used in the birth control pill.

- A.** How many sp^3 hybridised carbon atoms are present in compound (1)?
 (a) 8 (b) 9 (c) 10 (d) 11 (e) 12
- B.** How many sp^2 hybridised carbon atoms are present in compound (1) ?
 (a) 4 (b) 5 (c) 6 (d) 7 (e) 8
- C.** How many sp hybridised carbon atoms are present in compound (1) ?
 (a) 2 (b) 4 (c) 6 (d) 8 (e) 10
- D.** Which of the following functional group is contained in compound (1) ?
 (a) A ketone (b) An alcohol (c) A carboxylic acid (d) An ester
- E.** How many asymmetric (stereogenic) centres are present in compound (1) ?
 (a) 2 (b) 3 (c) 4 (d) 5

46. Match the column.

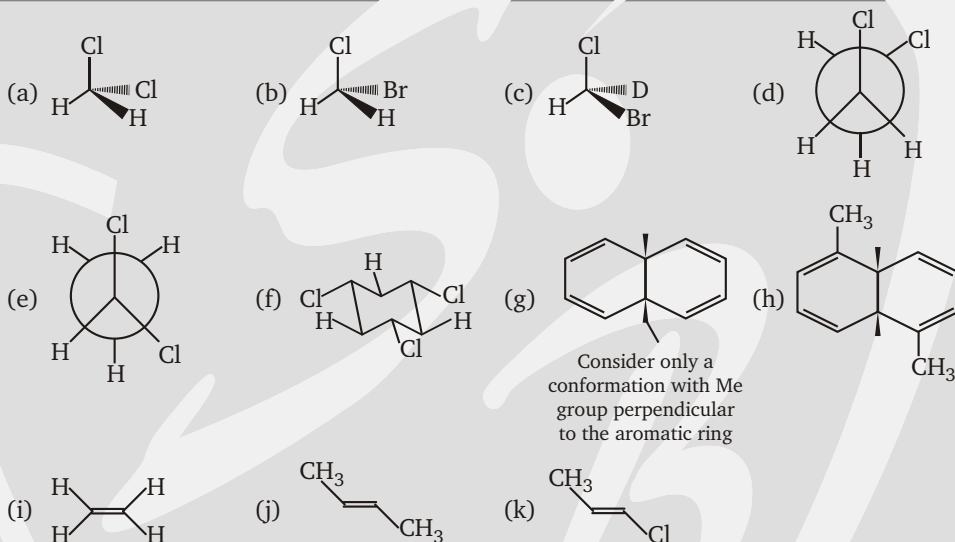
Column (I)		Column (II)	
(a)	C_2 -axis of symmetry	(p)	
(b)	C_3 -axis of symmetry	(q)	
(c)	Plane of symmetry	(r)	
(d)	Centre of symmetry	(s)	

(t)

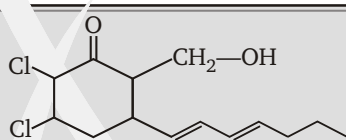


Subjective Problems

1. Number of chiral isomers are:

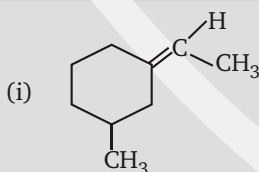


2.



Number of stereoisomer are

3.

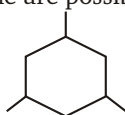


(C)

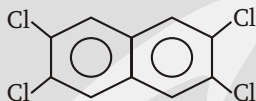
Sum of number of stereoisomer (C) Degree of unsaturations in (D).

4. How many 5 membered parent chain alkane are possible for C_7H_{16} ?

5. Theoretical possible geometrical isomer of



6. Total number of possible structural isomers of $C_5H_{11}Br$.
 7. Total number of plane of symmetry present in given compound is



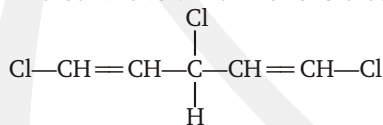
8. Total number of isomers for $C_4H_6Br_2$ containing cyclobutane ring are (including stereoisomer) ?
 9. Total number of structural isomers of C_9H_{18} containing cyclohexane ring.
 10. How many structural isomer are possible for $C_4H_{10}O$ (only alcohol).
 11. Number of structural isomer of C_6H_{14} is .

12. (a) (x) (Number of plane of symmetry)

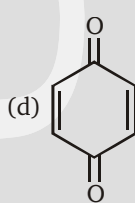
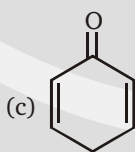
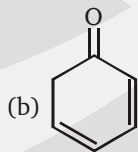
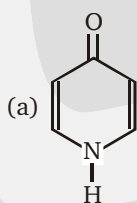
- (b) (y) (Number of mesoisomer of 1, 2-dichlorocyclopentane)

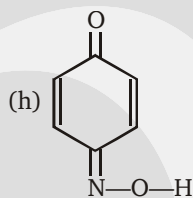
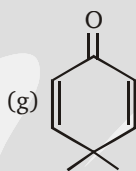
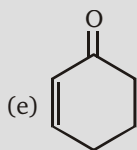
Sum of $(x+y=?)$

13. Find out the total number of stereocentre in the given compound.
 $CH_3-CH=CH-\underset{\substack{| \\ Br}}{CH}-\underset{\substack{| \\ Cl}}{CH}-CH_3$
14. Find out the total number of stereoisomers of the given following compound.



15. Find the total number of isomers of C_7H_{14} (only 5-membered ring).
 16. x number of compounds which undergoes Tautomerisation to form an Aromatic product.





17. If molecule is pyramidal, **X** stereoisomers are possible for :

C_{abcd}

find the value of **X**.

ANSWERS — LEVEL 2

1. a – q; b – p; c – r; d – s
2. a – s; b – r; c – q; d – p
3. a – p, s; b – q, r, s; c – q, r, s; d – p, s
4. a – p, q; b – p, q; c – p, q; d – p, r
5. a – r; b – r; c – p; d – s
6. a – p, r; b – q, s; c – q, r; d – p, s
7. a – q, r; b – q, s; c – p, q, r; d – q, s
8. a – r, s; b – p, q; c – r, s; d – p, r, s
9. a – q; b – q, s; c – p, q, s; d – q, s
10. a – p; b – q; c – s; d – r
11. a – p; b – q; c – r; d – s
12. a – q, r; b – p; c – p; d – q, r
13. a – r; b – s; c – r; d – p
14. a – s; b – r; c – q; d – p
15. a – q, r; b – r, s; c – q, r; d – p, q
16. (a – p – x); (b – q, r – y); (c – p – x); (d – q, r – w)
17. A – b, h; B – a, g; C – c, e; D – d, f
18. A – d; B – a
19. a b c d 13
20. A – b, d, e; B – a, c, f, h; C – i, k, p
21. A – e, f, j; B – a, c, d, g, h, i, b; C – None
22. w x y z 12
23. A – (c & e), (b & d); B – (a & b) or (a & d); C – (a & c) or (a & e), (b & c), (b & e), (c & d) and (d & e)
24. A – (a & c) (b & f); B – (a & d) or (c & d), (a & e) (c & d); C – (d & e)
25. a – True; b – True; c – True; d – False
26. b, e, f, g, h, i
27. A – d, h; B – d; C – f, h; D – h
28.

Compound	A	B	C
I	c	a	a
II	c	b	b
III	c	a	a
IV	c	b	b
V	c	a	a
VI	c	a	a
VII	c	a	a
VIII	e	b	b
IX	e	a	a
X	e	b	b
29. (a) (R) (b) (L)

30A. 1 - N; 2 - Z; 3 - E; 4 - Z; 5 - Z; 6 - E; 7 - N

B. (a) 1 - Z; 2 - E; 3 - E; 4 - Z; 5 - Z; 6 - E (b) 2^9 (c) 10

31. 1 - Z; 2 - N; 3 - E

32. (a) A - 10, B - 4, C - 2

(c) A - 4, B - 2, C - 0

(e) A - 5, B - 2, C - 1

(g) A - 4, B - 2, C - 0

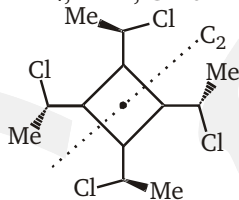
(b) A - 5, B - 0, C - 5

(d) A - 2, B - 1, C - 0

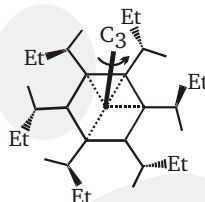
(f) A - 4, B - 1, C - 2

(h) A - 4, B - 1, C - 2

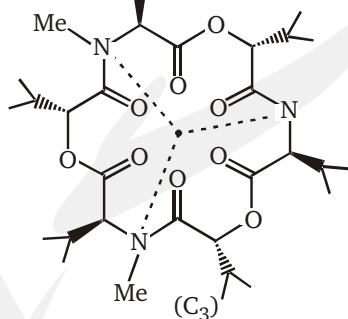
33. (1)



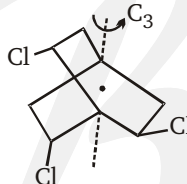
(2)



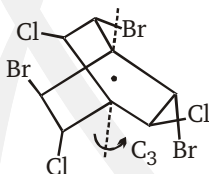
(3)



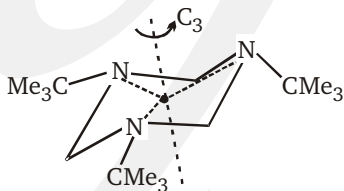
(4)



(5)



(6)

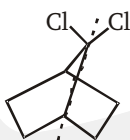


(7) C_3 axis, C_2 axis

(8) C_3 -axis

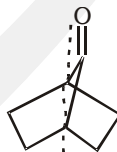
(9) C_3 axis

(11)



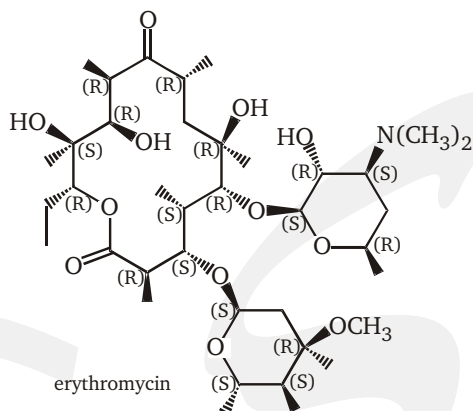
(12) C_2 -axis

(10)

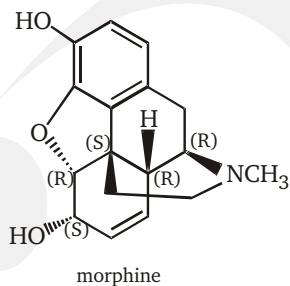


(13) C_2 -axis

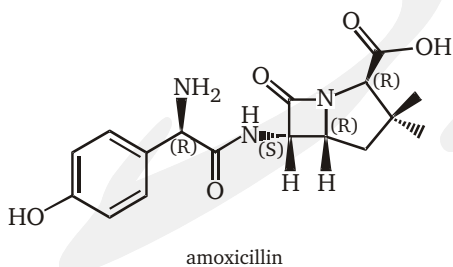
34. (1)



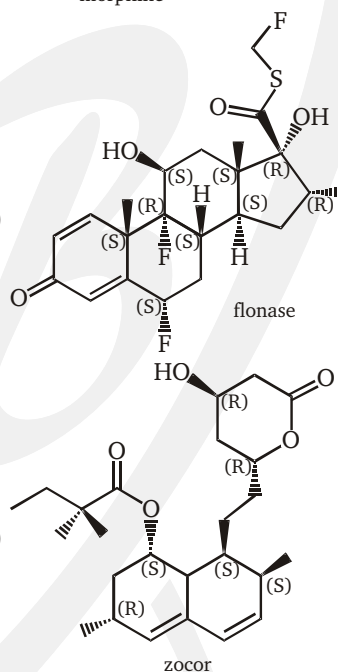
(2)



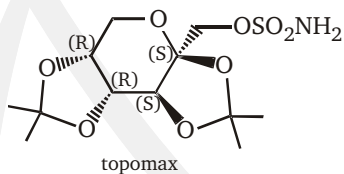
(3)



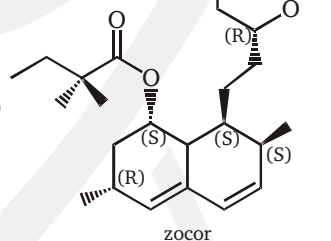
(4)



(5)



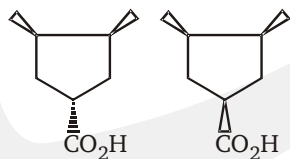
(6)



35.

Identical Enantiomer Diastereomer Constitutional Isomer

(1)



×

×

✓

×

(2)

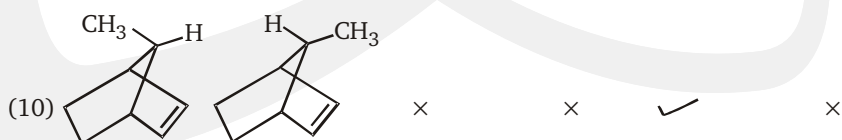
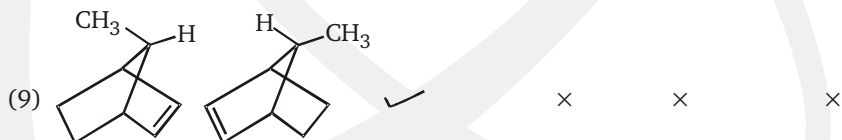
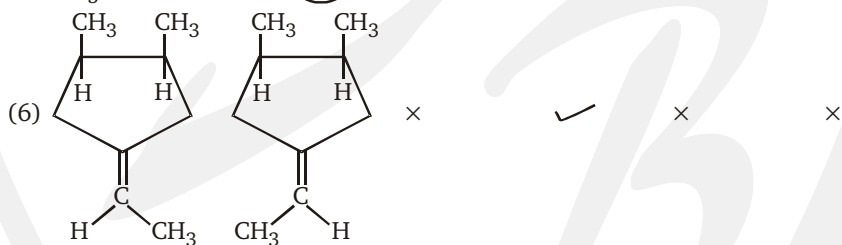


×

×

✓

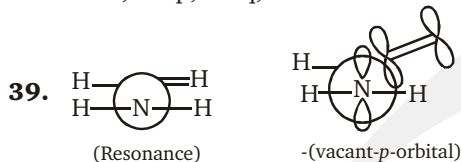
×



36. (A) - (d) (B) - (b)

37. a - p; b - q; c - r; d - s

38. a - s; b - p; c - q; d - r



40. (1) 16 (2) 4 (3) 16 (4) 4 (5) 4
 (6) 4 (7) 4 (8) 3 (9) 4 (10) 4
 (11) 2

41. a - 2, b - 4, c - 3, d - 4, e - 8, f - 2

42. a - s; b - p; c - q; d - r

43. a - 2^5 , b - 2^5 , c - 2^7 , 2^3 , d - 2^9

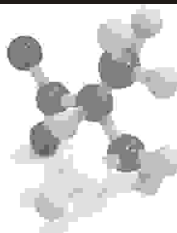
44. A - e; B - b; C - d; D - a

45. A - e; B - c; C - a; D - b; E - d

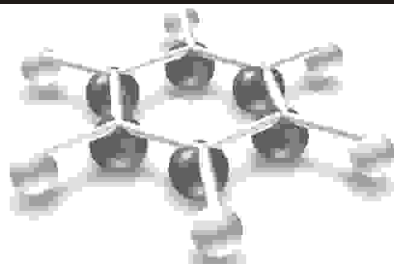
46. a - p, q, s, t; b - p, r; c - p, q, r, s, t; d - p

Subjective Problems

1. 3 (c, f, h) 2. 64 3. 5 4. 5 5. 2 6. 8. 7. 3
 8. 6 9. 12 10. 4 11. 5 12. 4 13. 4 14. 4 15. 8
 16. 4 (a, b, c, h) 17. 6



3

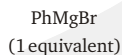
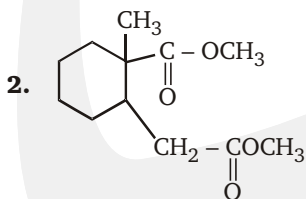
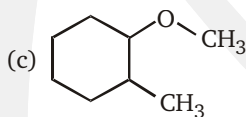
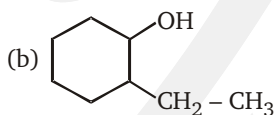
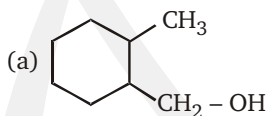
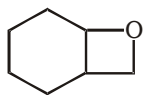


GRIGNARD REAGENT

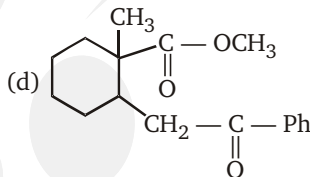
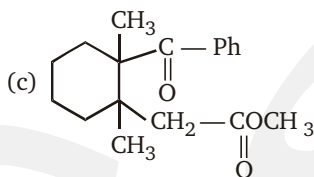
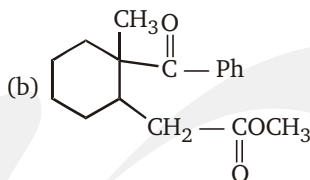
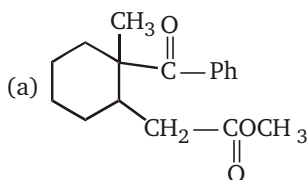


Level - 1

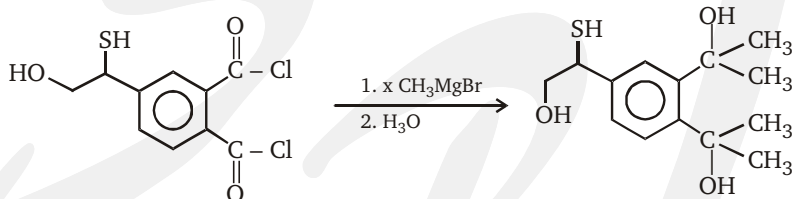
1. What is the major product of the following reaction ?



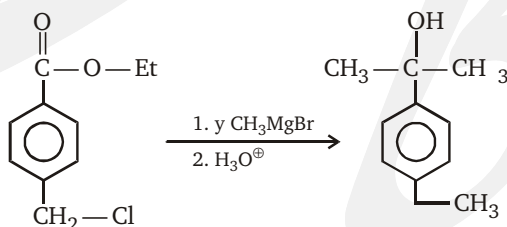
(P). Product (P) is :



3. Reaction- 1 ;



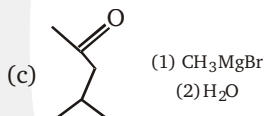
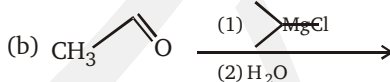
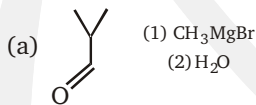
Reaction- 2 ;



What is the ratio of (x/y) in above problem ?

- (a) 1.5 (b) 2 (c) 2.5 (d) 3

4. In which of the following reaction 2° alcohol is obtained as a product ?

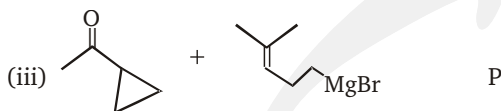
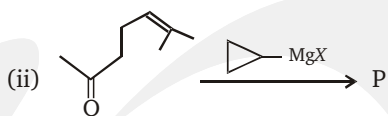
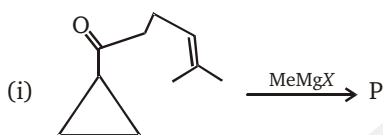


- (d) Both (a) and (b)

5. What product would you expect to obtain from Grignard reaction when an excess of phenylmagnesium bromide reacts with dimethyl carbonate $\text{CH}_3\text{OCOOCH}_3$?

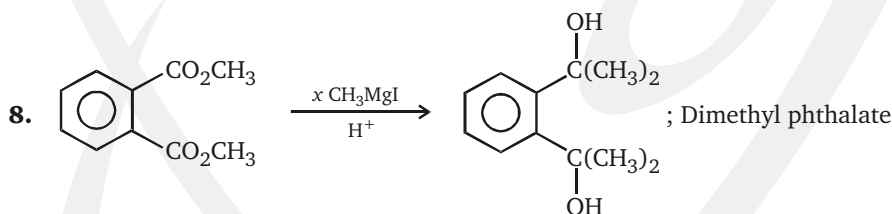
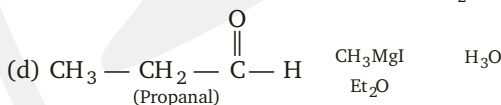
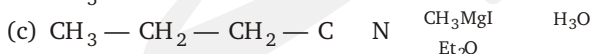
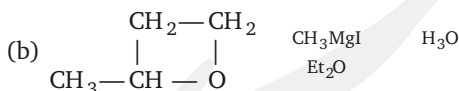
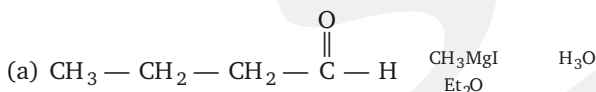
- (a) (b) (c) (d)

6. In which of the following reactions product formed is same ?



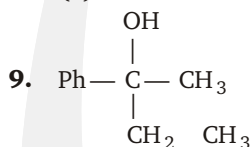
- (a) (i) and (ii) (b) (ii) and (iii) (c) (i) and (iii) (d) (i), (ii) and (iii)

7. Which of the following reaction sequences would be the best for synthesis of 2-pentanone ?

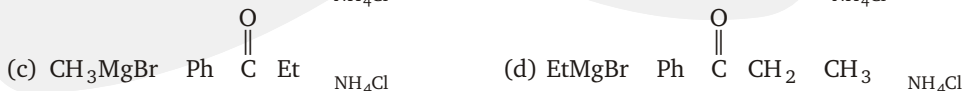


Number of moles (x) of Grignard reagent consumed in the above reaction is :

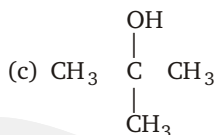
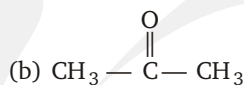
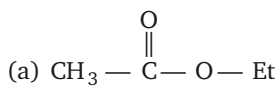
- (a) 2 (b) 3 (c) 4 (d) 5

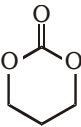


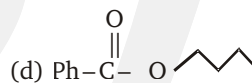
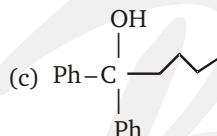
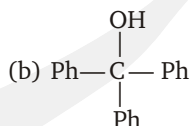
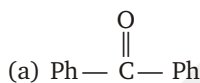
Which of the following combinations can not be used to prepare alcohol given above ?

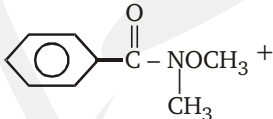


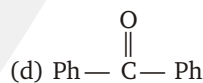
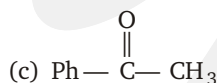
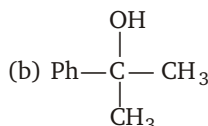
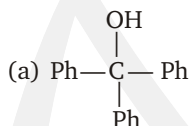
10. $\text{Et}-\text{O}-\overset{\text{O}}{\parallel}{\text{C}}-\text{O}-\text{Et}$ (1) CH_3MgBr (excess) (A), Product (A) is :
(2) H_2O



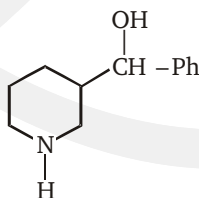
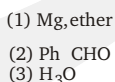
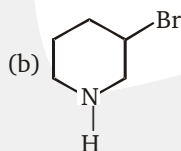
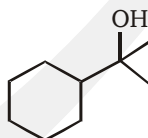
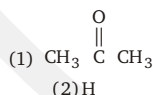
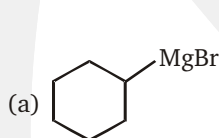
11. PhMgBr (excess)  H (A). Product (A) is :

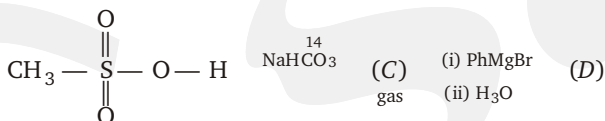
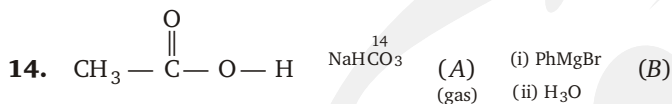
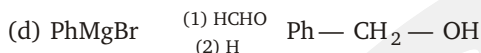
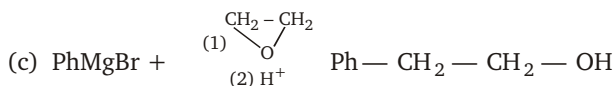


12.  + CH_3MgBr (Methyl magnesium bromide 75 equivalent) 5% HCl / H_2O (P). Product (P) is : 90%
- N-Methoxy-N-methyl benzamide 1-equivalent

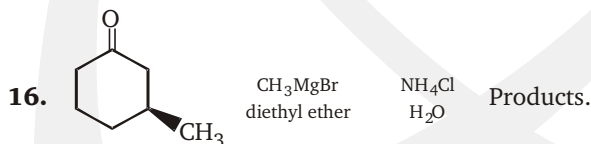
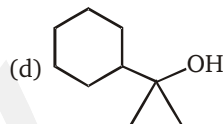
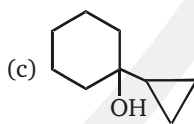
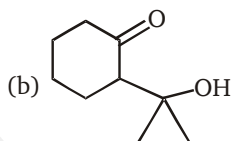
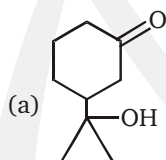
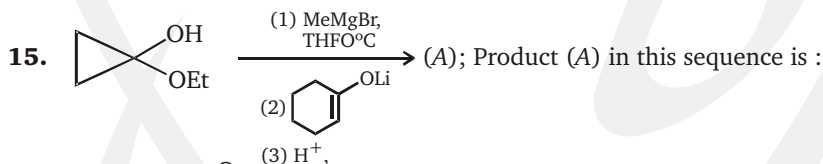
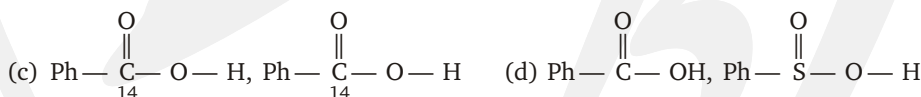
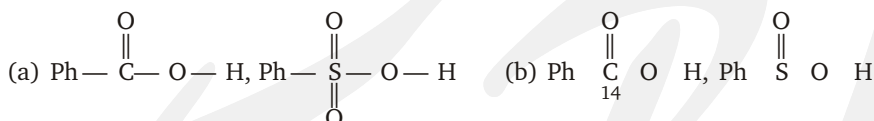


13. Point out the incorrect synthesis :





Product (B) and (D) in the above reaction are :



Comment on optical activity of the products. They are :

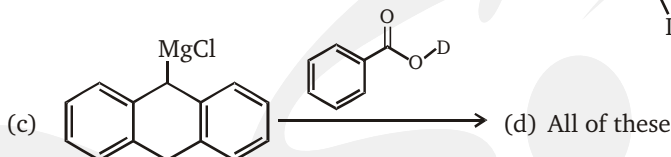
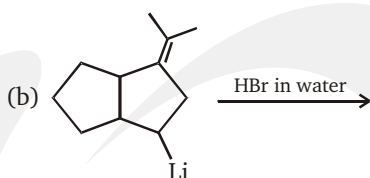
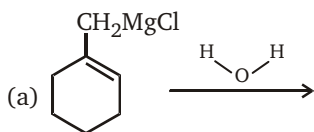
(a) racemic mixture

(b) diastereomers

(c) meso forms

(d) optically inactive due to absence of chiral centre

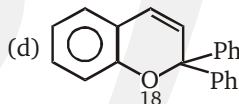
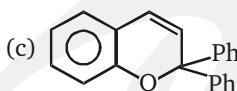
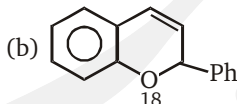
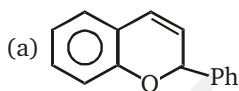
17. In which of the following reaction an acid-base reaction takes place ?



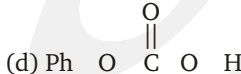
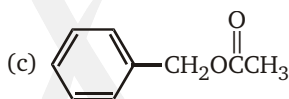
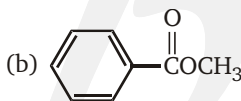
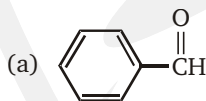
(d) All of these



18. (1) 2PhMgBr (2) H₂O A H₂SO₄ (B), Product (B) in this reaction is :



19. All of the following compounds react with ethylmagnesium bromide. Alcohols are formed from three of the compounds. Which one does not give an alcohol ?



20. A student was carrying out a Grignard reaction between PhMgBr and ethyl benzoate. She ran out of anhydrous ether just after the Grignard reagent was made. Which of the following solvents can still be used to dissolve the ethyl benzoate for its reaction with already formed PhMgBr ?

(a) acetone

(b) ethyl acetate

(c) absolute alcohol (d) benzene

21. (1) xPhMgBr (2) H₃O⁺

Number of equivalents of Grignard reagent (x) used in reaction (1) is :

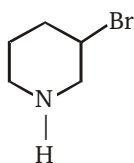
(a) 3 equivalent

(b) 4 equivalent

(c) 5 equivalent

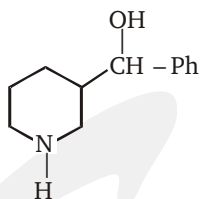
(d) 6 equivalent

22.



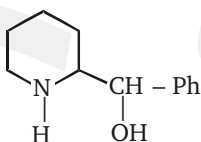
(1) Mg; ether

(2) Ph CHO

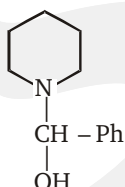
(3) H₃O⁺

The given product can not be obtained in the above reaction. Identify the correct product obtained.

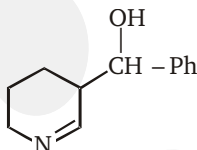
(a)



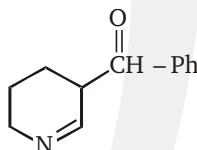
(b)



(c)

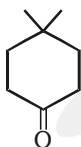


(d)

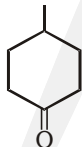


23. Which of the following gives two isomers of 3° alcohol, when treated with phenyl magnesium bromide ?

(a)



(b)



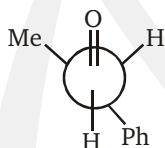
(c)



(d)



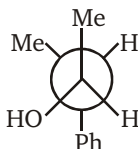
24.



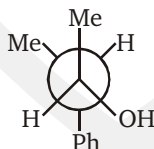
MeMgBr

Product of the reaction is :

(a)

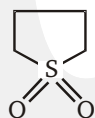


(b)



(c) Both (a) and (b) (d) None of these

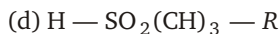
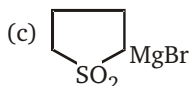
25.



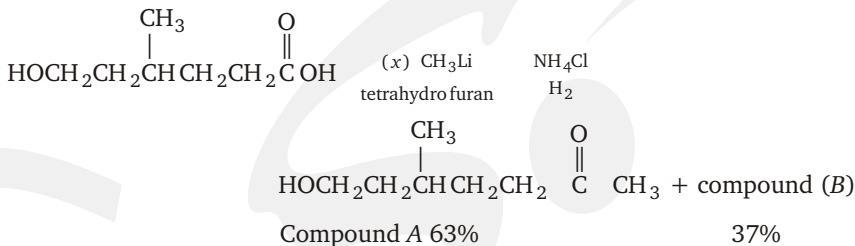
RMgBr

Product; The product of the reaction is :

(a) HSO₂ — CH₂ — CH₂ — CH₂ — CH₂ — R(b) H — SO₂(CH₂)₃ — R

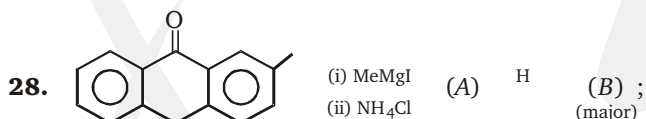
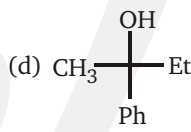
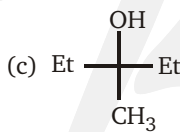
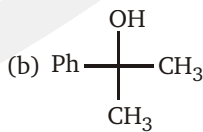
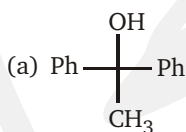


26. When carboxylic acid reacts with organolithium reagents to give ketones, side reaction sometimes occur. For example,

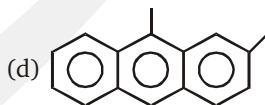
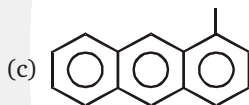
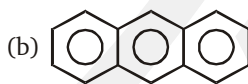
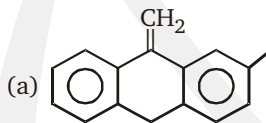


Value of (x) in above reaction is :

- (a) 2 (b) 3 (c) 4 (d) 5
27. Which of the following alcohol can not be prepared by the reaction of acid chloride with excess of Grignard reagent followed by acidification ?



Product (B) of the above reaction is :

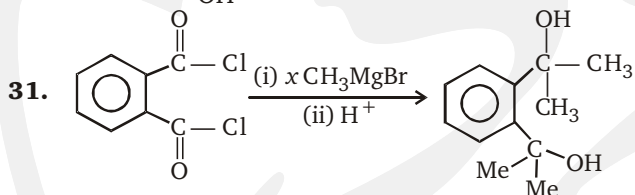
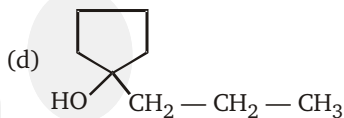
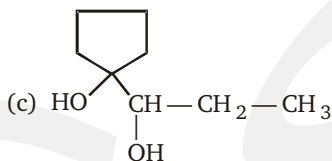
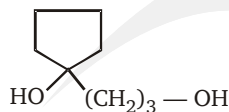
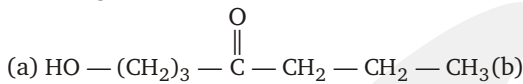
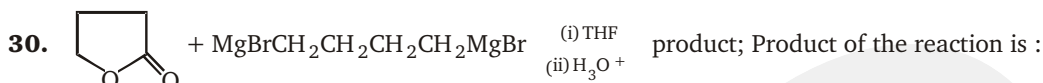


29. The reaction of elemental sulphur with Grignard reagent followed by acidification leads to the formation of

- (a) mercaptan (b) sulfoxide
(c) thioether (d) sulphonic acid



Grignard Reagent



Number of moles of CH_3MgBr consumed in above reaction is :

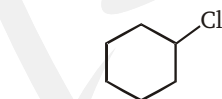
(a) 2

(b) 4

(c) 6

(d) 8

32. End product of the given reaction is :

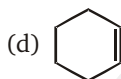
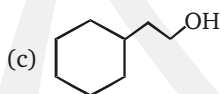
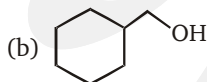


Mg
 Et_2O

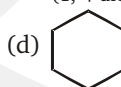
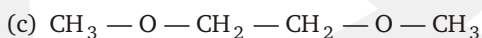
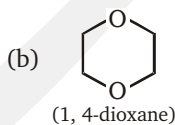
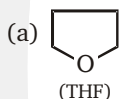
(A)

1. CH_2O
2. H_3O^+

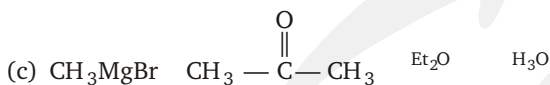
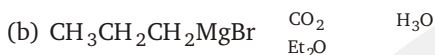
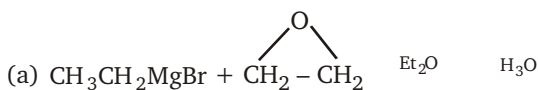
(B)



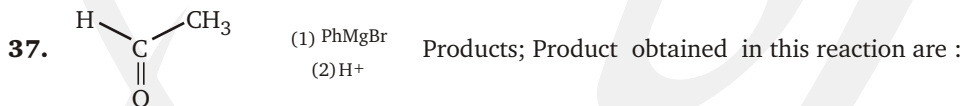
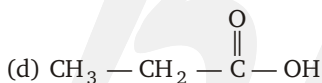
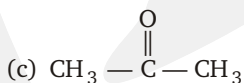
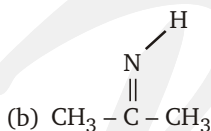
33. Which of the following compound is not a suitable solvent for Grignard reaction ?



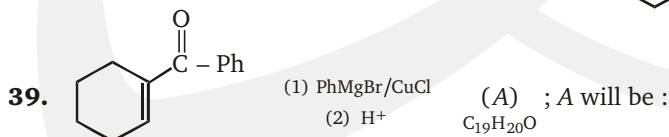
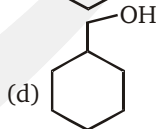
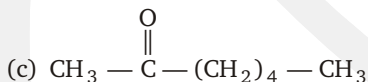
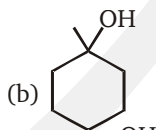
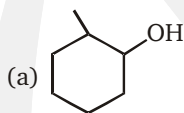
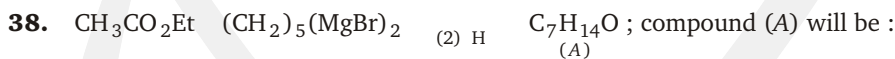
35. Which of the following reaction sequences would be the best for synthesis of t-butyl alcohol ?

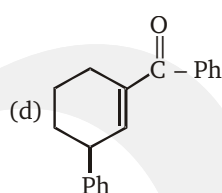
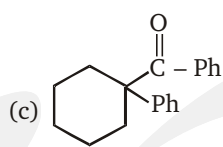
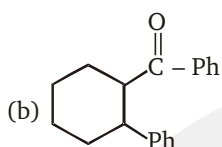
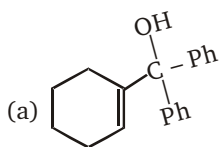


36. What is the major product of the following reaction ?

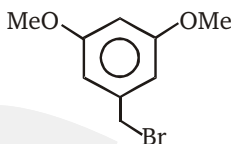


(a) diastereomers (b) racemic (c) pure enantiomer (d) meso



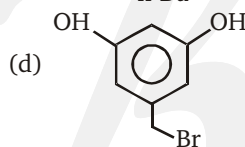
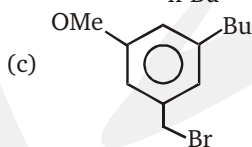
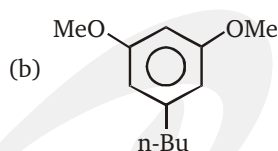
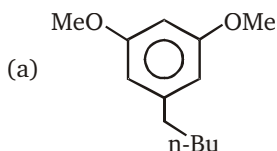


40.

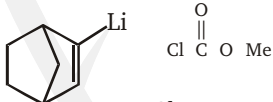


$n\text{-Bu}_2\text{CuLi}$
(n Bu n butyl group)

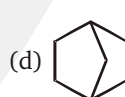
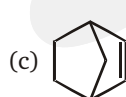
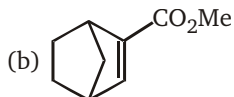
Product of the reaction will be :



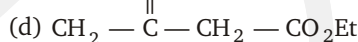
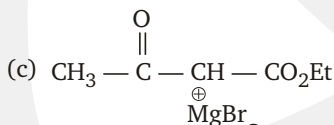
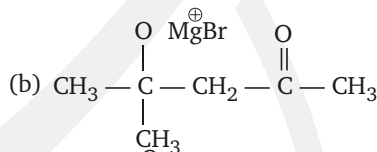
41.



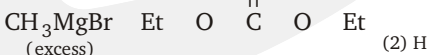
(?) ; Product of this reaction is :



42. Ethyl acetoacetate when reacts with one mole methyl magnesium iodide then product of reaction will be :



43.



(A); Product A is :



44. For the sequence of reactions, A $\xrightarrow[\text{ether}]{\text{C}_2\text{H}_5\text{MgI}}$ B $\xrightarrow{\text{H}_2\text{O/H}}$ tert-Pentyl alcohol. The compound A in the sequence is :
 (a) 2-Butanone (b) Acetaldehyde (c) Acetone (d) Propanal

45. $\text{PhMgBr} + \text{CH}_3\text{CN} \xrightarrow{\text{H}_3\text{O}^+} \text{(A)}$ $\text{Ph} - \text{C}(=\text{O}) - \text{OH} \xrightarrow[\text{(2) H}_2\text{O}]{\text{(1) excess CH}_3\text{Li}} \text{(A)}$

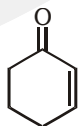
Same product (A) will form in both reactions. A is :

- (a) $\text{Ph} - \text{C}(\text{OH})(\text{CH}_3)_2$ (b) $\text{Ph} - \text{CHO}$ (c) $\text{Ph} - \text{C}(=\text{O}) - \text{CH}_3$ (d) $\text{Ph} - \text{CH}_2 - \text{CO}_2\text{H}$

46. Which of the following Grignard reagent can be prepared ?

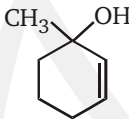
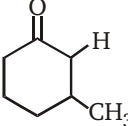
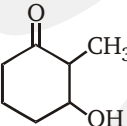
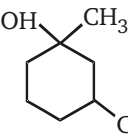
- (a) $\text{Br} - \text{Mg} - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{O} - \text{H}$ (b) $\text{Br} - \text{Mg} - \text{CH}_2 - \text{CH}_2 - \text{SH}$
 (c) $\text{BrMg} - \text{CH}_2 - \text{CH}_2 - \text{NH}_2$ (d) $\text{BrMg} - \text{CH}_2 - \text{CH}_2 - \text{N}(\text{CH}_3)_2$

47. In the reaction sequence :



- (i) $\text{CH}_3\text{MgBr}/\text{CuCl}$
 (ii) $\text{H}_2\text{O}/\text{H}^+$

(X), Product (X) will be :

- (a)  (b)  (c)  (d) 

48. $(\text{C}_2\text{H}_5\text{O})_2\text{CO} \xrightarrow[\text{H}_3\text{O}^+]{\text{CH}_3\text{MgBr}(\text{excess})}$ A. A (alcohol) can also be obtained by :

- (a) $\text{CH}_3\text{CH}_2\text{CHO} \xrightarrow[\text{H}_3\text{O}^+]{\text{CH}_3\text{MgBr}(2\text{mol})}$ (b) $\text{CH}_3\text{C}(=\text{O})\text{OC}_2\text{H}_5 \xrightarrow[\text{H}_3\text{O}^+]{\text{CH}_3\text{MgBr}(2\text{mol})}$
 (c) $\text{CH}_3\text{C}(=\text{O})\text{CH}_3 \xrightarrow[\text{H}_3\text{O}^+]{\text{CH}_3\text{MgBr}(1\text{mol})}$ (d) as in (b) and (c)

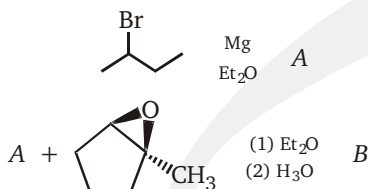
49. The principal product of the reaction between methyl butanoate and 2 moles of CH_3MgBr after hydrolysis is :

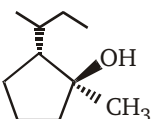
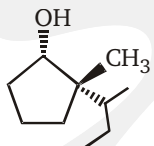
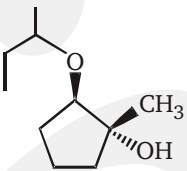
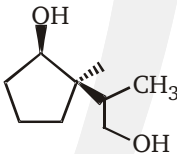
- (a) $\text{C}_3\text{H}_7\text{COCH}_3$ (b) $\text{C}_3\text{H}_7\text{C}(\text{OH})(\text{CH}_3)_2$
 (c) $\text{C}_3\text{H}_7\text{CHOHCH}_3$ (d) $\text{C}_3\text{H}_7\text{COCH}(\text{CH}_3)_2$

50. Which of the following compounds will form hydrocarbon on reaction with Grignard reagent ?

- (a) $\text{CH}_3\text{CH}_2\text{OH}$ (b) CH_3CHO (c) CH_3COCH_3 (d) $\text{CH}_3\text{CO}_2\text{CH}_3$

51. What is the product (B) of the following reaction sequence ?

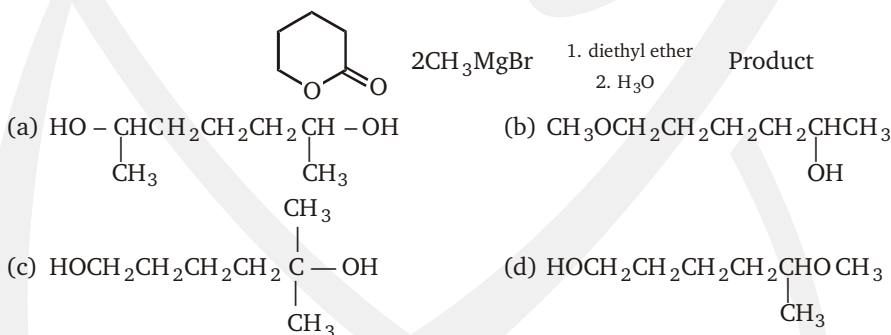


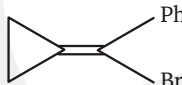
- (a) 
 (b) 
 (c) 
 (d) 

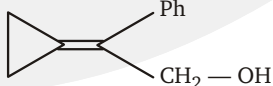
52. Which, if any, of the following pairs of reagents could be used to prepare 2-phenyl-2-butanol?

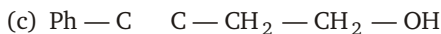
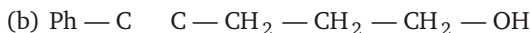
- (a) $\text{CH}_3\text{CH}_2\text{MgBr}$ $\text{Ph}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3$ (b) $\text{CH}_3\text{CH}_2\text{MgBr}$ $\text{C}_6\text{H}_5\text{CH}_2\overset{\text{O}}{\parallel}{\text{C}}\text{H}$
 (c) CH_3MgI $\text{C}_6\text{H}_5\text{CH}_2\overset{\text{O}}{\parallel}{\text{C}}\text{CH}_3$ (d) $\text{C}_6\text{H}_5\text{MgCl}$ $\text{CH}_3\overset{\text{O}}{\parallel}{\text{C}}\text{CH}_2\text{CH}_2\text{CH}_3$

53. What is the product of the following reaction?

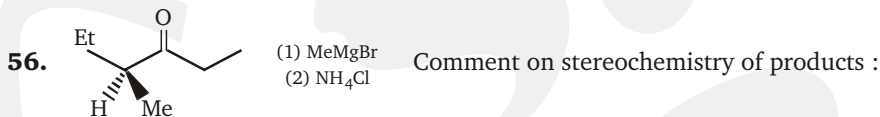


54.  $\xrightarrow[\text{ether}]{\text{Mg}}$ A $\xrightarrow[\text{2. H}^+]{\text{1. HCHO}}$ B ; Product (B) is :

- (a) 



55. What sequence of steps represents the best synthesis of 4-heptanol ($\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CHOH}$) ?
- (a) $\text{CH}_3\text{CH}_2\text{CH}_2\text{MgBr}$ (2 moles) + formaldehyde ($\text{H}_2\text{C}=\text{O}$) in diethyl ether followed by H_3O^+
- (b) $\text{CH}_3\text{CH}_2\text{CH}_2\text{MgBr}$ + butanol ($\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$) in diethyl ether followed by H_3O^+
- (c) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{MgBr}$ + acetone [$(\text{CH}_3)_2\text{C}=\text{O}$] in diethyl ether followed by H_3O^+
- (d) $(\text{CH}_3\text{CH}_2\text{CH}_2)_2\text{CHMgBr}$ + formaldehyde ($\text{H}_2\text{C}=\text{O}$) in diethyl ether followed by H_3O^+

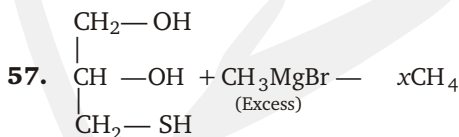


(a) diastereomers

(b) racemic

(c) single stereoisomer

(d) meso

What is the value of x in the above reaction ?

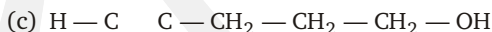
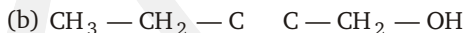
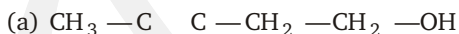
(a) 1

(b) 2

(c) 3

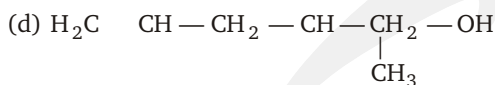
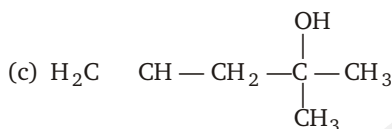
(d) 4

58. 0.40 g of an organic compound (A), (M.F.- $\text{C}_5\text{H}_8\text{O}$) reacts with x mole of CH_3MgBr to liberate 224 mL of a gas at STP. With excess of H_2 , (A) gives pentan-1-ol. The correct structure of (A) is :

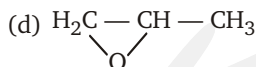
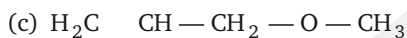
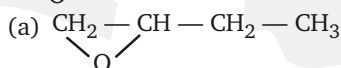


End product (X) of the above reaction is :

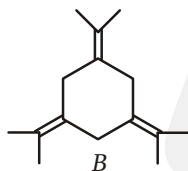




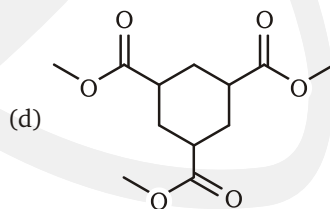
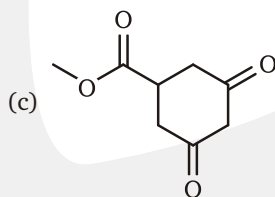
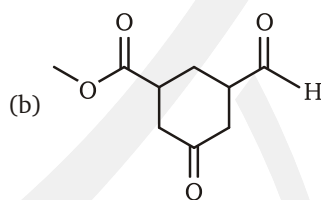
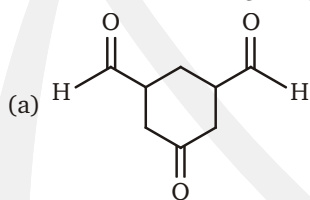
60. $\text{CH}_2 - \text{CH} - \text{CH}_2 - \text{Br}$ $\xrightarrow[\text{dry ethane}]{\text{Mg}}$ (A) $\xrightarrow{\text{CH}_3\text{I}}$ (B); Product (B) is :



61. Compound A was treated with a large excess of CH_3MgBr . The resulting product was exposed to $\text{POCl}_3/\text{pyridine}$ to give compound B, as one of many products :



Which of the following compounds can be A ?

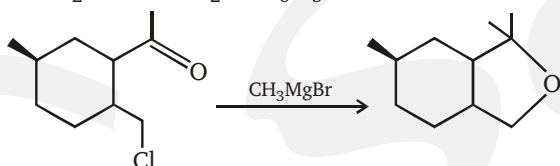


62. Identify product Z in the following reaction sequence :





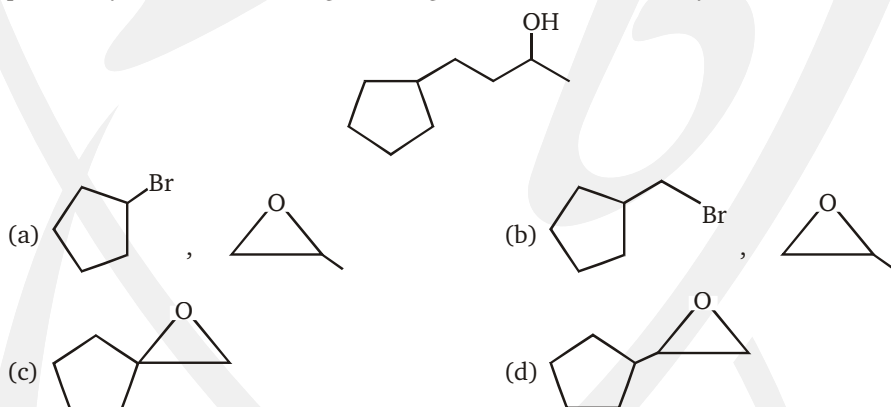
63.



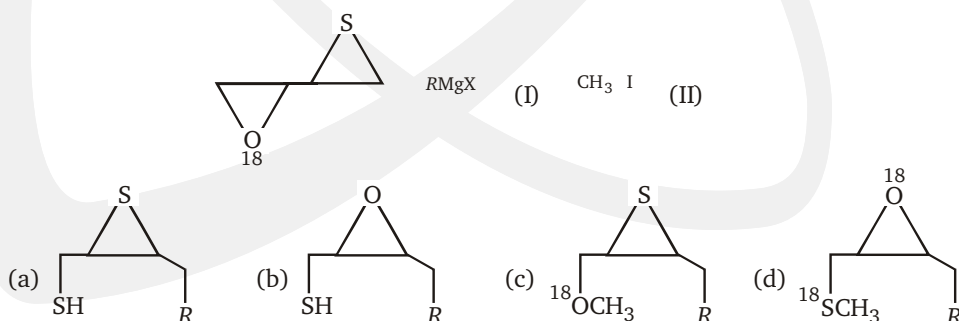
(Consider all steps and intermediate) correct statement is :

- (a) Nucleophilic addition (b) Nucleophilic substitution reaction
(c) Product obtained is chiral (d) All

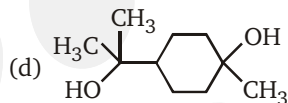
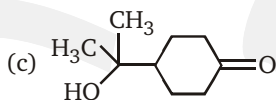
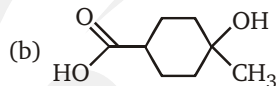
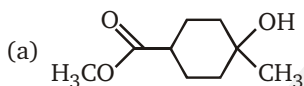
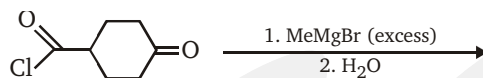
64. Which combination(s) of alkyl bromide and epoxide can be used to prepare the following product by addition of the Grignard reagent derived from the alkyl bromide to the epoxide ?



65. What will be the final major product ?



66. Give the expected product of the following reaction.



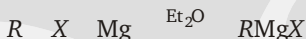
ANSWERS — LEVEL 1

1.	(b)	2.	(d)	3.	(b)	4.	(d)	5.	(c)	6.	(d)	7.	(c)	8.	(c)
9.	(d)	10.	(c)	11.	(b)	12.	(b)	13.	(b)	14.	(c)	15.	(b)	16.	(b)
17.	(d)	18.	(d)	19.	(d)	20.	(d)	21.	(c)	22.	(b)	23.	(b)	24.	(c)
25.	(c)	26.	(b)	27.	(d)	28.	(d)	29.	(a)	30.	(b)	31.	(b)	32.	(b)
33.	(d)	34.	(b)	35.	(c)	36.	(c)	37.	(b)	38.	(b)	39.	(b)	40.	(a)
41.	(b)	42.	(c)	43.	(b)	44.	(c)	45.	(c)	46.	(d)	47.	(b)	48.	(d)
49.	(b)	50.	(a)	51.	(a)	52.	(a)	53.	(c)	54.	(b)	55.	(b)	56.	(a)
57.	(c)	58.	(c)	59.	(b)	60.	(c)	61.	(d)	62.	(a)	63.	(d)	64.	(b)
65.	(c)	66.	(d)												

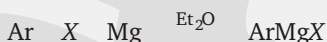
Level - 2

1. Comprehension

Grignard reagent is usually prepared by



Grignard reagent



Grignard reagent

Grignard reagent acts as a strong base. Grignard reagent carry out nucleophilic attack in absence of acidic hydrogen. Grignard reagent form complex with its ether solvent. Complex formation with molecule of ether is an important factor in the formation and stability of Grignard reagent.

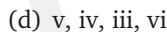
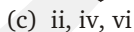
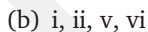
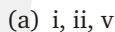
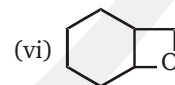
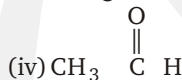
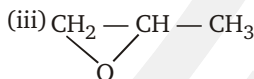
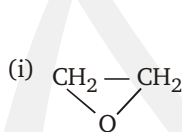
A. What is the correct order of reactivity of halides with magnesium ?



B. Which of the following will undergo acid-base reaction with Grignard reagent ?



C. Which of the following reactants give primary alcohol as a major product when reacts with $RMgX$ followed by acidification ?



D. $\xrightarrow[(2) NH_4Cl]{(1) xRMgX}$ 3 alcohol. Value of x is :



E. $\xrightarrow[(2) H]{(1) x PhMgBr}$ Ph, Value of x is :



F. Which of the following Grignard reagents is not possible ?

- (a) $\text{HS-CH}_2\text{-CH}_2\text{-CH}_2\text{MgBr}$ (b) $\text{HO-CH}_2\text{-CH}_2\text{-CH}_2\text{MgBr}$
 (c) $\text{NH}_2\text{-CH}_2\text{-CH}_2\text{-CH}_2\text{MgBr}$ (d) All of these

G. How many different Grignard reagents when react with EtOH, give *n*-butane as product (excluding stereoisomerism).

- (a) 1 (b) 2 (c) 3 (d) 4

2. Match the column I and II. (Matrix)

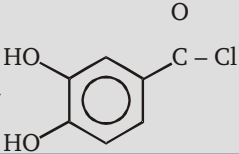
Column (I)			Column (II)	
Reactant			Product	
(a)	PhMgBr (excess)	$\begin{array}{c} \text{O} \\ \\ \text{Cl-C-O-Et} \\ \\ \text{H} \end{array}$	(p)	Ph-CH ₂ -OH
(b)	PhMgBr (excess)	$\begin{array}{c} \text{O} \\ \\ \text{H-C-O-Et} \\ \\ \text{H} \end{array}$	(q)	$\begin{array}{c} \text{Ph-CH-Ph} \\ \\ \text{OH} \end{array}$
(c)	PhMgBr (excess)	$\begin{array}{c} \text{O} \\ \\ \text{H-C-H} \\ \\ \text{H} \end{array}$	(r)	$\begin{array}{c} \text{OH} \\ \\ \text{Ph-C-Ph} \\ \\ \text{Ph} \end{array}$
(d)	PhMgBr (excess)	$\begin{array}{c} \text{O} \\ \\ \text{CH}_3\text{-C-O-Et} \\ \\ \text{H} \end{array}$	(s)	$\begin{array}{c} \text{OH} \\ \\ \text{Ph-C-Ph} \\ \\ \text{CH}_3 \end{array}$

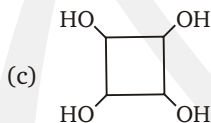
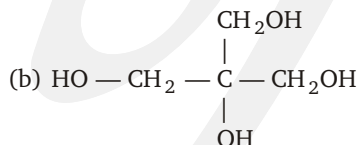
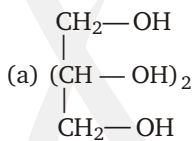
3. Match the column I and II. (Matrix)

Column (I)			Column (II)	
Reaction			Reactant	
(a)	PhMgBr + (A)	$\begin{array}{c} \text{H} \\ \\ \text{1}^\circ\text{alcohol} \end{array}$	(p)	$\begin{array}{c} \text{O} \quad \text{O} \\ \quad \\ \text{CH}_3\text{-C-CH}_2\text{-C-CH}_3 \end{array}$
(b)	PhMgBr + (B)	$\begin{array}{c} \text{H} \\ \\ \text{2}^\circ\text{alcohol} \end{array}$	(q)	$\begin{array}{c} \text{O} \\ \\ \text{CH}_3\text{-C-CH}_3 \end{array}$
(c)	PhMgBr + (C)	$\begin{array}{c} \text{H} \\ \\ \text{3}^\circ\text{alcohol} \end{array}$	(r)	$\begin{array}{c} \text{O} \\ \\ \text{CH}_3\text{-C-H} \end{array}$
(d)	PhMgBr + (D)	$\begin{array}{c} \text{H} \\ \\ \text{C}_6\text{H}_6 \end{array}$	(s)	$\begin{array}{c} \text{O} \\ \\ \text{H-C-H} \end{array}$

Match the missing reactant A, B, C, D

4. Match the column I and II. (Matrix)

Column (I)				Column (II)	
Reaction				Moles of PhMgBr used	
(a)	PhMgBr	Et O $\overset{\text{O}}{\parallel}$ C O Et H	3°alcohol	(p)	1
(b)	PhMgBr	HO CH ₂ $\overset{\text{O}}{\parallel}$ C CH ₃ H	3° alcohol	(q)	2
(c)	PhMgBr	CH ₃ $\overset{\text{O}}{\parallel}$ C CH ₃ H	3°alcohol	(r)	3
(d)	PhMgBr +		3°alcohol	(s)	4

5. When 20 g of a compound (A) (M.F. $\text{C}_4\text{H}_{10}\text{O}_4$) reacts with excess of CH_3MgBr , 14.6 L of CH_4 is obtained at STP. What is structural formula of (A) ?

(d) Both (a) & (b)

Subjective Problems

1.

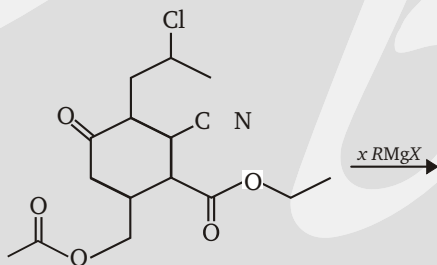


How many geometrical isomer of (X) is possible ?

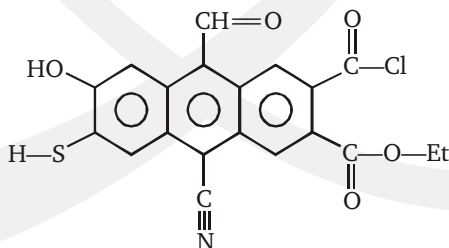
2. How many isomer of C_4H_8O when reacts with CH_3MgBr followed by acidification to give 2° alcohol (only consider carbonyl isomers)?
(including stereoisomer)

3.

Total number of $RMgX$ are consumed in the following reaction



4. How many isomers of $C_4H_{10}O$ reacts with CH_3MgBr to evolve CH_4 gas ? (Excluding stereoisomer)
5. How many carbonyl isomers of $C_5H_{10}O$ which reacts with $PhMgBr$ to give racemic mixture ?
6. How many moles of Grignard reagent will consume when it reacts with following compound?



ANSWERS — LEVEL 2

1. A – c; B – d; C – a; D – b; E – b; F – d; G – b
2. a – r; b – q; c – p; d – s
3. a – s; b – r; c – q; d – p
4. a – r; b – q; c – p; d – s
5. (d)

Subjective Problems

1. 4

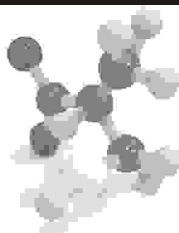
2. 2

3. 7

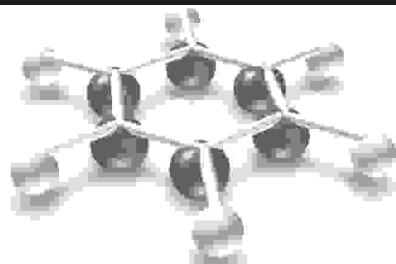
4. 4

5. 5

6. 8



4A



HYDROCARBONS (ALKANES)

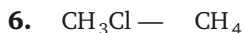


Level - 1

- On halogenation, an alkane gives only one monohalogenated product. The alkane may be :
 - 2-methyl butane
 - 2, 2-dimethyl propane
 - cyclopentane
 - both (b) and (c)
- Which of the following compounds can be best prepared by Wurtz-reaction ?
 - Iso-butane
 - n*-butane
 - n*-pentane
 - Iso-pentane
- A hydrocarbon A (V.D. 36) forms only one monochloro substitution product. A will be :
 - iso-pentane
 - neo-pentane
 - cyclohexane
 - methyl-cyclohexane
- Ethyl iodide and *n*-propyl iodide are allowed to undergo Wurtz reaction. The alkane which will not be obtained in this reaction is :
 - butane
 - propane
 - pentane
 - hexane
- $$\begin{array}{c} \text{CH}_3 - \text{CH} - \text{CH}_2 - \text{CH}_3 \\ | \\ \text{CH}_3 \end{array} \xrightarrow[\text{h}]{\text{Cl}_2}$$

Number of chiral centers generated during monochlorination in the above reaction :

- 1
- 2
- 3
- 4



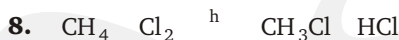
Above conversion can be achieved by :

- (a) Zn / H (b) LiAlH_4
 (c) $\text{Mg} / (\text{ether})$ then H_2O (d) all of these



Give the total number of monochloro products (including stereoisomers), which are possible in the above reaction.

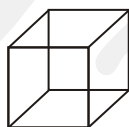
- (a) 2 (b) 3 (c) 4 (d) 5



To obtain high yields of CH_3Cl , the ratio of CH_4 to Cl_2 must be :

- (a) high (b) low
 (c) equal (d) can't be predicted

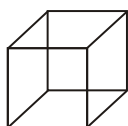
9. Double bond equivalent of cubane is :



Cubane

- (a) 4 (b) 5 (c) 6 (d) 7

10. How many bond cleavages are required to convert cubane into non-cyclic skeleton ?



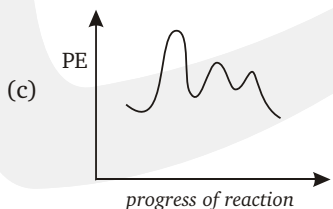
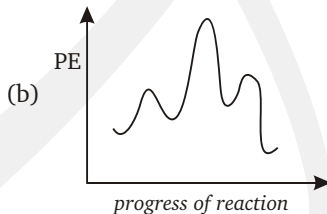
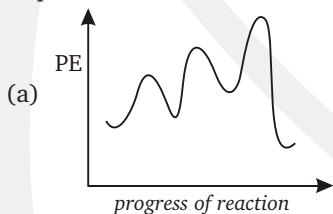
Cubane



Non-cyclic skeleton

- (a) 2 (b) 3 (c) 4 (d) 5

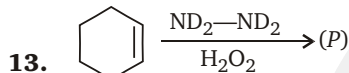
11. Draw an energy profile diagram for a three step reaction in which first step is slowest and last step is fastest. (Assume that reaction is exothermic)



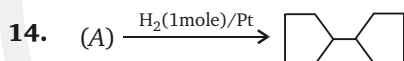
(d) None of these

12. $\text{CH}_3 - \underset{\text{CH}_3}{\underset{|}{\text{CH}}} - \text{CH}_2 - \text{CH}_3$ $\xrightarrow[h]{\text{Cl}_2}$ (x) = Number of monochloro product including stereoisomers.

(a) 4 (b) 5 (c) 6 (d) 7



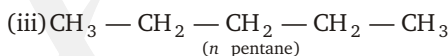
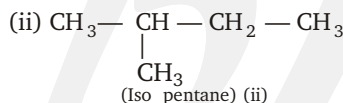
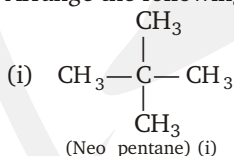
Product (P) is :



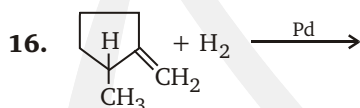
Double bond equivalent (degree of Unsaturation) of (A) is :

(a) 1 (b) 2 (c) 3 (d) 4

15. Arrange the following alkanes in decreasing order of their heats of combustion.

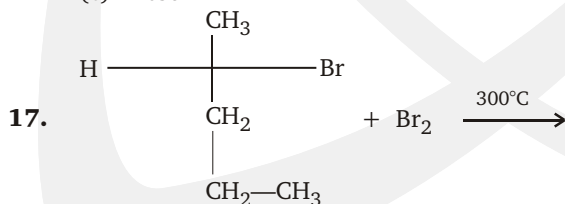


(a) (i) > (ii) > (iii) (b) (iii) > (i) > (ii)
(c) (iii) > (ii) > (i) (d) (i) > (iii) > (ii)

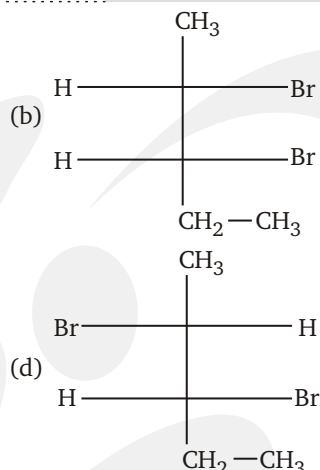
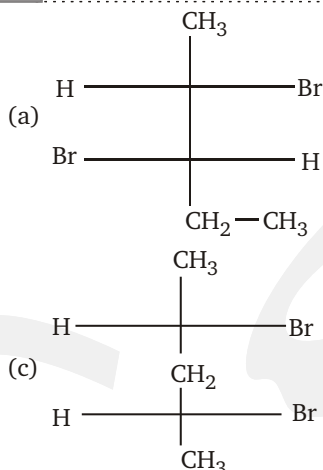


Product of the above reaction will be :

(a) Racemic mixture (b) Diastereomers
(c) Meso (d) Constitutional isomers



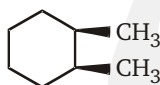
Which of the following compound will not be obtained as a product in the above reaction ?



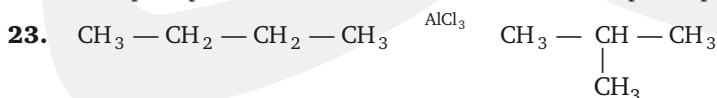
18. Following are the structures of four isomer of hexane. Among the names given below, which correctly identifies the fifth isomer ?



- (a) 2-Methyl pentane
(b) 2-Ethyl butane
(c) 2,3-Dimethyl butane
(d) 3-Methyl pentane
19. Which of the following describes the best relationship between the methyl groups in the chair conformation of the substance shown below ?



- (a) Trans
(b) Anti
(c) Gauche
(d) Eclipsed
20. Compare the stabilities of the following two compounds (A) and (B):
A : *cis*-1-ethyl-3-methyl cyclohexane
B : *trans*-1-ethyl-3-methyl cyclohexane
(a) A is more stable
(b) B is more stable
(c) A and B are of equal stability
(d) No comparison can be made
21. Which conformation of ethane has the lowest potential energy ?
(a) Eclipsed
(b) Skew
(c) Staggered
(d) All will have equal potential energy
22. Ethane is subjected to combustion process. During the combustion the hybrid state of carbon changes from :
(a) sp^2 to sp^3
(b) sp^3 to sp
(c) sp to sp^3
(d) sp^2 to sp^2



Above reaction is an example of :

- (a) isomerization (b) polymerization
(c) cracking (d) de-hydrogenation
24. Which of the following has highest chlorine content ?
(a) Pyrene (b) DDT (c) Chloral (d) Gammaxene
25. Pure methane can be prepared by :
(a) Wurtz reaction (b) Kolbe electrolysis method
(c) soda-lime de-carboxylation (d) reduction with H_2
26. Calcium carbide + heavy water — ?
The product of the above reaction is :
(a) C_2H_2 (b) CaD_2 (c) $Ca(OD)_2$ (d) CD_4
27. $CH_3 - CH_2 - \text{Cyclopentane}$ (I) $CH_3 - CH_2 - \text{Cyclohexane}$ (II) $CH_3 - CH_2 - \text{Cycloheptane}$ (III)
Ethyl cyclopentane Ethyl cyclohexane Ethyl cycloheptane
- Arrange the compounds I, II and III in decreasing order of their heats of combustion:
(a) $II > I > III$ (b) $I > II > III$
(c) $III > II > I$ (d) $III > I > II$
28. An alkane (mol. wt. 86) on bromination gives only two monobromo derivatives (excluding stereoisomers). The alkane is :
(a) $CH_3 - \underset{\substack{| \\ CH_3}}{CH} - CH_2 - CH_2 - CH_3$ (b) $CH_3 - \overset{\substack{CH_3 \\ |}}{\underset{\substack{CH_3 \\ |}}{C}} - CH_2 - CH_3$
(c) $CH_3 - \underset{\substack{| \\ CH_3}}{CH} - \underset{\substack{| \\ CH_3}}{CH} - CH_3$ (d) $CH_3 - \overset{\substack{CH_3 \\ |}}{\underset{\substack{CH_3 \\ |}}{C}} - CH_3$
29. Order of the bond strength of C — H bonds involving sp, sp^2 and sp^3 hybridized carbon atoms is :
(a) $sp \ sp^2 \ sp^3$ (b) $sp^3 \ sp^2 \ sp$
(c) $sp^2 \ sp^3 \ sp$ (d) $sp^2 \ sp \ sp^3$
30. (I) (II) (III)
Among the structures given, select the enantiomers :
(a) I and II (b) I and III
(c) II and III (d) I, II and III

31.



(I)



(II)

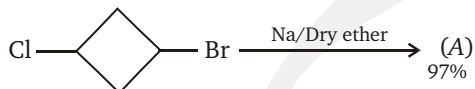


(III)

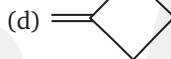
The correct order of reactivity of I, II & III towards addition reactions is :

- (a) I > III > II (b) I > II > III (c) III > II > I (d) III > I > II

32.



Product (A) of above reaction is :



33.

Which of the following reactants is suitable for preparation of methane and ethane by using one step only ?

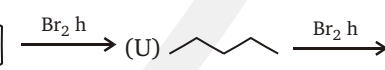
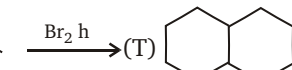
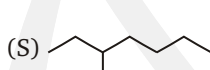
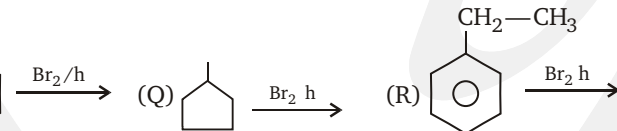
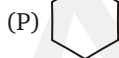
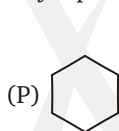
- (a) $\text{H}_2\text{C} \quad \text{CH}_2$ (b) CH_3OH
(c) $\text{CH}_3 - \text{Br}$ (d) $\text{CH}_3 - \text{CH}_2 - \text{OH}$

34.

How many carbon atoms does an alkane (not a cycloalkane) need before it is capable to exist in enantiomeric form ?

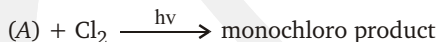
- (a) 4 (b) 5 (c) 6 (d) 7

35.

Among the following free radical bromination reactions, select those in which 2° halide is the major product —

- (a) P, Q, R, S (b) P, R, U (c) P, R, S, T (d) P, Q, R, S, T

36.



To maximise the yield of monochloro product in the above reaction ?

- (a) Cl_2 must be added in excess
(b) Reactant (A) must be added in excess
(c) Reaction must be carried out in dark
(d) Reaction must be carried out with equimolar mixture of Cl_2 and A

37.

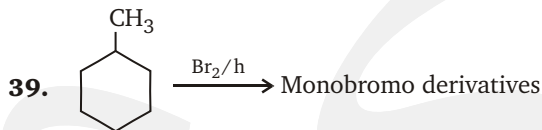


Major product in the above reaction is :

- (a) Racemic mixture (b) Meso
(c) Diastereomers (d) Constitutional isomers

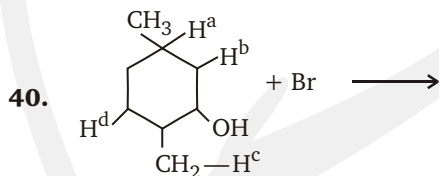
38. Select the chain propagation steps in the free-radical chlorination of methane.

- (1) $\text{Cl}_2 \longrightarrow 2\text{Cl}$ (2) $\text{Cl} + \text{CH}_4 \longrightarrow \text{CH}_3\text{Cl} + \text{H}$
 (3) $\text{Cl} + \text{CH}_4 \longrightarrow \text{CH}_3 + \text{HCl}$ (4) $\text{H} + \text{Cl}_2 \longrightarrow \text{HCl} + \text{Cl}$
 (5) $\text{CH}_3 + \text{Cl}_2 \longrightarrow \text{CH}_3\text{Cl} + \text{Cl}$
 (a) 2, 3, 5 (b) 1, 3, 6
 (c) 3, 5 (d) 2, 3, 4



The number of possible monobromo products is (excluding stereoisomers):

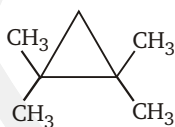
- (a) 4 (b) 5 (c) 8 (d) 10



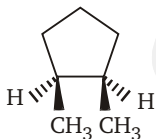
Br will abstract which of the hydrogen most readily ?

- (a) a (b) b (c) c (d) d

41. Arrange the following compounds in decreasing order of their heats of combustion :



(i)

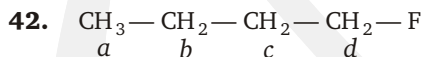


(ii)



(iii)

- (a) (iii) > (ii) > (i) (b) (ii) > (i) > (iii)
 (c) (iii) > (i) > (ii) (d) (i) > (ii) > (iii)



Arrange the hydrogens a, b, c, d, in decreasing order of their reactivities towards chlorination:

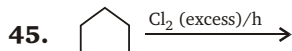
- (a) $a > b > c > d$ (b) $b > c > d > a$
 (c) $b > c > a > d$ (d) $c > b > a > d$

43. On catalytic reduction (H_2/Pt) how many alkenes will give n-butane ?

- (a) 1 (b) 2
 (c) 3 (d) 4

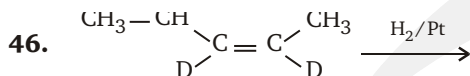
44. On catalytic reduction (H_2/Pt) how many alkenes will give 2-methylbutane ?

- (a) 1 (b) 2
 (c) 3 (d) 4



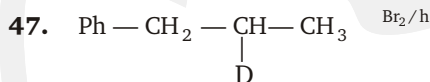
How many dichloro products are formed in the above reaction (including stereoisomers)?

- (a) 5 (b) 6
(c) 7 (d) 9



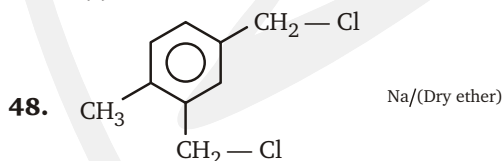
Product of the above reaction will be :

- (a) Racemic mixture (b) Diastereomers
(c) Meso (d) Constitutional isomers

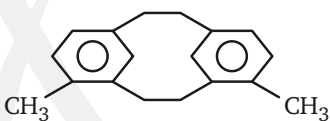
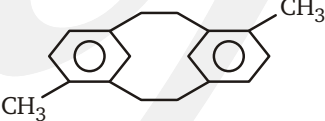
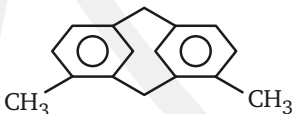


Product of the above reaction will be :

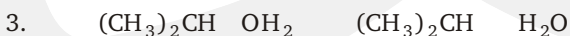
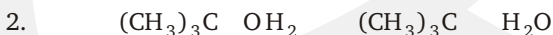
- (a) Diastereomers (b) Racemic mixture
(c) Meso (d) Constitutional isomers



Products obtained in above Wurtz reaction is :

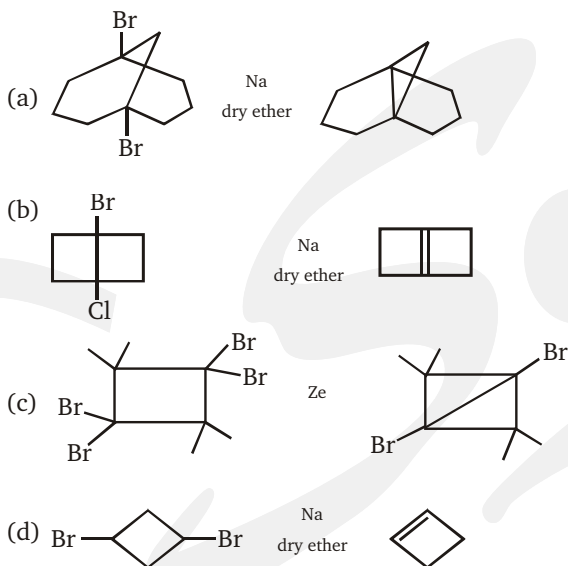
- (a)  (b) 
(c)  (d) Both (a) and (b)

49. Rank the transition states that occur during the following reaction steps in order of increasing stability (least most stable) :



- (a) $1 < 2 < 3$ (b) $2 < 3 < 1$
(c) $1 < 3 < 2$ (d) $2 < 1 < 3$

50. Which of the following does not represent major product of that reaction ?



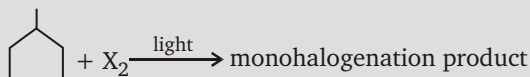
ANSWERS — LEVEL 1

1.	(d)	2.	(b)	3.	(b)	4.	(b)	5.	(b)	6.	(d)	7.	(b)	8.	(a)
9.	(b)	10.	(d)	11.	(c)	12.	(c)	13.	(b)	14.	(c)	15.	(c)	16.	(b)
17.	(d)	18.	(d)	19.	(c)	20.	(a)	21.	(c)	22.	(b)	23.	(a)	24.	(a)
25.	(c)	26.	(c)	27.	(c)	28.	(c)	29.	(a)	30.	(c)	31.	(b)	32.	(b)
33.	(c)	34.	(d)	35.	(b)	36.	(b)	37.	(a)	38.	(c)	39.	(b)	40.	(a)
41.	(d)	42.	(c)	43.	(c)	44.	(c)	45.	(c)	46.	(a)	47.	(a)	48.	(d)
49.	(c)	50.	(d)												

Level - 2

1. Comprehension

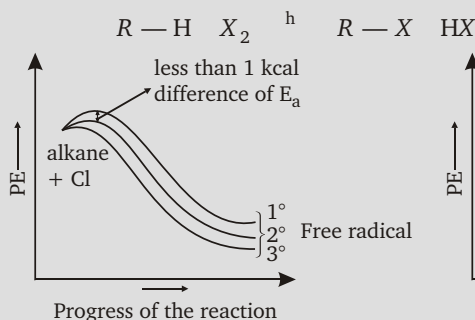
For the given question (1, 2, 3), consider the following reaction.



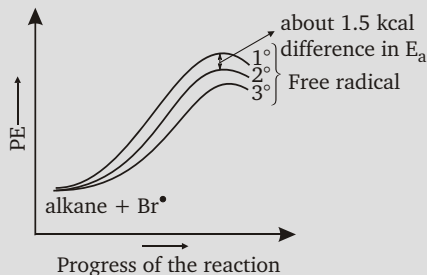
- A.** Light is involved in which step of the reaction :
- (a) Initiation only (b) Termination only
(c) Propagation only (d) Propagation and Termination
- B.** Which halogen will give the best yield of a single monohalogenation product ?
- (a) F_2 (b) Cl_2 (c) Br_2 (d) I_2
- C.** How many monohalo derivatives are possible (excluding stereoisomers) ?
- (a) 3 (b) 4 (c) 5 (d) 6

2. Comprehension

Halogenation is a substitution reaction, where halogen replaces one or more hydrogens of hydrocarbon.



Chlorination is exothermic and transition state resembles with products



Bromination is endothermic and transition state resembles with products

Chlorine free radical make 1° , 2° , 3° radicals with almost equal ease, whereas bromine free radicals have a clear preference for the formation of tertiary free radicals. So, bromine is less reactive, and more selective whereas chlorine is less selective and more reactive.

The relative rate of abstraction of hydrogen by $\text{Br}^\cdot$ is

$$3^\circ > 2^\circ > 1^\circ$$

(1600) (82) (1)

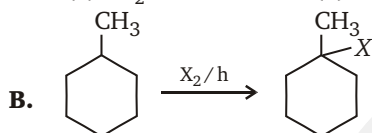
The relative rate of abstraction of hydrogen by $\text{Cl}^\cdot$ is :

$$3^\circ > 2^\circ > 1^\circ$$

(5) (3.8) (1)

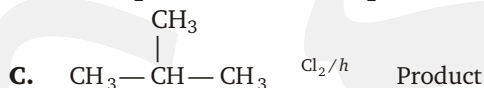
Consider the above argument and answer A to G :

- A.** 1-halo-2,3-dimethyl butane will be obtained in better yields, if halogen is :
 (a) Br_2 (b) Cl_2 (c) I_2 (d) Can't be predicted

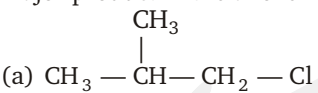
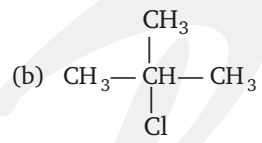
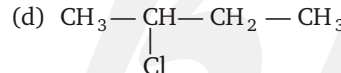


Above product will obtained in better yield if X is

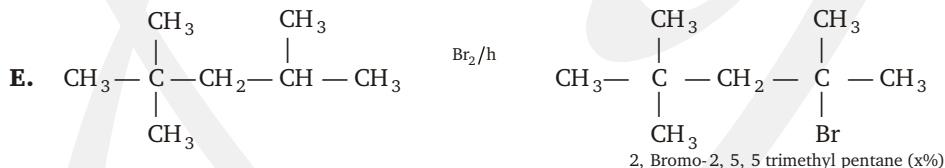
- (a) Cl_2 (b) I_2 (c) Br_2 (d) Can't be predicted



Major product in the above reaction is :

- (a)  (b) 
 (c) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{Cl}$ (d) 

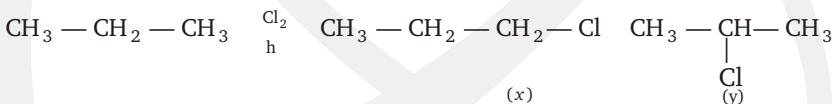
- D.** Which of the following will give five monochloro products, when allowed to react with Cl_2 in presence of sun light (excluding stereoisomers) ?
 (a) n -pentane (b) Iso-pentane (c) 2-methyl-pentane (d) 3-methyl pentane



What is the value of x (% yield of product)?

- (a) 18 % (b) 82 % (c) 90 % (d) 60 %

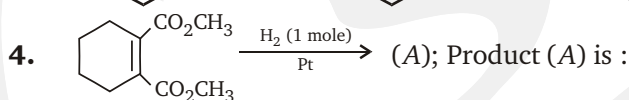
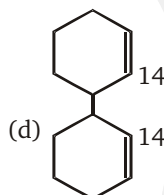
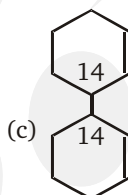
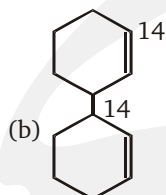
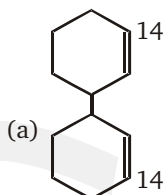
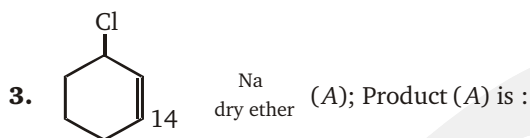
- F.** What would be the product ratio x/y in the chlorination of propane if all the hydrogen were abstracted at equal rate ?



- (a) $\frac{1}{3}$ (b) $\frac{3}{1}$ (c) $\frac{9}{1}$ (d) $\frac{1}{9}$

- G.** How many dichloro products (including stereoisomers) will be formed when R-2-chloropentane reacts with Cl_2 in presence of UV radiation ?

- (a) 5 (b) 6 (c) 7 (d) 8



(a) Meso compound

(b) Racemic mixture

(c) Diastereomers

(d) Optically active

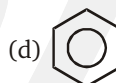


Product (A) is :

(a) $\text{Ph}-\text{CO}_2\text{H}$

(b) $\text{Ph}-\text{CH}_2-\text{OH}$

(c) $\text{Ph}-\text{CH}_3$



6. Match the column I with column II and with column III.

Column (I)		Column (II)		Column (III)	
Compound		Mono-chloro products		Monochloro products	
		(excluding stereoisomerism)		(including stereoisomerism)	
(a)		(p)	1	(w)	1
(b)	$\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\text{CH}_2-\text{CH}_3$	(q)	2	(x)	3
(c)	$\begin{array}{c} \text{CH}_3 \quad \text{CH}_3 \\ \quad \\ \text{CH}_3-\text{C}-\text{C}-\text{CH}_3 \\ \quad \\ \text{CH}_3 \quad \text{CH}_3 \end{array}$	(r)	3	(y)	5

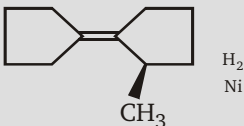
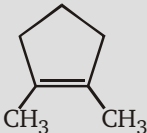
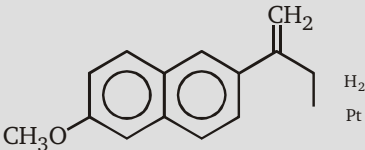
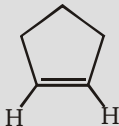
(d)	$\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CH}_3$	(s)	4	(z)	6
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7.

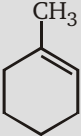
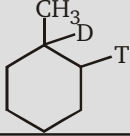
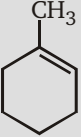
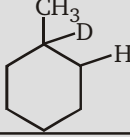
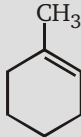
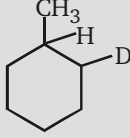
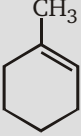
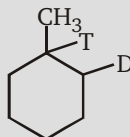
A.	R-2-chloropentane	Cl_2 h	Optically active di-chloro products (P)
B.		Cl_2 h	Optically active dichloro products (Q)
C.	R-2-chlorobutane	Cl_2 h	Optically active di-chloro products (R)

Sum P Q R is :

8. Match the column I and II.

Column (I)		Column (II)	
Reaction		Type of Reaction	
(a)		(p)	Meso compound
(b)		(q)	Diastereomers
(c)		(r)	Racemic
(d)		(s)	Optically inactive due to absence of chiral center

9. Match the column :

Column (I)		Column (II)	
Reaction		Product	
(a)	 (1) BD_3 ; THF (2) $\text{CH}_3\text{CO}_2\text{T}$	(p)	
(b)	 (1) BT_3 ; THF (2) $\text{CH}_3\text{CO}_2\text{D}$	(q)	
(c)	 (1) BD_3 ; THF (2) $\text{CH}_3\text{CO}_2\text{H}$	(r)	
(d)	 (1) BH_3 ; THF (2) $\text{CH}_3\text{CO}_2\text{D}$	(s)	

10. How many distinct monochlorinated products, (including stereoisomers) may be obtained when the alkane shown below is heated in the presence of Cl_2 ?



- (a) 1 (b) 2 (c) 3 (d) 4 (e) 6

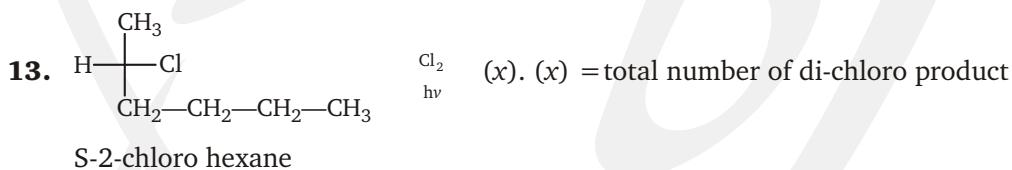
11. How many distinct monochlorinated products, (including stereoisomers) may be obtained when the alkane shown below is heated in the presence of Cl_2 ?



- (a) 2 (b) 4 (c) 5 (d) 6 (e) 8

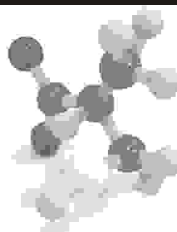
12. Match the column :

Column (I)			Column (II)	
	Wurtz reaction			Number of dimerization product
(a)	CH_3Cl	$\xrightarrow[\text{dry ether}]{\text{Na}}$	(p)	5
(b)	$\text{CH}_3\text{CH}_2\text{Cl}$	$\xrightarrow[\text{dry ether}]{\text{Na}}$	(q)	6
(c)	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Cl}$	$\xrightarrow[\text{dry ether}]{\text{Na}}$	(r)	3
(d)	$\text{H}_2\text{C}(\text{CH}_3)\text{CH}_2\text{CH}_2\text{CH}_2\text{Cl}$	$\xrightarrow[\text{dry ether}]{\text{Na}}$	(s)	1

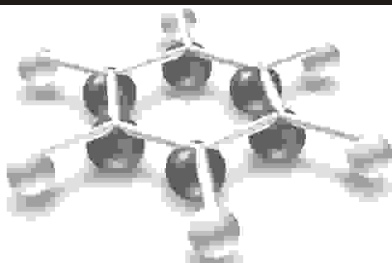


ANSWERS — LEVEL 2

1. A – a; B – c; C – b
2. A – b; B – c; C – a; D – c; E – c; F – b; G – c
3. a, b, c
4. a
5. c
6. a – q – x; b – s – z; c – p – w; d – q – x
7. $P + Q + R = 10$
8. a – q; b – p; c – r; d – s
9. a – p; b – s; c – q; d – r
10. a
11. e
12. a s; b r; c p; d q
13. 9



4B

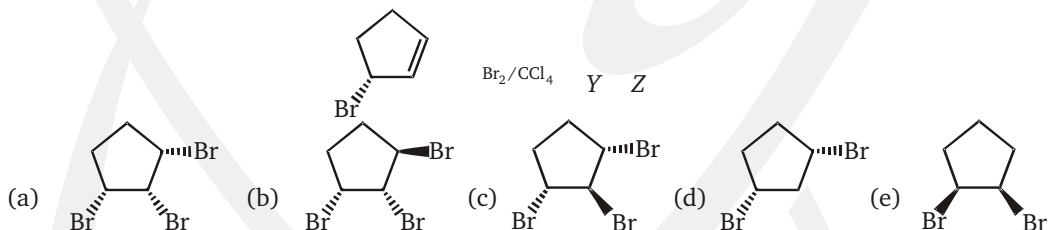


HYDROCARBONS (ALKENES)

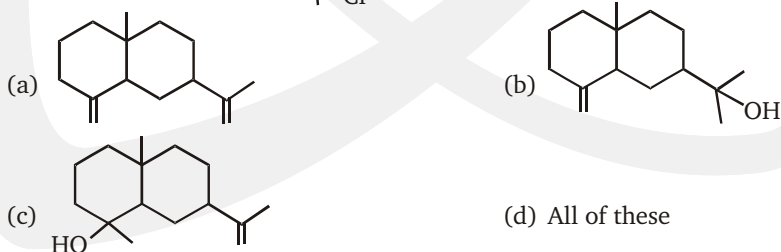


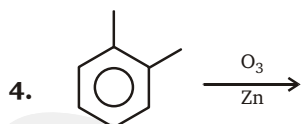
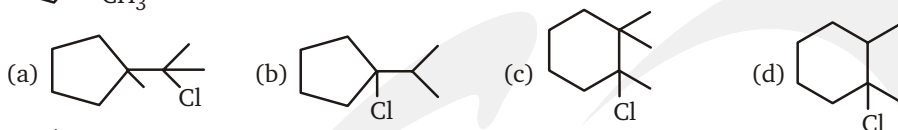
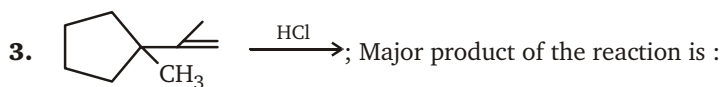
Level - 1

1. (R)-3-bromocyclopentene (shown below) reacts with Br_2/CCl_4 to form two products, Y and Z, Y is not optically active (does not rotate plane-polarized light). What is the structure of Y?

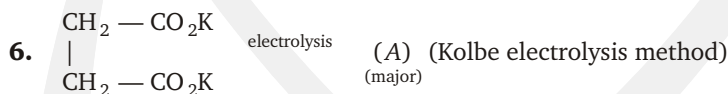
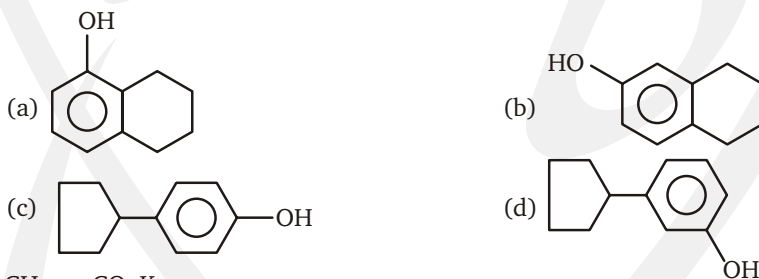


2. A $\xrightarrow{2\text{HCl}}$. Reactant (A) can be:

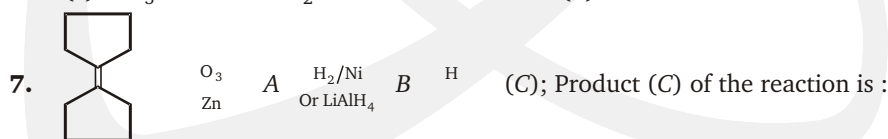


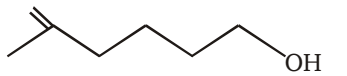


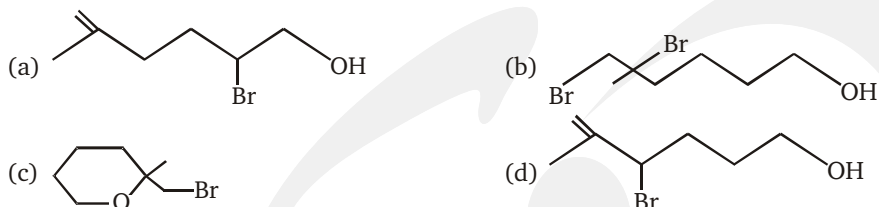
Which of the following products cannot be obtained in ozonolysis of o-xylene?



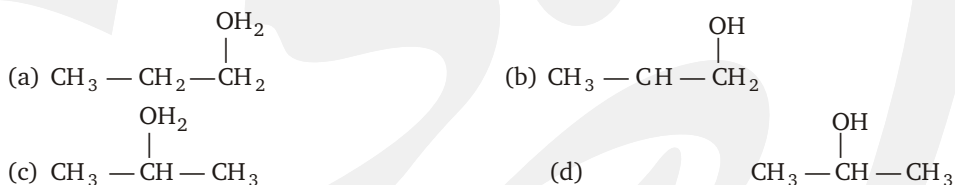
Product (A) of the reaction is :



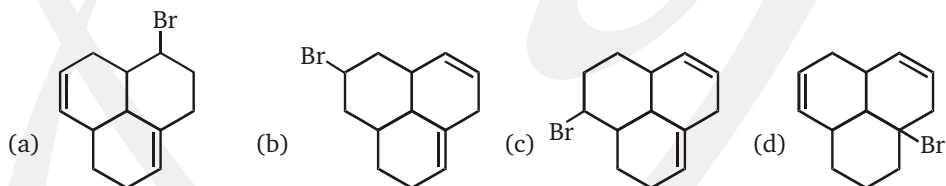
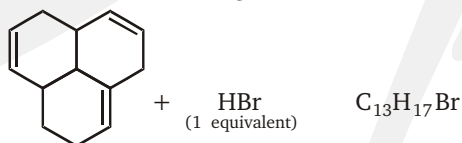
8.  Br_2
 CH_2Cl_2 W. Product W is :



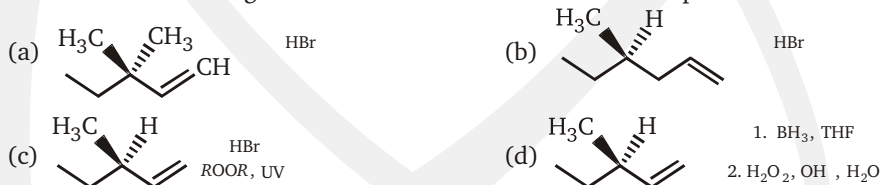
9. The reaction of propene with H_3O^+ will proceed with which of the following intermediates ?



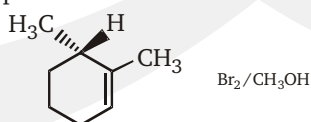
10. Which of the following bromides is the major product of the reaction shown below, assuming that there are no carbocation rearrangement ?

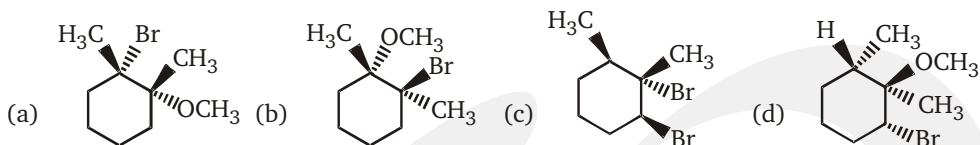


11. Which of the following reactions results in the formation of a pair of diastereomers ?

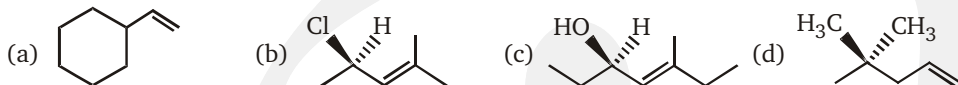


12. What is a likely product of the reaction shown ?

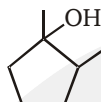




13. Which of the following, when undergoing addition of HBr, will form ONLY a pair of diastereomers ?

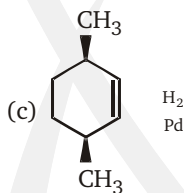
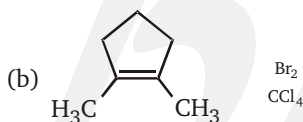
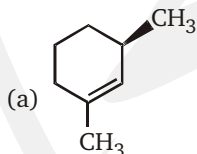


14. How many transition states and intermediates will be formed during the course of following reaction ?

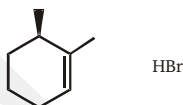


- (a) 3 transition states and 3 intermediates (b) 4 transition states and 3 intermediates
(c) 3 transition states and 2 intermediates (d) 5 transition states and 4 intermediates

15. Product of which of the following reactions, is racemic mixture ?



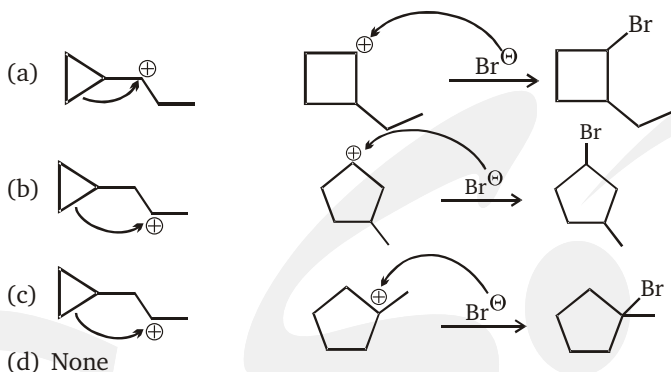
16. The product(s) of the following reaction can best be described as :



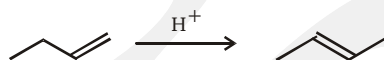
- (a) a racemic mixture (b) a single enantiomer
(c) a pair of diastereomers (d) an achiral molecule

17. Taking into account the stability of various carbocations and, as well as the rules governing mechanisms of carbocation rearrangements, which reaction is most likely to occur during the given reaction ?

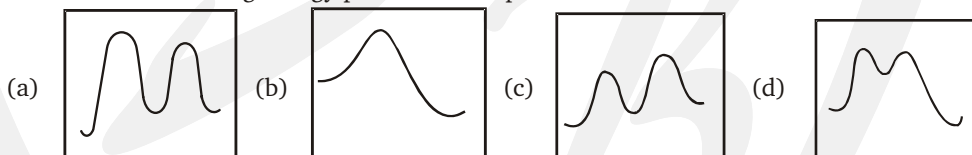




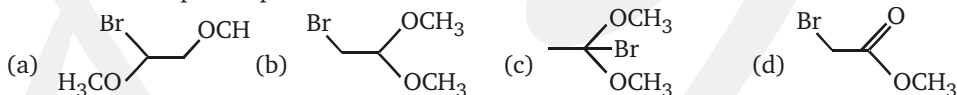
18. Consider the following reaction in which the intermediate carbocation loses H^+ to give the final product ?



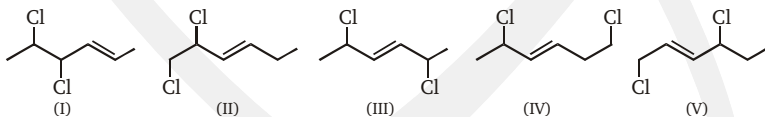
Which of the following energy profiles best represents the overall reaction ?



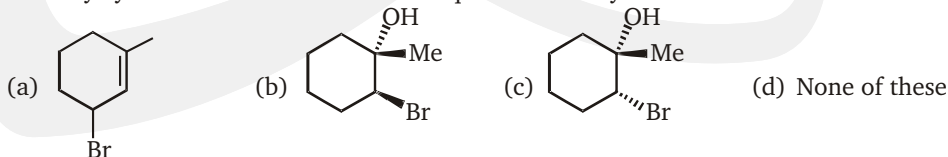
19. Methyl vinyl ether, $H_2C=CH-OCH_3$, reacts with Br_2/CH_3OH . If methanol is reacting as water would, and if this reaction follows a typical mechanism of electrophilic addition, what would be the expected product ?



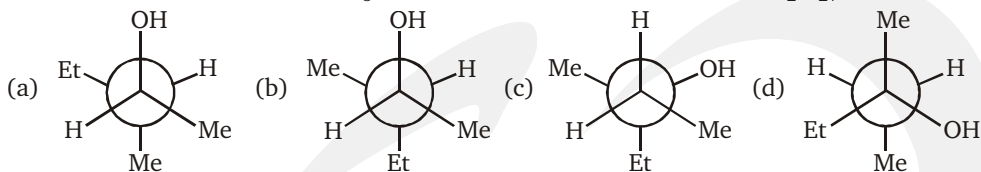
20. 2, 4-hexadiyne (C_6H_6) is allowed to react with Li in $NH_3(liq)$. The product obtained is treated with 1 equivalent of Cl_2 in CCl_4 . Which of the following constitutional isomers are possible products ?



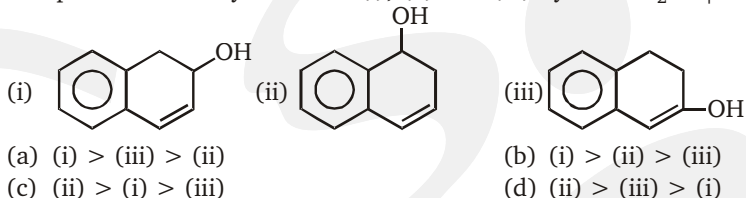
- (a) I and II
(b) II and III
(c) I and V
(d) I and III
21. Which of the following is the best stereochemical representation when reaction between 1-methylcyclohexene and NBS react in aqueous dimethyl sulfoxide ?



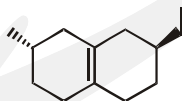
22. Which of the following is among the major products of the reaction of (E)-3-methyl-2-pentene with BH_3 in THF followed by the addition of $\text{H}_2\text{O}_2/\text{HO}^-$?



23. Compare rate of dehydration of (i), (ii) and (iii) by conc. H_2SO_4 .



24. How many products will be formed in this reaction?

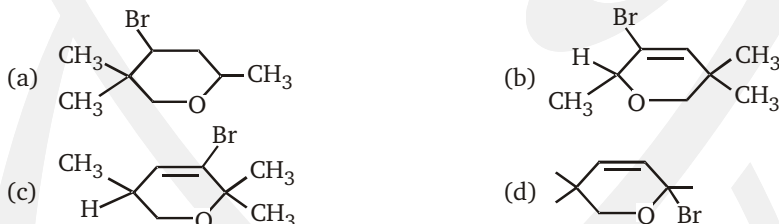


1. OsO_4
 2. NaHSO_3

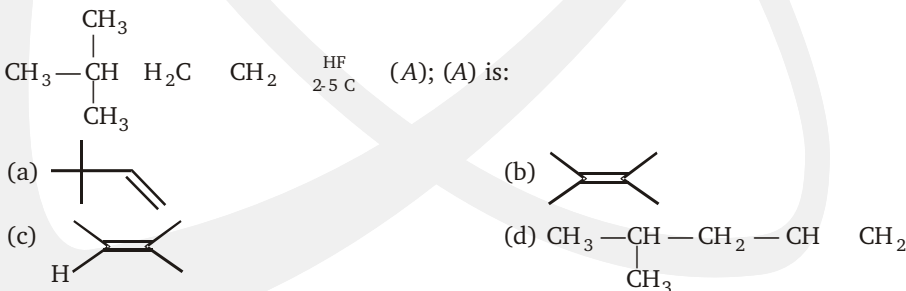
- (a) 10 (b) 2
 (c) 3 (d) 4

25. (A). Product (A) of the reaction is:

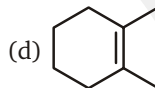
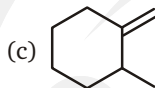
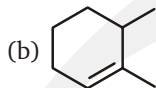
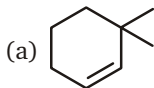
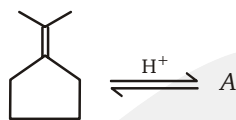
Br_2
 CCl_4



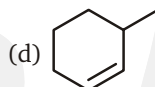
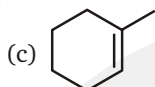
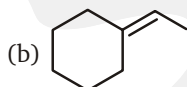
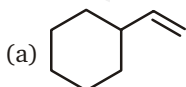
26. $\text{CH}_3 - \text{CH}(\text{CH}_3) - \text{CH}_2 - \text{CH}_2 \xrightarrow[2-5^\circ \text{C}]{\text{HF}}$ (A); (A) is:



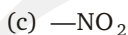
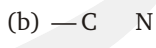
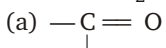
27. Predict the product (A) of the following reaction



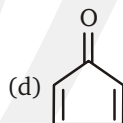
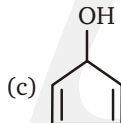
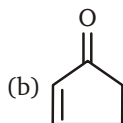
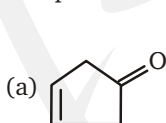
28. (A) Major-product (A) is:



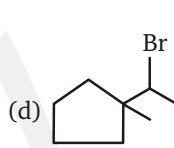
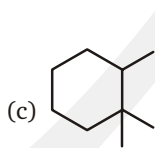
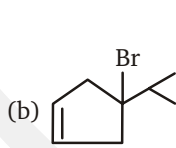
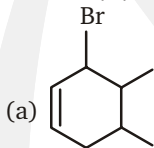
29. Di-imide (N_2H_4) is used to reduce double bond of:



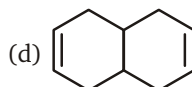
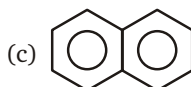
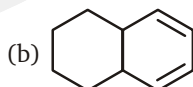
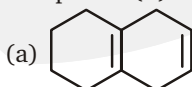
30. End product of the reaction is :

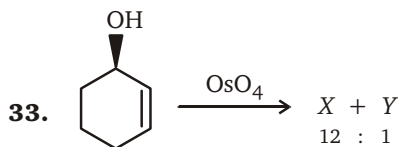


31. Product (A) is :

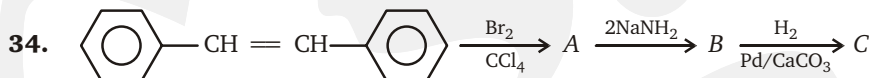
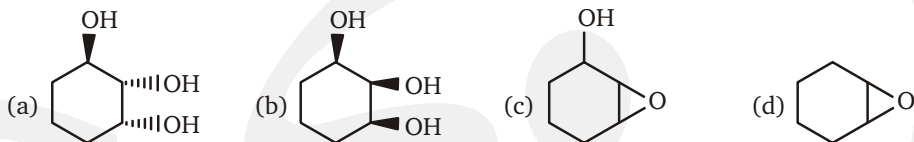


32. Compound (A) is :

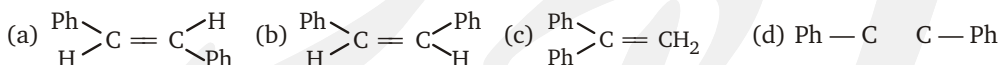




Product (X) will be :

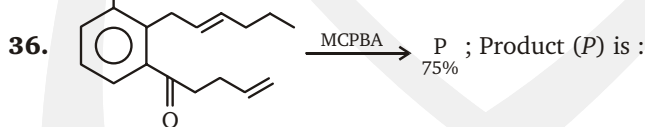
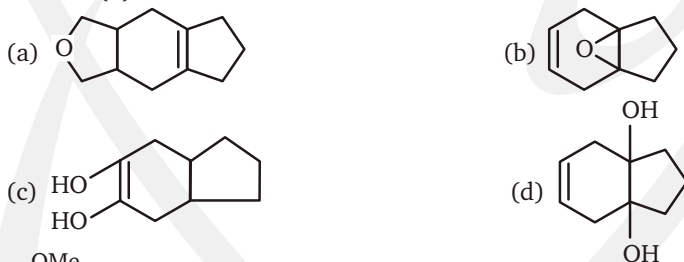


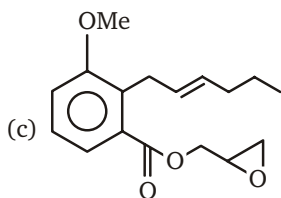
Product (C) is :



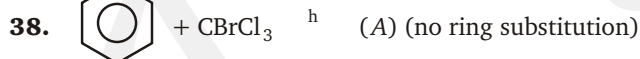
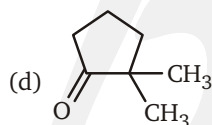
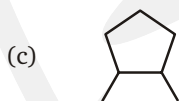
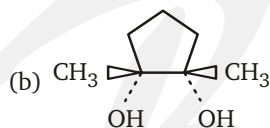
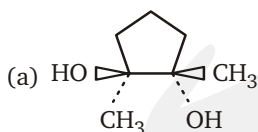
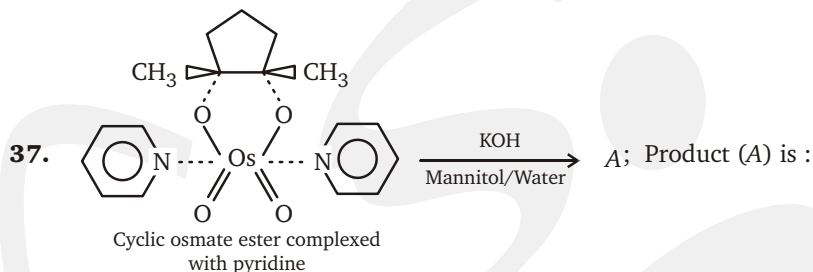
MMPP Magnesium mono peroxy phthalate.

Product (X) is :

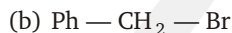
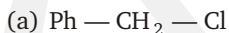




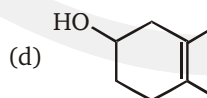
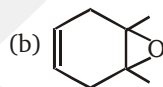
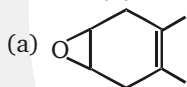
(d) None of these



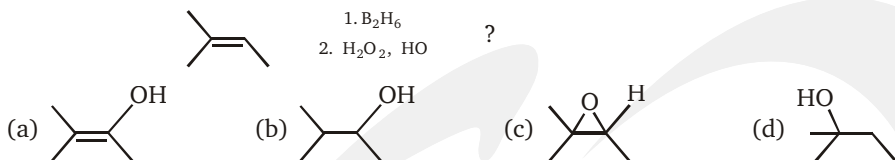
Product (A) is :



Product (A) of the above reaction is :



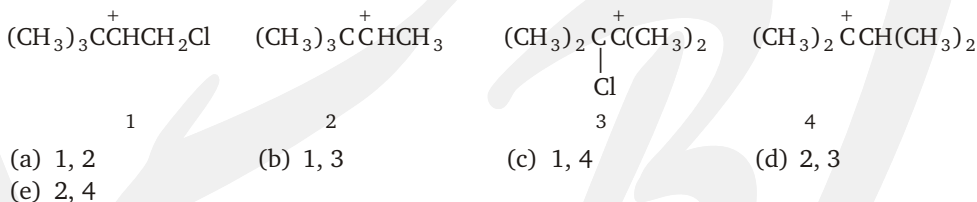
40. The major product of the following reaction sequence is :



41. Which one of the following compounds gives acetone $(\text{CH}_3)_2\text{C}=\text{O}$ as one of the product of its ozonolysis ?



42. Addition of HCl to 3, 3-dimethyl-1-butene yields two products, one of which has a rearranged carbon skeleton. Among the following carbocations, select the possible intermediates in that reaction ?



43. Conversion of cyclohexene to cyclohexanol can be conveniently achieved by :

- (a) $\text{NaOH} + \text{H}_2\text{O}$ (b) $\text{Br}_2 - \text{H}_2\text{O}$
 (c) hydroboration, oxidation (d) hydroboration hydrolysis

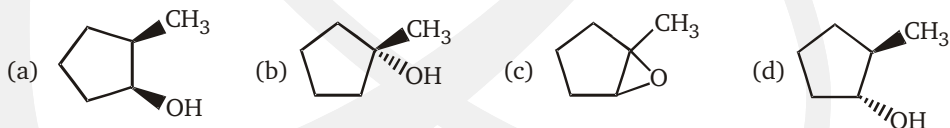
44. *Trans*-cyclohexane-1,2-diol can be obtained by the reaction of cyclohexene with :

- (a) KMnO_4 (b) OsO_4
 (c) peroxy formic acid / H_3O^+ (d) SeO_2

45. Bromination of (E)-2-butenedioic acid gives

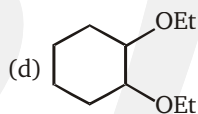
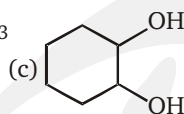
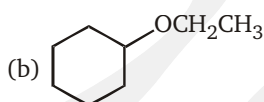
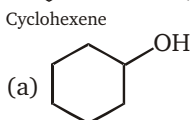
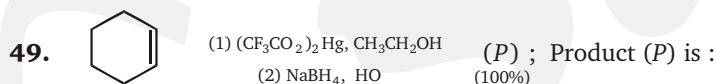
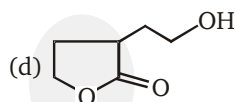
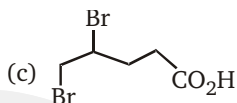
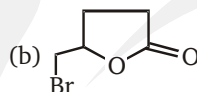
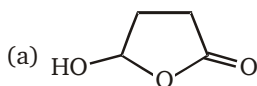
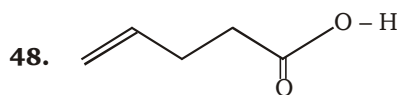
- (a) (2R, 3S)-2, 3-dibromosuccinic acid
 (b) (2R, 3R)-2, 3-dibromosuccinic acid
 (c) a mixture of (2R, 3R) and (2S, 3S)-2, 3-dibromosuccinic acid
 (d) (2S, 3S)-2, 3-dibromosuccinic acid

46. The major product formed during the reaction of 1-methyl cyclopentene with $\text{CH}_3\text{CO}_3\text{H}$ is

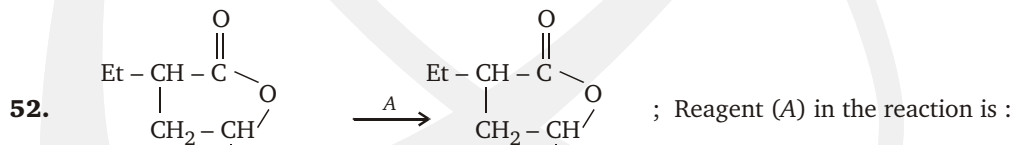
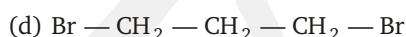
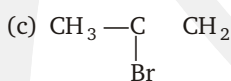
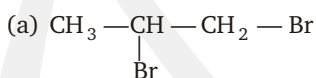
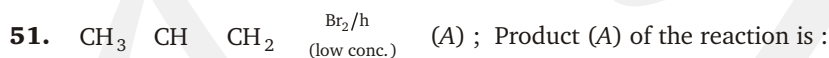
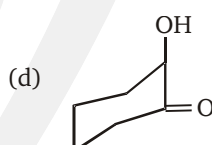
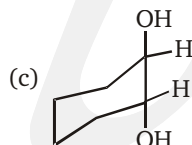
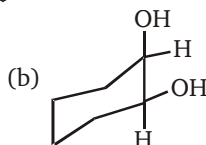
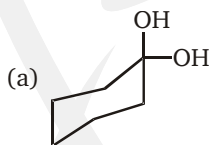


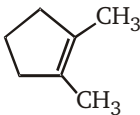
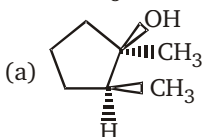
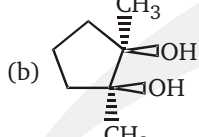
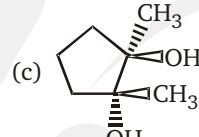
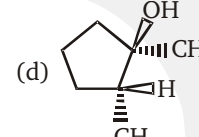
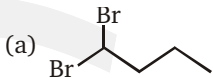
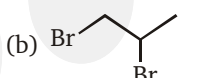
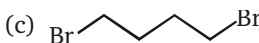
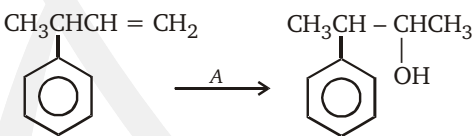
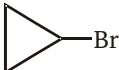
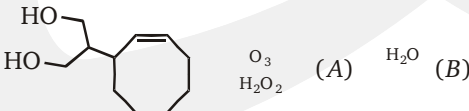
47. $\begin{array}{c} \text{CH} - \text{CO}_2\text{H} \\ || \\ \text{CH} - \text{CO}_2\text{H} \end{array}$ $\xrightarrow[\text{(two mole)}]{\text{NaOH}}$ (A) electrolysis (B); Product (B) of the reaction is :

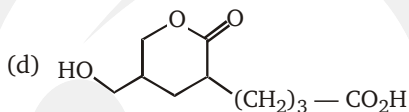
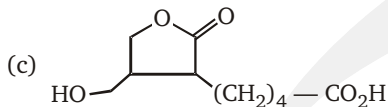
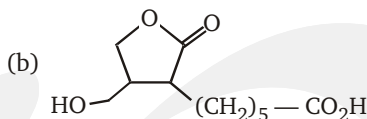
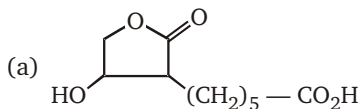
- (a) $\text{CH}_3 - \text{CH}_3$ (b) $\text{H}_2\text{C} = \text{CH}_2$
 (c) $\text{H} - \text{C} = \text{C} - \text{H}$ (d) $\text{CH}_2 = \text{CH} - \text{CH} = \text{CH}_2$



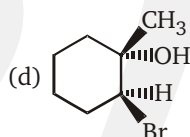
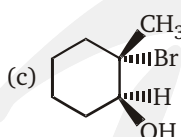
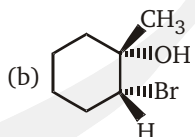
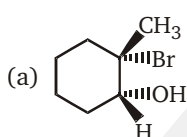
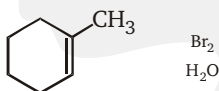
50. What is the major product expected from the following reaction ?



53.  $\xrightarrow[2. \text{NaHSO}_3]{1. \text{OsO}_4}$ Product of the reaction is:
- (a)  (b)  (c)  (d) 
54. Which compound is a possible product from addition of Br_2 to 1-butene ?
- (a)  (b)  (c)  (d) 
55. Addition of Br_2 to *cis*-2-butene would give a product which is:
- (a) achiral (b) racemic (c) meso (d) optically active
56. Addition of Br_2 to *trans*-2-butene would give a product which is:
- (a) achiral (b) racemic (c) meso (d) optically active
57. Addition of OsO_4 to cyclopentene would give a product which is:
- (a) achiral (b) racemic (c) meso (d) optically active
58. Addition of BH_3 followed by H_2O_2 to *trans*-2-butene would give a product which is:
- (a) achiral (b) racemic (c) meso (d) optically active
59.  ; Reagent A may be :
- (a) $\text{H}_2\text{O}/\text{H}$ (b) $\text{BH}_3 \cdot \text{THF}/\text{H}_2\text{O}_2$ (c) $\text{Hg}(\text{OCOCH}_3)_2 \cdot \text{H}_2\text{O}/\text{NaBH}_4 \cdot \text{NaOH}$ (d) All are possible
60. The major product of the following reaction is :
- $\text{CH}_3 - \text{CH} = \text{CH}_2 \xrightarrow{(\text{C}_6\text{H}_5\text{CO})_2\text{O}_2 \text{ peroxide}} \text{CH}_3 - \text{CH}(\text{Br}) - \text{CH}_3$
- (a) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{Br}$ (b) $\text{CH}_3\text{CH}(\text{Br}) - \text{CH}_3$ (c) $\text{BrCH}_2 - \text{CH} = \text{CH}_2$ (d) 
61.  Identify (B) :



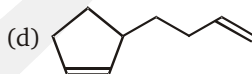
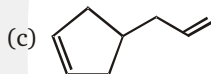
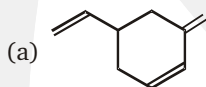
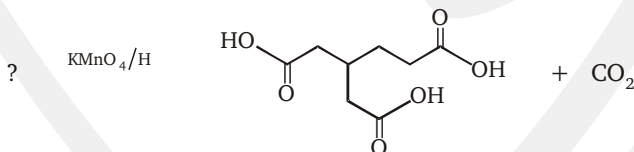
62. Which of the following is a major product of the reaction shown below?



63. In methyl alcohol solution, bromine reacts with ethylene (ethene) to yield $\text{BrCH}_2\text{CH}_2\text{OCH}_3$ in addition to 1, 2-dibromoethane because

- (a) the methyl alcohol solvates the bromine
- (b) the ion formed initially may react with Br^- or CH_3OH
- (c) this is a free radical reaction
- (d) the reaction follows Markovnikov's rule

64. Which of the following compound was the starting material for the oxidation shown below ?



65. Which series of reactions will achieve the following transformation ?



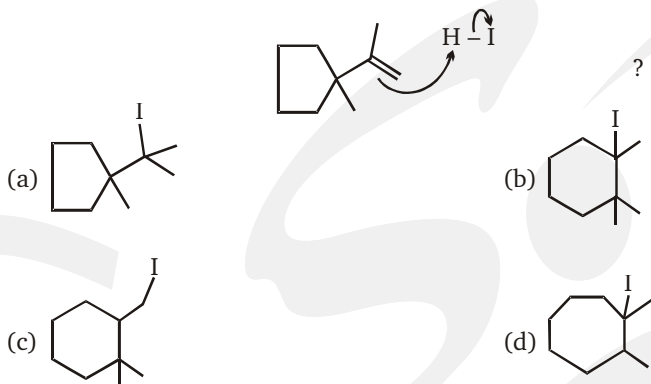
- 1
(a) Cl_2/CCl_4
(c) Cl_2/CCl_4

- 2
(b) Br_2
(d) $\text{NBS}/h\nu$

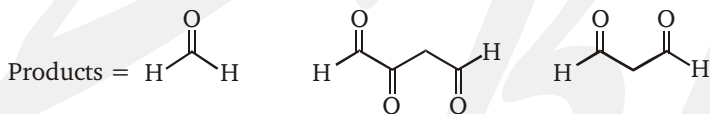
- 1
(b) HBr
(d) $\text{NBS}/h\nu$

- 2
(c) Cl_2/CCl_4
(d) Cl_2/CCl_4

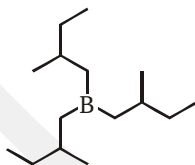
66. Taking into account the stability of various cycloalkanes and carbocations, as well as the rules governing mechanisms of carbocation rearrangements, what is the most likely product of this reaction ?



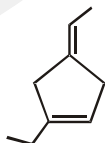
67. A triene is treated with ozone followed by zinc in acetic acid to give the following three products. What is the structure of the triene ?



68. Which of the following compound would yield trialkylborane shown below when treated with BH_3/THF ?

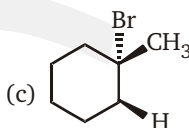
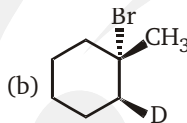
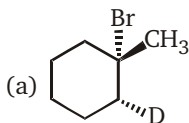


- (a) 2-methylbut-1-ene (b) 2-methylbut-2-ene
(c) 3-methylbut-1-ene (d) 3-methylbut-1-yne
69. If the following compound is treated with Pd/C in excess of hydrogen gas, how many stereoisomers of the product will be obtained ?



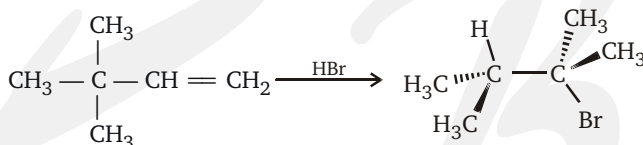
- (a) 1 (b) 2
(c) 3 (d) 4

70. Which is the most precise designation of stereochemistry for the products formed in the electrophilic addition of DBr to 1-methylcyclohexene ? (D ^2H , an isotope of hydrogen)



(d) both (a) and (b)

71. Consider the addition of HBr to 3,3-Dimethyl-1-butene shown below. What is the best mechanistic explanation for the formation of the observed product ?



- (a) Protonation of the alkene followed by a hydride shift and addition of bromide to the carbocation
(b) Double bond shift in the alkene following by the protonation and addition of bromide to the carbocation
(c) Addition of bromide to the alkene followed by a double bond shift and protonation
(d) Protonation of the alkene followed by a methyl shift and addition of bromide to the carbocation

72. Propene $\text{CH}_3\text{CH}=\text{CH}_2$ can be converted into 1-propanol by oxidation. Indicate which sets of reagents amongst the following is ideal to effect the above conversion ?

- (a) KMnO_4 (alkaline) (b) Osmium tetroxide ($\text{OsO}_4/\text{CH}_2\text{Cl}_2$)
(c) B_2H_6 and alk. H_2O_2 (d) O_3/Zn

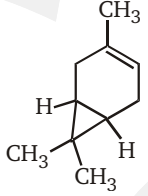
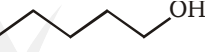
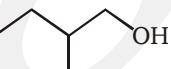
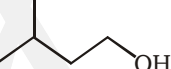
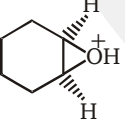
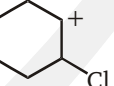
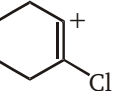
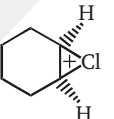
73. Which is the most suitable reagent among the following distinguish compound (3) from the others ?

- (1) $\text{CH}_3\text{C}(\text{CH}_3)_2\text{CH}_3$ (2) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$
(3) $\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)_2$ (4) $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}_3$
(a) Bromine in carbon tetrachloride (b) Bromine in acetic acid solution
(c) Alk. KMnO_4 (d) Ammonical silver nitrate

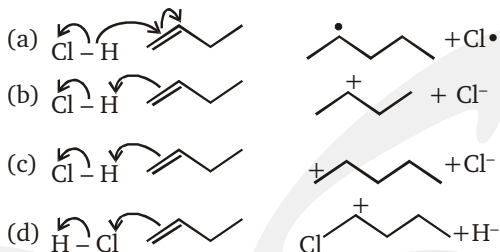
74. The principal organic product formed in the reaction given below is :



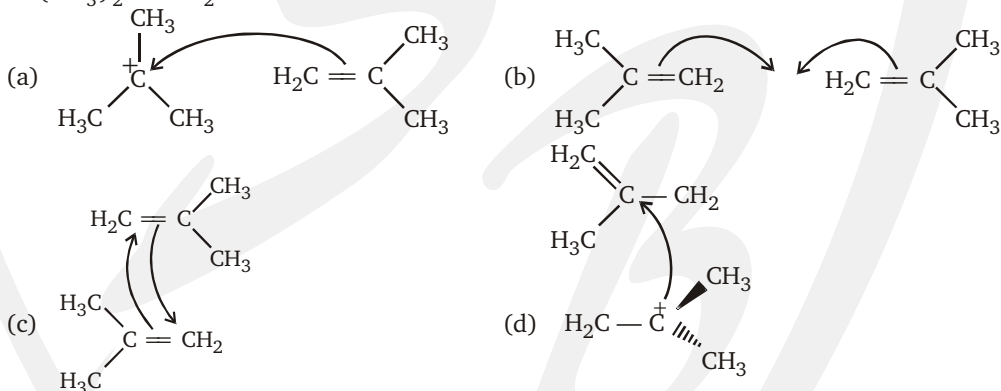
- (a) $\text{CH}_3(\text{CH}_2)_8\text{COOH}$ (b) $\text{CH}_3(\text{CH}_2)_8\text{COBr}$
(c) $\text{CH}_3\text{BrCH}_2(\text{CH}_2)_8\text{COOH}$ (d) $\text{CH}_3\text{BrCH}_2(\text{CH}_2)_7\text{CH}_2\text{COOH}$

75. When 2-butyne is treated with $\text{H}_2/\text{Pd} - \text{BaSO}_4$; the product formed will be :
 (a) *cis*-2-butene (b) *trans*-2-butene (c) 1-butene (d) 2-hydroxy butane
76. In the reaction, $\text{CH}_3\text{C} \equiv \text{C} - \text{CH}_3 \xrightarrow[\text{(ii) Zn/H}_2\text{O}]{\text{(i) X}} \text{CH}_3 - \text{C}(=\text{O}) - \text{C}(=\text{O}) - \text{CH}_3$, X is :
 (a) HNO_3 (b) O_2 (c) O_3 (d) KMnO_4
77. Which of the following alkene on catalytic hydrogenation given *cis* and *trans*-isomer ?
 (a)  (b) 
 (c)  (d) all of these
78. In the reaction of hydrogen bromide with an alkene (in the absence of peroxides), the first step of the reaction is the to the alkene.
 (a) fast addition of an electrophilic (b) slow addition of an electrophile
 (c) fast addition of a nucleophilic (d) slow addition of a nucleophile
79. Which of the following alcohols cannot be prepared from hydration of an alkene ?
 (a)  (b) 
 (c)  (d) 
80. Which of the species shown below is the most stable form of the intermediate in the electrophilic addition of Cl_2 in water to cyclohexene to form a halohydrin ?
 (a)  (b) 
 (c)  (d) 
81. The reaction, $(\text{CH}_3)_2\text{C} = \text{CH}_2 + \text{Br} \cdot \rightarrow (\text{CH}_3)_2\dot{\text{C}}\text{CH}_2\text{Br}$ is an example of a/an step in a radical chain reaction.
 (a) initiation (b) termination
 (c) propagation (d) heterolytic cleavage

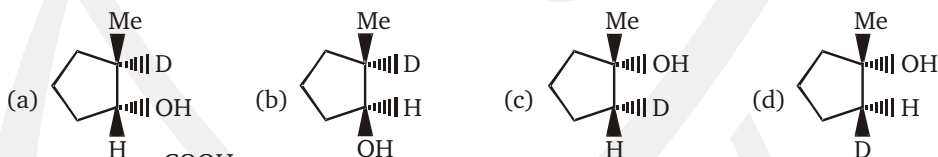
82. Which of the following most accurately describes the first step in the reaction of hydrogen chloride with 1-butene ?

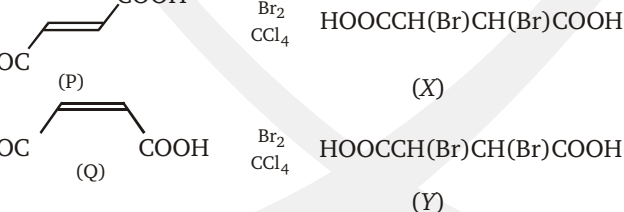


83. Which of the following best describes the flow of electrons in the acid-catalyzed dimerization of $(\text{CH}_3)_2\text{C}=\text{CH}_2$?



84. Hydroboration of 1-methylcyclopentene using B_2D_6 , followed by treatment with alkaline hydrogen peroxide, gives

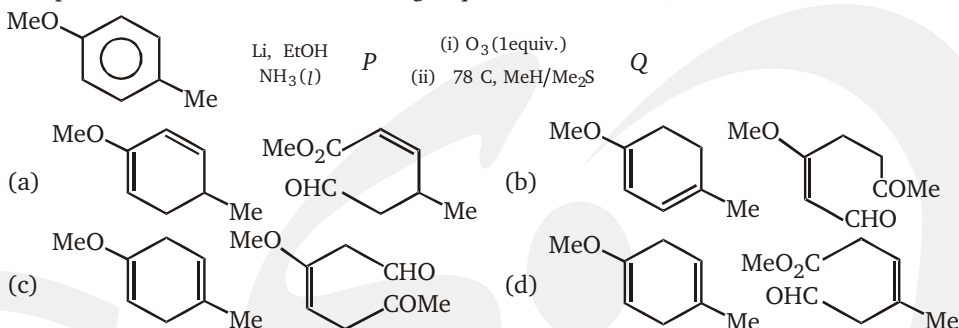


85.  $\text{HOOC}-\text{CH}=\text{CH}-\text{COOH} \xrightarrow{\text{Br}_2/\text{CCl}_4} \text{HOOCCH}(\text{Br})\text{CH}(\text{Br})\text{COOH}$

The correct statements with respect to the above pair of reactions are that

- (I) the reactions are stereospecific
 (II) (X) is erythro and (Y) is threo isomer
 (III) (X) is threo and (Y) is erythro isomer
 (IV) each of (P) and (Q) gives a mixture of (X) and (Y)
- (a) I and II (b) I and III (c) I and IV (d) II and IV

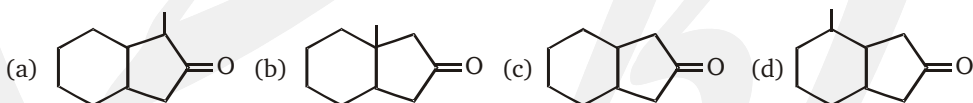
86. The products *P* and *Q* in the following sequence of reactions, are



87. 4-Pentenoic acid when treated with I_2 and $NaHCO_3$ gives :

- (a) 4, 5-diiodopentanoic acid (b) 5-iodomethyl-dihydrofuran-2-one
(c) 5-iodo-tetrahydropyran-2-one (d) 4-pentenoliodide

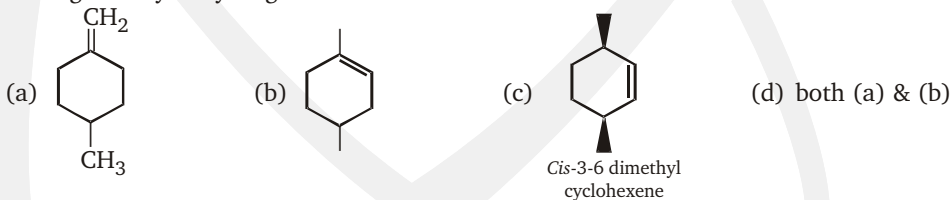
88. (A) HCl (B); Product (B) of the reaction is:



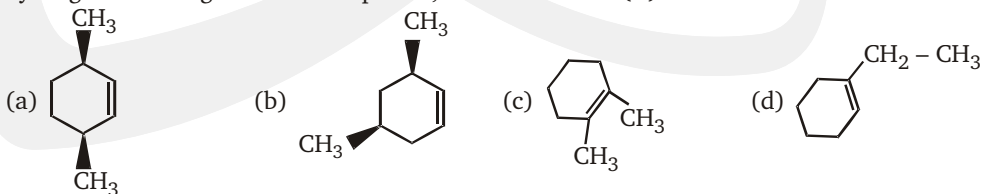
89. (A) (B) (C), Product (C) is :

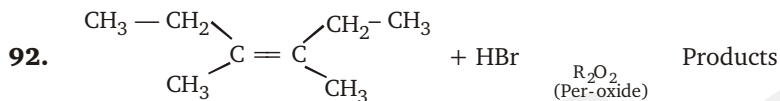


90. Which of the following will give a mixture of *cis* and *trans*-1,4-dimethyl cyclohexane, when undergo catalytic hydrogenation ?



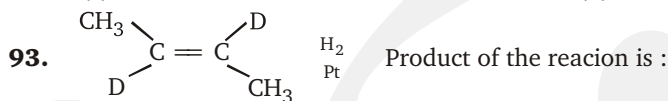
91. An optically active compound *A* with molecular formula C_8H_{14} undergoes catalytic hydrogenation to give meso compound, the structure of (A) is :





How many products will be formed in above reaction ?

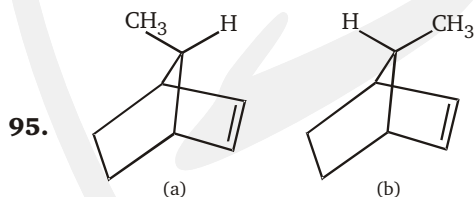
- (a) 2 (b) 4
(c) 3 (d) 6



- (a) Racemic (b) Diastereomers
(c) Meso (d) Pure enantiomers

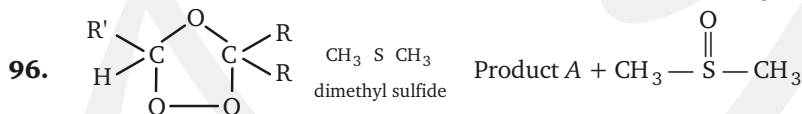


- (a) Racemic (b) Diastereomer
(c) Meso (d) *E* and *Z* isomer



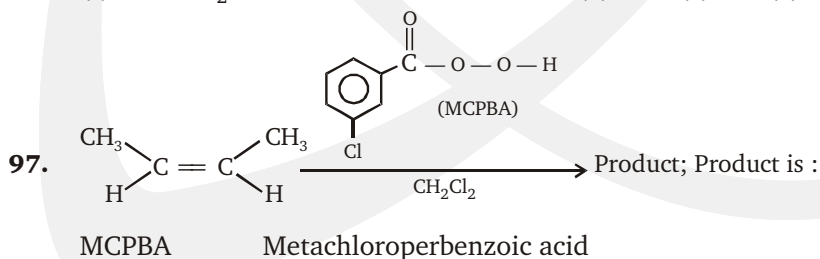
Rate of reaction towards reduction using (H_2/Pt) :

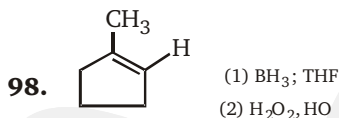
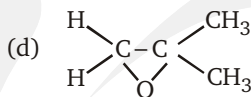
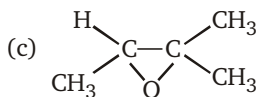
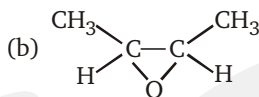
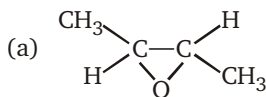
- (a) $a > b$ (b) $a = b$
(c) $b > a$ (d) Reduction of given molecule is not possible



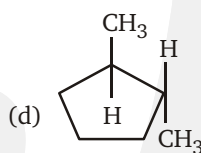
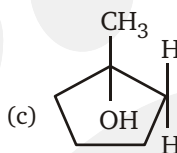
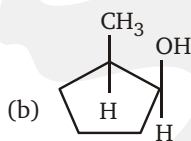
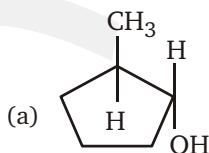
Product A of the above reaction is :

- (a) $\text{R} - \overset{\text{O}}{\parallel} \text{C} - \text{R}$ (b) $\text{R} - \text{CHO}$
(c) $\text{R} - \text{CO}_2\text{H}$ (d) both (a) and (b)

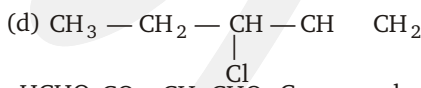
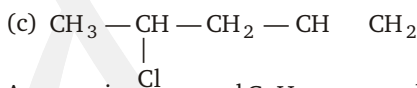
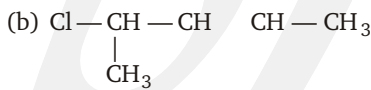
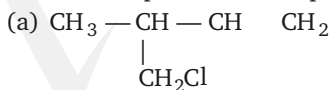




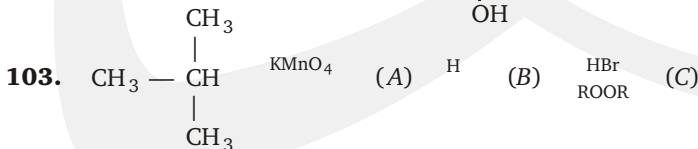
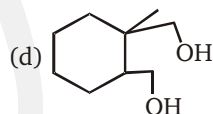
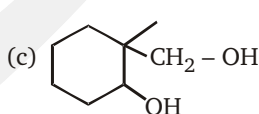
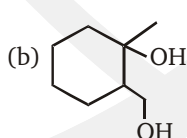
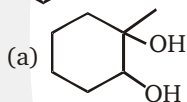
(A) ; Product of the reaction is :



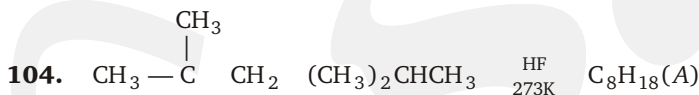
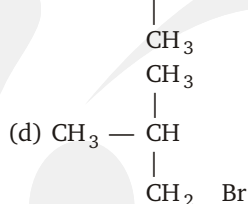
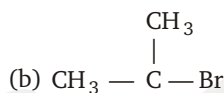
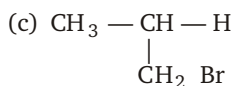
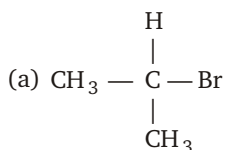
100. Optically active isomer (A) of $(\text{C}_5\text{H}_9\text{Cl})$ on treatment with one mole of H_2 gives an optically inactive compound (B) compound (A) will be :



101. An organic compound C_4H_6 on ozonolysis give HCHO , CO_2 , CH_3CHO . Compound will be :



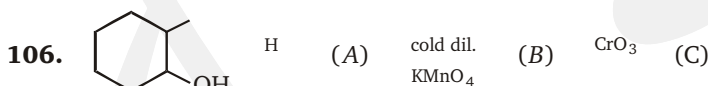
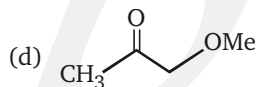
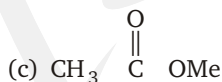
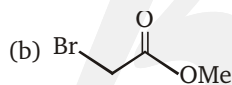
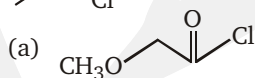
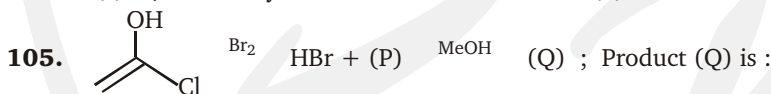
Product (C) in the above reactions is :



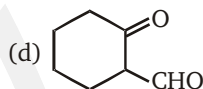
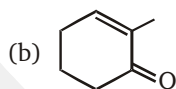
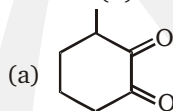
Unknown (A) in the above reaction is :

- (a) 2, 2, 3-trimethyl pentane
(c) 2, 2-dimethyl hexane

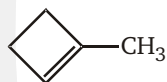
- (b) 2, 2, 4-trimethyl pentane
(d) *n*-octane



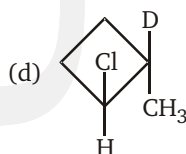
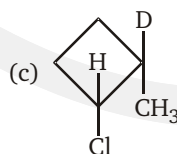
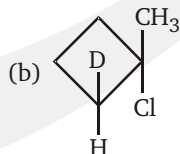
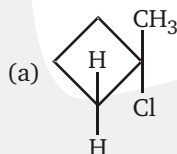
Product (C) of the reaction is:



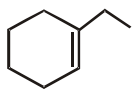
107. What is the major product expected from the following reaction ?



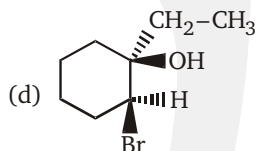
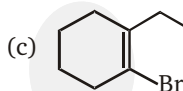
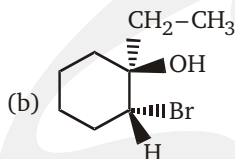
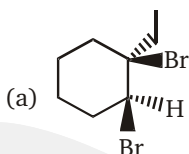
D Cl Product



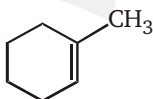
108. Choose the correct product of this reaction :



1. Br_2
2. H_2O

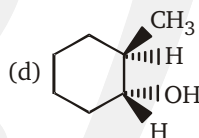
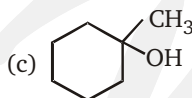
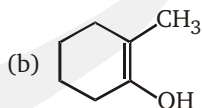
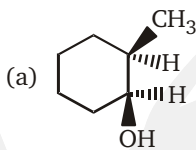


109.

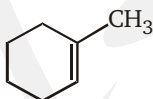


1. BH_3/THF
2. $\text{H}_2\text{O}_2/\text{OH}^-$

A; Product A is:

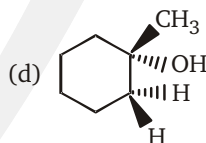
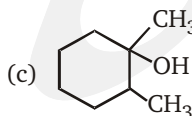
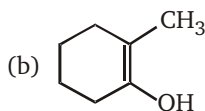
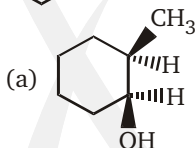


110.

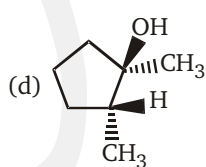
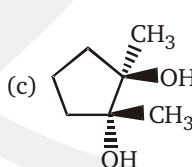
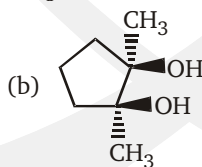
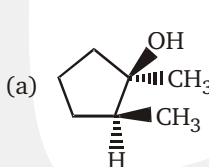
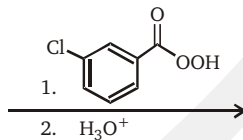
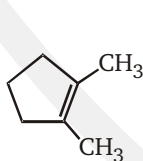


1. $\text{Hg}(\text{OAc})_2, \text{H}_2\text{O}$
2. NaBH_4

Product; Product is :



111. Choose the correct product of the following reactions :



112. How many stereoisomeric tetrabromides will be formed in the following reaction ?



2Br_2
 CCl_4

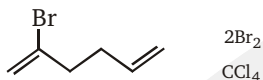
(a) 2

(b) 3

(c) 4

(d) 6

113. How many stereoisomeric pentabromides will be formed in the following reaction ?



(a) 2

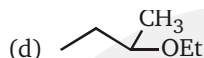
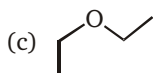
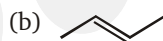
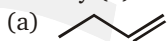
(b) 3

(c) 4

(d) None of these

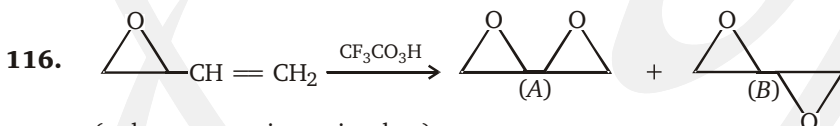
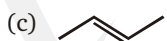
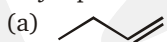
114. (A) EtONa (Z) (major)

Identify (Z) in the above sequence of reactions :



115. $\begin{array}{c} \text{CH}_3 - \text{CH} - \text{CO}_2\text{K} \\ | \\ \text{CH}_3 - \text{CH} - \text{CO}_2\text{K} \end{array}$ $\xrightarrow{\text{electrolysis}}$ (A) (Major)

Major product (A) of the above reaction :



(only one enantiomer is taken)

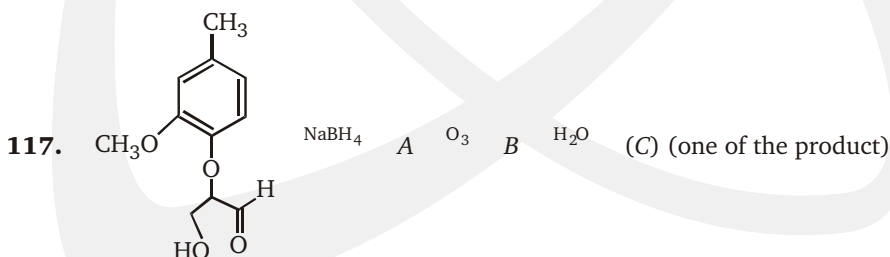
Which of the following statement is correct about A and B ?

(a) A and B are mixture of diastereomers

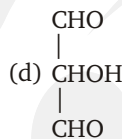
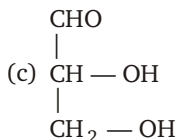
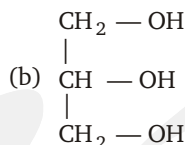
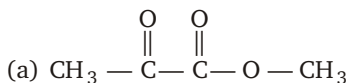
(b) A and B are mixture of enantiomers

(c) A and B are optically active

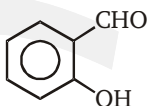
(d) B is racemic mixture



Identify the product (C):



118.



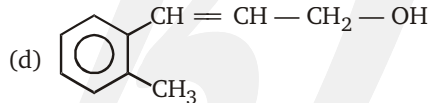
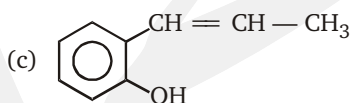
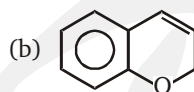
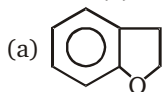
HO

(X)



(Y)

Product (Y) of the above reaction is :

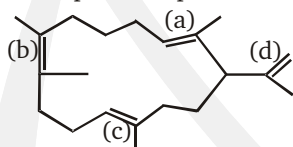


119.

In the reaction $\text{Me} - \text{C} \equiv \text{C} - \text{Et} \xrightarrow{\text{Na/liq.NH}_3} \text{P} \xrightarrow[\text{CCl}_4]{\text{Br}_2} \text{(Q)}$; then Q is :

- (a) A pure compound which is optically inactive due to internal compensation
 (b) A binary mixture which is optically inactive due to external compensation
 (c) A binary mixture which is optically active
 (d) A pure compound which is optically inactive due to absence of chiral centre

120.

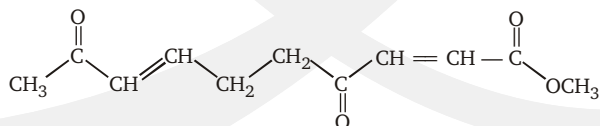


Which (-bond) will reduce first, when above compound undergoes catalytic hydrogenation ?

- (a) a (b) b (c) c (d) d

121.

Compound A, which is a degradation product of the antibiotic vermiculine has following structure



(A) $\text{C}_{11}\text{H}_{14}\text{O}_4$

H_2
Pd/C

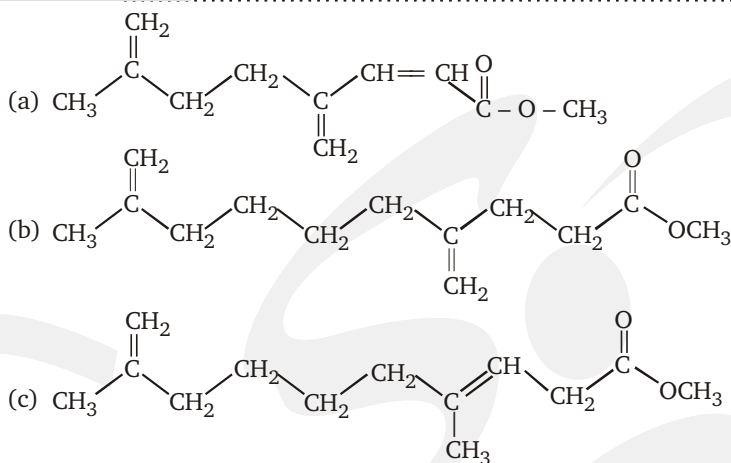
(B) $\text{C}_{11}\text{H}_{18}\text{O}_4$

$(\text{CH}_3)_2\text{S}$

O_3
 CH_2Cl_2

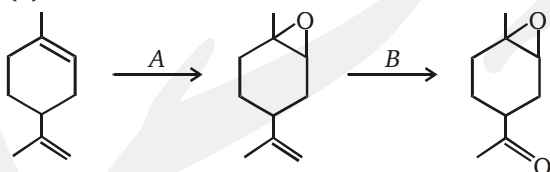
(C) $\text{C}_{11}\text{H}_{18}\text{O}_2$

. Unknown (C) is:



(d) None of these

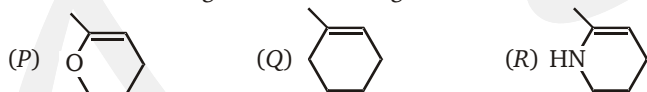
122.



Reagent (A) and (B) in above reaction are :

- (a) $A = \text{RCO}_3\text{H}$, $B = \text{H}_2\text{O}_2$ (b) $A = \text{RCO}_3\text{H}$, $B = \text{HIO}_4$
 (c) $A = \text{RCO}_3\text{H}$, $B = \text{O}_3$ (d) $A = \text{O}_3$, $B = \text{RCO}_3\text{H}$

123. Rank the following in the increasing order of rate of reaction with HBr .

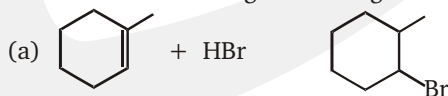


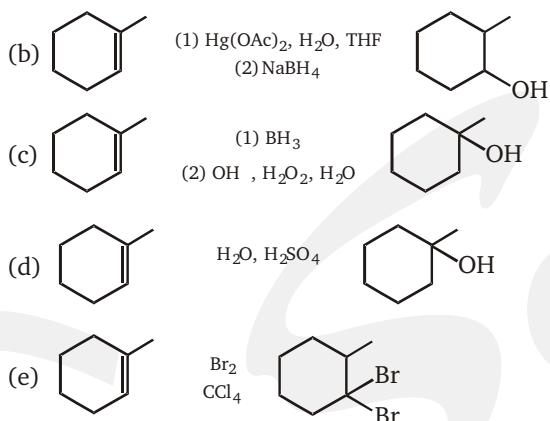
- (a) $R > P > Q$ (b) $R > Q > P$
 (c) $P > R > S$ (d) $P > S > R$

124. Select the reaction(s) that would result in the formation of 2-bromopropane.

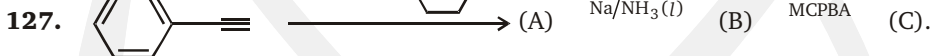
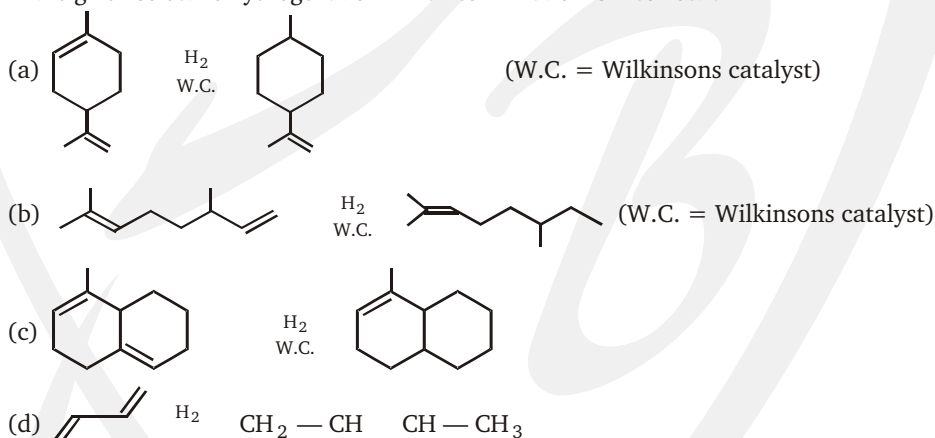
- (I) $\text{CH}_3\text{CH}=\text{CH}_2 \xrightarrow[\text{peroxide}]{\text{HBr}}$ (II) $\text{CH}_3\text{CH}=\text{CH}_2 \xrightarrow[\text{CCl}_4]{\text{HBr}}$
 (III) $\text{CH}_3\text{CH}_2\text{CH}_3 \xrightarrow[\text{h}]{\text{Br}_2}$ (IV) $\text{CH}_3\text{CH}=\text{CH}_2 \xrightarrow[\text{CCl}_4]{\text{Br}_2}$
 (a) I and III (b) II and III
 (c) I, II, and III (d) I, II and III

125. Which of the following reactions generates the major product ? Ignore stereoisomerism.

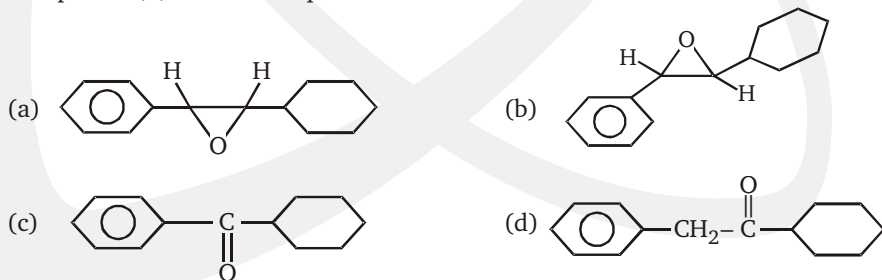


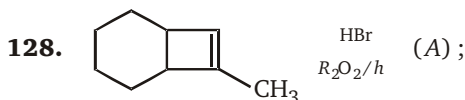


126. In the given selective hydrogenation which combination is incorrect ?

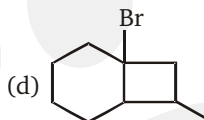
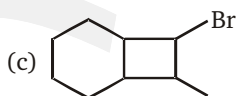
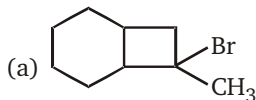


Compound (C) in above sequence of reaction is :

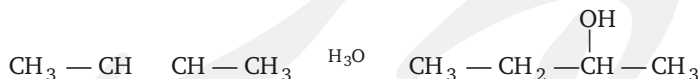




Major product (A) is :



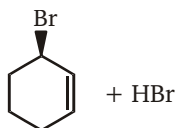
129. In the reaction given below, the product would be :



- (a) a mixture of diastereomers
(c) optically pure enantiomer

- (b) optically active
(d) a racemic mixture

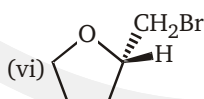
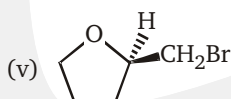
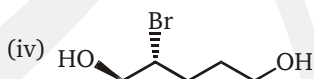
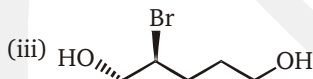
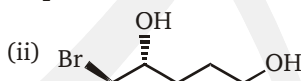
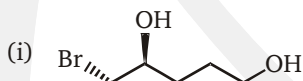
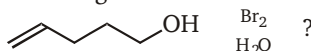
130. Surprisingly, the reaction shown below goes through classical carbocation. What is the major product of this reaction ?



- (a) *trans*-1, 3-dibromocyclohexane
(c) *trans*-1, 2-dibromocyclohexane

- (b) *cis*-1, 3-dibromocyclohexane
(d) *cis*-1, 2-dibromocyclohexane

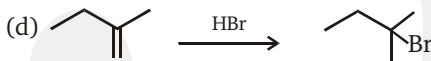
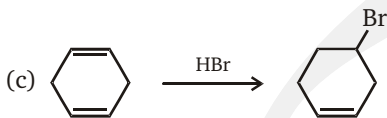
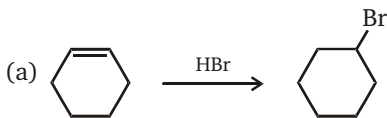
131. The major product of the reaction given below is :



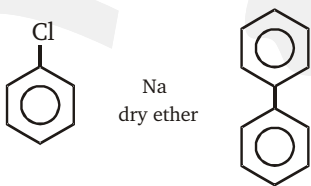
- (a) (i) and (ii)
(c) (v) and (vi)

- (b) (iii) and (iv)
(d) none of these

132. Which reaction will occur at the fastest rate ?



133.



Above reaction is known as :

- (a) Wurtz reaction
(c) Fittig reaction

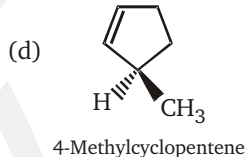
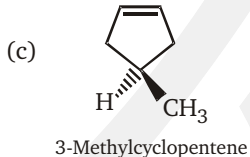
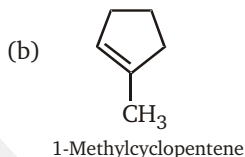
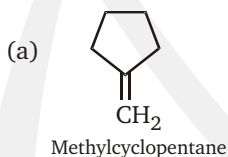
- (b) Wurtz fittig reaction
(d) Kolbe electrolysis

134. $\text{CH}_3 - \text{CH}_2 - \overset{\text{O}}{\parallel} \text{C} - \text{H} \xrightarrow[\text{HI}]{\text{Red P}} \text{A}$

Product A is :

- (a) propane (b) propanol (c) propanoic acid (d) propene

135. Which of the following compound give diastereomers when treated with Br_2 in CCl_4 ?



136. A mixture of C_2H_6 , C_2H_4 and C_2H_2 is bubbled through alkaline solution of copper (I) chloride, contained in Woulf's bottle. The gas coming out is :

- (a) original mixture (b) C_2H_6
(c) C_2H_6 and C_2H_4 mixture (d) C_2H_4 and C_2H_2

137.  $\xrightarrow{\text{Br}_2/\text{CCl}_4}$ Possible products (x) (y) products

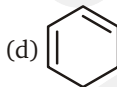
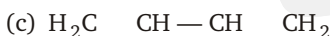
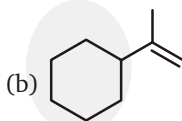
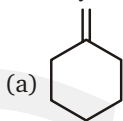
The number of possible products for x and y is :

- (a) 2, 4 (b) 3, 5
(c) 3, 6 (d) 3, 4

138. Select the incorrect statement :

- (a) Bromine is more selective and less reactive
- (b) Chlorine is less selective and more reactive
- (c) Benzyl free radical is more stable than 2° free radical
- (d) Vinyl free radical more stable than allyl free radical

139. Which of the following compound does not evolve CO_2 gas, when undergo oxidative ozonolysis ?



140. *cis*-3-hexene $\xrightarrow{(a)}$ meso 3,4-hexanediol
trans-3-hexene $\xrightarrow{(b)}$ meso 3,4-hexanediol.

Choose pair of reagent (a, b) for above conversions.

- (a) Cold KMnO_4 , OsO_4
- (b) Cold KMnO_4 , $\text{RCO}_3\text{H}/\text{H}_3\text{O}$
- (c) $\text{RCO}_3\text{H}/\text{H}_3\text{O}$, cold KMnO_4
- (d) None of these

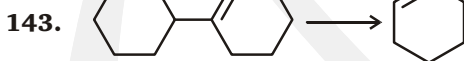
141.  $\xrightarrow[\text{Liq. NH}_3]{\text{Na}}$ (A) $\xrightarrow[\text{Zn}]{\text{O}_3}$ (B) $\xrightarrow{\text{Ph}_3\text{P}} \text{CH}_2(2\text{mole})$ (C)

Product (C) of the above reaction is :

- (a) 1,3-hexadiene
- (b) 1,4-pentadiene
- (c) 1,3-butadiene
- (d) 1,3-heptadiene

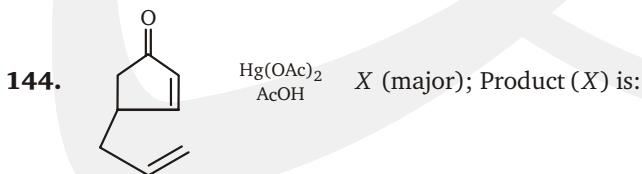
142. How many carbon-hydrogen bond orbitals are available for overlap with the vacant *p*-orbital in ethyl carbocation ?

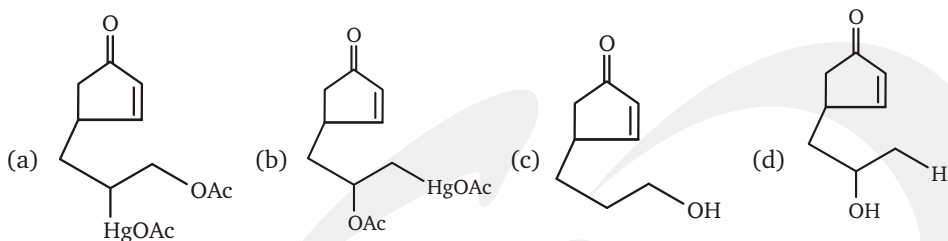
- (a) 0
- (b) 3
- (c) 5
- (d) 6



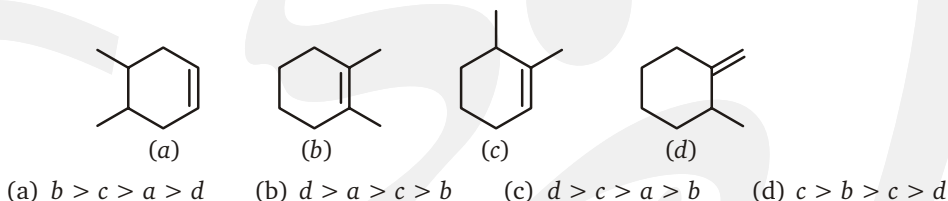
To achieve above conversion, the reagents used will be :

- (a) $\text{O}_3/\text{H}_2\text{O}_2$, HO^- /
- (b) HBr , alc. KOH , O_3 , LiAlH_4 , H^+ /
- (c) HBr , *t*- BuOK , O_3 , KMnO_4 , /
- (d) HCl , KMnO_4 (cold), H^+ /

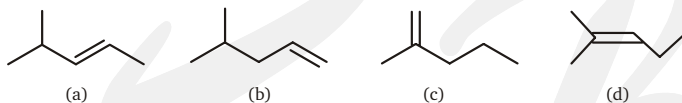




145. Decreasing order of heat evolved upon catalytic hydrogenation of given reactants with a H_2 (Pd/C) is :



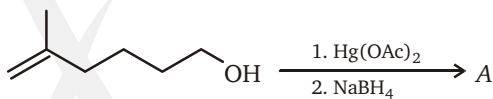
146.



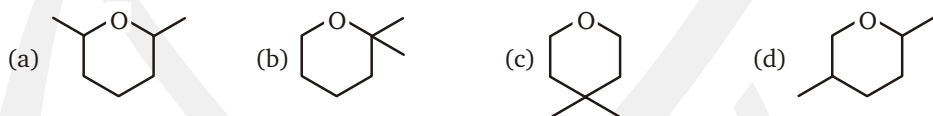
The correct order of heat of hydrogenation of given molecules is :

- (a) $d > c > a > b$ (b) $d > c > b > a$
 (c) $b > a > c > d$ (d) $d > a > c > b$

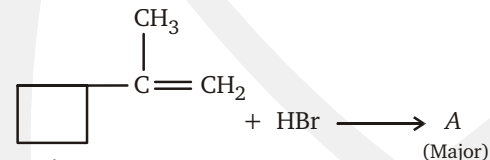
147.



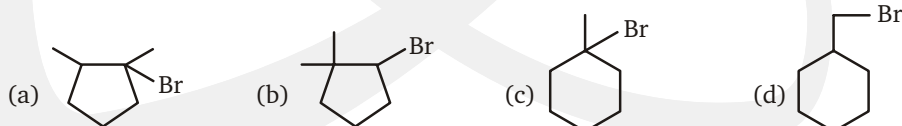
Product (A) of the above reaction is :



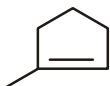
148.



Product (A) is :



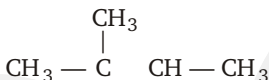
149.

HBr
CCl₄

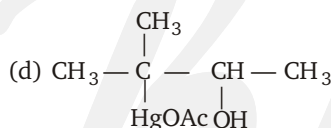
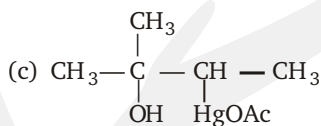
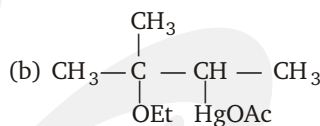
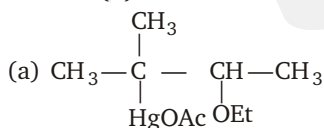
Product; Comment upon optical activity of the product.

- (a) Racemic mixture
 (b) Diastereomers
 (c) Meso
 (d) Optically inactive due to absence of chiral center

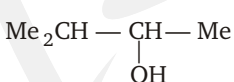
150.

Hg(OAc)₂/EtOH(A)
(90%)

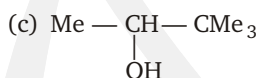
Product (A) of the above reaction is :



151.

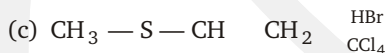
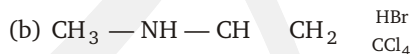
Al₂O₃
350 °C(A) (i) HI
(ii) AgOH (B)

Product (B) of above reaction :



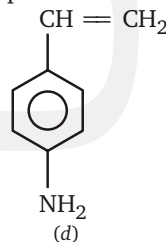
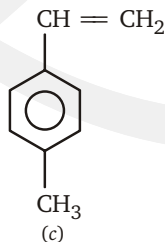
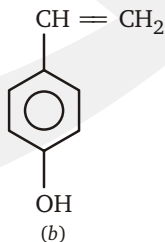
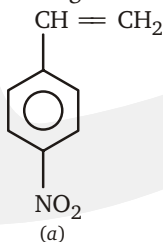
152.

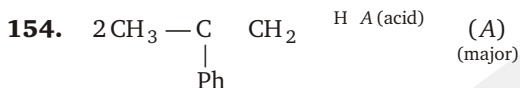
In which of the following reaction, Markownikoff's rule is violated ?



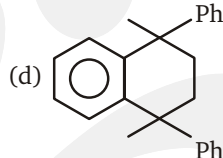
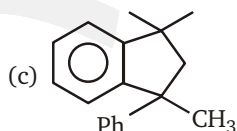
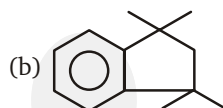
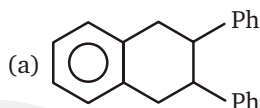
153.

Decreasing order of rate of reaction of molecules towards electrophilic addition reaction is :

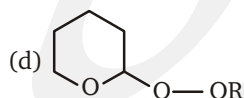
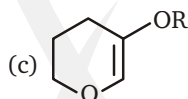
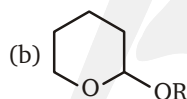
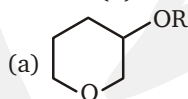


(a) $a > b > c > d$ (c) $d > b > c > a$ (b) $b > c > a > d$ (d) $b > d > c > a$ 

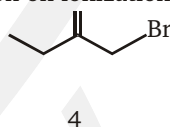
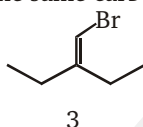
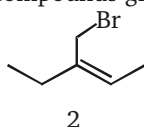
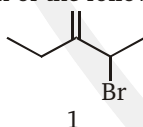
Product (A) is :



Product (B) of the above reaction is :



156. Which of the following compounds gives the same carbocation on ionization ?



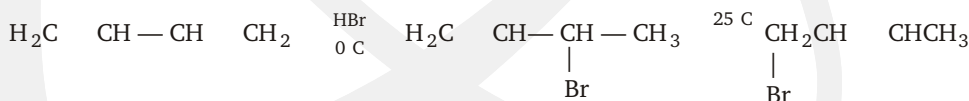
(a) 1 and 3

(b) 2 and 4

(c) 1 and 2

(d) 1 and 4

157. For the following reactions the major products are shown :



These provide an example of 1 control at low temperature and 2 control at higher temperature.

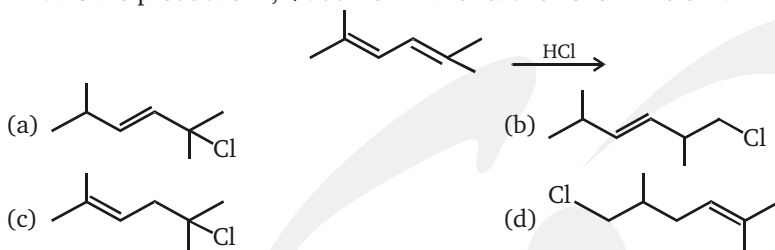
1 (a) kinetic
(c) kinetic

2 thermodynamic
kinetic

1 (b) thermodynamic
(d) thermodynamic

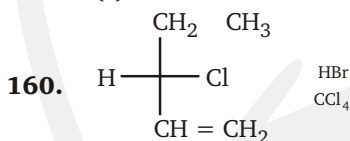
2 kinetic
thermodynamic

158. What is the product of 1, 4-addition in the reaction shown below ?



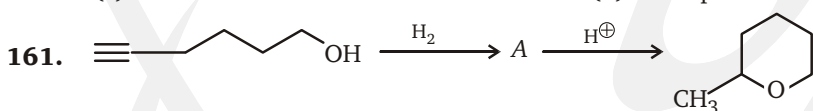
159. Dehydration of the above compound will give :

- (a) meso product (b) racemic mixture
(c) diastereomer (d) optically pure enantiomer



What is stereochemistry of product ?

- (a) Racemic mixture (b) Optically inactive
(c) Diastereomers (d) Meso product

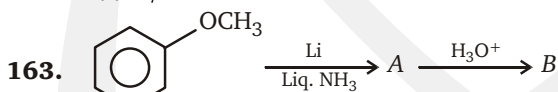


End product formed in the above reaction is :

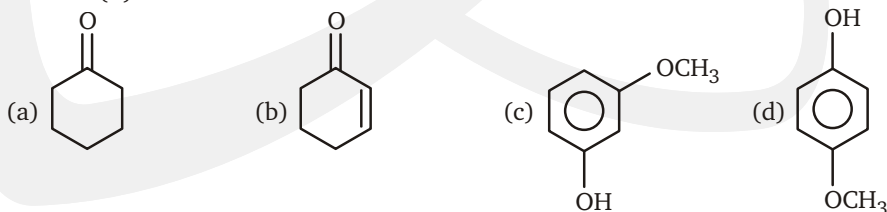
- (a) Optically active (b) Racemic (c) Meso (d) Diastereomer

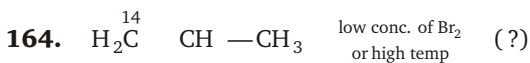
162. How many moles of BH_3 are needed to react completely with 2 mole of 1-pentene in hydroboration-oxidation reaction ?

- (a) 2 mole (b) 3 mole
(c) $2/3$ mole (d) $3/2$ mole



Product (B) in the above reaction is :



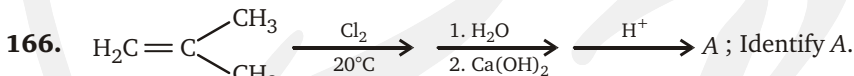


Product of the above reaction is :

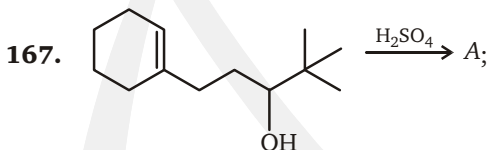
- (a) $\text{H}_2\text{C} \overset{14}{\text{CH}} - \text{CH}_2 - \text{Br}$ (b) $\text{H}_2\text{C} - \text{CH} \overset{14}{\text{CH}_2} - \text{Br}$
 (c) $\begin{array}{c} \text{CH}_2 - \text{CH} - \text{CH}_3 \\ | \quad | \\ \text{Br} \quad \text{Br} \end{array}$ (d) both (a) and (b)

165. In which of the following reactions 1,3-butadiene will be obtained as a major product ?

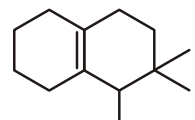
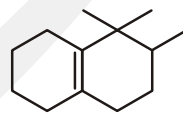
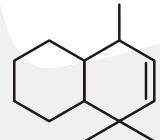
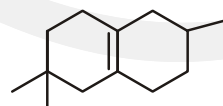
- (a) $\text{Br} - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{Br} \xrightarrow[(\text{CH}_3)_3\text{COH}]{(\text{CH}_3)_3\text{COK (2mole)}}$
 (b) $\text{HO} - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{OH} \xrightarrow[\text{Conc. H}_2\text{SO}_4]{\text{H}_2(1\text{mole})}$
 (c) $\text{H}_2\text{C} = \text{CH} - \text{C}(\text{CH}_3) = \text{CH}_2 \xrightarrow[\text{Ni}_2\text{B}]{\text{H}_2(1\text{mole})}$
 (d) All of these



- (a) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3 - \text{C} - \text{CH}_2 \\ | \quad \backslash \\ \text{O} \quad \text{O} \end{array}$ (b) $\begin{array}{c} \text{CH} \quad \text{CHO} \\ | \\ \text{CH}_3 \\ | \\ \text{CH}_3 \end{array}$
 (c) $\text{CH}_3 - \overset{\text{O}}{\parallel}{\text{C}} - \text{CH}_2 - \text{CH}_3$ (d) $\begin{array}{c} \text{CH} \quad \text{CH}_2 \\ | \\ \text{CH}_3 \\ | \\ \text{CH}_3 \end{array}$



Product (A) is :

- (a)  (b) 
 (c)  (d) 

(a) a (b) b (c) c (d) d

(a)  (b) 

$$\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3 - \text{C} - \text{C} - \text{CH}_2 - \text{CH} \\ | \quad \quad | \\ \text{CH}_3 \quad \quad \text{CH}_3 \end{array}
 \xrightarrow[\text{H}^+]{\text{Cl} - \text{C}_6\text{H}_4 - \text{COOH}}
 \text{(Product)}$$

(a)
$$\begin{array}{c} \text{CH}_3 \quad \text{CH}_3 \\ | \quad | \\ \text{CH}_3 - \text{C} - \text{C} - \text{CH}_2 - \text{CH} - \text{CH}_2 \\ | \quad \quad | \\ \text{O} \quad \text{O} \end{array}$$

(b)
$$\begin{array}{c} \text{CH}_3 \quad \text{CH}_3 \\ | \quad | \\ \text{CH}_3 - \text{C} - \text{C} - \text{CH}_2 - \text{CH} - \text{CH}_2 \\ | \quad \quad | \\ \text{O} \quad \text{O} \end{array}$$

(c)
$$\begin{array}{c} \text{CH}_3 \quad \text{CH}_3 \\ | \quad | \\ \text{CH}_3 - \text{C} - \text{C} - \text{CH}_2 - \text{CH} - \text{CH}_2 \\ | \quad \quad | \\ \text{OH} \quad \text{OH} \end{array}$$

(d)
$$\begin{array}{c} \text{CH}_3 \quad \text{CH}_3 \\ | \quad | \\ \text{CH}_3 - \text{C} - \text{C} - \text{CH}_2 - \text{CH} - \text{CH}_2 \\ | \quad \quad | \\ \text{OH} \quad \text{OH} \end{array}$$



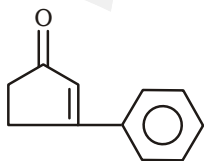
(a) Meso compound
(b) Enantiomeric pair
(c) Diastereomers
(d) Optically pure enantiomer

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Product (A) of above reaction is:

- (a) $\text{CH}_3\text{O}-\underset{\text{CO}_2\text{H}}{\text{CH}}-\text{CH}_2-\text{CH}_2-\text{CHO}$
- (b) $\text{CH}_3\text{O}-\text{CH}_2-\underset{\text{CO}_2\text{H}}{\text{CH}}-\text{CH}_2-\text{CO}_2\text{H}$
- (c) $\text{CH}_3\text{O}-\underset{\text{CO}_2\text{H}}{\text{CH}}-\text{CH}_2-\text{CH}_2-\text{CO}_2\text{H}$
- (d) $\text{CH}_3\text{O}-\underset{\text{CO}_2\text{H}}{\text{CH}}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CHO}$

173.

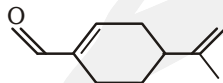


H_2 (2-3atm) (1 mole)
10% Pd/C

Products ; Comment up on optical activity of products.

- (a) Diastereomers (b) Racemic mixture (c) Meso (d) Optically pure enantiomer

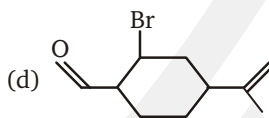
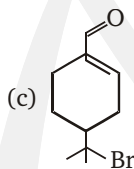
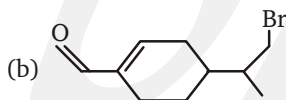
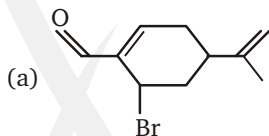
174.



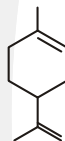
+ HBr
(1 mole)

Product

Addition of a mineral acid to an olefin bond leads to major product, Identify it:



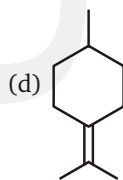
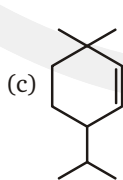
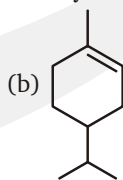
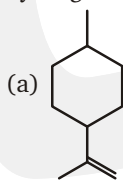
175.

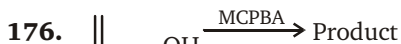


$\xrightarrow[\text{PtO}_2]{\text{H}_2 \text{ (one mole)}}$

Product

In polyenes that contain differently substituted (C=C) double bonds, it is possible to hydrogenate chemoselectively one (C=C) double bond. Product is :

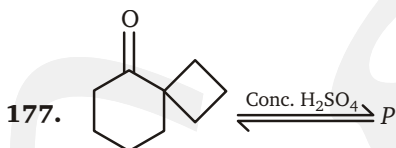




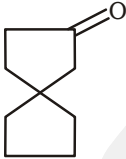
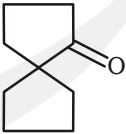
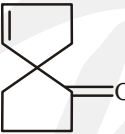
(MCPBA meta-chloro perbenzoic acid)

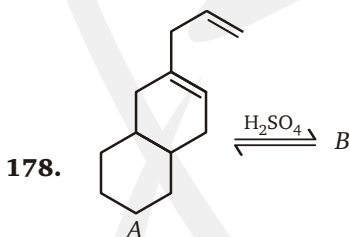
Stereochemistry of the product of above reaction is :

- (a) Meso (b) Racemic
(c) Diastereomers
(d) Optically inactive due to absence of chiral center.

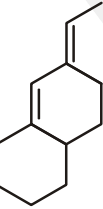
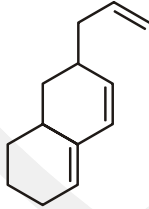
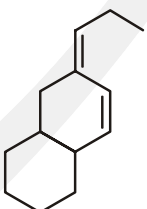
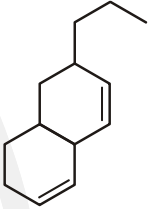


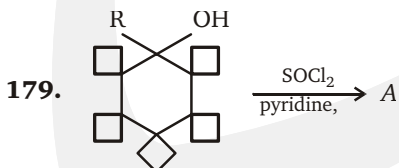
Identify product (P).

- (a)  (b) 
(c)  (d) 

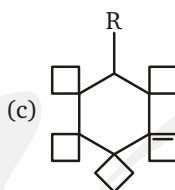
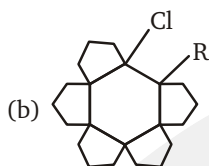
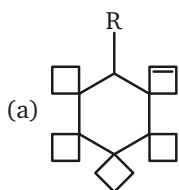


A isomerise to B on addition of traces of acid H_2SO_4 . Compound (B) is :

- (a)  (b) 
(c)  (d) 

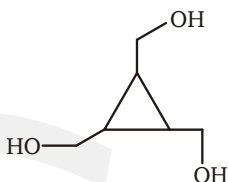


Product (A) of the reaction is :



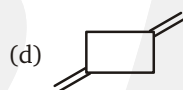
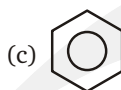
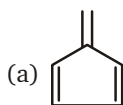
(d) None of these

180.



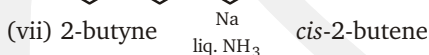
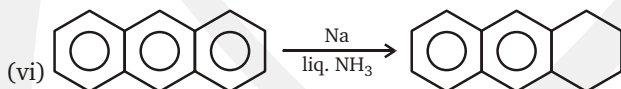
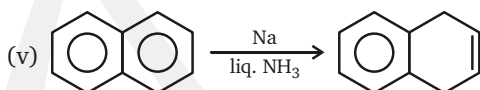
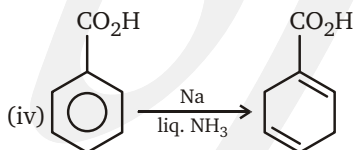
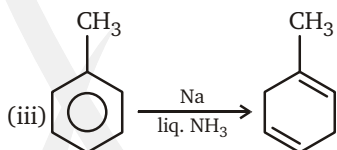
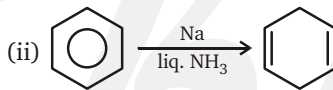
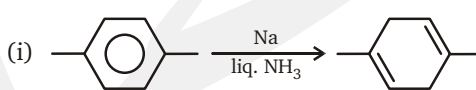
H (excess)

(A). Product (A) is :



181.

Which of the following reactions do not represent the major product of given Birch reductions ?



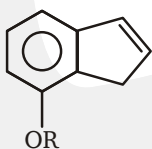
(a) (i), (iii), (vi)

(b) (iv), (vi), (vii)

(c) (iv), (v), (vi)

(d) (i), (ii), (v), (vii)

182.

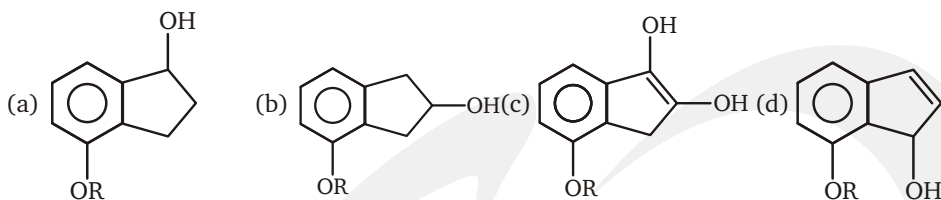


BH_3

$\text{H}_2\text{O}_2, \text{HO}$

(A)

Product (A) is:

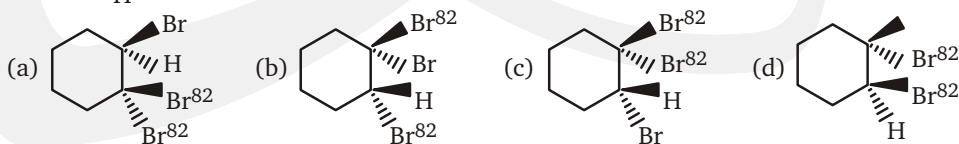
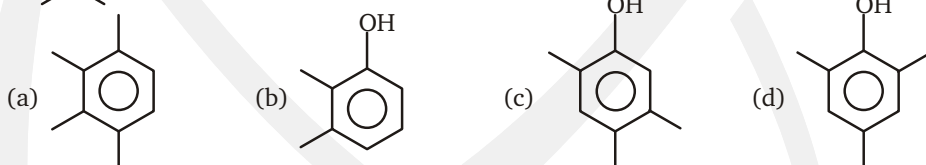
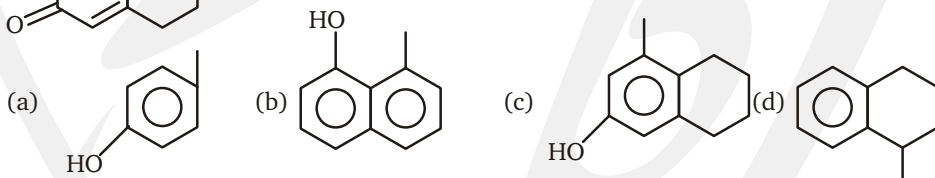


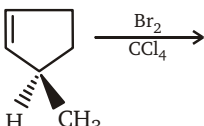
Hint : Think carefully about the relative stabilization of developing positive charge, when the double bond reacts with an electrophile.



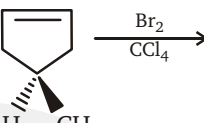
Correct statement about above reaction is:

- (a) A *cis*-2-chlorocyclohexanol, B cyclohexeneoxide
 (b) A *trans*-2-chloro cyclohexanol, B anti-diol
 (c) A *trans*-2-chlorocyclohexanol, B cyclohexeneoxide
 (d) A *cis*-2-chlorocyclohexanol, B anti-diol

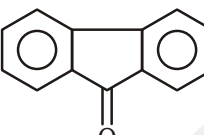


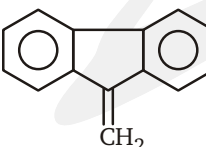
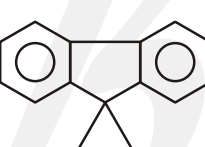
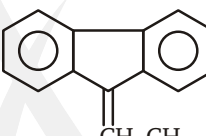
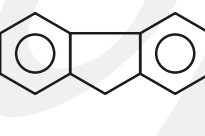
187.  stereochemistry of the product is:

- (a) Diastereomers (b) Racemic mixture
(c) Meso (d) Pure Enantiomers

188.  Product/s obtained is/are :

- (a) Diastereomers (b) Racemic
(c) Meso (d) Optically pure enantiomers

189.  (x) ; Product (x) is :

- (a)  (b) 
(c)  (d) 

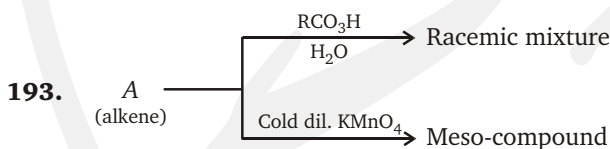
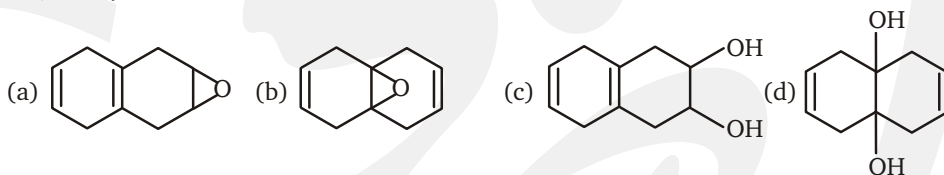
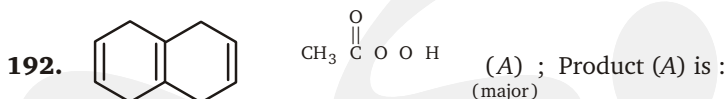
190. $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \underset{\text{CH}_3}{\overset{\text{CH}_3}{\text{C}}} - \text{CH}_2 - \text{OH} \xrightarrow{\text{H}^+} \text{(A)} ; \text{Product (A) is :}$
(major)

- (a) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \underset{\text{CH}_3}{\text{C}} - \text{CH} - \text{CH}_3$
(b) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{CH} = \text{C} \begin{matrix} \diagup \text{CH}_3 \\ \diagdown \text{CH}_3 \end{matrix}$
(c) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \underset{\text{CH}_3}{\text{C}} - \text{CH}_2$
(d) $\text{CH}_3 - \text{CH}_2 - \underset{\text{CH}_3}{\text{CH}} - \text{CH}_2 - \underset{\text{CH}_3}{\text{C}} - \text{CH}_2$



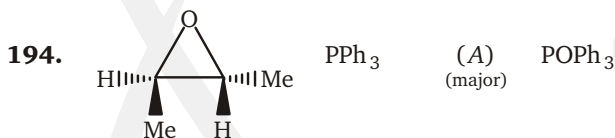
Comment on optical activity of the products:

- (a) Racemic (b) Diastereomer
(c) Meso (d) Optically pure enantiomer



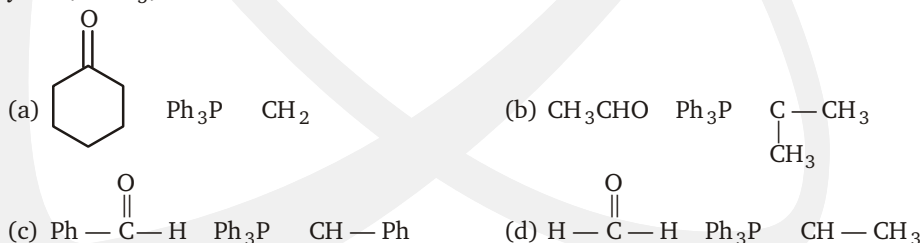
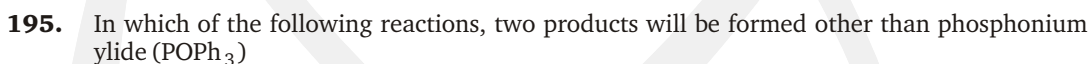
Alkene (A) will be :

- (a) *cis*-2-pentene (b) *cis*-2-hexene
(c) *cis*-4-octene (d) *trans*-2-hexene

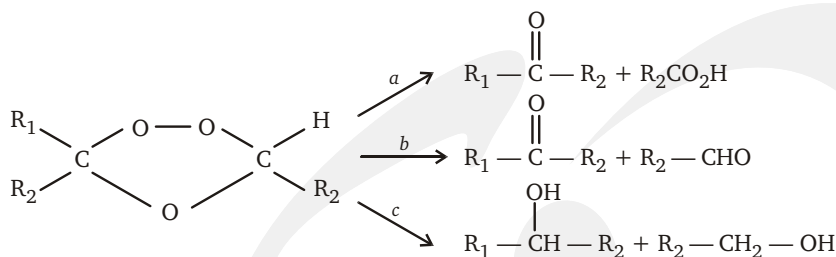


Product (A) is

- (a) *trans*-2-butene (b) *cis*-2-butene (c) 1-butene (d) Iso-butene

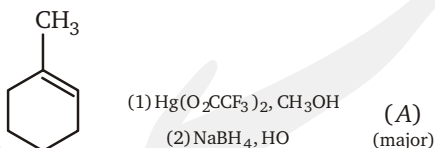


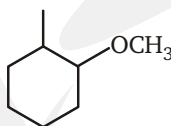
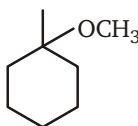
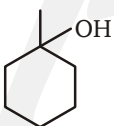
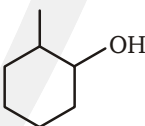
196. To carry out the given conversions, select the correct option:

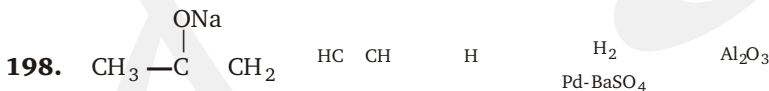


- (a) $a \text{ Ag}_2\text{O},$ $b \text{ Zn/CH}_3\text{CO}_2\text{H},$ $c \text{ LiAlH}_4$
 (b) $a \text{ H}_2\text{O}_2,$ $b \text{ CH}_3\text{SCH}_3,$ $c \text{ NaBH}_4$
 (c) Both (a) and (b)
 (d) None of these

197. The product (A) of given alkoxymercuration de-mercuration is :



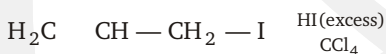
- (a)  (b)  (c)  (d) 



End product of the reaction is :

- (a) $\text{H}_2\text{C} - \text{CH} - \overset{\text{CH}_3}{\underset{\text{CH}_2}{\text{C}}} - \text{CH}_2$ (b) $\text{CH}_3 - \text{CH} - \text{CH} - \text{CH} - \text{CH}_2$
 (c) $\text{H}_2\text{C} - \text{CH} - \text{CH} - \text{CH}_2$ (d) $\text{H}_2\text{C} - \text{CH} - \text{CH}_2 - \text{CH} - \text{CH}_2$

199. Major product of the given reaction is :

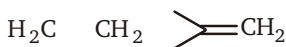


- (a) $\text{CH}_3 - \underset{\text{I}}{\text{CH}} - \underset{\text{I}}{\text{CH}_2}$ (b) $\text{CH}_3 - \underset{\text{I}}{\text{CH}} - \text{CH}_3$
 (c) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{I}$ (d) $\text{I} - \text{CH}_2 - \text{CdH}_2 - \text{CH}_2 - \text{I}$

200. The rate constant for a reaction can be increased by a the stability of the reactant or by b the stability of the transition state. Select the correct choice for *a* and *b*.

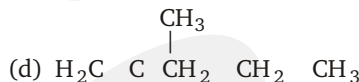
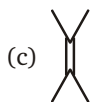
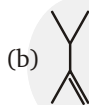
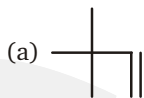
- (a) decreasing, decreasing (b) increasing, decreasing
(c) decreasing, increasing (d) increasing, increasing

201. Major product of the given reaction is :



H

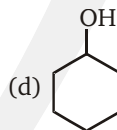
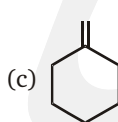
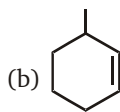
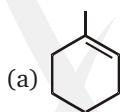
Product



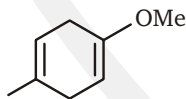
202. $\text{Ph}_3\text{P} \quad \text{CH}_2$

(A)
Major

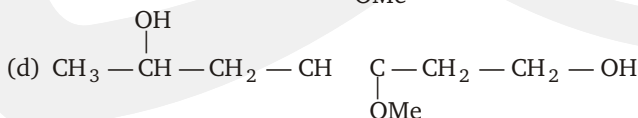
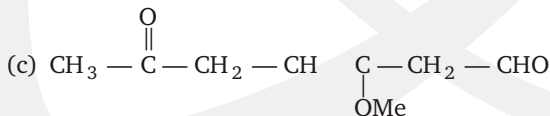
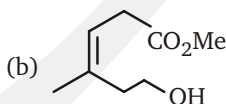
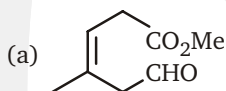
Major product (A) is :

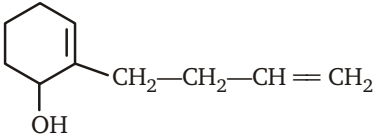


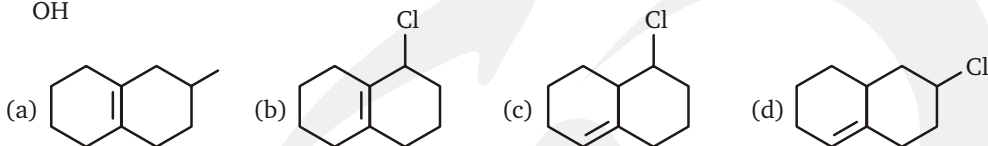
203. In the given reaction, only one alkene undergo preferential oxidation by electrophilic ozone. Identify product (P) of the given reaction:

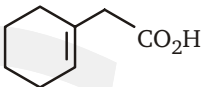


$\text{Me}_2\text{S}, \quad \text{O}_3, \quad 78^\circ\text{C} \text{ then } \text{NaBH}_4$ (P)
(52%)

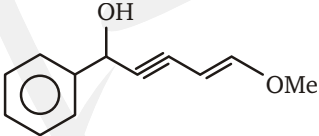


204.  $\xrightarrow{\text{HCl}}$ (P) ; Product (P) is:
(major)

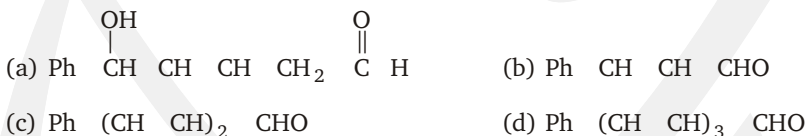


205.  $\xrightarrow[\text{NaHCO}_3]{\text{I}_2}$ (A) ; Major product of the reaction is :

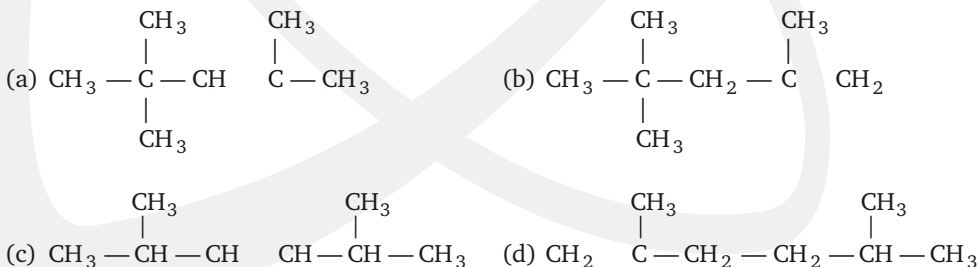


206.  $\xrightarrow[\text{Pd BaSO}_4]{\text{H}_2}$ (A) H_2O (B)

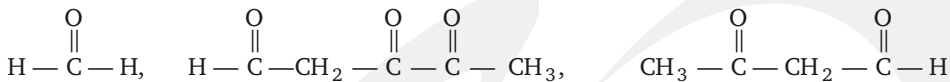
Product (B) is :



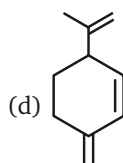
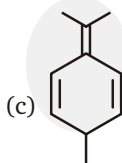
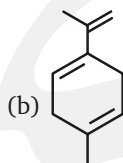
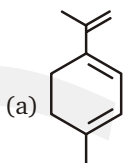
207. Isobutene, in the presence of H_2SO_4 , forms a mixture of two isomeric alkene (C_8H_{16}). The major alkene is :



- 208.** An unknown alkene (A) reacts with 3 mole of H_2 gas in presence of platinum catalyst to form 1-isopropyl-4-methyl cyclohexane. When unknown alkene (A) is ozonized and reduced, following product are obtained

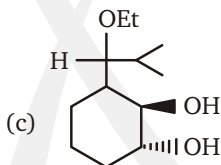
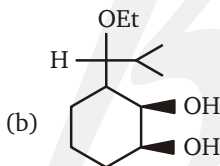
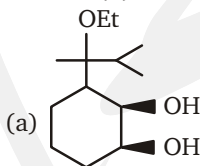


The alkene (A) is :



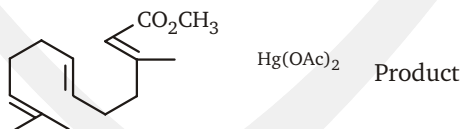
- 209.** (1) NBS
(2) Mg/ether

Product (C) is :

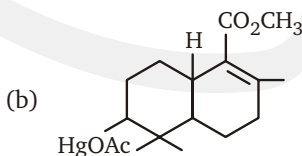
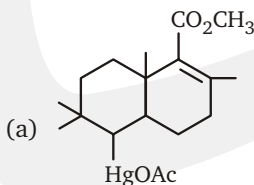


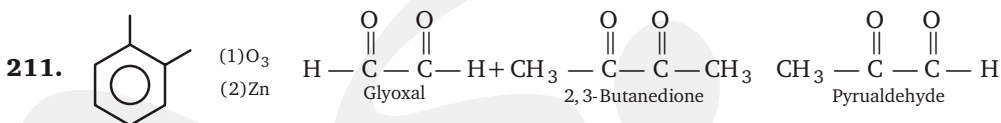
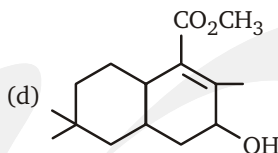
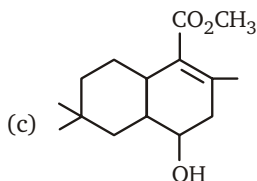
(d) Both (a) and (b)

- 210.** The following reaction take place in high yields.



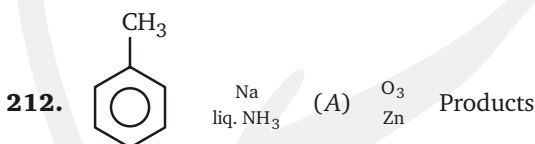
Use your knowledge of alkene chemistry to predict a product even though you have never seen this reaction before





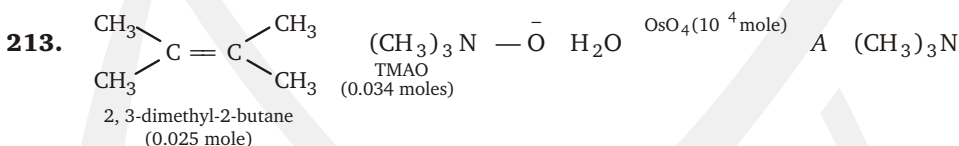
What is the ratio of glyoxal to pyruvaldehyde obtained in the above reaction ?

- (a) 1 : 3 (b) 3 : 1
(c) 3 : 2 (d) 2 : 3



Which of the following product cannot be obtained in above reaction ?

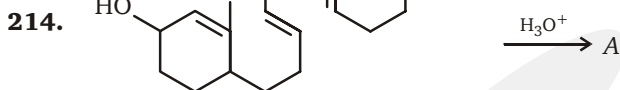
- (a) $\text{H}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_2-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$ (b) $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_2-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$
(c) $\text{CH}_3-\underset{\text{CHO}}{\underset{|}{\text{CH}}}-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$ (d) None of these



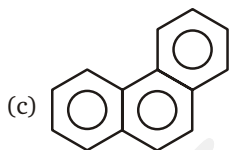
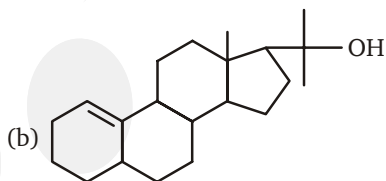
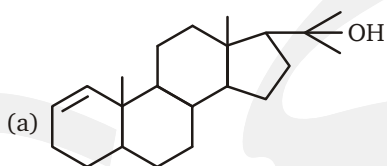
(TMAO trimethyl amine N oxide)

Product (A) is :

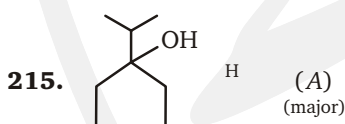
- (a) (b)
(c) $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3$ (d) $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{C}(\text{CH}_3)_3$



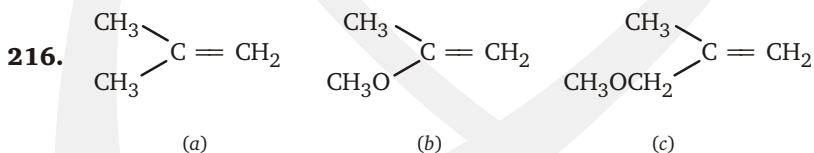
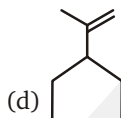
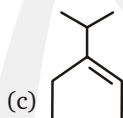
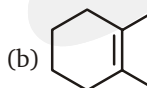
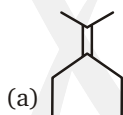
Product (A) of the reaction is :



(d) None of these



Product (A) is :



Arrange the above in the decreasing order of reactivity towards HBr :

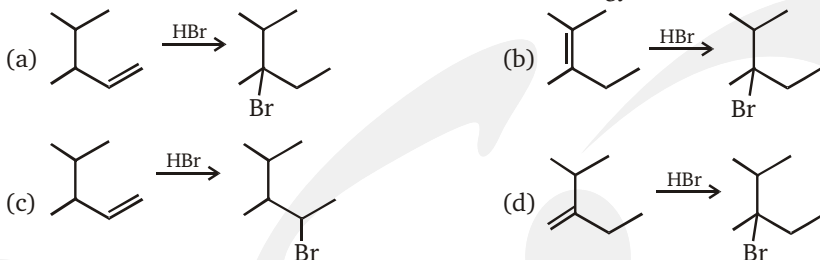
(a) $a > b > c$

(b) $b > a > c$

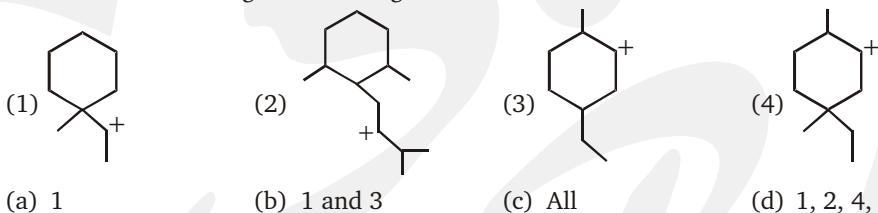
(c) $b > c > a$

(d) $a > c > b$

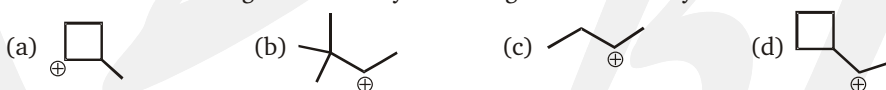
217. Which reaction has the lowest $G^\ddagger$ or (Activation-Energy)?



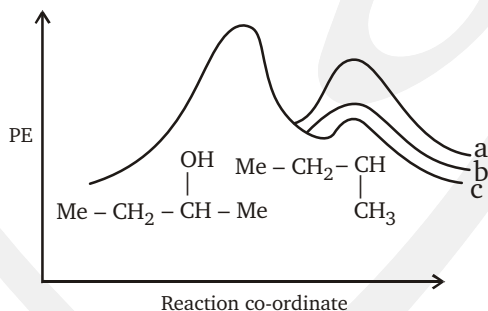
218. Which of the following will rearrange ?



219. Which of the following is most likely to undergo a favorable hydride shift ?



220. Energy profile diagram for dehydration of 2-butanol using conc. H_2SO_4 is given below :



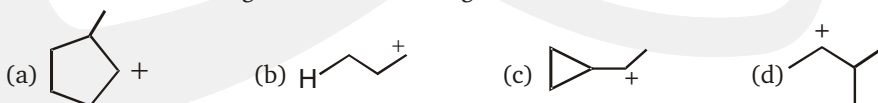
Product (b) of above reaction is :

- (a) 1-butene (b) *cis*-2-butene (c) *trans*-2-butene (d) *iso*-butene

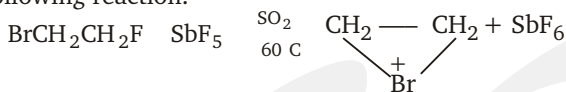
221. How many alkene on catalytic hydrogenation given isopentane as a product ?

- (a) 2 (b) 3 (c) 4 (d) 5

222. Which of the following would not rearrange to a more stable form ?



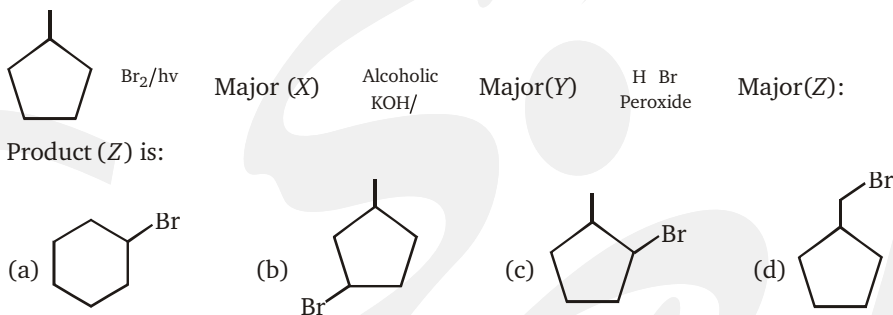
223. Consider the following reaction.



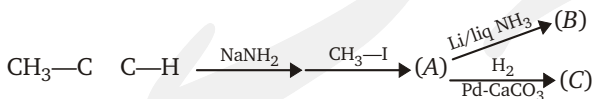
In this reaction SbF_5 acts as:

- (a) an acid (b) a base (c) a nucleophile (d) an electrophile

224.



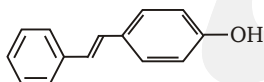
225.



Relation between (B) and (C) is:

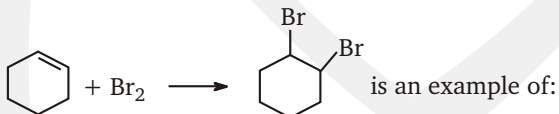
- (a) Enantiomer (b) Diastereomer
(c) Geometrical isomer (d) Meso

226. The reaction of HBr with the following compound would produce :



- (a) (b)
(c) (d)

227.

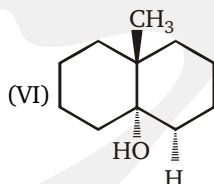
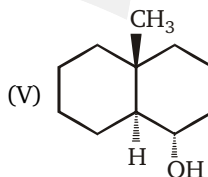
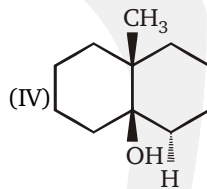
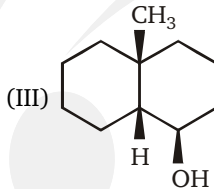
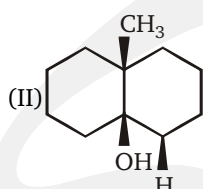
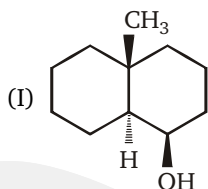
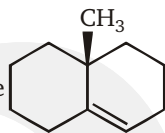


- (a) Nucleophilic addition (b) Nucleophilic substitution
(c) Electrophilic addition (d) Electrophilic substitution
(e) Free radical substitution

228. Olefins can be hydrogenated by :

- (a) Zinc and HCl (b) Nascent hydrogen
(c) Raney Ni and H (d) Lithium hydride in ether

229. What are the products obtained on hydroboration-oxidation of the given alkene

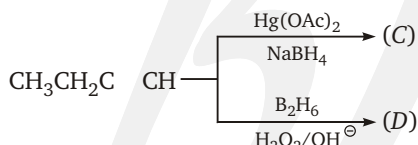
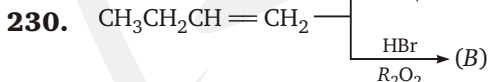


(a) I and III

(b) II and IV

(c) II and VI

(d) III and V



Relation between A and B, C and D are :

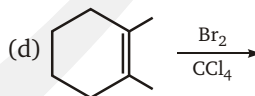
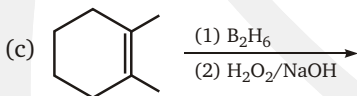
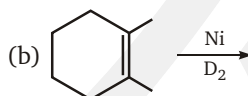
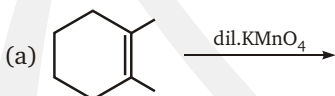
(a) Position, chain

(b) Position, Functional

(c) Chain, Identical

(d) Metamer, Functional

231. In which reaction syn addition doesn't take place.



232. Number of olefin of C_4H_8 $\xrightarrow{\text{HBr}}$ Number of Markonikow product (including stereo) (y)

x y is :

(a) 5

(b) 6

(c) 7

(d) 8

ANSWERS — LEVEL 1

1.	(c)	2.	(d)	3.	(c)	4.	(d)	5.	(b)	6.	(b)	7.	(c)	8.	(c)
9.	(c)	10.	(d)	11.	(b)	12.	(d)	13.	(c)	14.	(b)	15.	(b)	16.	(c)
17.	(d)	18.	(d)	19.	(b)	20.	(d)	21.	(b)	22.	(a)	23.	(b)	24.	(b)
25.	(b)	26.	(b)	27.	(d)	28.	(b)	29.	(d)	30.	(b)	31.	(c)	32.	(b)
33.	(a)	34.	(b)	35.	(b)	36.	(b)	37.	(b)	38.	(b)	39.	(b)	40.	(b)
41.	(d)	42.	(e)	43.	(c)	44.	(c)	45.	(a)	46.	(c)	47.	(c)	48.	(b)
49.	(b)	50.	(b)	51.	(b)	52.	(a)	53.	(b)	54.	(d)	55.	(b)	56.	(c)
57.	(c)	58.	(b)	59.	(c)	60.	(a)	61.	(b)	62.	(d)	63.	(a)	64.	(b)
65.	(d)	66.	(b)	67.	(d)	68.	(a)	69.	(c)	70.	(d)	71.	(d)	72.	(c)
73.	(d)	74.	(c)	75.	(a)	76.	(c)	77.	(d)	78.	(b)	79.	(d)	80.	(d)
81.	(c)	82.	(b)	83.	(a)	84.	(a)	85.	(a)	86.	(d)	87.	(b)	88.	(b)
89.	(c)	90.	(d)	91.	(b)	92.	(b)	93.	(a)	94.	(a)	95.	(a)	96.	(d)
97.	(b)	98.	(a)	99.	(c)	100.	(d)	101.	(b)	102.	(b)	103.	(d)	104.	(b)
105.	(b)	106.	(c)	107.	(b)	108.	(b)	109.	(d)	110.	(d)	111.	(c)	112.	(b)
113.	(a)	114.	(b)	115.	(c)	116.	(a)	117.	(b)	118.	(b)	119.	(b)	120.	(d)
121.	(b)	122.	(c)	123.	(a)	124.	(b)	125.	(d)	126.	(a)	127.	(b)	128.	(c)
129.	(d)	130.	(a)	131.	(c)	132.	(d)	133.	(c)	134.	(a)	135.	(d)	136.	(c)
137.	(b)	138.	(d)	139.	(d)	140.	(b)	141.	(b)	142.	(b)	143.	(b)	144.	(b)
145.	(b)	146.	(c)	147.	(b)	148.	(a)	149.	(d)	150.	(b)	151.	(a)	152.	(d)
153.	(c)	154.	(c)	155.	(b)	156.	(c)	157.	(a)	158.	(a)	159.	(b)	160.	(c)
161.	(b)	162.	(c)	163.	(b)	164.	(d)	165.	(d)	166.	(b)	167.	(b)	168.	(a)
169.	(c)	170.	(b)	171.	(b)	172.	(d)	173.	(b)	174.	(c)	175.	(b)	176.	(b)
177.	(b)	178.	(c)	179.	(b)	180.	(c)	181.	(b)	182.	(b)	183.	(c)	184.	(c)
185.	(c)	186.	(b)	187.	(a)	188.	(b)	189.	(b)	190.	(b)	191.	(a)	192.	(b)
193.	(c)	194.	(b)	195.	(c)	196.	(c)	197.	(b)	198.	(a)	199.	(b)	200.	(c)
201.	(c)	202.	(c)	203.	(b)	204.	(d)	205.	(b)	206.	(c)	207.	(b)	208.	(b)
209.	(b)	210.	(b)	211.	(c)	212.	(c)	213.	(b)	214.	(a)	215.	(b)	216.	(b)
217.	(d)	218.	(c)	219.	(a)	220.	(b)	221.	(b)	222.	(c)	223.	(d)	224.	(c)
225.	(b,c)	226.	(b)	227.	(c)	228.	(c)	229.	(d)	230.	(b)	231.	(d)	232.	(c)

* For question 56 and 57, option (a) is also correct.

Level - 2

1.

Reagents

A. HCl	B. Br ₂	C. Hg(OAc) ₂ in H ₂ O	D. B ₂ H ₆ (BH ₃) in ether
E. H ₂ O ₂	F. KMnO ₄ in H ₂ O	G. HOBr	H. NaBH ₄

In each reagent box write a letter designating the best reagent and condition selected from the above list of reagents.

Reactant	Reagent		Product
$(\text{CH}_3)_2\text{CHCH}=\text{CH}_2$ 3-methyl-1-butene	(i)	☐	$(\text{CH}_3)_2\text{CHCH}(\text{Cl})\text{CH}_3$ 2-Chloro-3-methyl butane
	(ii)	☐	$(\text{CH}_3)_2\text{CHCHBrCH}_2\text{Br}$ 1, 2-dibromo-3-methyl butane
	(iii)	☐	$(\text{CH}_3)_2\text{CHCHOHCH}_2\text{Br}$ 1, bromo-3-methyl 2 butanol
	(iv)	☐	$(\text{CH}_3)_2\text{CHCH}(\text{OH})\text{CH}_3$ 3-methyl-2-butanol
	(v)	☐	$(\text{CH}_3)_2\text{CHCH}(\text{OH})\text{CH}_2\text{OH}$ 3-methyl-1, 2-butanediol

2. Propene ($\text{CH}_3 - \text{CH} = \text{CH}_2$) can be transformed to compounds (a to j) listed in the left-hand column.

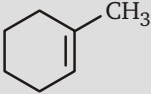
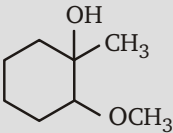
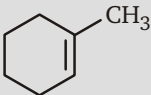
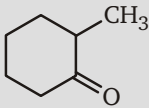
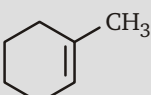
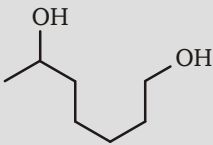
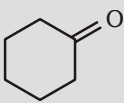
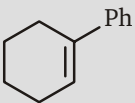
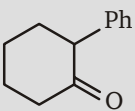
Write letter designating the reagent, you believe will achieve desired transformation. In the case of a multi step sequence write the reagent in the order they are to be used.

Desired Product	No. of Steps	Write options	Reagent List	
a. $\text{CH}_3\text{CHBrCH}_2\text{Br}$	one	☐	A.	Hg(OAc) ₂ in H ₂ O
b. $(\text{CH}_3)_2\text{CHOH}$	two	☐	B.	B ₂ H ₆ in THF

c.	$\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$	two		C.	NaBH_4 in alcohol
d.	CH_3COCH_3	three		D.	Br_2 in CH_2Cl_2
e.	$\text{CH}_3\text{CH}_2\text{CHO}$	three		E.	H_2O_2 in aqueous base
f.	$\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{Br}$	one		F.	HOBr (NBS in aqueous acetone)
g.	$(\text{CH}_3)_2\text{CHBr}$	one		G.	HBr in CH_2Cl_2
h, k.	$\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{OH}$	two		H.	OsO_4 in ether
i.	$\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$	three		I.	Thionyl chloride (SOCl_2)
j.	$\text{CH}_3\text{C}(\text{CH}_3)=\text{CH}_2$	two		J.	NaHSO_3 in aqueous acetone
				K.	NaOH in alcohol and reflux
				L.	NaNH_2 (strong base)

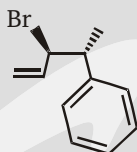
3. In each reaction box write a single letter designating the best reagent and condition selected from the list at bottom of the page.

(F.S., first step, S.S. second step, T.S. third step)

Reaction	Reactant	Options	Product
1.		$\xrightarrow[\text{S.S. } \square]{\text{F.S. } \square}$	
2.		$\xrightarrow[\text{T.S. } \square]{\text{S.S. } \square}$ F.S. ☐	
3.		$\xrightarrow[\text{S.S. } \square]{\text{F.S. } \square}$	
4.		$\xrightarrow[\text{S.S. } \square]{\text{F.S. } \square}$	
5.		$\xrightarrow[\text{T.S. } \square]{\text{S.S. } \square}$ F.S. ☐	

A. $\text{NaBH}_4/\text{alcohol}$	B. $\text{Ph CO}_3\text{H}/\text{CH}_2\text{Cl}_2$	C. PCC	D. $\text{CH}_3\text{ONa}/\text{CH}_3\text{OH}$
E. B_2H_6 in THF	F. $\text{H}_2\text{O}_2/\text{aq. NaOH}$	G. H_3PO_4 & heat	H. $\text{AlCl}_3/\text{C}_6\text{H}_6$
I. O_3 in CH_2Cl_2	J. Br_2 in CH_2Cl_2	K. 20% KOH & heat	L. $\text{Ph Li}/\text{ether}$

4. Match the reagents a-j with products A-J. There is one best product for each reaction.



(x)

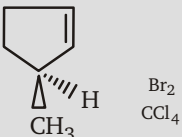
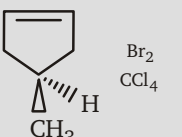
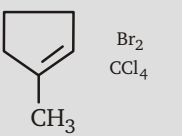
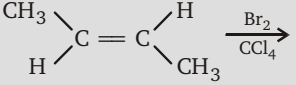
The molecule (x) is the starting material for all reactions in problem. Do the ones you know first and then tackle the rest by deductive reasoning

Products		Reagents		Option
A 	B 	(a)	H_2O heat, pH 7	
		(b)		
C 		(c)	tBuOK , polar aprotic solvent	
		(d)	(1) O_3 , ether (2) H_2O , NaOH , H_2O_2	
D 	E 	(e)	Br_2 , CCl_4	
F 	G 	(f)	NBS, $h\nu$, CCl_4	
H 	I 	(g)	(1) H_3O^+ (2) NaOH , H_2O	
J 		(h)	(1) BH_3 , ether (2) H_2O_2	
		(i)	(1) OsO_4 (2) NaOH , H_2O	
		(j)	H_2 / Pd/C (EtOH)	

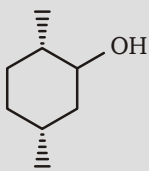
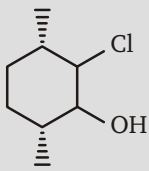
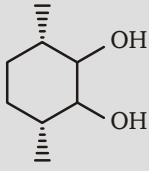
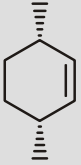
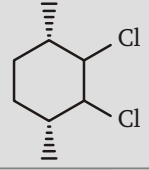
5. Match the column:

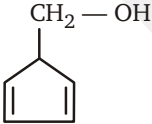
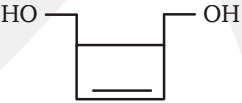
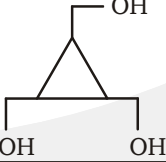
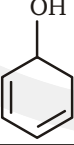
Column (I)		Column (II)	
(a)	$\text{CH}_3 - \text{C}(\text{C})(\text{CH}_3) - \text{CH}_3$	(p)	cis-product with $\text{H}_2 / \text{Pd} - \text{BaSO}_4$
(b)	$\text{CH}_3 - \text{CH}_2 - \text{C}(\text{CH}_3) - \text{CH}_3$	(q)	Trans-product with $\text{Na} / \text{liq. NH}_3$
(c)	$\text{CH}_3 - \text{C}(\text{CH}_3) = \text{CH}_2$	(r)	White with amm. AgNO_3
(d)	$\text{CH}_3 - \text{C}(\text{CH}_3) = \text{CH} - \text{Et}$	(s)	H_2 gas with Na

6. Match the column I with column II and with column III (Matrix).

Column-I		Column- II		Column- III	
Reaction		Nature of product formed		Number of chiral center present in product. (Consider only one isomer in case of racemic mixture or Diastereomer)	
(a)		(p)	Racemic mixture	(w)	0
(b)		(q)	Meso	(x)	1
(c)		(r)	Diastereomer	(y)	2
(d)		(s)	Vicinal dihalide	(z)	3

7. Match the column I and II.

Column (I)		Column (II)	
	Reaction		Product
(a)	 (1) OsO ₄ (2) NaOH, H ₂ O	(p)	
(b)	 (1) BH ₃ /ether (2) H ₂ O ₂ , NaOH, H ₂ O	(q)	
(c)	 Cl ₂ , H ₂ O	(r)	
(d)	 Cl ₂ /CCl ₄	(s)	

8. (1)  H (A) (2)  H (B)
- (3)  H (C) (4)  H (D)

Sum of molecular mass of A, B, C, D (i.e. A B C D) is equal to :

9. (1) C_2FCIBrI $\xrightarrow[\text{Ni}]{\text{H}_2}$ (A) (exclude stereoisomer)
(all isomers)

(2) C_4H_8 (alkene) $\xrightarrow[\text{Ni}]{\text{H}_2}$ (B) (exclude stereoisomer)
(all isomers)

Total number of products A and B (i.e. A + B) is equal to :

10.

Reaction 1	Reaction 2
$ \begin{array}{c} \text{CH}_3 \\ \\ \text{H} - \text{C} - \text{Br} \\ \\ \text{CH} \\ \quad (cis) \\ \text{CH} \\ \\ \text{H} - \text{C} - \text{Br} \\ \\ \text{CH}_3 \end{array} \quad \xrightarrow[\text{CCl}_4]{\text{Br}_2} \quad (P) $	$ \begin{array}{c} \text{CH}_3 \\ \\ \text{H} - \text{C} - \text{Br} \\ \\ \text{CH} \\ \quad (trans) \\ \text{CH} \\ \\ \text{H} - \text{C} - \text{Br} \\ \\ \text{CH}_3 \end{array} \quad \xrightarrow[\text{CCl}_4]{\text{Br}_2} \quad (Q) $
Reaction 3	Reaction 4
$ \begin{array}{c} \text{CH}_3 \\ \\ \text{H} - \text{C} - \text{Br} \\ \\ \text{CH} \\ \quad (cis) \\ \text{CH} \\ \\ \text{CH}_3 \end{array} \quad \xrightarrow[\text{CCl}_4]{\text{Br}_2} \quad (R) $	$ \begin{array}{c} \text{CH}_3 \\ \\ \text{H} - \text{C} - \text{Br} \\ \\ \text{CH} \\ \quad (trans) \\ \text{CH} \\ \\ \text{CH}_3 \end{array} \quad \xrightarrow[\text{CCl}_4]{\text{Br}_2} \quad (S) $

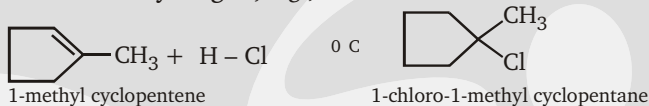
Sum of products P, Q, R, S (i.e. P + Q + R + S) is equal to :

11. Comprehension

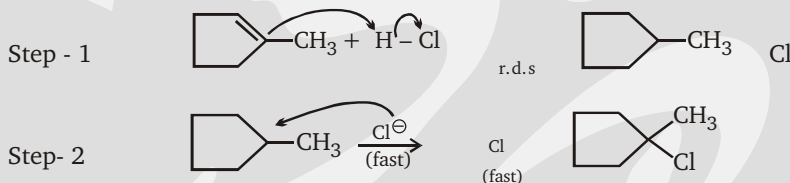
Vladimir Markovnikov rule :

Alkenes undergo electrophilic addition reactions. It is triggered by the acid acting as a electrophile toward π -electrons of the double bond.

Markovnikov's rule states that when an unsymmetrically substituted alkene reacts with a hydrogen halide, the hydrogen atom adds to the carbon that has the greater number of hydrogen, e.g.,



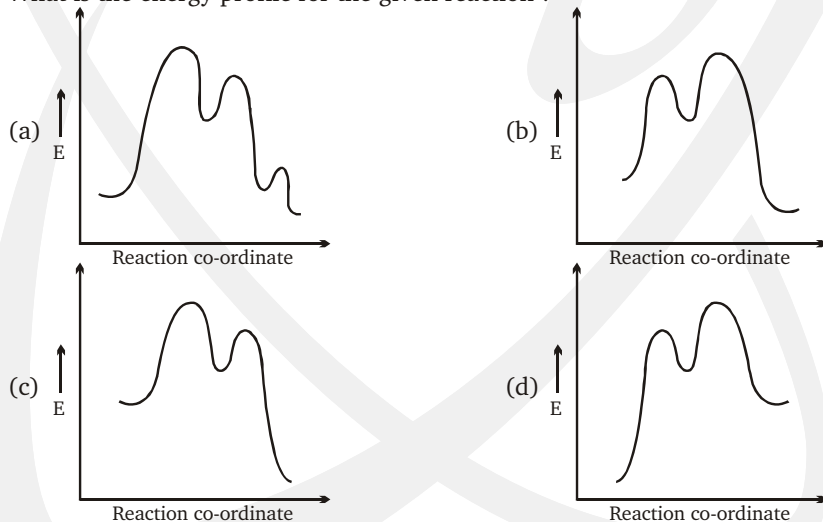
Mechanism :



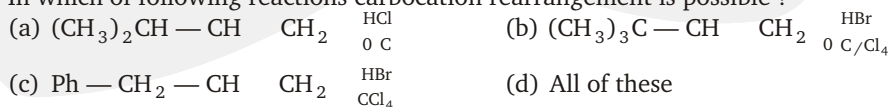
A. Which of the following is most reactive toward Markovnikov addition ?



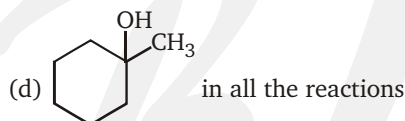
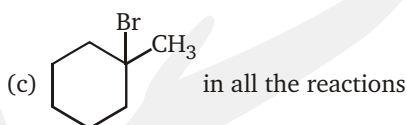
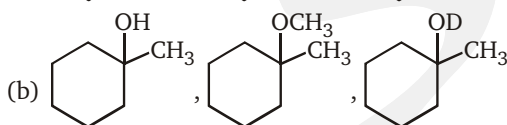
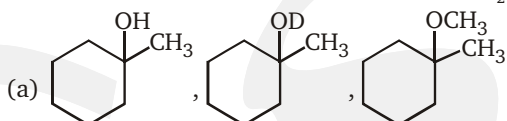
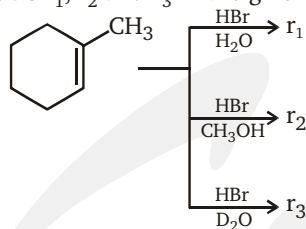
B. What is the energy profile for the given reaction ?



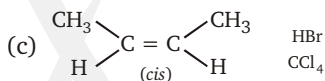
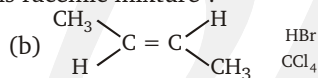
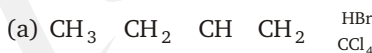
C. In which of following reactions carbocation rearrangement is possible ?



D. Identify the major products r_1 , r_2 and r_3 in the given reactions.

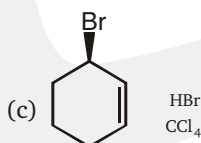
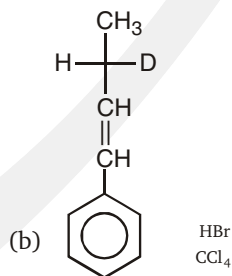
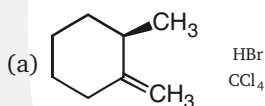


E. In which of the following reactions, product is racemic mixture ?

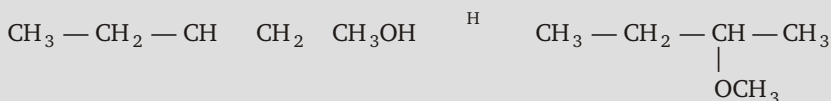


(d) All of these

F. In which of the following reactions, diastereomers will be formed ?



(d) All of these



A. What is electrophile in first step ?

- (a) CH_3 (b) H
(c) $\text{CH}_3 - \text{CH}_2 - \text{CH} - \text{CH}_3$ (d) HO

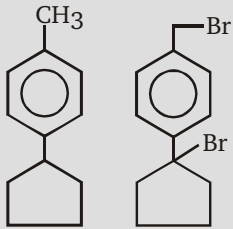
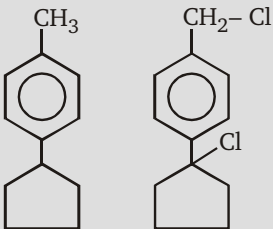
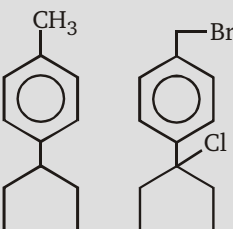
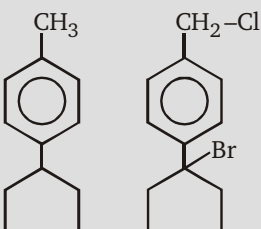
(a) CH_3OH (b) 1-butene
(c) H_2O (d) CH_3OCH_3

(a) CH_3 (b) H
(c) $\text{CH}_3 - \text{CH}_2 - \text{CH} - \text{CH}_3$ (d) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CH}_2$

(a) $\text{CH}_3 - \text{CH}_2 - \text{CH} \quad \text{CH}_2$ (b) CH_3OH
(c) H_2O (d) $\text{CH}_3 - \text{O} - \text{CH}_3$

(a) attack of nucleophile CH_3OH (b) attack of electrophile H^+
(c) attack of nucleophile H_2O (d) attack of electrophile CH_3^+

13. Match the column I and II :

Column (I)		Column (II)	
Conversion		Reagent	
(a)		(p)	$\text{SO}_2\text{Cl}_2 / h\nu$ (2 equivalent)
(b)		(q)	NBS (2 equivalent)
(c)		(r)	NBS then $\text{SO}_2\text{Cl}_2/h\nu$
(d)		(s)	$\text{SO}_2 \text{Cl}_2 / h\nu$ then NBS

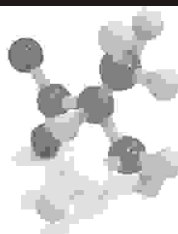
ANSWERS — LEVEL 2

1. (i) – A; (ii) – B; (iii) – G; (iv) – C; (v) – F
2. a – D; b – A, C; c – B, E; d – A, C, F; e – B, E, F; f – F; g – G; h – I, K; i – B, E, I; j – D, L
3. Reaction 1 : B, D; Reaction 2 : E, F, C Reaction 3 : I, A
Reaction 4 : L, G Reaction 5 : B, L, C
4. a – C; b – D; c – A; d – F; e – I; f – J; g – E; h – H; i – B; j – G
5. a – p, q; b – r, s; c – r, s; d – p, q
6. a – r, s – z; b – p, s – y; c – p, s – y; d – q, s – y
7. a – r; b – p; c – q; d – s
8.

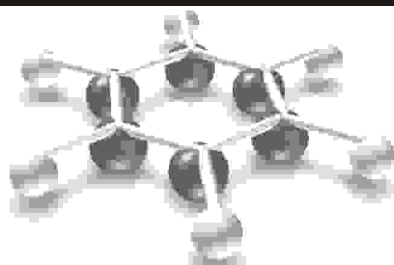
A	B	C	D	312
P	Q	R	S	8
9.

A	B	5
---	---	---
10.

P	Q	R	S	8
---	---	---	---	---
11. A – b; B – c; C – d; D – b; E – d; F – d
12. A – b; B – b; C – c; D – b; E – b
13. a-q; b-p; c-s; d-r



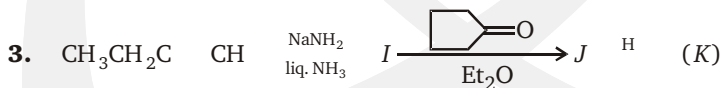
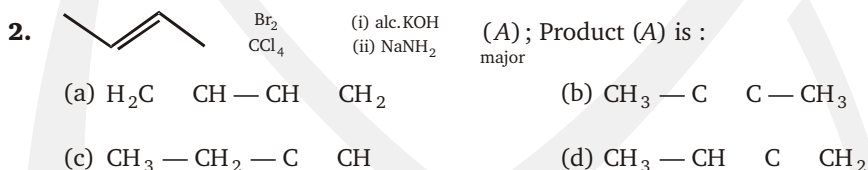
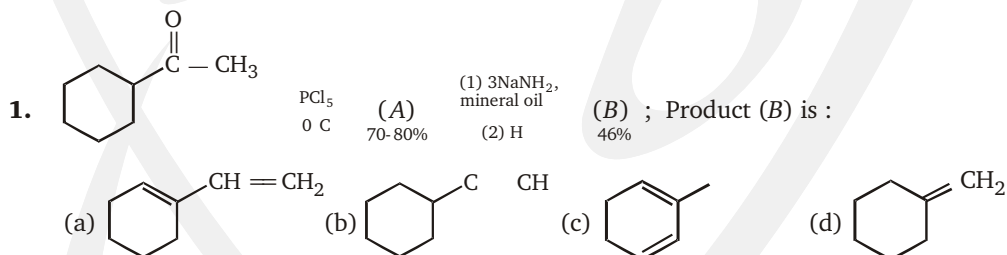
4c



HYDROCARBONS (ALKYNES)

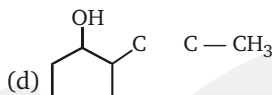
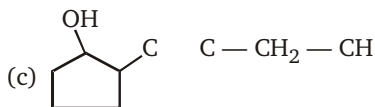


Level - 1

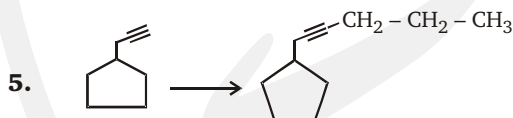
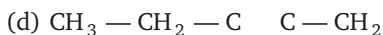
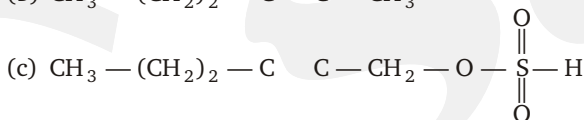
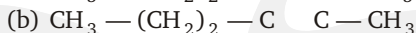
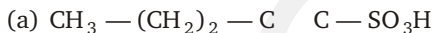


Product (K) of the above reaction is :

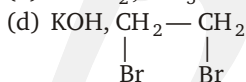
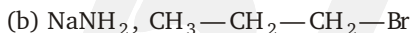
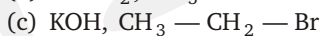




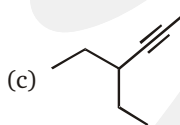
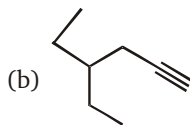
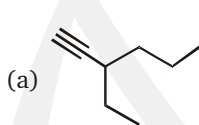
Give the structural formula of compound (B) :



; This conversion can be achieved by :

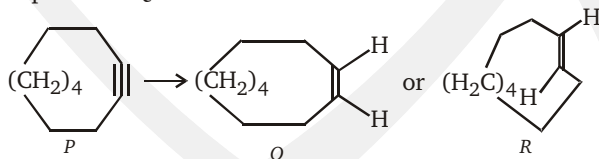


6. Which alkyne will give 3-ethylhexane on catalytic hydrogenation ?

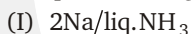


(d) All of these

7. Reactant P gives products Q or R.



The possible reagents are :



The correct statement with respect to the above conversion is/are :

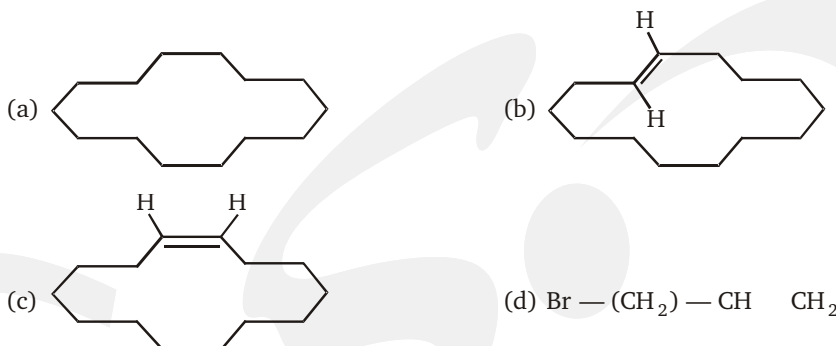
(a) Q is obtained on treatment with reagent (I)

(b) R and Q are obtained on treatment with reagent (II)

(c) R is obtained on treatment with reagent (I)

(d) R is obtained on treatment with reagent (II)

8. $\text{Br} - (\text{CH}_2)_{12} - \text{C} \equiv \text{CH} \xrightarrow{\text{NaNH}_2} \text{(A)} \xrightarrow{\text{Lindlar Catalyst}} \text{(B)}$; Product (B) is :



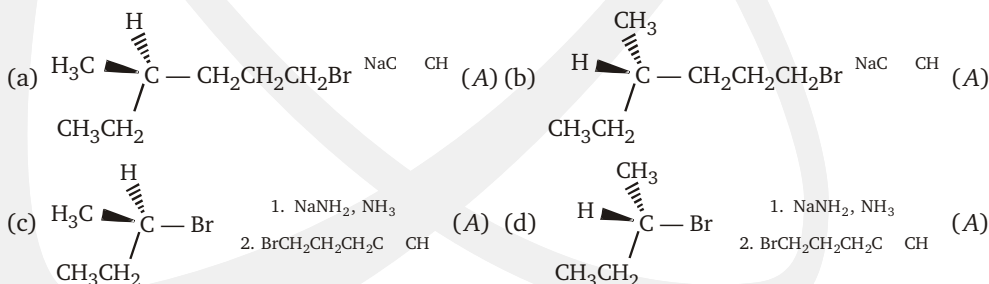
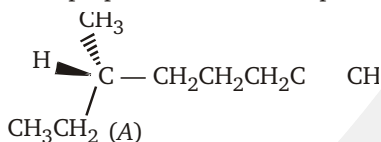
9. $\text{Ph} - \text{C} \equiv \text{CH} \xrightarrow[\text{MeOH}]{\text{MeO}^-}$ Major product of the reaction is :



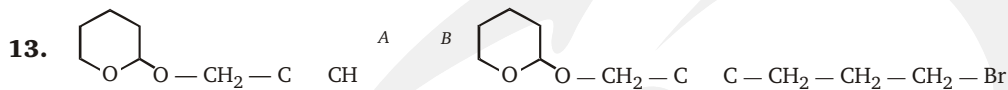
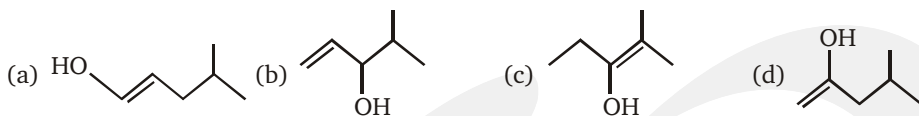
10. $\text{Ph} - \text{C}(\text{Cl})_2 - \text{CH}_3 \xrightarrow{3\text{NaNH}_2} \text{(A)}$; Product (A) is :



11. Which combination is best for preparation of the compound (A) shown below ?

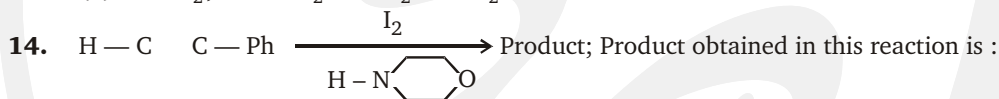


12. Which one of the following is the intermediate in the preparation of a ketone by hydration of an alkyne in the presence of sulfuric acid and mercury (II) sulphate ?

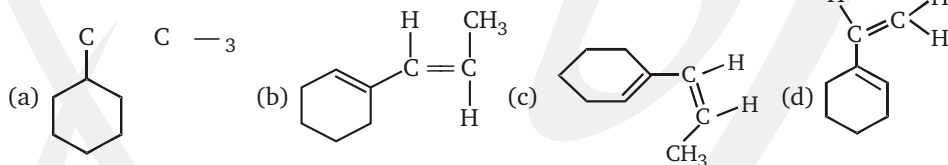
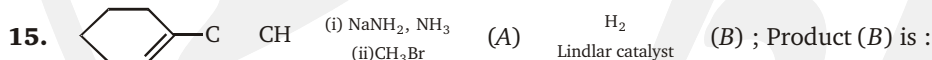


To carry out above conversion, (A) and (B) respectively, are :

- (a) NaNH_2 , $\text{Cl}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{Br}$
 (b) NaNH_2 , $\text{F}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{Br}$
 (c) NaNH_2 , $\text{I}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{Br}$
 (d) NaNH_2 , $\text{I}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{I}$

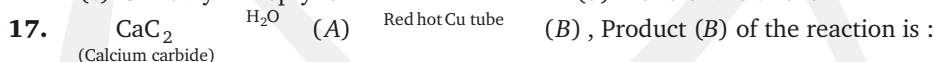


- (a) $\text{Ph}-\text{C}(\text{CH}_3)_2-\text{CH}_2-\text{I}$ (b) $\text{Ph}-\text{CH}(\text{I})-\text{CH}_2-\text{I}$
 (c) $\text{Ph}-\text{C}(\text{I})_2-\text{C}-\text{I}$ (d) $\text{I}-\text{C}(\text{I})-\text{C}-\text{H}$



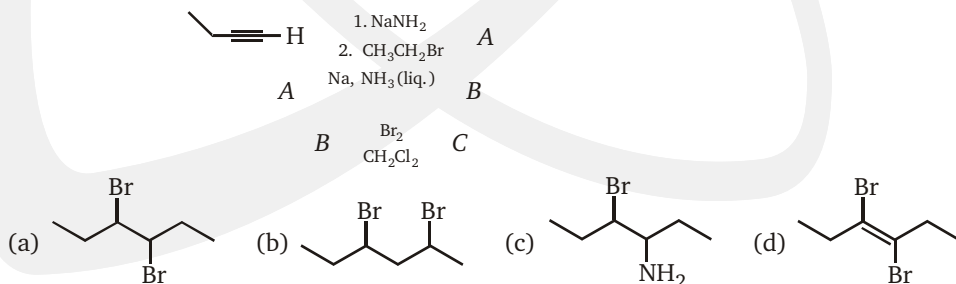
16. Which of the following alkyne on treatment with H_2 (2 mole)/ Pt gives an optically inactive compound ?

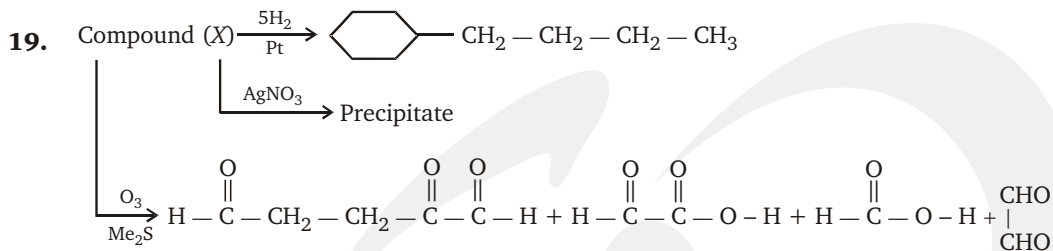
- (a) 3-Methyl-1-pentyne (b) 4-Methyl-1-hexyne
 (c) 3-Methyl-1-heptyne (d) None of the above



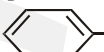
- (a) Toluene (b) Ethyl-benzene (c) Benzene (d) Butyne

18. What is the final product, C, of the following reaction sequence ?





Compound (X) will be :

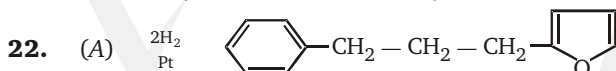
- (a)  $-\text{CH} = \text{CH} - \text{C} \begin{array}{c} \text{CH} \\ \text{CH} \end{array}$ (b)  $=\text{CH} - \text{CH}_2 - \text{C} \begin{array}{c} \text{CH} \\ \text{CH} \end{array}$
- (c)  $=\text{CH} - \text{CH}_2 - \text{C} \begin{array}{c} \text{CH} \\ \text{CH} \end{array}$ (d)  $=\text{C} = \text{CH} - \text{C} \begin{array}{c} \text{CH} \\ \text{CH} \end{array}$

20. Choose the sequence of steps that describes the best synthesis of 1-butene from ethanol :

- (a) (1) $\text{NaC} \begin{array}{c} \text{CH} \\ \text{CH} \end{array}$; (2) H_2 , Lindlar Pd
 (b) (1) $\text{NaC} \begin{array}{c} \text{CH} \\ \text{CH} \end{array}$; (2) Na , NH_3
 (c) (1) HBr , heat ; (2) $\text{NaC} \begin{array}{c} \text{CH} \\ \text{CH} \end{array}$; (3) H_2 , Lindlar Pd
 (d) (1) HBr , heat ; (2) $\text{KOC}(\text{CH}_3)_3$, DMSO ; (3) $\text{NaC} \begin{array}{c} \text{CH} \\ \text{CH} \end{array}$; (4) H_2 , Lindlar catalyst

21. Which alkyne yields butanoic acid ($\text{CH}_3\text{CH}_2\text{CH}_2\text{CO}_2\text{H}$) as the only organic product on treatment with ozone followed by the hydrolysis ?

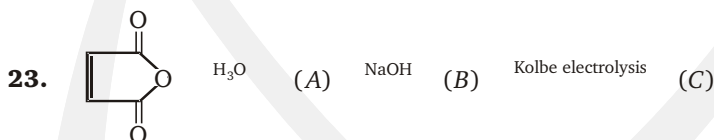
- (a) 1-Butyne (b) 4-Octyne (c) 1-Pentyne (d) 2-Hexyne



Carlina oxide

Unit of unsaturation in compound (A) ?

- (a) 7 (b) 8 (c) 9 (d) 10



Product (C) of above reaction is:

- (a) $\text{H}_2\text{C} \begin{array}{c} \text{CH}_2 \\ \text{CH} \end{array}$ (b) $\text{CH}_3 - \text{C} \begin{array}{c} \text{C} \\ \text{CH} \end{array} - \text{CH}_3$
 (c) $\text{HC} \begin{array}{c} \text{CH} \\ \text{CH} \end{array}$ (d) $\text{CH}_3 - \text{CH} \begin{array}{c} \text{CH} \\ \text{CH} \end{array} - \text{CH}_3$

24. To convert 1-butyne to 1-D-butanol, one would carry out the following steps :

- (I) Sodium amide, then D_2O
 (II) Disiamylborane, then hydrogen peroxide/sodium hydroxide
 (III) The transformation can not be carried out with the indicated reagents.
 (a) I, followed by II (b) II, followed by I (c) III (d) II

25. An unknown compound (A) has a molecular formula C_4H_6 . When (A) is treated with excess of Br_2 a new substance (B) with formula $C_4H_6Br_4$ is formed. (A) forms a white ppt. with ammonical silver nitrate solution. (A) may be :
- (a) But-1-yne (b) But-2-yne
(c) But-1-ene (d) But-2-ene
26. One mole of 1,2-dibromopropane on treatment with X moles of $NaNH_2$ followed by treatment with ethyl bromide gave a pentyne. The value of X is :
- (a) One (b) Two (c) Three (d) Four

27.
$$\begin{array}{c} CH_3 \\ | \\ CH_3 - CH - C \quad CH \end{array} \xrightarrow{\text{excess HBr}}$$
- The product of the above reaction is :

- (a)
$$\begin{array}{c} CH_3 & Br & Br \\ | & | & | \\ CH_3 - CH - CH - CH_2 \end{array}$$
- (b)
$$\begin{array}{c} CH_3 & Br \\ | & | \\ CH_3 - CH - C = CH_2 \end{array}$$
- (c)
$$\begin{array}{c} CH_3 & Br \\ | & | \\ CH_3 - CH - C - CH_3 \\ | \\ Br \end{array}$$
- (d)
$$\begin{array}{c} CH_3 & Br \\ | & | \\ CH_3 - CH - CH_2 - CH \\ | \\ Br \end{array}$$

28. $CH_3 - C \quad C - CH_3 \xrightarrow{\text{Cold } KMnO_4} (A)$
- Product (A) is:

- (a)
$$CH_3 - CH_2 - \overset{\overset{O}{||}}{C} - \overset{\overset{O}{||}}{C} - H$$
- (b)
$$CH_3 - \overset{\overset{O}{||}}{C} - \overset{\overset{O}{||}}{C} - CH_3$$
- (c)
$$\begin{array}{c} OH & OH \\ | & | \\ CH_3 - CH - CH - CH_3 \end{array}$$
- (d) $O \quad CH - CH_2 - CH_2CH \quad O$

29. In which reaction last product is $Ph - C \quad CH$?

- (a)
$$\begin{array}{c} Br \\ | \\ C_6H_5 - C - CH_3 \\ | \\ Br \end{array} \xrightarrow[\text{Mineral oil, heat}]{3NaNH_2} \xrightarrow{NH_4Cl}$$
- (b)
$$C_6H_5CH \quad CH_2 \xrightarrow[CCl_4]{Br_2} \xrightarrow[\text{Mineral oil, heat}]{3NaNH_2} \xrightarrow{NH_4Cl}$$
- (c)
$$C_6H_5 - \overset{\overset{O}{||}}{C} - CH_3 \xrightarrow{PCl_5} \xrightarrow[\text{Mineral oil, heat}]{3NaNH_2} \xrightarrow{NH_4Cl}$$
- (d) All

30. Predict the product of the following reaction sequence.

ethyne

(1) excess NaNH_2

(2) excess $\text{I}-\text{CH}_2-(\text{CH}_2)_2-\text{CH}_3$

(3) H^+

(a) 6-iodo-1-hexyne

(b) 1-hexyne

(c) 5-decyne

(d) 1-iodo-1-hexene

31. The best sequence of reactions to prepare 2-heptanone is

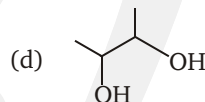
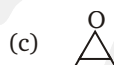
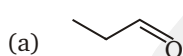
(a) propyne $\xrightarrow{\text{NaNH}_2}$ X $\xrightarrow{n\text{-C}_4\text{H}_9\text{Br}}$ Y $\xrightarrow[\text{H}_2\text{SO}_4]{\text{H}_2\text{O, Hg}^{2+}}$

(b) ethyne $\xrightarrow{\text{NaNH}_2}$ X $\xrightarrow{n\text{-C}_5\text{H}_{12}\text{Br}}$ Y $\xrightarrow[\text{H}_2\text{SO}_4]{\text{H}_2\text{O, Hg}^{2+}}$

(c) 1-hexyne $\xrightarrow{\text{NaNH}_2}$ X $\xrightarrow{\text{CH}_2\text{Br}}$ Y $\xrightarrow[\text{H}_2\text{SO}_4]{\text{H}_2\text{O, Hg}^{2+}}$

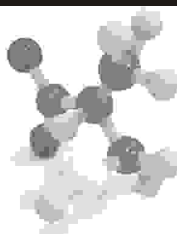
(d) 1-pentyne $\xrightarrow{\text{NaNH}_2}$ X $\xrightarrow{\text{C}_2\text{H}_5\text{Br}}$ Y $\xrightarrow[\text{H}_2\text{SO}_4]{\text{H}_2\text{O, Hg}^{2+}}$

32. The major product of the reaction of 2-butene with cold alkaline KMnO_4 , is

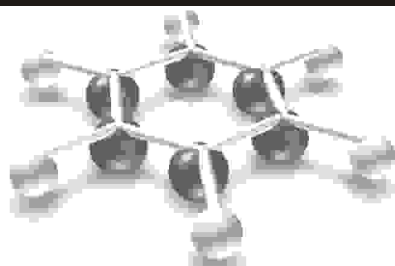


ANSWERS — LEVEL 1

1.	(b)	2.	(b)	3.	(b)	4.	(b)	5.	(b)	6.	(d)	7.	(c)	8.	(c)
9.	(b)	10.	(d)	11.	(b)	12.	(d)	13.	(c)	14.	(c)	15.	(c)	16.	(a)
17.	(c)	18.	(a)	19.	(a)	20.	(c)	21.	(b)	22.	(c)	23.	(c)	24.	(c)
25.	(a)	26.	(c)	27.	(c)	28.	(b)	29.	(d)	30.	(c)	31.	(b)	32.	(d)



5A



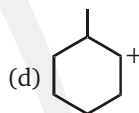
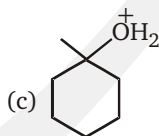
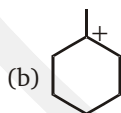
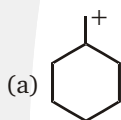
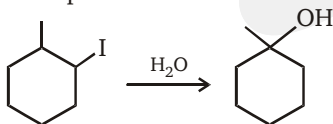
ALKYL HALIDES

Substitution Reactions (S_{N1} , S_{N2} , S_{Ni})

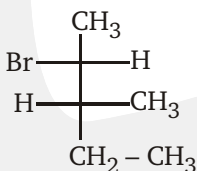


Level - 1

1. Which of the following is not expected to be intermediate of the following reaction ?



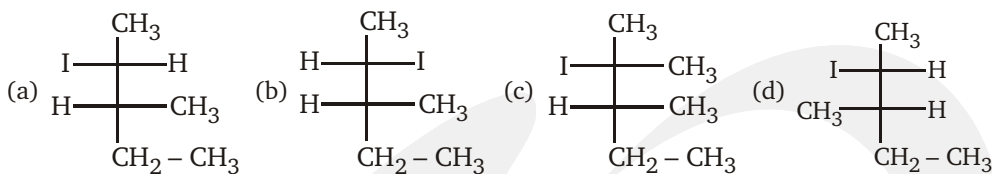
2.



NaI

Acetone

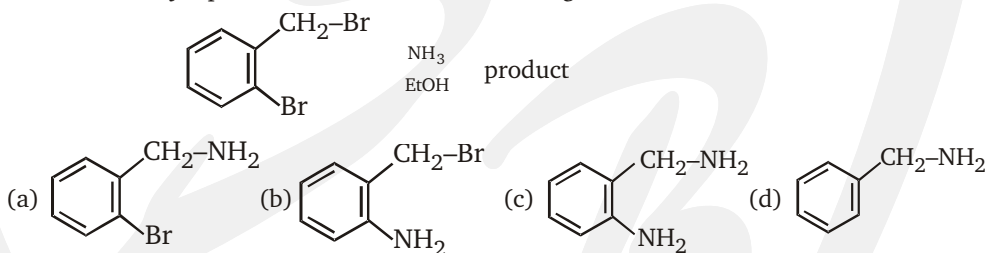
product; S_{N2} product of the reaction is :



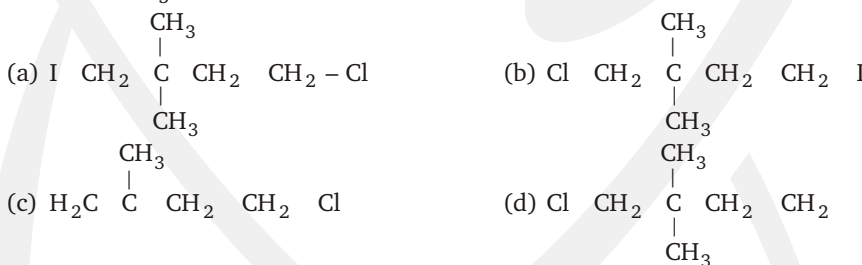
3. Rate of S_N2 will be negligible in :



4. What is the major product obtained in the following reaction ?



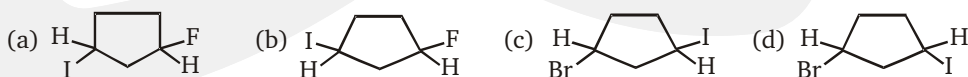
5. product; Major product of this reaction is:



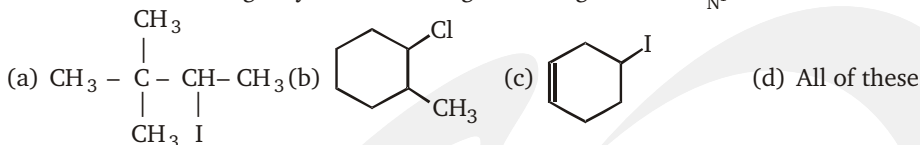
6. Which of the following expressions is representative of the rate law for a S_N2 reaction ?

- (a) Rate = k [electrophile]
 (b) Rate = k [electrophile] [nucleophile]
 (c) Rate = k [nucleophile]²
 (d) Rate = k [electrophile]²

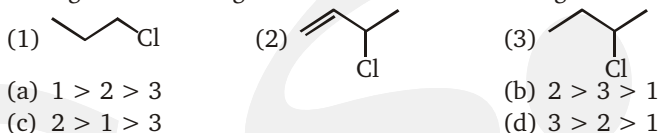
7. (A) ; Major product of this reaction is : (major)



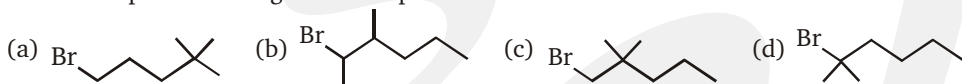
8. Which of the following alkyl halide undergo rearrangement in S_N1 reaction ?



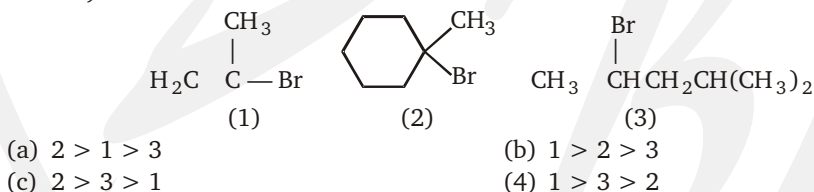
9. Arrange the following three chlorides in decreasing order towards S_N1 reactivity.



10. Which compound undergoes nucleophilic substitution with NaCN at the fastest rate ?



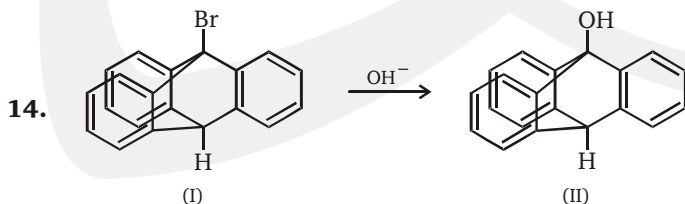
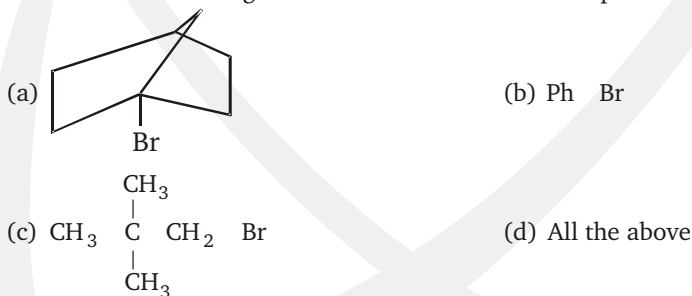
11. Rank the following in order of decreasing rate of solvolysis with aqueous ethanol (fastest to slowest)



12. The reaction of 4-bromobenzyl chloride with sodium cyanide in ethanol leads to the formation of :



13. Which of the following reactant will not favour nucleophilic substitution reaction ?



Conversion of I to II :

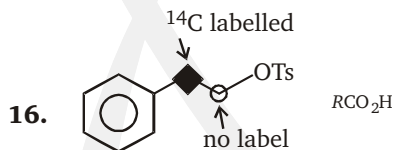
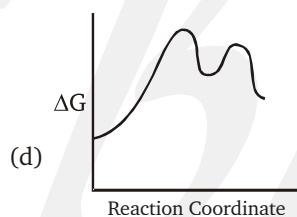
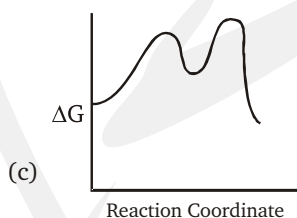
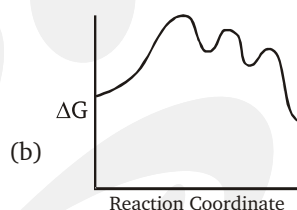
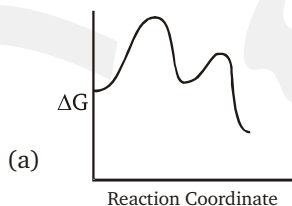
(a) takes place by S_N1

(b) takes place by S_N2

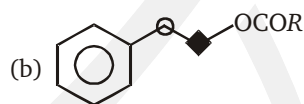
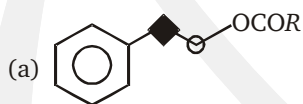
(c) takes place both by S_N1 and S_N2

(d) does not take place

15. Which is the correct reaction coordinate diagram for the following solvolysis reaction ?

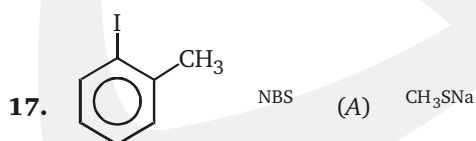


product; Product of this reaction is :

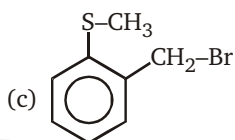
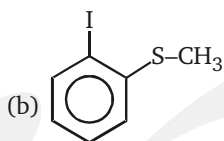
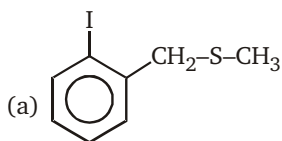


(c) both (a) and (b)

(d) None of these

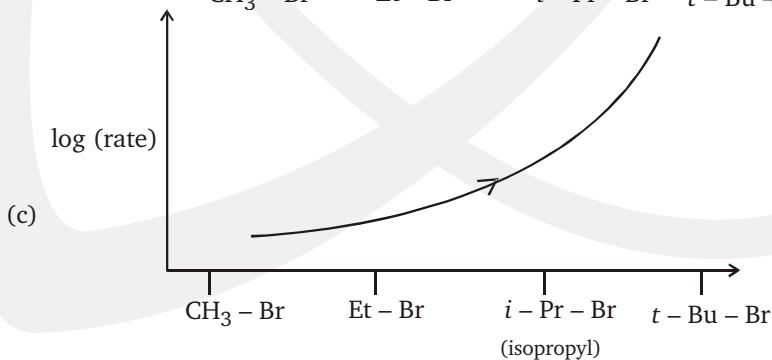
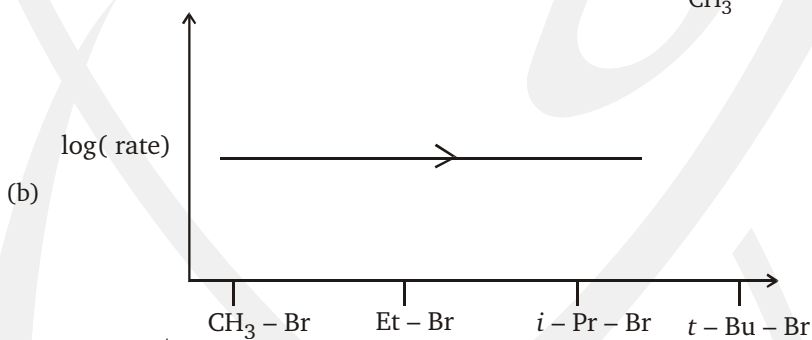
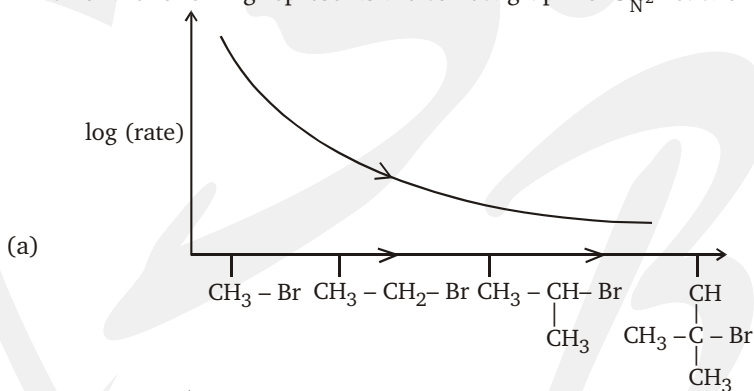


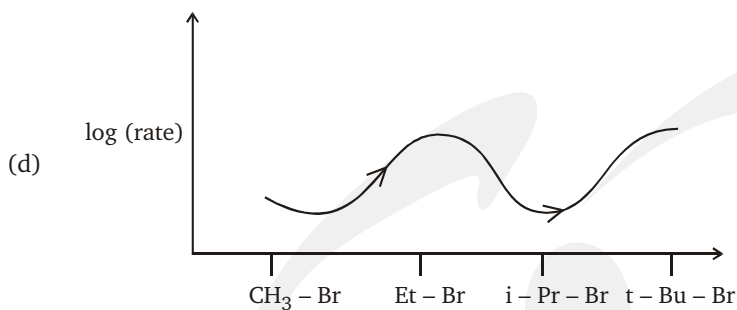
(B) , Product (B) is :



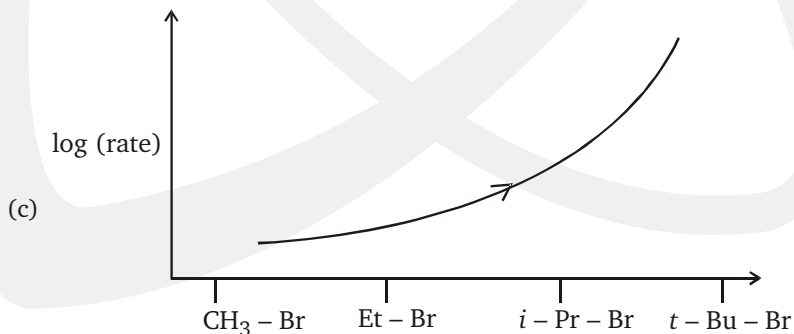
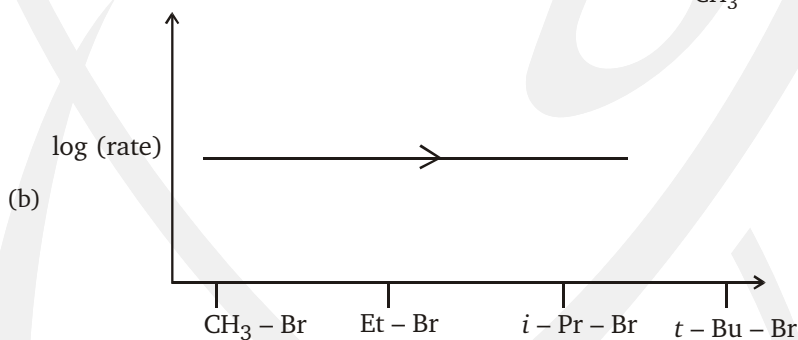
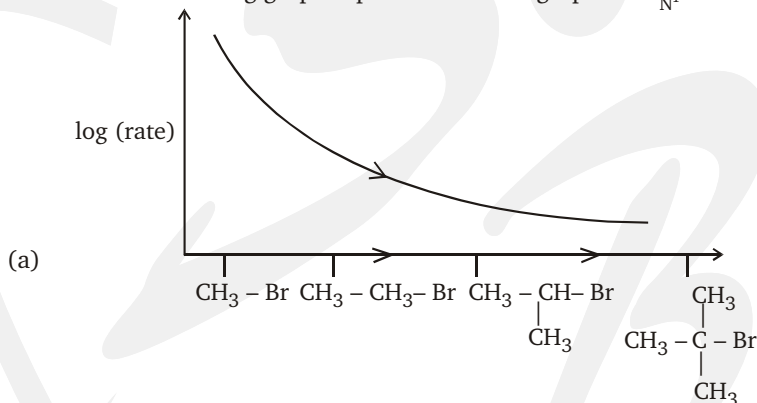
(d) None of these

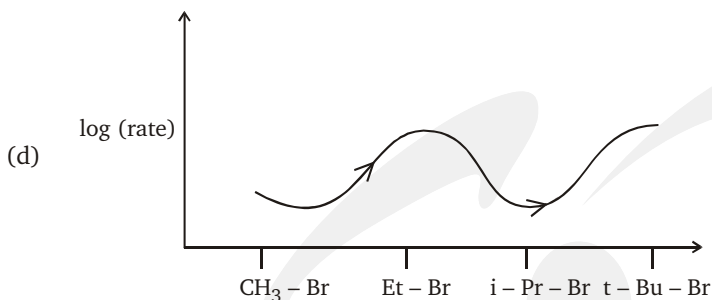
18. Which of the following represents the correct graph for S_N2 reaction ?



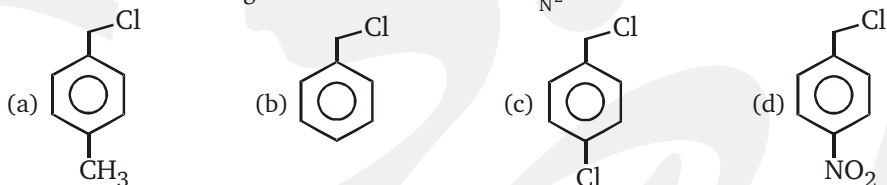


19. Which of the following graph represents correct graph for S_{N1} reaction :

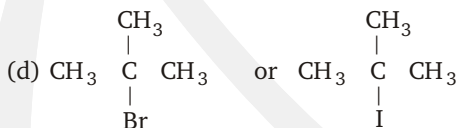
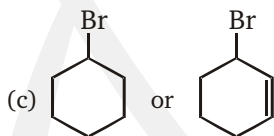
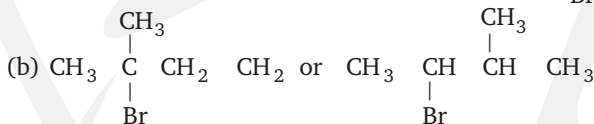
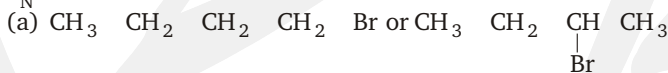




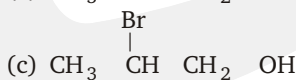
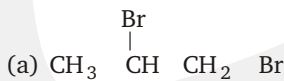
20. Which of the following is most reactive toward S_N2 reaction ?



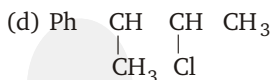
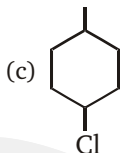
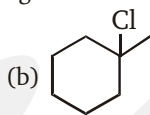
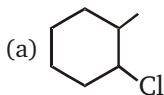
21. Among the given pairs, in which pair first compound reacts faster than second compound in S_N1 reaction ?



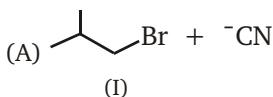
22. What is the major product of the following reaction ?



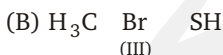
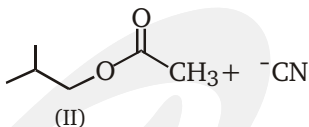
23. S_N1 and S_N2 products are same with (excluding stereoisomer) :



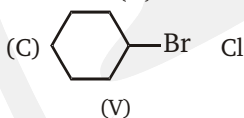
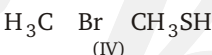
24. Consider the nucleophilic attacks given below. Select in each pair that shows the greater S_N2 reaction rate.



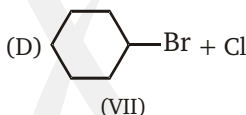
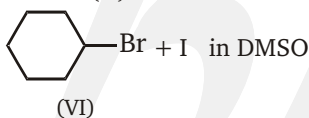
or



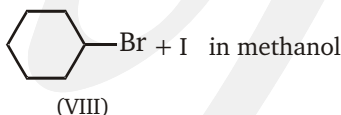
or



or



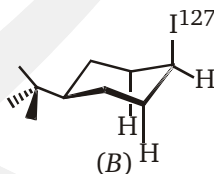
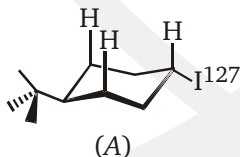
or



- A B C D
(a) (I) ; (IV) ; (VI) ; (VIII)
(c) (I) ; (III) ; (V) ; (VIII)

- A B C D
(b) (II) ; (III) ; (V) ; (VIII)
(d) (I) ; (III) ; (V) ; (VII)

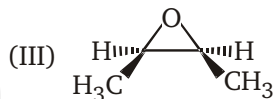
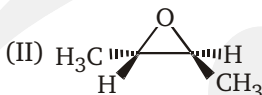
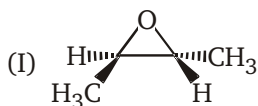
25. Which of the two stereoisomers of 4- t -butylcyclohexyl iodide (^{127}I) will undergo S_N2 substitution with ^{128}I faster, and why ?



- (a) A will react faster because it is the more stable of the two isomers
(b) A will react faster because it will yield a more stable product, and the transition state for both reactions is of the same energy
(c) A will react faster because the approach of ^{128}I can depart unhindered.

(d) *B* will react faster because it is less stable than *A*, and the transition state for both reactions is of the same energy

26. (*Z*)-2-Butene reacts with $\text{Br}_2/\text{H}_2\text{O}$. The resulting bromohydrin when treated with methoxide in methanol undergoes an intramolecular $\text{S}_{\text{N}}2$ reaction. Taking into consideration the stereochemical consequences of the reaction mechanism involved, choose the final product(s) of these transformations.



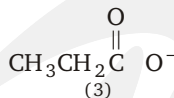
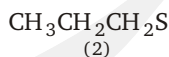
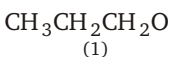
(a) (I) only

(b) (II) only

(c) (III) only

(d) Equal amounts of (I) and (II)

27. Rank the following species in order of decreasing nucleophilicity in a polar protic solvent (most least nucleophilic) :



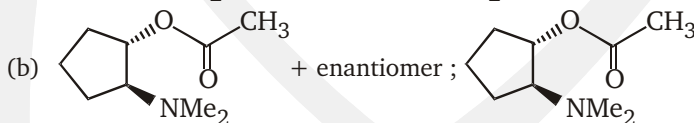
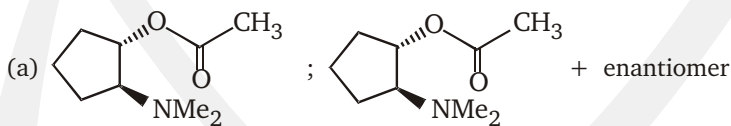
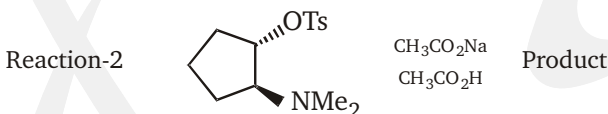
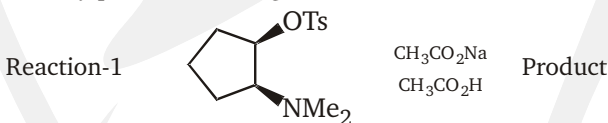
(a) 3 1 2

(b) 2 3 1

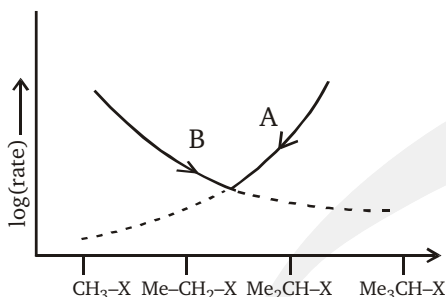
(c) 1 3 2

(d) 2 1 3

28. Identify products of the given reactions :



29.



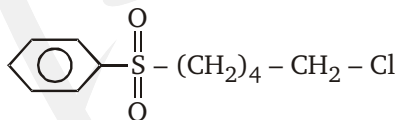
Which of the following is true about given graphs A and B ?

- (a) A S_N1 B S_N2 (b) A S_N2 , B S_N1
 (c) A & B E_1 (d) A & B E_2

30. In each of the following groups, which is the strongest (best) nucleophile ?

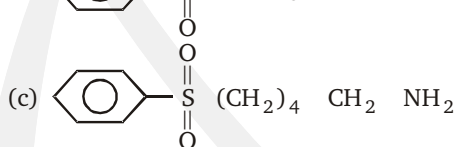
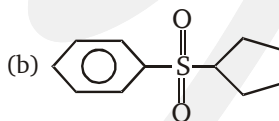
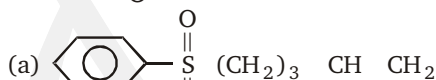
- (I) (1) H_3C-O^- (2) $CH_3CH_2O^-$ (3) H_3C-S^- in CH_3OH
 (II) (1) OH^- (2) H_2O (3) NH_2 in DMF
 (III) (1) $(CH_3)_2CHCH_2O^-$ (2) $(CH_3)_3CO^-$ (3) CH_3O^- in DMSO
 (a) I,3 ; II,3 ; III,2 (b) I,2 ; II,1 ; III,3
 (c) I,1 ; II,2 ; III,1 (d) I,3 ; II,1 ; III,3

31.



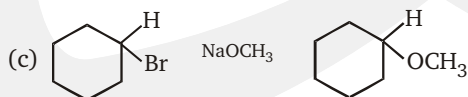
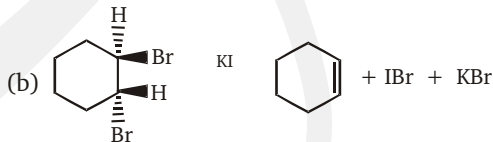
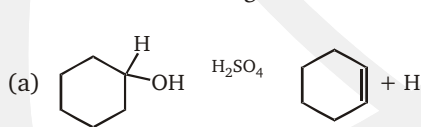
$NaNH_2$
dimethoxy ethane

(A) ; Product (A) is :

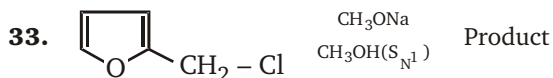


(d) None of these

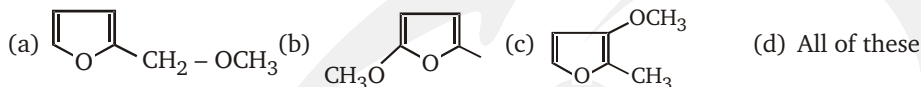
32. Which of the following reaction is an elimination reaction ?



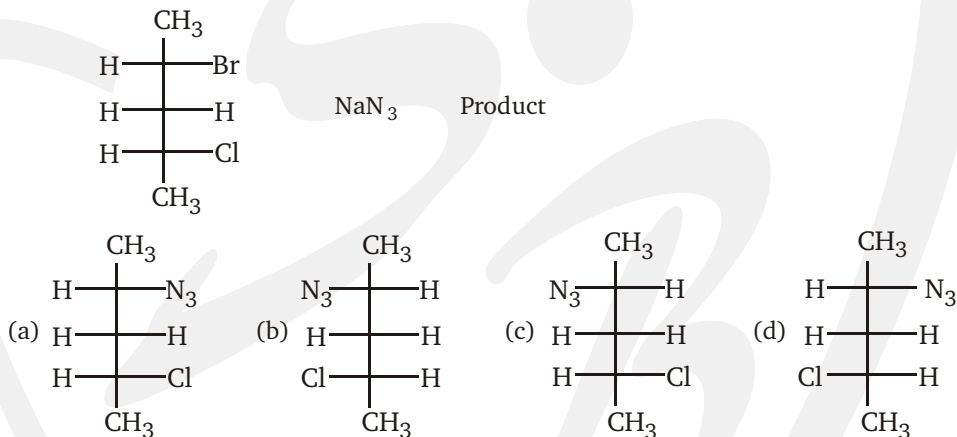
(d) both (a) and (b)



Which of the following products can be obtained from above reaction ?



34. What is the principal product of the following reaction ?



35. What would be the effect of increasing solvent polarity on the rate of each of the following reactions ? (Nu = neutral nucleophile)

- (A) Nu R L Nu R L
- (a) increases (b) decreases (c) constant (d) can not be predicted
- (B) R L R :L
- (a) increases (b) decreases (c) constant (d) cannot predict

36. Which of the following is most reactive toward $\text{S}_{\text{N}}2$ reaction ?



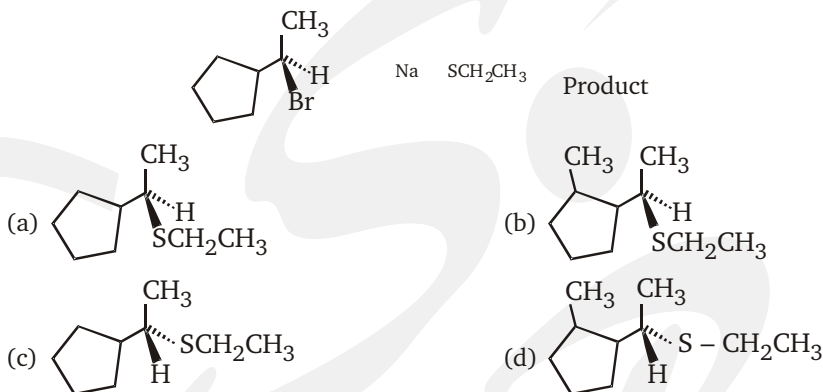
37. 4-chloro-1-butanol + NaOH (B)
Product (B) of the above reaction is :



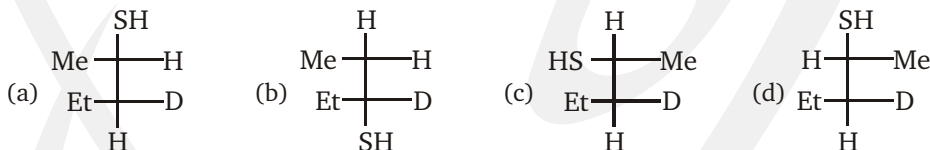
- 38.** In the given pairs of alkyl-halide, in which pair the first compound is more reactive than second compound toward S_N2 reaction ?
- (a) $(CH_3)_2CHBr$ or $CH_3CH_2CH_2Br$
 (b) $CH_3CH_2CH_2Br$ or $CH_3CH_2CH_2I$
 (c) $PhBr$ or $CH_3CH_2CH_2Br$
 (d) $CH_2=CHCH_2Cl$ or $H_2C=CHCl$
- 39.** In the given pair of reaction in which pair the second reaction is more reactive than first toward S_N2 reaction ?
- (a) CH_3CH_2Cl $CH_3CH_2O^-$ Et O Et (or)
 $CH_2=CHCl$ CH_3CH_2OH Et O Et
 (b) CH_3CH_2Cl EtO Et O Et (or)
 CH_3CH_2Cl EtS^- CH_3CH_2S Et
 (c) $\underset{(1m)}{Et}Cl$ $\underset{(2m)}{CH_3O}$ Et O CH_3 (or)
 $\underset{(2m)}{Et}Cl$ $\underset{(1m)}{CH_3O}$ Et O CH_3
 (d) Et Br Ph_3P Et PPh_3 (or)
 Et Br Ph_3N Et NPh_3
- 40.** Among the following pair of reactions in which pair the second reaction is more reactive than first in S_N1 reaction ?
- (a) Me_3CCl H_2O Me_3COH (or) Me_3CBr H_2O Me_3COH
 (b) Me_3CCl CH_3OH Me_3C-OCH_3 (or) $Me_3C-Cl + H_2O$
 $\downarrow$
 Me_3C-OH
 (c) $\underset{(1M)}{Me_3CCl}$ H_2O (or) $\underset{(2M)}{Me_3CCl}$ H_2O
 (d) All of these
- 41.** Which is a true statement concerning the transition state of an S_N2 reaction ?
- (a) Closely resembles a carbocation intermediate
 (b) The electrophile is responsible for the reaction
 (c) Lower is energy than the starting materials
 (d) Involves both the nucleophile and electrophile
- 42.** Increasing the concentration of a nucleophile in a typical S_N2 reaction by a factor of 10 will cause the reaction rate to :
- (a) increase by a factor of 10 (b) increase by a factor of 10^2
 (c) decrease by a factor of 10 (d) remain about the same
- 43.** Decreasing the concentration of an electrophile in a typical S_N2 reaction by a factor of 3 will cause the reaction ratio to :
- (a) increase by a factor of 3 (b) increase by a factor of 3^2
 (c) decrease by a factor of 3 (d) remain about the same

44. Increasing the concentration of an electrophile in a typical S_N2 reaction by a factor of 3 and the concentration of the nucleophile by a factor of 3 will change the reaction rate to :
- (a) increase by a factor of 6 (b) increase by a factor of 9
(c) decrease by a factor of 3 (d) remain about the same

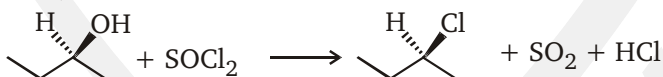
45. Consider the following reaction and select the best choice that represents the reaction.



46. $\xrightarrow{\text{KSH}}$ Product; Identify the product.



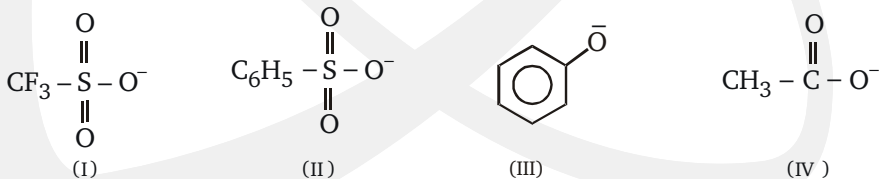
47. The reaction ,



proceeds by the..... mechanism.

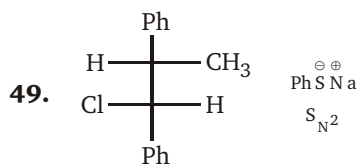
- (a) S_Ni (b) S_N2 (c) S_E2 (d) S_N1

48. Consider the following anions.

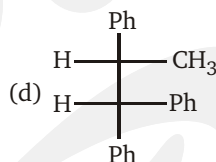
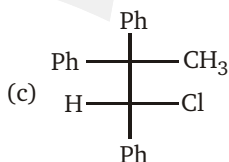
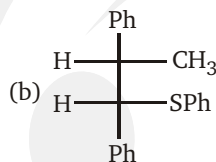
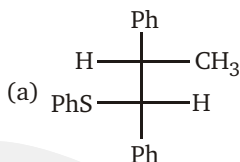


When attached to sp^3 -hybridized carbon, their leaving group ability in nucleophilic substitution reaction decreases in the order :

- (a) $I > II > III > IV$ (b) $I > II > IV > III$ (c) $IV > I > II > III$ (d) $IV > III > II > I$



Principal organic product of the reaction will be :



50. Reaction of *R*-2-butanol with *p*-toluenesulphonyl chloride in pyridine followed by reaction with LiBr gives:

- (a) *R*-2-butyl bromide
(c) *R*-2-butyl tosylate

- (b) *S*-2-butyl tosylate
(d) *S*-2-butyl bromide

51. The compound which undergoes $\text{S}_{\text{N}}1$ reaction most rapidly is :



52. Addition of KI accelerates the hydrolysis of primary alkyl halides because :

- (a) KI is soluble in organic solvents
(b) the iodide ion is a weak base and a poor leaving group
(c) the iodide ion is a strong base
(d) the iodide ion is a powerful nucleophile as well as a good leaving group

53. Which of the following phrases are not correctly associated with $\text{S}_{\text{N}}1$ reaction ?

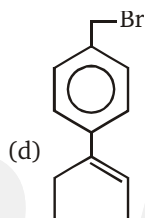
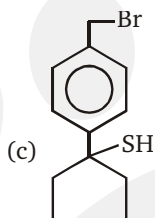
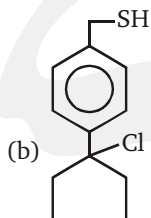
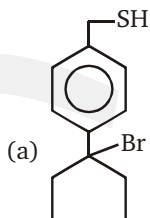
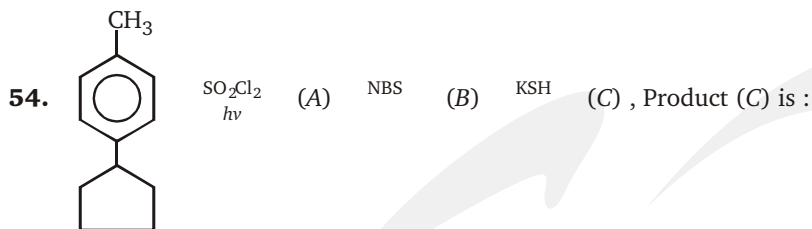
- (1) Rearrangement is possible
(2) Rate is affected by polarity of solvent
(3) The strength of the nucleophile is important in determining rate
(4) The reactivity series is tertiary > secondary > primary
(5) Proceeds with complete inversion of configuration

(a) 3, 5

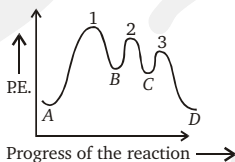
(b) 5 only

(c) 2, 3, 5

(d) 3 only

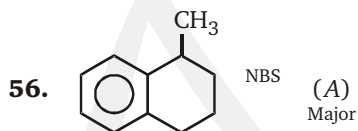


55. Energy profile diagram for an exothermic reaction, $A \xrightarrow{1} B \xrightarrow{2} C \xrightarrow{3} D$, is given below.

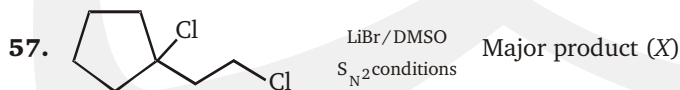
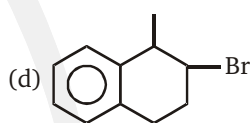
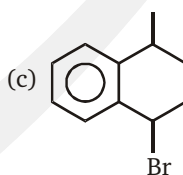
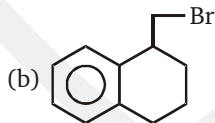
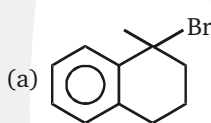


The rate determining step of the reaction is :

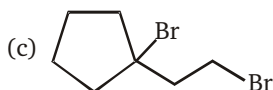
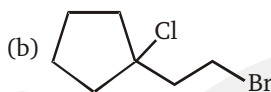
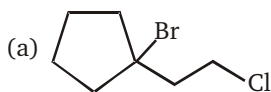
- (a) A $\rightarrow$ B (b) B $\rightarrow$ C (c) C $\rightarrow$ D (d) can not predict



Major product (A) is :



The product X is :



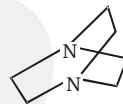
58. Relative rate of reaction of the following amine with methyl iodide is:



(A)



(B)



(C)

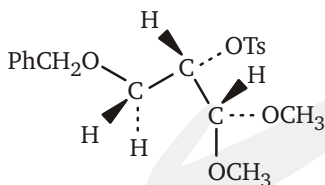
(a) $A > B > C$

(b) $A > C > B$

(c) $B > C > A$

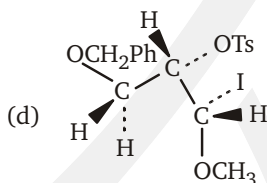
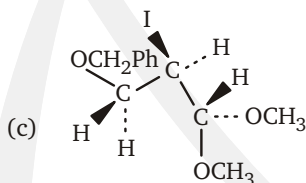
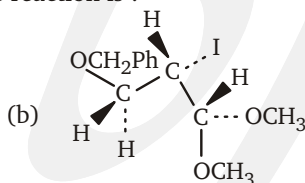
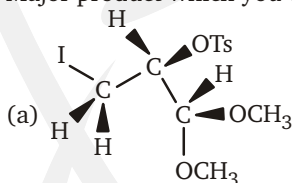
(d) $B > A > C$

59.

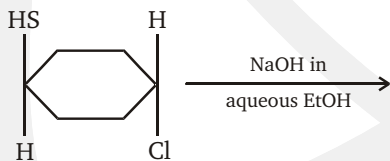


NaI
Dimethyl sulfoxide

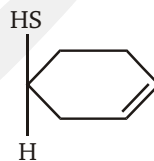
Major product which you expect in the above reaction is :



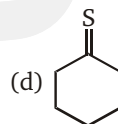
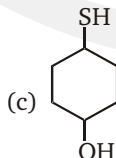
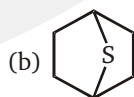
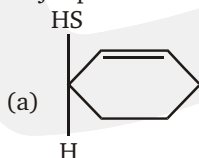
60.

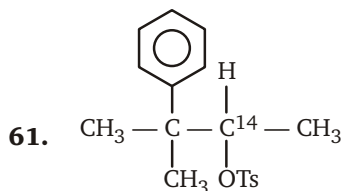


(A)
(major)
 $C_6H_{10}S$

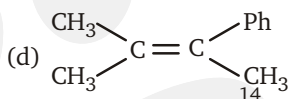
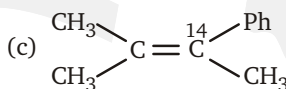
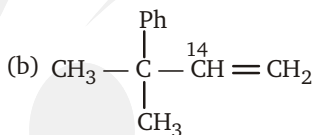
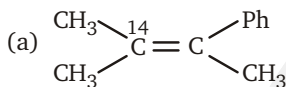


Major product of the above reaction is :

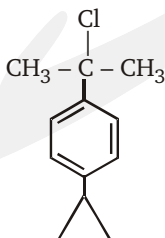




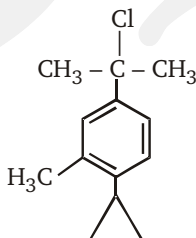
Major product of the reaction is:



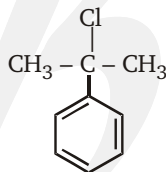
62. The decreasing order of reactivity of the compounds given below towards solvolysis under identical conditions is :



(I)



(II)



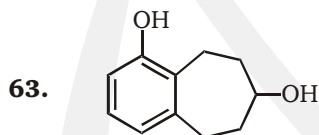
(III)

(a) II > III > I

(b) I > II > III

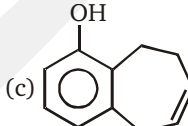
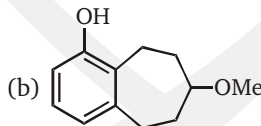
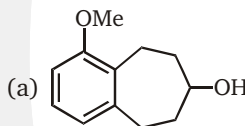
(c) III > II > I

(d) II > I > III



1. One Eq. NaOH
2. MeBr

(A); Product (A) is :



(d) None of these

64. (R)-2-octyl tosylate is solvolyzed in water under ideal S_N1 conditions. The product(s) will be:

(a) R-2-octanol and S-2-octanol in a 1 : 1 ratio

(b) R-2-octanol and S-2-octanol in a 1.5 : 1 ratio

(c) R-2-octanol only

(d) S-2-octanol only

65. From each of the following pairs select the compound that will react faster with sodium iodide in acetone :

Pair-A: (1) 2 - Chloropropane

(2) 2 - Bromopropane

Pair-B: (3) 1 - Bromobutane

(4) 2 - Bromobutane

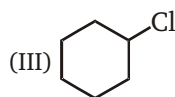
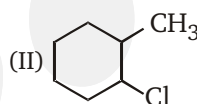
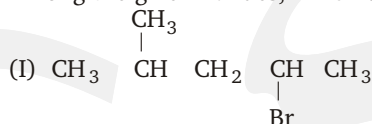
(a) 1,3

(b) 1,4

(c) 2,3

(d) 2,4

66. Among the given halides, which one will give same product in both S_N1 and S_N2 reactions.



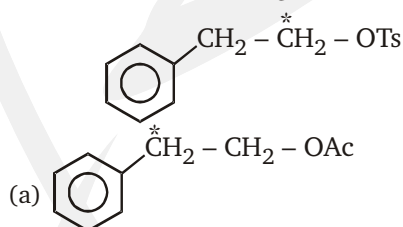
(a) (III) only

(b) (I) & (II)

(c) (III) & (IV)

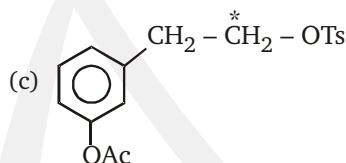
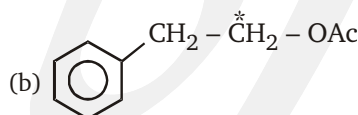
(d) (I), (III) & (IV)

67. Product(s) formed during this reaction is/are :



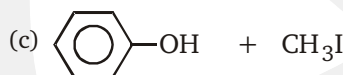
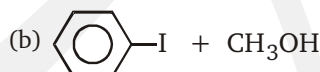
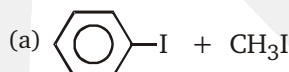
AcONa
AcOH

Product ? [C^* isotopic carbon]

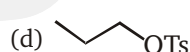
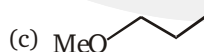
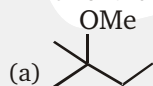


(d) Both (a) & (b)

68. Anisole $\xrightarrow[\text{reflux}]{\text{excess HI (conc.)}}$ Product



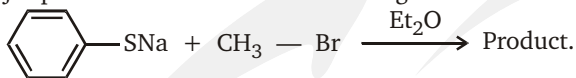
69. Which of the following compounds would react faster with NaCN in an S_N2 reaction ?

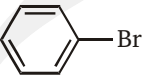
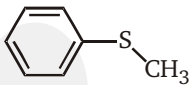


70. $\text{HC} \quad \text{CNa} \quad \text{Cl} - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{I}$ (A); Major product (A) is :

- (a) $\text{H} - \text{C} \quad \text{C} - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{I}$ (b) $\text{CH}_2 \quad \text{CH} - \text{CH}_2 - \text{I}$
 (c) $\text{H} - \text{C} \quad \text{C} - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \text{Cl}$ (d) $\text{CH}_2 \quad \text{CH} - \text{CH}_2 - \text{Cl}$

71. What is the major product obtained in the following reaction ?

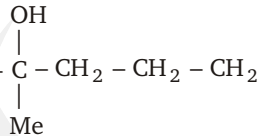
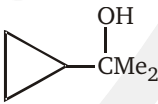


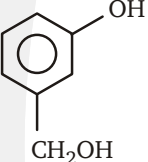
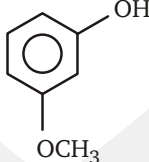
- (a)  (b)  (c)  (d) 

72.  $\text{OH} \quad \text{S}_{\text{N}}2$ A; The product A is :

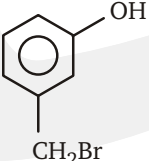
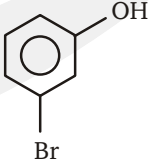
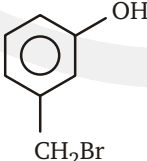
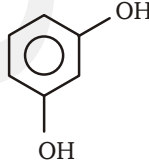
- (a)  (b) 
 (c) Both (a) and (b) are correct (d) None is correct

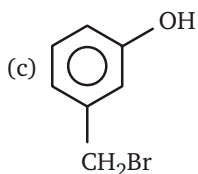
73. $\text{Me}_2\text{C} \quad \text{CH} - \text{CH}_2 - \text{CH}_2 - \text{Cl} \xrightarrow[\text{CaCO}_3]{\text{H}_2\text{O}}$ (X); Major product of the reaction is :

- (a)  (b) $\text{Me}_2\text{C} \quad \text{CH} - \text{CH}_2 - \text{CH}_2 - \text{OH}$
 (c) $\text{Me}_2\text{C} \quad \text{CH} - \underset{\text{OH}}{\text{CH}} - \text{CH}_2 - \text{OH}$ (d) 

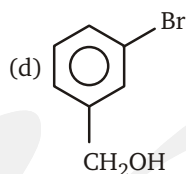
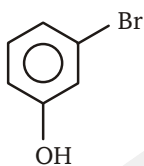
74.  HBr (A) ,  HBr (B)

Product (A) and (B) respectively are :

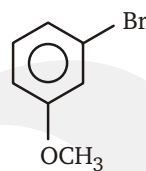
- (a)  and  (b)  and 



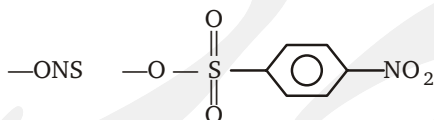
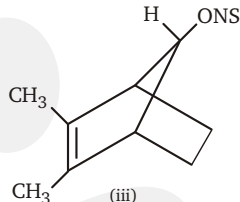
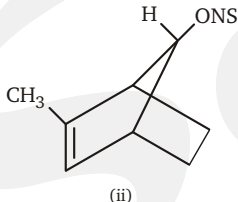
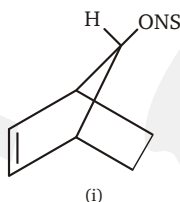
and



and



75. Relative rate of reaction with H_2O .



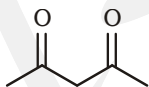
(a) (i) > (ii) > (iii)

(c) (iii) > (ii) > (i)

(b) (ii) > (i) > (iii)

(d) (iii) > (i) > (ii)

76.

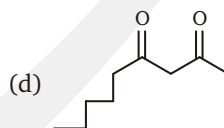
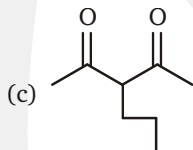
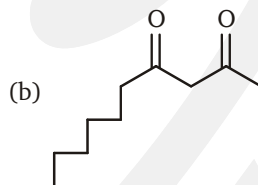
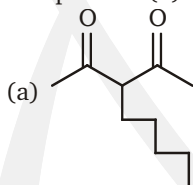


2 eq. KNH_2
 $\text{NH}_3(l)$

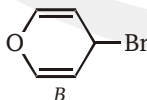
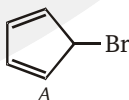
$n\text{—C}_4\text{H}_9\text{—Br}$
then H_3O^+

(P) ;

End product (P) of the above reaction is :

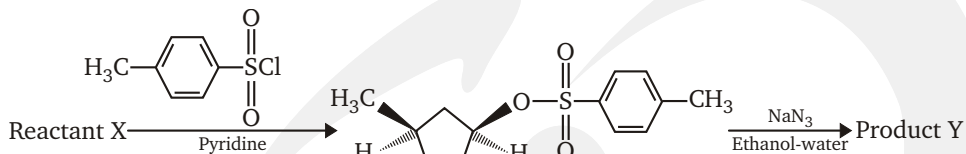


77. Which of the following statements is correct regarding the rate of hydrolysis of the compounds (A) and (B) by $\text{S}_{\text{N}}1$ reaction ?



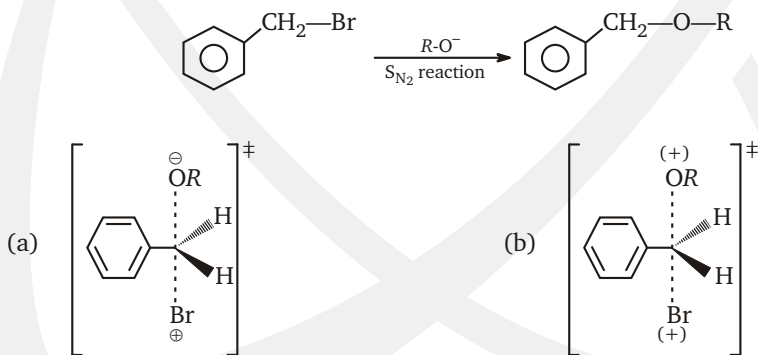
- (a) A reacts faster than B (b) B reacts faster than A
 (c) Both A and B reacts at the same rate (d) Neither A nor B reacts

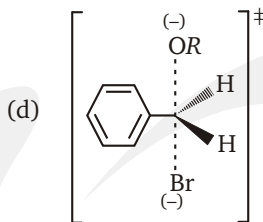
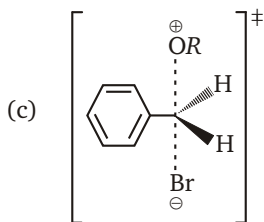
78. What are reactant X and product Y in the following sequence of reactions ?



- Reactant X Product Y
 (a)
 (b)
 (c)
 (d)

79. Transition state of given S_N2 is :

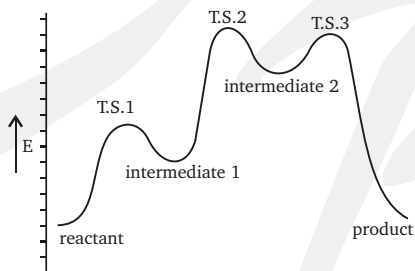




80. $\text{C}_6\text{H}_{13}\text{Br} \xrightarrow{\text{OH}^-} \text{C}_6\text{H}_{13}\text{OH} + \text{Br}^-$ is an example of:

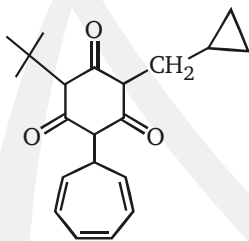
- (a) Nucleophilic addition (b) Nucleophilic substitution
(c) Electrophilic addition (d) Electrophilic substitution
(e) Free radical substitution

81. Transition state 2 is structurally most likely as :



- (a) intermediate 1 (b) transition state 3
(c) intermediate 2 (d) product

82.



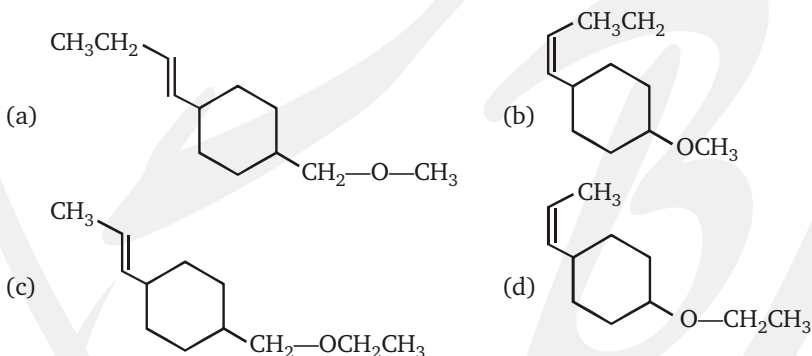
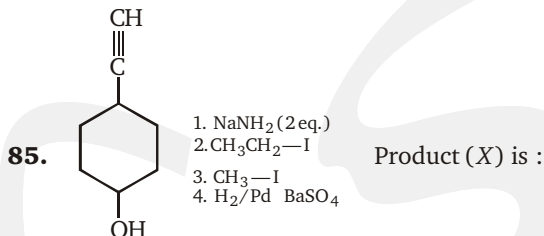
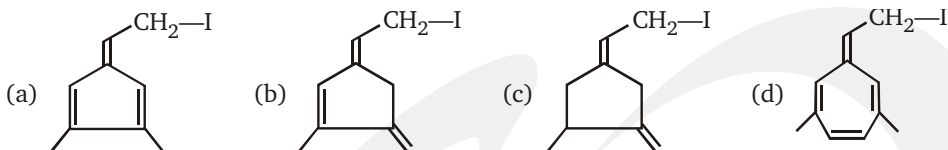
x Number of aromatic compound obtained when above compound undergo complete acidic hydrolysis.

- (a) 1 (b) 2 (c) 3 (d) 4

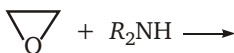
83. $\text{S}_{\text{N}}1$ and $\text{S}_{\text{N}}2$ reactions are

- (a) Both stereospecific
(b) Both stereoselective
(c) Stereoselective and stereospecific respectively
(d) Stereospecific and stereoselective respectively

84. Most reactive compound toward S_N1 is :

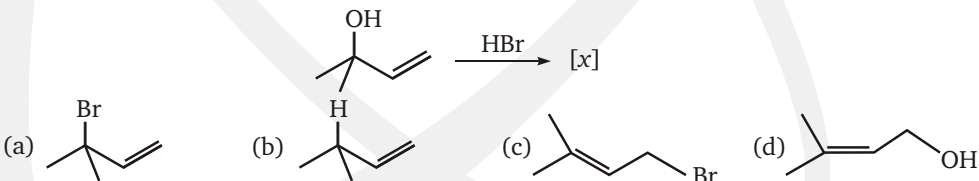


86. Following reaction is an example of :

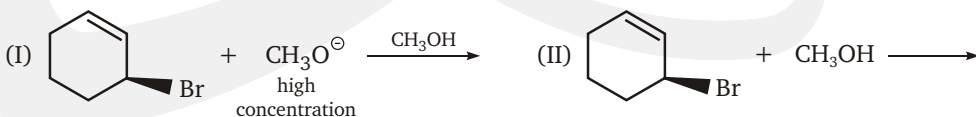


- (a) S_N2 Reaction (b) S_N1 Reaction
 (c) Electrophilic Addition (d) S_N - NGP

87. The major product of the following reaction :

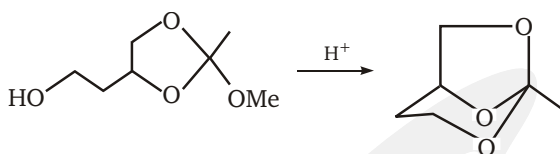


88. Choose the suitable option for the correct mechanism for the following reactions.

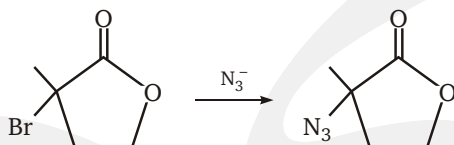


(a) S_N1, S_N1 (b) S_N1, S_N2 (c) S_N2, S_N1 (d) S_N2, S_N2

89.



Reaction 1

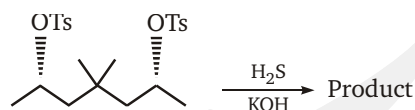


Reaction 2

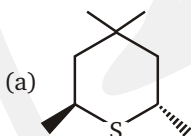
Types of mechanism followed by reaction 1 and reaction 2 respectively.

(a) S_N1, S_N1 (b) S_N2, S_N2 (c) S_N1, S_N2 (d) S_N2, S_N1

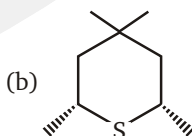
90.



Product of above reaction will be :



(a)



(b)



(c)



(d)

ANSWERS — LEVEL 1

1.	(a)	2.	(b)	3.	(c)	4.	(a)	5.	(b)	6.	(b)	7.	(b)	8.	(d)
9.	(b)	10.	(a)	11.	(c)	12.	(a)	13.	(d)	14.	(d)	15.	(b)	16.	(c)
17.	(a)	18.	(a)	19.	(c)	20.	(d)	21.	(b)	22.	(a)	23.	(c)	24.	(c)
25.	(d)	26.	(d)	27.	(d)	28.	(a)	29.	(a)	30.	(d)	31.	(b)	32.	(d)
33.	(d)	34.	(c)	35.	A(a)	35.	B(b)	36.	(d)	37.	(b)	38.	(d)	39.	(b)
40.	(d)	41.	(d)	42.	(a)	43.	(c)	44.	(b)	45.	(c)	46.	(d)	47.	(a)
48.	(b)	49.	(b)	50.	(d)	51.	(b)	52.	(d)	53.	(a)	54.	(b)	55.	(a)
56.	(a)	57.	(b)	58.	(c)	59.	(c)	60.	(b)	61.	(c)	62.	(d)	63.	(a)
64.	(b)	65.	(c)	66.	(d)	67.	(d)	68.	(c)	69.	(d)	70.	(c)	71.	(c)
72.	(b)	73.	(d)	74.	(b)	75.	(c)	76.	(d)	77.	(b)	78.	(b)	79.	(d)
80.	(b)	81.	(c)	82.	(b)	83.	(b,c)	84.	(d)	85.	(b)	86.	(a)	87.	(c)
88.	(c)	89.	(c)	90.	(b)										

Level - 2

1. **Statement-1** : Nucleophilicity order in polar-protic solvent is $I < Br < Cl < F$

Statement-2 : Due to bigger size of I it is less solvated in polar-protic solvent.

- (a) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (b) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (c) Statement-1 is true, statement-2 is false.
 (d) Statement-1 is false, statement-2 is true.

2. **Statement - 1** : CH_3CH_2Cl NaI ^{Acetone} CH_3CH_2I $NaCl$

Statement- 2 : Acetone is polar-protic solvent and solubility order of sodium halides decreases dramatically in order $NaI > NaBr > NaCl$. The last being virtually insoluble in this solvent and a 1° and 2° chloro alkane in acetone is completely driven to the side of Iodoalkane by the precipitation reaction.

- (a) Statement-1 is true, Statement-2 is true and Statement-2 is correct explanation for statement-1.
 (b) Statement-1 is true, Statement-2 is true and Statement-2 is Not the correct explanation for statement-1.
 (c) Statement-1 is true, Statement-2 is false.
 (d) Statement-1 is false, Statement-2 is true.

3. Encircle whichever of the following :

- (a) is the stronger nucleophile (aprotic solvent) : F^- or I^-
 (b) is the stronger nucleophile (protic solvent) : F^- or I^-
 (c) is the stronger base : F^- or I^-
 (d) is the stronger nucleophile (protic solvent) : NH_3 or NH_2NH_2
 (e) is the better leaving group : CH_3COO^- or $CH_3SO_3^-$

4. Encircle whichever of the following :

- (a) undergoes an S_N2 reaction more rapidly, CH_3CH_2Br or $CH_3CH_2CH_2Br$
- (b) undergoes an S_N1 reaction more rapidly, CH_3CH_2Br or $CH_3CH_2CH_2Br$
- (c) undergoes an E_2 reaction to give (Z)-1,2-diphenylpropene :
- $$\begin{array}{c} \text{Ph} \\ | \\ \text{H} - \text{C} - \text{Br} \\ | \\ \text{H} - \text{C} - \text{CH}_3 \\ | \\ \text{Ph} \end{array}$$

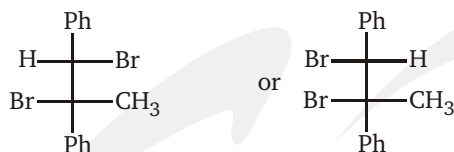
or

$$\begin{array}{c} \text{Ph} \\ | \\ \text{Br} - \text{C} - \text{H} \\ | \\ \text{H} - \text{C} - \text{CH}_3 \\ | \\ \text{Ph} \end{array}$$
- $$\begin{array}{c} \text{Br} \\ | \\ \text{CH} \\ | \\ \text{CH} \\ | \\ \text{Ph} \end{array} \quad \text{CH}_3$$

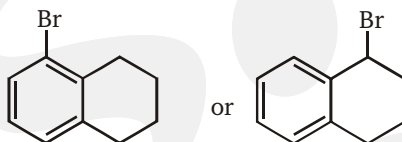
$$\begin{array}{c} \text{Br} \\ | \\ \text{CH} \\ | \\ \text{CH} \\ | \\ \text{Ph} \end{array} \quad \text{CH}_3$$

$$\begin{array}{c} \text{Ph} \quad \text{Ph} \\ \diagdown \quad \diagup \\ \text{C} = \text{C} \\ \diagup \quad \diagdown \\ \text{H} \quad \text{CH}_3 \end{array}$$

(d) reacts with NaI to give (Z)-1,2-diphenylpropene :

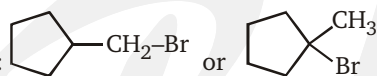


(e) undergoes an $\text{S}_{\text{N}}1$ reaction more rapidly,

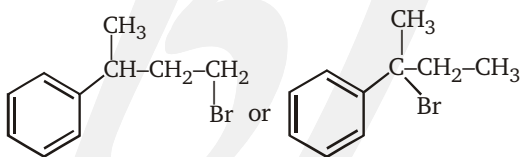


5. Encircle whichever of the following :

(a) undergoes an $\text{S}_{\text{N}}2$ reaction more rapidly :



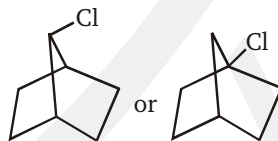
(b) undergoes an E_1 reaction more rapidly :



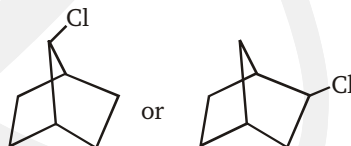
(c) undergoes an $\text{S}_{\text{N}}1$ reaction more rapidly:



(d) undergoes an $\text{S}_{\text{N}}2$ reaction more rapidly:



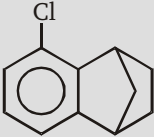
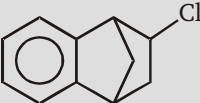
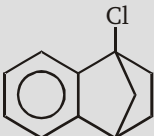
(e) undergoes an E_2 reaction more rapidly :



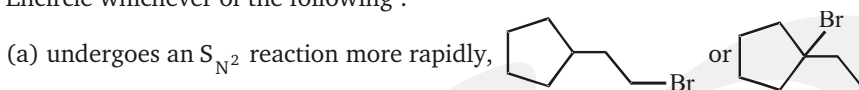
6. Match the column :

Alkyl halide			Relative rate (S_{N1})		Relative rate (S_{N2})
(a)	$\text{CH}_3 - \text{Br}$	(p)	1	(w)	1200
(b)	$\text{CH}_3 - \text{CH}_2 - \text{Br}$	(q)	1.05	(x)	40
(c)	$\begin{array}{c} \text{CH}_3 \quad \text{CH} \quad \text{Br} \\ \\ \text{CH}_3 \end{array}$	(r)	11	(y)	16
(d)	$\begin{array}{c} \text{CH}_3 \\ \\ \text{CH}_3 - \text{C} - \text{Br} \\ \\ \text{CH}_3 \end{array}$	(s)	1,200,000	(z)	1

7. Matrix :

Column (I)		Column (II)	
Compound		Type of reaction	
(a)		(p)	S_{N1} reaction can take place
(b)		(q)	S_{N2} reaction can take place
(c)		(r)	S_{N1} is not possible
(d)		(s)	S_{N2} is not possible

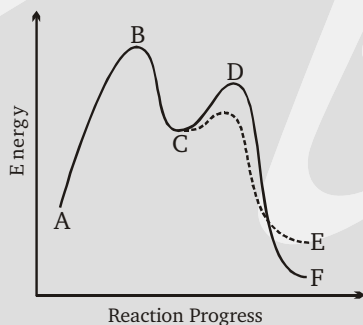
8. Encircle whichever of the following :



9. Reactivity : Circle the reaction that reacts FASTER by S_N2 in each pair :



10. Consider the potential energy diagram given below



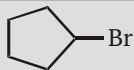
(X) Name the positions A-D

(Y) Answer the following questions .

- (i) Both reaction pathways are : EXOTHERMIC or ENDOTHERMIC
 (ii) Which step is the rate determining step (RDS) ? B or D
 (iii) Which product is most stable ? E or F
 (iv) In accordance with Hammonds postulate, exothermic reactions tend to have
 (a) early transition states that are reactant - like
 (b) late transition states that are reactant-like
 (c) early transition states that are product-like
 (d) late transition states that are product-like.

11. Select whether the following combinations of reactants will react by substitution (S_{N1} or S_{N2} mechanism), elimination (E_1 or E_2 mechanism)

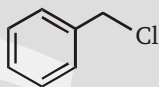
A.



NaI in acetone
25 C

(a) S_{N1} (b) S_{N2} (c) E_1 (d) E_2

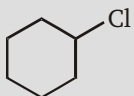
B.



NaOCH_3 in methanol
50 C

(a) S_{N1} (b) S_{N2} (c) E_1 (d) E_2

C.



NaOCH_3 in methanol
25 C

(a) S_{N1} (b) S_{N2} (c) E_1 (d) E_2 D. $(\text{CH}_3)_3\text{C}-\text{OH}$

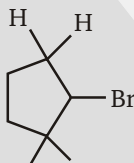
HBr 48% in H_2O
25 C

(a) S_{N1} (b) S_{N2} (c) E_1 (d) E_2 E. $(\text{CH}_3)_2\text{CH}-\text{Br}$

NaCN in ethanol
25 C

(a) S_{N1} (b) S_{N2} (c) E_1 (d) E_2

F.



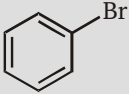
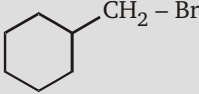
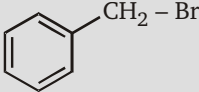
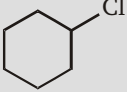
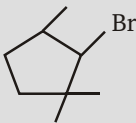
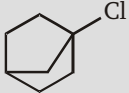
NaCN in ethanol
25 C

(a) S_{N1} (b) S_{N2} (c) E_1 (d) E_2 G. $(\text{CH}_3)_2\text{CHCH}_2\text{CH}_2-\text{OH}$

HBr 48% in H_2O
50 C

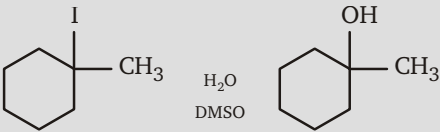
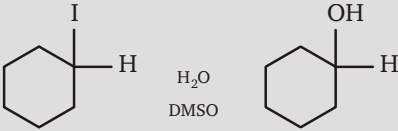
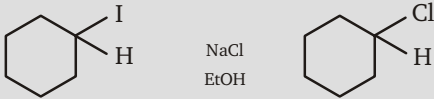
(a) S_{N1} (b) S_{N2} (c) E_1 (d) E_2

12. Examine the ten structural formulas shown in fig. & select that satisfy each of the following conditions. Write one or more (a through j) in each answer box.

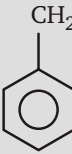
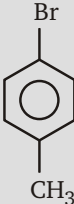
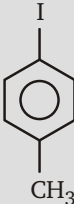
(a)		(b)	$\begin{array}{c} \text{CH}_3 \\ \\ \text{H}_3\text{C} - \text{C} - \text{Cl} \\ \\ \text{CH}_3 \end{array}$	(c)	
(d)	$\text{CH}_3 \quad \text{I}$	(e)		(f)	
(g)	$\begin{array}{c} \text{CH}_3 \\ \\ \text{H}_3\text{C} - \text{C} - \text{CH}_2 - \text{Cl} \\ \\ \text{CH}_3 \end{array}$	(h)	$\begin{array}{c} \text{H}_2\text{C} \\ \diagdown \\ \text{C} = \text{CH}_2 - \text{Cl} \\ \\ \text{CH}_3 \end{array}$	(i)	
(j)					

- A. Which compounds give an $\text{S}_{\text{N}}2$ substitution reaction on treatment with alcoholic NaSH ?
 B. Which compounds give an E_2 elimination reaction on treatment with alcoholic KOH ?
 C. Which compounds do not react under either of the previous reaction conditions ?

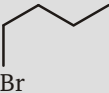
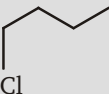
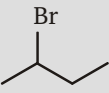
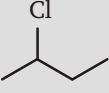
13. Select which reaction from the following reaction pairs will occur faster.

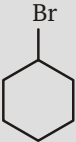
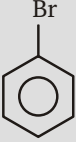
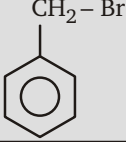
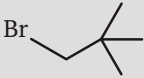
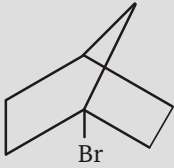
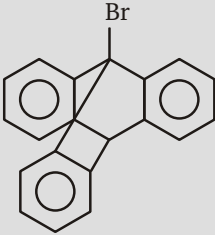
PART - 1	
Reaction A	
Reaction B	
PART - 2	
Reaction C	
Reaction D	
PART - 3	
Reaction E	
Reaction F	
PART - 4	
Reaction G	
Reaction H	

PART - 5

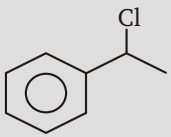
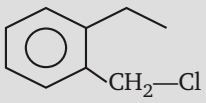
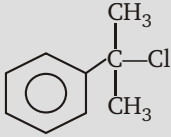
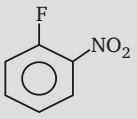
Reaction I	 
Reaction J	 

14. Tick your answer in the given box.

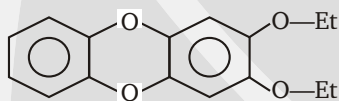
Alkyl Halide		2-D Structure		Expect S_N2 (at a reasonable rate)	
(a)	1-Bromobutane				Yes
					No
(b)	1-Chlorobutane				Yes
					No
(c)	2-Bromobutane				Yes
					No
(d)	2-Chlorobutane				Yes
					No
(e)	2-Chloro-2-methyl propane				Yes
					No

(f)	Bromocyclohexane			Yes
				No
(g)	Bromobenzene			Yes
				No
(h)	Benzyl bromide			Yes
				No
(i)	1-Bromo-2,2-dimethyl propane			Yes
				No
(j)	Bicyclo compound			Yes
				No
(k)	1-bromotriptycene			Yes
				No

15. Match the column

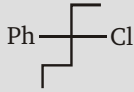
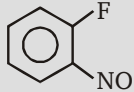
Column-I		Column-II	
(a)		(p)	It will undergo Nucleophilic Substitution reaction
(b)		(q)	It will undergo E_2 reaction
(c)		(r)	It will undergo E_1 reaction
(d)		(s)	It will undergo S_{N2} reaction
		(t)	It will undergo S_{N1} reaction

16.

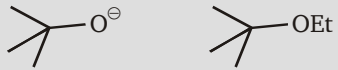
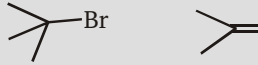
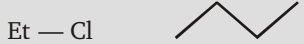
 $x\text{HI}$

How many (x) moles of HI consumed?

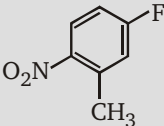
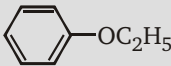
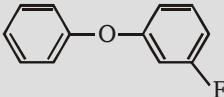
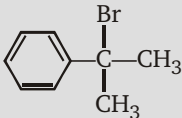
17.

Column (I)		Column (II)	
(a)	 H_2O	(p)	$\text{S}_{\text{N}}1$
(b)	 Ph-SNa	(q)	$\text{S}_{\text{N}}2$
(c)	 KOH	(r)	Carbocation is intermediate
(d)	 Br_2 CCl_4	(s)	Carbanion is intermediate

18.

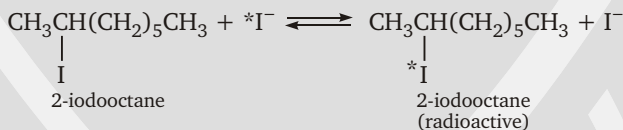
Column (I)		Column (II)	
	(Reaction sequence)		(Reagent required)
(a)		(p)	$\text{EtO}^\ominus$
(b)		(q)	EtBr
(c)		(r)	$\text{EtOH}/\text{H}^\oplus$
(d)		(s)	$\text{Et-Cl}/\text{Na ether}$

19. Choose the one compound within each set that meets the indicated criterion :

Column (I)		Column (II)	
(a)	The compound that reacts with alcoholic KOH to liberate Halide ion through substitution reaction.	(p)	
(b)	The compound that cannot be prepared by a Williamson ether synthesis.	(q)	
(c)	The compound that gives an acidic solution when allowed to stand in aqueous ethanol.	(r)	
(d)	The ether that cleaves more rapidly in HI.	(s)	

20. Comprehension

The first demonstration of the stereochemistry of the S_N2 reaction was carried out in 1935 by Prof. E.D. Hughes and his colleagues at the University of London. They allowed (R)-2-iodooctane to react with radioactive iodide ion ($^*I^-$).



The rate of substitution (rate constant K_s) was determined by measuring the rate of incorporation of radioactivity into the alkyl halide. The rate of loss of optical activity from the alkyl halide (rate constant K_o) was also determined under the same conditions.

What ratio K_o / K_s is predicted for each of the following stereochemical scenarios :

A. For inversion reaction :

- (a) $\frac{K_o}{K_s} = 1$ (b) $\frac{K_o}{K_s} = 1$ (c) $\frac{K_o}{K_s} = 1$ (d) can not be predicted

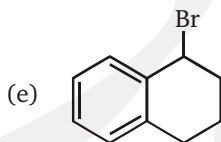
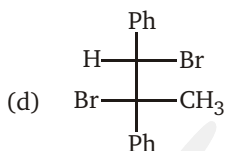
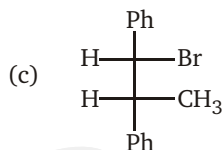
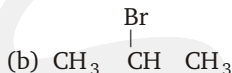
B. For equal amounts of both retention and inversion ?

- (a) $\frac{K_o}{K_s} = 1$ (b) $\frac{K_o}{K_s} = 1$ (c) $\frac{K_o}{K_s} = 1$ (d) can not be predicted

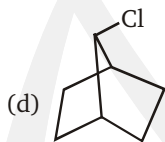
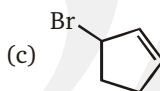
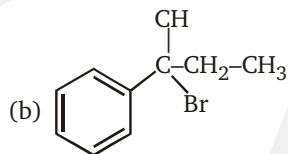
ANSWERS — LEVEL 2

1. d

2. c The reaction is Finkelstein reaction.

3. (a) F^- ; (b) I^- ; (c) F^- ; (d) $\text{NH}_2\text{—NH}_2$; (e) CH_3SO_3 4. (a) CH_3 Br

5. (a)



6. a - p, w; b - q, x; c - r, y; d - s, z

7. a - r, s; b - p, q; c - r, s; d - r, s

8. (a)

(b) $(\text{CH}_3)_3\text{C I}$ 

9. (1)

(2)

(3)

10. (X) A- reactants, B-transition state, C-Inter mediate, D- transition state

(Y) (i) exothermic (ii) B (iii) F (iv) a

11. A – b; B – b; C – d; D – a; E – b; F – b; G – b

12. A – c, d, e, f, h; B – b, c, f, i; C – a, g, j

13. Part – 1 2 3 4 5

Reaction– A D E G I

14. Yes – a, b, c, d, f, h,

No – e, g, i, j, k

15. a–p, q, r, s, t; b–p,s,t; c–p, q, r t; d–p

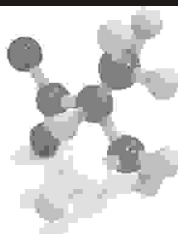
16. 2

17. a – p, r; b – q; c – s; d – r

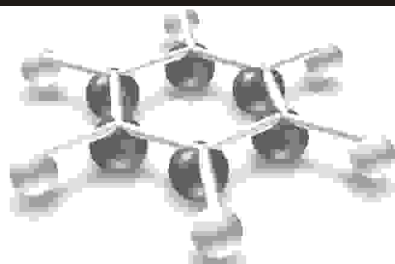
18. a – q; b – p; c – r; d – q

19. a – p; b – r; c – s; d – s

20. A – c; B – a



5B



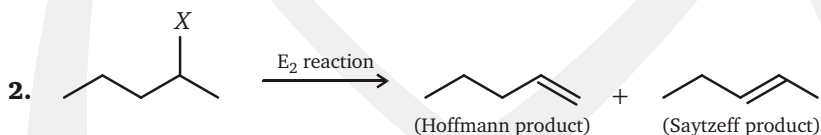
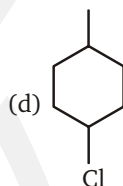
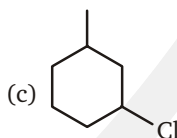
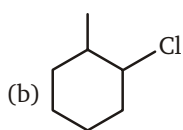
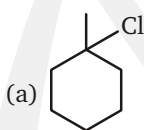
ALKYL HALIDES

Elimination Reactions (E_1, E_2, E_{1CB}, E_i)

Level - 1

1. Which of the following alkyl halide gives only one product (excluding stereoisomer) when undergo E_2 reaction ?

(E_2 elimination bi-molecular)



(A) In the above reaction, maximum Saytzeff product will obtained when :

- (a) X = I (b) X = Cl (c) X = Br (d) X = F

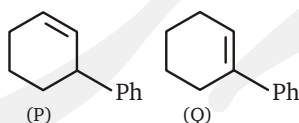
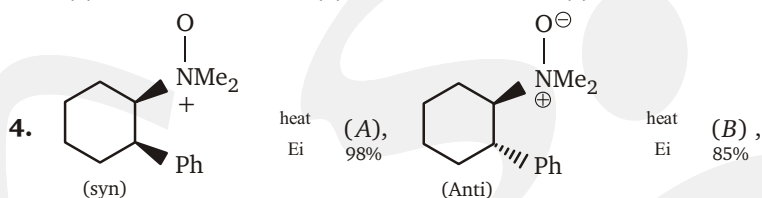
(B) In the above reaction Hoffmann product is major when X is :

- (a) I (b) - Cl (c) - Br (d) - F

Alkyl Halides (Elimination)

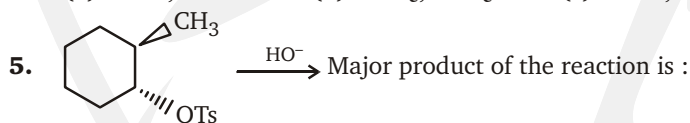


when (P) undergoes Hoffmann exhaustive methylation (twice) then the product obtained will be :

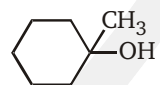
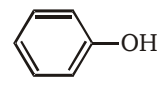
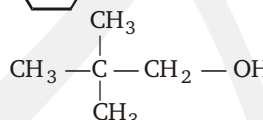


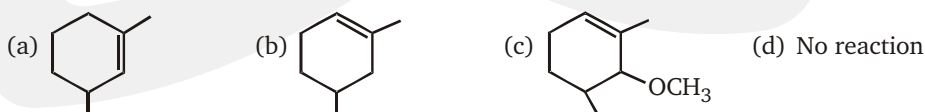
Product (A) & (B) of the above reaction is :

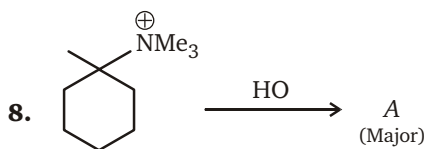
- (a) A P, B P (b) A Q, B Q (c) A P, B Q (d) A Q, B P



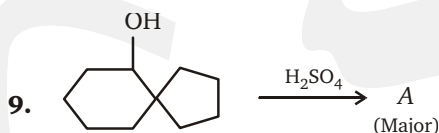
6. Which of these dehydrates most easily ?

- (a) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{OH}$ (b) 
 (c)  (d) 

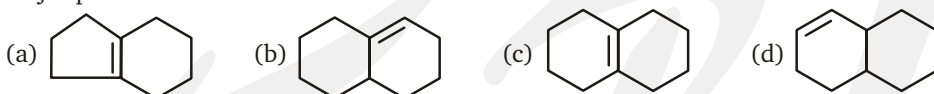




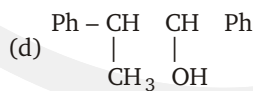
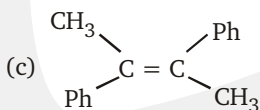
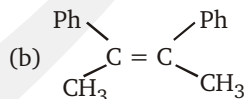
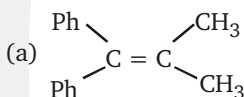
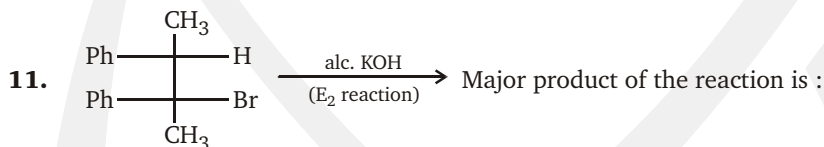
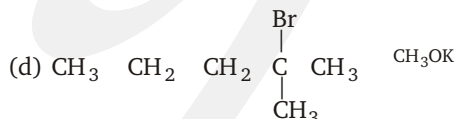
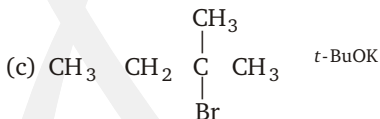
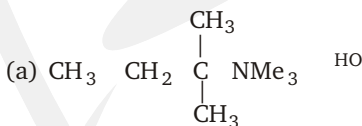
Major product A is :



Major product A is :



10. In which of the following reaction Saytzeff alkene is major product ?

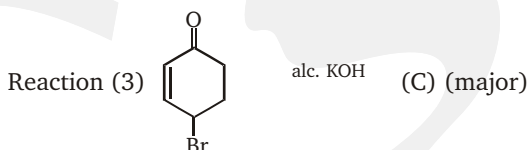
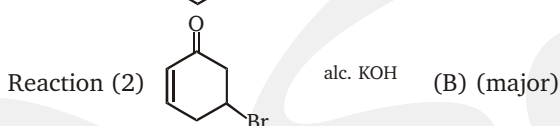
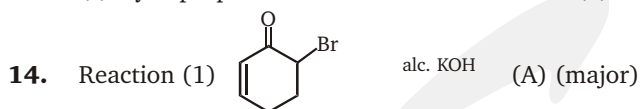


12. The conversion of 2, 3-dibromobutane to 2-butene with Zn is :

- (a) Redox reaction
(c) -Elimination

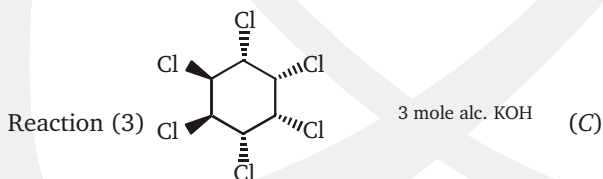
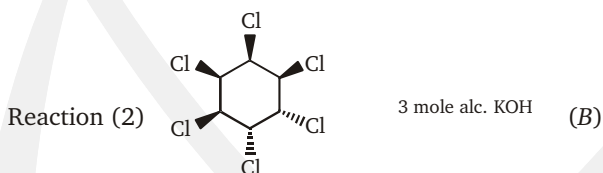
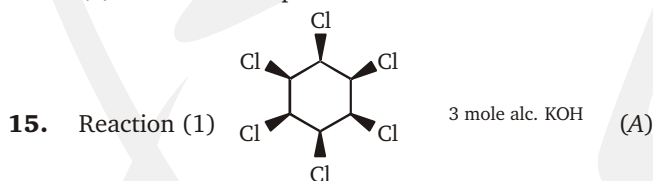
- (b) -Elimination
(d) Both -elimination and redox reaction

13. 1, 3-Dibromopropane is heated with zinc dust in ether. The product formed is :
 (a) propene (b) propane
 (c) cyclopropane (d) 3-bromopropane



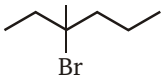
Product obtained in above reactions (1), (2) & (3) is :

- (a) A = B but C is different
 (b) A = C, but B is different
 (c) B = C, but A is different
 (d) A = B = C all product are identical



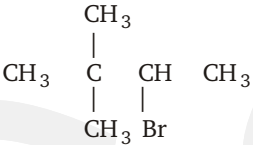
Product obtained in above reactions (1), (2) & (3) is :

- (a) A = B, C is different (b) A = C, B is different
 (c) B = C, A is different (d) A = B = C is same

16.  alc. KOH (x)
 x is number of E_2 product obtained (including stereoisomers)

Find (x) .

- (a) 3 (b) 4 (c) 5 (d) 6

17.  EtOH (A)

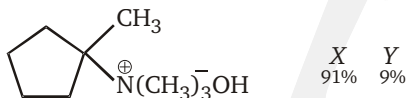
Major product (A) is :

- (a)  (b)  (c)  (d) 

18. Which one of the following compound will be least susceptible to elimination of hydrogen bromide ?

- (a) $\text{Br}-\text{CH}_2-\text{CH}_2-\text{NO}_2$ (b) $\text{Br}-\text{CH}_2-\text{CH}_2-\text{CH}_3$
 (c) $\text{Br}-\text{CH}_2-\text{CH}_2-\text{CN}$ (d) $\text{Br}-\text{CH}_2-\text{CH}_2-\text{CO}_2\text{Et}$

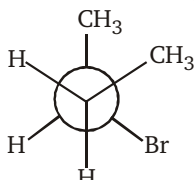
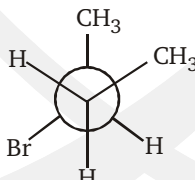
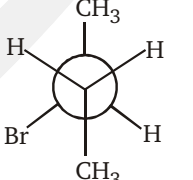
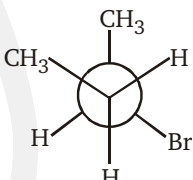
19. Two alkenes, X(91% yield) and Y(9% yield) are formed when the following compound is heated.

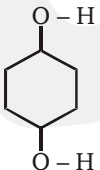


The structures of X and Y, respectively are :

- (a)  CH_3 and  CH_2 (b)  CH_3 and  CH_3
 (c)  CH_2 and  CH_3 (d)  CH_3 and  CH_2

20. In the dehydrohalogenation of 2-bromobutane; which conformation leads to the formation of *cis*-2-butene ?

- (a)  (b)  (c)  (d) 

21.  $\xrightarrow[\text{Pyridine}]{\text{CH}_3-\text{C}(=\text{O})-\text{Cl} \text{ (2 mole)}}$ (A) $\xrightarrow{\text{(Et)}}$ (B) + $\text{CH}_3\text{CO}_2\text{H}$

Product (B) of given reaction is:



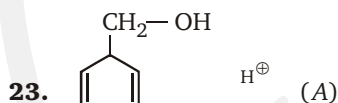
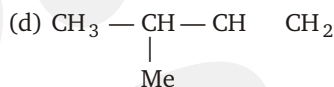
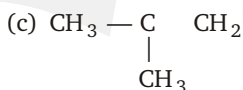
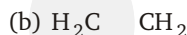
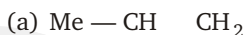
22. What product will be formed from Hoffmann exhaustive methylation of following compound?



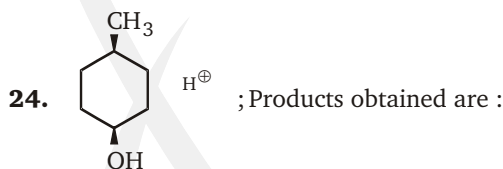
(i) CH_3I (excess)

(ii) Ag_2O
(iii)

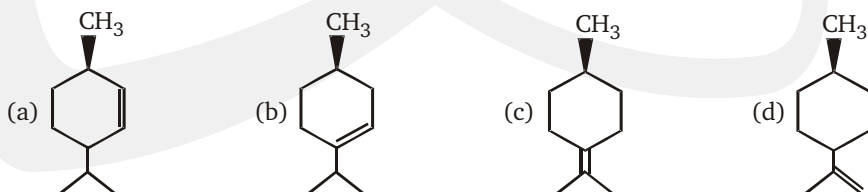
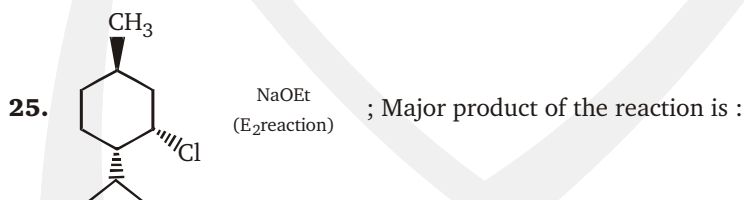
Product

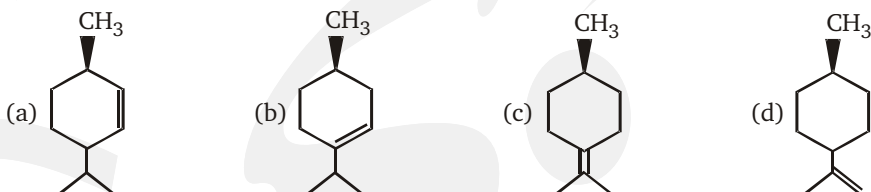
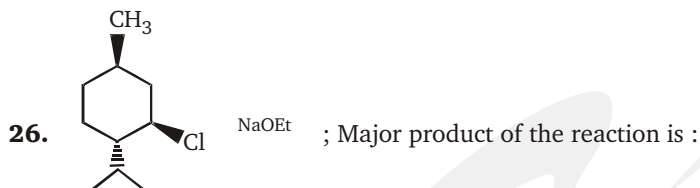


Product (A) is :

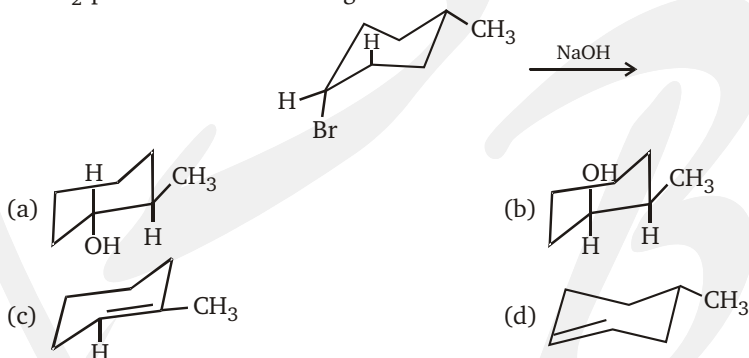


(a) Racemic (b) Diastereomers (c) G.I (d) Positional isomers

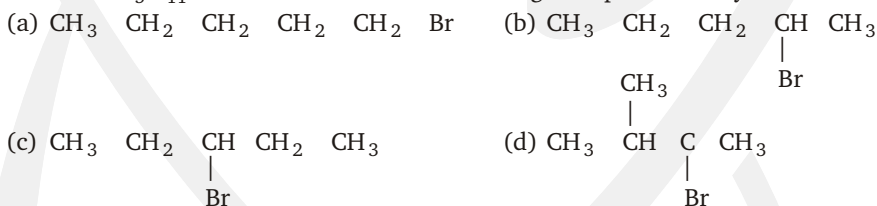




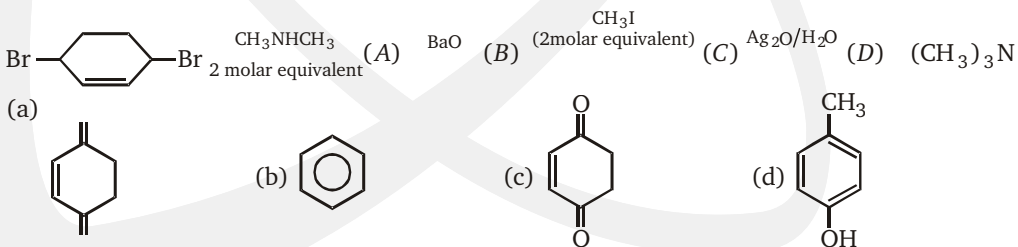
27. The E_2 product of the following reaction will be ?



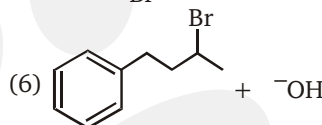
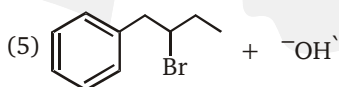
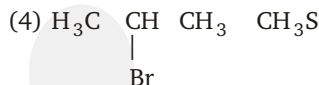
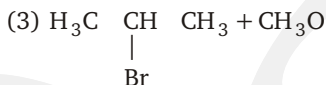
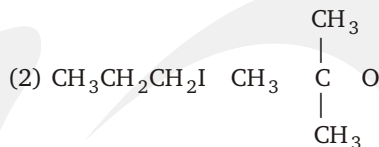
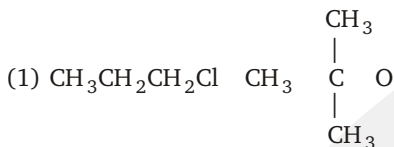
28. An halide $C_5H_{11}Br$ on treatment with alc. KOH give 2-pentene only. The halide will be :



29. End product (D) in the given sequence is:



30. For each of the following pairs of E_2 reaction, select the one that occurs with the greater rate constant.

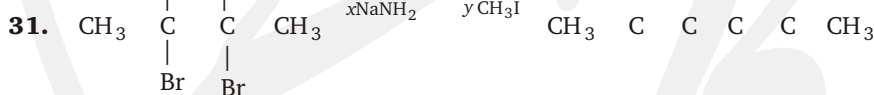


(a) 2, 4, 6

(b) 1, 3, 5

(c) 2, 3, 5

(d) 2, 4, 5



x and y mole consumed.

Value of $x + y =$

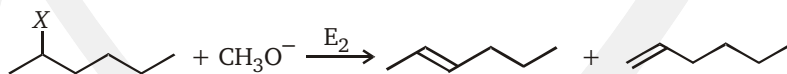
(a) 5

(b) 6

(c) 7

(d) 8

32. The following bimolecular elimination reaction (E_2) is carried out with different halogen leaving groups. The per cent yield of the two products (2-hexene and 1-hexene) for each leaving group is listed below.

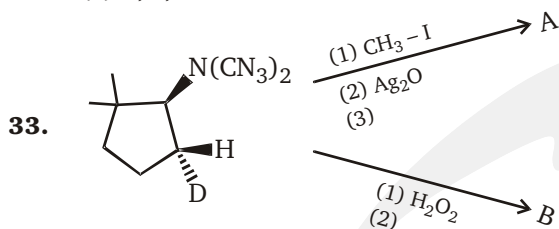


Leaving group	Conj. Acid $\text{p}K_a$	%-yield of 2-hexene	%-yield of 1-hexene
$\text{X} = \text{I}$	-10	81%	19%
$\text{X} = \text{Br}$	-9	72%	28%
$\text{X} = \text{Cl}$	-7	67%	33%
$\text{X} = \text{F}$	3.2	30%	70%

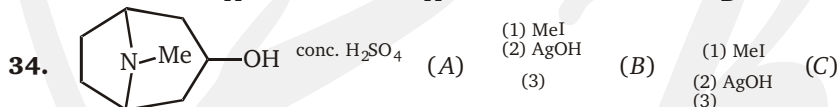
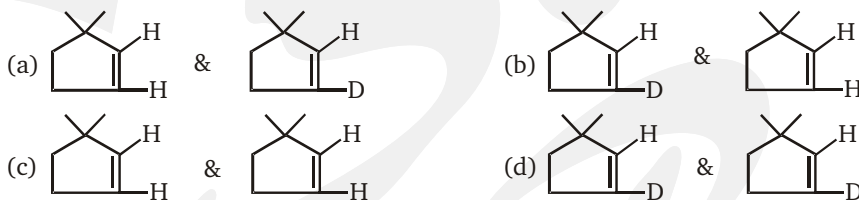
Which of the following statement is (are) true concerning this series of E_2 reactions ?

- (a) Based on the $\text{p}K_a$'s of the conjugate acid, I is the best leaving group and F is the poorest leaving group
- (b) When I, Br and Cl are used as leaving groups, Zaitsev's rule is followed
- (c) F is the strongest base (and therefore the poorest leaving group) and the transition state for reaction with fluoride as the leaving group has the least double bond character

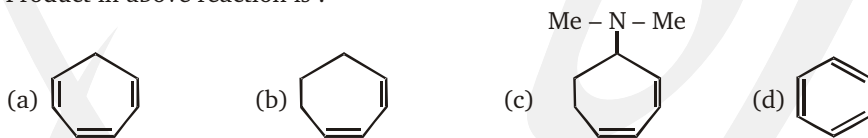
(d) a, b, c are true



Product (A) & (B) respectively are :



Product in above reaction is :

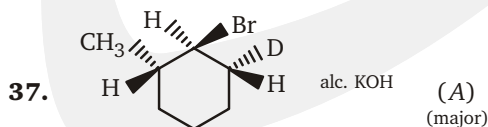


35. Major product obtained in the reaction of 1-phenyl-2-bromobutane with NaOMe is :

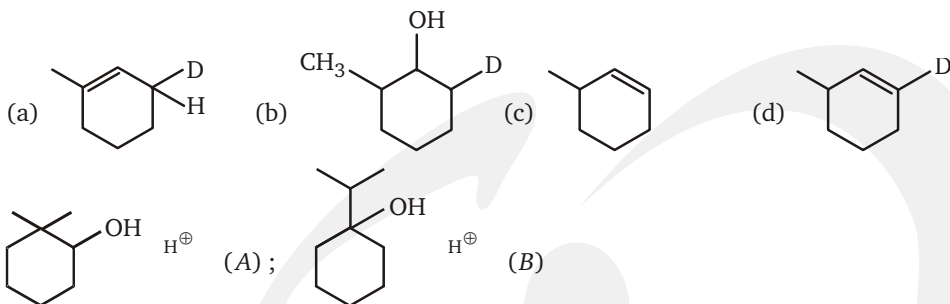
- (a) (E)-1-phenylbut-1-ene (b) (E)-1-phenylbut-2-ene
 (c) 1-phenyl-2-ethoxybutane (d) (Z)-1-phenylbut-2-ene

36. Which of the following alkyl halides give most complex mixture of alkene in an E_2 reaction ?

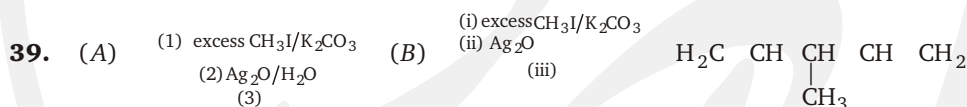
- (a) $\text{CH}_3 \text{ CH}_2 \text{ CH}_2 \text{ CH}_2 \text{ Br}$ (b) $\text{CH}_3 \text{ CH}_2 \text{ CH}_2 \text{ CH}(\text{Br}) \text{ CH}_3$
 (c) $\text{CH}_3 \text{ CH}_2 \text{ CH}(\text{Br}) \text{ CH}_2 \text{ CH}_3$ (d) $\text{CH}_3 \text{ C}(\text{CH}_3)(\text{Br}) \text{ CH}_2 \text{ CH}_3$



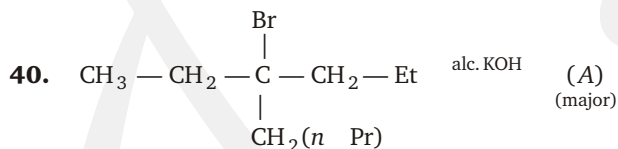
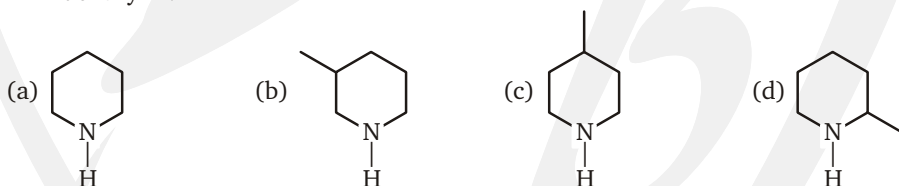
Product (A) is :

Sum of number of β -hydrogen present in compound A + B is :

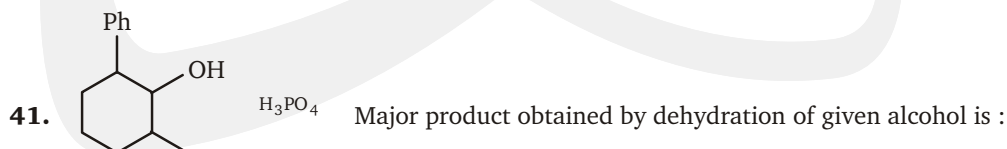
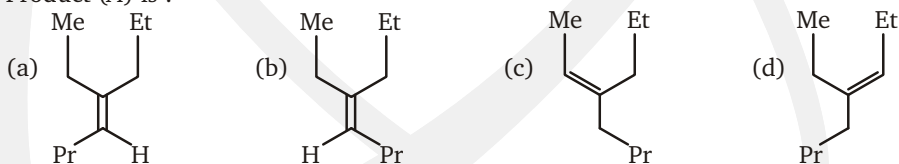
- (a) 18 (b) 19 (c) 20 (d) 21

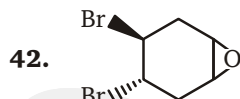
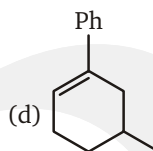
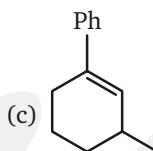
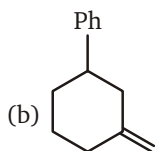
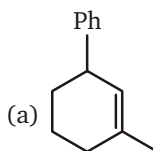


Identify A :

 n - Pr n - propyl

Product (A) is :





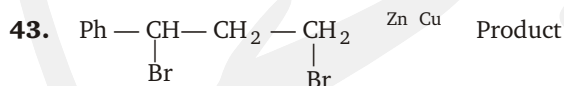
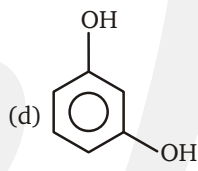
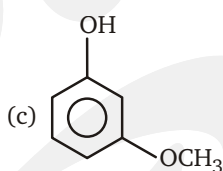
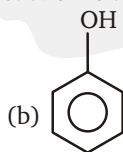
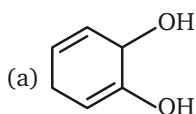
NaOCH_3 (2 mole)
 CH_3OH

(A)

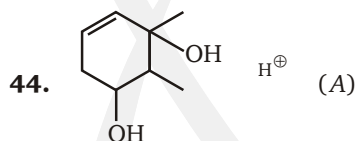
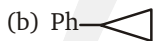
 H_3O^+

(B)

Product (B) of above reaction is :

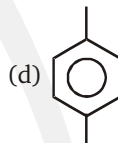
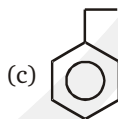
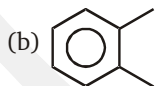
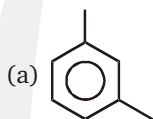


Product of the above reaction is :

 H^+

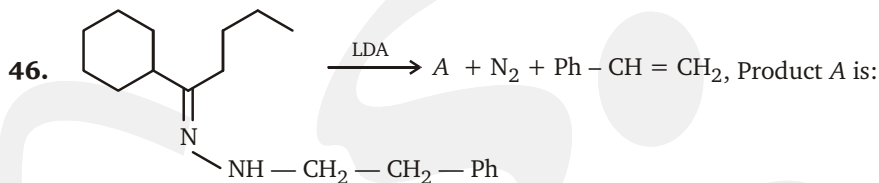
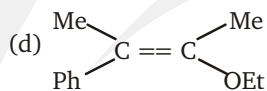
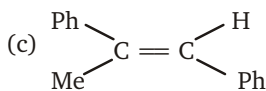
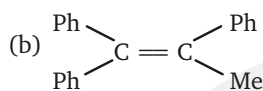
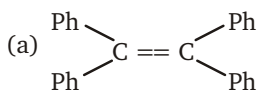
(A)

Product (A) is :

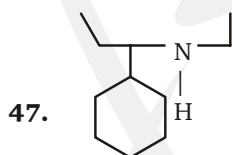
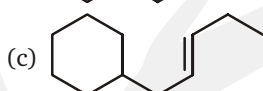
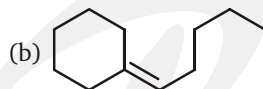
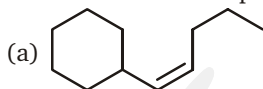
 NaOEt

Major product of the above reaction is :

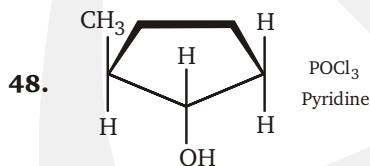
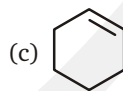
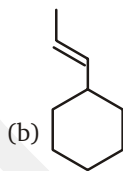
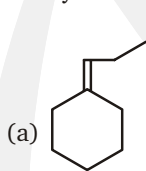
Alkyl Halides (Elimination)



LDA = Lithium di-isopropyl amide

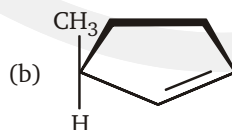
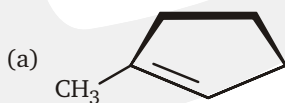


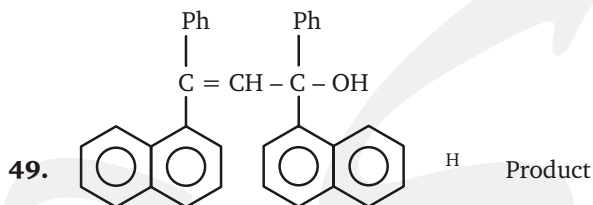
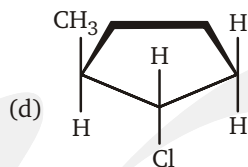
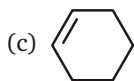
Major product of the reaction, when the given compound undergoes Hoffmann exhaustive methylation is :



POCl_3
Pyridine

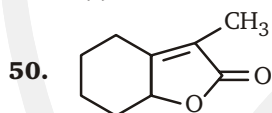
Major product of the reaction is :



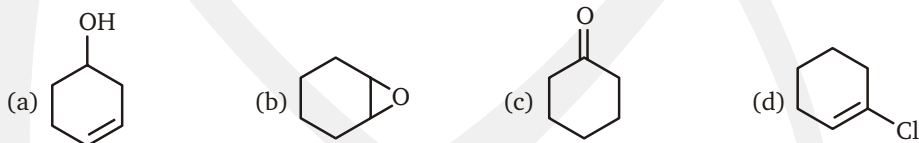
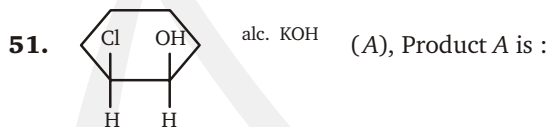
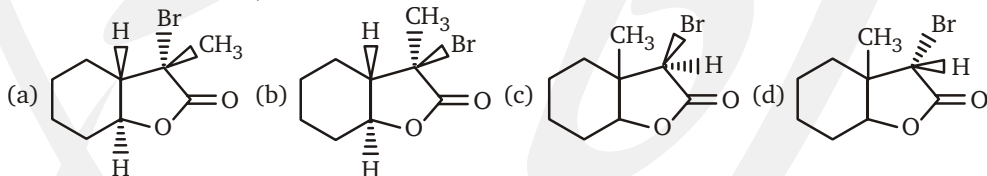


Stereochemistry of the product is :

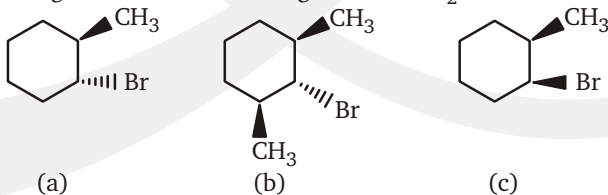
- (a) Meso compound
(b) Racemic mixture
(c) Diastereomer
(d) Optically pure enantiomers



Which of the following reactant is used to obtain above compound (A). (Assume that EtO is used in all the reaction)

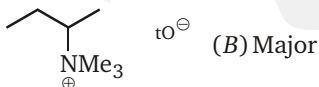
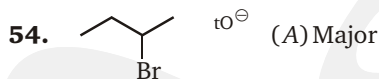
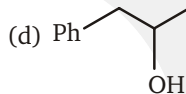
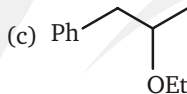
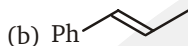
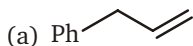
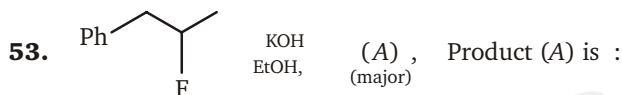


52. Rank the following in order of decreasing rate in an E_2 reaction:



- (a) $a > b > c$
(b) $c > a > b$
(c) $c > b > a$
(d) $b > a > c$

Alkyl Halides (Elimination)



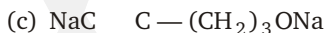
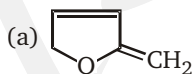
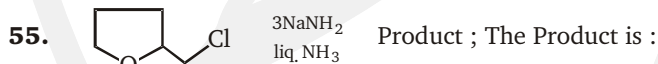
Relation between (A) and (B) is :

(a) G.I.

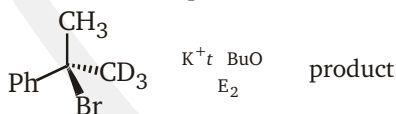
(c) Enantiomer

(b) Positional isomer

(d) Chain isomer



56. Which best describes the product of the following reaction ?



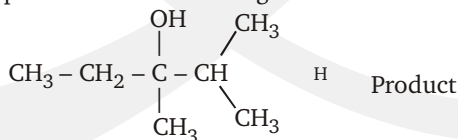
(a) Absolute configuration has been inverted

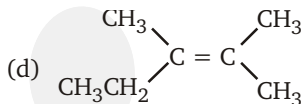
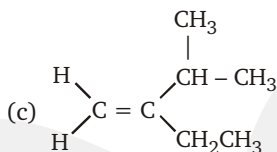
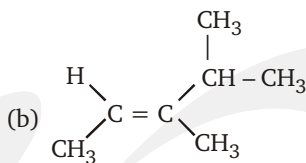
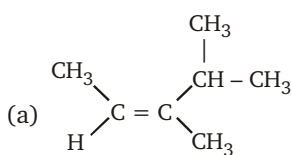
(b) Absolute configuration has been retained

(c) Racemization (loss of absolute configuration) has occurred

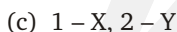
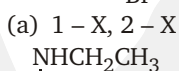
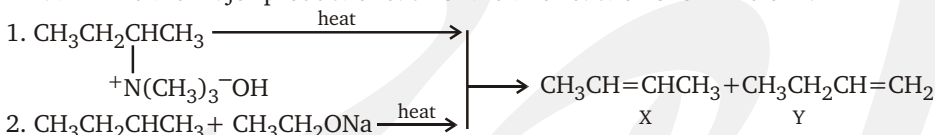
(d) Loss of chirality has occurred (the product is achiral)

57. What is the major product of the following reaction ?





58. What will be the major product of each of the two reaction shown below ?



59. + CH_3I (excess) product; The product is :

(a) a primary amine

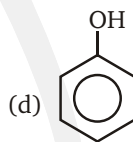
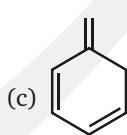
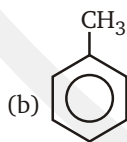
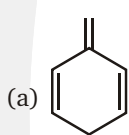
(b) a tertiary amine

(c) a secondary amine

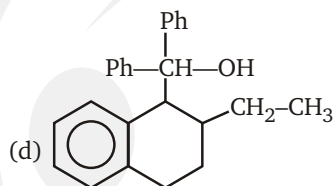
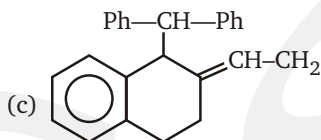
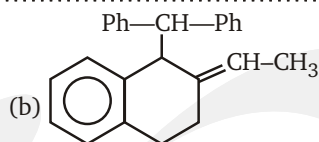
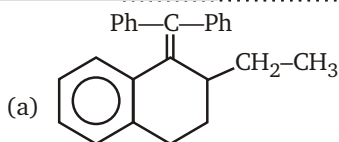
(d) a quaternary ammonium salt

60. $\xrightarrow{\text{H}}$ (A) Isomerisation (B)

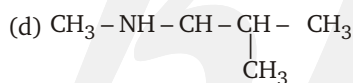
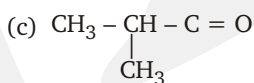
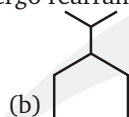
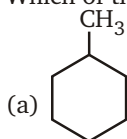
(A) on heating isomerizes to (B). What is the structure of (B) ?



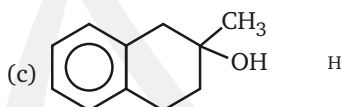
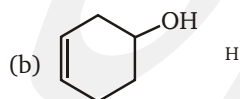
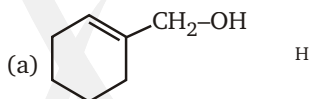
61. $\xrightarrow{\text{H}}$ (A) , major product (A) is :
(major)



62. Which of the following carbocation will undergo rearrangement ?

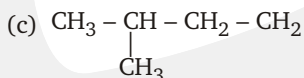
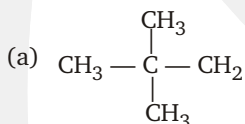


63. In which of the following reaction resonance stabilized product will form ?



(d) All of these

64. In which of following reaction rearrangement take place with change in carbon skeleton ?

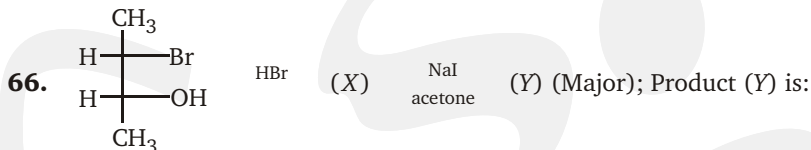


65. Consider the following reaction :



Which response contains all the correct statement about this process ?

- (1) Dehydration (2) E₂ mechanism
 (3) Carbon skeleton migration (4) Most stable alkene will form
 (5) Single-step reaction
- (a) 1, 3 (b) 1, 2, 3 (c) 1, 2, 5 (d) 1, 3, 4

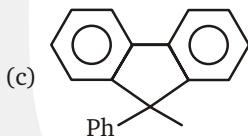
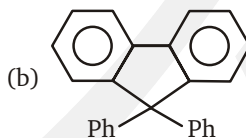
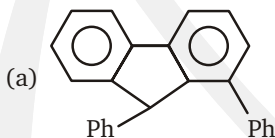
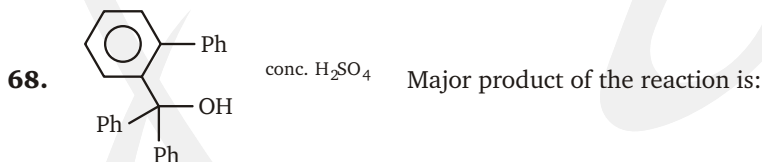


- (a) *cis*-2-butene (b) *trans*-2-butene
 (c) 1-butene (d) Iso-butene



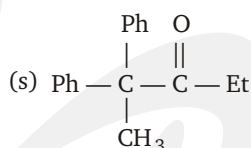
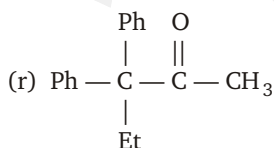
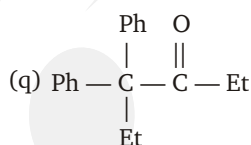
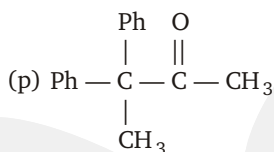
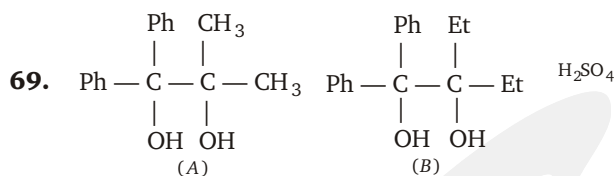
Above reaction is an example of 1,4-elimination. Predict the product.

- (a) $\text{CH}_3 - \text{CH} = \text{C} = \text{CH}_2$ (b) $\text{CH}_3 - \text{C} = \text{C} - \text{CH}_3$
 (c) $\text{CH}_3 - \text{CH}_2 - \text{C} = \text{CH}$ (d) $\text{H}_2\text{C} = \text{CH} - \text{CH} = \text{CH}_2$



(d) None of these

Alkyl Halides (Elimination)



When (A) and (B) reacts with H_2SO_4 products obtained are :

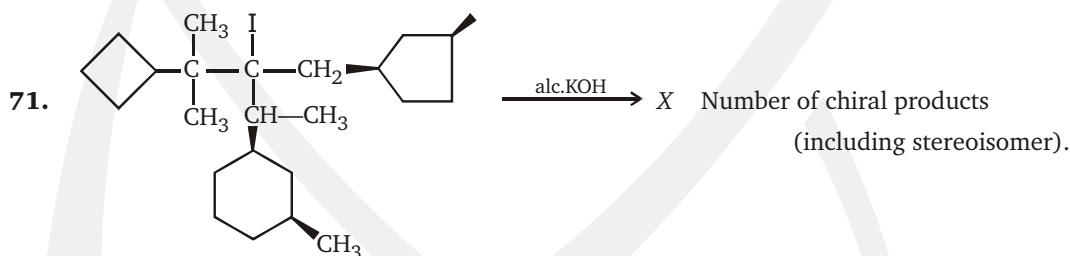
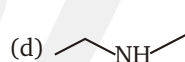
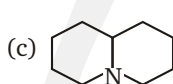
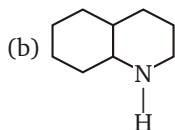
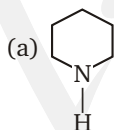
(a) p, q, r, s

(b) p, q

(c) p, q, r

(d) p, q, s

70. Which of the following compound gives even number of Hoffmann's exhaustive methylation and elimination?

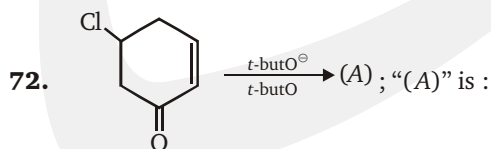


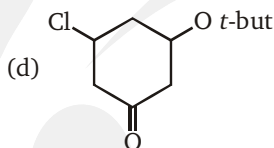
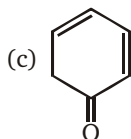
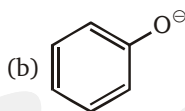
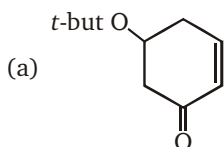
(a) 2

(b) 4

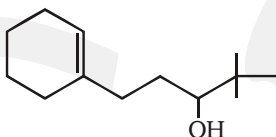
(c) 6

(d) 8

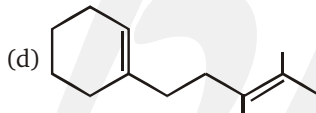
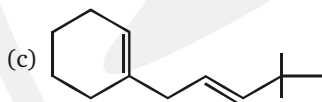
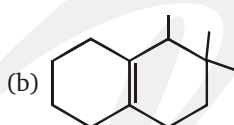
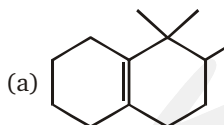




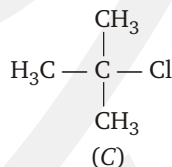
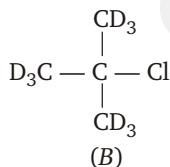
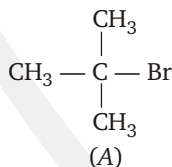
73.

Conc. H_2SO_4 A H_2O
(Major)

which of the following is product "A" in the above reaction?



74. Which of the following is true regarding increasing reactivity in the E_2 reaction with $\text{NaC}_2\text{H}_5\text{O}$?



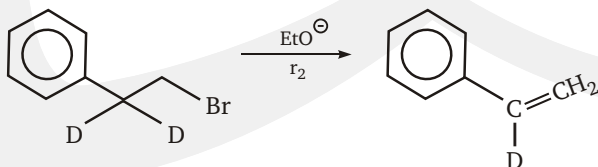
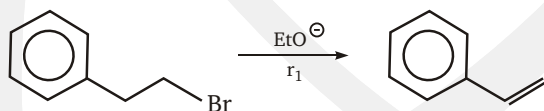
(a) A is more reactive than B

(b) B is more reactive than A

(c) A is more reactive than C

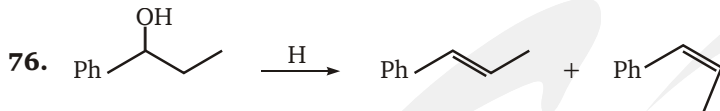
(d) B is less reactive than C

75.



Compare rate of Reactions.

- (a) r_1 r_2 (b) r_1 r_2 (c) r_2 r_1 (d) r_1 $\frac{r_2}{2}$



Which statement is not true for above reaction ?

- (a) It is dehydration reaction.
 (b) The mechanism followed by reaction is E_1
 (c) The reaction is stereo selective.
 (d) The major product is Z-alkene.

ANSWERS — LEVEL 1

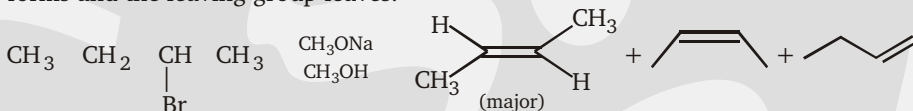
1.	(d)	2.	A B	a d	3.	(a)	4.	(c)	5.	(b)	6.	(b)	7.	(d)	8.	(b)
9.	(c)	10.	(d)		11.	(c)	12.	(c)	13.	(c)	14.	(d)	15.	(d)	16.	(c)
17.	(b)	18.	(b)		19.	(c)	20.	(a)	21.	(b)	22.	(a)	23.	(b)	24.	(a)
25.	(b)	26.	(a)		27.	(d)	28.	(c)	29.	(b)	30.	(c)	31.	(d)	32.	(d)
33.	(a)	34.	(a)		35.	(a)	36.	(b)	37.	(c)	38.	(c)	39.	(c)	40.	(c)
41.	(c)	42.	(b)		43.	(b)	44.	(b)	45.	(c)	46.	(a)	47.	(d)	48.	(b)
49.	(b)	50.	(a)		51.	(c)	52.	(b)	53.	(b)	54.	(b)	55.	(c)	56.	(d)
57.	(d)	58.	(b)		59.	(d)	60.	(b)	61.	(a)	62.	(b)	63.	(d)	64.	(a)
65.	(d)	66.	(b)		67.	(d)	68.	(b)	69.	(b)	70.	(a,b)	71.	(b)	72.	(b)
73.	(a)	74.	(a,c,d)		75.	(a)	76.	(d)								

Level - 2

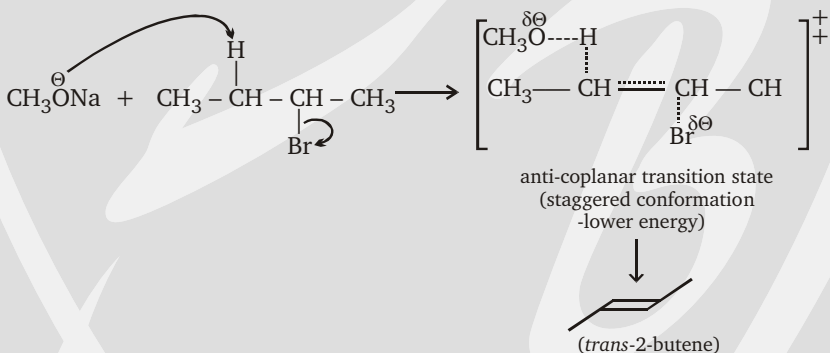
1. Comprehension

E_2 reaction Elimination bimolecular

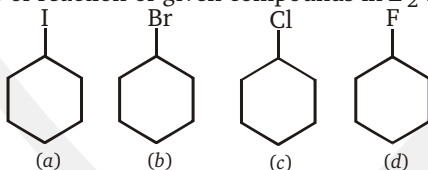
In the general mechanism of the E_2 reaction a strong base abstracts a proton on a carbon atom adjacent to the one of the leaving group. As the base abstracts a proton, a double bond forms and the leaving group leaves.



Mechanism :

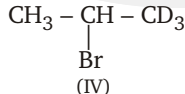
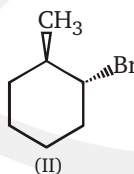
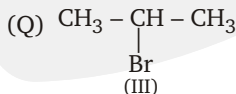
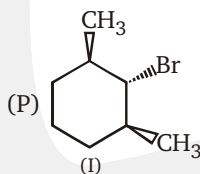


A. Identify the rate of reaction of given compounds in E_2 reaction:

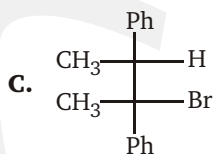
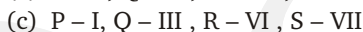
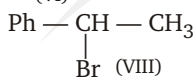
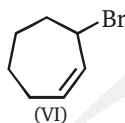
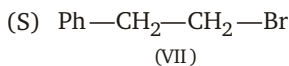
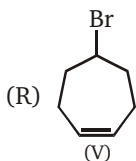


(a) $a > b > c > d$ (b) $a > c > b > d$ (c) $b > a > c > d$ (d) $b > d > a > c$

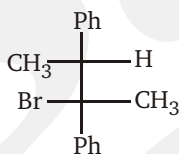
B. In given pairs, which compound is more reactive toward E_2 reaction:



Alkyl Halides (Elimination)

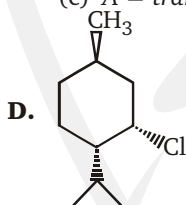
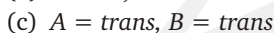
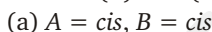


alc.KOH (A),

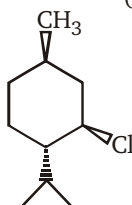


alc.KOH (B)

Product (A) and (B) are :

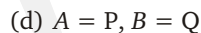
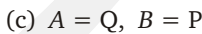
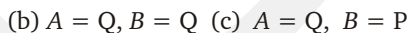
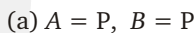
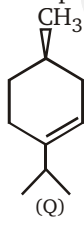
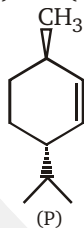


NaOEt (A),

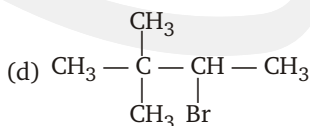
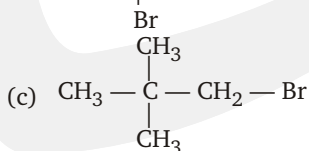
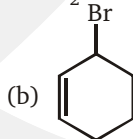


NaOEt (B)

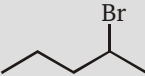
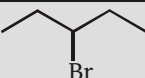
Select the products (A) and (B) from the compounds (P) and (Q) given below:



E. Which of the following compound is inert toward E_2 reaction.

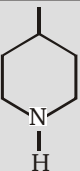
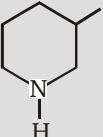
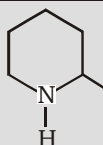
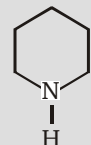


2. Match the column :

Column (I)		Column (II)	
E_2 reaction (elimination bimolecular)		No. of possible products. (including stereoisomerism)	
(a)	 alc. KOH	(p)	0
(b)	 alc. KOH	(q)	1
(c)	 alc. KOH	(r)	2
(d)	 alc. KOH	(s)	3

3. Match the Column :

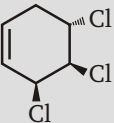
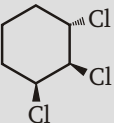
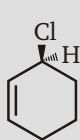
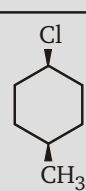
HEM = Hoffmann exhaustive methylation followed by elimination.

Column (I)		Column (II)	
Reaction		Product	
(a)	 HEM HEM	(p)	$\text{H}_2\text{C} \quad \text{CH} - \text{CH}_2 - \text{CH} \quad \text{CH}_2$
(b)	 HEM HEM	(q)	$\text{H}_2\text{C} \quad \text{CH} - \text{CH}_2 - \text{CH}_2 - \text{CH} \quad \text{CH}_2$
(c)	 HEM HEM	(r)	$\text{H}_2\text{C} \quad \text{CH} - \text{CH}_2 - \overset{\text{CH}_3}{\underset{ }{\text{C}}} \quad \text{CH}_2$
(d)	 HEM HEM	(s)	$\text{H}_2\text{C} \quad \text{CH} - \overset{\text{CH}_3}{\underset{ }{\text{CH}}} - \text{CH} \quad \text{CH}_2$

4. Match the column :

Column (I)		Column (II)	
(a)	 H^+	(p)	Product are Diastereomers
(b)	 alc. KOH	(q)	Carbocation is intermediate
(c)	 H^+	(r)	2nd order reaction
(d)	 alc. KOH	(s)	1st order reaction

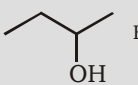
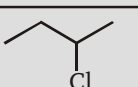
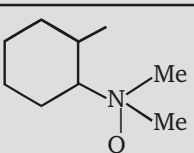
5. Match the column :

Column (I)		Column (II)	
(a)	 alc. KOH	(p)	Optically active product
(b)	 alc. KOH	(q)	Optically inactive product
(c)	 aq. KOH	(r)	2nd order reaction
(d)	 aq. KOH	(s)	unimolecular reaction

6. Match the column :

Column (I)		Column (II)	
E₂ reactions (elimination bimolecular)		Number of products (including stereoisomerism)	
(a)	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$ alc. KOH	(p)	1
(b)	$\text{CH}_3\text{CH}(\text{Br})\text{CH}_2\text{CH}_3$ alc. KOH	(q)	2
(c)	$\text{CH}_3\text{C}(\text{Br})(\text{CH}_3)\text{CH}_2\text{CH}_3$ alc. KOH	(r)	3
(d)	$\text{PhCH}_2\text{CH}(\text{Br})\text{CH}_2\text{CH}_3$	(s)	4

7. Match the column :

Column (I)		Column (II)	
(a)	 (A)	(p)	E ₁
(b)	 NaNH ₂	(q)	E ₂
(c)	$\text{CH}_3 - \text{C}(\text{O}) - \text{CH}_2 - \underset{\text{Br}}{\text{CH}} - \text{CH}_3$ EtONa	(r)	Ei (elimination intramolecular)
(d)		(s)	E _{1CB}

8. Match the column :

Column (I)		Column (II)	
Reaction		Product	
(a)		(p)	
(b)		(q)	
(c)		(r)	
(d)		(s)	

9. (a) alc. KOH (X) products
- (b) alc. KOH (Y)
- (c) alc. KOH (Z)
- (d) alc. KOH (P)

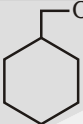
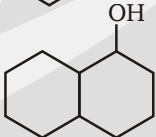
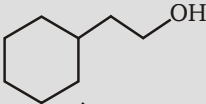
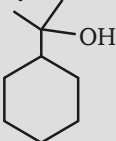
Sum of X Y Z P

- Reaction-1 : $\text{Ph}-\overset{\text{O}}{\underset{\text{||}}{\text{C}}}-\text{CH}_3 \xrightarrow{\text{PCl}_5} \boxed{x} \text{NaNH}_2 \xrightarrow{\text{CH}_3\text{I}} \text{Ph}-\overset{\text{O}}{\underset{\text{||}}{\text{C}}}-\text{CH}_2-\text{CH}_3$
- Reaction-2 : $\text{CH}_3-\overset{\text{Br}}{\underset{\text{|}}{\text{CH}}}-\overset{\text{Br}}{\underset{\text{|}}{\text{CH}}}-\text{CH}_3 \xrightarrow{\boxed{y} \text{NaNH}_2} \text{CH}_3-\overset{\text{O}}{\underset{\text{||}}{\text{C}}}-\text{CH}_2-\text{CH}_3$
- Reaction-3 : $\text{Ph}-\overset{\text{Cl}}{\underset{\text{|}}{\text{CH}}}-\overset{\text{Cl}}{\underset{\text{|}}{\text{CH}}}_2 \xrightarrow{\boxed{z} \text{NaNH}_2} \text{Ph}-\overset{\text{O}}{\underset{\text{||}}{\text{C}}}-\text{CH}_2-\text{CH}_2-\text{Et}$

x, y, z are moles used.

Sum of $[x \ y \ z]$

11. Sum of β -hydrogen in major product of the reaction.

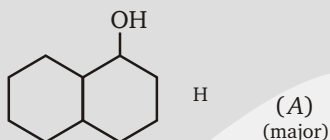
- Reaction-1  H (A)
- Reaction-2  H (B)
- Reaction-3  H (C)
- Reaction-4  H (D)

Sum of β -hydrogen is $A + B + C + D =$

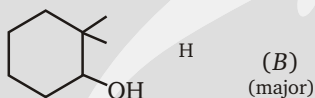
12. Reaction-1 $\text{CH}_3-\overset{\text{CH}_3}{\underset{\text{OH}}{\text{C}}}-\overset{\text{CH}_3}{\text{CH}}-\text{CH}_3$ H (A)
- Reaction-2 $\text{CH}_3-\overset{\text{CH}_3}{\text{CH}}-\text{CH}_2-\text{CH}_2-\text{OH}$ H (B)
- Reaction-3 $\text{CH}_3-\overset{\text{CH}_3}{\text{CH}}-\overset{\text{CH}_3}{\text{CH}}-\text{CH}_2-\text{OH}$ H (C)

Sum of β -hydrogen is $(A + B + C =)$

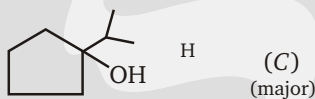
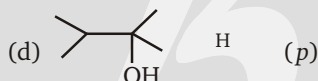
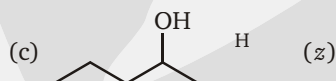
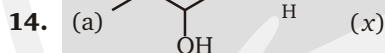
13. Reaction-1



Reaction-2



Reaction-3

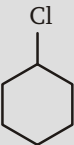
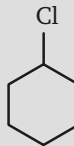
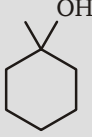
Sum of β -hydrogen (A + B + C) =

Total number of products obtained in above reactions including minor products is (including stereoisomer)

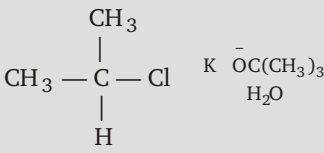
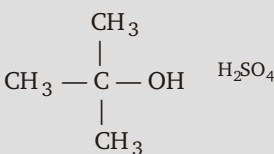
15. Match the column (I) and (II).

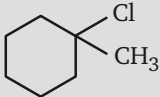
Column (I)		Column (II)	
Reaction		Type of Reaction	
(a)	R - 2 - chlorobutane $\xrightarrow[\text{acetone}]{\text{KSH}}$	(p)	S_N1
(b)	R - 2 - chlorobutane $\xrightarrow[\text{EtOH}]{\text{EtO}^- \text{ Na}^+}$	(q)	S_N2
(c)	2 - bromo- 2 - methyl propane $\xrightarrow{\text{H}_2\text{O}}$	(r)	E_1
(d)	2- butanol $\xrightarrow{\text{H}_2\text{SO}_4}$	(s)	E_2

16. Match the column (I) and (II).

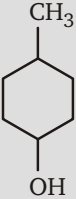
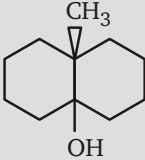
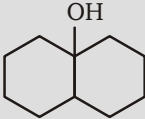
Column (I)		Column (II)	
Reaction		Type of Reaction	
(a)	 <chem>ClC1CCCCC1</chem> aq. KOH	(p)	S_N1
(b)	 <chem>ClC1CCCCC1</chem> alc. KOH	(q)	S_N2
(c)	 <chem>ClC1(C)CCCCC1</chem> H_2O	(r)	E_1
(d)	 <chem>OC1(C)CCCCC1</chem> H	(s)	E_2

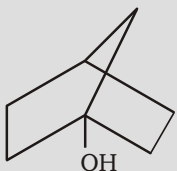
17. Select whether the following reagent combination will result in elimination or substitution reactions leading to the major product.

Reaction		Substitution	Elimination
(a)	 <chem>CC(C)Cl</chem> $K^+ \text{ } ^-\text{OC(CH}_3)_3$ H_2O		
(b)	 <chem>CC(C)(C)O</chem> H_2SO_4		

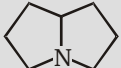
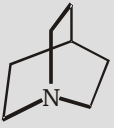
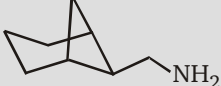
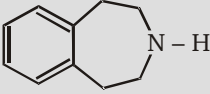
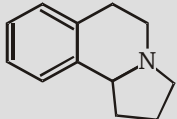
(c)	$\begin{array}{c} \text{Cl} \\ \\ \text{CH}_3 - \text{CH} - \text{CH}_2 - \text{CH}_3 \end{array} \quad \text{alc-KOH}$		
(d)	$\begin{array}{c} \text{CH}_3 \\ \\ \text{CH}_3 - \text{C} - \text{I} \\ \\ \text{H} \end{array} \quad \text{NaN}_3^-$		
(e)	 EtO		
(f)	$\begin{array}{c} \text{CH}_3 \\ \\ \text{CH}_3 - \text{C} - \text{Cl} \\ \\ \text{CH}_3 \end{array} \quad \text{H}_2\text{O}$		

18. Match the Column (I) and (II) (Matrix).

Column (I)		Column (II)	
Reaction		Comment on product	
(a)		(p)	Racemic mixture
(b)		(q)	Major product consist of even number of -hydrogen
(c)		(r)	Will not undergo dehydration

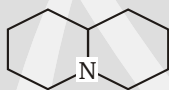
(d)		(s)	Major product consist of odd number of -hydrogen
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19. For each of the following amines (A through D), exhaustive methylation (treatment with excess methyl iodide), followed by Hoffmann elimination (heating with AgOH), repeated as necessary, removes the nitrogen atom in the form of trimethylamine. Indicate the number of repetitive Hoffmann eliminations required to remove the nitrogen by a number (1 to 4) in the designated answer sheet.

A.		B.		C.	
D.		E.		F.	

a.		b.		c.	
d.		e.		f.	

20.



x is total number of HEM (Hoffman Exhaustive Methylation and eliminations) to remove nitrogen from given compound.

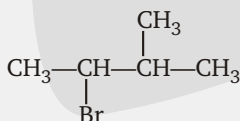


alc.KOH

y is total number of possible E_2 product (including stereoisomer)

Sum of $x+y=?$

21.



EtOH

(x)

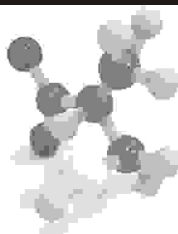
 (S_N1) (E_1)

products. (including stereoisomer)
consider all products

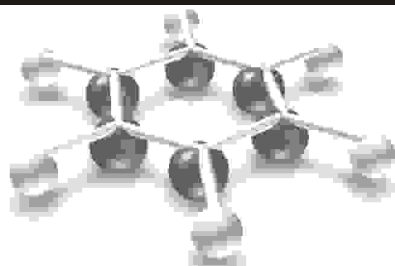
Total number of products are :

ANSWERS — LEVEL 2

1. A – a; B – a; C – b; D – c; E – c;
2. a – s; b – r; c – q; d – p
3. a – s; b – r; c – q; d – p
4. a – p, q, s; b – p, r; c – q, s; d – r
5. a – p, r; b – p, r; c – p, r; d – q, r
6. a – p; b – r; c – q; d – s
7. a – p; b – q; c – q; d – r
8. a – p; b – q; c – r; d – s
9. X 3, Y 3, Z 2, P 0 3 3 2 0 8
10. x 3, y 2, z 3 3 2 3 8
11. 32
12. 33
13. 28
14. x 3, y 1, z 3, p 2
Sum 9
15. (a – q), (b – s), (c – p), (d – r)
16. (a – q), (b – s), (c – p), (d – r)
17. Substitution – d, f
Elimination – a, b, c, e
18. a – p, q; b – p, q; c – q; d – r
19. a – 3; b – 2; c – 3; d – 1; e – 2; f – 3
20. 6
21. 6



5c

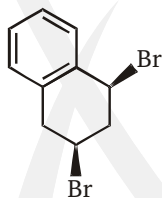


ALKYL HALIDES



Level - 1

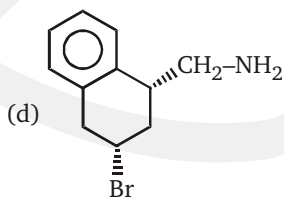
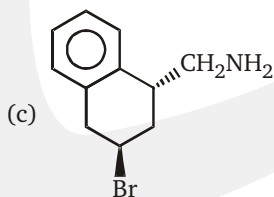
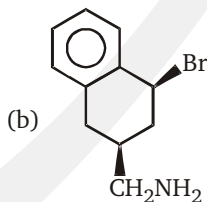
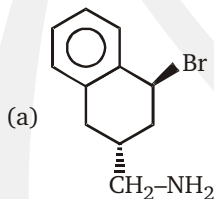
1.



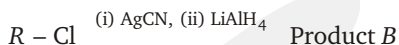
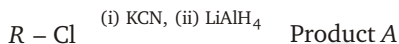
1. One eq. NaCN

2. LiAlH₄

(A); Product (A) is :



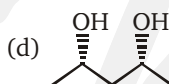
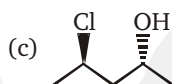
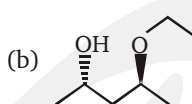
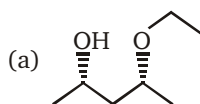
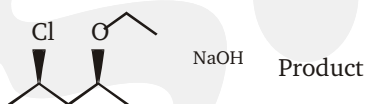
2. In the reactions given below,



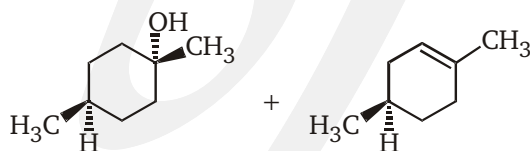
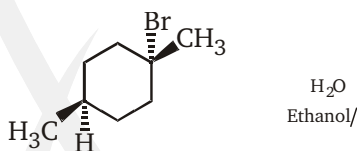
the compounds A and B are :

- (a) chain isomers (b) position isomers
(c) functional isomers (d) metamers

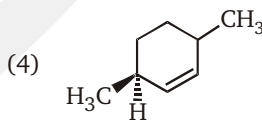
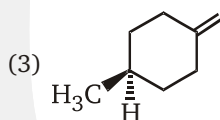
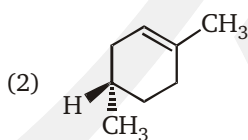
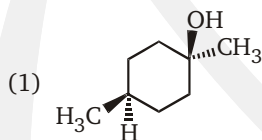
3. Which is the major product expected from the following S_N2 reaction ?



4. Consider the following E_1/S_N1 reaction :



The missing product(s) is(are) :



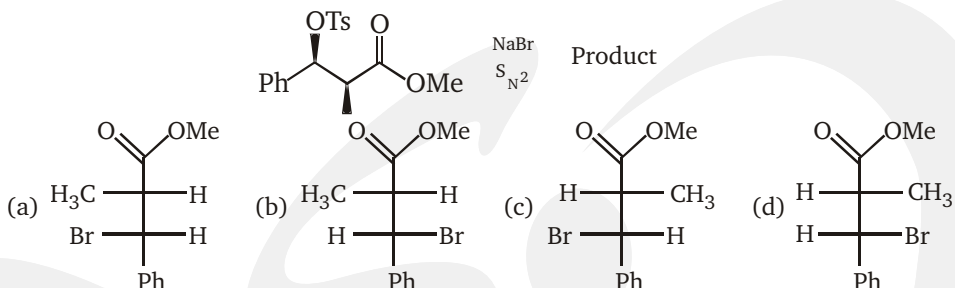
(a) 1, 2 and 3

(b) 3 and 4

(c) 2 and 3

(d) 1, 2, 3 and 4

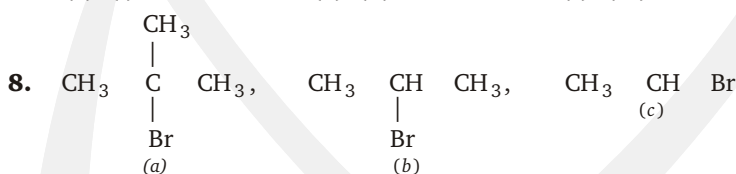
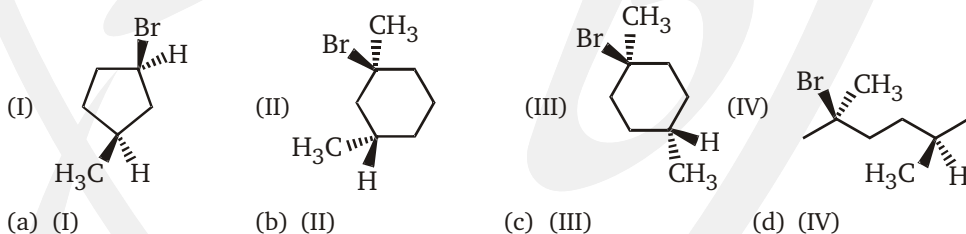
5. What is the product of the following S_N2 reaction?



6. Select the reagent that will yield the greater amount of substitution on reaction with $\text{CH}_3\text{CH}_2\text{Br}$:

- $\text{CH}_3\text{CH}_2\text{OK}$ in dimethyl sulfoxide (DMSO)
- $(\text{CH}_3)_3\text{COK}$ in dimethyl sulfoxide (DMSO)
- Both (a) and (b) will give comparable amounts of substitution
- Neither (a) nor (b) will give any amount of substitution

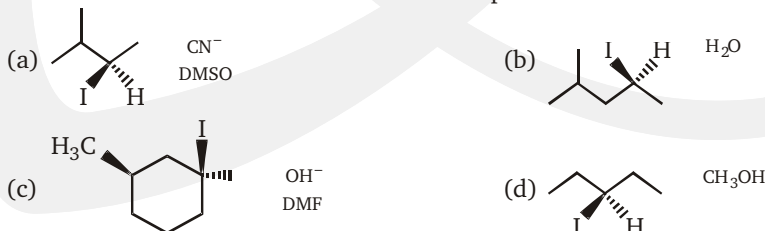
7. Under the specified conditions, substrate X undergoes substitution and elimination reactions to give products A – D . A and B are stereoisomers, but not enantiomers. C and D are enantiomers. A is not an isomer of C . Which of the following could be the starting material X ?



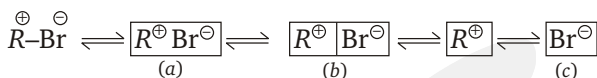
Compare rate of E_2 reaction:

- $c > b > a$
- $a > b > c$
- $b > a > c$
- $c > a > b$

9. Which reaction results in the formation of a pair of enantiomers?

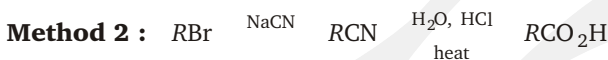


10. Rate limiting S_N1 follows the sequence

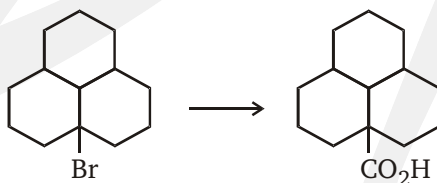


True statement about sequence on the basis of assumption that R contains 3 different groups is :

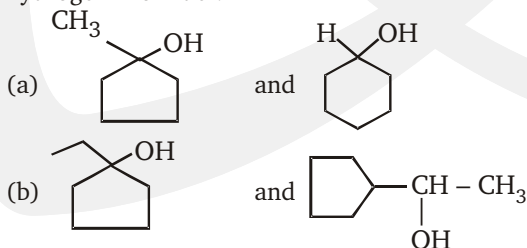
- (a) more stable carbocation, greater is in the proportion of racemization
 (b) the more nucleophilic the solvent greater in the proportion of inversion
 (c) In above sequence (b) represent separately solvated, pair of ions
 (d) All of these
11. Compare the two methods shown for the preparation of carboxylic acids :

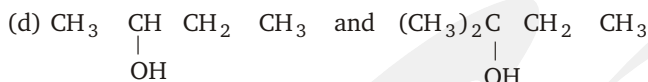
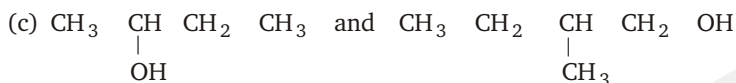


Which one of the following statements correctly describes this conversion ?

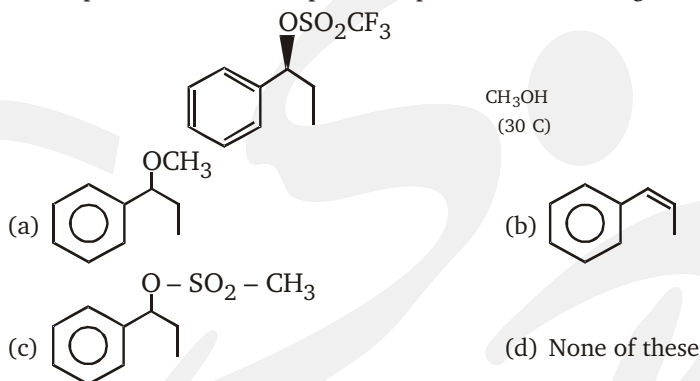


- (a) Both method 1 and method 2 are appropriate for carrying out this conversion
 (b) Neither method 1 nor method 2 is appropriate for carrying out this conversion
 (c) Method 1 will work well, but method 2 is not appropriate
 (d) Method 2 will work well, but method 1 is not appropriate
12. Which of the following statements is true ?
- (a) CH_3CH_2S is both a stronger base and more nucleophilic than CH_3CH_2O
 (b) CH_3CH_2S is a stronger base but is less nucleophilic than CH_3CH_2O
 (c) CH_3CH_2S is a weaker base but is more nucleophilic than CH_3CH_2O
 (d) CH_3CH_2S is both a weaker base and less nucleophilic than CH_3CH_2O
13. In the given pair of alcohols, in which pair second alcohol is more reactive than first towards hydrogen bromide ?

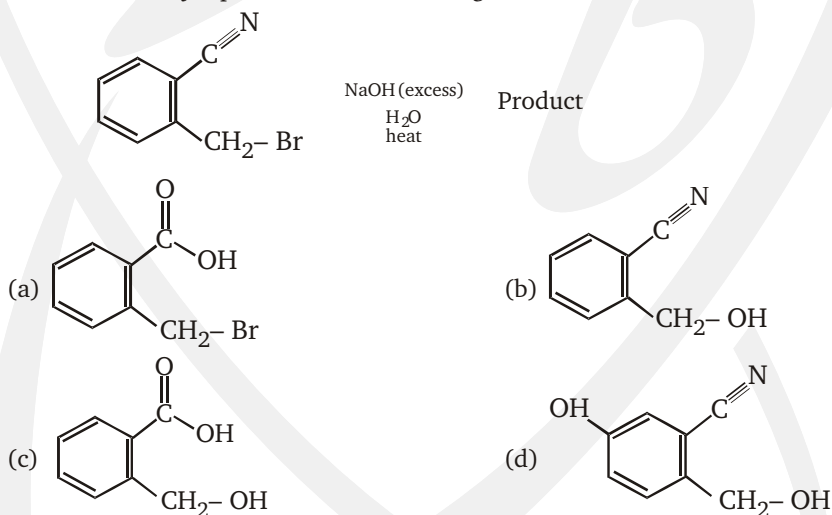




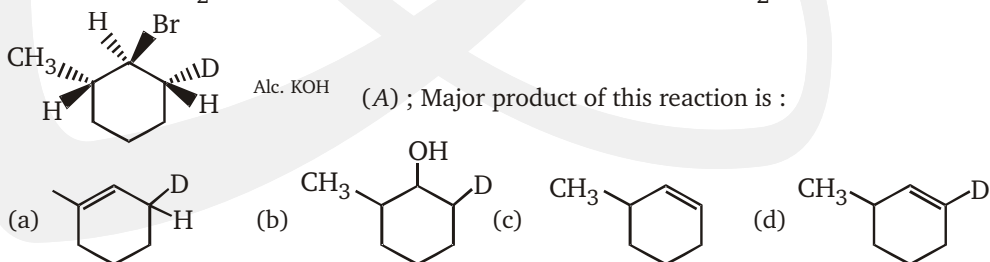
14. Which product would be expected to predominate in the given reaction ?



15. Which is the major product of the following reaction ?



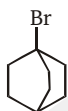
16. (A) ; Major product of this reaction is :



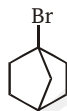
17. Rate of S_N2 reaction is :



(A)



(B)



(C)

(a) $(B) > (A) > (C)$

(c) $(A) > (B) > (C)$

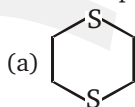
(b) $(C) > (A) > (B)$

(d) $(A) > (C) > (B)$

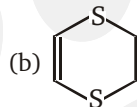
18. 1,2-dichloro ethane + $\text{NaSCH}_2\text{CH}_2\text{SNa}$

$\text{C}_4\text{H}_8\text{S}_2$ (P)

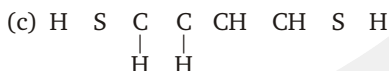
Unknown product (P) of the above reaction is :



(a)



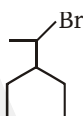
(b)



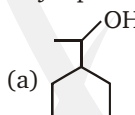
(c)



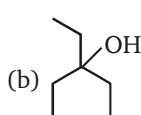
(d)

19.  Moist Ag_2O (A) product

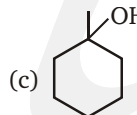
Major product (A) is :



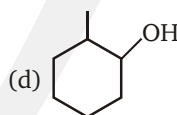
(a)



(b)

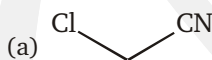


(c)

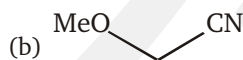


(d)

20.  KCN Product of reaction is :



(a)



(b)

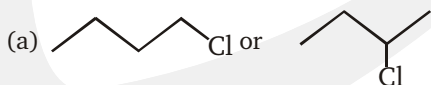


(c)

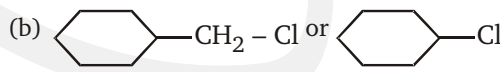


(d)

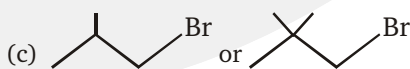
21. In the given pair of compound, in which pair the second compound is more reactive than first toward S_N2 reaction ?



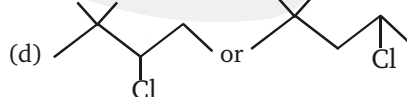
(a)



(b)

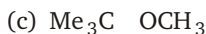


(c)



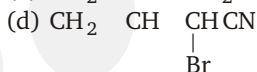
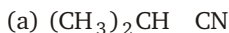
(d)

22. Which compound might be synthesized by the S_N2 displacement of an alkyl-halide ?

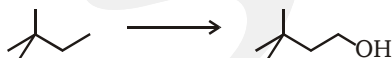


(d) All of these

23. Identify C in the following series $\text{C}_3\text{H}_7\text{I}$ $\xrightarrow[\text{alc.}]{\text{KOH}}$ A $\xrightarrow{\text{NBS}}$ B $\xrightarrow[\text{alc.}]{\text{KCN}}$ C.



24. What sequence of reagents is required to accomplish the following transformation ?



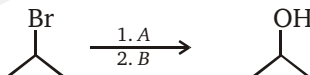
(a) (1) NBS, ROOR (2) $\text{CH}_3\text{CH}_2\text{O}$ (3) 2HBr (4) NH_2 (5) disiamyl borane (6) $\text{H}_2\text{O}_2, \text{OH}$

(b) (1) $\text{Cl}_2, h\nu$ (2) OH^- , heat; (3) 2HCl (4) OH^- , heat (5) $\text{HgSO}_4, \text{H}_2\text{SO}_4$

(c) (1) NBS, ROOR; OH^- , DMSO

(d) (1) $\text{Br}_2, h\nu$ (2) *t*-butoxide (3) BH_3 , THF (4) $\text{H}_2\text{O}_2, \text{OH}^-$

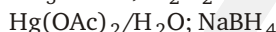
25. Which of the reagents shown below would accomplish the following transformations?



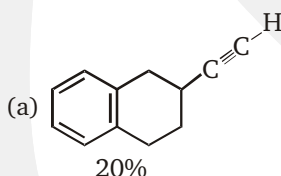
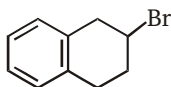
A



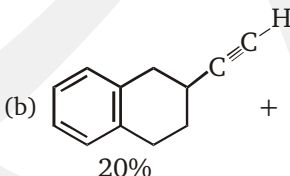
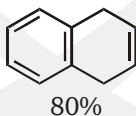
B



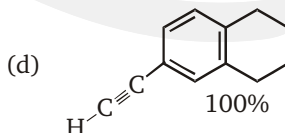
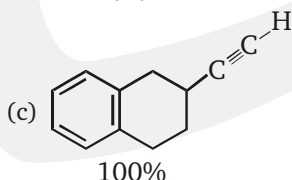
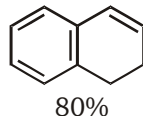
26. What are the products obtained from the following reaction ?



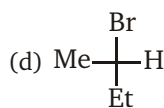
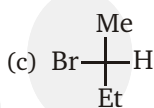
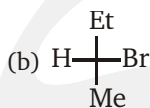
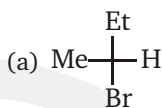
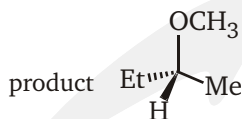
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27. The back-side attack on 2-bromobutane by methoxide (CH_3O^-) gives the product shown below. Which Fischer projection represents 2-bromobutane used as the reactant in this reaction?



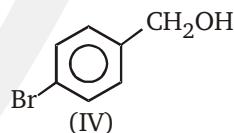
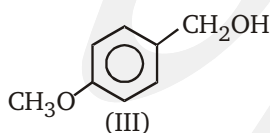
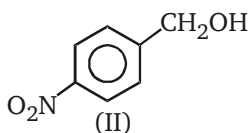
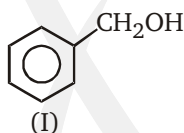
28. Consider the following statements :

- (1) Bridgehead halides are inert towards both $\text{S}_{\text{N}}1$ and $\text{S}_{\text{N}}2$ reactions (till one of the ring size is eight member ring)
- (2) The first step in both $\text{S}_{\text{N}}1$ and E_1 reactions is the same
- (3) $\text{S}_{\text{N}}2$ reactions proceed with total retention of configuration
- (4) E_2 eliminations are by the use of a solvent of low polarity and high concentration of a strong base

Which of the above statements are correct?

- (a) 1, 2 and 4
- (b) 1 and 3
- (c) 2, 3 and 4
- (d) 1, 2, 3 and 4

29. Consider the following alcohols :



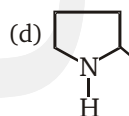
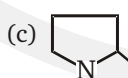
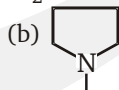
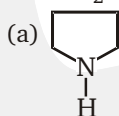
The order of decreasing reactivities of these alcohols towards substitution with HBr is :

- (a) $\text{III} > \text{I} > \text{IV} > \text{II}$
 - (b) $\text{III} > \text{I} > \text{II} > \text{IV}$
 - (c) $\text{I} > \text{III} > \text{IV} > \text{II}$
 - (d) $\text{I} > \text{III} > \text{II} > \text{IV}$
30. In solvolysis of 1,2-dimethyl propyl *p*-toluene sulfonate in acetic acid at 75°C , how many (alkene substitution) products will be formed ?
- (a) 2
 - (b) 3
 - (c) 4
 - (d) 5

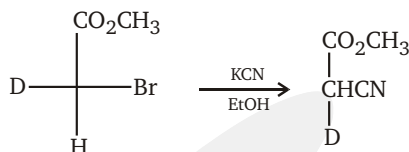
31. Benzotrichloride reacts with milk of lime to form :

- (a) Benzal
- (b) Benzoic acid
- (c) Benzyl alcohol
- (d) Phenol

32. $\text{Br}-\text{CH}_2-(\text{CH}_2)_2-\text{CH}_2-\text{Br}$ CH_3NH_2 Product of the reaction is :



33. The configurations of the reactant and the product in the following reaction, respectively, are:



(a) R, R

(b) R, S

(c) S, R

(d) S, S

34. 1-4-dichlorohexane (1 mole) + NaI (1 mole) $\xrightarrow{\text{Acetone}}$ Product of the reaction is :

(a) $\text{Cl} - \text{CH}_2 - \text{CH}_2 - \underset{\text{I}}{\text{CH}} - \text{CH}_2 - \text{CH}_3$ (b) $\text{I} - \text{CH}_2 - \text{CH}_2 - \underset{\text{Cl}}{\text{CH}} - \text{CH}_2 - \text{CH}_3$ (c) $\text{H}_2\text{C} = \text{CH} - \underset{\text{Cl}}{\text{CH}} - \text{CH}_2 - \text{CH}_3$ (d) $\text{I} - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \underset{\text{Cl}}{\text{CH}} - \text{CH}_2 - \text{CH}_3$

35. Alkyl halides can be obtained by all methods except :

(a) $\text{CH}_3\text{CH}_2\text{OH} \quad \text{HCl/ZnCl}_2$ (b) $\text{CH}_3 - \text{CH}_2 - \text{CH}_3 - \text{CH}_2 \quad \text{Cl}_2/\text{UV light}$ (c) $\text{C}_2\text{H}_5\text{OH} \quad \text{NaCl}$ (d) $\text{CH}_3\text{COOAg} \quad \text{Br}_2/\text{CCl}_4$

36. In order to prepare 1-chloropropane, which of the following reactants can be employed ?

(a) Propene and HCl in the presence of peroxide

(b) Propene and Cl_2 followed by treatment with aq. KOH(c) Propanol-1 and SOCl_2 /pyridine

(d) Any of the above can be used

37. Which alkyl halide has maximum density ?

(a) $\text{C}_3\text{H}_7\text{I}$ (b) $\text{C}_2\text{H}_5\text{I}$ (c) CH_3I (d) CH_3Br

38. Which of the following molecules would have a carbon-halogen bond most susceptible to nucleophilic substitution ?

(a) 2-fluorobutane

(b) 2-chlorobutane

(c) 2-bromobutane

(d) 2-iodobutane

39. When benzyl chloride is treated with ethanolic KCN, the major product formed is :

(a) benzyl ethyl ether

(b) benzyl alcohol

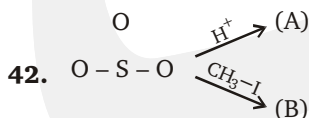
(c) benzyl cyanide

(d) benzyl isocyanide

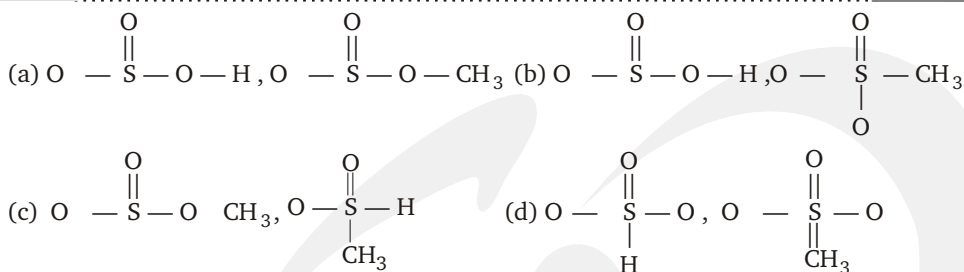
40. Which of the following is most reactive towards nucleophilic substitution reaction ?

(a) $\text{CH}_2 = \text{CH} - \text{Cl}$ (b) $\text{C}_6\text{H}_5\text{Cl}$ (c) $\text{CH}_3\text{CH} = \text{CHCl}$ (d) $\text{ClCH}_2 - \text{CH} = \text{CH}_2$

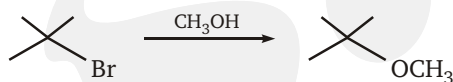
41. Which of the following reaction will not give ether as a major product ?

(a) $\text{CH}_3\text{CH}_2\text{Cl} \quad \text{Ag}_2\text{O}(\text{dry})$ (b) $(\text{CH}_3)_3\text{CCl} \quad \text{CH}_3\text{CH}_2\text{O}^- \text{Na}^+$ (c) $\text{CH}_3\text{CH}_2\text{Cl} \quad \text{Na}^+ \text{O}^- - \text{C}_6\text{H}_5$ (d) $\text{CH}_3\text{Cl} \quad \text{Na}^+ \text{O}^- - \text{C}(\text{CH}_3)_3$ 

Product (A) and (B) in above reaction is :

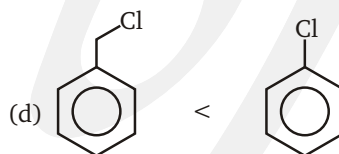
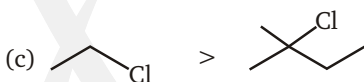


43. Which of the following would be true for the reaction shown ?



- (a) The rate of the reaction depends only on the alkyl bromide concentration.
 (b) The rate of the reaction depends only on the methanol concentration.
 (c) The rate of the reaction depends on both the alkyl halide concentration and the methanol concentration.
 (d) The rate of the reaction depends on the concentration of neither reactant.

44. The correct order of rate of $\text{S}_{\text{N}}1$ is :



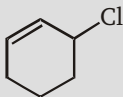
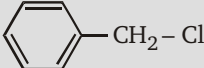
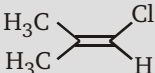
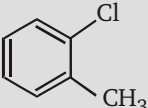
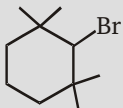
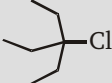
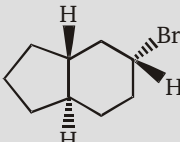
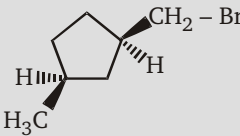
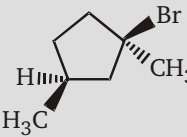
ANSWERS — LEVEL 1

1.	(c)	2.	(c)	3.	(b)	4.	(a)	5.	(a)	6.	(a)	7.	(c)	8.	(b)
9.	(b)	10.	(d)	11.	(c)	12.	(c)	13.	(d)	14.	(a)	15.	(c)	16.	(c)
17.	(c)	18.	(a)	19.	(c)	20.	(b)	21.	(d)	22.	(d)	23.	(b)	24.	(d)
25.	(d)	26.	(b)	27.	(d)	28.	(a)	29.	(a)	30.	(d)	31.	(b)	32.	(b)
33.	(d)	34.	(d)	35.	(c)	36.	(c)	37.	(a)	38.	(d)	39.	(c)	40.	(d)
41.	(b)	42.	(b)	43.	(a)	44.	(b)								

Level - 2

1. The following organic halide derivatives (A to J) are reacted in ethanol solution with each of the nucleophiles : acetate, methylthiolate, cyanide and hydroxide anions. Six possible results from these combinations of reactants are designated (1) through (6) below :

Write the number corresponding to your best estimate of the outcome of each reaction in the appropriate answer box below.

		$\text{CH}_3 - \text{I}$		
A	B	C	D	E
				
F	G	H	I	J

Possible Outcome :

- | | |
|--------------------------------------|-------------------------------------|
| (1) No reaction | (2) Substitution |
| (3) Elimination | (4) Substitution and elimination |
| (5) No reaction or slow substitution | (6) No reaction or slow elimination |

Compound		A	B	C	D	E	F	G	H	I	J
(i)	$\text{CH}_3\text{CO}_2\text{Na}$										
(ii)	CH_3SNa										
(iii)	NaCN										
(iv)	NaOH										

2. In each of the following sections three organic halogen compounds are listed. In the box given enter a number (1 to 3) indicating the order of reactivity of the designated (1 is most reactive and 3 is least).

(a) S_N2 substitution by NaOCOCH_3 in methanol:

1. $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$ ☐ 2. $(\text{CH}_3)_2\text{CHBr}$ ☐ 3. $\text{CH}_2=\text{CHCH}_2\text{Br}$ ☐

(b) S_N2 substitution by NaI in acetone:

1. $\text{C}_6\text{H}_5\text{Cl}$ ☐ 2. $\text{C}_6\text{H}_5\text{CH}_2\text{Cl}$ ☐ 3. $\text{C}_6\text{H}_5\text{CHClCH}_3$ ☐

(c) S_N2 substitution by NaCN in methanol:

1. $\text{CH}_3\text{CH}_2\text{Cl}$ ☐ 2. $\text{CH}_3\text{CH}_2\text{F}$ ☐ 3. $\text{CH}_3\text{CH}_2\text{I}$ ☐

(d) S_N2 substitution by NaSCH_3 in methanol:

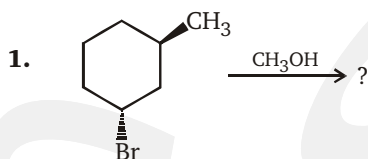
1. $(\text{CH}_3)_2\text{CHCH}_2\text{CH}_2\text{Br}$ ☐ 2. $\text{CH}_3\text{CH}_2\text{CHBrCH}_2\text{CH}_3$ ☐ 3. $(\text{CH}_3)_3\text{CCH}_2\text{Br}$ ☐

3. Isobutyl alcohol (2-methyl-1-propanol), $(\text{CH}_3)_2\text{CHCH}_2\text{OH}$, can be transformed to each of the compounds (a through l) listed in the left-hand column. In each case the number of steps needed to accomplish the change is noted, and an answer box is provided for your reagent selections. Fourteen reagents (designated A through N) are listed in the right-hand column.

Write letters designating the reagent or reagents you believe will achieve the desired transformation in the box to the right of the product formula. In the case of a multi-step sequence write the reagents in the order they are to be used. In some cases you may wish to use a previously prepared compound as a reactant. If so, write the number (a to l) corresponding to the desired compound.

Desired product		No. of Steps	Write Options	Reagent List	
a.	$(\text{CH}_3)_2\text{CHCH}_2\text{Br}$	one		A.	$\text{Hg}(\text{OAc})_2$ in H_2O
b.	$(\text{CH}_3)_2\text{C}=\text{CH}_2$	one		B.	PBr_3 & heat
c.	$(\text{CH}_3)_2\text{CHCH}=\text{O}$	one		C.	NaBH_4 in alcohol
d.	$(\text{CH}_3)_2\text{CHCO}_2\text{H}$	one		D.	LiAlH_4 in THF (aqueous workup)
e.	$(\text{CH}_3)_3\text{CBr}$	two		E.	NaCN in alcohol
f.	$(\text{CH}_3)_2\text{CHCH}_2\text{C}\equiv\text{N}$	two		F.	PCC in CH_2Cl_2
g.	$(\text{CH}_3)_2\text{CHCH}_2\text{OCOCH}_3$	one		G.	Jones' reagent (CrO_3 in H_3O^+)
h.	$(\text{CH}_3)_2\text{CHCO}_2\text{C}_2\text{H}_5$	two		H.	HBr in CH_2Cl_2
i.	$(\text{CH}_3)_2\text{CHCH}_2\text{OCH}_2(\text{CH}_3)$	two		I.	H_3PO_4 and heat
j.	$(\text{CH}_3)_3\text{COH}$	three		J.	$(\text{CH}_3\text{CO})_2\text{O}$ + pyridine
k.	$(\text{CH}_3)_2\text{CHCH}_2\text{NH}_2$	three		K.	NaN_3 in aqueous alcohol
l.	$(\text{CH}_3)_2\text{CHCH}_2\text{CH}_2\text{NH}_2$	two		L.	$\text{C}_6\text{H}_5\text{CO}_3\text{H}$ in CH_2Cl_2 (peracid)
				M.	NaH in ether and heat
				N.	$\text{C}_2\text{H}_5\text{OH}$ + acid catalyst & heat

Subjective Problems



X Total number of substitution and elimination product(s). Find the value of X.

ANSWERS — LEVEL 2

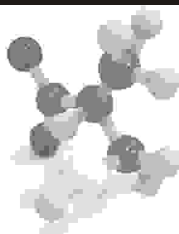
1.	A	B	C	D	E	F	G	H	I	J
(i)	2	2	2	1	1	1	6	2	2	6
(ii)	2	2	2	1	1	5	6	2	2	6
(iii)	2	2	2	1	1	1	3	3	2	3
(iv)	4	2	2	1	1	5	3	3	4	3

2. $a - 3 > 1 > 2$; $b - 2 > 3 > 1$; $c - 3 > 1 > 2$; $d - 1 > 2 > 3$

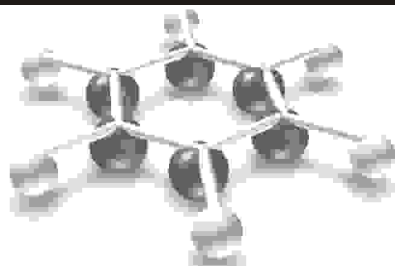
3. $a - B$; $b - I$; $c - F$; $d - G$; $e - I, H$ or $2 H$; $f - B, E$ or $1, E$; $g - J$; $h - G, N$ or $4N$
 $i - N, j - I, A, C$ or $2AC$ or ILD or $2LD$; $k - B, K, D$ or $1KD$; $l - B, E, D$ or $1ED$ or $6D$

Subjective Problems

1. 4



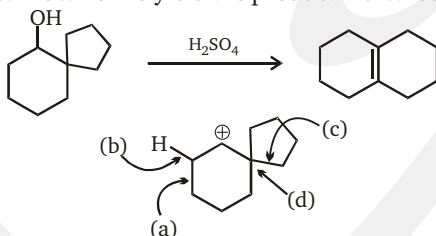
6



ALCOHOL, ETHERS AND EPOXIDES

Level - 1

1. The following transformation involves a carbocation rearrangement. The carbocation is generated by protonation of the hydroxyl group, followed by the loss of water. Which bond has to migrate in the carbocation to yield the product indicated (after the deprotonation) ?



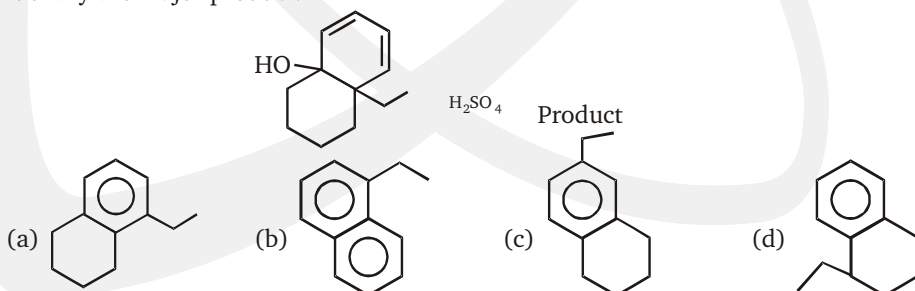
(a) a

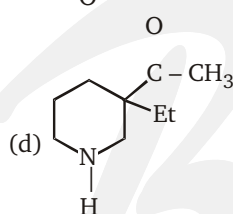
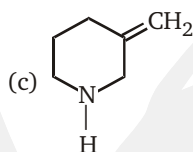
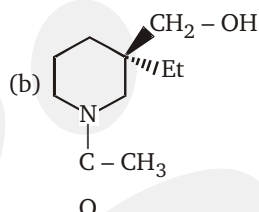
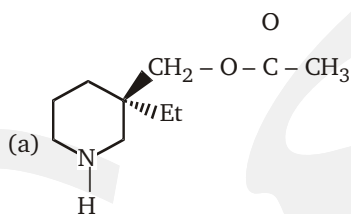
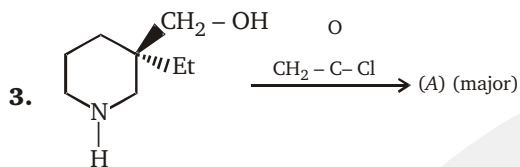
(b) b

(c) c

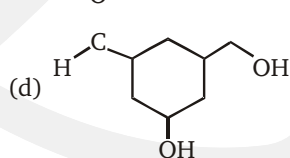
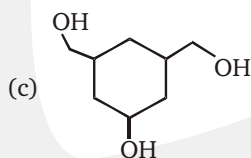
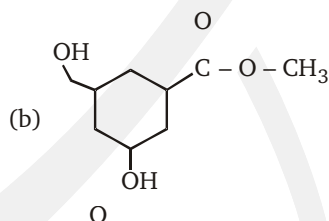
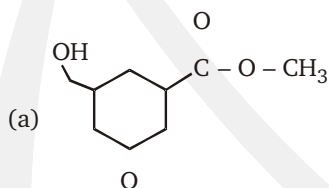
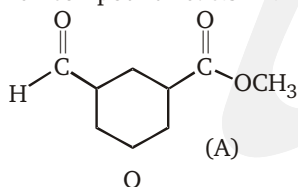
(d) d

2. Identify the major product.

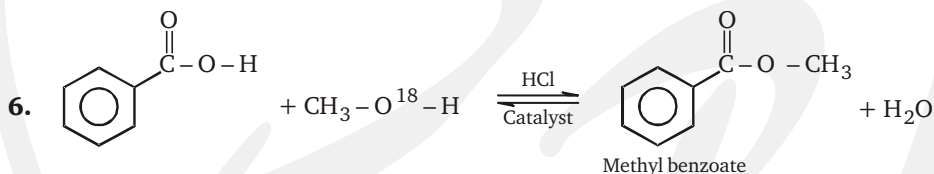
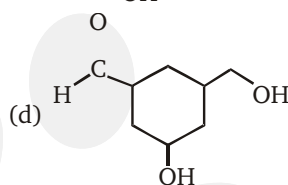
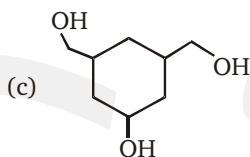
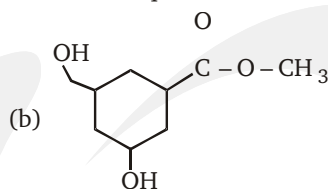
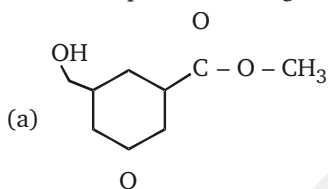




4. Predict the product when given compound reacts with LiAlH_4 :

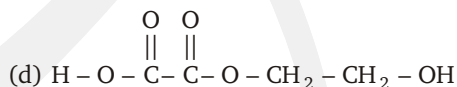
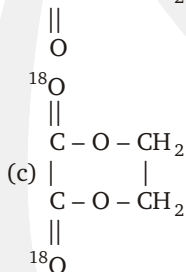
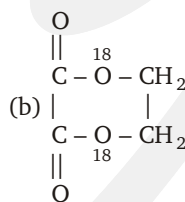
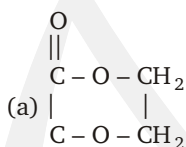
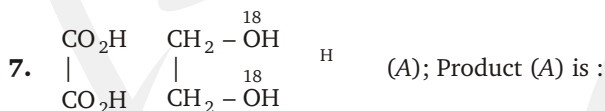


5. Predict the product when given compound (A, in the above question 4) reacts with NaBH_4 .

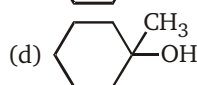
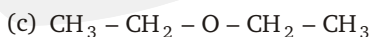


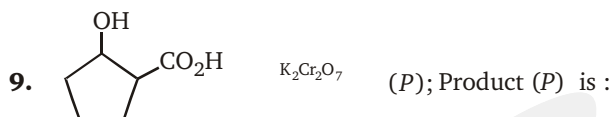
The labelled $-\text{O}^{18}$ will be in :

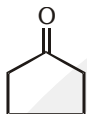
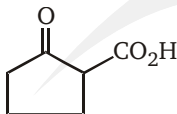
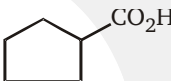
- (a) H_2O (b) Methyl benzoate
(c) Both (a) and (b) (d) Benzoic acid



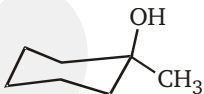
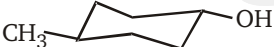
8. Which is oxidized most easily ?

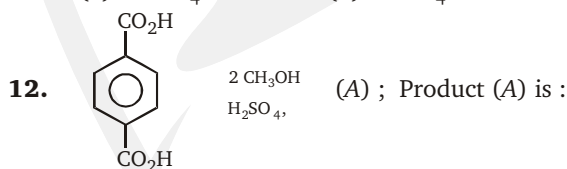
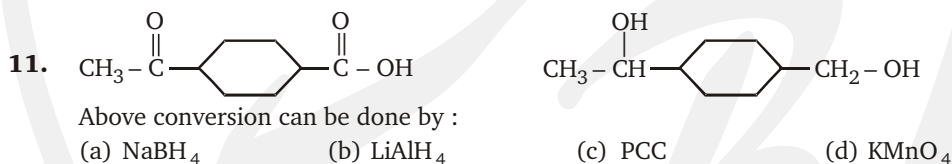


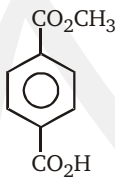
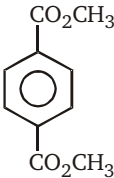
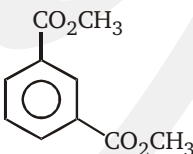
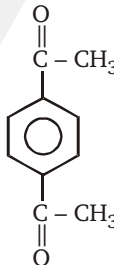


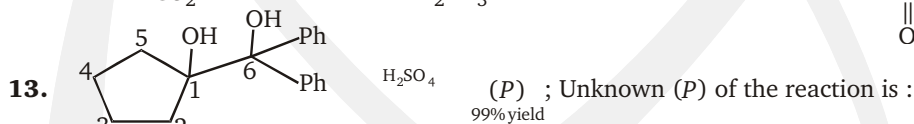
- (a)  (b)  (c)  (d) 

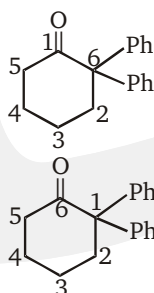
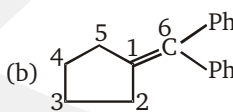
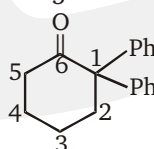
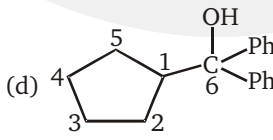
10. Which of the following react with HBr at faster rate ?

- (a)  (b) 
 (c)  (d) 

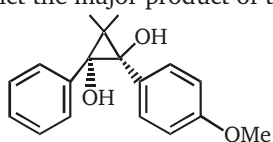


- (a)  (b) 
 (c)  (d) 

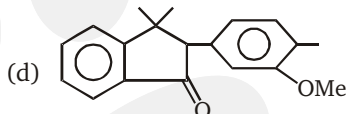
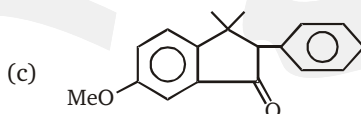
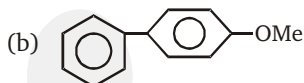
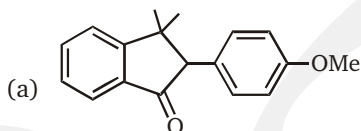


- (a)  (b) 
 (c)  (d) 

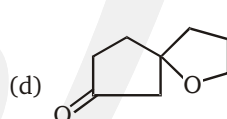
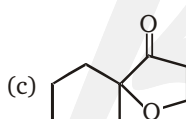
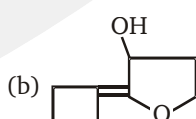
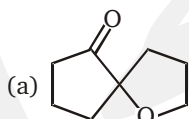
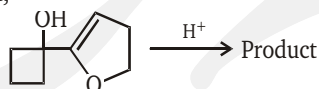
14. Predict the major product of the given reaction.



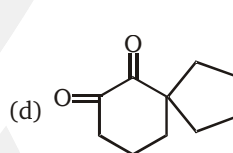
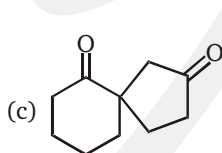
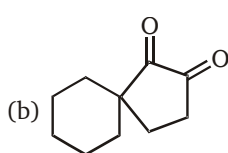
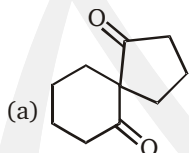
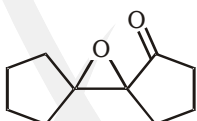
(A); Product (A) is :



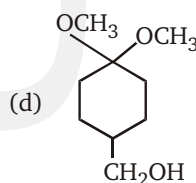
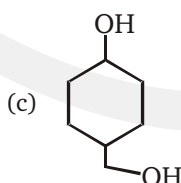
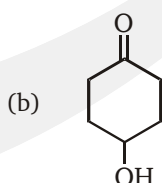
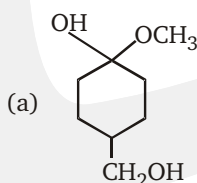
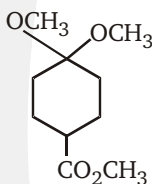
15. Identify the major product,

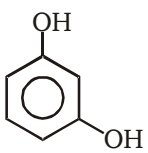


16. (A); Product (A) is :

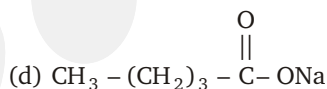
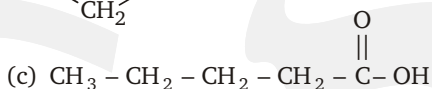
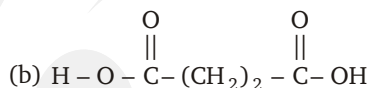
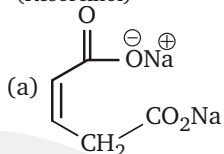


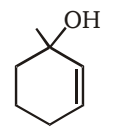
17. (A) Major; product (A) is :



18.  $\xrightarrow{\text{I}_2, \text{NaOH}}$ CHI_3 (Iodoform) (A); Compound (A) is :

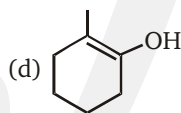
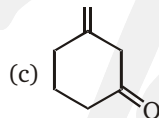
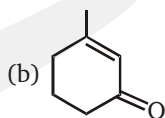
(Resorcinol)

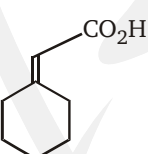


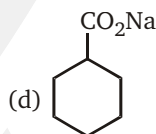
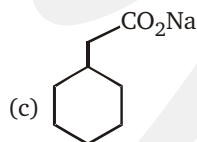
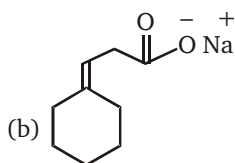
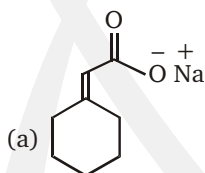
19.  $\xrightarrow{\text{H}_2\text{CrO}_4}$ (A). Product (A) is :

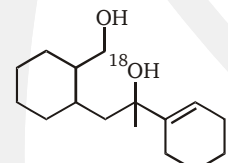
(3° alcohol)

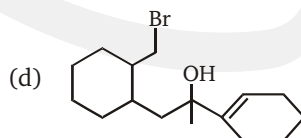
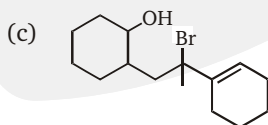
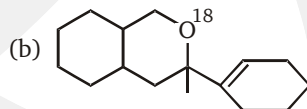
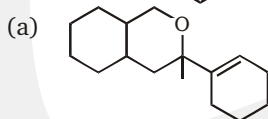
(a) No reaction



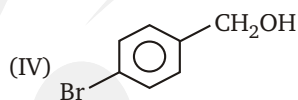
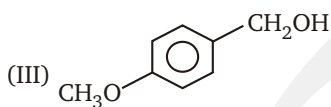
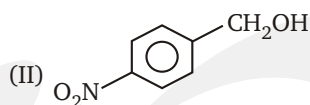
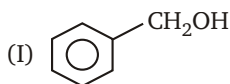
20.  $\xrightarrow[\text{(2) H}]{\text{(1) CH}_3 \text{ Li(excess)}}$ (A) $\xrightarrow[\text{NaOH}]{\text{I}_2}$ (B) CHI_3 ; Compound (B) is :



21.  $\xrightarrow{\text{H Br, CCl}_4}$ Major product obtained in the reaction is :



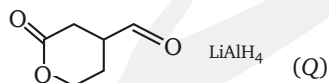
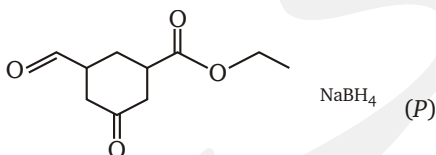
22. Consider the following alcohols,



The order of decreasing reactivities of these alcohols towards nucleophilic substitution with HBr is:

- (a) III I IV II (b) III I II IV (c) I III IV II (d) I III II IV

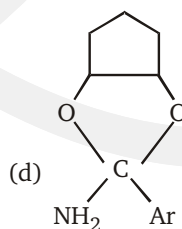
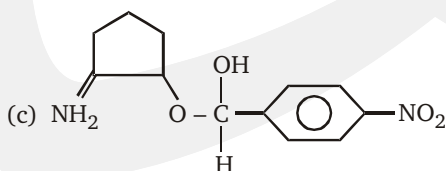
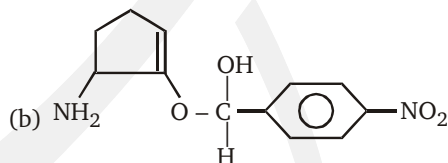
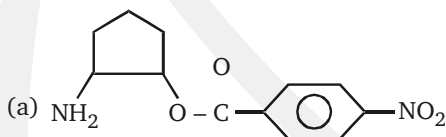
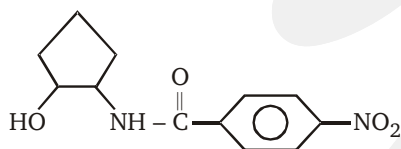
23.

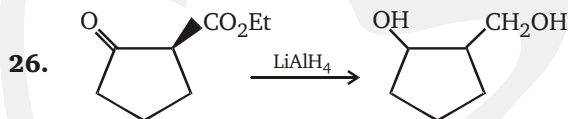
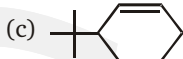
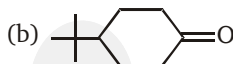
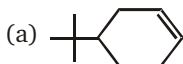
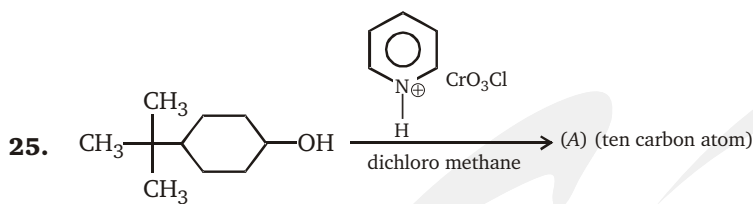


Sum of number of alcoholic groups in product (P) and (Q) is:

- (a) 1 (b) 2 (c) 3 (d) 5

24. In presence of dil. HCl, compound A is converted to a constitutional isomer (B), compound B is:

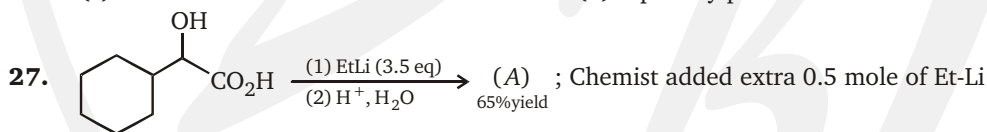




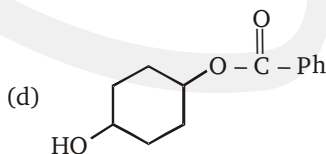
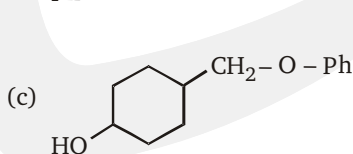
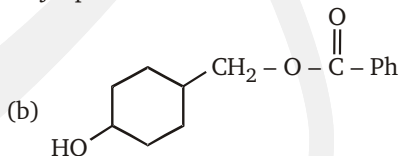
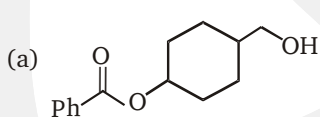
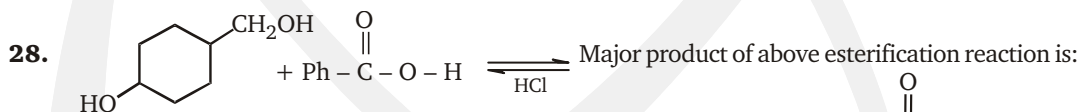
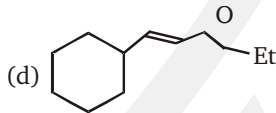
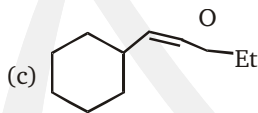
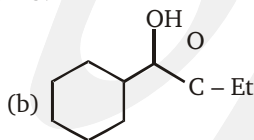
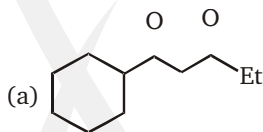
; Product of the reaction is:

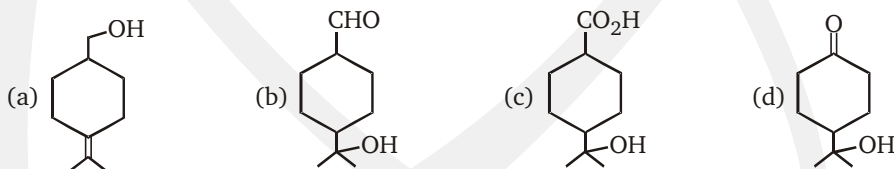
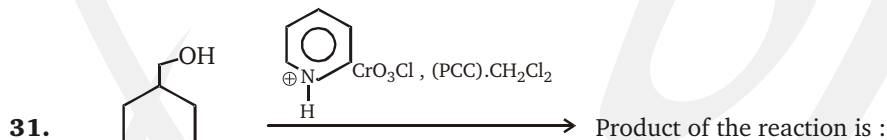
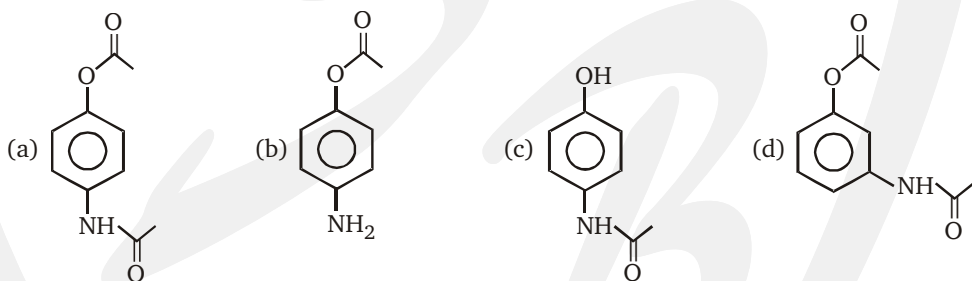
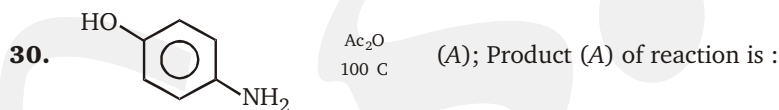
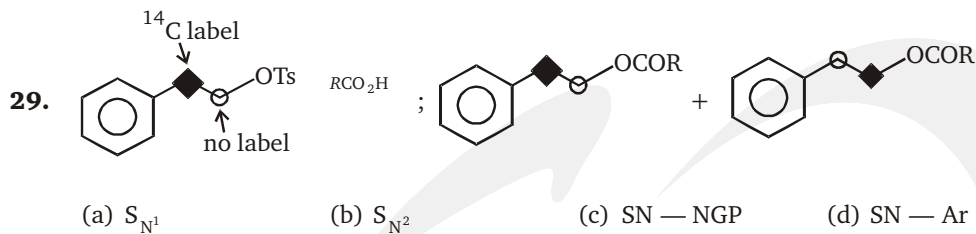
- (a) Racemic
(c) Meso

- (b) Diastereomer
(d) Optically pure

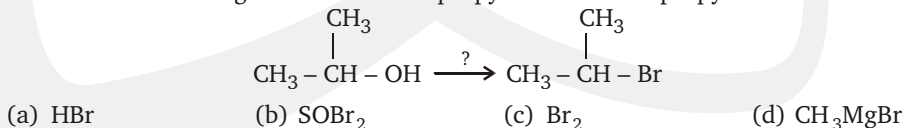


in above reaction to obtain product (A), which is?

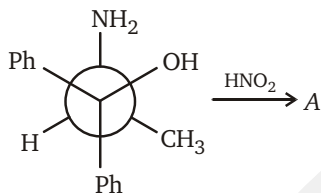




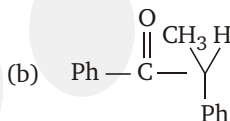
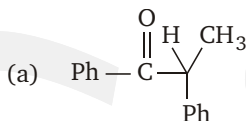
32. Which is the best reagent to convert isopropyl alcohol to isopropyl bromide ?



33.



Major product obtained in the above reaction is :



(c) Racemic

(d) Diastereomers

34.

 HIO_4

(A)

(1) LiAlH_4 (excess)
(2) H^+

(B)

Total number of stereoisomers of product (B) will be:

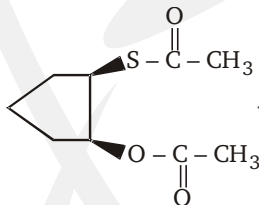
(a) 2

(b) 3

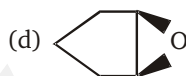
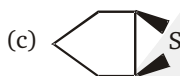
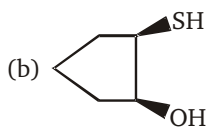
(c) 4

(d) 5

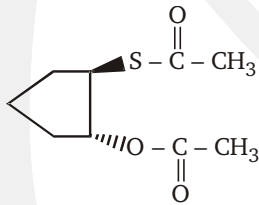
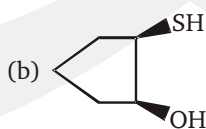
35.

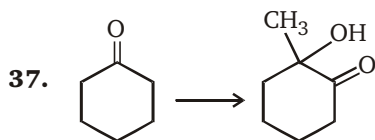
1. HO^-
2. H^+

Major product of the reaction is :

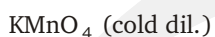


36.

 HO^- 



(1)



(2)



(3)



(4)

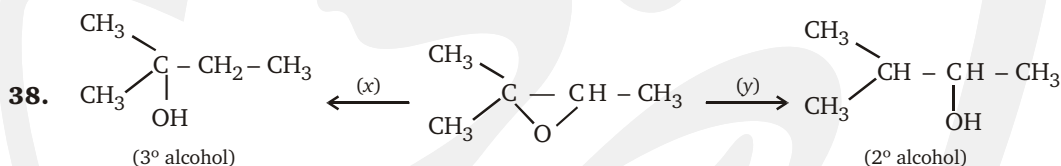
For the above conversion the correct order of reagents used is :

(a) 1 2 3 4

(b) 1 4 3 2

(c) 1 4 2 3

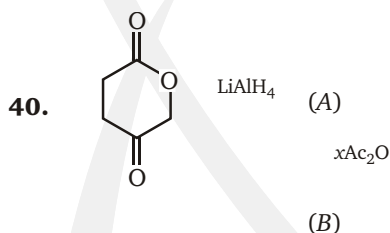
(d) 2 3 4 1



Find missing reagents.

(a) $x = \text{LiAlH}_4, y = \text{NaBH}_4$ (b) $x = \text{LiAlH}_4/\text{AlCl}_3, y = \text{LiAlH}_4$ (c) $x = \text{LiAlH}_4, y = \text{LiAlH}_4/\text{AlCl}_3$ (d) $x = \text{H}_2/\text{Ni}, y = \text{H}_2/\text{Pt}$

39. In solvolysis of 1, 2-dimethyl propyl p-toluene sulfonate in acetic acid at 75°C , (alkene + substitution products) will be formed by mechanism ?

(a) $\text{S}_{\text{N}}2, \text{E}_2$ (b) $\text{S}_{\text{N}}2, \text{E}_1$ (c) $\text{S}_{\text{N}}1, \text{E}_2$ (d) $\text{S}_{\text{N}}1, \text{E}_1$  x = moles of anhydride consumed

(a) 1

(b) 2

(c) 3

(d) 4

41. Identify product when (*R*) – and (*S*) – 2 – butanol reacts with (*R,R*) tartaric acid in acidic medium.

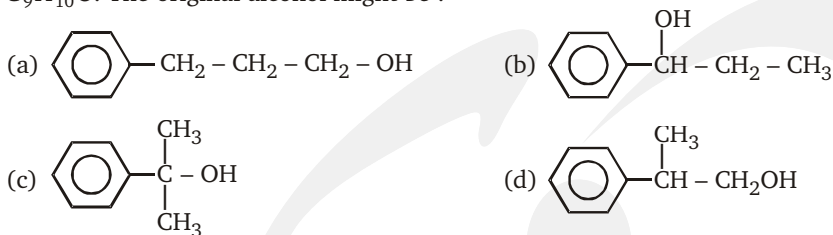
(a) Racemic

(b) Diastereomer

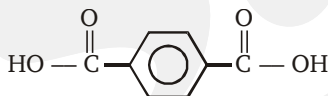
(c) Meso

(d) Pure enantiomer

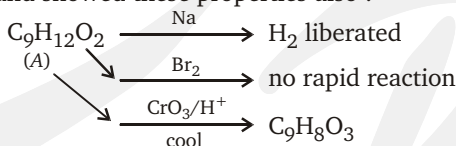
42. An alcohol of formula $C_9H_{12}O$ reacts with $Na_2Cr_2O_7$ to form a compound having formula $C_9H_{10}O$. The original alcohol might be :



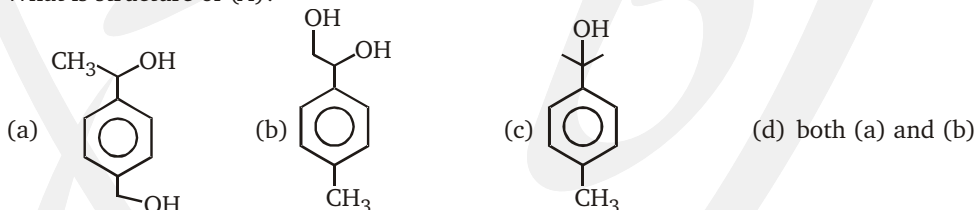
43. An optically active alcohol of formula $C_9H_{12}O_2$ produced the following compound when refluxed with $KMnO_4$.



The original compound showed these properties also :



What is structure of (A)?



44. Which are not cleaved by HIO_4 ?

I : glycerol

III : 1, 3-propenediol

(a) I, II, III, IV

(c) II, III

II : glycol

IV : methoxy-2-propanol

(b) I, II

(d) III, IV

45. Which of the following reactions require an oxidising agent ?

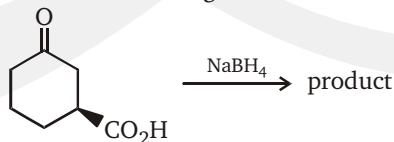
(a) $CH_3 - CH - CH_2$ $CH_3 - CH_2 - CH_3$

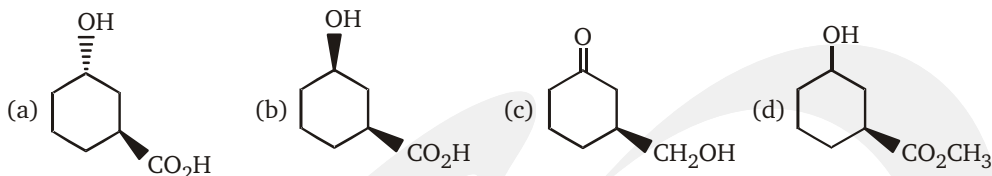
(b) $CH_3 - CH_2OH$ CH_3CHO

(c) $CH_3 - CH_2Cl$ $CH_3 - CH_3$

(d) $CH_3 - CH_2OH$ $CH_3 - CH_2Cl$

46. What is the major product of the following reaction ?





47. Which of the esters shown, after reduction with LiAlH_4 and aqueous workup, will yield two molecules of only a single alcohol ?

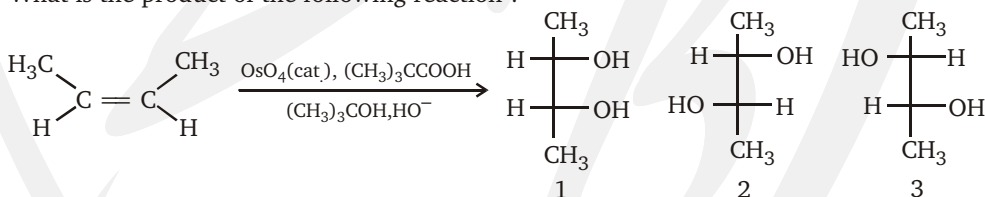
- (a) $\text{CH}_3\text{CH}_2\text{CO}_2\text{CH}_2\text{CH}_3$ (b) $\text{C}_6\text{H}_5\text{CO}_2\text{CH}_2\text{C}_6\text{H}_5$
(c) $\text{C}_6\text{H}_5\text{CO}_2\text{C}_6\text{H}_5$ (d) None of these

48. For the following reaction, select the statement that best describes the change.



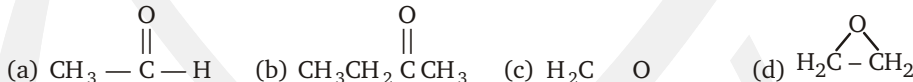
- (a) The alcohol is oxidized to an acid, and the Cr(VI) is reduced
(b) The alcohol is oxidized to an aldehyde, and the Cr(VI) is reduced
(c) The alcohol is reduced to an aldehyde, and the Cr(III) is oxidized
(d) The alcohol is oxidized to a ketone, and the Cr(VI) is reduced

49. What is the product of the following reaction ?

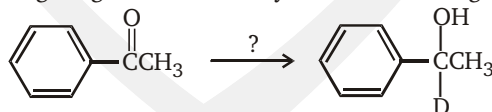


- (a) Only 1 (b) 1 : 1 mixture of 2 and 3
(c) Only 2 (d) 1 : 1 : 1 mixture of 1, 2, and 3

50. An organic compound *B* is formed by the reaction of ethylmagnesium iodide ($\text{CH}_3\text{CH}_2\text{MgI}$) with a substance *A*, followed by treatment with dilute aqueous acid. Compound *B* does not react with PCC in dichloromethane. Identify *A* ?

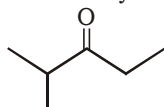


51. Which of the following reagents would carry out the following transformation ? ($\text{D} = {}^2\text{H}$)



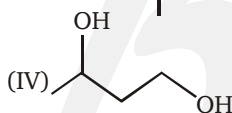
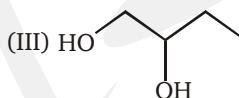
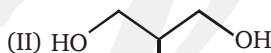
- (a) NaBD_4 in CH_3OH (b) LiAlH_4 , then D_2O
(c) NaBD_4 in CH_3OD (d) LiAlD_4 , then D_2O

52. Which sequence of steps describes the best synthesis of 2-methyl-3-pentanone ?



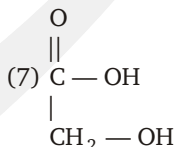
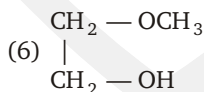
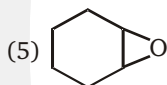
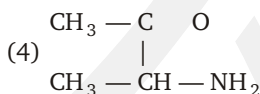
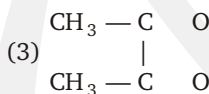
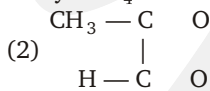
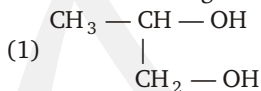
- (a) (1) 1-Propanol + $(\text{CH}_3)_2\text{CHMgBr}$, diethyl ether
 (2) H_3O^+
 (3) PCC , CH_2Cl_2
 (b) (1) 1-Propanol + $\text{Na}_2\text{Cr}_2\text{O}_7$, H_2SO_4 , H_2O , heat
 (2) SOCl_2
 (3) $(\text{CH}_3)_2\text{CHCl}$, AlCl_3
 (c) (1) 1-Propanol + PCC , CH_2Cl_2
 (2) $(\text{CH}_3)_2\text{CHLi}$, diethyl ether
 (3) H_3O^+
 (4) $\text{Na}_2\text{Cr}_2\text{O}_7$, H_2SO_4 , H_2O , heat
 (d) (1) 2-Propanol + $\text{Na}_2\text{Cr}_2\text{O}_7$, H_2SO_4 , H_2O , heat
 (2) $\text{CH}_3\text{CH}_2\text{CH}_2\text{Li}$, diethyl ether
 (3) H_3O^+
 (4) PCC , CH_2Cl_2

53. Diols (I-IV) which react with CrO_3 in aqueous H_2SO_4 and yield products that readily undergo decarboxylation on heating, are :

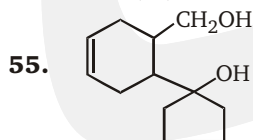


- (a) I and II (b) II and III (c) II and IV (d) I and IV

54. Which of the following compounds are not oxidized by HIO_4 ?



- (a) 5, 6, 7 (b) 4, 5, 6, 7 (c) 6, 7 (d) 3, 4, 5, 6, 7

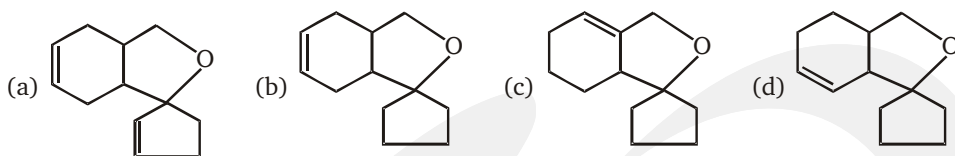


TsCl
Pyridine
 0°C

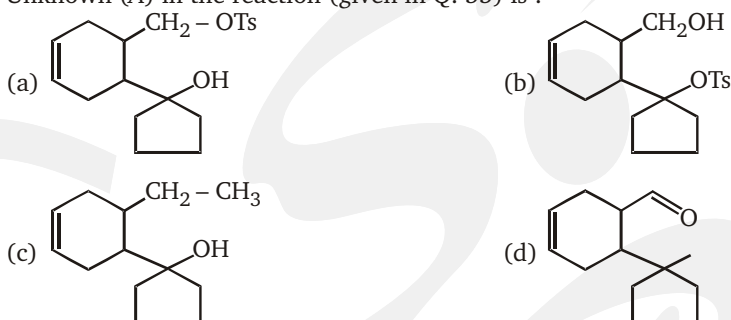
(A)

80°C
(HMPT Solvent)

(B) 84%; Final product (B) will be :



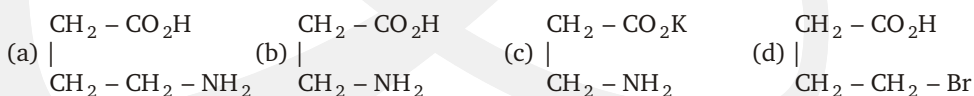
56. Unknown (A) in the reaction (given in Q. 55) is :



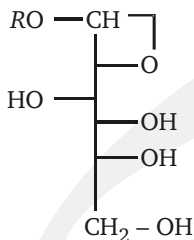
57. In the given table, identify the incorrect option. The digit in box indicate the moles of that substance.

Reactant	HIO_4 consumed	HCO_2H formed	HCHO formed
(a) $\text{HO}-\text{CH}_2-\overset{\text{OH}}{\underset{ }{\text{CH}}}-\text{CH}_2-\text{OH}$	2	1	2
(b) $\text{R}-\overset{\text{OH}}{\underset{ }{\text{CH}}}-\overset{\text{OH}}{\underset{ }{\text{CH}}}-\overset{\text{OH}}{\underset{ }{\text{CH}}}-\text{CH}_2-\text{OH}$	3	2	1
(c) $\text{HO}-\text{CH}_2-\overset{\text{OCH}_3}{\underset{ }{\text{CH}}}-\text{CH}_2\text{OH}$	0	0	0
(d) $\text{HO}-\text{CH}_2-\overset{\text{OH}}{\underset{ }{\text{CH}}}-\overset{\text{OCH}_3}{\underset{ }{\text{CH}_2}}$	1	1	1

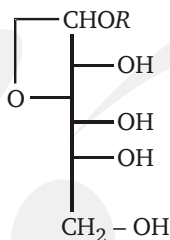
58. Succinic acid (A) $\xrightarrow{\text{NH}_3}$ (B) $\xrightarrow[\text{KOH}]{\text{Br}_2}$ (C); Product (C) will be :



- 59A.** Given are the structures of cyclic D-glucoside. Moles of HIO_4 consumed with X and Y are respectively:

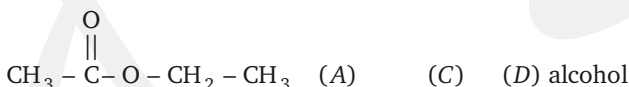


(X)



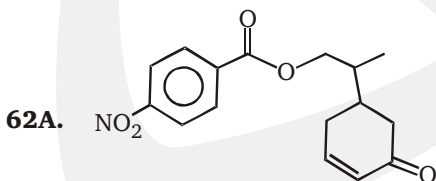
(Y)

- (a) 2, 2
(b) 3, 3
(c) 2, 3
(d) 3, 2
- B.** Moles of formic acid formed in X and Y respectively are:
(a) 1, 2
(b) 2, 1
(c) 2, 3
(d) 3, 2
- C.** Moles of HCHO formed are:
(a) 1, 1
(b) 2, 2
(c) 1, 2
(d) 2, 1
- 60.** In which of the following group, each member gives positive iodoform test ?
(a) methanol, ethanol, propanone
(b) ethanol, isopropanal, methanal
(c) ethanol, ethanal, isopropyl alcohol
(d) propanal, propanol-2, propanone
- 61.** H_2O^{18} Na (A) (B)
(base)

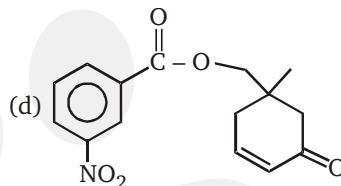
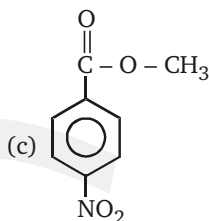
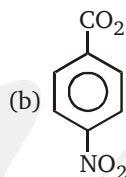
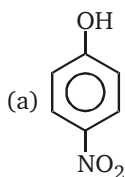


Product (C) of the reactions is:

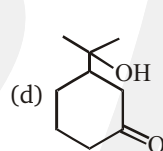
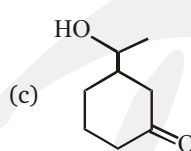
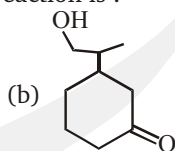
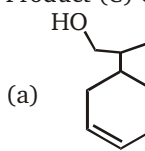
- (a) $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{O}-\text{H}^{18}$
(b) $\text{CH}_3-\overset{\text{O}^{18}}{\parallel}{\text{C}}-\text{O}-\text{H}$
(c) $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{O}^{18}-$
(d) $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{O} \text{ Na}$



Product (B) of the reaction is :



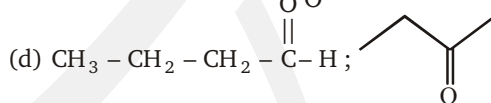
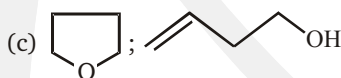
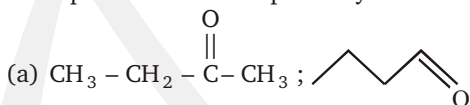
B. Product (C) of above reaction is :



63. Two unknown compounds X and Y, both having molecular formula C_4H_8O , give following results with four chemical tests.

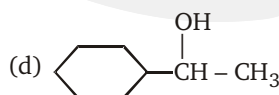
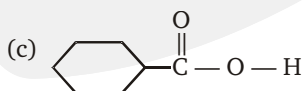
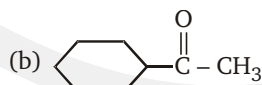
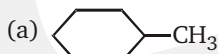
	Bromine	Na metal	Chromic acid	Lucas reagent
Compound X	decolourises	bubbles	Orange to Green	No reaction
Compound Y	No reaction	No reaction	No reaction	No reaction

Compound X and Y respectively are :

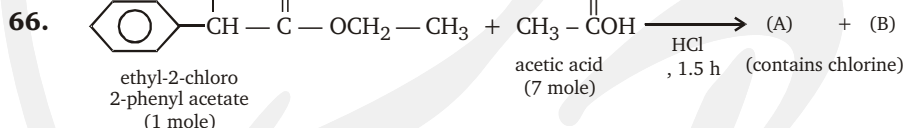
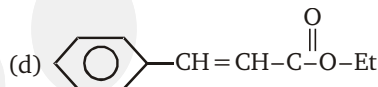
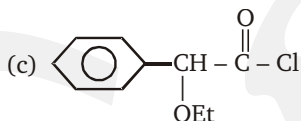
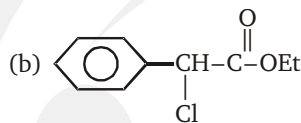
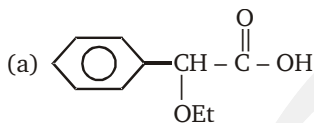
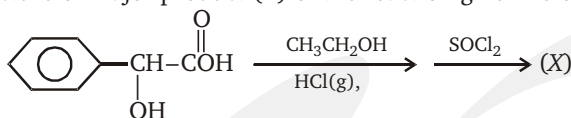


64. $\xrightarrow[\text{dimethoxy ethane}]{LiH}$ $\xrightarrow[L]{CH_3Br}$ $\xrightarrow[M]{H_3O^+}$ $N + CH_3OH$

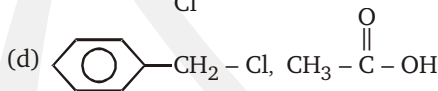
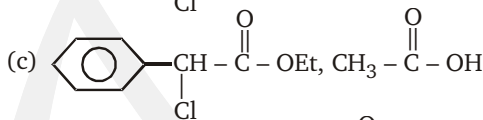
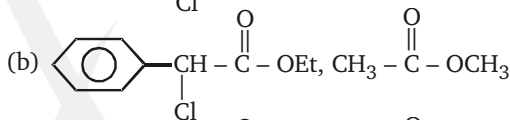
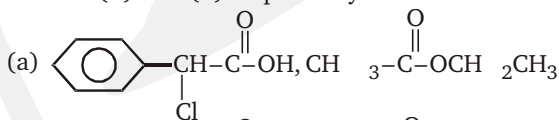
Product (N) is :



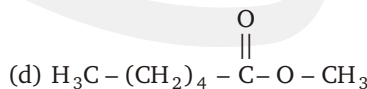
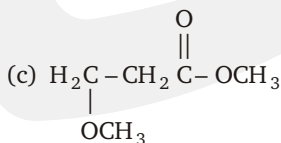
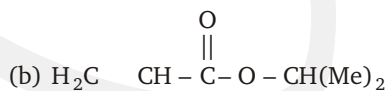
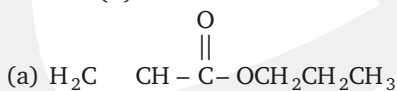
65. Assign the structure of major product (X) of the reaction given below.

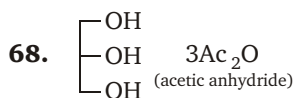


Product (A) and (B) respectively in the above reaction are :



Product (A) of above reaction is :

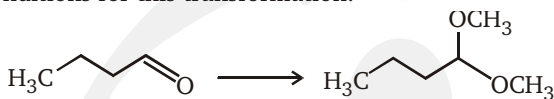




In above reaction molecular formula of glycerol increases by :

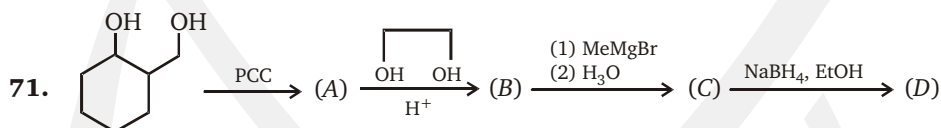
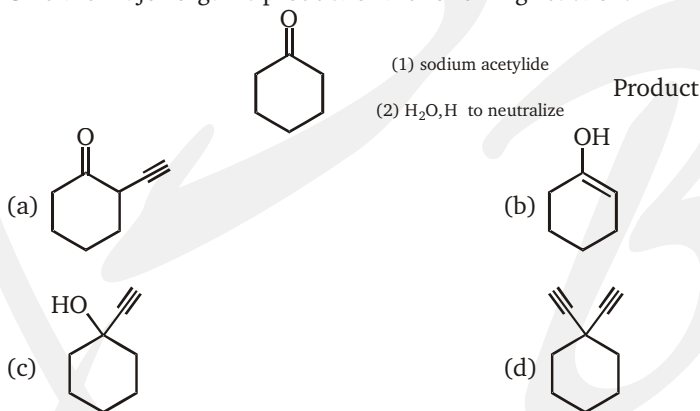
- (a) $\text{C}_4\text{H}_4\text{O}_2$ (b) $\text{C}_6\text{H}_6\text{O}_6$ (c) $\text{C}_6\text{H}_6\text{O}_2$ (d) $\text{C}_6\text{H}_6\text{O}_3$

69. Give the best conditions for this transformation:



- (a) CH_3OH , H (cat.), heat (b) H_2O , H (cat.), heat
(c) Mg , ether, CH_3OH (d) SOCl_2 , CH_3OH

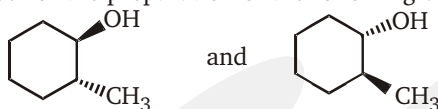
70. Give the major organic product of the following reaction.



Product (D) in above reaction is :

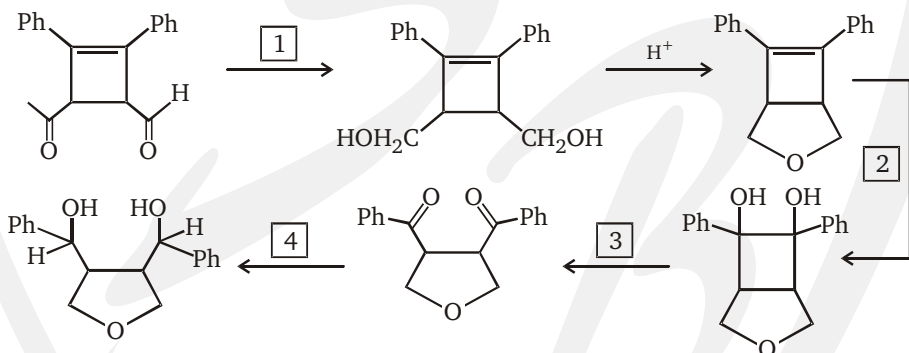


72. Select the best method for the preparation of the following compounds :



(MCPBA = Metachloro per benzoic acid)

- (a) reaction of cyclohexanone with CH_3Li
 (b) reaction of 1-methylcyclohexene with $\text{Hg}(\text{OAc})_2$ followed by NaBH_4
 (c) reaction of cyclohexene with BH_3 ; $\text{NaOH}/\text{H}_2\text{O}_2$, following by CH_3Br
 (d) reaction of cyclohexene with MCPBA, followed by CH_3MgBr
73. Identify the reagents (1-4), required for the transformations shown and arrange them in correct order.



(1) LAH (LiAlH_4)

(3) NaIO_4

(a) 1 3 4 2

(c) 2 1 3 4

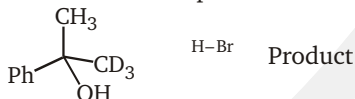
(2) OsO_4

(4) NaBH_4

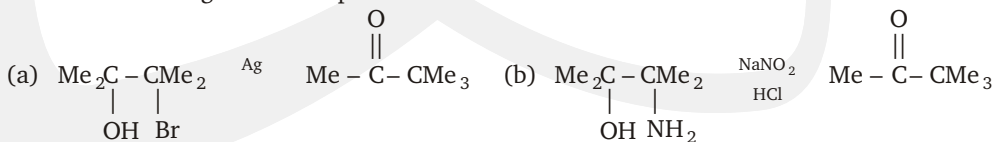
(b) 2 3 1 4

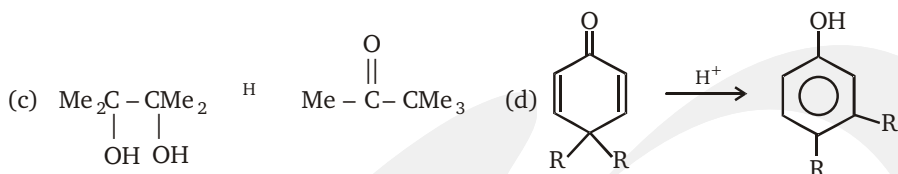
(d) 1 2 3 4

74. Which describes the best stereochemical aspects of the following reaction ?

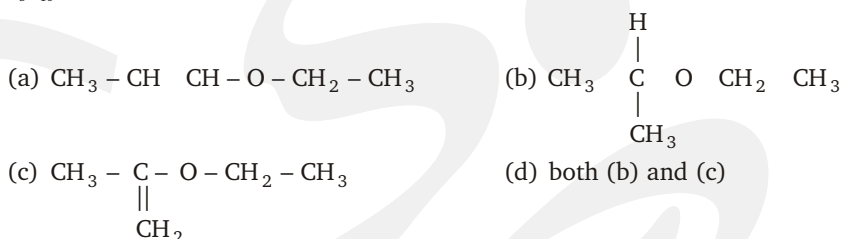


- (a) Inversion of configuration occurs at the carbon undergoing substitution.
 (b) Retention of configuration occurs at the carbon undergoing substitution.
 (c) Racemization (loss of configuration) occurs at the carbon undergoing substitution.
 (d) The carbon undergoing substitution is not stereogenic
75. Which of following is an example of Pinacol-Diazotization ?

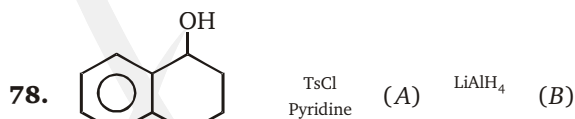




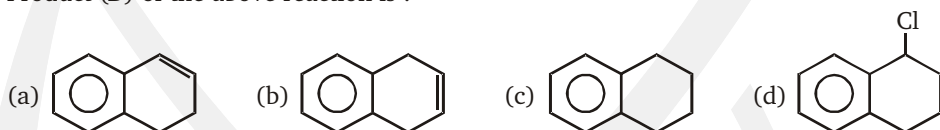
76. (A) $\text{C}_5\text{H}_{10}\text{O}$ $\xrightarrow{\text{H}_3\text{O}^+}$ B C ; (B) and (C) both give +ve iodoform test. Compound (A) is :



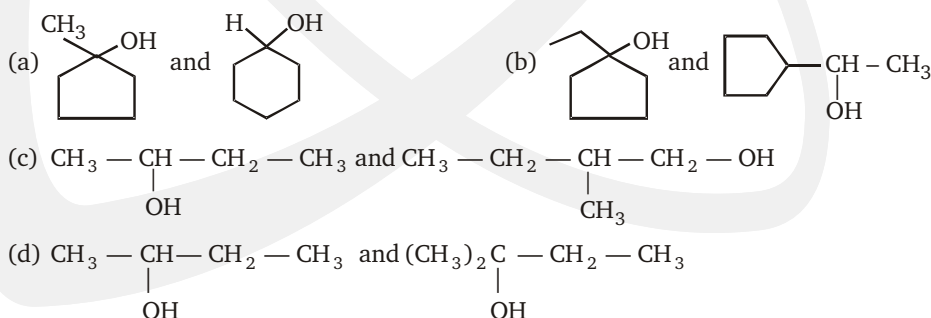
77. A solution of $\text{Ph}_3\text{CCO}_2\text{H}$ in conc. H_2SO_4 gives (X) when poured into methanol X is :



Product (B) of the above reaction is :



79. In the given pair of alcohol, in which pair second alcohol is more reactive than first towards hydrogen bromide?



80. Rank the transition states that occur during the following reaction steps in order of increasing stability (least most stable)

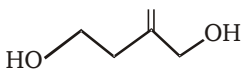


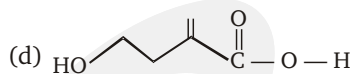
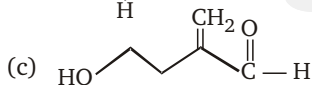
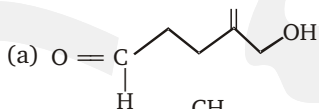
(a) $1 < 2 < 3$

(b) $2 < 3 < 1$

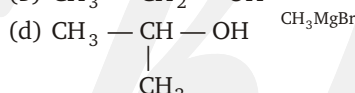
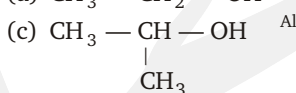
(c) $1 < 3 < 2$

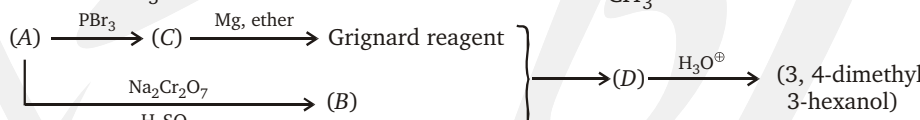
(d) $2 < 1 < 3$

81.  (A), Product (A) is :

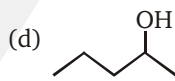
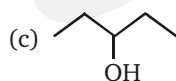
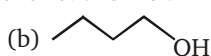
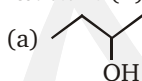


82. In which of the following reactions hydrogen gas will not be evolved ?

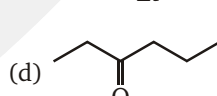
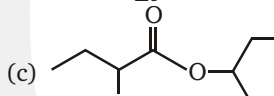
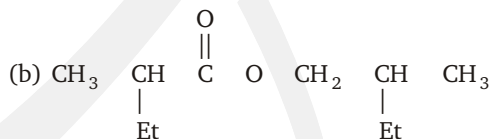
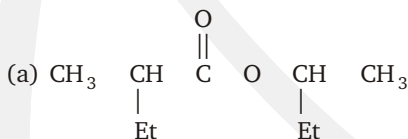


83. 

When Grignard reagent reacts with (B) product (D) will be obtained. Reactant (A) of the above reaction is :



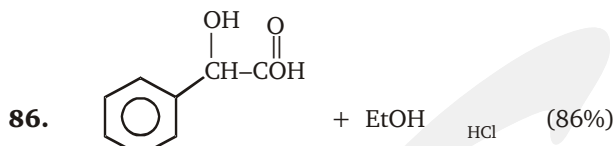
84. (A) $\xrightarrow{\text{LiAlH}_4} 2(\text{B})$; structure of (A) is :
(Chiral alcohol only)



85. 

Above conversion can be achieved by :

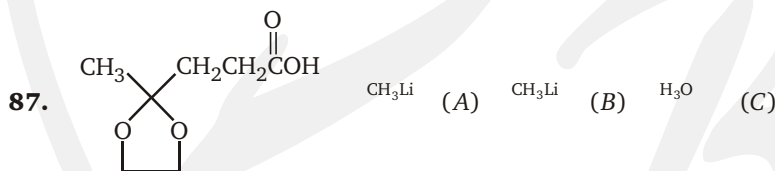
- (a) LiAlH_4 (b) NaBH_4 (c) H_2/Ni (d) CrO_3



(Mandelic acid)

Identify product of above Fischer esterification reaction :

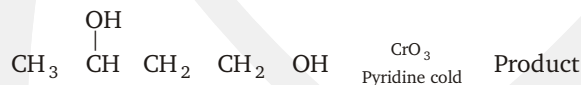
- (a) $\text{Ph} - \text{CH}(\text{OH}) - \text{CO}_2\text{Et}$ (b) $\text{Ph} - \text{CH}(\text{OEt}) - \text{CO}_2\text{Et}$
 (c) $\text{Ph} - \text{CH}(\text{OH}) - \text{CO}_2\text{Et}$ (d) $\text{Ph} - \text{CH}(\text{OH}) - \text{COEt}$



Product (C) of the above reaction is :

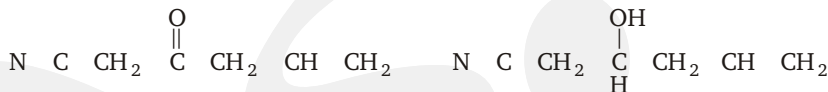
- (a) $\text{CH}_3 - \text{C}(=\text{O}) - \text{CH}_2 - \text{CH}_2 - \text{C}(\text{OH})(\text{CH}_3)_2$ (b) $\text{CH}_3 - \text{C}(\text{OH})(\text{CH}_3) - \text{CH}_2 - \text{CH}_2 - \text{C}(\text{OH})(\text{CH}_3)_2$
 (c) $\text{CH}_3 - \text{C}(=\text{O}) - \text{CH}_2 - \text{CH}_2 - \text{C}(=\text{O}) - \text{CH}_3$ (d) $\text{CH}_3 - \text{C}(=\text{O}) - \text{CH}_2 - \text{C}(=\text{O}) - \text{CH}_3$

88. What is the major product of the following reaction ?

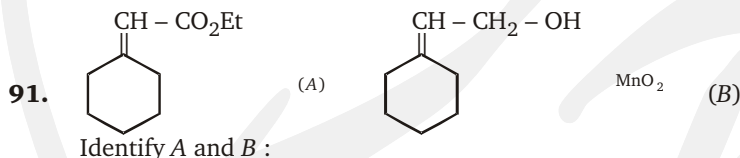


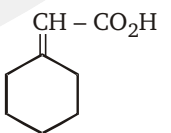
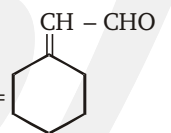
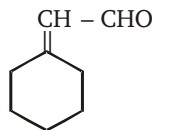
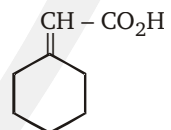
- (a) $\text{CH}_3 - \text{CH}(\text{OH}) - \text{CH}_2 - \text{C}(=\text{O}) - \text{H}$ (b) $\text{CH}_3 - \text{C}(=\text{O}) - \text{CH}_2 - \text{C}(=\text{O}) - \text{H}$
 (c) $\text{CH}_3 - \text{C}(=\text{O}) - \text{CH}_2 - \text{C}(=\text{O}) - \text{OH}$ (d) $\text{CH}_3 - \text{CH}(\text{OH}) - \text{CH}_2 - \text{C}(=\text{O}) - \text{OH}$

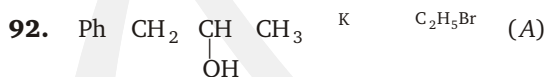
89. The major reason that phenol is a better Bronsted acid than cyclohexanol is that :
- it is a better proton donor.
 - the cyclohexyl group is an electron donating group by induction, which destabilizes the anion formed in the reaction by resonance.
 - phenol is able to stabilize the anion formed in the reaction.
 - the phenyl group is an electron withdrawing group by induction, which stabilizes the anion formed in the reaction.
90. Which of these reagents would accomplish the following reduction ?



- NaBH_4
- LiAlH_4
- 1 mole H_2 , poisoned catalyst, low pressure
- H_3O^+



- A NaBH_4 , B = 
- A NaBH_4 , B = 
- A LiAlH_4 , B = 
- A LiAlH_4 , B = 

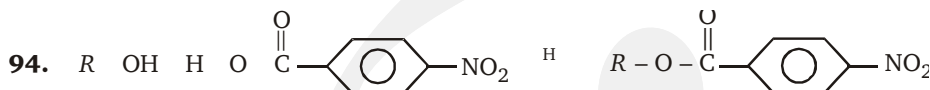
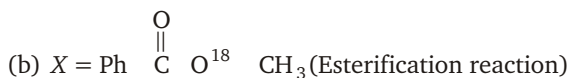


Product (A) in above reaction is:

- $\text{Ph} \quad \text{CH}_2 \quad \underset{\text{OEt}}{\text{CH}} \quad \text{CH}_3$, (inversion)
- $\text{Ph} \quad \text{CH}_2 \quad \underset{\text{OEt}}{\text{CH}} \quad \text{CH}_3$, (retention)
- $\text{Ph} \quad \text{CH}_2 \quad \underset{\text{OEt}}{\text{CH}} \quad \text{CH}_3$, (racemic)
- $\text{Ph} \quad \text{CH} \quad \text{CH} \quad \text{CH}_3$



- $\text{X} = \text{Ph} \quad \overset{\text{O}}{\parallel} \text{C} \quad \text{O}^{18} \quad \text{CH}_3$ (Trans esterification)

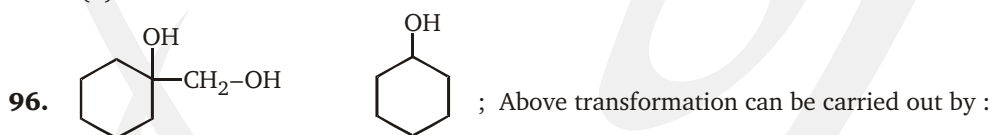


Fastest rate of reaction will be when R is :

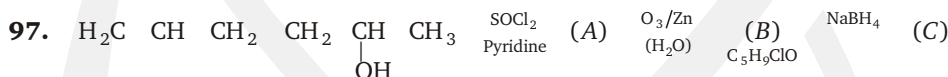
- (a) CH_3 (b) $\text{CH}_3 \text{CH}_2$ (c) $\text{CH}_3 \begin{array}{c} \text{CH} \\ | \\ \text{CH}_3 \end{array}$ (d) $\text{CH}_3 \begin{array}{c} \text{CH}_3 \\ | \\ \text{C} \\ | \\ \text{CH}_3 \end{array}$

95. Select the correct statement.

- (a) Solvolysis of $(\text{CH}_3)_2\text{C} \begin{array}{c} \text{CH} \\ | \\ \text{CH}_2 \end{array} \text{Cl}$ in ethanol is over 6000 times greater than alkyl chloride (25°C)
 (b) $\text{CH}_3 \begin{array}{c} \text{CH} \\ | \\ \text{CH} \\ | \\ \text{CH}_2 \end{array} \text{OH}$ when reacts with HBr give a mixture of 1-bromo-2-butene and 3-bromo 1-butene
 (c) When solution of 3-buten-2-ol in aqueous sulphuric acid is allowed to stand for one week, it was found to contain both 3-buten-2-ol and 2-buten-1-ol
 (d) All of these



- (a) $\text{H} /$, Zn(Hg) , HCl (b) HIO_4 , LiAlH_4
 (c) HIO_4 , $\text{H} /$ (d) $\text{H} /$, HIO_4



Compound (C) is :

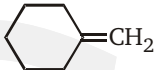
- (a) $\text{CH}_3 \begin{array}{c} \text{OH} \\ | \\ \text{CH} \end{array} \text{CH}_2 \begin{array}{c} \text{CH} \\ | \\ \text{Cl} \end{array} \text{CH}_3$
 (b) $\text{HOCH}_2 \text{CH}_2 \text{CH}_2 \text{CH}_2 \text{CH}_2 \text{CH}_2 \text{Cl}$
 (c) $\text{HO} \text{CH}_2 \text{CH}_2 \text{CH}_2 \begin{array}{c} \text{Cl} \\ | \\ \text{CH} \end{array} \text{CH}_3$
 (d) $\text{HO} \text{CH}_2 \text{CH}_2 \begin{array}{c} \text{CH} \\ | \\ \text{Cl} \end{array} \text{CH}_2 \text{CH}_3$

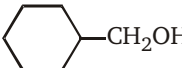
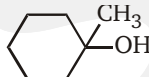
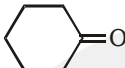
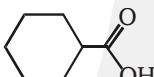
98. Iodoform can be obtained on warming NaOH and iodine with :

- (a) $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$ (b) $(\text{CH}_3)_2\text{CH}\overset{\text{O}}{\parallel}\text{C}\text{C}_2\text{H}_5$
 (c) $\text{CH}_3-\overset{\text{O}}{\parallel}\text{C}-\text{OCH}_3$ (d) $(\text{CH}_3)_2\text{CHCH}_2\text{OH}$

99. Which of these is a reducing agent ?

- (a) CrO_3/H (b) KMnO_4
 (c) LiAlH_4 (d) O_3

100.  (i). $(\text{BH}_3)_2$ (P); Product (P) in the reaction is:
 (ii). $\text{H}_2\text{O}_2/\text{OH}$

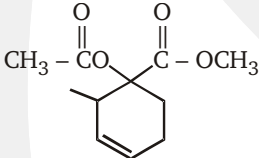
- (a)  (b)  (c)  (d) 

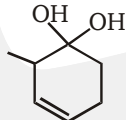
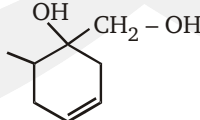
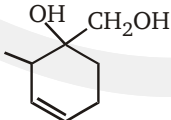
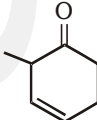
101. $\text{CH}_3-\overset{\text{CH}_3}{\underset{\text{OH}}{\text{C}}}-\text{CH}_3$ $\xrightarrow[\text{cool}]{\text{Na}_2\text{Cr}_2\text{O}_7}$ (P); Product (P) in the reaction is:

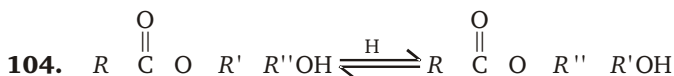
- (a) $\text{CH}_3-\overset{\text{CH}_3}{\text{C}}-\text{CH}_2$ (b) $\text{CH}_3-\overset{\text{CH}_3}{\text{CH}}-\text{CH}_3$
 (c) $\text{CH}_3-\overset{\text{CH}_3}{\underset{\text{CH}_3}{\text{C}}}-\text{O}-\overset{\text{CH}_3}{\underset{\text{CH}_3}{\text{C}}}-\text{CH}_3$ (d) No reaction

102. 1, 2, 3 - butanetriol undergoes oxidative cleavage of HIO_4 . During this process

- (a) 1 equivalent of HIO_4 consumed & HCO_2H & $\text{H}_3\text{C}-\overset{\text{O}}{\parallel}\text{C}-\text{CO}_2\text{H}$ are formed
 (b) 2 equivalents of HIO_4 consumed & HCO_2H , $\text{HCH}=\text{O}$ & $\text{CH}_3-\text{CH}=\text{O}$ are formed
 (c) 3 equivalents of HIO_4 consumed & HCO_2H (2 eq.) & 1 eq. of $\text{CH}_3\text{CO}_2\text{H}$ are formed
 (d) 2 equivalents of HIO_4 consumed & 2 eq. of HCO_2H & 1 eq. of $\text{CH}_3\text{CH}=\text{O}$ is formed

103.  (i) LiAlH_4 (A); Product (A) of the reaction is :
 (ii) H_2O (96%)

- (a)  (b)  (c)  (d) 



Above reaction is/an example of :

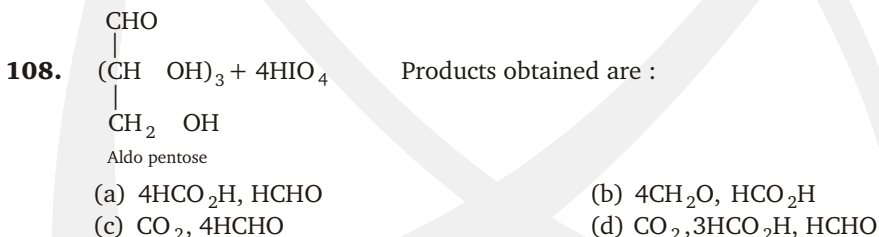
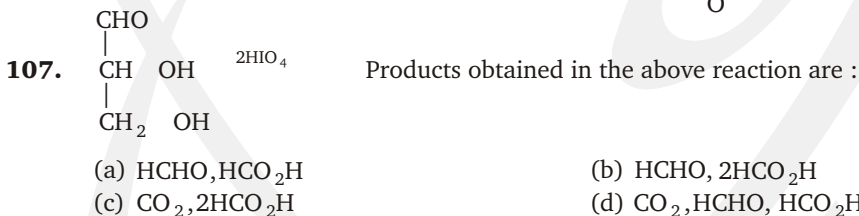
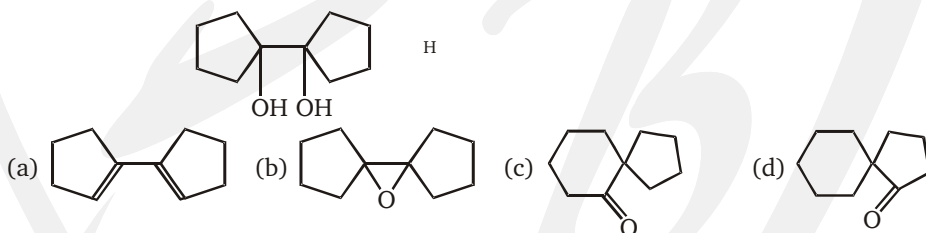
- (a) esterification (b) saponification
(c) *trans*-esterification (d) hydrolysis

105. What is the major organic product of the following sequence of reactions ?



- (a) $(\text{CH}_3)_2\text{CHCH}(\text{OH})\text{CH}_2\text{CH}_3$ (b) $(\text{CH}_3)_2\text{CHCH}_2\text{CH}_2\text{OH}$
(c) $(\text{CH}_3)_2\text{CHCH}_2\text{CH}(\text{OH})\text{CH}_3$ (d) $(\text{CH}_3)_2\text{CHCH}_2\text{CH}_2\text{CH}_2\text{OH}$

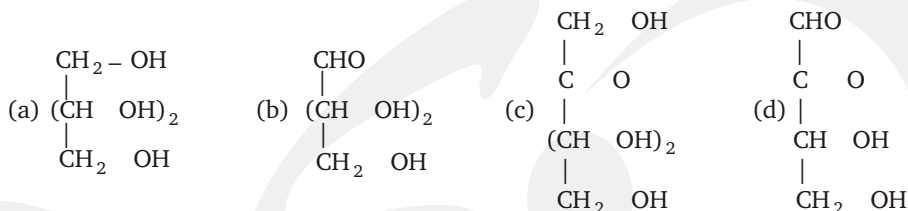
106. The structure of the product formed in the reaction given below is :



Ratio of moles of formic acid obtained in reaction (i) and reaction (ii) is :

- (a) 3/4 (b) 4/5 (c) 1 (d) 5/4

110. Which of the following compound gives 2HCHO , CO_2 , HCO_2H when oxidised by periodic acid?



111. Hydration of 3-phenylbut-1-ene in dil. H_2SO_4 will give mainly :

- (a) 3-Phenylbutan-1-ol (b) 3-Phenylbutan-2-ol
(c) 2-Phenylbutan-2-ol (d) 2-Phenylbutan-1-ol

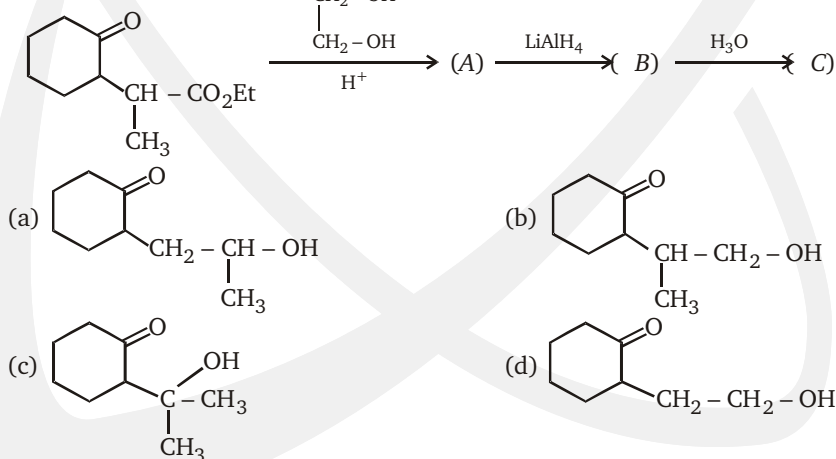
112. Decarboxylation of sodium salicylate with soda lime forms :

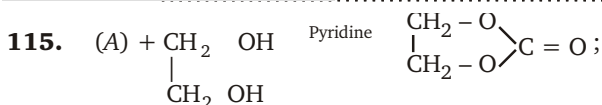
- (a) Salicylic acid (b) Phenol (c) Benzene (d) None of these

113. $\text{MeO} - \text{CH} - \text{CH} - \text{OH} - \text{CH} - \text{OH} - \text{CH} - \text{OH} - \text{HC} - \text{CH}_2 - \text{OH}$ $\xrightarrow{x\text{HIO}_4}$ What is the maximum value of (x) ?

- (a) 1 (b) 2 (c) 3 (d) 4

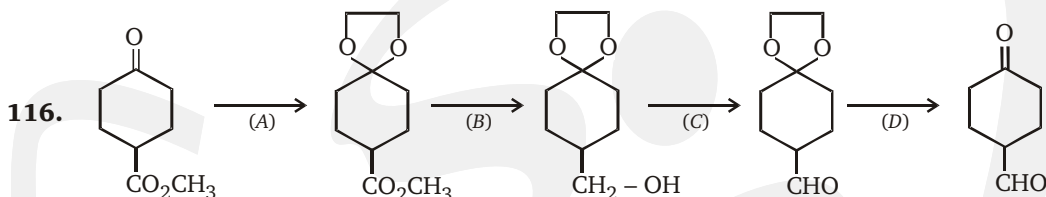
114. $\text{Cyclohexanone} - \text{CH}(\text{CH}_3) - \text{CO}_2\text{Et} \xrightarrow[\text{H}^+]{\text{CH}_2 - \text{OH}, \text{CH}_2 - \text{OH}} (\text{A}) \xrightarrow{\text{LiAlH}_4} (\text{B}) \xrightarrow{\text{H}_3\text{O}^+} (\text{C})$. Product C is :





Reactant A of the above reaction is :

- (a) $\text{CH}_3 \text{C}(=\text{O})\text{CH}_3$ (b) COCl_2 (c) $\text{CH}_3 \text{C}(=\text{O})\text{Cl}$ (d) $\text{CH}_3 \text{C}(=\text{O})\text{OEt}$



Identify correct combination :

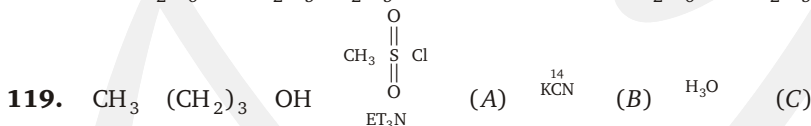
- (a) (A) = $\begin{array}{c} \text{CH}_2 \text{ OH} \\ | \\ \text{CH}_2 \text{ SH} \end{array}$ (b) (B) = NaBH_4 (c) (C) = KMnO_4 (d) (D) = H_3O^+

117. In the Libermann's nitroso reaction, sequential changes in the colour of phenol occurs as :

- (a) Brown or red green deep blue (b) Red deep blue green
(c) Red green white (d) White red green

118. Ethanol when reacts with PCl_5 gives A, POCl_3 and HCl . A reacts with dry Ag_2O to form B (major product) and AgCl . A and B respectively are :

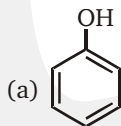
- (a) $\text{C}_2\text{H}_5\text{Cl}$ and $\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$ (b) C_2H_4 and $\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$
(c) C_2H_6 and $\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$ (d) C_2H_6 and $\text{C}_2\text{H}_5\text{NO}_2$



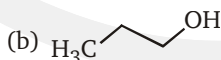
Product (C) is :

- (a) $\text{CH}_3(\text{CH}_2)_3\text{CO}_2\text{H}$ (b) $\text{CH}_3(\text{CH}_2)_3^{14}\text{CO}_2\text{H}$
(c) $\text{CH}_3\text{CO}_2\text{H}$ (d) $\text{CH}_3\text{C}(=\text{O})\text{O}^{14}\text{H}$

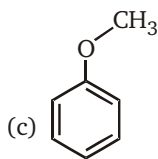
120. Choose the best synthesis of phenyl *n*-propyl ether.



1. Na metal
2. $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$ /
polar, aprotic solvent



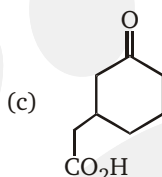
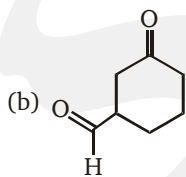
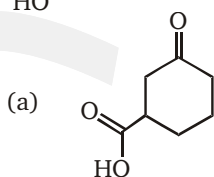
1. Na metal
2. Ph Br



1. Na metal
2. $\text{CH}_3\text{CH}_2\text{Br}$

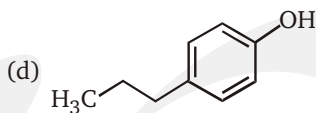
$\text{Na}_2\text{Cr}_2\text{O}_7/\text{H}_2\text{SO}_4$

The product obtained is :



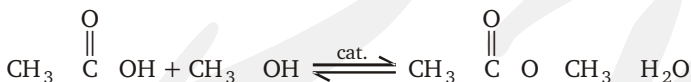
(d) None of these

121.



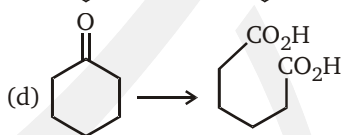
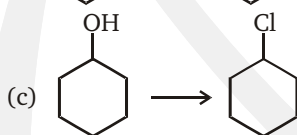
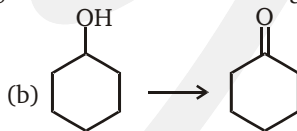
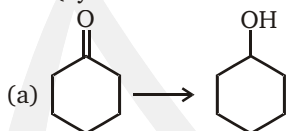
1. Na metal
2. HBr

122. What is true for the equilibrium reaction ?



- (a) The use of equimolar quantities of CH_3OH and CH_3COOH will give the greatest yield of the ester at equilibrium
 (b) Removal of water will increase the amount of ester at equilibrium
 (c) Addition of $\text{CH}_3\text{COOCH}_3$ will cause the formation of equal an equal number of moles of water
 (d) Application of pressure increases the amount of ester at equilibrium

123. PCC (Pyridinium chloro chromate) is a good reagent for which of the following transformations ?



124. How many primary alcohols (including stereoisomers) are possible with formula $\text{C}_5\text{H}_{12}\text{O}$?

- (a) Two
 (b) Three
 (c) Four
 (d) Five

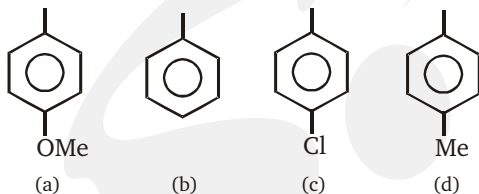
125. 1-Phenylethanol can be prepared by the reaction of benzaldehyde with the product obtained in the reaction between:

- (a) CH_3I and Mg
 (b) $\text{C}_2\text{H}_5\text{I}$ and Mg
 (c) CH_3Br and AlCl_3
 (d) CH_3OH and ZnCl_2

126. 0.092 g of a compound with the molecular formula $C_3H_8O_3$ on reaction with an excess of CH_3MgI gives 67.00 mL of methane at STP. The number of active hydrogen atoms present in a molecule of the compound is :

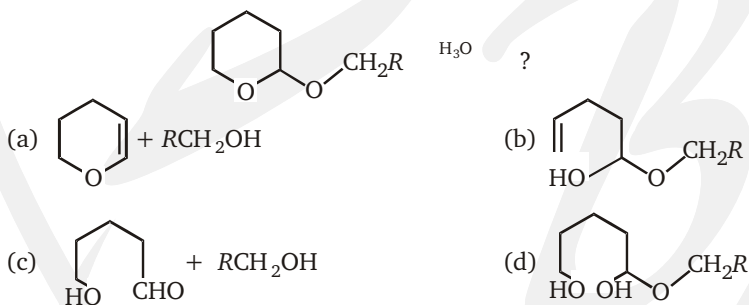
- (a) one (b) two
(c) three (d) four

127. Migratory aptitude of the following in decreasing order is :



- (a) $a > c > b > d$ (b) $a > d > b > c$
(c) $a > d > c > b$ (d) $b > c > a > b$

128. The major product formed in the reaction is :



129. Reaction of *R*-2-butanol with *p*-toluenesulphonyl chloride in pyridine then LiBr gives :

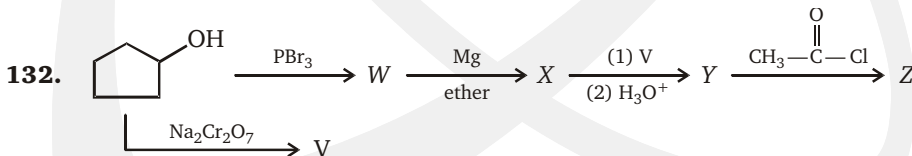
- (a) *R*-2-butyl bromide (b) *S*-2-butyl tosylate
(c) *R*-2-butyl tosylate (d) *S*-2-butyl bromide

130. Optically active 2-octanol rapidly loses its optical activity when exposed to :

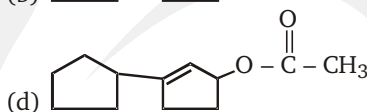
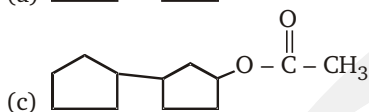
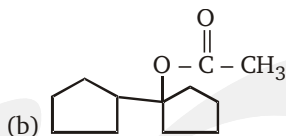
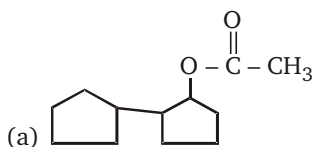
- (a) dilute acid (b) dilute base (c) light (d) humidity

131. If $(\pm)$ 2-methyl butanoic acid were esterified by reaction with $(\pm)$ 2-butanol, how many optically active compounds would be present in the final equilibrium reaction mixture ?

- (a) 2 (b) 3 (c) 4 (d) 6

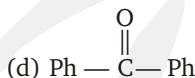
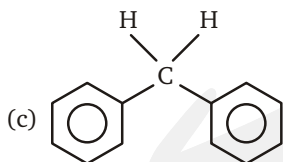
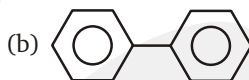


Product Z of above reaction is :



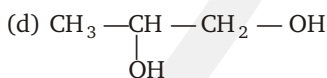
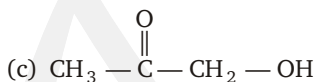
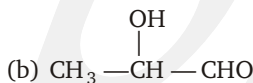
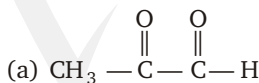
133. $\xrightarrow[\text{AlCl}_3]{\text{LiAlH}_4}$ (A) ; Identify the product :

(a) No reaction



134. $(\text{EtO})_2\text{CHCHO} \xrightarrow[\text{H}_3\text{O}^+]{\text{CH}_3\text{MgI}}$ (A)

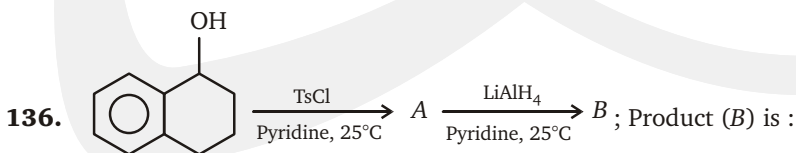
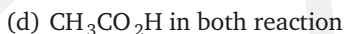
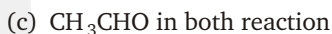
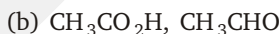
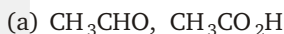
Product obtained in the above reaction is :

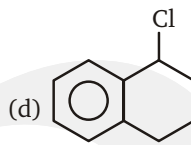
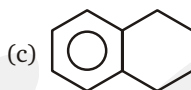
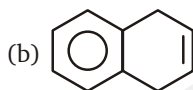
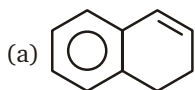


135. **Reaction - (1):** $\text{CH}_3-\text{CH}(\text{OH})-\text{CH}_3 \xrightarrow[\text{Cold}]{\text{KMnO}_4}$ (A) $\xrightarrow{\text{NaIO}_4}$ (B) 2 mole

Reaction - (2): $\text{CH}_3-\text{CH}(\text{OH})-\text{CH}_3 \xrightarrow{\text{KMnO}_4/\text{NaIO}_4}$ (C) 2 mole

Product (B) and (C) respectively are :





137. $\text{HO} - (\text{CH}_2)_6 - \text{OH}$, this conversion can be achieved by

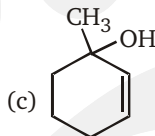
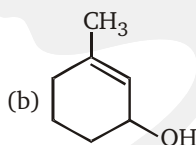
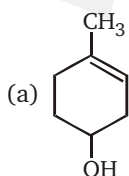
(a) O_3 , Zn, then LiAlH_4

(b) $\text{O}_3/\text{H}_2\text{O}_2$, then LiAlH_4

(c) cold dil. KMnO_4 , HIO_4 , then LiAlH_4

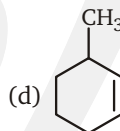
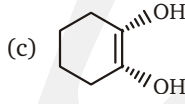
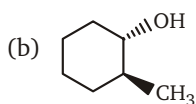
(d) All of these

138. Which of the following alcohol on treatment with HCl give 3-chloro-3-methyl cyclohexene as a product ?

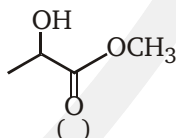
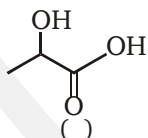


(d) All of these

139. (a) RCO_3H (A); Product of the reaction is :
(b) CH_3MgBr (c) $\text{H}_2, \text{H}_2\text{O}$



140. Esterification (shown below) is a reaction converting a carboxylic acid to its ester. It involves only the carbonyl carbon. Esterification of () -lactic acid with methanol yields (+)-methyl lactate. Assuming that there are no side reactions, what is true about this reaction ?



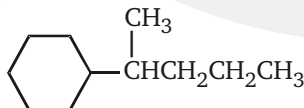
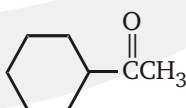
(a) An $\text{S}_{\text{N}}2$ process has occurred, inverting the absolute configuration of the chiral center

(b) An $\text{S}_{\text{N}}1$ reaction at the chiral center has inverted the optical rotation

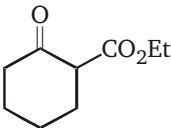
(c) A diastereomer has been produced; diastereomers have different physical properties including optical rotation

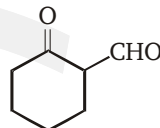
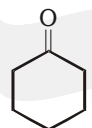
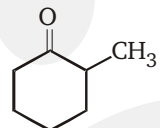
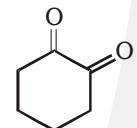
(d) Optical rotation is not directly related to absolute configuration, so the change in sign of rotation is merely a coincidence

141. Which of the following sets of reagents, used in the order shown, would successfully accomplish the conversion shown ?

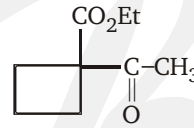
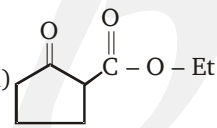


- (a) $\text{CH}_3\text{CH}_2\text{CH}_2\text{MgBr}$; H_3O^+ ; PCC, CH_2Cl_2
 (b) $\text{CH}_3\text{CH}_2\text{CH}_2\text{MgBr}$; H_3O^+ ; H_2SO_4 , heat PCC, CH_2Cl_2
 (c) $(\text{C}_6\text{H}_5)_3\text{P}^- \text{CHCH}_2\text{CH}_3$, B_2H_6 ; $\text{CH}_3\text{CO}_2\text{H}$
 (d) $(\text{C}_6\text{H}_5)_3\text{P}^- \text{CHCH}_2\text{CH}_3$; H_2O

142.  (1) H_3O^+
 (2) Product; Product of the reaction is :

- (a)  (b)  (c)  (d) 

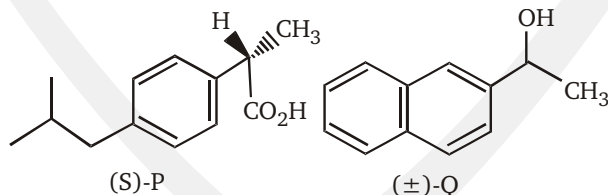
143. Which of the following compound on hydrolysis followed by heating gives a product, which gives positive iodoform test?

- (a) $\text{CH}_3\text{CH}(\text{CO}_2\text{Et})\text{C}(=\text{O})\text{CH}_2\text{CH}_3$ (b) 
 (c) $\text{CH}_3\text{CH}(\text{CO}_2\text{Et})\text{CO}_2\text{Et}$ (d) 

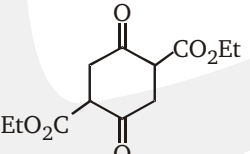
144. Treatment of a 2° OH with $\text{CrO}_3/\text{H}_2\text{SO}_4$ yields an/a :

- (a) aldehyde (b) carboxylic acid (c) ester (d) ketone

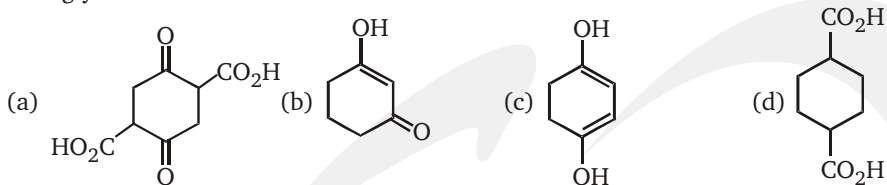
145. Esterification of the acid **P** with the alcohols **Q** will give :

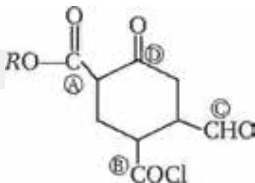


- (a) only one enantiomer (b) a mixture of diastereomers
 (c) a mixture of enantiomers (d) only one diastereomer

146.  $\text{H}_2\text{O}/$ product

Identify major product of the reaction, when the given compound is hydrolysed and heated strongly:

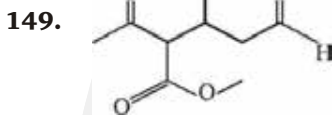
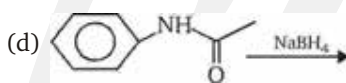
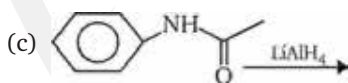
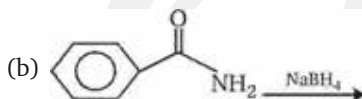
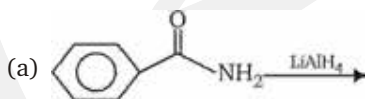


147.  find out the reactivity order with LiAlH_4 :

- (a) A B C D
(c) D C B A

- (b) B C D A
(d) B D C A

148. Find out the reaction in which obtained product give positive isocyanide test:



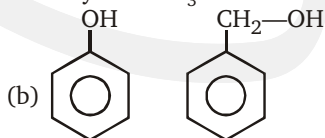
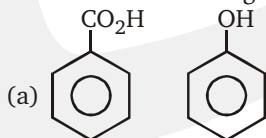
In the above given compound how many functional group reduced by LAH (Lithium aluminium hydride) and SBH (sodium borohydride) respectively?

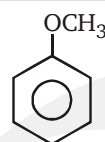
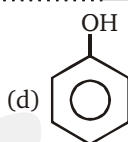
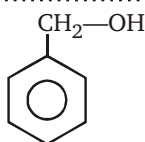
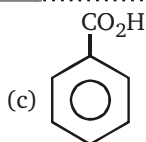
- (a) 4, 4 (b) 4, 3 (c) 3, 4 (d) 4, 2

150. An unknown compound (A) (molar mass 180) on acylation gives a product (molar mass 390) then find the number of hydroxyl group present in compound (A).

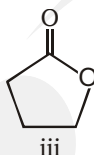
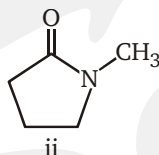
- (a) 5 (b) 6 (c) 10 (d) 1

151. Which of the following compound is differentiated by NaHCO_3 as well as by NaOH ?





152. Arrange the following compounds in order of their reactivity toward LiAlH_4 .



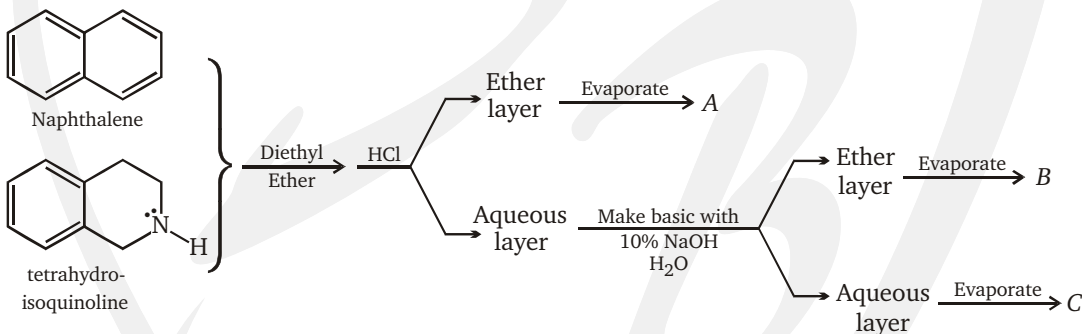
(a) i ii iii

(c) ii i iii

(b) i iii ii

(d) ii iii i

153. Choose the statement that is true about A, B and C in the following separation scheme.



(a) A tetrahydroisoquinoline, B naphthalene and C inorganic ions such as Na^+ and Cl^-

(b) A naphthalene, B tetrahydroisoquinoline and C inorganic ions such as Na^+ and Cl^-

(c) A inorganic ions such as Na^+ and Cl^- , B naphthalene and C tetrahydroisoquinoline

(d) A inorganic ions such as Na^+ and Cl^- , B naphthalene and C tetrahydroisoquinoline

154. value of x is :

(a) 2

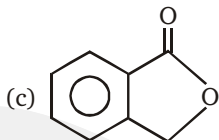
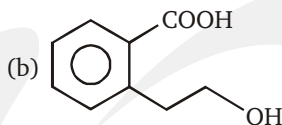
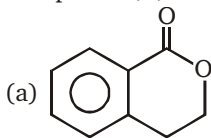
(b) 3

(c) 4

(d) 5

155. NaBH_4 (A) H^+ (B) Cyclic

Compound (B) is :

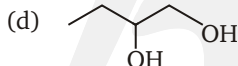
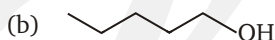
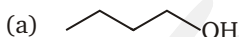


(d) (a) and (c) both

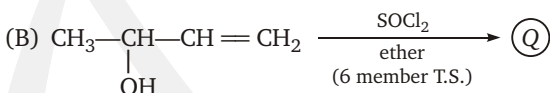
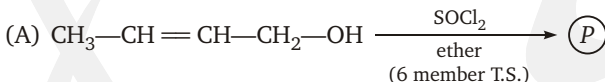
156. 1-Phenoxypropane is treated with excess of conc. HI at 0 °C and the mixture of products is treated with thionyl chloride. The products formed are

- (a) *n*-propanol Chlorobenzene (b) Phenol *n*-propyl iodide
(c) *n*-propyl chloride Chlorobenzene (d) *n*-propyl chloride Phenol

157. Amongst the following compounds, the compound having the lowest boiling point is



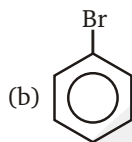
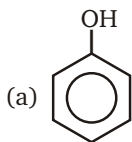
158. Which statement/s are true about products P & Q.



- (a) P & Q are position isomers
(b) P show geometrical isomerism but Q not



Which of the following is/are possible end product of the above reaction?



(d) both (a) and (c)

ANSWERS — LEVEL 1

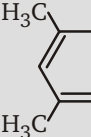
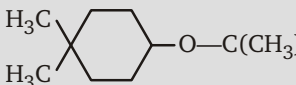
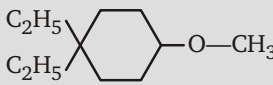
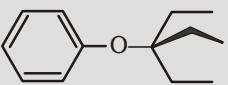
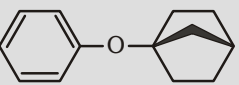
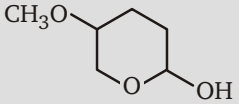
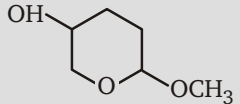
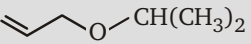
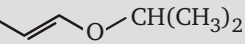
1.	(c)	2.	(a)	3.	(b)	4.	(c)	5.	(b)	6.	(b)	7.	(b)	8.	(a)
9.	(b)	10.	(b)	11.	(b)	12.	(b)	13.	(a)	14.	(a)	15.	(a)	16.	(a)
17.	(d)	18.	(a)	19.	(b)	20.	(a)	21.	(a)	22.	(a)	23.	(d)	24.	(a)
25.	(b)	26.	(b)	27.	(b)	28.	(b)	29.	(c)	30.	(c)	31.	(b)	32.	(b)
33.	(a)	34.	(b)	35.	(b)	36.	(c)	37.	(c)	38.	(c)	39.	(d)	40.	(c)
41.	(b)	42.	(b)	43.	(d)	44.	(d)	45.	(b)	46.	(a)	47.	(b)	48.	(b)
49.	(a)	50.	(b)	51.	(a)	52.	(c)	53.	(c)	54.	(c)	55.	(b)	56.	(a)
57.	(d)	58.	(c)	59.	A-d	59.	B-b	59.	C-a	60.	(c)	61.	(c)	62.	A-b
62.	B-b	63.	(b)	64.	(c)	65.	(b)	66.	(a)	67.	(a)	68.	(d)	69.	(a)
70.	(c)	71.	(b)	72.	(d)	73.	(d)	74.	(c)	75.	(b)	76.	(d)	77.	(c)
78.	(c)	79.	(d)	80.	(c)	81.	(c)	82.	(d)	83.	(a)	84.	(b)	85.	(b)
86.	(c)	87.	(c)	88.	(b)	89.	(d)	90.	(a)	91.	(c)	92.	(b)	93.	(b)
94.	(a)	95.	(d)	96.	(b)	97.	(c)	98.	(a)	99.	(c)	100.	(a)	101.	(d)
102.	(b)	103.	(c)	104.	(c)	105.	(d)	106.	(c)	107.	(b)	108.	(a)	109.	(c)
110.	(d)	111.	(c)	112.	(b)	113.	(b)	114.	(b)	115.	(b)	116.	(d)	117.	(b)
118.	(a)	119.	(b)	120.	(a)	121.	(a)	122.	(b)	123.	(b)	124.	(d)	125.	(a)
126.	(c)	127.	(b)	128.	(c)	129.	(d)	130.	(a)	131.	(c)	132.	(b)	133.	(c)
134.	(b)	135.	(a)	136.	(c)	137.	(d)	138.	(d)	139.	(b)	140.	(d)	141.	(c)
142.	(b)	143.	(b)	144.	(d)	145.	(b)	146.	(c)	147.	(b)	148.	(a)	149.	(d)
150.	(a)	151.	(c)	152.	(d)	153.	(b)	154.	(b)	155.	(a)	156.	(b)	157.	(c)
158.	(a,c)	159.	(d)												

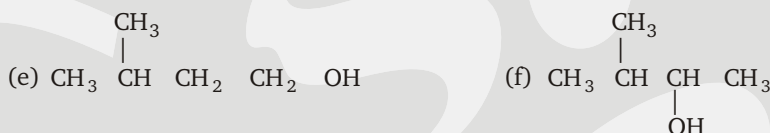
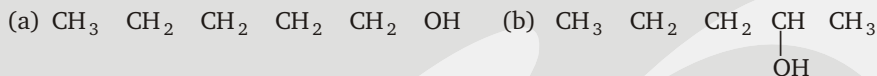
Level - 2

1. Consider the pairs of ethers, numbered I through V, shown below. To the right of each pair is a description of reaction conditions to be applied to each. One compound of the pair will react more rapidly than the other.

Which ether of the two will be more rapidly cleaved?

Write your answer in box.

	(A)	Ether Pairs	(B)	Cleavage Conditions
I.		$\text{O}-\text{CH}(\text{CH}_3)_2$ & 		Treated with HBr in CH_3CN , 40°C
II.		$\text{O}-\text{C}(\text{CH}_3)_3$ & 		Treated with H_2SO_4 in CH_3CN , 40°C
III.		& 		Treated with H_2SO_4 in CH_3CN , 40°C
IV.		& 		Treated with 5% aqueous H_2SO_4 , 25°C
V.		& 		Treated with 5% aqueous H_2SO_4 , 25°C

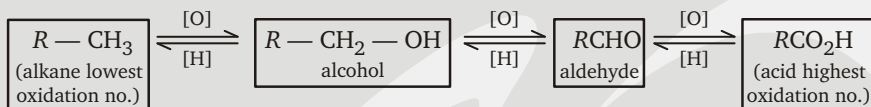
2. Comprehension

Above compounds (a) to (h) are isomers of $\text{C}_5\text{H}_{12}\text{O}$.

Based on the above isomer answer the following (A to F).

- A.** Which isomer is most reactive towards dehydration by conc. H_2SO_4 ?
- B.** Which isomer will undergo rearrangement when treated with conc. H_2SO_4 ?
- C.** Which isomers on dehydration with conc. H_2SO_4 give alkene which is capable to show geometrical isomerism?
- D.** Which isomer is least acidic?
- E.** Which isomers on dehydration give most stable alkene?
- F.** Which isomer on dehydration with conc. H_3PO_4 undergo maximum rearrangement?

3. Comprehension



[O] = Oxidation

[H] = Reduction

Consider the above sequence and answer A to F.

- A.** Conversion $(CH_3 - CH_3 \rightarrow CH_3 - CH_2 - OH)$ alkane to alcohol is achieved by:
- (a) $Br_2/h\nu$, alc. KOH (b) $Br_2/h\nu$, aq. KOH
 (c) Br_2/CCl_4 , $LiAlH_4$ (d) Br_2/CCl_4 , $NaBH_4$
- B.** Conversion $R - CH_2 - OH \rightarrow R - CHO$ can be done by:
- (a) PCC/ CH_2Cl_2 (b) Cu, $300^\circ C$
 (c) CrO_3 (d) All of these
- C.** Conversion $R - CHO \rightarrow R - CO_2H$ can be done by:
- (a) $KMnO_4$ (b) H_2CrO_4
 (c) $K_2Cr_2O_7$ (d) All of these
- D.** Conversion $R - CO_2H \rightarrow R - CHO$ can be done by:
- (a) $LiAlH_4$ (b) $NaBH_4$
 (c) DIBAL - H (d) All of these
- E.** Conversion $R - CHO \rightarrow R - CH_2 - OH$ can be done by:
- (a) $LiAlH_4$ (b) $NaBH_4$
 (c) H_2/Ni (d) All of these
- F.** Reduction $R - CH_2 - OH \rightarrow R - CH_3$ can be done by:
- (a) $LiAlH_4$ (b) $NaBH_4 - AlCl_3$
 (c) $H_2 - Ni$ (d) Red P + HI

4. Which of the following is true for 3-methylbutanal ?

a.	This compound may be classified as an aldehyde.	
b.	This compound may be classified as a ketone	
c.	An aldol reaction takes place on treatment with NaOH solution.	
d.	There is no reaction with $LiAlH_4$ in ether solution.	
e.	An excess of CH_3MgBr in ether reacts to give 4-methyl-2-pentanol.	
f.	Wolff-Kishner reduction gives butane.	
g.	This compound is an isomer of 3-pentanone.	

5. This problem is an introduction to the planning of multistep syntheses.

For use, you have six reactant compounds (**A** through **F**) ; and eight reagents (**1** through **8**), shown below.

Following these lists, five multistep syntheses are outlined. For each of these, certain reactants or reagents must be identified by writing an appropriate letter or number in designated answer boxes. Write a single letter or number, indicating your choice of the best reactant or reagent, in each answer box.

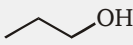
Reactant Compounds :



(A)



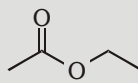
(B)



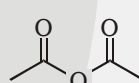
(C)



(D)



(E)



(F)

Reagents :

(1) Jones' reagent [$\text{Na}_2\text{Cr}_2\text{O}_7$ in H_3O^+]

(2) PCC [CrO_3 in pyridine + HCl]

(3) Sodium hydride NaH

(4) Sodium borohydride NaBH_4

(5) Thionyl chloride SOCl_2

(6) Phosphorus tribromide PBr_3

(7) Aluminium trichloride AlCl_3

(8) Magnesium turnings in ether

1. (1) Reagent ☐
(2) reactant ☐
(3) $\text{H}_3\text{O}^{(+)}$ $\text{C}_9\text{H}_{12}\text{O}_2$ Reagent ☐

2. Reactant ☐ C ☐ (1) Reagent ☐
Reagent ☐ $\text{C}_3\text{H}_6\text{O}_2$ (2) Reactant ☐
heat

3. (1) Reagent ☐
(2) Reagent ☐ $\text{C}_6\text{H}_{10}\text{O}_2$ (1) Reactant ☐
(3) CO_2 heating with
(4) $\text{H}_3\text{O}^{(+)}$ catalytic acid

4. Reactant ☐ C ☐ (1) NaOH solution
Reagent ☐ $\text{C}_3\text{H}_6\text{O}$ (2) Reagent ☐
(3) Reactant ☐

5. Reactant ☐ B ☐ (1) Reagent ☐
(2) Reagent ☐ $\text{C}_3\text{H}_7\text{Br}$ (2) Reactant ☐
(3) $\text{H}_3\text{O}^{(+)}$

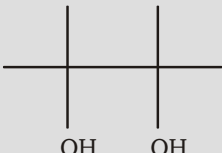
6. Which of the following is true for 3-methyl-2-butanone?

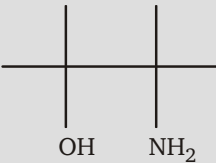
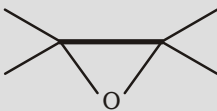
a.	It may be prepared by CrO_3 oxidation of 2-methyl-2-butanol.	
b.	Its reaction with NaBH_4 gives a secondary alcohol.	
c.	It may be prepared by acidic Hg^{2+} catalyzed hydration of 3-methyl-1-butyne.	
d.	It forms a silver mirror on treatment with $[\text{Ag}(\text{NH}_3)_2]$.	
e.	This compound is an isomer of 4-penten-1-ol.	

7. Which of these methods would serve to prepare 1-phenyl-2-propanol?

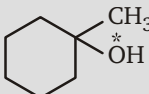
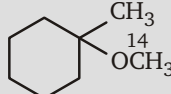
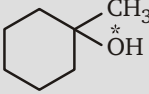
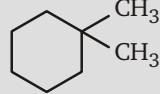
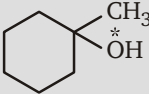
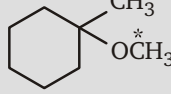
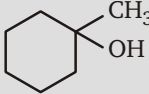
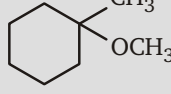
a.	Addition of benzyl Grignard reagent to acetaldehyde (ethanal).	
b.	Addition of phenyl lithium to propylene oxide (methyloxirane).	
c.	Addition of phenyl Grignard reagent to acetone (2-propanone).	
d.	Acid-catalyzed hydration (addition of water to) of 2-phenyl-1-propene.	
e.	Addition of methyl Grignard reagent to acetophenone (methyl phenyl ketone).	
f.	Addition of methyl Grignard reagent to phenylacetaldehyde.	

8. Match the Column (I) and (II).

Column (I)		Column (II)	
Reaction		Name of Reaction	
(a)		(p)	Pinacol-Pinacolone rearrangement

(b)	 NaNO_2 2HCl	(q)	Semi-Pinacol reaction
(c)	 H	(r)	Pinacolic-Diazotization
(d)	 $(1) \text{TsCl},$ $(2) \text{Et}_3\text{N},$	(s)	Pinacol fashion reaction

9. Match the Column (I) and (II).

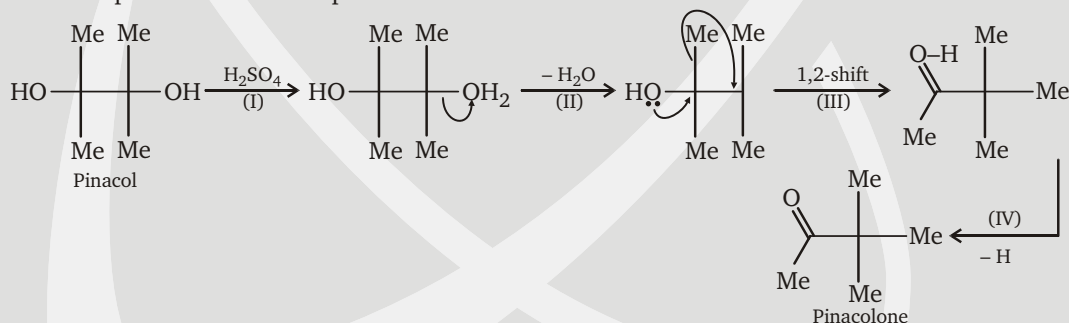
Column (I)		Column (II)	
Reactant		Products	
(a)	 CH_3OH $\text{H}_2\text{SO}_4 \text{ Conc.}$	(p)	
(b)	 $(1) \text{NaH}$ $(2) \text{CH}_3\text{I}$	(q)	
(c)	 $(1) \text{HBr}$ $(2) \text{Mg}$ $(3) \text{CH}_3\text{I}$	(r)	
(d)	 $(1) \text{Na}$ $(2) \text{C}^{14}\text{H}_5\text{I}$	(s)	

Reaction 1.	$ \begin{array}{c} \text{CH}_2 - \text{OH} \\ \\ \text{C} = \text{O} \\ \\ \begin{array}{c} \text{HO} - \text{---} - \text{H} \\ \text{H} - \text{---} - \text{OH} \\ \text{H} - \text{---} - \text{OH} \end{array} \\ \\ \text{CH}_2\text{OH} \end{array} \xrightarrow{5\text{HIO}_4} $
Reaction 2.	$ \begin{array}{c} \text{CHO} \\ \\ \begin{array}{c} \text{HO} - \text{---} - \text{H} \\ \text{H} - \text{---} - \text{OH} \\ \text{H} - \text{---} - \text{OH} \\ \text{H} - \text{---} - \text{OH} \end{array} \\ \\ \text{CH}_2\text{OH} \end{array} \xrightarrow{5\text{HIO}_4} $

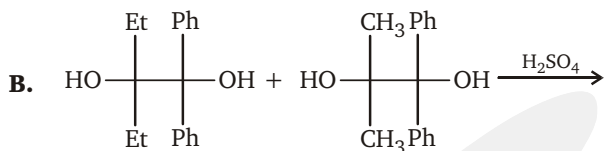
Ratio of moles of formaldehyde obtained in the reaction (1) and reaction (2) ?

11. Comprehension

Di-tert-glycols rearrange in the presence of acid to give α -tertiary ketones. The trivial name of the simplest glycol of this type is pinacol, and this type of reaction therefore is named pinacol rearrangement (in this specific case, the reaction is called a pinacol-pinacolone rearrangement). The rearrangement involves 4 steps. one of the hydroxyl groups is protonated in the first step. A molecule of water is eliminated in the second step and a tertiary carbocation is formed. The carbocation rearranges in the third step into a more stable carboxonium ion via a [1, 2] rearrangement. In the last step, the carboxonium ion is deprotonated and the product ketone is obtained.

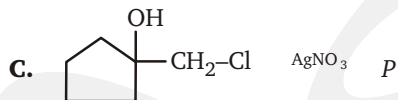


- A.** What is R.D.S. of pinacol-pinacolone rearrangement ?
- (a) I step (b) II step
(c) III step (d) IV step

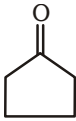
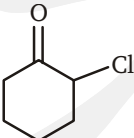
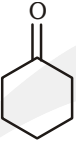
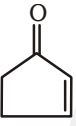


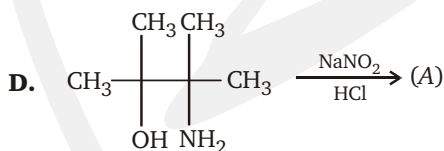
How many products obtained in above reaction ?

- (a) 1 (b) 2 (c) 3 (d) 4



Product 'P' is :

- (a)  (b)  (c)  (d) 

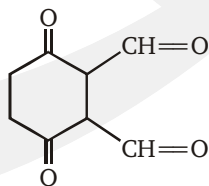


Product (A) is :

- (a)
$$\begin{array}{c} \text{O} \\ || \\ \text{CH}_3-\text{C}-\text{C}-\text{CH}_3 \\ | \quad | \\ \text{CH}_3 \quad \text{CH}_3 \end{array}$$
 (b)
$$\begin{array}{c} \text{CH}_3 \quad \text{CH}_3 \\ | \quad | \\ \text{CH}_3-\text{C}-\text{C}-\text{CH}_3 \\ | \quad | \\ \text{OH} \quad \text{OH} \end{array}$$
- (c)  (d) None of these

Subjective Problems

1. (A) 4 moles of PCC

Maximum number of moles of Ac_2O consumed by reactant (A) is :

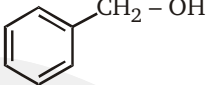
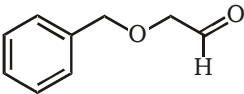
ANSWERS — LEVEL 2

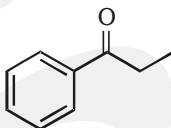
1. I – B; II – A; III – A; IV – B; V – B

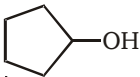
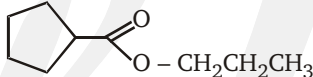
2. A – d; B – a, c, e, f, g, h; C – a, b, c; D – d; E – d, e, f, g, h; F – e

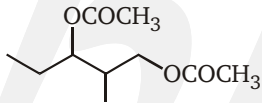
3. A – b; B – d, C – d, D – c, E – d; F – d

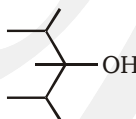
4. a, c, e, g

5.  (1) Reagent [3]
 (2) reactant [d]
 (3) $\text{H}_3\text{O}^{(+)}$ $\text{C}_9\text{H}_{12}\text{O}_2$ Reagent [2] 

Reactant [C] Reagent [1] $\text{C}_3\text{H}_6\text{O}_2$ (1) Reagent [5]
 (2) Reactant [A] Reagent [7] heat + 

 (1) Reagent [6]
 (2) Reagent [8]
 (3) CO_2
 (4) $\text{H}_3\text{O}^{(+)}$ $\text{C}_6\text{H}_{10}\text{O}_2$ (1) Reactant [C]
 heating with catalytic acid 

Reactant [C] Reagent [2] $\text{C}_3\text{H}_6\text{O}$ (1) NaOH solution
 (2) Reagent [4]
 (3) Reactant [F] 

Reactant [B] (1) Reagent [4]
 (2) Reagent [6] $\text{C}_3\text{H}_7\text{Br}$ (1) Reagent [8]
 (2) Reactant [C or F]
 (3) $\text{H}_3\text{O}^{(+)}$ 

6. b, c, e

8. a – p, b – r, c – s, d – q

10. Ratio of reaction I and II = 2

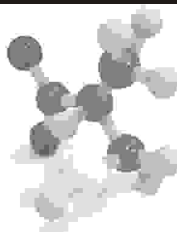
7. a, b, f

9. a – s, b – r, c – q, d – p

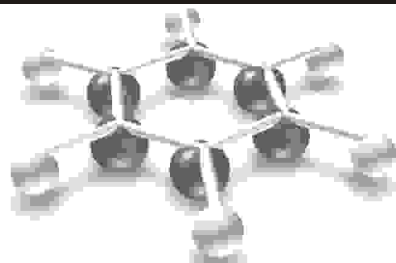
11. A – b; B – b; C – c; D – a

Subjective Problems

1. 4



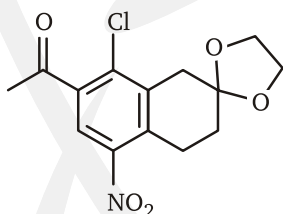
7



ALDEHYDES AND KETONES

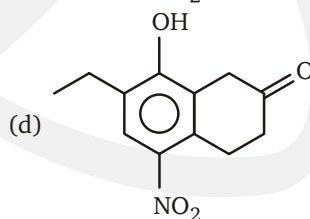
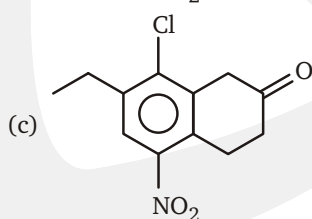
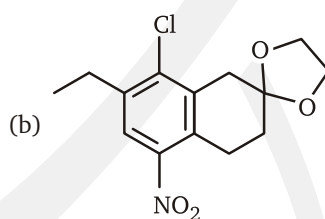
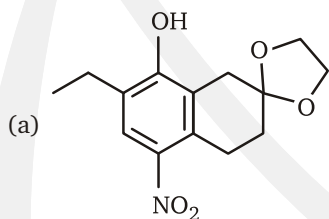
Level - 1

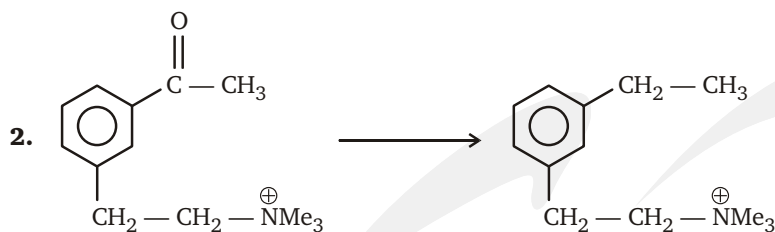
1.



$\text{N}_2\text{H}_4/\text{KOH}/\text{H}_2\text{O}$
(Wolff-Kishner reduction)

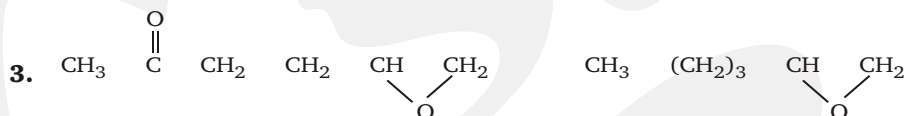
(A) ; Product A is :
major





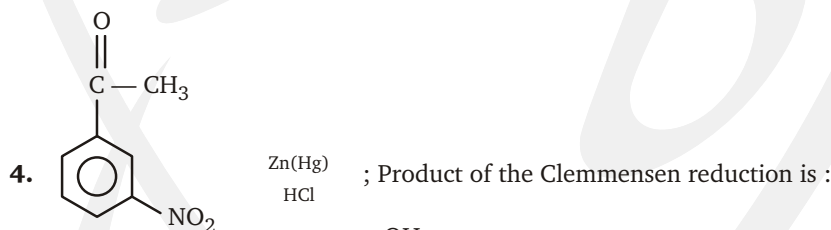
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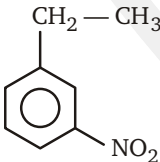
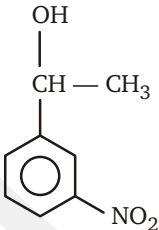
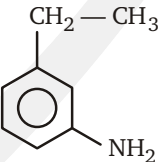
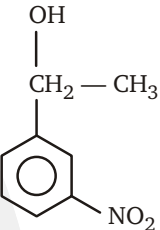
- (a) Wolf-Kishner reduction (b) Clemmensen reduction
(c) LiAlH_4 (d) NaBH_4

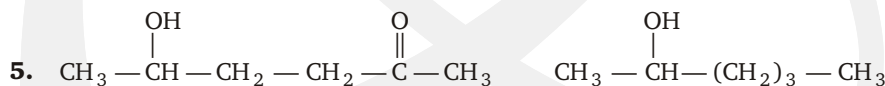


Above conversion can be achieved by :

- (a) Wolff-Kishner reduction
(b) Clemmensen reduction
(c) $\text{HS}-\text{CH}_2-\text{CH}_2-\text{SH}$, following by Raney Ni
(d) None of these

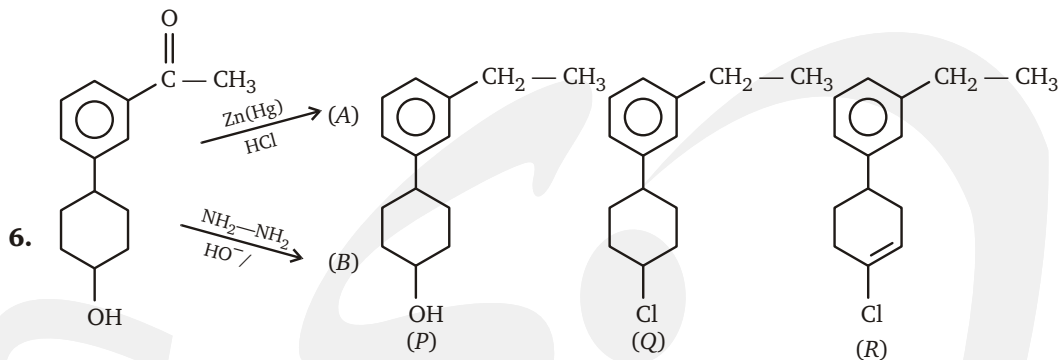


- (a)  (b)  (c)  (d) 



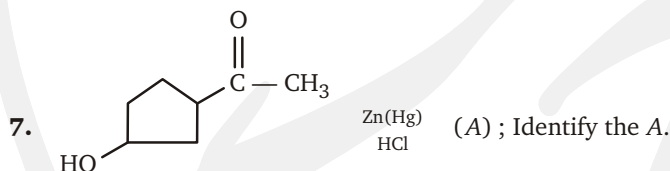
Above conversion can be achieved by :

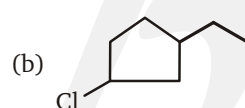
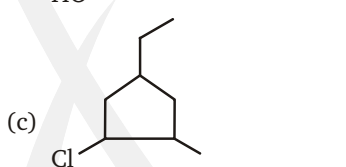
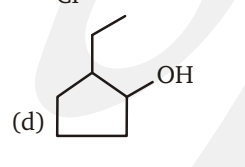
- (a) Wolff-Kishner reduction (b) Clemmensen reduction
(c) LiAlH_4 (d) NaBH_4

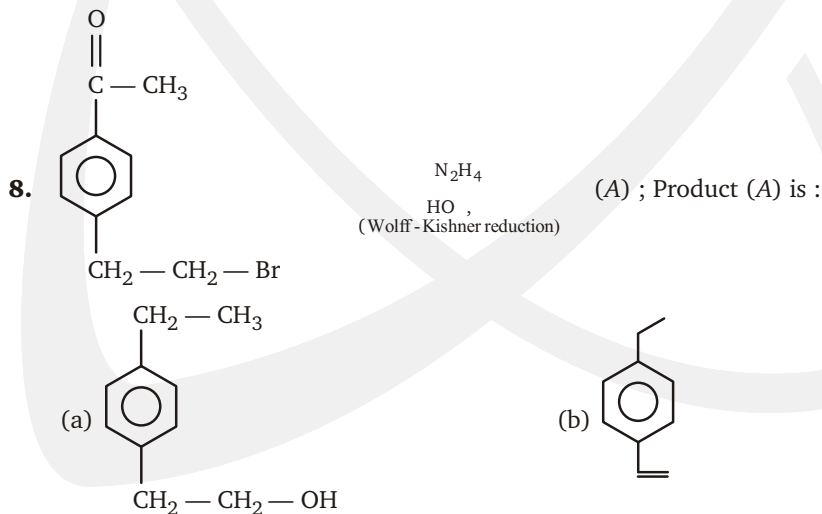


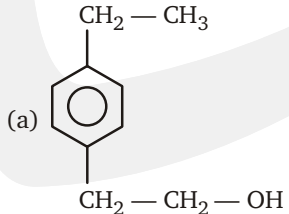
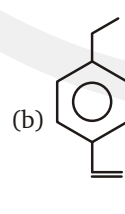
Identify product (A) & (B) from the given product P, Q, R:

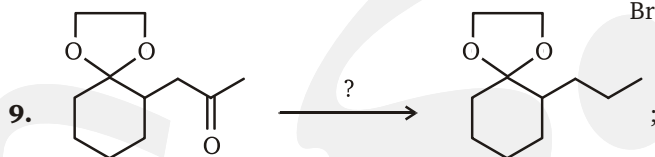
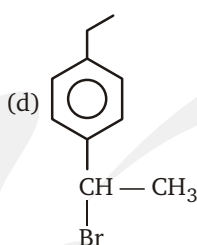
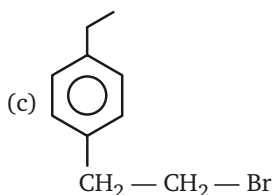
- (a) A P, B Q (b) A Q, B R (c) A Q, B P (d) A R, B P



- (a)  (b) 
 (c)  (d) 



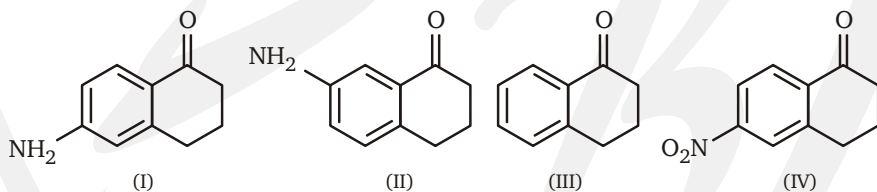
- (a)  (b) 



Above conversion can be carried out by :

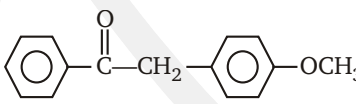
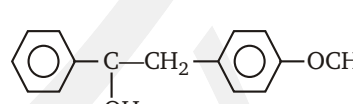
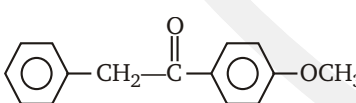
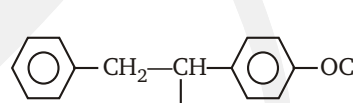
- (a) Clemmensen reduction (b) Wolff-Kishner reduction
 (c) LiAlH_4 (d) NaBH_4

10. Increasing order of equilibrium constants for the formation of a hydrate:

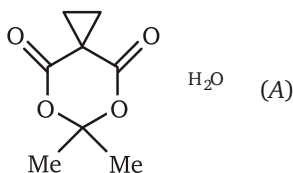


- (a) $\text{IV} < \text{III} < \text{II} < \text{I}$ (b) $\text{IV} < \text{III} < \text{I} < \text{II}$
 (c) $\text{I} < \text{II} < \text{III} < \text{IV}$ (d) $\text{II} < \text{III} < \text{I} < \text{IV}$

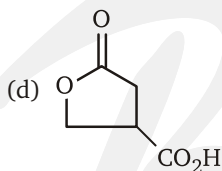
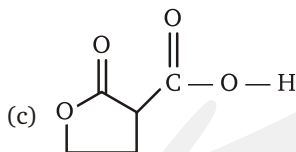
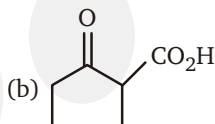
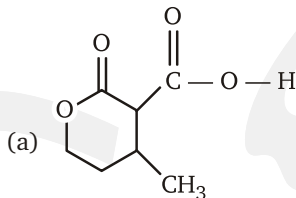
11.  (A) Product (A) is:

- (a)  (b) 
 (c)  (d) 

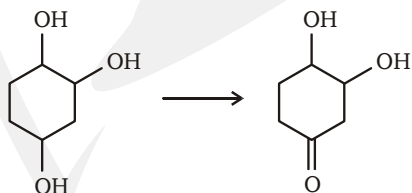
12.



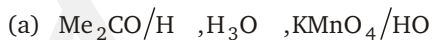
Predict the product of hydrolysis of the above molecule.



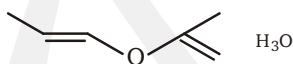
13.



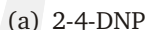
, This conversion can be achieved by :



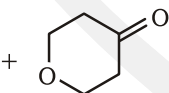
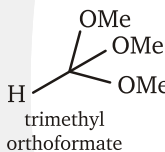
14.



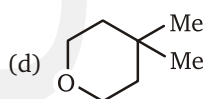
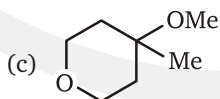
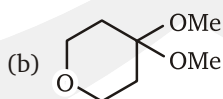
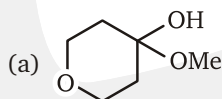
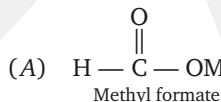
A + B. Compound (A) & (B) can be differentiated by:



15.



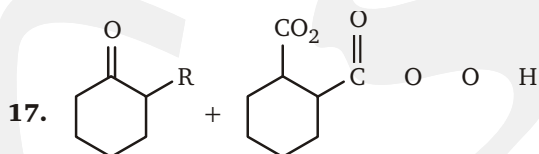
$\text{HOCH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$ / catalyst
20 °C, 15 min



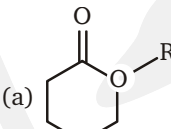
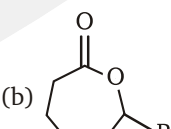
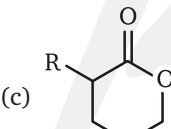
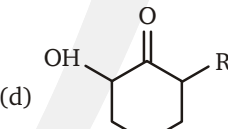


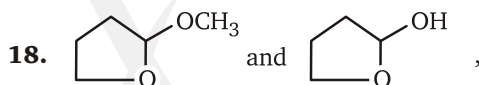
Reagents to carry out above conversion, P, Q, R respectively are :

- (a) $\text{H}_2\text{C}=\text{CH}-\text{CH}_2-\text{Br}$, $(\text{HO})_2\text{C}=\text{CH}_2$, $[\text{HO}-\text{CH}_2-\text{CH}_2-\text{OH}]$, Wacker-process
 (b) $\text{H}_2\text{C}=\text{CH}-\text{CH}_2-\text{Br}$, $(\text{HO})_2\text{C}=\text{CH}_2$, Wacker-process, $\text{HO}-\text{CH}_2-\text{CH}_2-\text{OH}$
 (c) Wacker process, $\text{H}_2\text{C}=\text{CH}-\text{CH}_2-\text{Br}$, $(\text{HO})_2\text{C}=\text{CH}_2$, $\text{HO}-\text{CH}_2-\text{CH}_2-\text{OH}$
 (d) Wacker process, $(\text{HO})_2\text{C}=\text{CH}_2$, $\text{H}_2\text{C}=\text{CH}-\text{CH}_2-\text{Br}$, $(\text{HO})_2\text{C}=\text{CH}_2$



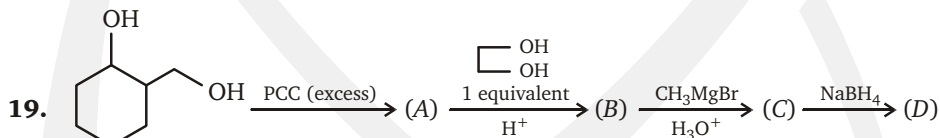
Above reaction is a Baeyer Villiger rearrangement of an asymmetric ketone with magnesium mono peroxy phthalate hexahydrate (in the drawing, Mg^{2+} is omitted for clarity) Identify major product.

- (a)  (b) 
 (c)  (d) 

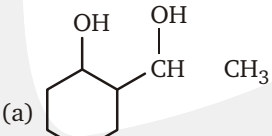
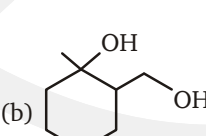
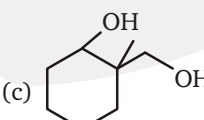
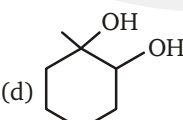


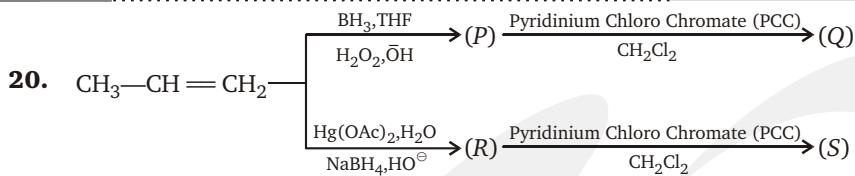
Above compounds can be differentiated by following reagent:

- (a) 2-4 DNP (Brady reagent) (b) Tollen's reagent
 (c) Lucas reagent (d) NaHSO_3



Product (D) will be :

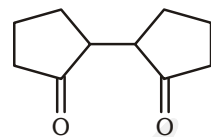
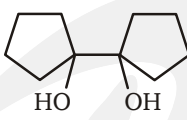
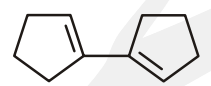
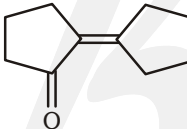
- (a)  (b) 
 (c)  (d) 



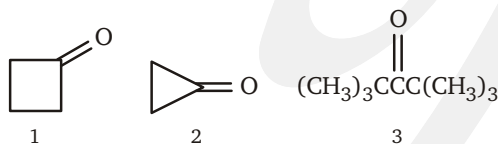
Relationship between products (Q) and (S) is:

- (a) Positional isomer (b) Chain isomer
(c) Stereoisomer (d) Functional isomer

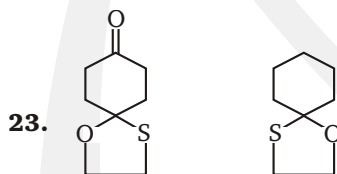


- (a)  (b) 
(c)  (d) 

22. Rank the following in order of increasing value of the equilibrium constant for hydration, K_{hyd} . (smallest value first).



- (a) $1 < 2 < 3$ (b) $3 < 1 < 2$ (c) $2 < 1 < 3$ (d) $2 < 3 < 1$



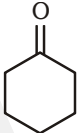
Above conversion can be achieved by:

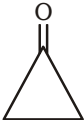
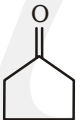
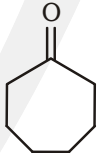
- (a) $\text{Zn}(\text{Hg}), \text{HCl}$ (b) $\text{NH}_2-\text{NH}_2/\text{KOH}/$
(c) LiAlH_4 (d) H_2/Ni

24. Which sequence represents the best synthesis of hexanal ?

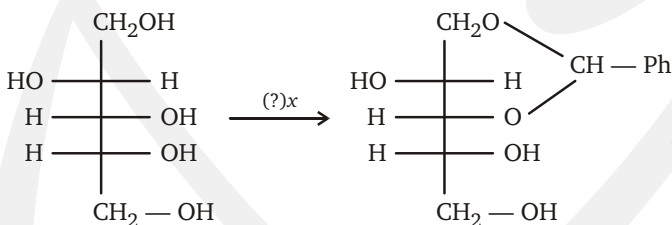


- (a) 1. $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$ $\text{NaC}\equiv\text{CH}$ (b) 1. $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}=\text{CH}_2$ $\text{CH}_3\text{C}(=\text{O})\text{OOH}$
 2. H_2O , H_2SO_4 , HgSO_4 2. CH_3MgBr , diethyl ether
 3. H_3O^+
 4. PCC, CH_2Cl_2
- (c) 1. $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{C}(=\text{O})\text{CH}_3$ (d) 1. $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{MgBr} + \text{H}_2\text{C}=\text{CH}_2$
 2. $\text{CH}_3\text{C}(=\text{O})\text{OOH}$ 3. LiAlH_4 2. H_3O^+ 3. PCC, CH_2Cl_2
 4. H_2O 5. PCC, CH_2Cl_2

25.  HNO_3 (A) $\text{Ca}(\text{OH})_2$ (B), Product (B) in this reaction is :

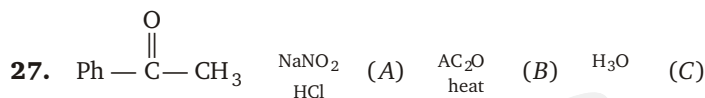
- (a)  (b)  (c)  (d) 

26.



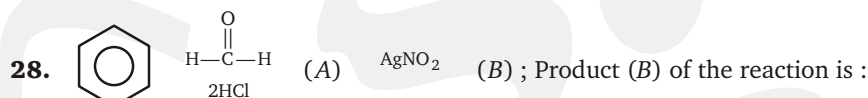
Compound (x) in the above reaction is :

- (a) $\text{Ph} - \text{C}(=\text{O}) - \text{CH}_3$ (b) $\text{Ph} - \text{C}(=\text{O}) - \text{H}$
 (c) $\text{Ph} - \text{CH}_2 - \text{C}(=\text{O}) - \text{H}$ (d) $\text{Ph} - \text{CH}_2 - \text{C}(=\text{O}) - \text{CH}_3$

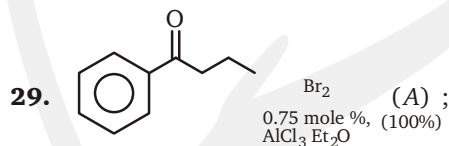


Product (C) of the above reaction is :

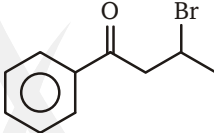
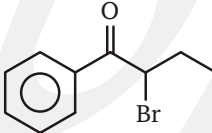
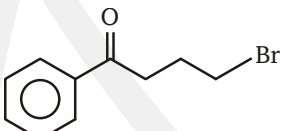
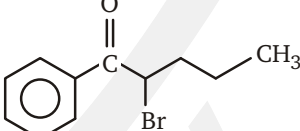
- (a) $\text{Ph}-\text{CO}_2\text{H}$ (b) $\text{Ph}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CO}_2\text{H}$
 (c) $\text{Ph}-\overset{\text{O}}{\parallel}{\text{C}}-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$ (d) $\text{Ph}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_2\text{OH}$

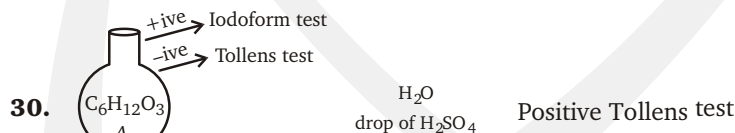


- (a) $\text{Ph}-\text{CH}_2-\text{NO}_2$ (b) $\text{Ph}-\text{CH}_2-\text{ONO}$
 (c) $\text{Ph}-\text{CHO}$ (d) $\text{Ph}-\text{O}-\text{N}=\text{O}$



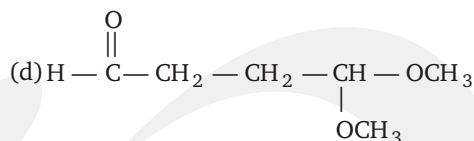
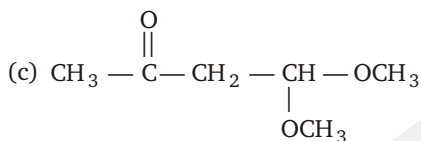
Product (A) of the above reaction is (bromination occur not in the benzene ring) :

- (a)  (b) 
 (c)  (d) 

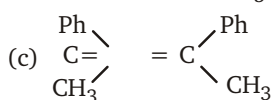
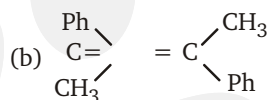
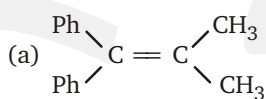


Compound (A) is :

- (a) $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\underset{\text{OCH}_3}{\underset{|}{\text{CH}}}-\underset{\text{OCH}_3}{\underset{|}{\text{CH}_2}}$ (b) $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\underset{\text{OCH}_3}{\underset{|}{\text{C}}}(\text{OCH}_3)-\text{CH}_3$

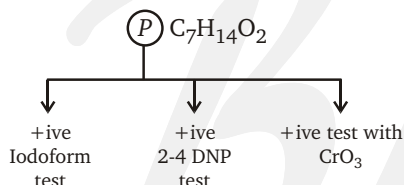
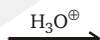
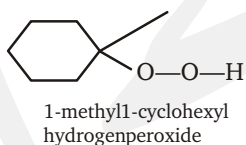


31. $\text{C}_{16}\text{H}_{16}$ (A) $\xrightarrow{\text{O}_3}$ (B) $\text{C}_8\text{H}_8\text{O}$ $\xrightarrow[\text{HO}]{\text{NH}_2-\text{NH}_2}$ ; Reactant (A) in this reaction is :

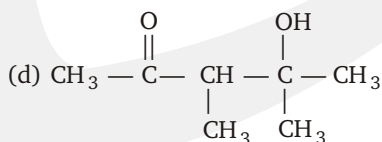
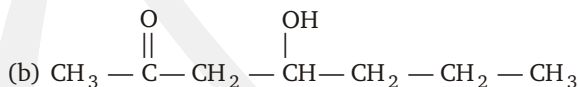
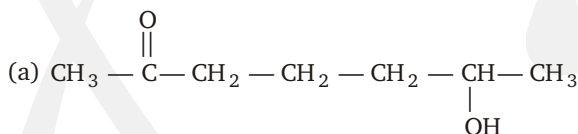


(d) both (b) and (c)

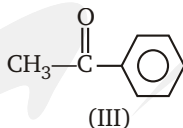
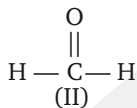
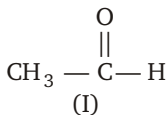
32.



Compound (P) is :



33. Correct order of reactivity of following compounds towards Grignard reagent?



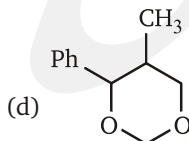
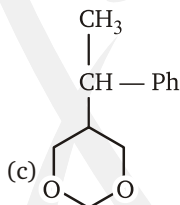
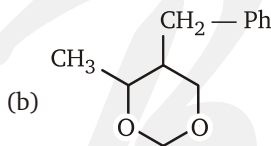
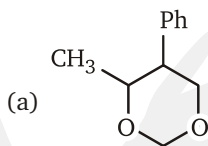
(a) I II III

(b) II I III

(c) II III I

(d) I III II

34. $\text{CH}_3 - \underset{\text{OH}}{\underset{|}{\text{CH}}} - \underset{\text{CH}_2}{\underset{|}{\text{CH}}} - \text{C}_6\text{H}_5$ $\xrightarrow[\text{TsOH}]{\text{HCHO}}$ (A) ; Product (A) is : 93%



35. $(\text{CH}_3)_3\text{CO} - \overset{\text{O}}{\parallel} \text{C} - \text{CH}_2 - \text{CH}_2 - \underset{\text{OH}}{\underset{|}{\text{CH}}} - \text{C}_6\text{H}_2(\text{OCH}_3)_2(\text{I}) - \text{OCH}_2\text{CH}_2\text{CH}_3$ $\xrightarrow[\text{H}_2\text{O}]{\text{KOH}}$

Total number of products obtained in above reaction is :

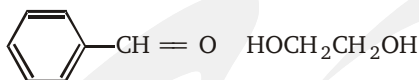
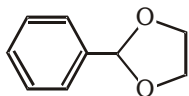
(a) 2

(b) 3

(c) 4

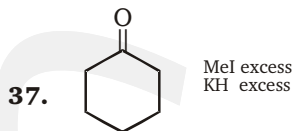
(d) 5

36. What reagent and/or reaction conditions would you choose to bring about the following conversion?

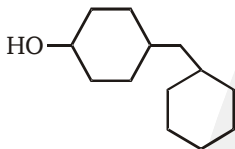
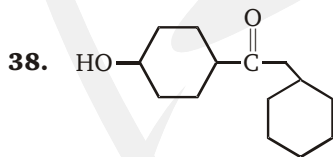
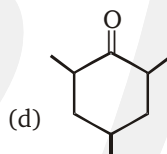
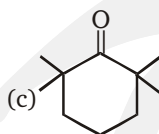
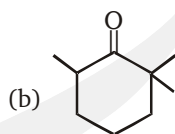
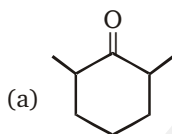


- (a) 1. LiAlH_4 , 2. H_2O
(c) H_2O , NaOH , heat

- (b) H_2O , H_2SO_4 , heat
(d) PCC , CH_2Cl_2

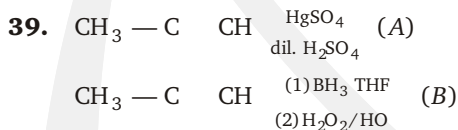


81% yield ; Product of the reaction is :



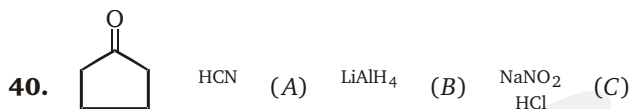
The above reduction can be best carried out by :

- (a) Clemmensen reduction
(b) Wolff-Kishner reduction
(c) NaBH_4
(d) None of these

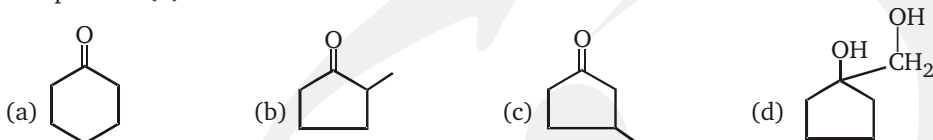


Product (A) and (B) is differentiated by:

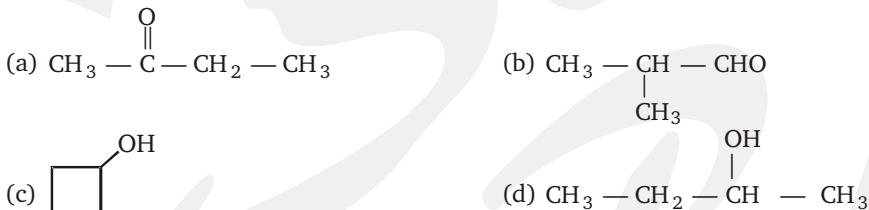
- (a) 2-4-DNP
(b) NaOI
(c) Na-metal
(d) NaHSO_3



End product (C) in above reaction is :



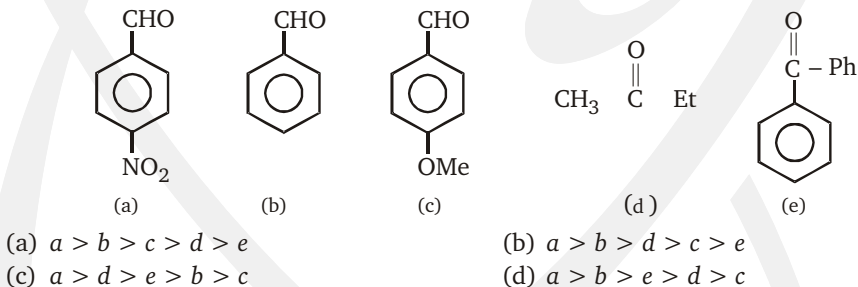
41. Compound (X) $\text{C}_4\text{H}_8\text{O}$, which reacts with 2, 4-DNP derivative and gives negative haloform test is :

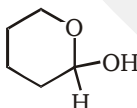


42. When a nucleophile encounters a ketone, the site of attack is :

- (a) the carbon atom of the carbonyl
 (b) the oxygen atom of the carbonyl
 (c) both the carbon and oxygen atoms, with equal probability
 (d) no attack occurs as ketones do not react with nucleophiles

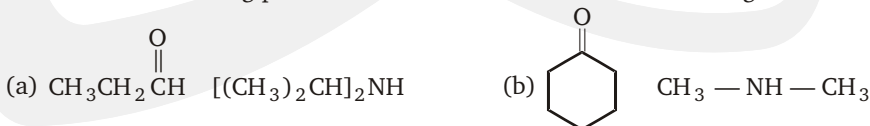
43. The correct order of rate of reaction toward nucleophilic addition reaction:

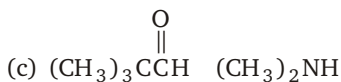


44. The structure  would be best classified as a(an) :

- (a) Acetal (b) Hemiacetal (c) Hydrate (d) Cyanohydrin

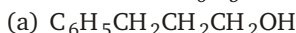
45. Which of the following pairs of reactants is most effective in forming an enamine ?

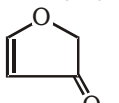


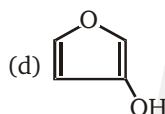
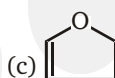
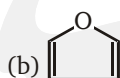
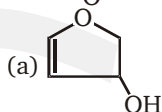


(d) None of these form an enamine.

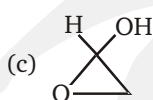
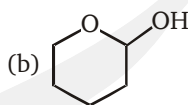
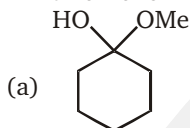
46. The reaction of $\text{C}_6\text{H}_5\text{CH}=\text{CHCHO}$ with LiAlH_4 gives :



47.  (A) H^+ (B) ; Product (B) of the reaction is :

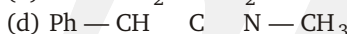
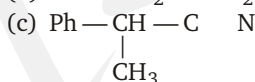
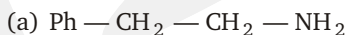


48. Which of following compound is hemiacetal ?



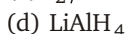
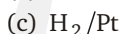
(d) all of these

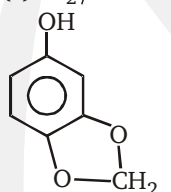
49. $\text{Ph}-\text{CH}_2-\text{C}(=\text{N})-\text{N}(\text{CH}_3)_2$ $\xrightarrow[\text{THF}]{\text{LDA}}$ CH_3I 71% ; End product of the reaction will be :

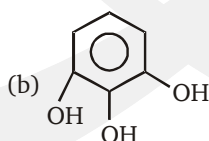
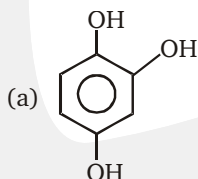


50. $\text{Ph}-\text{CH}=\text{CH}-\text{C}(=\text{O})-\text{CH}_3 \xrightarrow{\text{H}_2\text{O}}$ $\text{Ph}-\text{CH}(\text{OH})-\text{CH}_2-\text{CO}_2\text{H}$

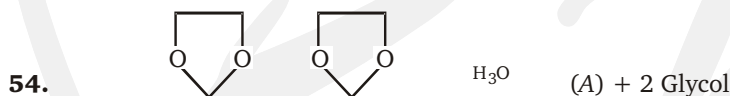
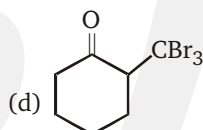
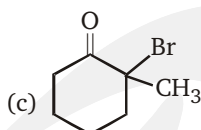
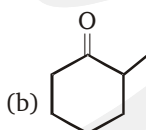
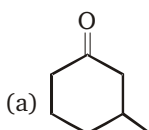
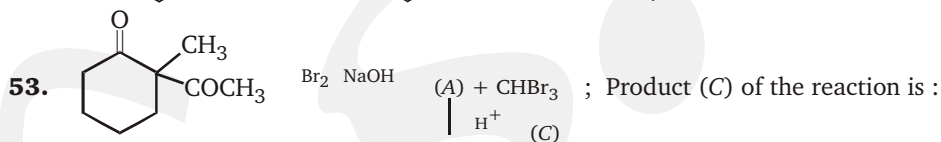
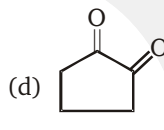
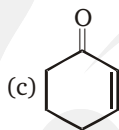
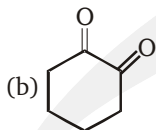
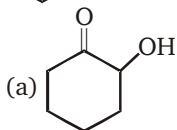
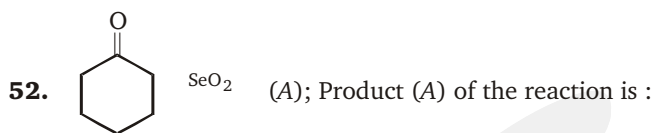
Above conversion can be achieved by :



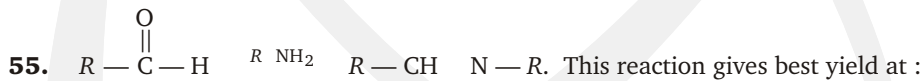
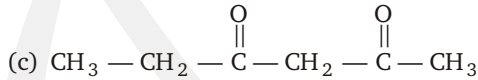
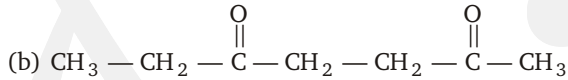
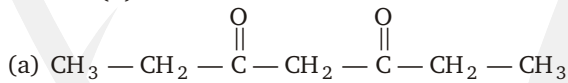
51.  H_3O^+ Products ; Product of the reaction is/are :



(d) Both (a) and (c)



Product (A) of the reaction will be :

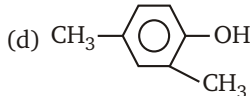
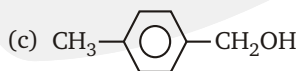
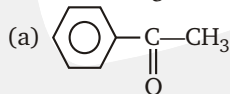
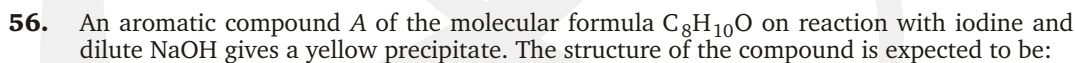


(a) pH 1 - 2

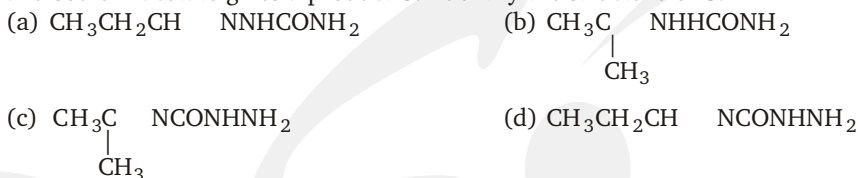
(b) pH 4 - 5

(c) pH 10 - 11

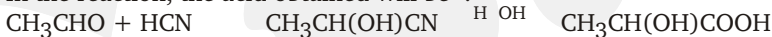
(d) pH 13 - 14



57. Compound A (molecular formula C_3H_8O) is treated with acidified potassium dichromate to form a product B (molecular formula C_3H_6O). B forms a shining silver mirror on warming with ammoniacal silver nitrate, B when treated with an aqueous solution of $NH_2NHCONH_2$ and sodium acetate gives a product C. Identify the structure of C.

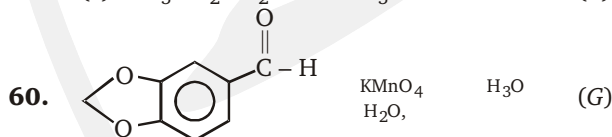
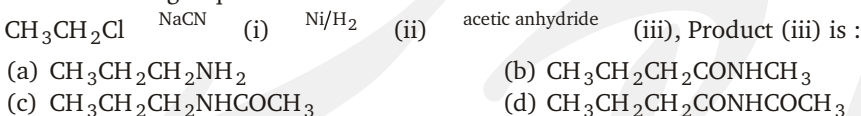


58. In the reaction, the acid obtained will be :

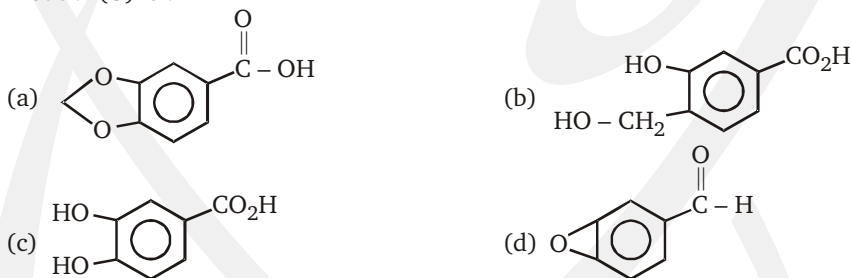


- (a) D-isomer (b) L-isomer
 (c) (80%D + 20%L) mixture (d) (50%D + 50%L) mixture

59. In the following sequence :



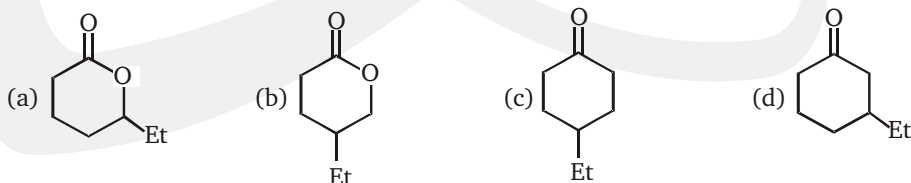
Product (G) is :

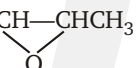


61. Carbonyl compounds can generally be converted to hydrocarbons by :

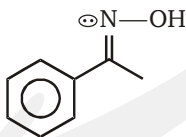


62. (A); Product (A) is :

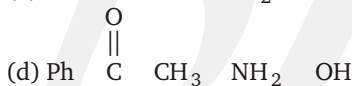
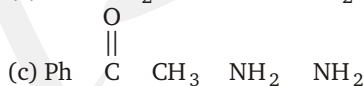


63. Which statement about the aldol condensation is correct ?
 (a) A Lewis acid is commonly used as a catalyst
 (b) The initial step is probably the formation of a carbanion
 (c) A Lewis base is employed to induce carbocation formation
 (d) The carbon chain is lengthened through the elimination of 1 mole of water
64. A compound gives a positive test with $I_2/NaOH$ and is extracted from benzene by saturated $NaHSO_3$. It may be :
 (a) $CH_3(CH_2)_4CHO$
 (b) $CH_3(CH_2)_3COCH_3$
 (c) $CH_3CH_2COCH_2CH_3$
 (d) $CH_3(CH_2)_4CH_2OH$
65. Which of the following compounds on reaction with excess CH_3MgBr and subsequent hydrolysis will give a tertiary alcohol?
 (a) C_2H_5CHO (b) $C_2H_5CO_2CH_3$ (c) C_2H_5COOH (d) $CH_3CH-CHCH_3$


66. $A + B$

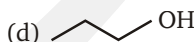
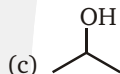
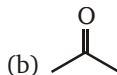
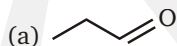


Reactant (A) and (B) is :

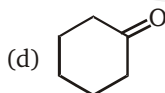
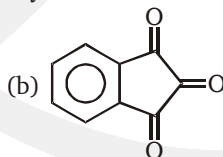
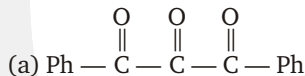


67. $CH_3-C(=O)-OH \xrightarrow{Ca(OH)_2} (A)$

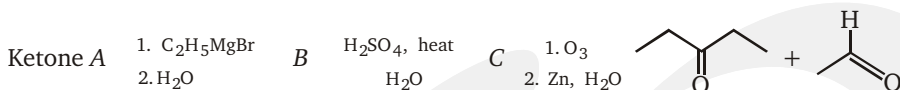
Product (A) is :



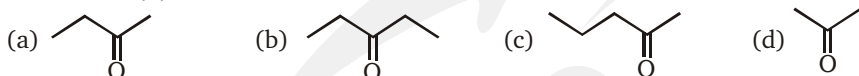
68. Which of the following does not form a stable hydrate on addition of H_2O ?



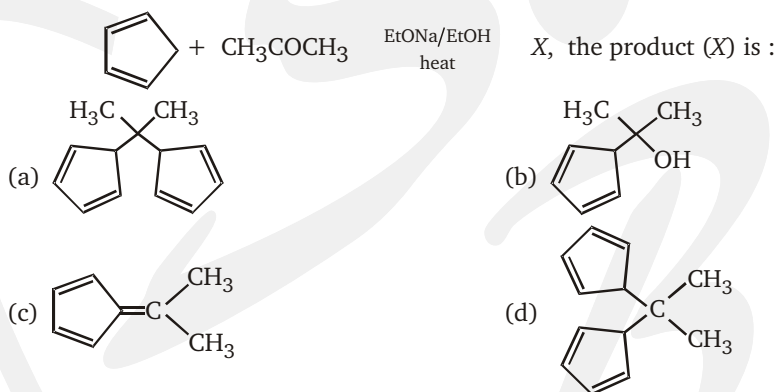
69. Consider the following sequence of reactions.



The ketone (A) is :



70. In the reaction,



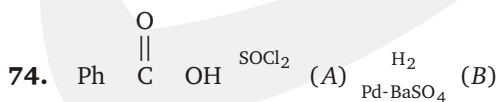
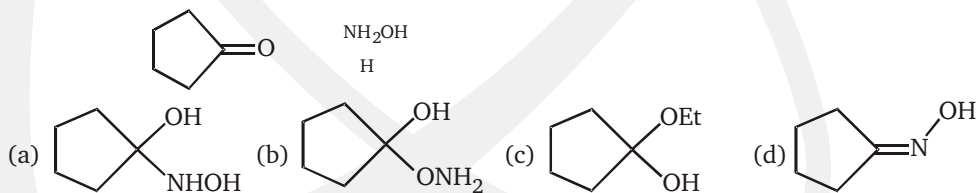
71. The conversion of acetophenone into benzoic acid can be achieved by its reaction with :

- (a) sodium hydroxide followed by acidification
 (b) iodine and sodium hydroxide, followed by acidification
 (c) hydroxylamine followed by reaction with H_2SO_4
 (d) *m*-chloroperoxybenzoic acid

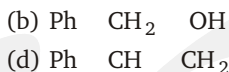
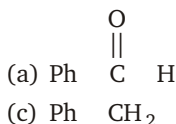
72. In which of the following compounds the methylenic hydrogens are the most acidic ?

- (a) $\text{CH}_3\text{COCH}_2\text{CH}_3$ (b) $\text{CH}_3\text{CH}_2\text{COOC}_2\text{H}_5$
 (c) $\text{CH}_3\text{CH}_2\text{CH}(\text{COOC}_2\text{H}_5)_2$ (d) $\text{CH}_3\text{COCH}_2\text{CN}$

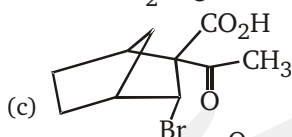
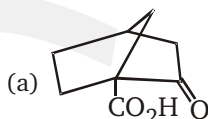
73. Which is the major product of the following reaction ?



Product (B) is :



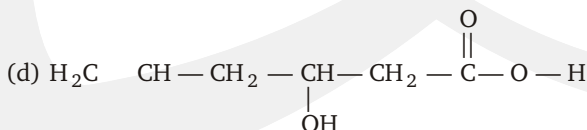
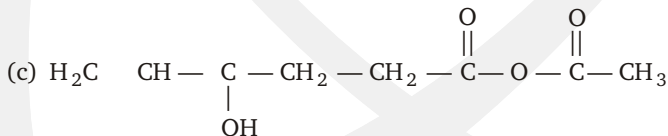
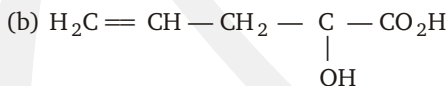
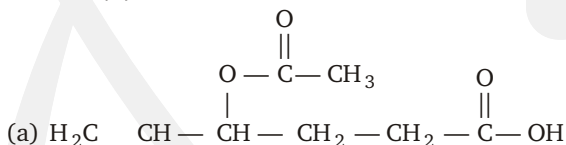
75. The presence of unsaturation in organic compounds can be tested with :
 (a) Schiff's reagent (b) Tollens' reagent (c) Fehling's reagent (d) Baeyer's reagent
76. Which of the following gives iodoform test ?
 (a) $\text{CH}_3\text{CH}_2\text{OH}$ (b) $\text{C}_2\text{H}_5\text{CHO}$ (c) $(\text{CH}_2\text{OH})_2$ (d) None of these
77. Which of the following α -keto carboxylic acid does not undergo decarboxylation on heating ?



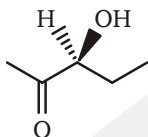
(d) None of these

78. $\text{HOCH}_2\text{CH}_2\text{CH}_2 - \text{C}(=\text{O}) - \text{OCH}_2\text{CH}_3$ $\xrightarrow{\text{PCC}}$ (A) $\text{H}_2\text{C} = \text{CHMgBr}$ (1 molar equivalent) (B) $\text{NH}_4\text{Cl}/\text{H}_2\text{O}$ (C) KOH H_2O (D) $(\text{CH}_3 - \text{C}(=\text{O}))_2\text{O}$ Pyridine

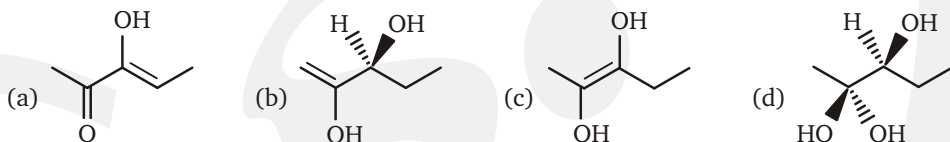
Product (D) is :



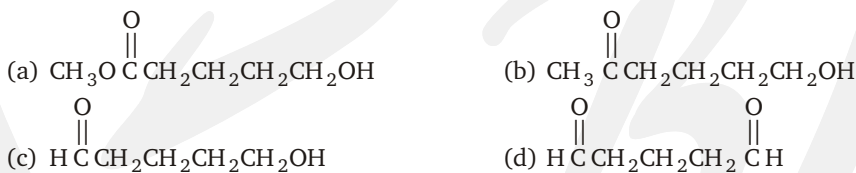
79. The compound shown in the below undergoes racemization on reaction with aqueous acid.



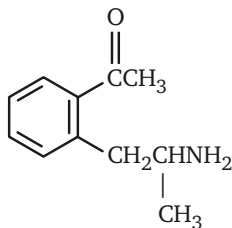
Which of the following structures best represents the intermediate responsible for this process?



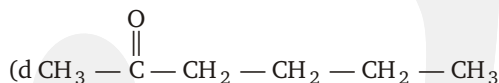
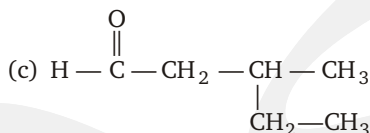
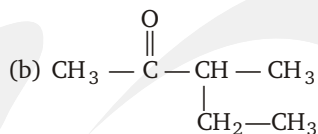
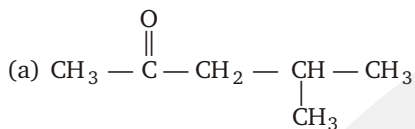
80. The final product of the following sequence of reaction is :



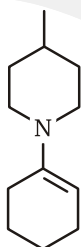
81. The amino ketone shown below undergoes a spontaneous cyclization on standing. What is the major product of this intramolecular reaction ?



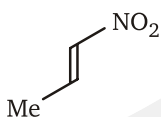
82. Compound (A) $C_6H_{12}O$ is optically active. Compound (A) give negative Tollens test and positive test with 2-4-di-nitro phenyl hydrazine. Identify A.



83.

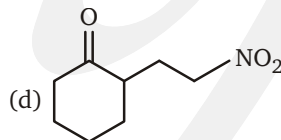
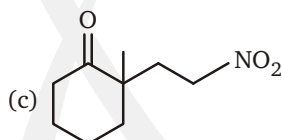
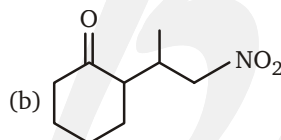
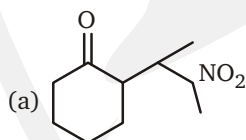


+

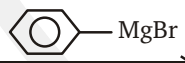
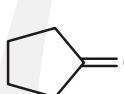


(1) Et_2O
(2) H_3O^+

(A) ; Product (A) of the reaction is :
99%



84.



(N)

NH_4Cl
 H_2O

(O)

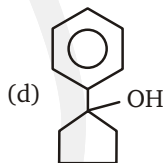
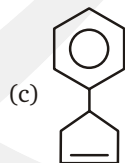
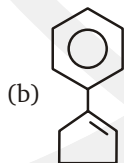
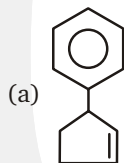
HCl (conc.)
cold

(P)

KOH (4 molar)
ethanol,

(Q)

Product (Q) will be :



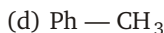
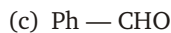
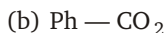
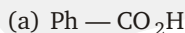
85.

 CrO_2Cl_2

(A)

conc. KOH 

Product (B) of above the reaction is :



86. $\begin{array}{c} \text{CHO} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{CH}_2 - \text{OH} \end{array}$ (i) KCN (ii) H Product ; Product obtained in the reaction is :

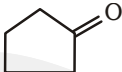
D-(+)-Glyceraldehyde

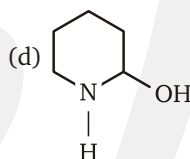
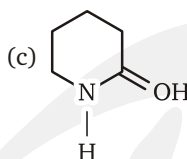
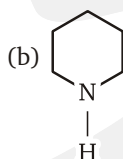
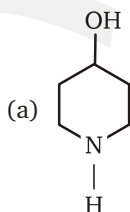
(a) Diastereomer

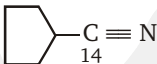
(c) Meso

(b) Racemic

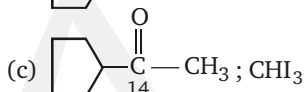
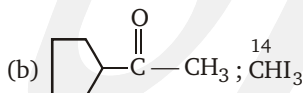
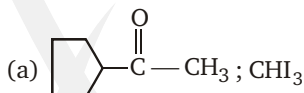
(d) Optically pure enantiomer

87.  NH_2OH (A) H (B) LAH (C); Product (C) of the reaction is :



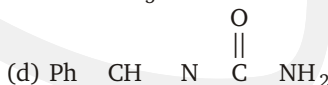
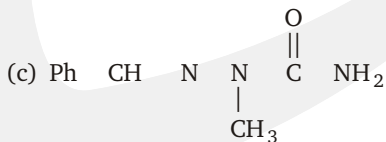
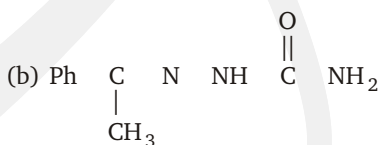
88.  CH_3MgBr H_3O^+ (A) NaOI (B) (C)

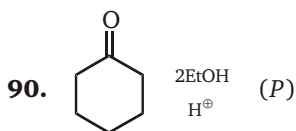
Product (A) and (C) is :



89. $\text{Ph} - \text{CH}(\text{OH}) - \text{CH}_3$ PCC (A) $\text{NH}_2 - \text{NH} - \text{C}(=\text{O}) - \text{NH}_2$ (B)

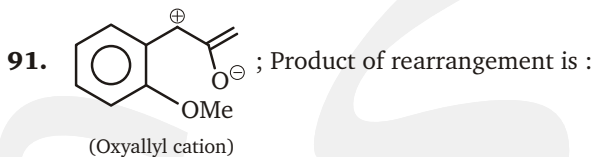
Product (B) is :

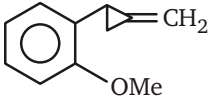
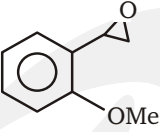
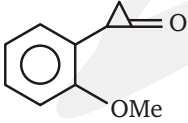
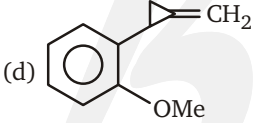


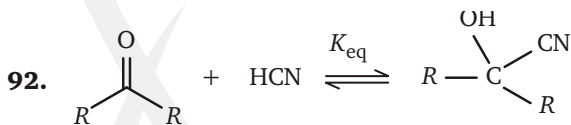


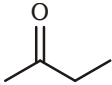
Product (P) is :

- (a) Hemiacetal (b) Acetal (c) Alcohol (d) Alkane



- (a)  (b) 
(c)  (d) 

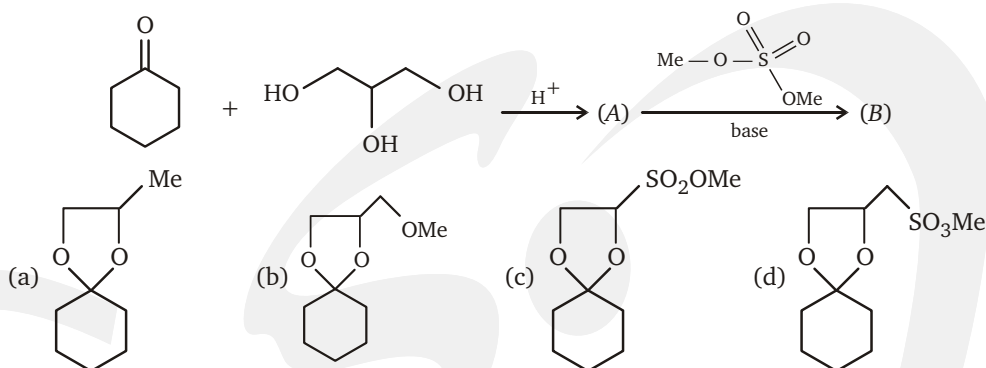


Reactant	$K_{\text{eq.}}$
PhCHO	<i>a</i>
	<i>b</i>
$\text{Ph}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3$	<i>c</i>
$\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{H}$	<i>d</i>

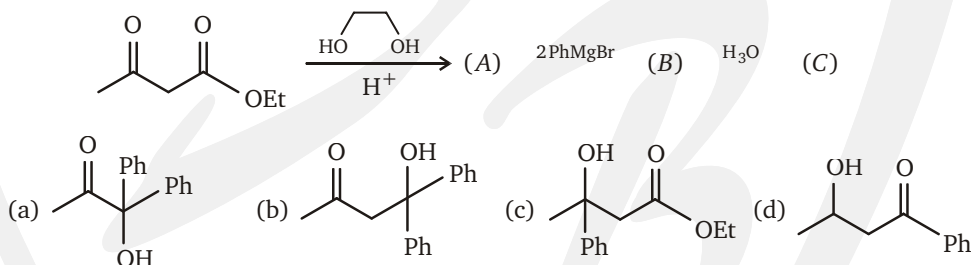
The correct order of decreasing value of $K_{\text{eq.}}$ is :

- (a) $a > b > c > d$ (b) $d > a > b > c$
(c) $d > b > a > c$ (d) $d > a > c > d$

93. Product (B) of the given reaction is :



94. End product (C) of the reaction is :



95. (A) $\xrightarrow{\text{O}_3}$ does not undergo self aldol condensation
 $\text{C}_{11}\text{H}_{18}\text{O} \xrightarrow{\text{O}_3} \text{Ph}-\text{CHO} + 2\text{b} \xrightarrow{\text{Ag}^+} \text{oxalic acid}$

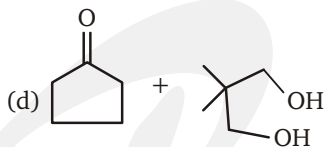
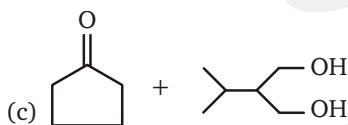
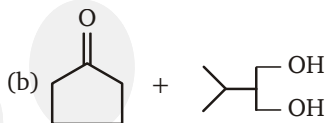
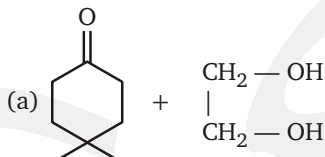
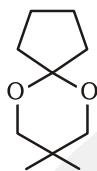
Compound (A) will be :

- (a) $\text{Ph}-\text{C}(\text{CH}_3)_2-\text{CH}_2-\text{CHO}$ (b) $\text{Ph}-\text{C}(\text{CH}_3)_2-\text{CH}_2-\text{CH}_2-\text{CHO}$
 (c) $\text{Ph}-\text{CH}(\text{CH}_3)-\text{CH}_2-\text{CH}_2-\text{CHO}$ (d) $\text{Ph}-\text{CH}(\text{CH}_3)-\text{CH}_2-\text{CH}_2-\text{CO}_2\text{H}$

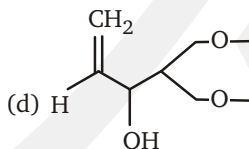
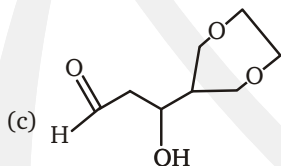
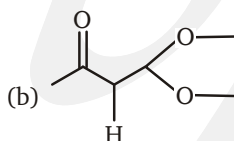
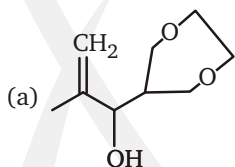
96. Product ; Product of the reaction is :



97. Which pair of reactants compounds may be used to make given acetal ?



98. (A) $\xrightarrow{\text{H}}$ (B) ; (A) & (B) are isomers ; Isomer (B) is :



99. PCC (A)

PCC (B)

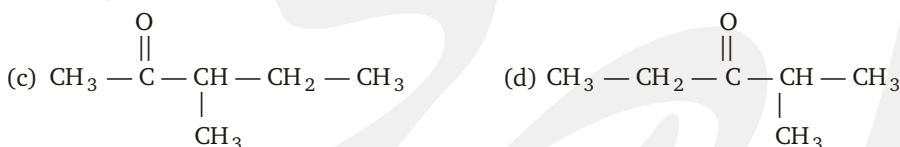
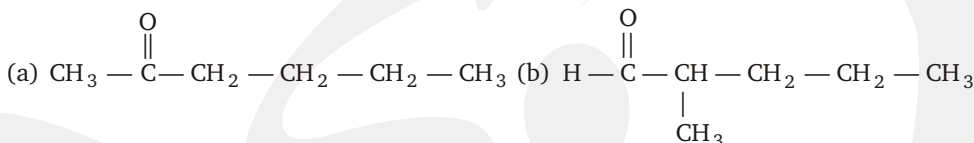
(A) and (B) is differentiated by :

- (a) NaH (b) 2-4 DNA (c) Tollen's reagent (d) NaHSO_3

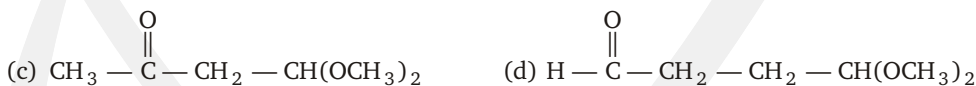
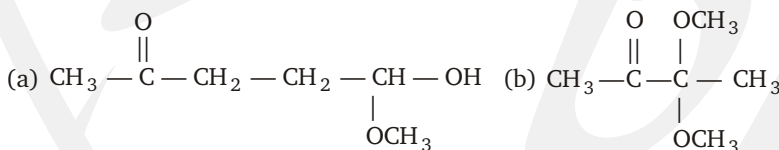
100. Which of the following pairs cannot be differentiated by Tollens' reagent ?

- (a) Benzaldehyde and benzyl alcohol (b) Hexanal and 2-hexanone
(c) 2-Hexanol and 2-hexanone (d) Pentanal and diethyl ether

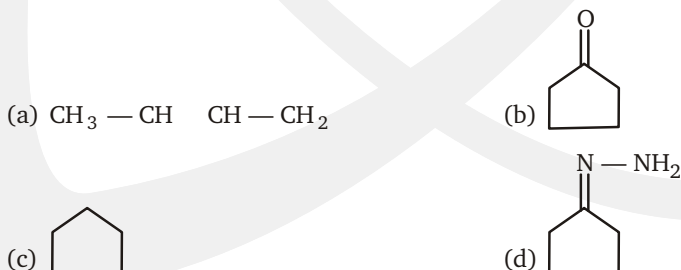
101. An optically active compound $C_6H_{12}O$ gives positive test with 2, 4-dinitrophenyl hydrazine, but negative with Tollens' reagent, what is the structure of the compound ?



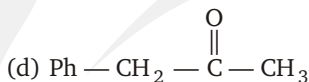
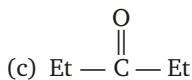
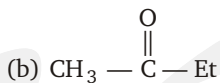
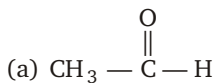
102. Compound (A) $C_6H_{12}O_3$, when treated with I_2 in aqueous sodium hydroxide gives yellow precipitate. When A is treated with Tollens reagent no reaction occur. When A is hydrolysed and then treated with Tollens reagent, a silver mirror is formed in test tube. Compound (A) will be :

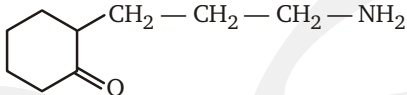


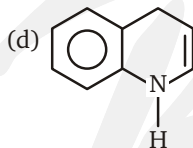
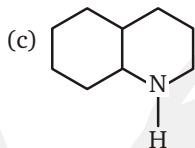
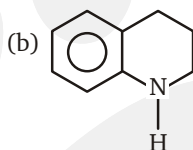
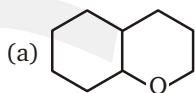
103. $\begin{array}{c} CH_2 - CH_2 - CO_2H \\ | \\ CH_2 - CH_2 - CO_2H \end{array} \xrightarrow{BaCO_3} A \xrightarrow{NH_2-NH_2} B \xrightarrow[heat]{KOH} (C)$, Product (C) obtained is :

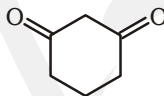


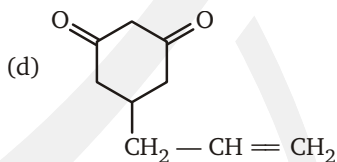
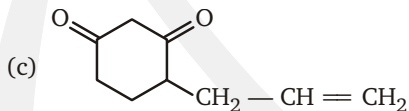
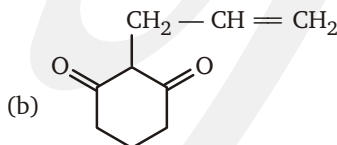
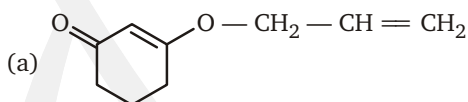
104. Which of following does not react with NaHSO_3 (sodium bisulphite)?

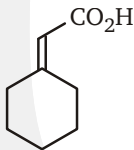


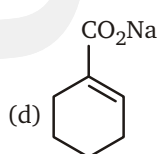
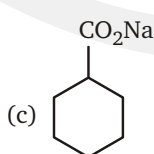
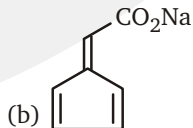
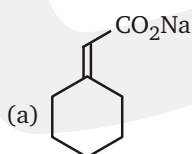
105.  $\xrightarrow[\text{Raney Ni}]{\text{H}_2}$ (A) ; Product (A) is :

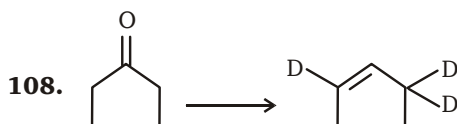


106.  + $\text{CH}_2 = \text{CH} - \text{CH}_2 - \text{Br} \xrightarrow{\text{KOH}}$ (A) ; Product (A) is : (75%)



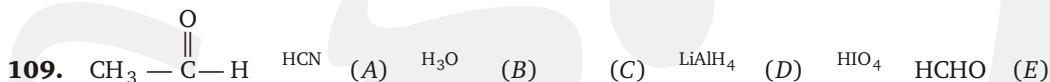
107.  $\xrightarrow[\text{(2) HCl/H}_2\text{O}]{\text{(1) Me-Li(excess)}}$ (A) $\xrightarrow[\text{NaOH}]{\text{I}_2}$ (B) CHI_3 ; Product (B) in this reaction is :





Arrange the following reagent in the correct order in which above transformation is carried out :

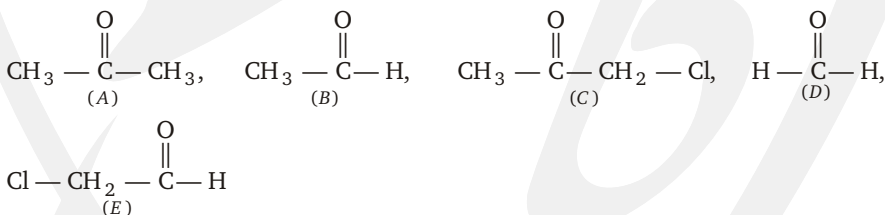
- (a) $\text{KOD}/\text{D}_2\text{O}, \text{H} / , \text{LiAlH}_4$ (b) $\text{H} / , \text{KOD}/\text{D}_2\text{O}, \text{LiAlH}_4$
 (c) $\text{KOD}/\text{D}_2\text{O}, \text{LiAlH}_4, \text{H} /$ (d) $\text{LiAlH}_4, \text{H} / , \text{KOD}/\text{D}_2\text{O}$



Compound (C) can show geometrical isomerism. Product (E) of the reaction will be :

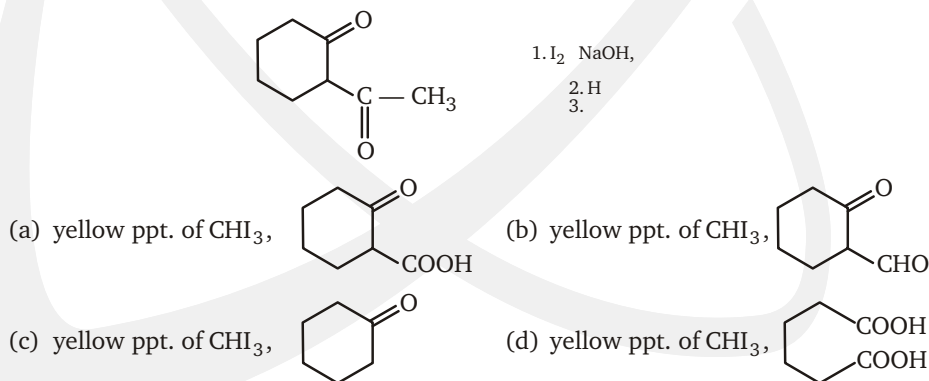
- (a) $\text{CH}_3 - \overset{\text{O}}{\parallel} \text{C} - \text{CH}_3$ (b) $\text{CH}_3 - \text{CH}_2 - \overset{\text{O}}{\parallel} \text{C} - \text{H}$
 (c) $\text{CH}_3 - \text{CHO}$ (d) HCHO

110. Arrange in their increasing order of equilibrium constants for hydration ?

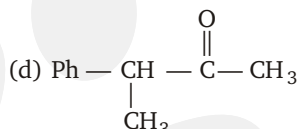
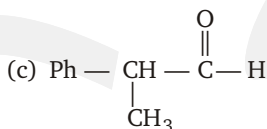
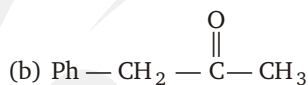
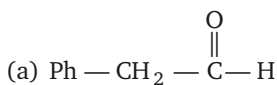


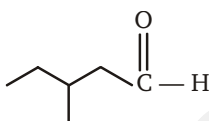
- (a) $A < B < C < D < E$ (b) $A < C < B < E < D$
 (c) $A < C < E < B < D$ (d) $C < A < B < E < D$

111. End products of the following sequence of reactions are :



112. $\text{Ph}-\text{CH}_2-\text{CN}$ $\xrightarrow{(1) \text{EtONa}}$ (P) ; Product (P) of the reaction will be :
 $\xrightarrow{(2) \text{CH}_3-\text{C}(=\text{O})-\text{Cl}}$
 $\xrightarrow{(3) \text{H}_3\text{O}^+}$

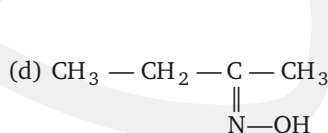
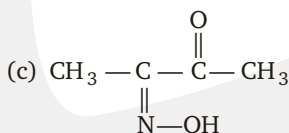
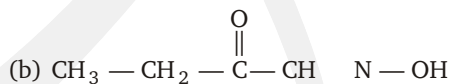
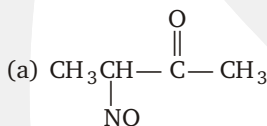


113.  $\xrightarrow{(i) \text{NaCN}}$ Products. Products of the reaction are :
 $\xrightarrow{(ii) \text{H}_2\text{SO}_4}$
- (a) Racemic mixture
 (b) Diastereomers
 (c) Meso
 (d) Mixture of meso compound and optically active compound

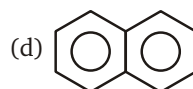
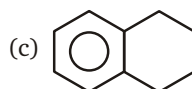
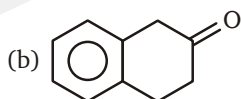
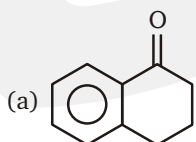
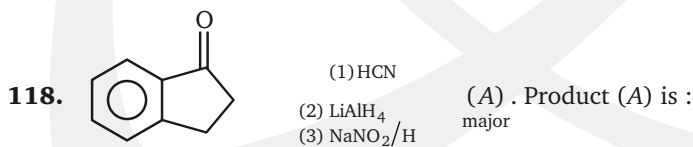
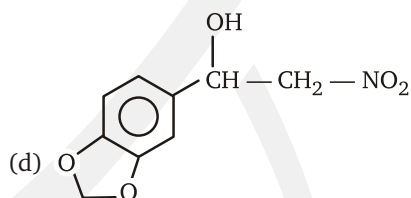
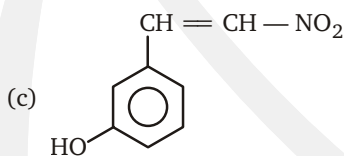
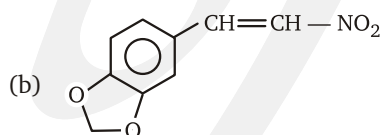
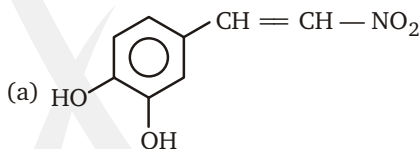
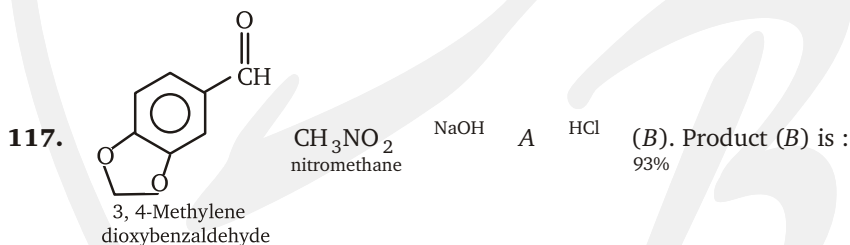
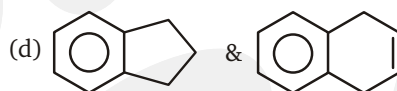
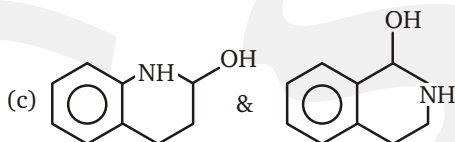
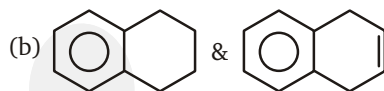
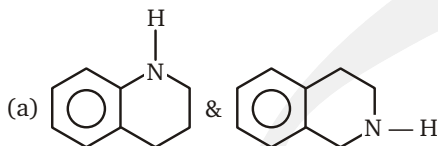
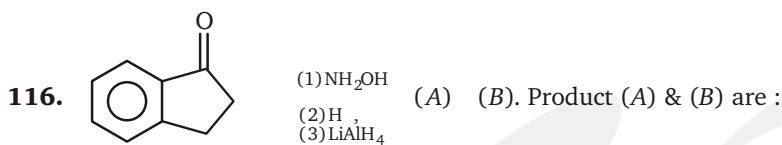
114. (A) $\xrightarrow{\text{HgSO}_4, \text{dil. H}_2\text{SO}_4}$ (B) $\xrightarrow{\text{LiAlH}_4}$ (C)
 reactant (A) is :
 (C) is a racemic mixture

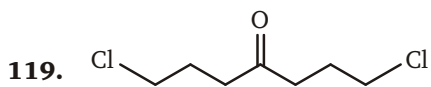


115. $\text{CH}_3\text{CH}_2-\text{C}(=\text{O})-\text{CH}_3 \xrightarrow[\text{HCl}]{\text{NaNO}_2}$; Major product of this reaction is :



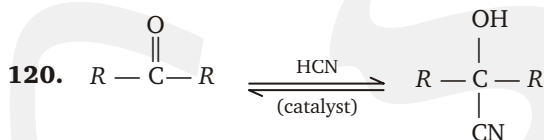
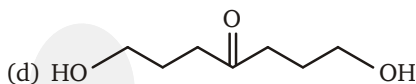
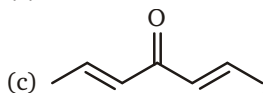
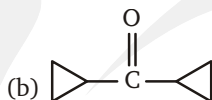
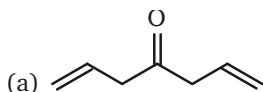
Aldehydes and Ketones





alc. KOH
(2 mole)

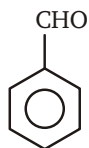
(A). Product (A) is :



Which of following can be used as a catalyst in the above reaction?

- (a) Cl (b) $CH_3 - \overset{\overset{O}{\parallel}}{C} - O$ (c) Et — O (d) HSO_4

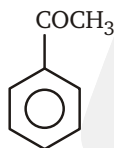
121. Arrange the following carbonyl compounds in decreasing order of their reactivity in nucleophilic addition reaction.



(i)



(ii)



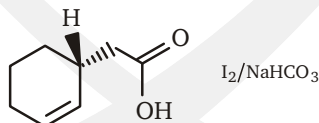
(iii)



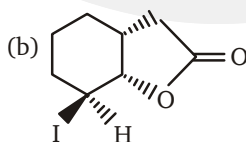
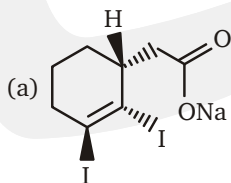
(iv)

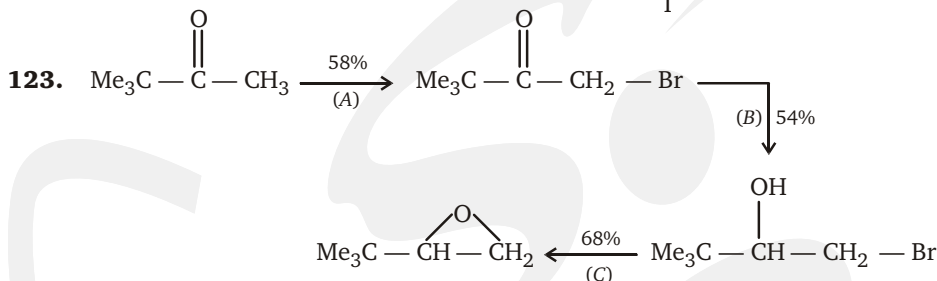
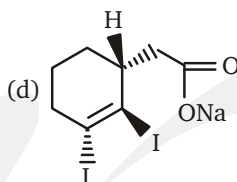
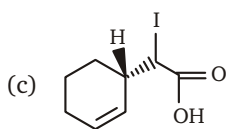
- (a) ii > iii > i > iv (b) ii > i > iv > iii
(c) iii > ii > i > iv (d) iii > i > iv > ii

122. The following reaction were carried out.



The final product formed in the above reaction sequence is :



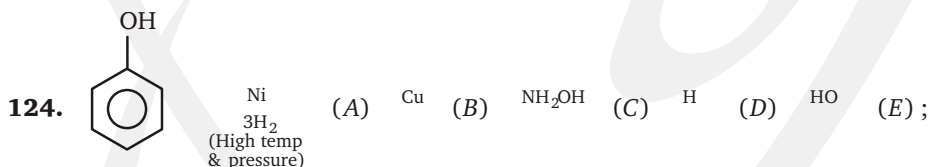


A. Yield of each step as actually carried out in the laboratory is given above. What is overall yield of reaction?

- (a) 42% (b) 31%
(c) 21% (d) 60%

B. What is the appropriate reagent to carry out above synthesis, i.e., A, B, C respectively are :

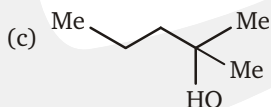
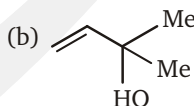
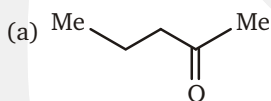
- (a) Br_2/H^+ , $\text{LiAlH}_4, \text{H}^+$ (b) Br_2/H^+ , $\text{NaBH}_4, \text{HO}^-$
(c) NBS, $\text{AlCl}_3, \text{HO}^-$ (d) Br_2/HO^- , BF_3, HO^-



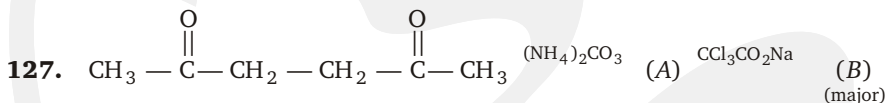
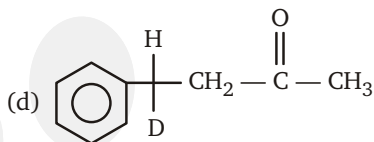
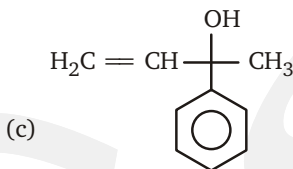
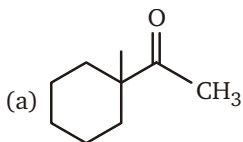
Product (E) is :

- (a) Nylon 66 (b) Nylon 6 (c) Styrene (d) Polystyrene

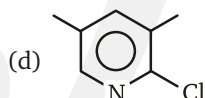
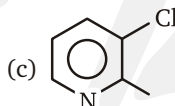
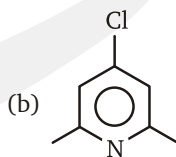
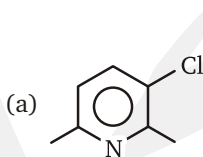
125. Methyl vinyl ketone on reaction with LiCuMe_2 gives a major product, whose structure is :



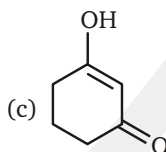
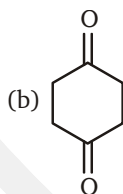
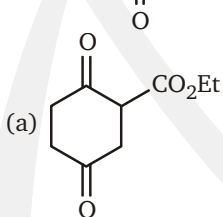
126. Which of following is incapable to show iodoform test ?



Product (B) of above reaction is :



128. $\xrightarrow{\text{H}_3\text{O}^+}$ A ; Product obtained is :

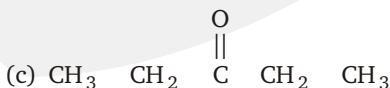
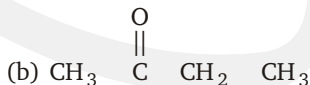
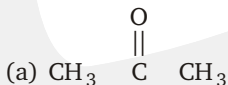


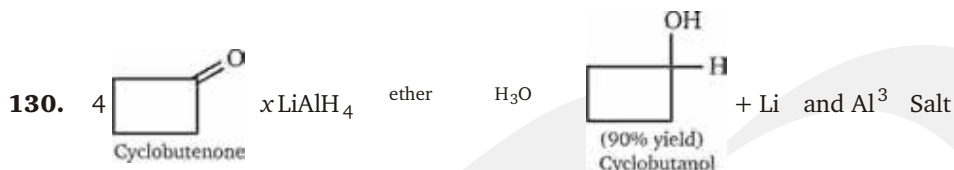
(d) None of these

129. (A) $\xrightarrow{\text{LiAlH}_4}$ (B) $\xrightarrow{\text{H}^+}$ Diastereomers

Symmetrical Ketone

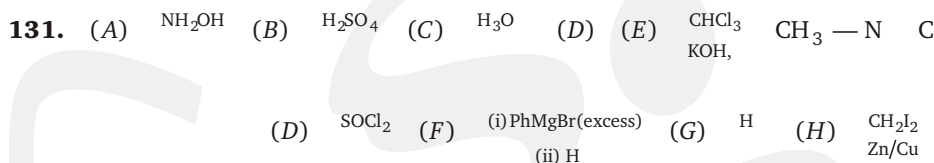
Reactant (A) is :





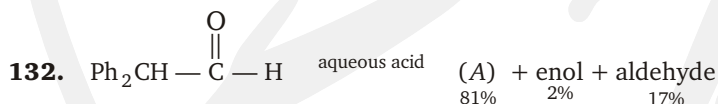
Value of x in above reaction is :

- (a) 1 (b) 2 (c) 3 (d) 4



Molecular weight of compound (A) is :

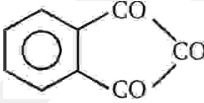
- (a) 58 (b) 120
(c) 60 (d) 182



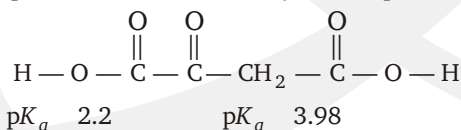
Product (A) of above reaction will be :

- (a) $\text{Ph}-\text{C}(\text{Ph})_2-\text{CH}_2-\text{OH}$ (b) $\text{Ph}_2\text{CH}-\text{CH}_2-\text{OH}$
(c) $\text{Ph}_2\text{CH}-\text{CH}(\text{OH})-\text{OH}$ (d) $\text{Ph}_2\text{CH}-\text{C}(=\text{O})-\text{CH}_3$

133. Which of the following will form stable hydrate ?

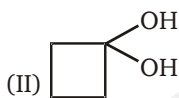
- (a) CCl_3CHO (Chloral) (b)  (Ninhydrin)
(c) $(\text{CF}_3)_2\text{CO}$ (d) All of these

134. The pH at which maximum hydrate is present in an solution of oxaloacetic acid:



- (a) $\text{pH} = 0$ (b) $\text{pH} = 12$
(c) $\text{pH} = 4$ (d) $\text{pH} = 6$

135. Arrange their stabilities of given gem-diols in decreasing order.



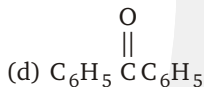
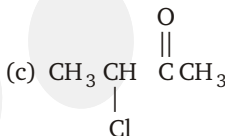
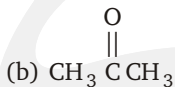
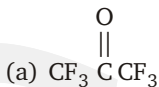
(a) I > II > III

(b) III > II > I

(c) I > III > II

(d) III > I > II

136. Maximum hydration takes place of :



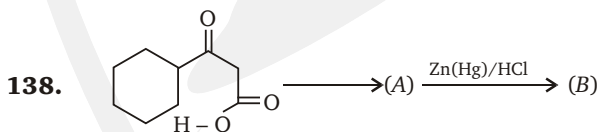
137. The conversion, $\text{PhCN} \rightarrow \text{PhCOCH}_3$, can be achieved most conveniently by reaction with:

(a) CH_3MgBr followed by hydrolysis

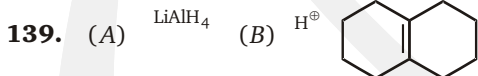
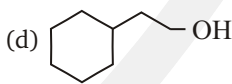
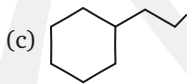
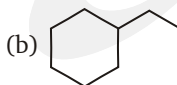
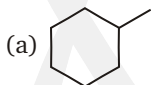
(b) $\text{I}_2 - \text{NaOH}$, CH_3I

(c) dil. H_2SO_4 followed by reaction with CH_2N_2

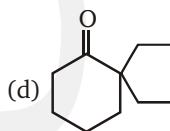
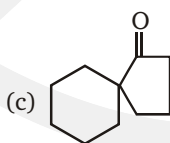
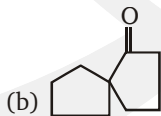
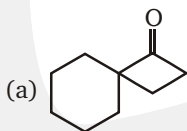
(d) LAH followed by reaction with CH_3I

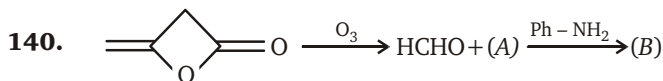


In the above reaction, product (B) is:

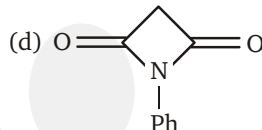
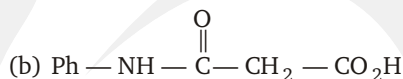


Structure of A is :





Product (B) is :



To carry out above conversion, arrange the following reagents in correct order.

O_3/Zn
(1)

$\text{EtONa} / \text{EtOH}/$
(2)

NaOCl
(3)

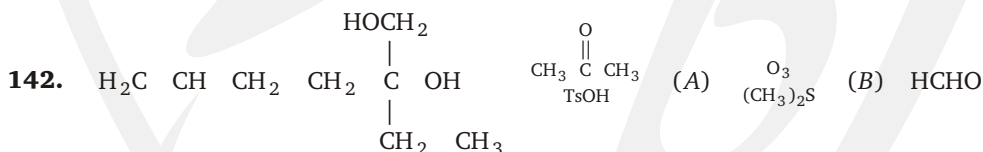
H
(4)

(a) 1 3 2 4

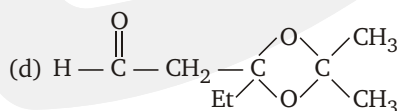
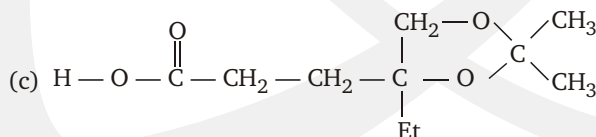
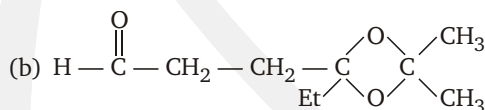
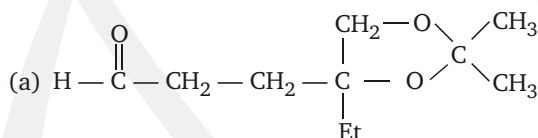
(b) 1 2 4 3

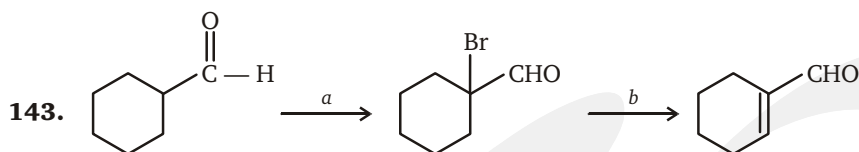
(c) 1 3 4 2

(d) 1 2 3 4



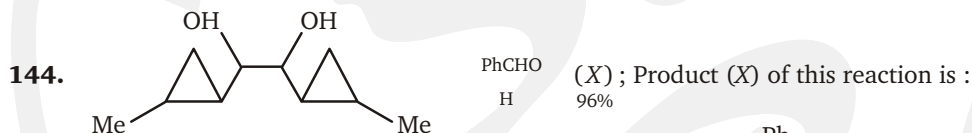
Product (B) is:

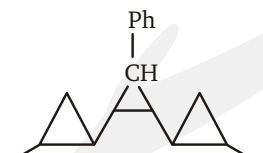
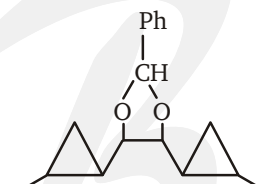




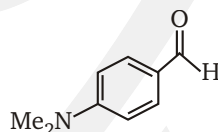
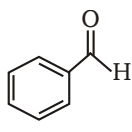
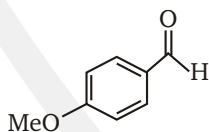
Identify appropriate reagents for the above reaction:

- (a) a Br_2/CCl_4 , b aq.KOH
 (b) a Br_2/H , b aq. KOH
 (c) a Br_2/H , b alc.KOH
 (d) a Br_2/HO , b aq.KOH

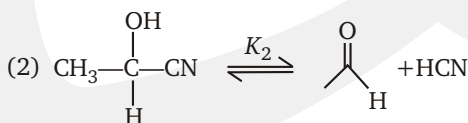
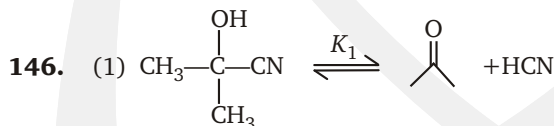


- (a) 
 (b) 
 (c) 
 (d) 

145. The K_{eq} values in HCN addition to following aldehydes are in the order :



- (a) $\text{I} > \text{II} > \text{III}$ (b) $\text{II} > \text{III} > \text{I}$ (c) $\text{III} > \text{I} > \text{II}$ (d) $\text{II} > \text{I} > \text{III}$



relation between K_1 and K_2 is :

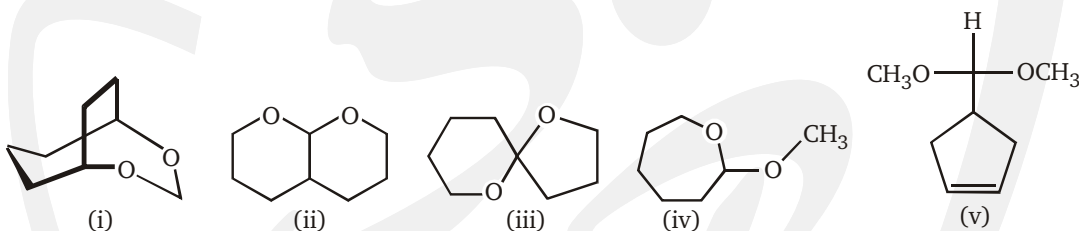
- (a) $K_1 < K_2$ (b) $K_1 > K_2$ (c) $K_2 < K_1$ (d) $K_1 > K_2$ 1

147. Which of the following is correct for the reaction ?



- (a) A is cyanohydrin
 (b) Nucleophilic-addition reaction
 (c) The above reaction is not shown by alkenes
 (d) All of these

148. Which of the following compounds (i through v) should not be classified as an acetal ?



- (a) ii and iii
 (b) iv
 (c) i
 (d) none (they are all acetals)

149. In which of reactions final product is NOT a ketone :

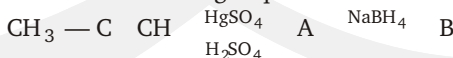
- (a) $\text{CH}_3 - \text{C} \equiv \text{C} - \text{H} \xrightarrow{\text{NaNH}_2} \text{CH}_3 - \text{I}$ (A) $\text{CH}_3 - \text{I} \xrightarrow[\text{H}_2\text{SO}_4]{\text{HgSO}_4}$ (C)
 (b) $\text{H} - \text{C} \equiv \text{C} - \text{H} \xrightarrow{\text{NaNH}_2} \text{CH}_3 - \text{CH}_2 - \text{I}$ (D) $\text{CH}_3 - \text{CH}_2 - \text{I} \xrightarrow[\text{NaBH}_4, \text{H}]{\text{Hg}(\text{OAc})_2, \text{H}_2\text{O}}$ (E)
 (c) $\text{R} - \text{C}(=\text{O}) - \text{OH} \xrightarrow{\text{NaOH}} \text{R} - \text{C}(=\text{O}) - \text{I}$ (A) $\text{CH}_3 - \text{I} \xrightarrow{\text{NaOH}}$ (B)

- (d) 1-butyne $\xrightarrow{\text{NaNH}_2} \text{CH}_3 - \text{I} \xrightarrow[\text{(2) H}_2\text{O}_2/\text{HO}^\ominus]{\text{(1) BH}_3 \text{ THF}}$ (C)

150. The reaction of ethyl methyl ketone with $\text{Cl}_2/\text{excess OH}^-$ gives the following major product

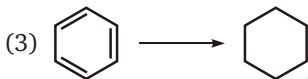
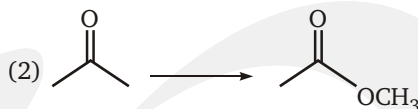
- (a) $\text{ClCH}_2\text{CH}_2\text{COCH}_3$ (b) $\text{CH}_3\text{CH}_2\text{COCCl}_3$
 (c) $\text{ClCH}_2\text{CH}_2\text{COCH}_2\text{Cl}$ (d) $\text{CH}_3\text{CCl}_2\text{COCH}_2\text{Cl}$

151. The product obtained from the following sequence of reactions is



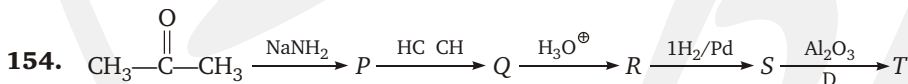
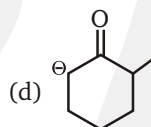
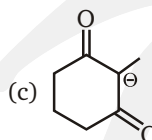
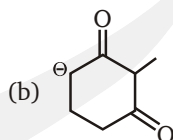
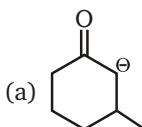
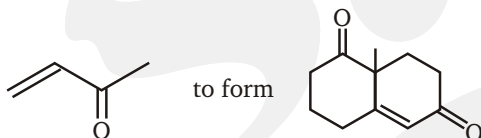
- (a) propanol (b) 2-propanol (c) 1-propanol (d) propanhe

152. Which of the following shows oxidation of reactant.

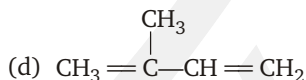
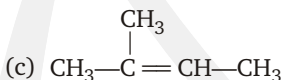
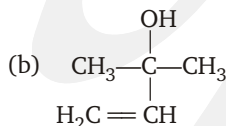
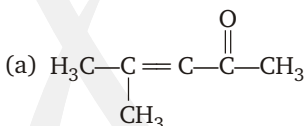


- (a) 1 (b) 1, 2 (c) 1, 2, 3 (d) All of the above

153. The enolate ion that reacts with



What is the final product T.



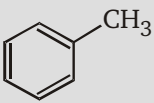
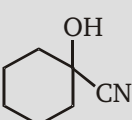
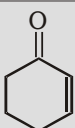
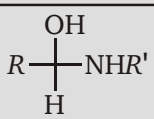
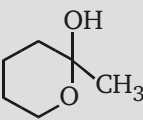
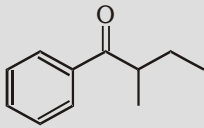
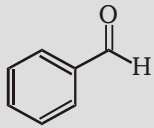
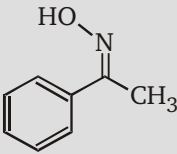
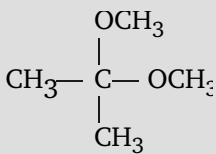
ANSWERS — LEVEL 1

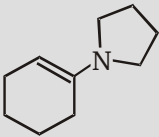
1.	(a)	2.	(b)	3.	(d)	4.	(c)	5.	(a)	6.	(c)	7.	(b)	8.	(b)
9.	(b)	10.	(c)	11.	(c)	12.	(c)	13.	(b)	14.	(b)	15.	(b)	16.	(b)
17.	(b)	18.	(b)	19.	(b)	20.	(d)	21.	(b)	22.	(b)	23.	(b)	24.	(d)
25.	(c)	26.	(b)	27.	(b)	28.	(a)	29.	(b)	30.	(c)	31.	(d)	32.	(c)
33.	(b)	34.	(b)	35.	(a)	36.	(b)	37.	(c)	38.	(d)	39.	(b)	40.	(a)
41.	(b)	42.	(a)	43.	(a)	44.	(b)	45.	(b)	46.	(a)	47.	(b)	48.	(d)
49.	(c)	50.	(b)	51.	(d)	52.	(b)	53.	(b)	54.	(c)	55.	(b)	56.	(b)
57.	(a)	58.	(d)	59.	(c)	60.	(c)	61.	(c)	62.	(a)	63.	(b)	64.	(b)
65.	(b)	66.	(d)	67.	(b)	68.	(d)	69.	(b)	70.	(c)	71.	(b)	72.	(d)
73.	(d)	74.	(a)	75.	(d)	76.	(a)	77.	(a)	78.	(a)	79.	(c)	80.	(c)
81.	(d)	82.	(b)	83.	(b)	84.	(b)	85.	(b)	86.	(a)	87.	(b)	88.	(c)
89.	(b)	90.	(b)	91.	(c)	92.	(b)	93.	(b)	94.	(b)	95.	(c)	96.	(c)
97.	(d)	98.	(b)	99.	(c)	100.	(c)	101.	(c)	102.	(c)	103.	(c)	104.	(c)
105.	(c)	106.	(b)	107.	(a)	108.	(c)	109.	(c)	110.	(b)	111.	(c)	112.	(b)
113.	(b)	114.	(c)	115.	(c)	116.	(a)	117.	(a)	118.	(a)	119.	(b)	120.	(c)
121.	(b)	122.	(b)	123.	A-c	123.	B-b	124.	(b)	125.	(a)	126.	(c)	127.	(a)
128.	(b)	129.	(c)	130.	(a)	131.	(a)	132.	(c)	133.	(d)	134.	(a)	135.	(a)
136.	(a)	137.	(a)	138.	(b)	139.	(d)	140.	(b)	141.	(d)	142.	(a)	143.	(c)
144.	(b)	145.	(d)	146.	(b)	147.	(d)	148.	(d)	149.	(c)	150.	(b)	151.	(b)
152.	(b)	153.	(c)	154.	(d)										

Level - 2

1. Select the best choice for example (A to L) from the examples (a to n) given below. Write your choice in the box given.

A.	An acetal derivative of a ketone.	
B.	A chiral ketone.	
C.	An aldehyde that gives a aldol condensation with itself.	
D.	An oxime derivative	
E.	A reagent that reduces aldehydes to 1°- alcohols.	
F.	An α,β -unsaturated ketone.	
G.	A reagent that oxidizes aldehydes to carboxylic acids.	
H.	A reagent that reduces ketones to alkanes.	
I.	An enamine derivative of a ketone.	
J.	An intermediate in imine formation.	
K.	A cyclic hemiacetal.	
L.	A cyanohydrin derivative.	

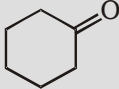
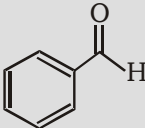
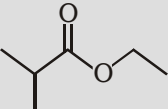
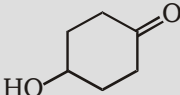
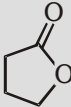
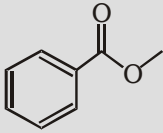
(a)		(b)		(c)	
(d)		(e)		(f)	$\text{Zn(Hg)H}_3\text{O}^{(+)}$
(g)		(h)	NaBH_4 aq. alcohol	(i)	
(j)	$\text{Ag(NH}_3)_2^+ \text{OH}^-$	(k)		(l)	

(m)		(n)	$\text{CH}_3-\text{CH}_2-\text{C}(=\text{O})\text{H}$	
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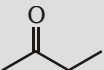
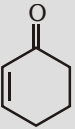
2. The following questions refer to the compounds (A to G) shown below :

i.	Which compounds are reduced by sodium borohydride ?	ii.	Which compounds are hydrolyzed by hot aqueous acid ?	iii.	Which compound are oxidized by $\text{CrO}_3/\text{pyridine}$?
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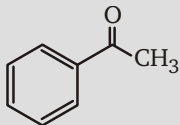
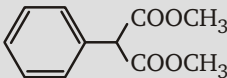
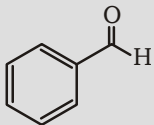
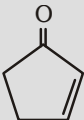
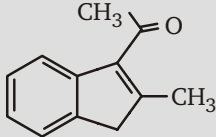
A		E		A		E	
B		F		B		F	
C		G		C		G	
D		H		D		H	

A.		B.		C.		D.	
E.		F.		G.		H.	

3. Match the column:

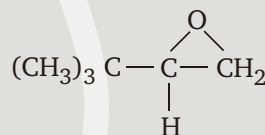
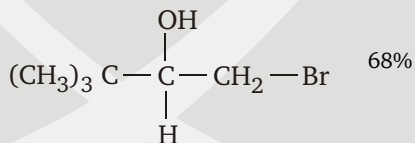
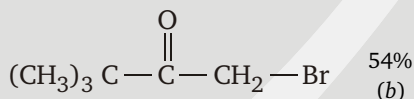
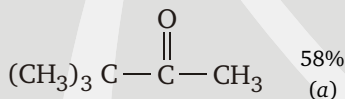
Column (I)		Column (II)	
(a)	 LiAlH_4	(p)	racemic mixture
(b)	 (1) KCN (2) H^+	(q)	Diastereomers
(c)	$\text{Ph}-\text{CH}_2-\text{Cl}$ KCN	(r)	Nu-addition reaction
(d)	 (1) CH_3MgBr (2) H^+	(s)	Nu-Substitutions reaction

4. Complete the following table.

	REACTANT	REAGENT(S)/ CONDITIONS	MAJOR ORGANIC PRODUCTS
a.		$\text{H}_2/\text{Pd} - \text{C}$ in ethanol (solvent)	A
b.		$\text{H} / \text{H}_2\text{O} /$	B
c.		$(\text{CH}_3)_2 \bar{\text{C}} - \text{P}^+(\text{C}_6\text{H}_5)_3$	C
d.		1. $\text{Li} [(\text{CH}_3)_2\text{Cu}]$ in dry ether 2. $\text{H} / \text{H}_2\text{O}$	D
e.	E	$\text{OH} / \text{ethanol} /$	

5. Comprehension

Consider the following reactions and answer A and B.



A. Suggest a reagent appropriate step (a) the synthesis.

(a) HO / Br_2 (1mole)

(b) H / Br_2 (1 mole)

(c) both (a) and (b)

(d) None of these

B. Yield of each step as actually carried out in laboratory is given above each arrow. What is overall yield of the reaction ?

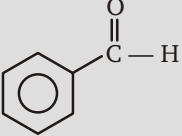
(a) 60%

(b) 21%

(c) 40%

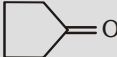
(d) 68%

6.

Reaction 1.	 $\xrightarrow[(2) \text{H}_3\text{O}^+]{(1) \text{Ac}_2\text{O}, \text{AcONa}}$ (A)
Reaction 2.	$\text{Ph}-\text{CH}=\text{CH}-\text{C}(=\text{O})-\text{CH}_3 \xrightarrow[\text{CH}_3-\text{CH}(\text{OH})-\text{CH}_3]{\text{Al}(\text{OCHMe}_2)_3}$ (B)
Reaction 3.	$\text{Ph}-\text{CH}=\text{CH}-\text{CH}(\text{OH})-\text{CH}_3 \xrightarrow[(2) \text{H}]{(1) \text{NaOH}}$ (C)

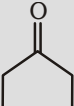
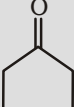
Degree of unsaturation present in compound (A + B + C) is ?

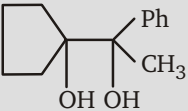
7. Within each set, which compound should be more reactive toward carbonyl addition reaction ?

	A	B
Set (1)	$\text{CH}_3-\text{C}(=\text{O})-\text{CH}_2-\text{Br}$	$\text{CH}_3-\text{C}(=\text{O})-\text{CH}_2-\text{CH}_2-\text{Br}$
Set (2)	$\text{CH}_3-\text{C}(=\text{O})-\text{C}(=\text{O})-\text{CH}_3$	$\text{CH}_3-\text{C}(=\text{O})-\text{CH}_2-\text{CH}_3$
Set (3)	$\text{CH}_3\text{O}-\text{C}_6\text{H}_4-\text{CHO}$	$\text{O}_2\text{N}-\text{C}_6\text{H}_4-\text{CHO}$
Set (4)	$\text{C}_6\text{H}_5-\text{CHO}$	CH_3-CHO
Set (5)		

Set (6)		
Set (7)		
Set (8)		
Set (9)		
Set (10)		

8. Match the Column (I) and Column (II). (Matrix)

Column (I)						Column (II)		
(A)		HCN traces of KOH	(A)	LiAlH ₄	(B)	NaNO ₂ HCl	(p)	Formation of six member ring takes place
(B)		NH ₂ OH	(A)	H	(B)	LAH	(q)	Final product is Ketone

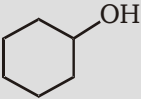
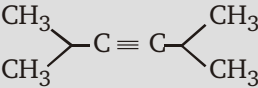
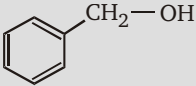
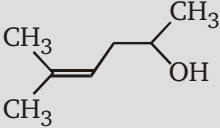
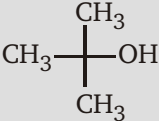
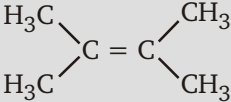
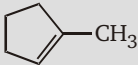
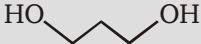
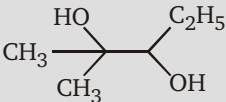
(C)	$\text{CH}_3 - \overset{\text{O}}{\parallel} \text{C} - \text{CH}_2 - \text{CH}_2 - \text{CH}_2 - \overset{\text{O}}{\parallel} \text{C} - \text{H} \quad \text{HO}^- \quad (\text{A})$	(r)	Final product formed will give positive Tollens test
(D)	 $\text{H} \quad (\text{A})$	(s)	Final product formed will react with 2,4-DNP. (2,4-di-nitrophenyl hydrazine)

9. Consider reactions A through F. Those carbon atoms undergoing change, as part of a functional group, are marked as C^{12} , C^{14} or starred. In the cases shown, each carbon atom has either been reduced or oxidized. Your job is to identify the change in oxidation state that has occurred for each of the marked carbon.

Reaction		C^{12}		C^{14}	
A.	$\text{CH}_3\text{CH}_{12} - \text{CH}_2_{14} \xrightarrow{\text{Br}_2} \text{CH}_3\text{CHBrCH}_2\text{Br}$	Reduced		Reduced	
		Oxidized		Oxidized	
B.	$\text{CH}_3\text{CH}_{12} - \text{CH}_2_{14} \xrightarrow[\text{(ii) H}_2\text{O}_2, \text{NaOH}]{\text{(i) B}_2\text{H}_6} \text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$	Reduced		Reduced	
		Oxidized		Oxidized	
C.	$\text{CH}_3\text{CH}_2\text{CH}^* - \text{O} \xrightarrow{\text{NaBH}_4} \text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$	Reduced			
		Oxidized			
D.	$\text{CH}_3\text{CH}_2\text{CH}^* - \text{O} \xrightarrow[\text{H}_2\text{O, pH 8}]{\text{Ag}^+} \text{CH}_3\text{CH}_2\text{CO}_2\text{H}$	Reduced			
		Oxidized			
E.	$\text{CH}_3\text{COCH}_{12} - \text{CH}_2_{14} - \text{CO}_2\text{H} \xrightarrow{\text{Heat}} \text{CH}_3\text{COCH}_3$	Reduced		Reduced	
		Oxidized		Oxidized	
F.	$\text{H}_2\text{C}_{12} - \text{C}(\text{OH})\text{C}_2\text{H}_5_{14} \xrightarrow{\text{tautomerization}} \text{H}_3\text{CCOC}_2\text{H}_5$	Reduced		Reduced	
		Oxidized		Oxidized	

10. Consider the possible formation of an aldehyde or ketone product when each of the ten compounds in the column on the left is treated with each of the reagents shown in the top row. Check the designated answer box if you believe an aldehyde or ketone will be formed.

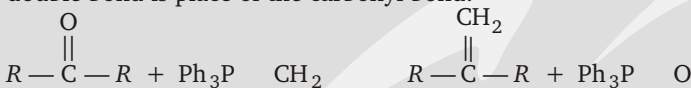
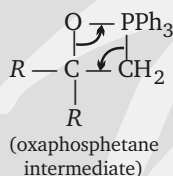
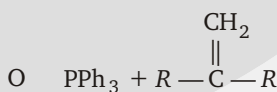
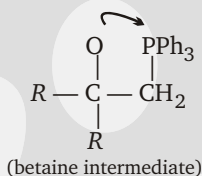
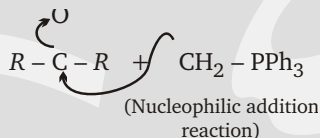
Assume that the reagents may be present in excess. For each checked reaction, try to draw the structure of the major product (s).

Starting	PCC $\text{C}_5\text{H}_5\text{NHCrO}_3\text{Cl}$	Jone's Reagent CrO_3 in aq. acid	$\text{Pb}(\text{OAc})_4$ [for HIO_4]	(i) O_3 , (ii) Zn dust	H_3O^+	(i) BH_3 in THF (ii) $\text{H}_2\text{O}_2 + \text{NaOH}$
						
						
						
						
						
						
						
						
						
						

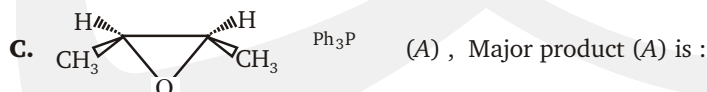
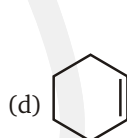
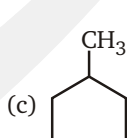
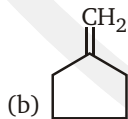
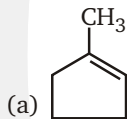
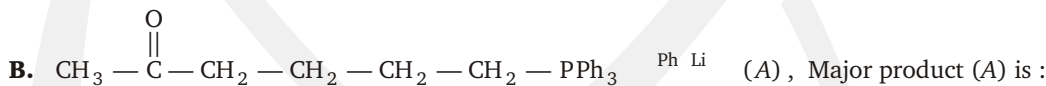
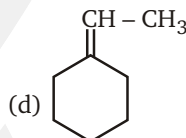
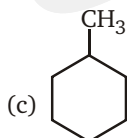
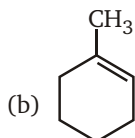
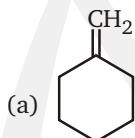
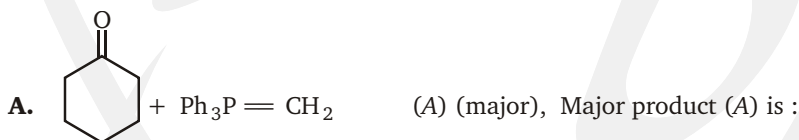
11. Comprehension

Wittig reaction :

The reaction of a phosphorus ylide with an aldehyde (or) ketone introduces a carbon-carbon double bond in place of the carbonyl bond.

**Mechanism :**

Driving force of the reaction is high bond energy of (P=O) . (H = -ve)



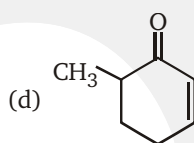
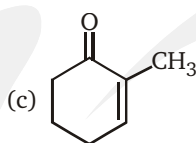
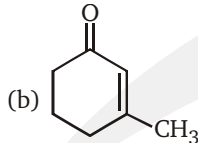
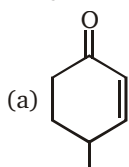
(a) cis-2-butene

(b) trans-2-butene

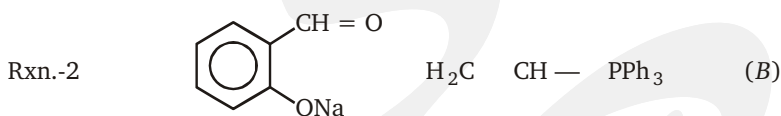
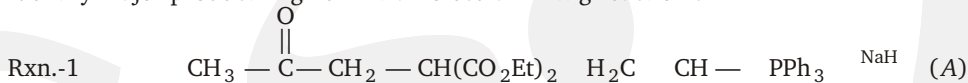
(c) iso-butene

(d) 1-butene

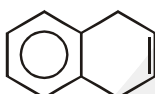
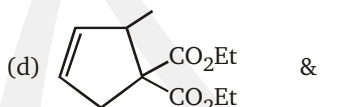
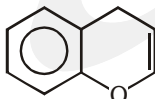
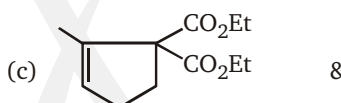
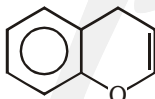
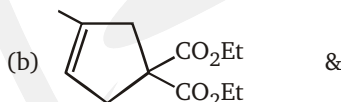
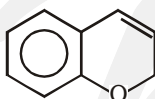
D. $\text{CH}_3\text{C}(=\text{O})(\text{CH}_2)_3\text{C}(=\text{O})\text{CH}_2\text{P}(\text{OEt})_2$ $\xrightarrow{\text{NaH}}$ (A) (cyclic). Product (A) is :

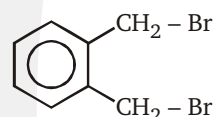


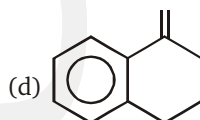
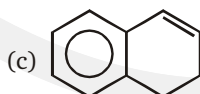
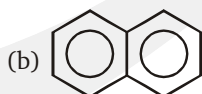
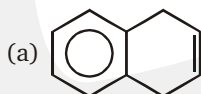
E. Identify major product in given intramolecular Wittig reaction :



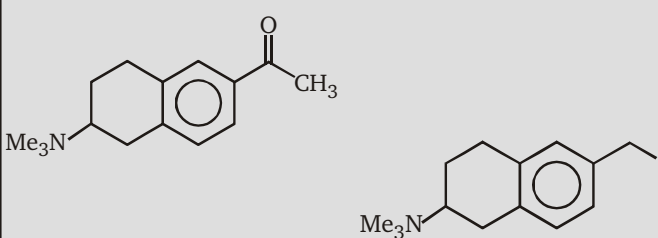
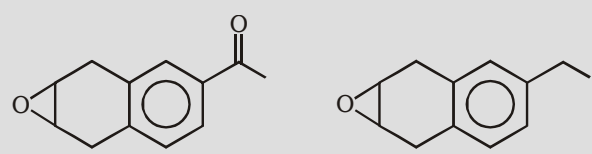
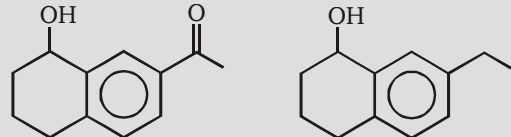
Product (A) and (B) respectively are :

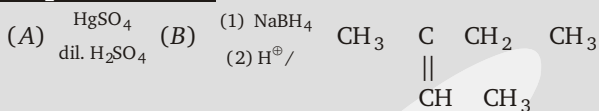
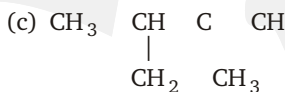
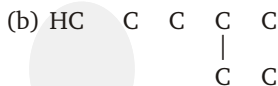


F.  $\xrightarrow[\text{(3) CHO}]{\text{(1) Ph}_3\text{P (2 mole), (2) 2PhLi}}$ (A) ; product (A) is :

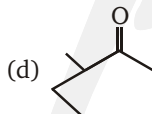
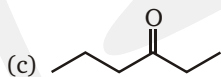
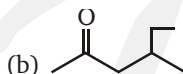


12. Match the column :

Column (I)		Column (II)	
Conversion		Reagent	
(a)		(p)	$\text{NH}_2/\text{NH}_2/\text{HO}$, (Wolff-Kishner reduction)
(b)		(q)	Zn(Hg), HCl (Clemmensen reduction)
(c)		(r)	LiAlH_4
(d)		(s)	None


$$(a) \begin{array}{ccccccc} \text{CH}_3 & & \text{C} & & \text{C} & & \text{CH} & & \text{CH}_3 \\ & & & & & & | & & \\ & & & & & & \text{CH}_3 & & \end{array}$$


(a) 



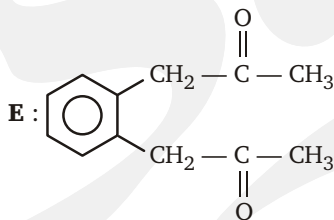
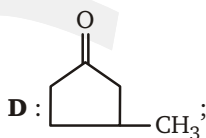
ANSWERS — LEVEL 2

1. A – l; B – g; C – n; D – k; E – h; F – c; G – j; H – f; I – m; J – d; K – e; L – b

2. i – A, B, C, E, F; ii – D, G, H; iii – B, E, F

3. a – p, r; b – r; c – s; d – p, r

4. A : $\text{Ph} - \overset{\text{OH}}{\underset{|}{\text{CH}}} - \text{CH}_3$; B : $\text{Ph} - \text{CH}_2 - \text{COOH}$; C : $\text{Ph} - \text{CH} - \text{C} \begin{matrix} \diagup \text{CH}_3 \\ \diagdown \text{CH}_3 \end{matrix}$;



5. A – b; B – b

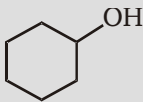
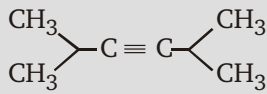
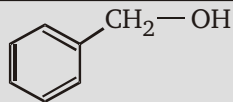
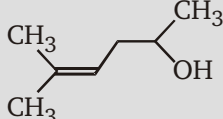
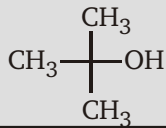
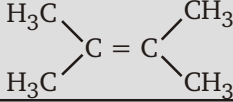
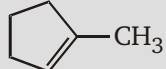
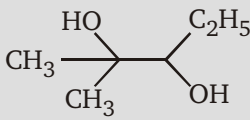
6. A B C 17

7. set 1 – A; set 2 – A; set 3 – B; set 4 – B; set 5 – A; set 6 – B; set 7 – B;
set 8 – B; set 9 – A; set 10 – B

8. A – p, q, s; B – p; C – p, q, s; D – p, q, s

9. A : both are oxidized; B : C^{12} is reduced, C^{14} is oxidized; C : reduced; D : oxidized
E : C^{12} is reduced, C^{14} is oxidized; F : C^{12} is reduced, C^{14} is oxidized

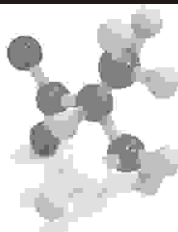
10.

Compound	PCC $\text{C}_5\text{H}_5\text{NHCrO}_3\text{Cl}$	Jone's Reagent CrO_3 in aq. acid	$\text{Pb}(\text{OAc})_4$ [or HIO_4]	(i) O_3 , (ii) Zn dust	H_3O^+	(i) BH_3 in THF (ii) $\text{H}_2\text{O}_2 + \text{NaOH}$
	✓	✓	✗	✗	✗	✗
	✗	✗	✗	✓	✓	✓
	✓	✓	✗	✓	✗	✗
	✗	✗	✗	✓	✗	✗
	✓	✓	✗	✓	✓	✓
	✗	✗	✗	✗	✗	✗
	✗	✗	✗	✓	✓	✓
	✗	✗	✗	✓	✓	✓
	✓	✓	✗	✗	✗	✗
	✓	✓	✓	✗	✗	✗

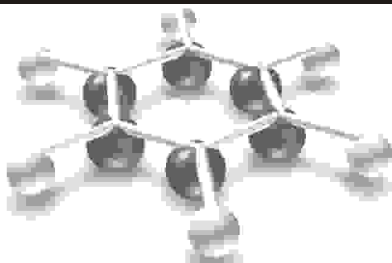
11. A – a; B – a; C – b; D – b; E – a; F – b

12. a – q; b – s; c – r; d – p

13. A. (c) B. (d)



8

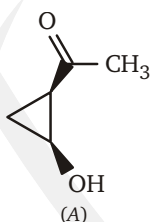


ALDOL AND CANNIZARO REACTION

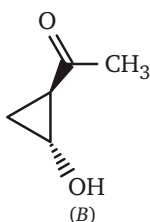


Level - 1

1. Compound A and B, both were treated with NaOH, producing a single compound C.

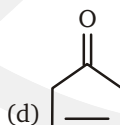
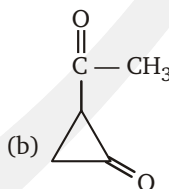
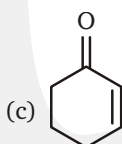
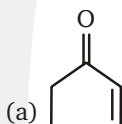


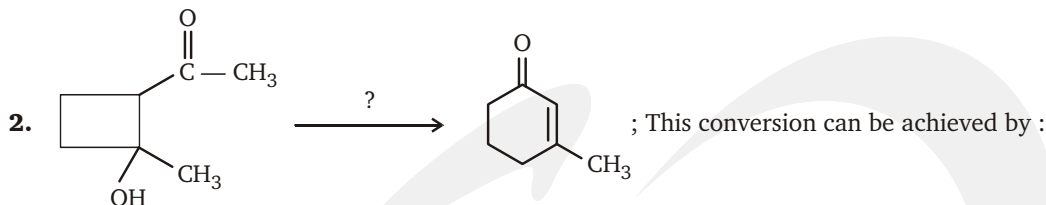
+



HO
heat

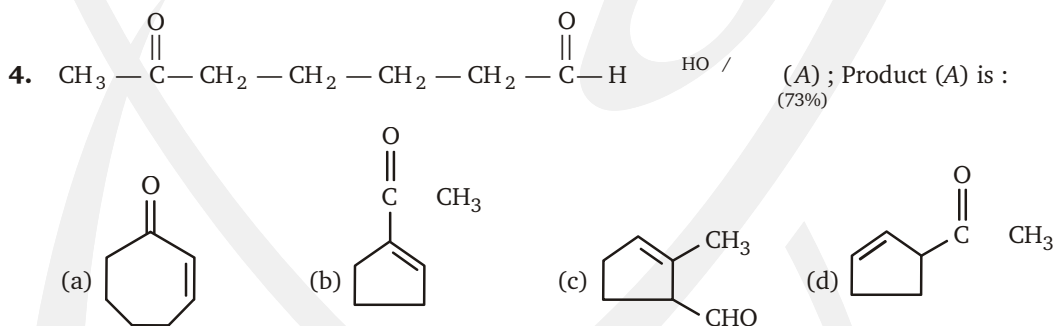
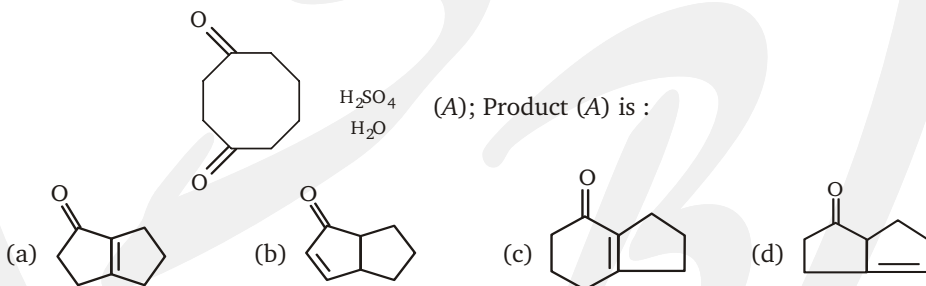
C. Compound (C) is :



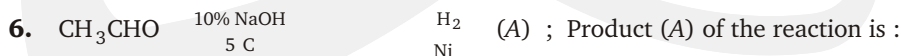


- (a) Dehydration, Hydrolysis
 (b) Retro aldol and further condensation
 (c) Perkin condensation & Clemmensen reduction
 (d) Clemmensen and Perkin condensation

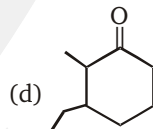
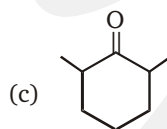
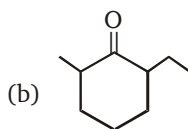
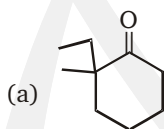
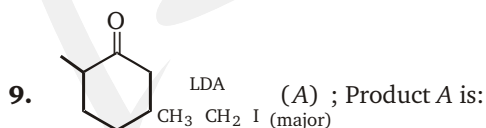
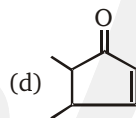
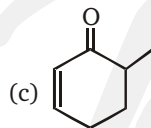
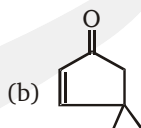
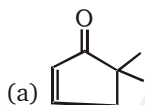
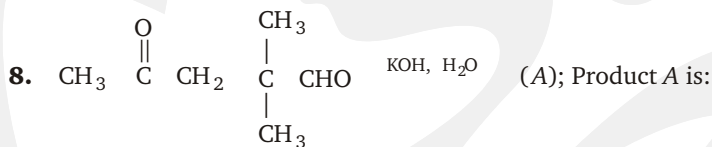
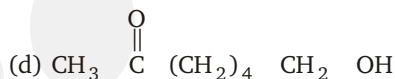
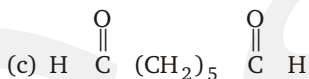
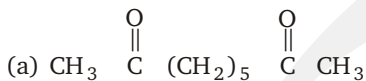
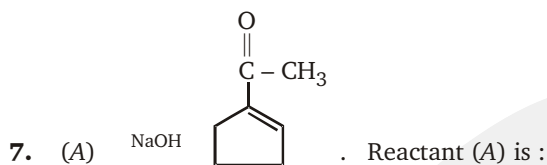
3. This is an example of an intramolecular aldol reaction :



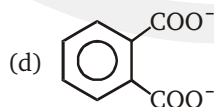
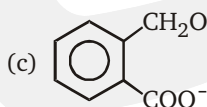
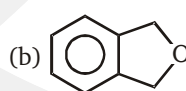
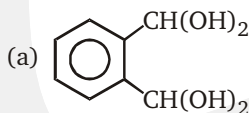
- (a) $\text{Ph} - (\text{CH} = \text{CH})_2 - \text{CHO}$ (b) $\text{Ph} - (\text{CH} = \text{CH})_3\text{CHO}$
 (c) $\text{Ph} - (\text{CH} = \text{CH})_4\text{CHO}$ (d) $\text{Ph} - \text{CH} = \text{CH} - \text{CH} = \text{CH} - \text{CH} = \text{CH}_2$



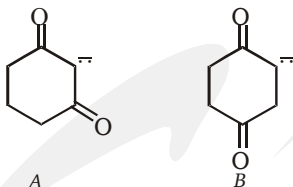
- (a) propanol (b) ethanol (c) butanol (d) pentanol



10. The reaction ,



11. Compare enolate *A* with enolate *B*.



Which of the following statements is true ?

- (a) *A* is more stable than *B* (b) *A* and *B* have the same stability
 (c) *B* is more stable than *A* (d) No comparison of stability can be made
12. Benzalacetone is the product of mixed aldol condensation between benzaldehyde ($\text{C}_6\text{H}_5\text{CH}=\text{O}$) and acetone [$(\text{CH}_3)_2\text{C}=\text{O}$]. What is its structure ?

- (a) $\text{C}_6\text{H}_5\text{CH}=\text{CH}-\text{C}(\text{CH}_3)=\text{CH}_2$ (b) $\text{C}_6\text{H}_5\text{CH}=\text{C}(\text{CH}_3)_2$
 (c) $\text{C}_6\text{H}_5\text{C}(\text{CH}_3)=\text{CH}-\text{CH}_2\text{CH}_3$ (d) $\text{C}_6\text{H}_5\text{CH}_2-\text{C}(\text{CH}_3)=\text{CH}-\text{CH}_2\text{CH}_3$

13. Identify the major product *P* in the following reaction:

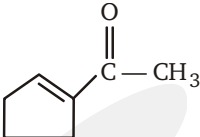


- (a) (b)
 (c) (d)

14. (A) $\xrightarrow{\text{dil. NaOH}}$ (B)

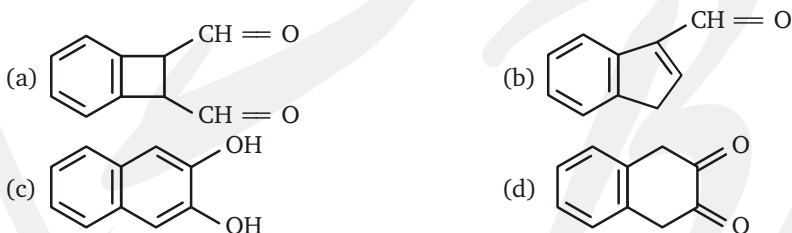
Product (*B*) is:

- (a) (b) (c) (d)

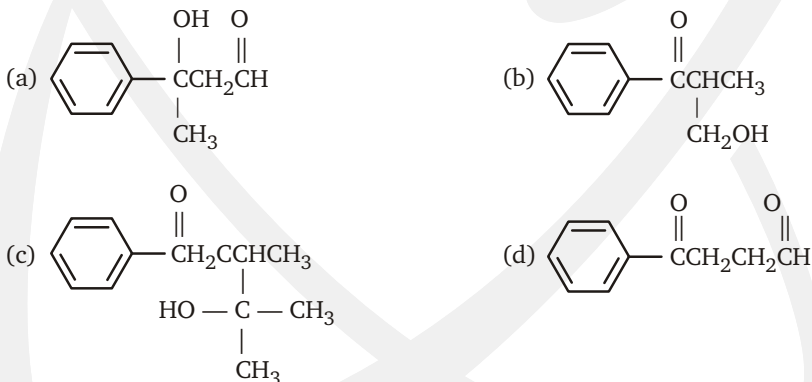
15. (A) $\xrightarrow[\text{(ii) Zn, H}_2\text{O}]{\text{(i) O}_3}$ (B) $\xrightarrow{\text{NaOH}}$  the reactant (A) will be :



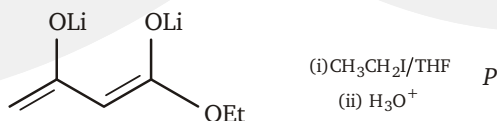
16. Identify the principal product of the following reaction?

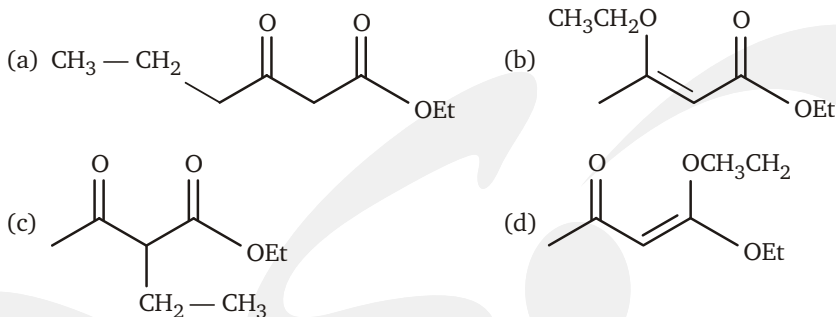


17. Which one of the following compounds is the best choice for being prepared by an efficient mixed aldol addition reaction?

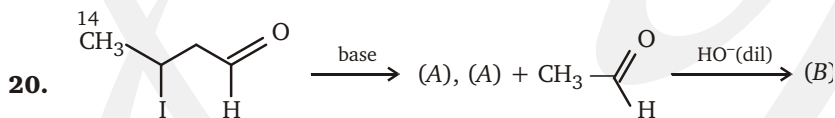
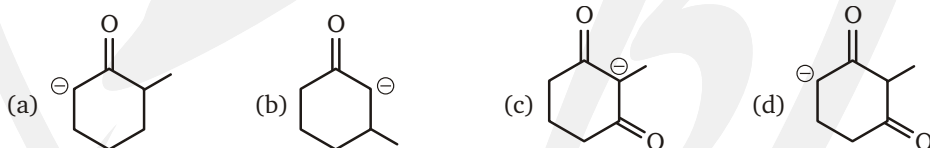
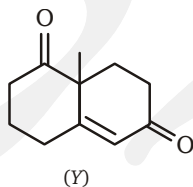


18. Identify the major product *P* in the following reaction:

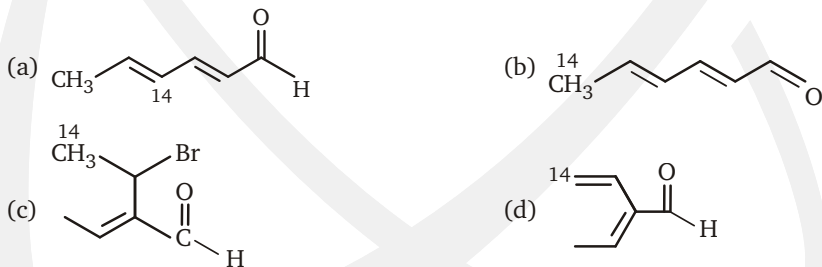




19. The enolate ion that reacts with 3-buten-2-one to form (Y) is :



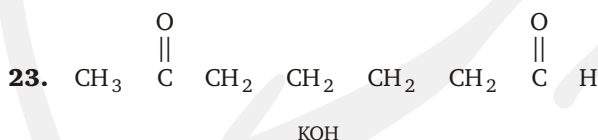
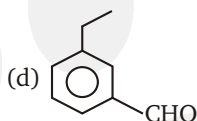
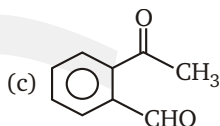
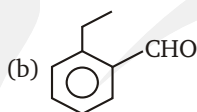
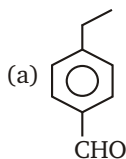
Product (B) in the above reaction is :



21. $\text{H} - \text{C}(=\text{O}) - \text{D} \xrightarrow[\text{(50\%)}]{\text{HO}}$; Product of this Cannizzaro reaction is :

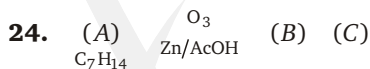


- 22.** An organic compound with the molecular formula $C_9H_{10}O$ forms a 2,4-DNP derivative, reduces Tollen's reagent and undergoes Cannizzaro reaction, on vigorous oxidation it gives 1,2-benzenedicarboxylic acid. Structure of organic compound is:



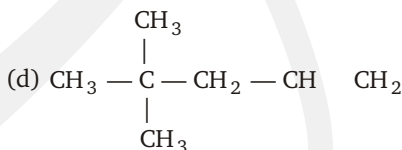
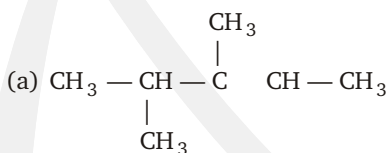
Number of intramolecular aldol condensation product is :

- (a) 1 (b) 2 (c) 3 (d) 4

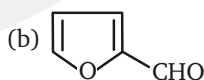
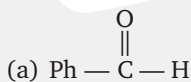


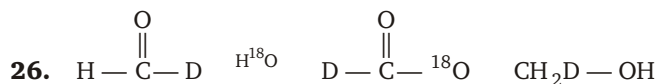
Compound (A) exist in geometrical isomers and (B) gives Cannizzaro reaction.

(A) will be :



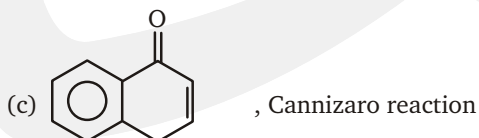
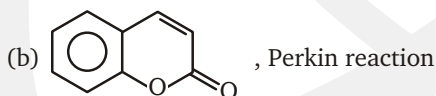
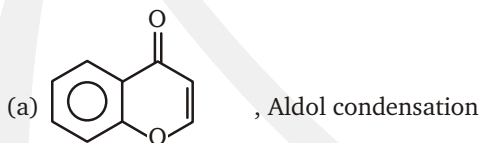
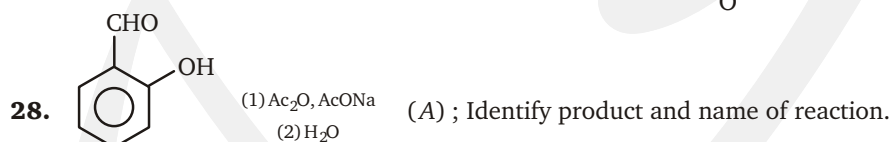
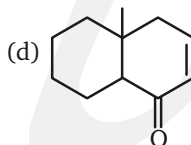
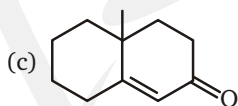
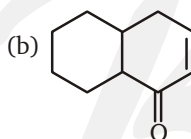
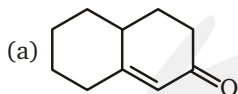
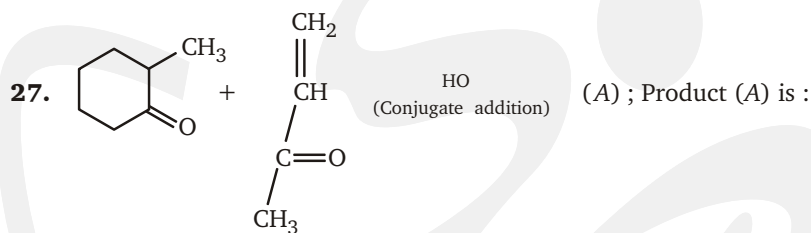
- 25.** Which of the following compounds will not undergo Cannizzaro reaction, when treated with 50% aqueous alkali?

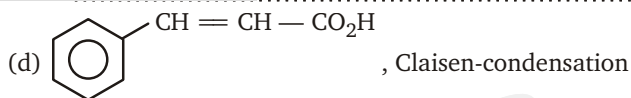




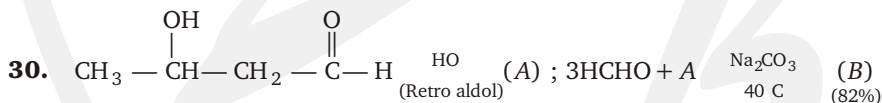
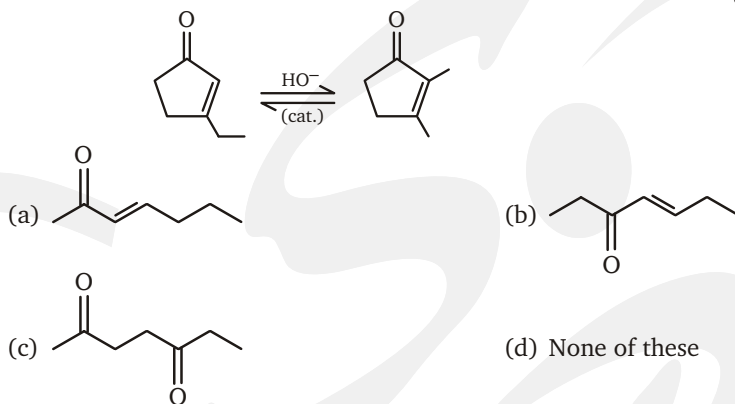
Above reaction is known as :

- (a) Cannizzaro reaction, Disproportionation reaction
- (b) Tischenko reaction, Disproportionation reaction
- (c) Cross Cannizzaro reaction, Redox reaction
- (d) Tischenko reaction, Redox reaction

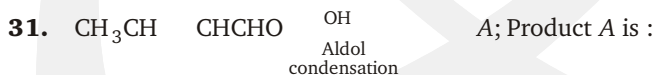
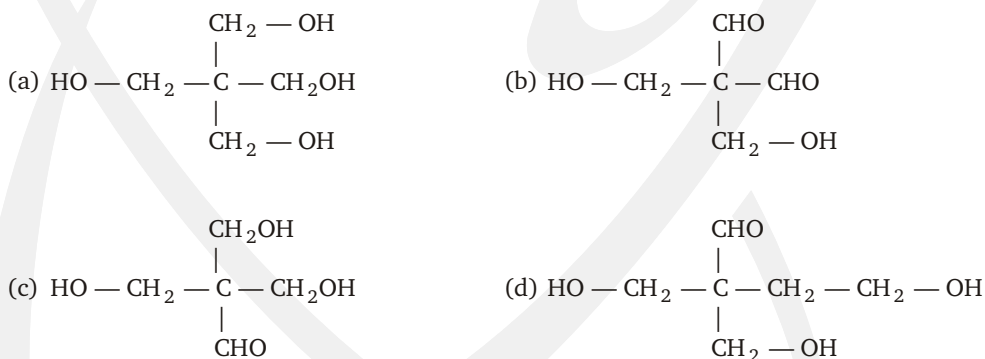


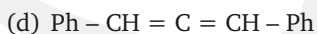
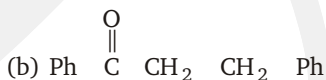
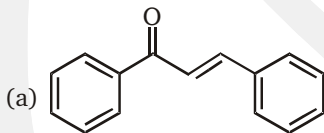
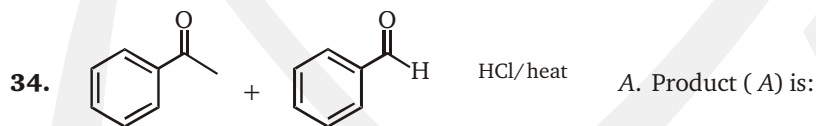
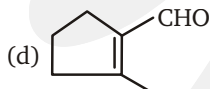
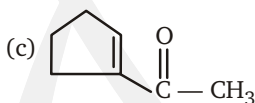
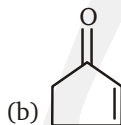
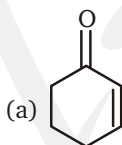
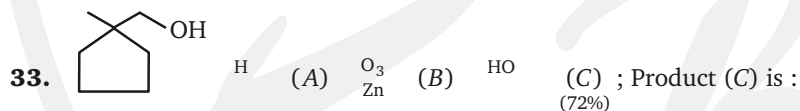
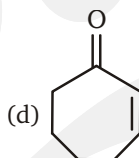
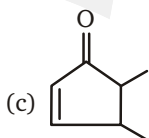
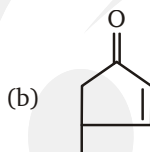
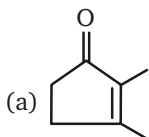
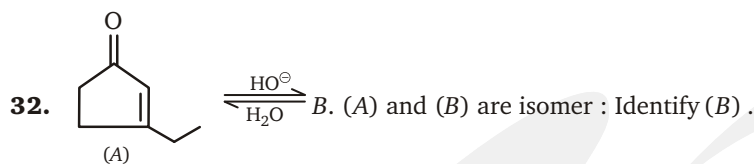


29. Choose the most reasonable reaction intermediate for the following reaction.

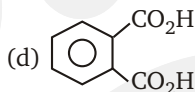
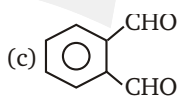
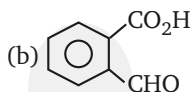
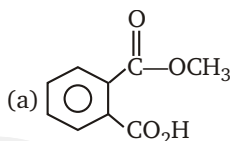
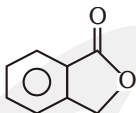


Product (B) of the above reaction is :

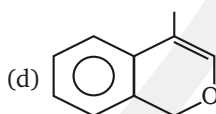
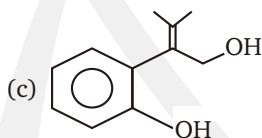
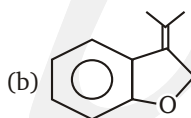
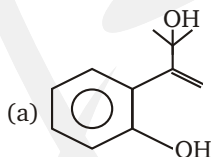




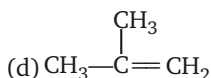
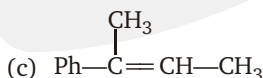
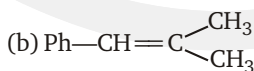
35. Which of the following reactant on reaction with conc. NaOH followed by acidification gives the following lactone as the product ?



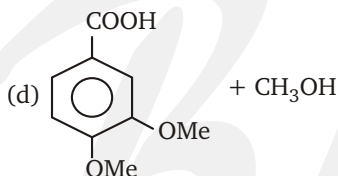
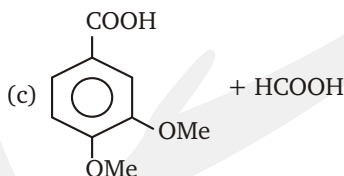
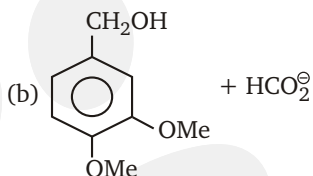
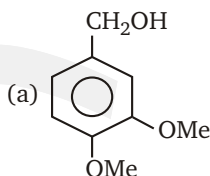
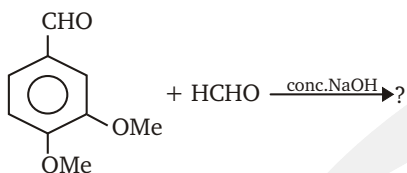
36. $\xrightarrow{\text{NaOH}}$ Product; Product is:



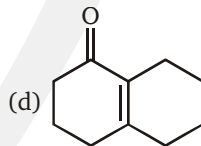
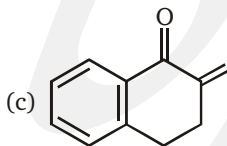
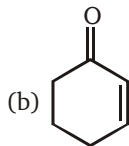
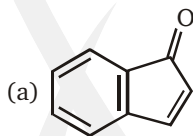
37. (P) KOH
 (Q) KOH $\text{Ph}-\text{CH}_2-\text{OH} + \text{Ph}-\text{CO}_2^-$
 (R) O_3 P + Q, Structure of (R) is :



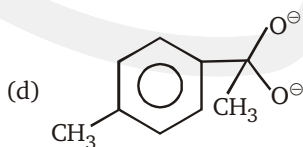
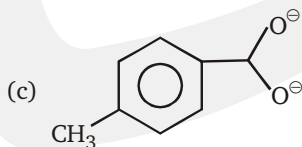
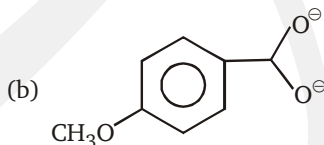
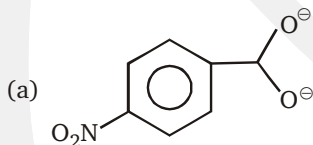
38. The following reaction gives:



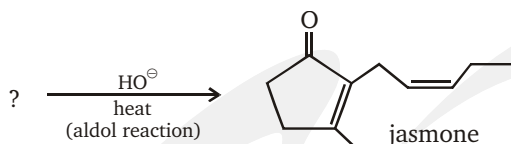
39. Which of the following is not the product of an intramolecular aldol condensation ?



40. x no. of compound better hydride donor than $\text{Ph}-\overset{\overset{\text{O}^-}{|}}{\underset{\underset{\text{H}}{|}}{\text{C}}}-\text{O}^-$

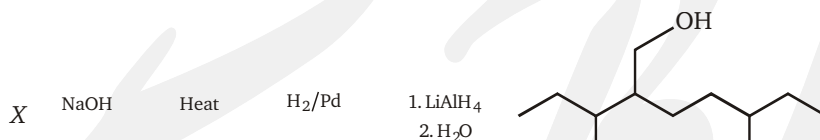


41. Choose the reactant whose aldol reaction would give jasmine.



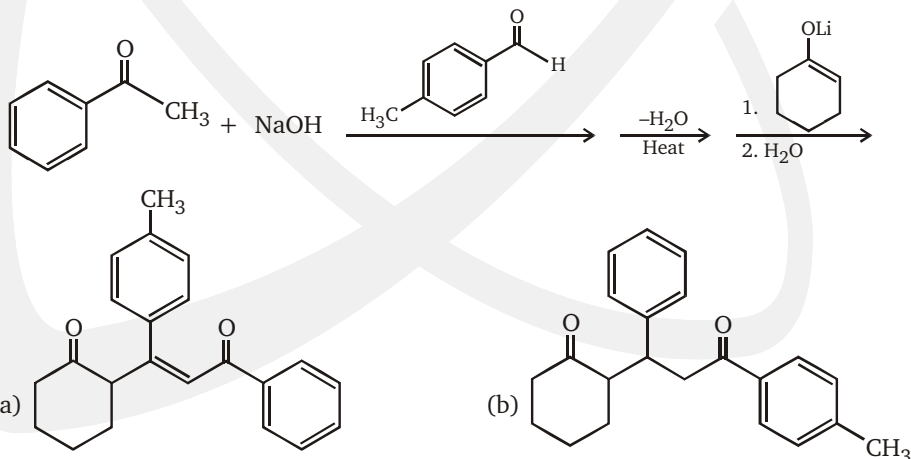
- (a)
- (b)
- (c)
- (d)

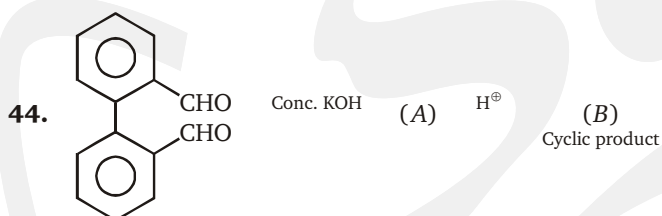
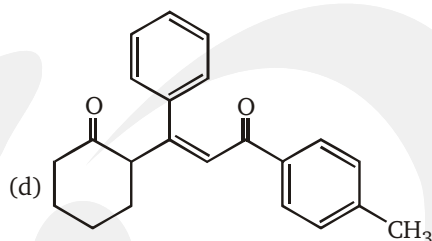
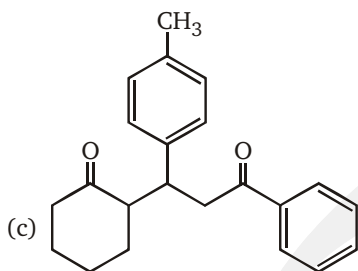
42. Compound X undergoes the following reaction sequence. What is the structure of compound X ?



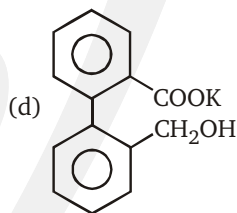
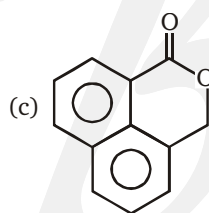
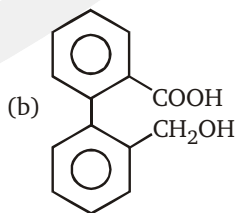
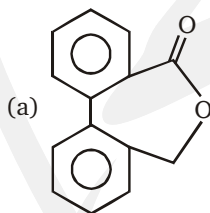
- (a)
- (b)
- (c)
- (d)

43. Predict the major product of the following reaction sequence

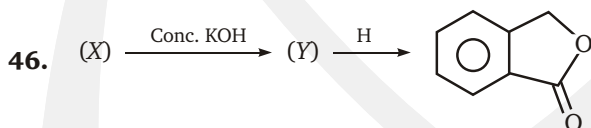
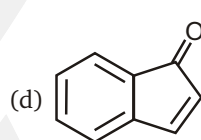
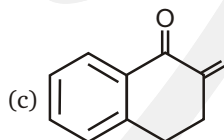
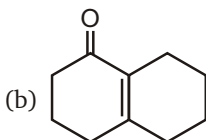
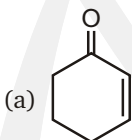




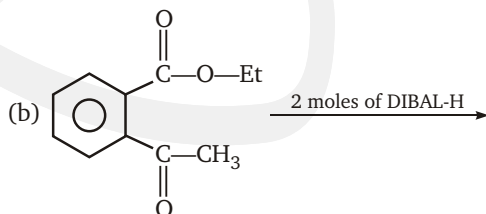
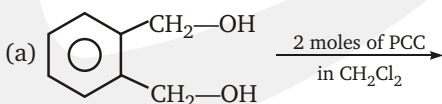
Structure of (B) is :

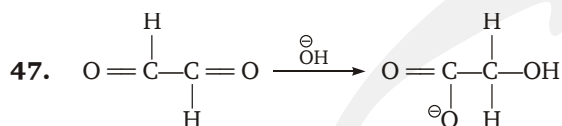
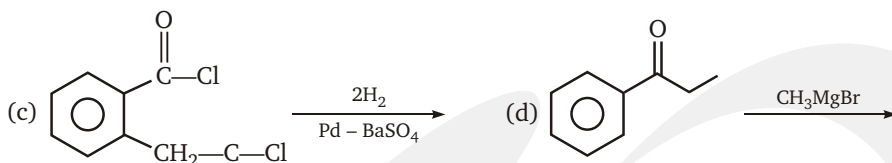


45. Which of the following is not the product of an intramolecular aldol condensation ?



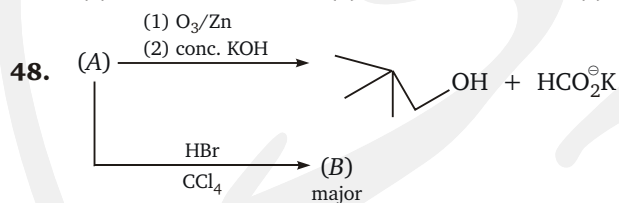
Compound (X) is obtained by which of the following reaction ?



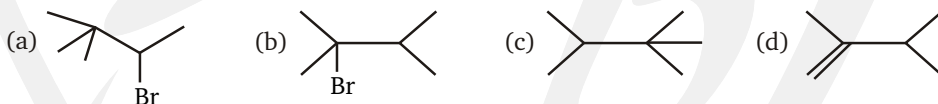


Given reaction is Cannizzaro reaction at normal concentration of base. Predict the order of above reaction

- (a) 1 (b) 2 (c) 3 (d) 4



Product (B) is :



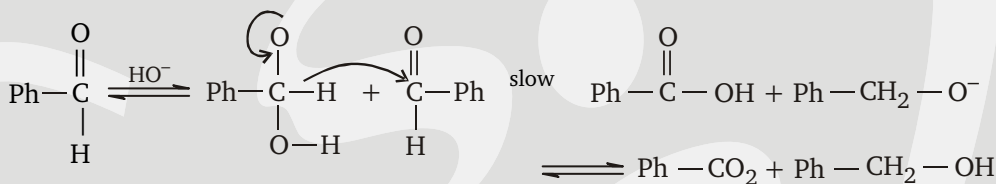
ANSWERS — LEVEL 1

1.	(a)	2.	(b)	3.	(a)	4.	(b)	5.	(b)	6.	(c)	7.	(b)	8.	(b)
9.	(b)	10.	(c)	11.	(a)	12.	(a)	13.	(a)	14.	(b)	15.	(a)	16.	(b)
17.	(b)	18.	(a)	19.	(c)	20.	(a)	21.	(c)	22.	(b)	23.	(c)	24.	(c)
25.	(d)	26.	(a)	27.	(c)	28.	(b)	29.	(c)	30.	(c)	31.	(a)	32.	(a)
33.	(c)	34.	(a)	35.	(c)	36.	(b)	37.	(b)	38.	(b)	39.	(c)	40.	(b,c)
41.	(d)	42.	(d)	43.	(c)	44.	(a)	45.	(c)	46.	(a)	47.	(b)	48.	(b)

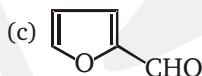
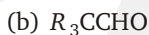
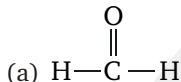
Level - 2

1. Comprehension

Mechanism of Cannizzaro's reaction of benzaldehyde is



A. Which of the following reactants can undergo Cannizzaro's reaction.?

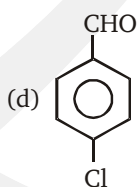
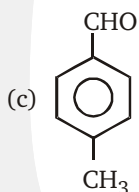
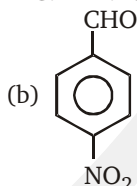
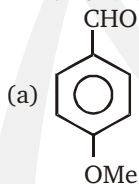


B. Order of the above reaction is:

- (a) 1
(c) 3

- (b) 2
(d) 4

C. Which of the following is best hydride donor in Cannizzaro's reaction?

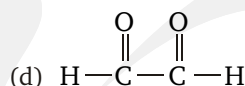
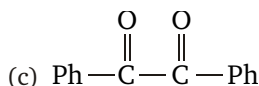
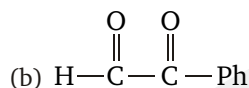
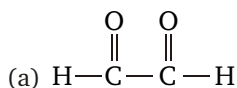


D. Cannizzaro's reaction is:

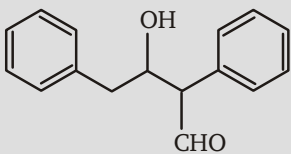
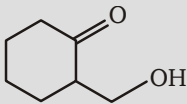
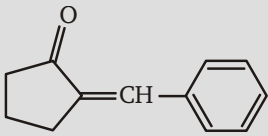
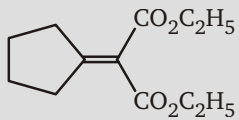
- (a) Reduction
(c) Oxidation

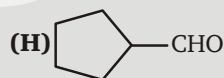
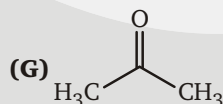
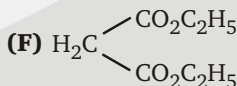
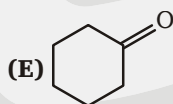
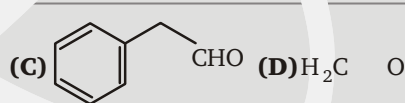
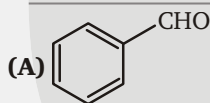
- (b) Disproportionation reaction
(d) Ion - exchange reaction

E. Which of the following cannot undergo intramolecular Cannizzaro's reaction?



2. Aldol condensation proceeds by carbon-carbon bond formation between an enolate donor and a carbonyl acceptor. For each of the following aldol products (a through e) select a donor and an acceptor compound from the list at the bottom of the page (compounds A through H). Write the letter corresponding to your selection in the appropriate answer box.

	Aldol Product	Donor	Acceptor
a.			
b.			
c.			
d.	$(\text{CH}_3)_2\text{C}(\text{OH})\text{CH}_2\text{COCH}_3$		
e.			



3. Comprehension

During an experimental workup procedure, a chemist treated a starting material with NaOH in the solvent acetone [$(\text{CH}_3)_2\text{C}=\text{O}$]; however, the starting material was recovered unreacted. Instead, the chemist isolated a small amount of Product A (shown below).

Product A

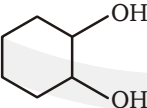
The chemist determined that Product A resulted from the aldol self-condensation of acetone. Product A was identified based on the following observations.

Observations about Product A

1. Elemental analysis of Product A indicated that it consisted only of carbon, hydrogen, and oxygen.
2. product A had a molecular weight of 116 g/mol.
3. Product A was a methyl ketone because it gave a positive iodoform test.
4. When product A was treated with Br_2 in CCl_4 , the red bromine colour persisted, because no carbon-carbon double bonds were present to react with the bromine.

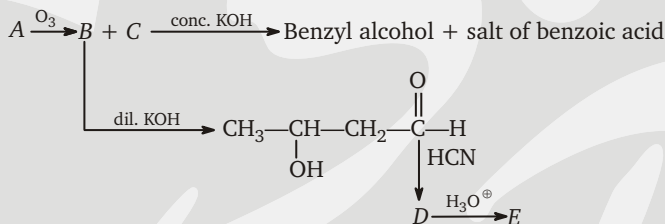
The structure of Product A was further confirmed when treatment with hot sulfuric acid resulted in the corresponding dehydration product, Product B.

- A.** What is the molecular weight of a compound that undergoes an aldol self-condensation reaction to result in a β -hydroxy ketone with a molecular weight of 144 ?
- (a) 70 g/mol (b) 72 g/mol
(c) 74 g/mol (d) 76 g/mol
- B.** The aldol self-condensation of acetone is an equilibrium that favours acetone over its condensation product. Which of the following experimental modifications is most likely to shift the position of equilibrium toward Product A ?
- (a) Using only a catalytic amount of NaOH
(b) Using only a catalytic amount of acetone
(c) Removing Product A as it is formed
(d) Increasing the reaction temperature to the boiling point of acetone
- C.** Based only on observation 1 and 2, which of the following compounds could have been Product A ?

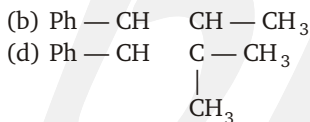
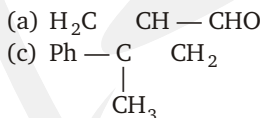
- (a) $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$
- (b) $\text{CH}_3\text{C}(=\text{O})\text{CH}_2\text{CH}(\text{OH})\text{CH}_2\text{CH}_2\text{CH}_3$
- (c) $\text{CH}_2=\text{CHCH}_2\text{OCH}_2\text{CH}_2\text{CH}_3$
- (d) 

- D.** When a drop of Br_2 in CCl_4 is added to Product B, the resulting solution will be :
- colourless, because Product B does not contain a carbon-carbon double bond
 - colourless, because Product B contains a carbon-carbon double bond
 - red, because Product B does not contain a carbon-carbon double bond
 - red, because Product B contains a carbon-carbon double bond
- E.** Which of the following compounds from the passage will give a positive iodoform test ?
- Product A only
 - Product A and Product B
 - Product A and acetone only
 - Product A, Product B, and acetone

4. Comprehension



A. Structure of A is :

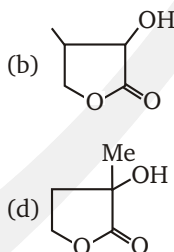
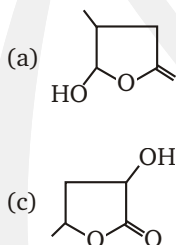


B. Structure of (B) and (C) differentiated by :

- Tollen's reagent
- 2,4-DNP

- Fehling solution
- NaHSO_3

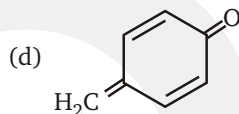
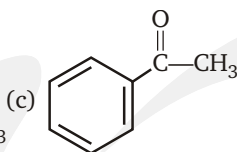
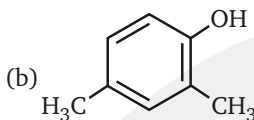
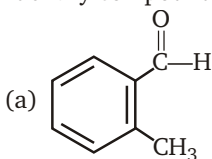
C. Structure of E is :



5. Comprehension

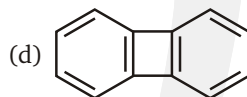
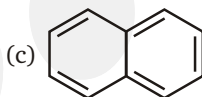
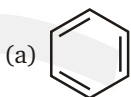
An organic compound 'P' with molecular formula $\text{C}_8\text{H}_8\text{O}$ forms an organic red ppt with 2, 4-DNP gives yellow ppt. on heating with I_2 in presence of NaOH . It neither reduces Tollen's or Fehling reagent nor does it decolourise Br_2 water or Baeyer's reagent. On drastic oxidation with chromic acid (H_2CrO_4) it gives an acid (Q) having molecular formula $\text{C}_7\text{H}_6\text{O}_2$.

A. Identify compound *P* :



B. *Q* $\xrightarrow{\text{NaOH/CaO}}$ *R* (Major product)

R will be :



C. *P* $\xrightarrow{\text{dil. } \text{OH}^-}$ *S* $\xrightarrow[\text{H}_3\text{O}^+]{\text{CH}_3\text{MgBr (1mole)}}$

No. of stereoisomer in the final product *T* :

(a) 2

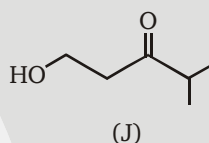
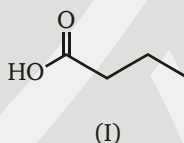
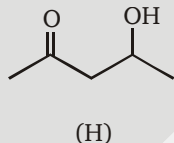
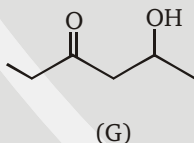
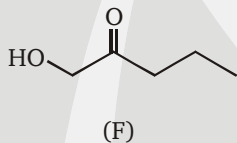
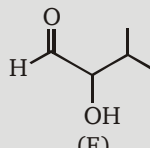
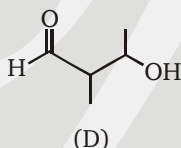
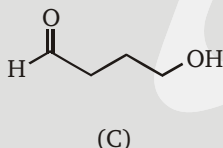
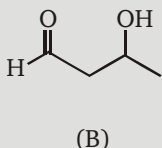
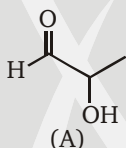
(b) 3

(c) 4

(d) 6

Subjective Problems

1.

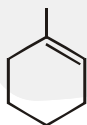


X Number of compound obtained by aldol reaction

Y Number of compounds react with NaHCO_3

Sum of *X* *Y* is

2. In the scheme given below, the total number of intramolecular aldol condensation products formed from '*Y*' is :



1. O_3
2. $\text{Zn, H}_2\text{O}$

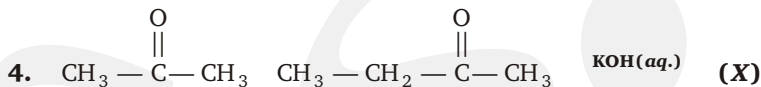
Y

1. NaOH(aq)
2. heat



x moles of HCHO consumed.

Value of (x) will be



X number of aldol condensation product (including stereoisomer).

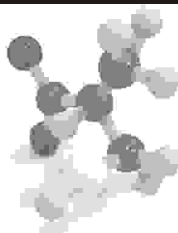
Find out the value of (X).

ANSWERS — LEVEL 2

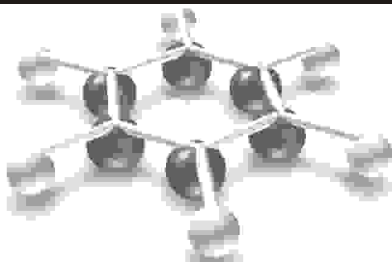
1. A – d; B – c; C – a; D – b; E – c
2. a – Donor = C, Acceptor = C; b – Donor = E, Acceptor = D;
c – Donor = B, Acceptor = A; d – Donor = G, Acceptor = G; e – Donor = F, Acceptor = B
3. A – b; B – c; C – d; D – b; E – d
4. A – b, B – b, C – c
5. A – c, B – a, C – c

Subjective Problems

1. 6 2. 3 3. 6 4. 9



9



CARBOXYLIC ACID AND THEIR DERIVATIVES



Level - 1

1. Identify C in the following sequence of reactions :



SOCl_2

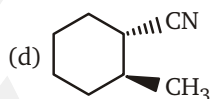
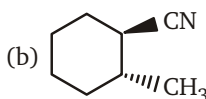
A

NH_3

B

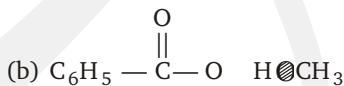
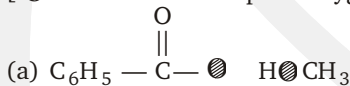
P_4O_{10}
heat

$\text{C}(\text{C}_8\text{H}_{13}\text{N})$

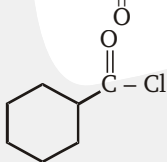


2. Saponification (basic hydrolysis) of $\text{C}_6\text{H}_5\text{C}(=\text{O})\text{CH}_3$ will yield :

[$\otimes$ = mass - 18 isotope of oxygen]



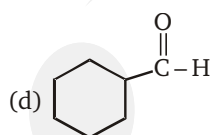
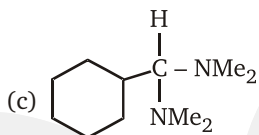
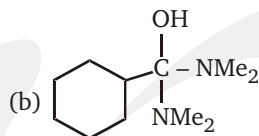
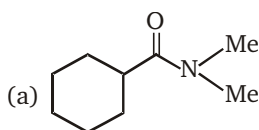
- 3.



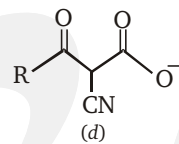
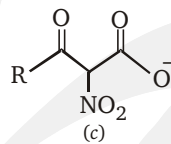
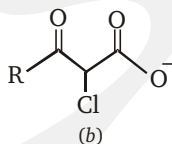
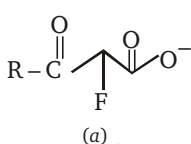
Me_2NH
0 $^\circ\text{C}$, 2h

(X)
86-89% (yield);
(3 equivalent)

Product (X) of the reaction is :



4. Which of the following is the correct order of decarboxylation of α -keto carboxylate anion ?

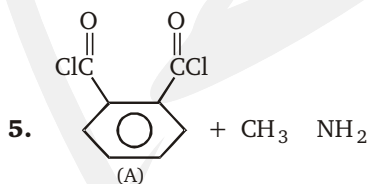


(a) $a > b > c > d$

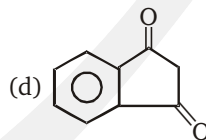
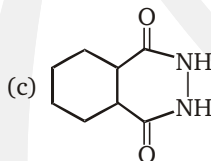
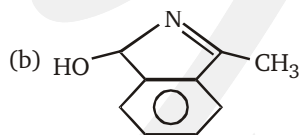
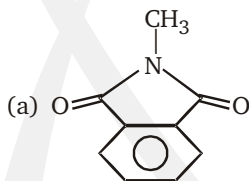
(b) $c > d > a > b$

(c) $c > d > b > a$

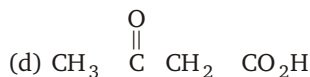
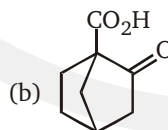
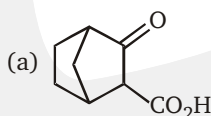
(d) $d > c > a > b$



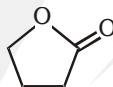
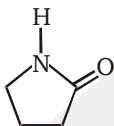
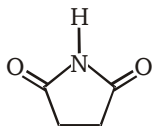
Product of the reaction is :



6. Which α -keto acid shown will not undergo decarboxylation ?



7. Choose the response that matches the correct functional group classification with the following group of structural formulas.



(a) Anhydride

Lactam

Lactone

(b) Lactam

Imide

Lactone

(c) Imide

Lactone

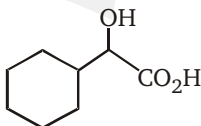
Anhydride

(d) Imide

Lactam

Lactone

8.

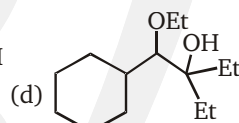
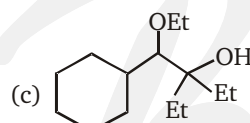
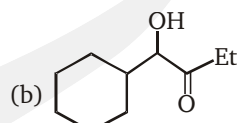
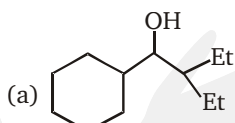


(1) Et Li(3 eq.)

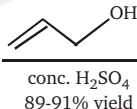
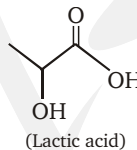
(P) ; Product (P) of the reaction is :

(2) H₂, H₂O

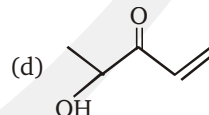
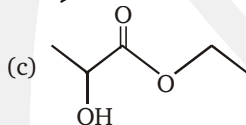
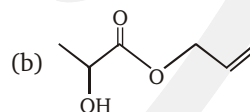
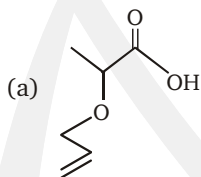
(65% yield)



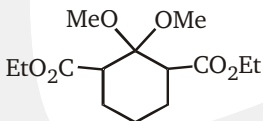
9.

(X) + H₂O

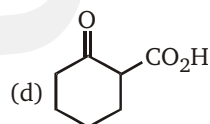
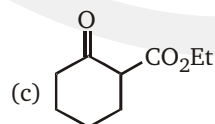
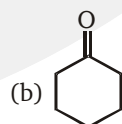
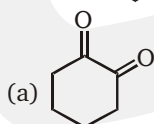
Product (X) of the reaction is :



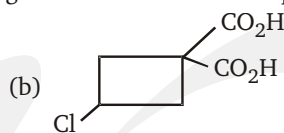
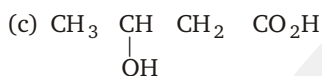
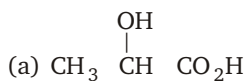
10.

H₃O⁺

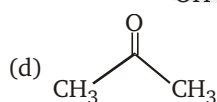
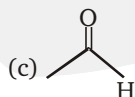
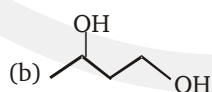
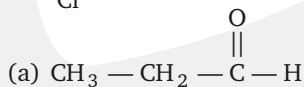
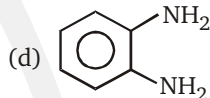
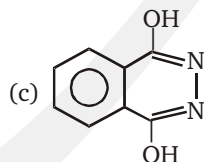
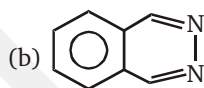
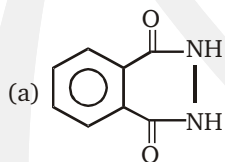
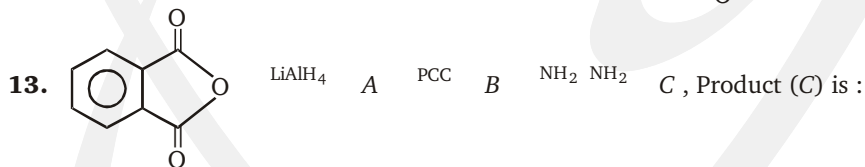
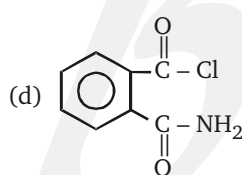
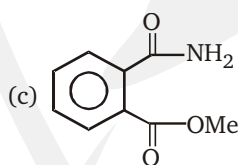
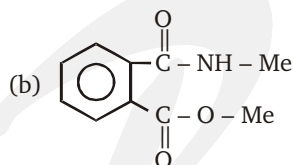
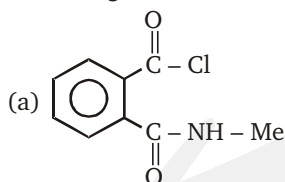
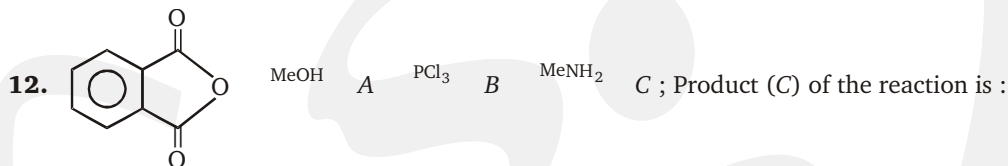
(A) , Product (A) obtained is :

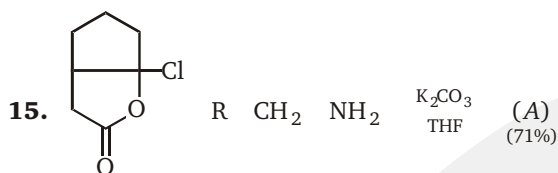


11. Which of the following acid on heating gives geometrical isomers as a product ?

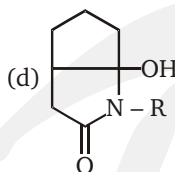
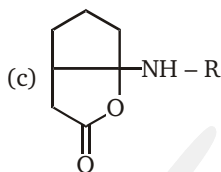
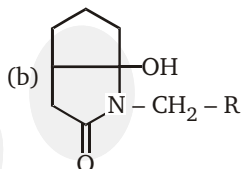
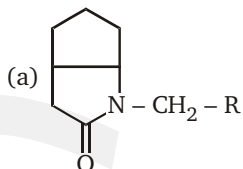


(d) All of these

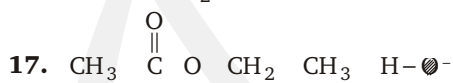
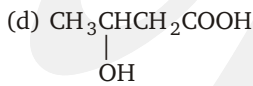
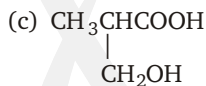
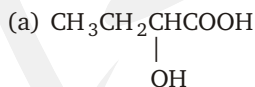




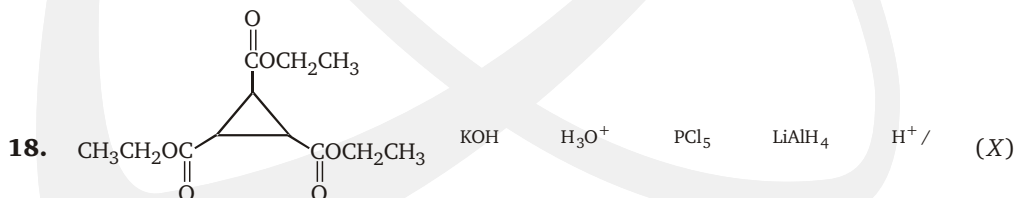
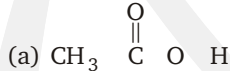
In above reaction identify major product (A) of the reaction:



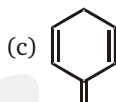
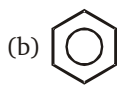
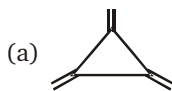
16. An optically active compound 'X' has molecular formula $\text{C}_4\text{H}_8\text{O}_3$. It evolves CO_2 with NaHCO_3 . 'X' reacts with LiAlH_4 to give an achiral compound. 'X' is :



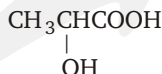
($\text{O}^- = \text{O}^{18}$) One of the product of the reaction is :



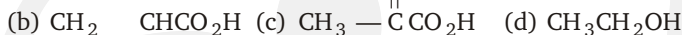
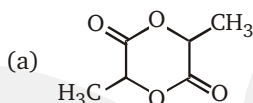
Product (X) is :



19. Identify final product in the following reaction;



Product



20. Select the final product from this sequence of reactions.

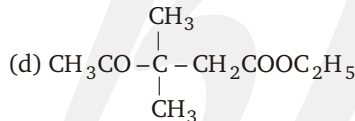
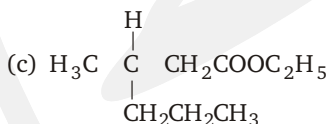
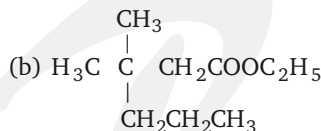
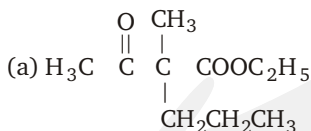
Acetoacetic ester

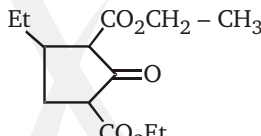
NaOEt

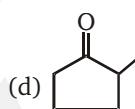
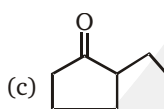
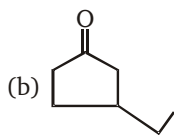
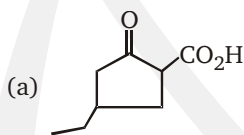
CH_3I

NaOEt

$\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$

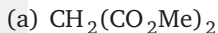


21.  $\xrightarrow[\text{heat}]{\text{H}_2\text{O}, \text{H}_2\text{SO}_4}$ (A) ; Product (A) will be :

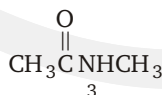
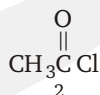
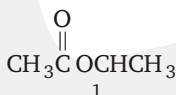


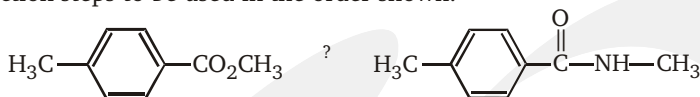
22. $\text{CH}_2(\text{CO}_2\text{Me})_2 \xrightarrow[\text{(ii) AcOH}]{\text{(i) Na}} ?$ $\text{CH}(\text{CO}_2\text{Me})_3$

Which of the following reactants will complete the above reaction ?



23. Arrange the following in order of increasing reactivity (least most) towards nucleophile

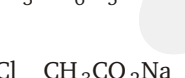
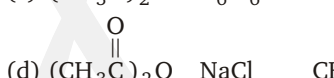
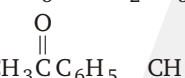
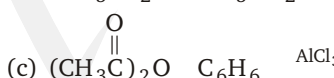
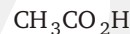
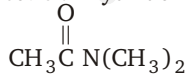




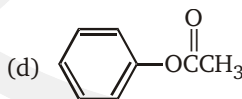
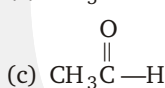
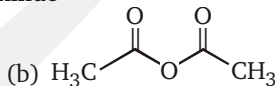
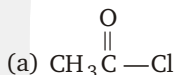
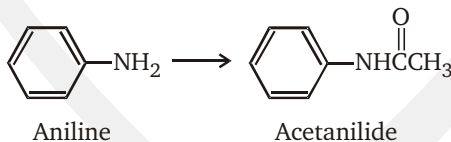
- (a) H_3O^+ to $\text{CH}_3\text{C}(=\text{O})\text{NH}_2$



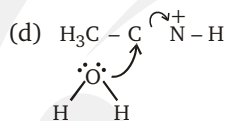
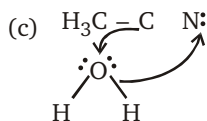
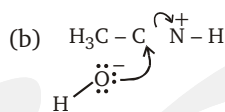
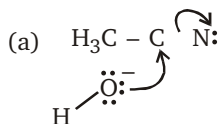
- $$(a) \quad (\text{CH}_3\overset{\text{O}}{\parallel}\text{C})_2\text{O} + 2\text{HN}(\text{CH}_3)_2$$



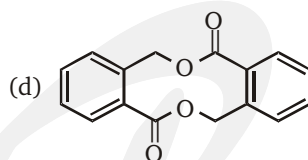
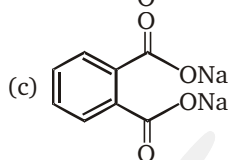
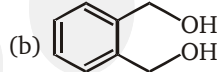
27. All but one of the following compounds react with aniline to give acetanilide. Which one does not?



28. Which of the following best describes the nucleophilic addition step in the acid-catalyzed hydrolysis of acetonitrile (CH_3CN) ?



29. The major product expected, when Phthalamide is treated with NaOH, is :



30. Which of following acid remains unaffected on heating ?

(a) malonic acid

(b) maleic acid

(c) Fumaric acid

(d) Succinic acid

31. $\text{Br}-\text{CH}_2(\text{CH}_2)_n-\text{Br} + \text{CH}_2(\text{CO}_2\text{Et})_2 \xrightarrow[\text{EtOH}]{\text{NaOEt}}$ cyclic product

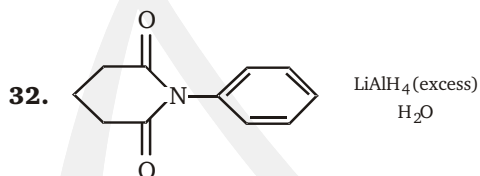
At which value of n the formation of six membered ring takes place ?

(a) $n = 2$

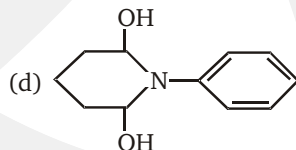
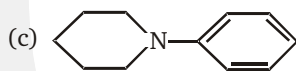
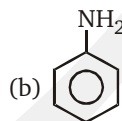
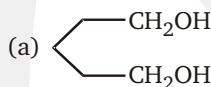
(b) $n = 3$

(c) $n = 5$

(d) $n = 6$



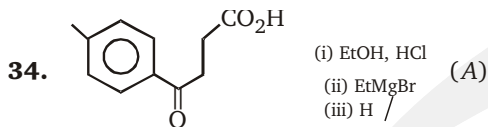
Product of the reaction is :



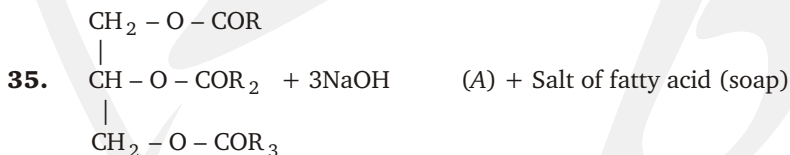
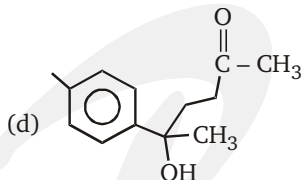
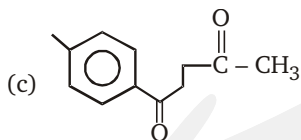
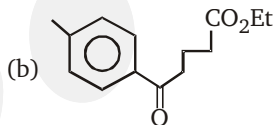
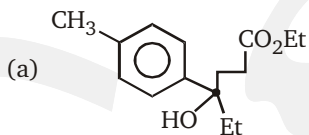
33. $\xrightarrow[\text{(H}_2\text{O)}]{\text{(CO}_2\text{)}}$, Product of the reaction is :

- (a) *cis*-anhydride
(c) both (a) & (b)

- (b) *trans*-anhydride
(d) mono-basic acid



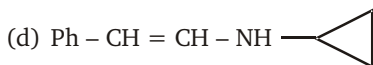
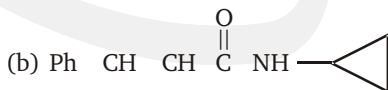
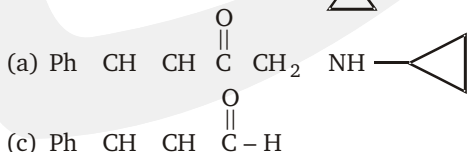
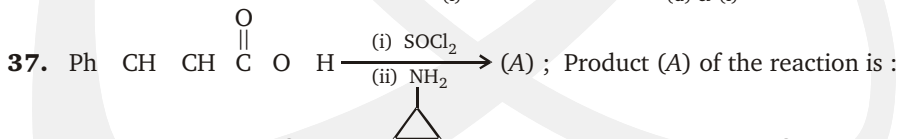
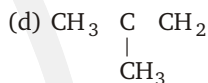
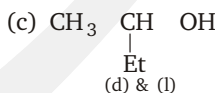
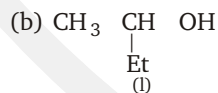
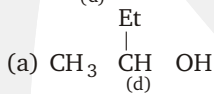
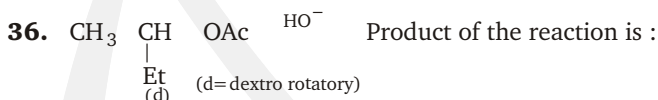
Product (A) of the reaction is :

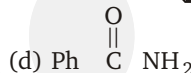
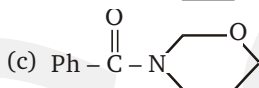
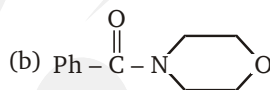
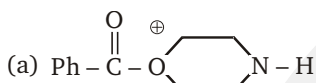
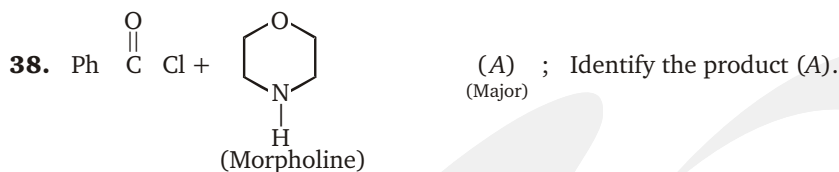


Product (A) of the reaction is :

- (a) Ethylene glycol
(c) Glyceryltrinitrate (explosive)

- (b) Glycerol
(d) Cumene hydrogen peroxide





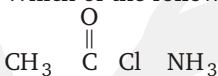
(a) Esterification

(b) Saponification

(c) Hydrolysis

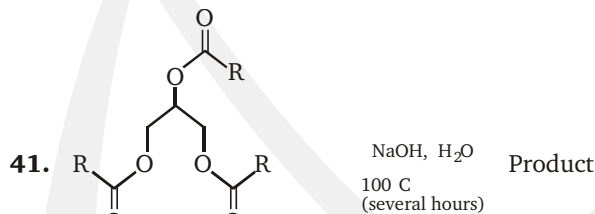
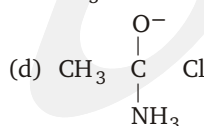
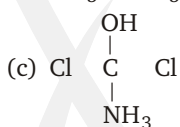
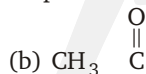
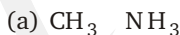
(d) Trans Esterification

40. Which of the following is an intermediate formed in the reaction shown below ?



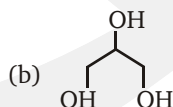
Intermediate

product.



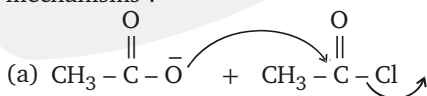
(Principal component of coconut oil.)

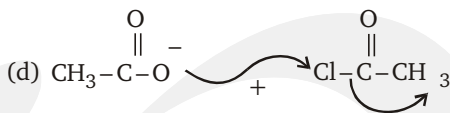
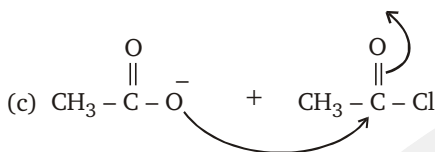
Product is obtained in the above reaction is :



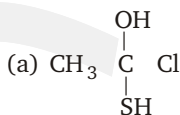
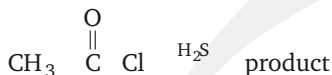
(c) Both (a) and (b) (d) None of these

42. The reaction of sodium acetate with acetyl chloride proceeds through which of the following mechanisms ?

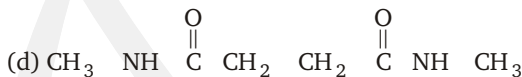
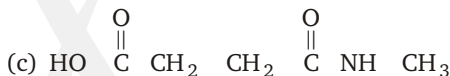
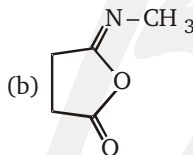
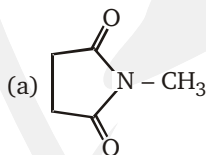
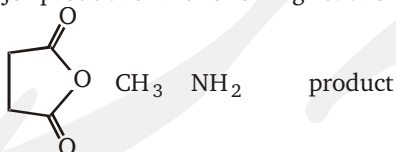




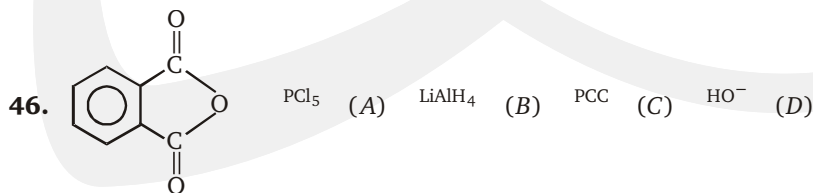
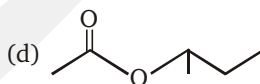
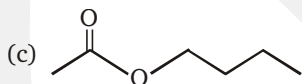
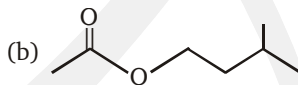
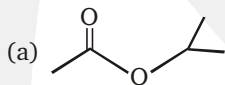
43. Which is the major product of the following reaction ?

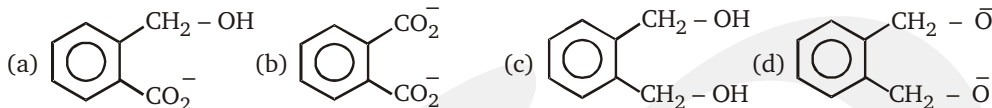


44. Which is the major product of the following reaction ?

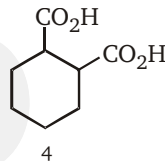
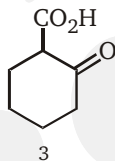
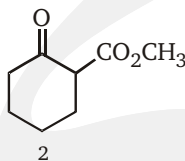
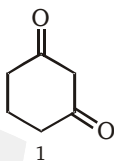


45. Ethanoic acid + 3-methyl-1-butanol $\xrightleftharpoons[\text{H}_2\text{SO}_4]{\text{traces}}$ (A); Compound (A) is :



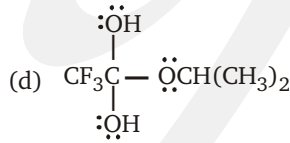
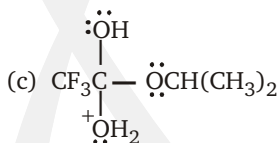
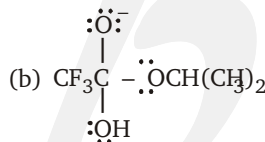
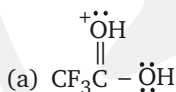


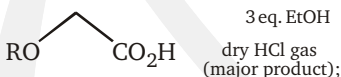
47. Which of the following compounds will undergo decarboxylation on heating ?



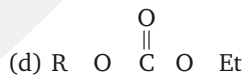
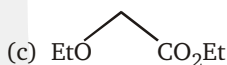
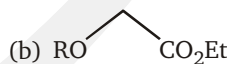
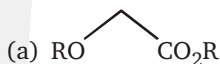
- (a) 2 and 3 (b) 3 and 4 (c) 3 only (d) 1 and 4

48. Which one of the following is not an intermediate in the generally accepted mechanism for the reaction shown below ?



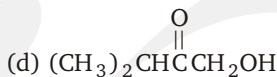
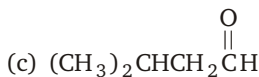
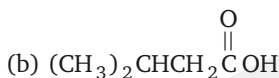
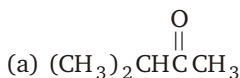
49. 

(A); Product A is :

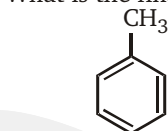


50. Identify the compound C in the following sequence :





51. What is the final product (B) of this sequence ?

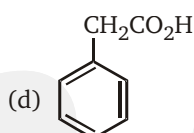
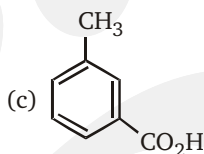
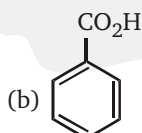
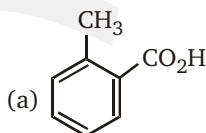


Br_2
light

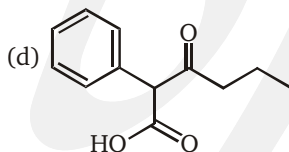
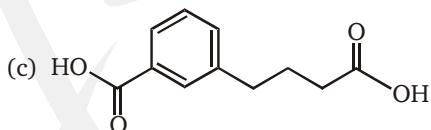
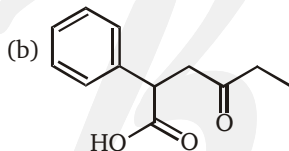
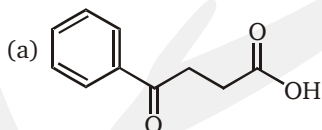
A

1. KCN
2. H_3O^+ , heat

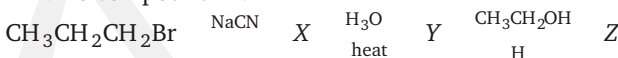
B



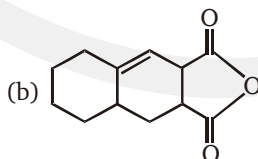
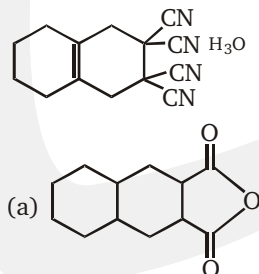
52. Which of the following undergoes decarboxylation most readily on being heated ?

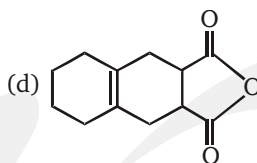
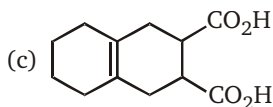


53. What is compound Z ?

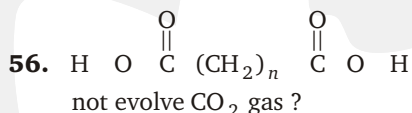
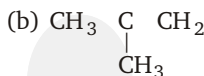


54. (A); Product (A) of the reaction is :

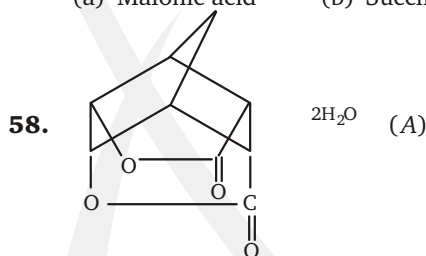
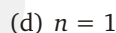
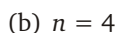
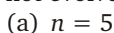




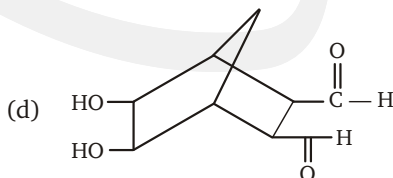
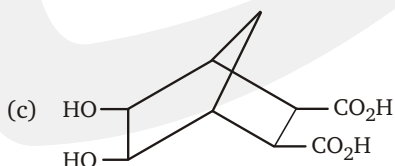
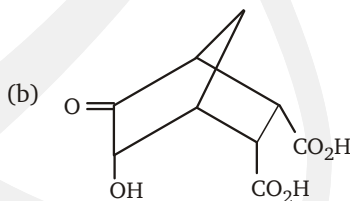
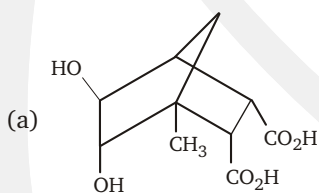
(X) (major); Product (X) is :

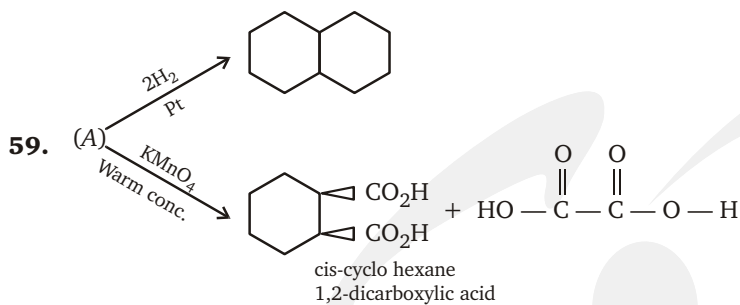


product , At what value of (n) given compound will

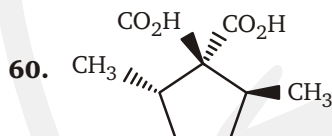
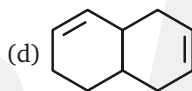
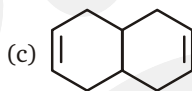
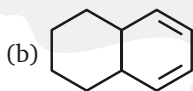
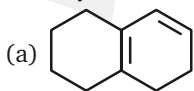


Product (A) of the above reaction is :





Identify (A).



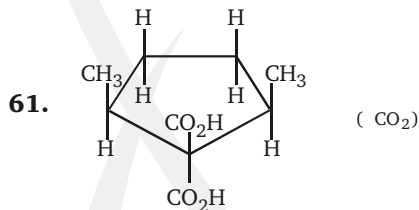
How many product will be formed when above compound undergo de-carboxylation?

(a) 0

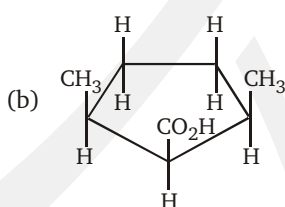
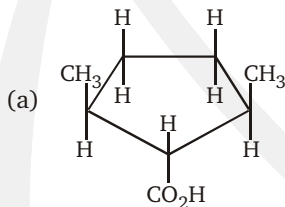
(b) 1

(c) 2

(d) 3



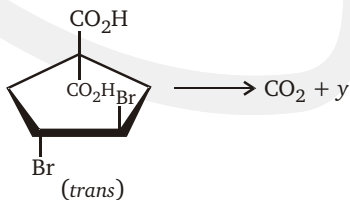
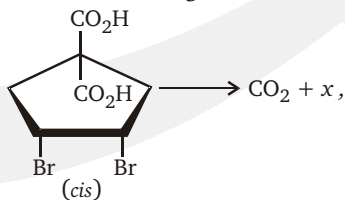
Product of the reaction is :



(c) Both (a) and (b)

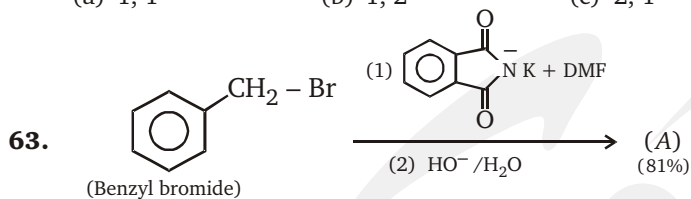
(d) none of these

62. Products obtained in the given reactions are shown below.



The number of possible products for x and y are :

- (a) 1, 1 (b) 1, 2 (c) 2, 1 (d) 2, 2

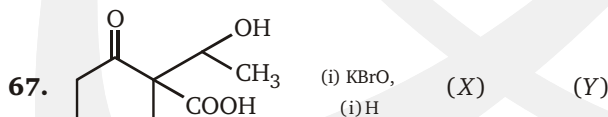
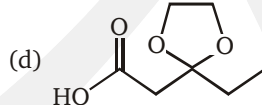
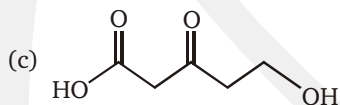
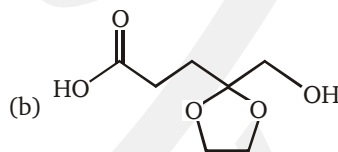
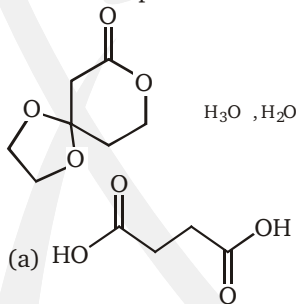


Product (A) of the above reaction is :

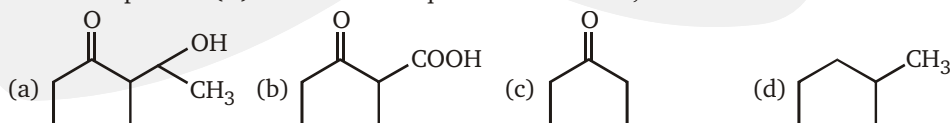
- (a) Ph NH₂ (b) Ph CH₂ NH₂
 (c) Ph CH₂ NH CO₂H (d) Ph CH₂ NH CHO
64. Which of the following pair is C₂-epimer ?
 (a) D-Glucose, D-Maltose (b) D-Glucose, D-Mannose
 (c) D-Allose, D-Ribose (d) D-Glucose, D-Arabinose
65. Total number of enol possible for the compound formed during given reaction will be (including stereoisomer):

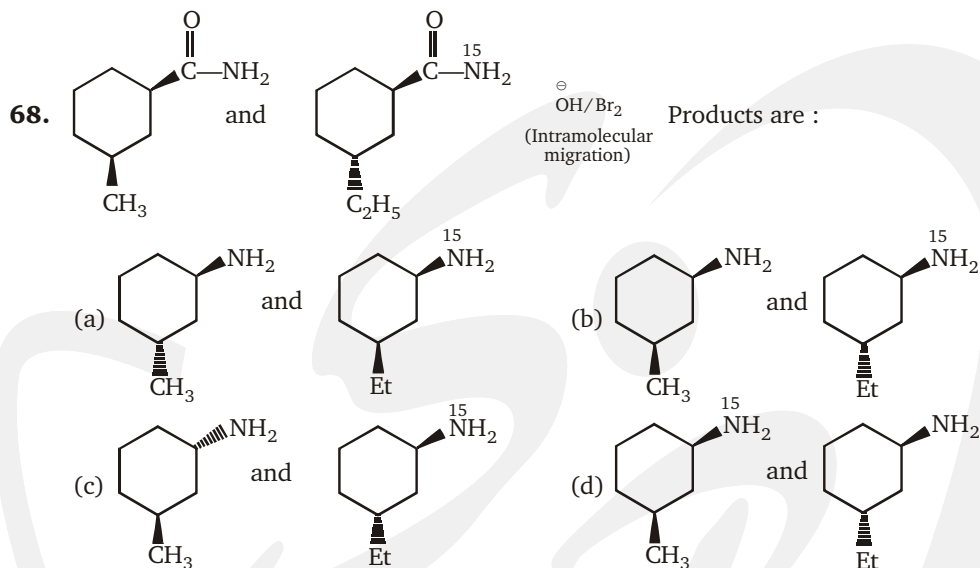


- (a) 2 (b) 3 (c) 4 (d) 5
66. What is the product of the following reaction ?



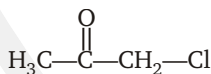
Hence the product (Y) in the above sequence of reactions, is :



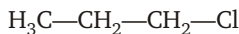


In the above reaction, if the reactant alcohol is a pure R-isomer the product would.

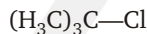
- (a) have configuration inverted at the chiral atom
 (b) be a racemic mixture
 (c) have the same configuration at the chiral atom
 (d) be optically inactive
70. The order of $\text{S}_\text{N}1$ reactivity in aqueous acetic acid solution for the compounds



(1)



(2)



(3)

(a) 1 2 3

(b) 1 3 2

(c) 3 2 1

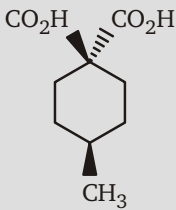
(d) 3 1 2

ANSWERS — LEVEL 1

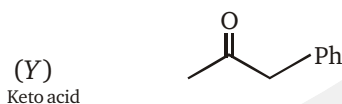
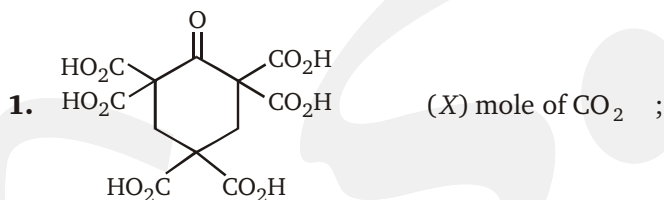
1.	(b)	2.	(b)	3.	(a)	4.	(c)	5.	(a)	6.	(b)	7.	(d)	8.	(b)
9.	(b)	10.	(b)	11.	(d)	12.	(b)	13.	(b)	14.	(a)	15.	(b)	16.	(c)
17.	(c)	18.	(b)	19.	(a)	20.	(a)	21.	(b)	22.	(c)	23.	(b)	24.	(a)
25.	(b)	26.	(d)	27.	(c)	28.	(d)	29.	(c)	30.	(c)	31.	(b)	32.	(c)
33.	(a)	34.	(a)	35.	(b)	36.	(a)	37.	(b)	38.	(b)	39.	(d)	40.	(d)
41.	(c)	42.	(c)	43.	(b)	44.	(c)	45.	(b)	46.	(a)	47.	(c)	48.	(b)
49.	(b)	50.	(c)	51.	(d)	52.	(d)	53.	(d)	54.	(d)	55.	(c)	56.	(c)
57.	(c)	58.	(c)	59.	(b)	60.	(b)	61.	(c)	62.	(c)	63.	(b)	64.	(b)
65.	(b)	66.	(c)	67.	(c)	68.	(b)	69.	(c)	70.	(c)				

Level - 2

1. Match the Column (I) and (II). (Matrix)

Column (I)		Column (II)	
Reaction		Products formed	
(a)	$ \begin{array}{c} \text{CH}_3 \\ \\ \text{HO}_2\text{C} - \text{C} - \text{CO}_2\text{H} \\ \quad \\ \text{H} \quad \text{D} \\ \\ \text{Ph} \end{array} $	(p)	Diastereomers
(b)	$ \begin{array}{c} \text{CH}_3 \\ \\ \text{HO}_2\text{C} - \text{C} - \text{CO}_2\text{H} \\ \\ \text{Et} \end{array} $	(q)	Racemic mixture
(c)		(r)	Meso compound
(d)		(s)	CO ₂ gas will evolve

Subjective Problems



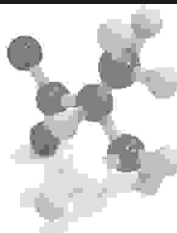
(Y) is including stereoisomers. Value of (X + Y) will be

ANSWERS — LEVEL 2

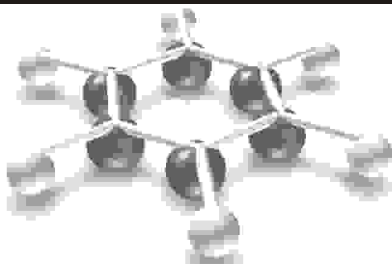
1. a – p, s; b – q, s; c – p, s; d – r

Subjective Problems

1. 8



10

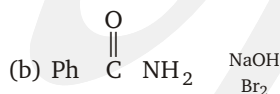
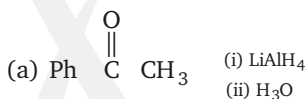


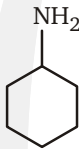
AMINES



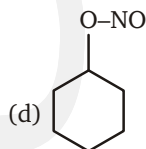
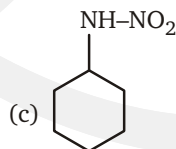
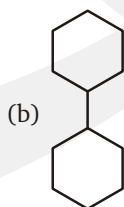
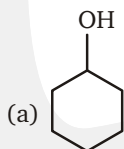
Level - 1

1. In which of the following reaction cyanide will be obtained as a major product ?

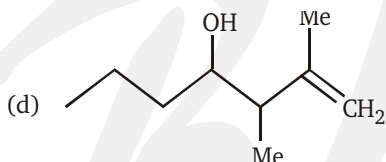
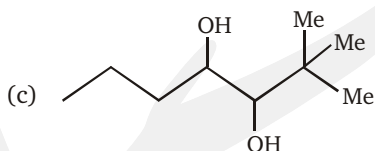
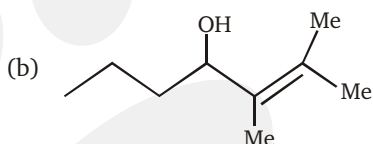
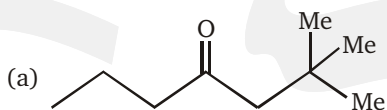
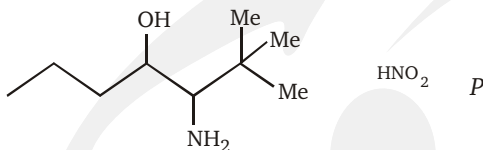


2.  HNO_2 A (Major) :

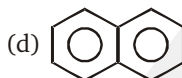
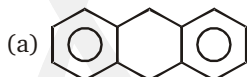
Product (A) is :



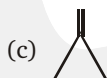
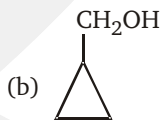
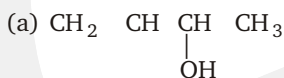
3. Which of the following alkene cannot be prepared by de-amination of $n\text{-Bu}-\text{NH}_2$ with NaNO_2/HCl ?
 (a) 1-butene (b) *cis*-2-butene (c) *trans*-2-butene (d) Iso-butene
4. Predict the major product P in the following reaction.



5. (A); Product of this reaction is :

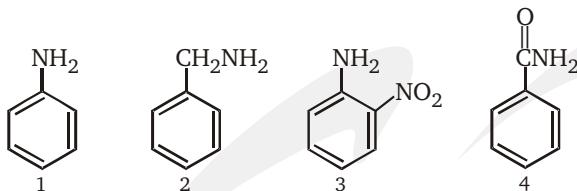


6. + $\text{CH}_2=\text{CH}-\text{CH}_2-\text{CH}_2-\text{OH}$
- A will be :



7. Which of the following isomers of $\text{C}_8\text{H}_9\text{NO}$ is the weakest base ?
 (a) *o*-Aminoacetophenone (b) *p*-Aminoacetophenone
 (c) *m*-Aminoacetophenone (d) Acetanilide

8. Rank the following compounds in order of increasing basic strength. (weakest strongest) :

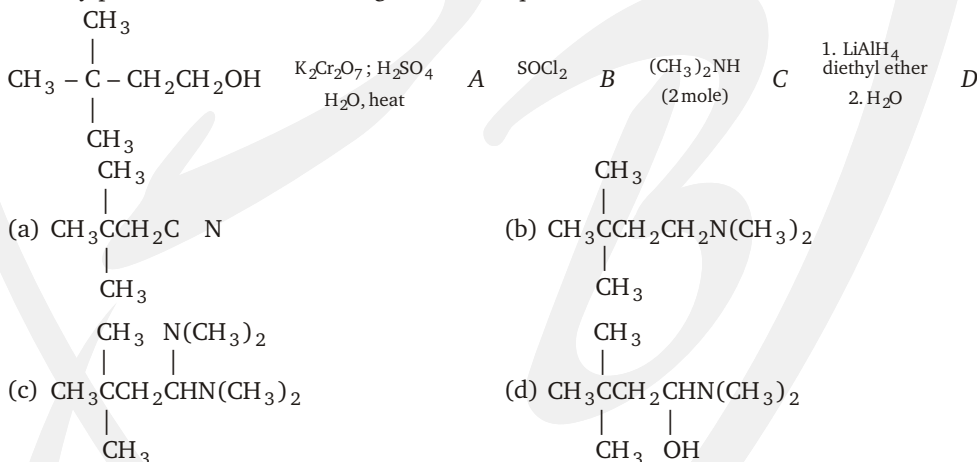


- (a) $4 < 2 < 1 < 3$ (b) $4 < 3 < 1 < 2$ (c) $4 < 1 < 3 < 2$ (d) $2 < 1 < 3 < 4$

9. Which of the following arylamines will not form a diazonium salt on reaction with sodium nitrite in hydrochloric acid ?

- (a) *m*-Ethylaniline (b) *p*-Aminoacetophenone
(c) 4-Chloro-2-nitroaniline (d) *N*-Ethyl-2-methylaniline

10. Identify product *D* in the following reaction sequence :

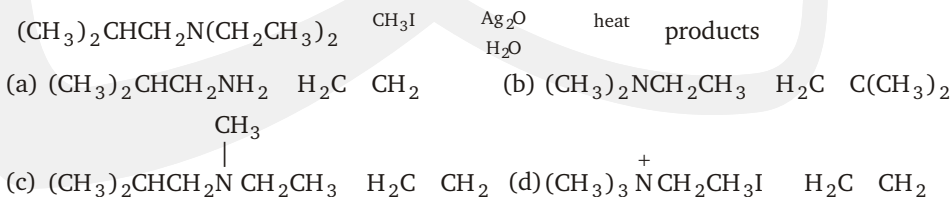


11. Which one of the following is best catalyst for the reaction shown below ?

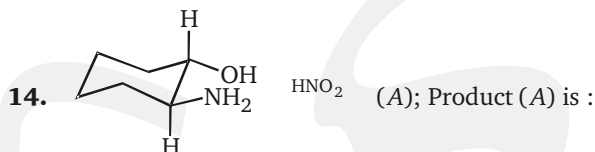
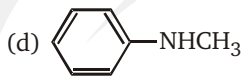
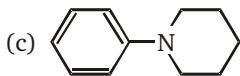
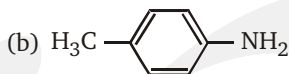
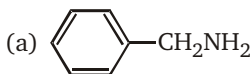


- (a) (b)
(c) (d)

12. The major products obtained from the following sequence of reactions are :



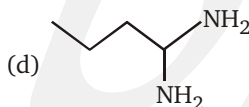
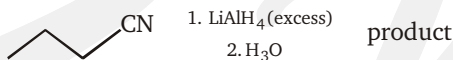
13. Which amine yields *N*-nitroso amine after treatment with nitrous acid ($\text{NaNO}_2, \text{HCl}$) ?



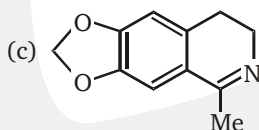
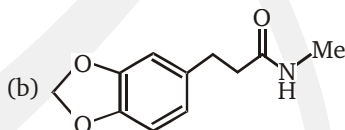
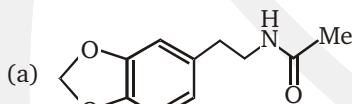
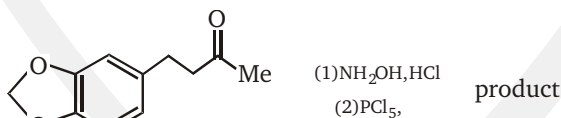
- (a) cyclopentane carboxyaldehyde
(c) 2-aminocyclohexene

- (b) cyclohexane-1, 2-diol
(d) cyclohex-2-enol

15. Choose the appropriate product for this reaction.



16. Which of the following product will be obtained in the given (consider minor product also) Beckmann-type rearrangement ?



(d) all of these

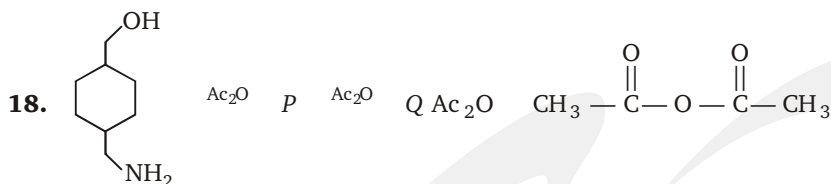
17. Deamination (or) diazotization of *n*-Bu-NH₂ with NaNO_2/HCl gives isomeric butene.

(a) 2

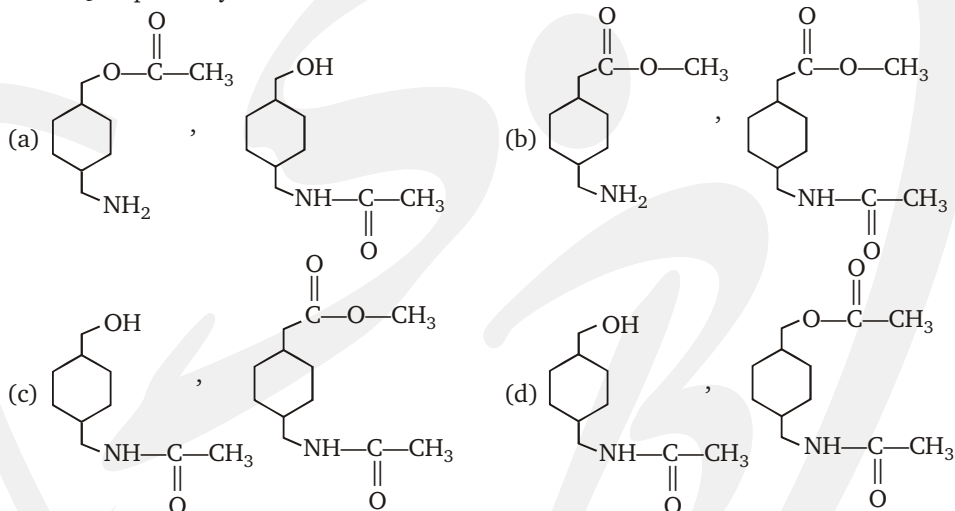
(b) 3

(c) 4

(d) 5



P and Q respectively are :



19. A nitrile X is treated with LiAlH_4 to obtain compound Y ($\text{C}_2\text{H}_7\text{N}$). In a separate reaction X is hydrolyzed in an acid medium to obtain Z. The product obtained after mixing Y and Z will be



20. The compound X ($\text{C}_7\text{H}_9\text{N}$) reacts with benzenesulfonyl chloride to give Y ($\text{C}_{13}\text{H}_{13}\text{NO}_2\text{S}$) which is insoluble in alkali. The compound X is:

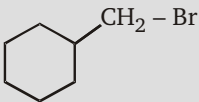
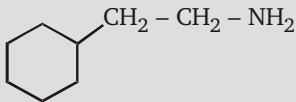
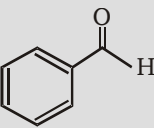
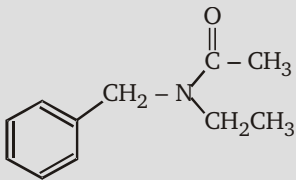
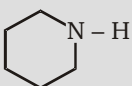
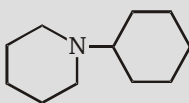
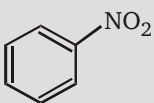
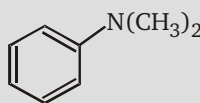
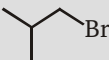
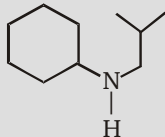


ANSWERS — LEVEL 1

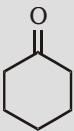
1.	(c)	2.	(a)	3.	(d)	4.	(a)	5.	(b)	6.	(b)	7.	(d)	8.	(b)
9.	(d)	10.	(b)	11.	(c)	12.	(c)	13.	(d)	14.	(a)	15.	(b)	16.	(d)
17.	(b)	18.	(d)	19.	(c)	20.	(a)								

Level - 2

1. Five amine syntheses are outlined below. In each reaction box enter a single letter designating the best reagent and conditions selected from the list at the bottom of the page.

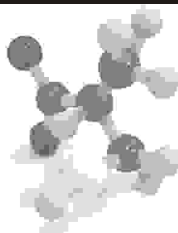
A.		First Step Second Step →	
B.		First Step Second Step Third Step →	
C.		First Step Second Step →	
D.		First Step Second Step →	
E.		First Step Second Step Third Step Fourth Step →	

(a)	(i) LiAlH_4 in ether (ii) H_2O & base	
(b)	$\text{C}_2\text{H}_5\text{NH}_2$ (cat. H^+)	
(c)	NaCN in alcohol	
(d)	H_2 & Ni catalyst or H_2 & Pd catalyst	
(e)	NaN_3 in alcohol	
(f)	$(\text{CH}_3\text{CO})_2\text{O}$ & pyridine	
(g)	$\text{C}_2\text{H}_5\text{Br}$	

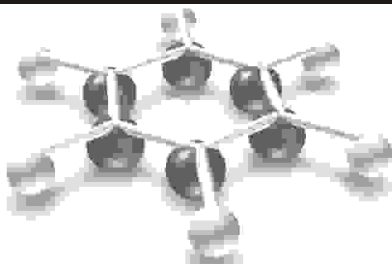
(h)	 , $\text{H}^{\oplus}$	
(i)	$2\text{CH}_3\text{I}$ & pyridine	
(j)	KOH in H_2O	

ANSWERS — LEVEL 2

1. A – c, a or c, d; B – b, d, f; C – h, d; D – d, i or a, i; E – e, a, h, a



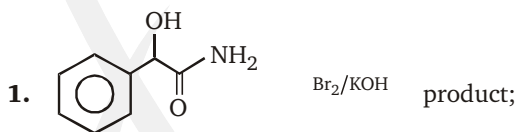
11



CARBENE AND NITRENE

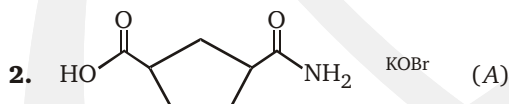
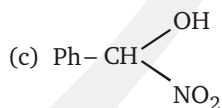
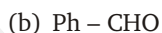
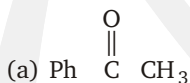


Level - 1

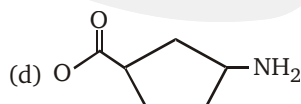
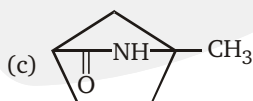
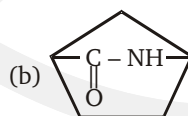
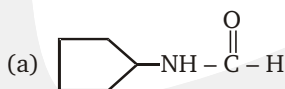


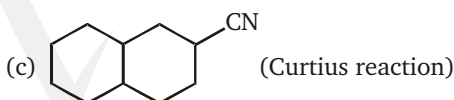
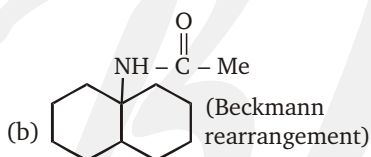
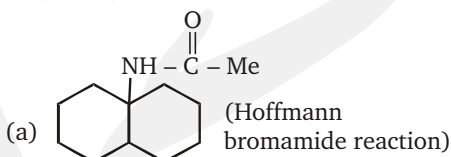
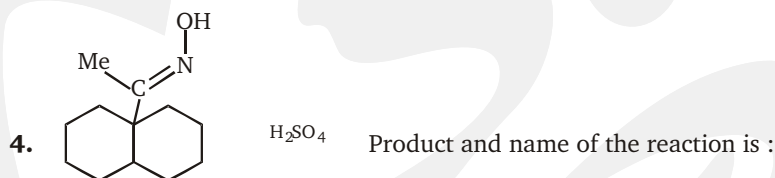
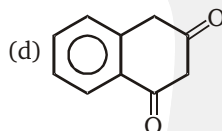
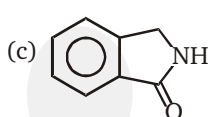
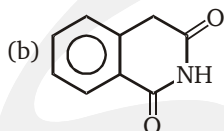
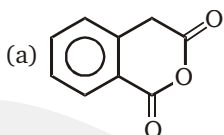
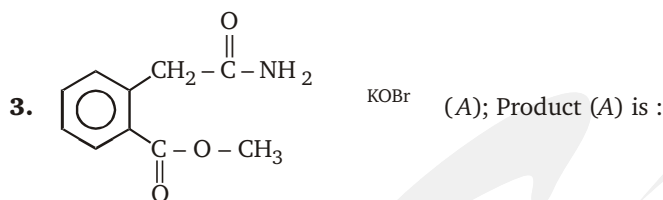
(-hydroxy amide)

Product of this Hoffmann bromamide reaction is :

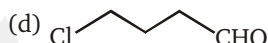
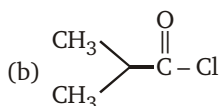
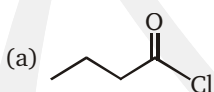


(B). Compound (B) is :

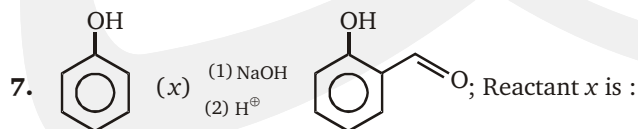
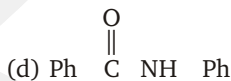
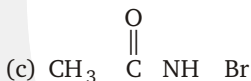
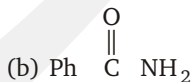
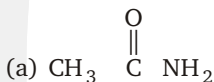


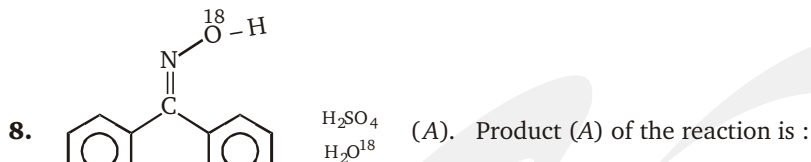


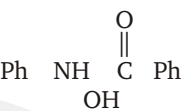
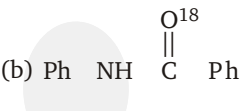
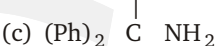
(d) None of these

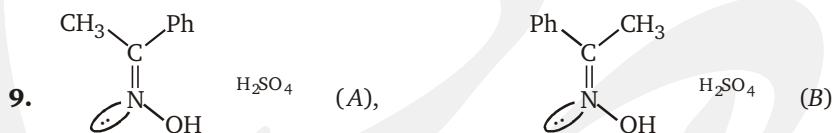


6. Which of the following will not give Hoffmann bromamide reaction ?

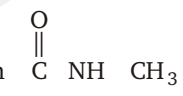
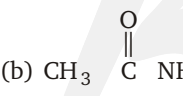
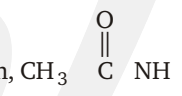


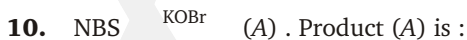


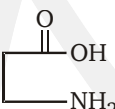
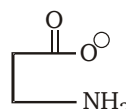
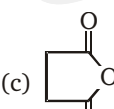
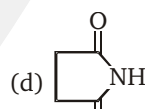
- (a)  (b) 
 (c)  (d) 



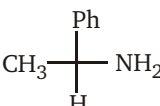
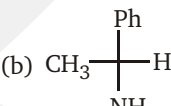
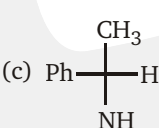
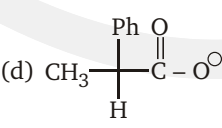
Product (A) & (B) respectively in the above reaction are :

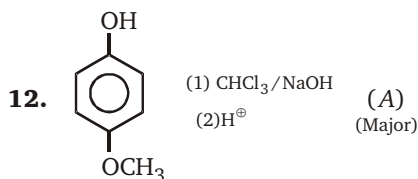
- (a)   (b)  
 (c)   (d)  



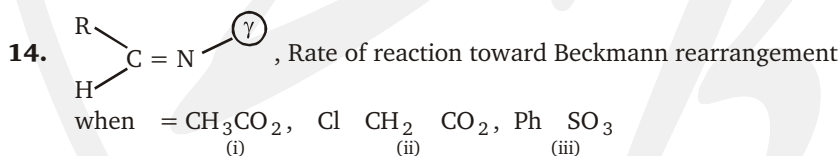
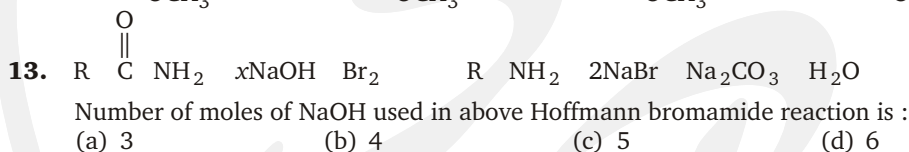
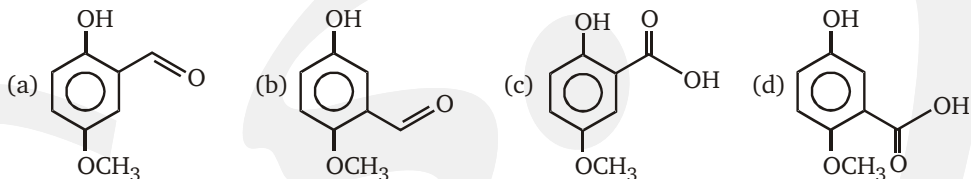
- (a)  (b) 
 (c)  (d) 



- (a)  (b) 
 (c)  (d) 



Product (A) is :



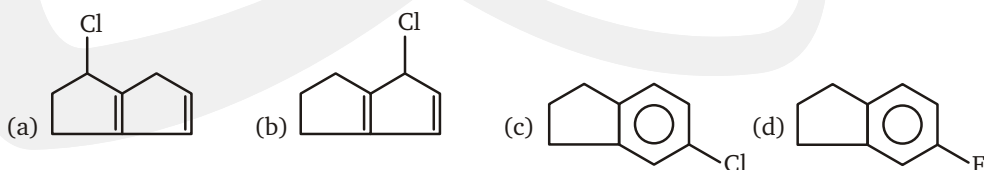
- (a) (i) > (ii) > (iii) (b) (ii) > (i) > (iii)
(c) (iii) > (ii) > (i) (d) (iii) > (i) > (ii)

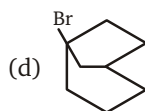
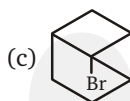
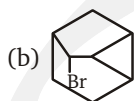
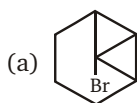
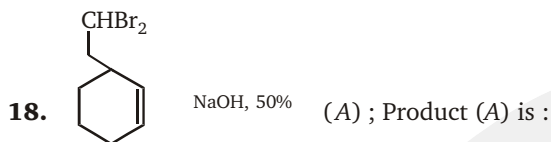
15. When primary amine reacts with chloroform in ethanolic KOH, then product is :

- (a) an isocyanide (b) an aldehyde
(c) a cyanide (d) an alcohol

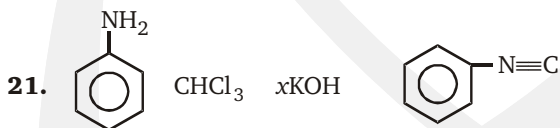
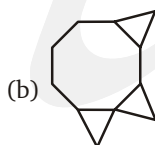
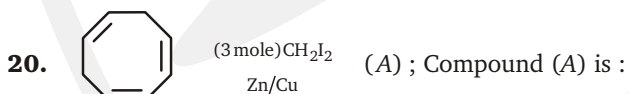
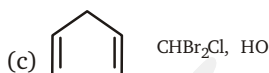
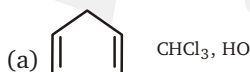
16. The reaction of chloroform with alcoholic KOH and *p*-toluidine forms :

17. What is the product (Q) of the following reaction ?





19. Which of the following reaction, does not give chloro benzene as a product ?



x = moles of KOH consumed is :

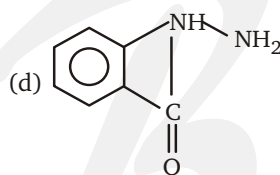
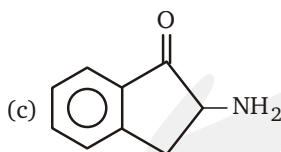
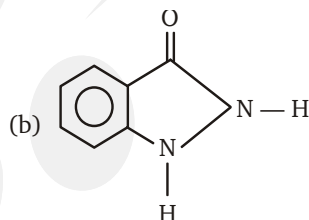
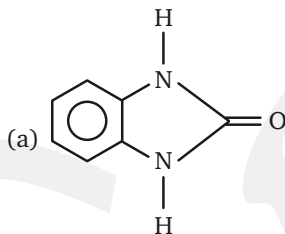
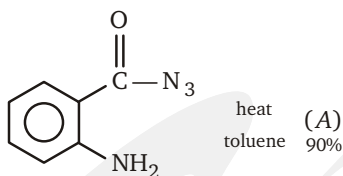
(a) 1

(b) 2

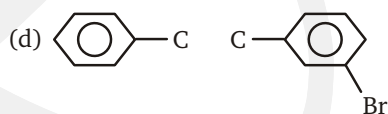
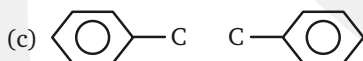
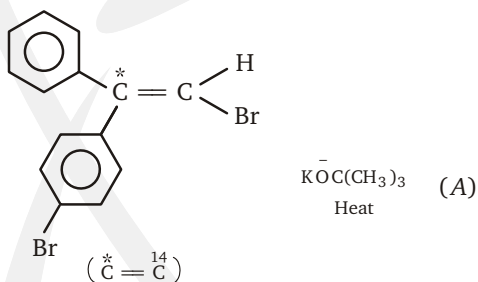
(c) 3

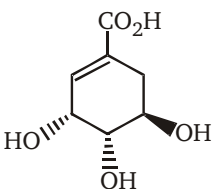
(d) 4

22. Heating the acyl azide in dry toluene under reflux for 3-hours give a 90% yield for a heterocyclic product. Identify the product (A).

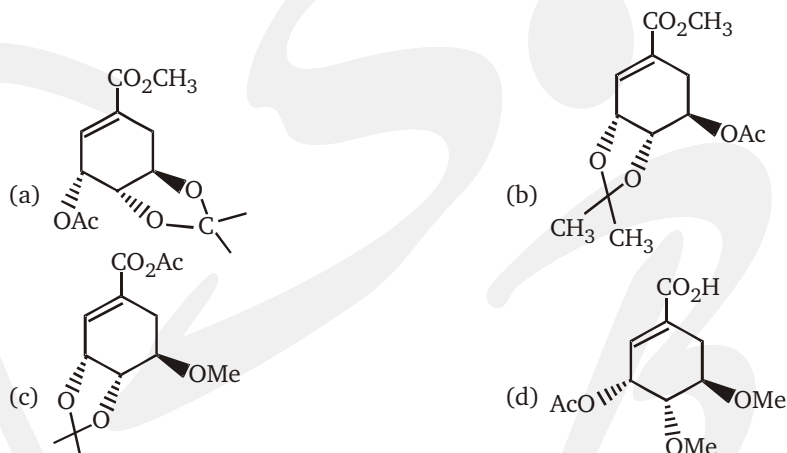


23.

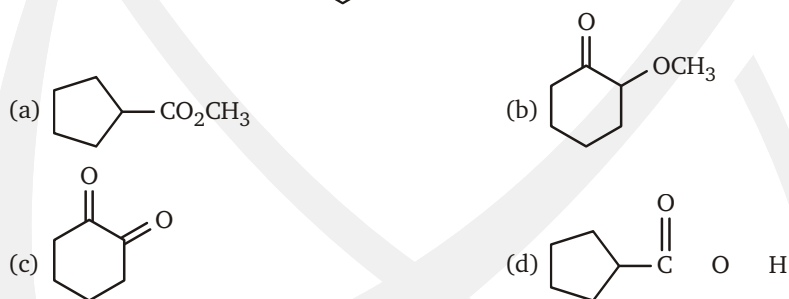
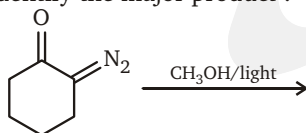


24.  (1) CH_2N_2 (A) (de-colourises Br_2 water)
 (2) acetone/ H^+
 (3) Ac_2O , Acetic anhydride

Product (A) of the above reaction is :

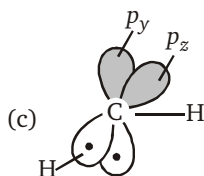


25. A rather interesting example of the Wolff rearrangement with 2-diazocyclohexanone in methanol is given below. Identify the major product :



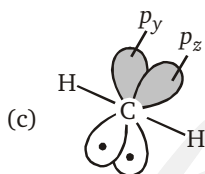
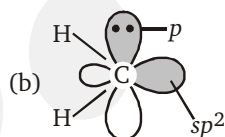
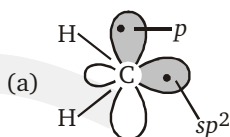
26. The orbital picture of a singlet carbene ($:\text{CH}_2$) can be drawn as :





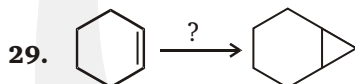
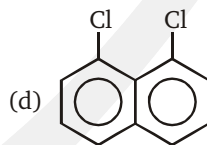
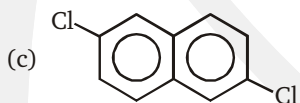
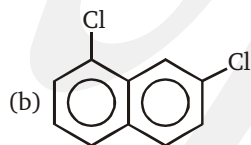
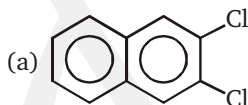
(d) none of these

27. The orbital picture of a triplet carbene can be drawn as :



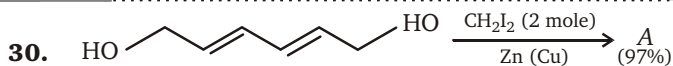
(d) none of these

28. (A) (B) ; Product (B) is :



Select the suitable reagent for above conversion.

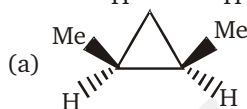
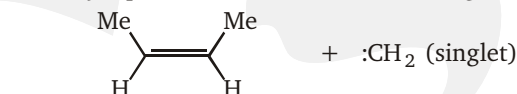
(a) CH_2N_2 /(b) CBr_4 / RLi (c) H_2C CH_2 (d) $t\text{-BuOK}$



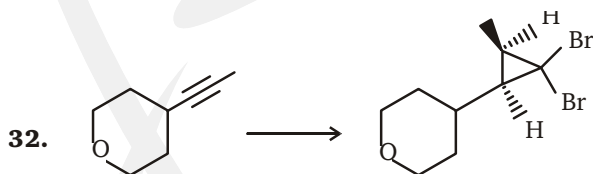
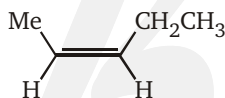
Product (A) will be :



31. The major product formed in the following reaction is



(c) 50 : 50 mixture of above two compounds



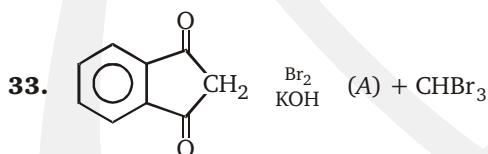
To carry out above conversion reagent used in decreasing order.

(a) Na/liq. NH_3 , $\text{CHBr}_3/\text{NaOH}$ ()

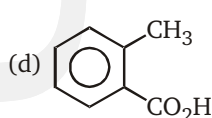
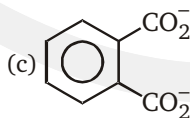
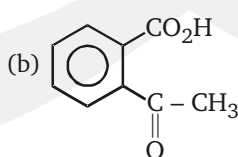
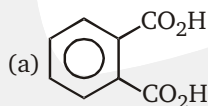
(b) $\text{H}_2/\text{Pd} - \text{CaCO}_3$, $\text{CHBr}_3/\text{NaOH}$ ()

(c) Na/liq. NH_3 , $\text{CHCl}_3/\text{NaOH}$

(d) $\text{H}_2/\text{Pd} - \text{CaCO}_3$, $\text{CHCl}_3/\text{NaOH}$



Product (A) of the reaction is :



O=[N+]([O-])c1ccc(C#N)cc1

(2) HO, Br₂

(a)  [O-]C(=O)c1ccc(N)cc1

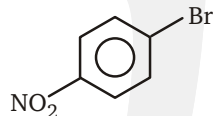
(b)



(c)

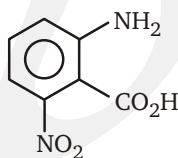


(d)

O=C1NC(=O)c2ccccc2C1[N+](=O)[O-]
$$\text{NaOCl}$$
$$\text{H}_2\text{O}$$

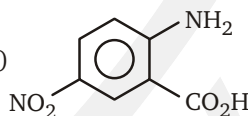
(X) ; Product X will be :
(80%)

(a) 



(c) 

(d)

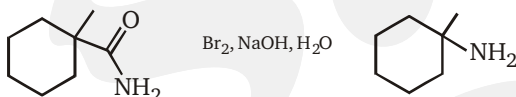
 $\text{CH}_2\text{N}_2/$

(d) 3-Methyl pentane

37. $\text{CH}_3-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\text{NH}_2 \xrightarrow{\text{KOH}} (A)$
 $\xrightarrow{\text{LiAlH}_4} (B)$

(d) Positional isomers

38. If we use pyrene (CCl_4) in the Riemer-Tiemann reaction in place of chloroform, the product formed is :
- (a) Salicylaldehyde (b) Phenolphthalein (c) Salicylic acid (d) Cyclohexanol
39. When ethyl amine is heated with chloroform and alcoholic KOH, a compound with offensive smell is obtained. This compound is :
- (a) A secondary amine (b) An isocyanide
(c) A cyanide (d) An acid
40. Which of the following species would not be involved in the Hoffmann rearrangement shown below ?



- (a)
- (b)
- (c)
- (d) All of the above are involved in the reaction.
41. In which of the following reactions migration of alkyl group from carbon to oxygen is observed ?
- (a) Pinacol-pinacolone rearrangement
(b) Bayer-villiger oxidation.
(c) Preparation of phenol from cumene hydroperoxide.
(d) Both (b) & (c)

ANSWERS — LEVEL 1

1.	(b)	2.	(b)	3.	(c)	4.	(b)	5.	(a)	6.	(d)	7.	(c)	8.	(b)
9.	(c)	10.	(b)	11.	(a)	12.	(a)	13.	(b)	14.	(c)	15.	(a)	16.	(b)
17.	(d)	18.	(a)	19.	(d)	20.	(a)	21.	(c)	22.	(a)	23.	(b)	24.	(b)
25.	(a)	26.	(a)	27.	(c)	28.	(c)	29.	(a)	30.	(b)	31.	(a)	32.	(b)
33.	(c)	34.	(c)	35.	(b)	36.	(d)	37.	(c)	38.	(c)	39.	(b)	40.	(d)
41.	(d)														

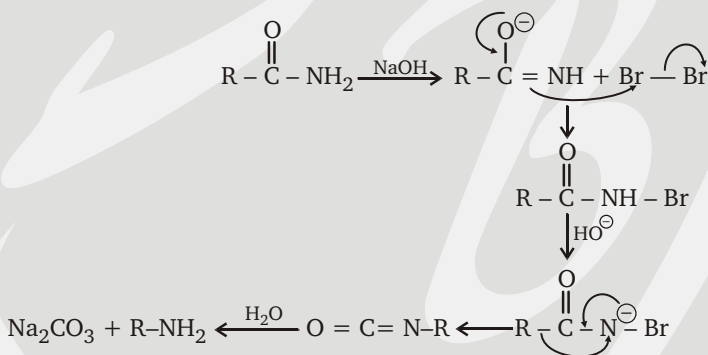
Level - 2

1. Comprehension

Hoffmann bromamide reaction involves conversion of a carboxylic acid amide into an amine with a loss of a carbon atom on treatment with aqueous sodium hypobromite. Thus Hoffmann result in shortening of a carbon chain.

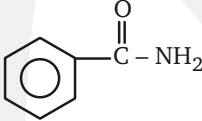


Mechanism of the reaction is :



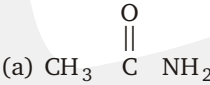
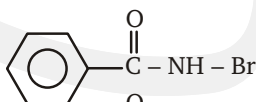
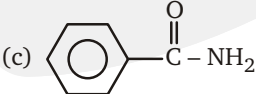
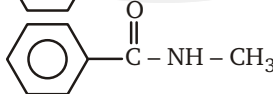
A. Number of moles of NaOH consumed in above reaction.

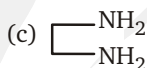
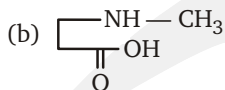
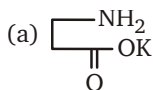
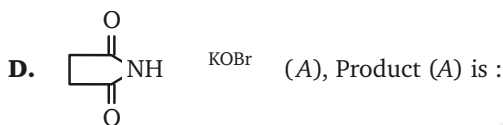
- (a) 1 (b) 2 (c) 3 (d) 4

B.  $\xrightarrow[\text{KOH}]{\text{Br}_2}$ (A) ; Product (A) :

- (a) Ph - NH₂ (b) Ph - CH₂ - NH₂ (c) Ph - NH - CH₃ (d) Ph - N $\begin{array}{l} \nearrow \text{CH}_3 \\ \searrow \text{CH}_3 \end{array}$

C. Which of the following will not give Hoffmann bromamide reaction.

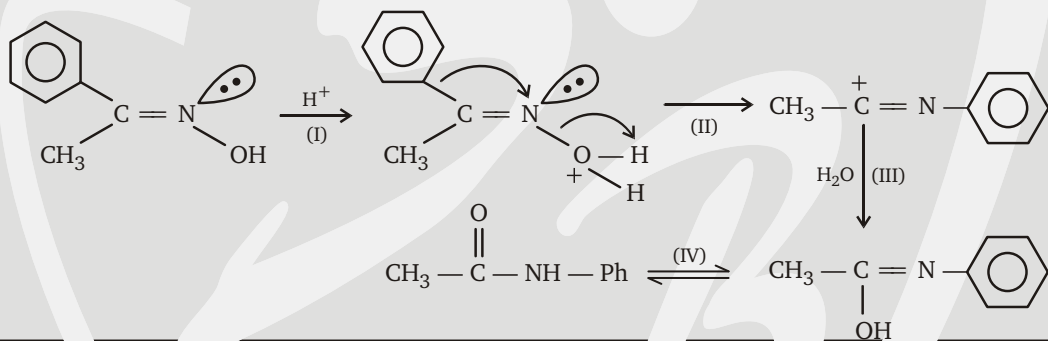
- (a)  (b) 
 (c)  (d) 



(d) None of these

2. Comprehension

Given is mechanism of Beckmann rearrangement.



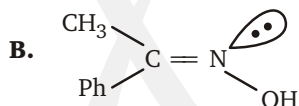
A. Rate determining step in Beckmann rearrangement :

(a) I

(b) II

(c) III

(d) IV



On treatment H_2SO_4 followed by hydrolysis in acidic medium above compound gives.

(a) $\text{CH}_3\text{CO}_2\text{H}$, PhNH_2

(b) CH_3NH_2 , PhCO_2H

(c) PhCH_2NH_2 , PhCO_2H

(d) PhCO_2H , $\text{CH}_3\text{CO}_2\text{H}$

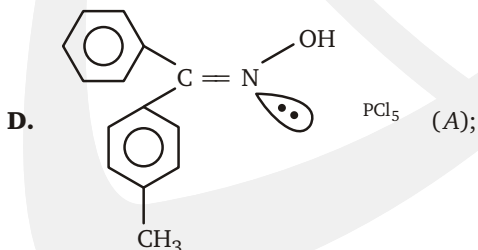
C. Which of the following reagent cannot used in Beckmann rearrangement ?

(a) TsOH

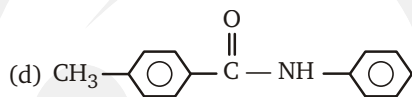
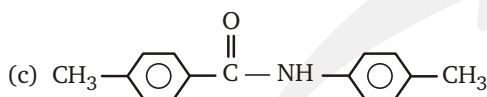
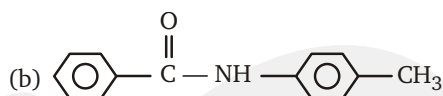
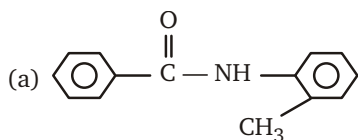
(b) $\text{R-SO}_2\text{Cl}$

(c) BF_3

(d) Ph-Li



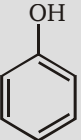
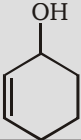
Product (A) of the above reaction is :



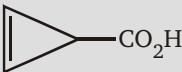
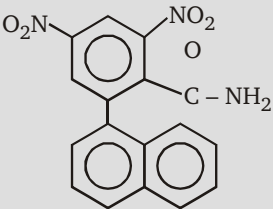
3. Match the column I and II.

Column (I)				Column (II)	
(a)		aq. KOH	(A) H (B) CHCl_3 (C) KOH	(p)	D.B.E. = even for product (Double bond equivalent)
(b)		H	(A) CHCl_3 (B) KOH	(q)	D.B.E. = odd for product
(c)		CHCl_3 KOH	(A) CHCl_3 (B) KOH	(r)	Ring expansion takes place
(d)		H	(A) CHFCIBr (B)	(s)	Carbene will formed

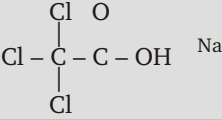
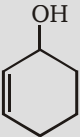
4. Match the column I and II.

Column (I)		Column (II)	
(a)	 CHCl_3 KOH	(p)	Reimer Tiemann reaction
(b)	 CHCl_3 KOH	(q)	Reimer Tiemann expansion (or) Abnormal RNT reaction
(c)	 CCl_3COONa	(r)	Simman-smith reaction.
(d)	 CH_2I_2 Zn	(s)	Increase in carbon takes place

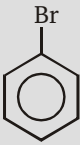
5. Match the column I and II.

Column (I)		Column (II)	
(a)	 SOCl_2 NH_3 KOH HNO_2	(p)	Aromatic compound will formed
(b)		(q)	Migration take place from carbon to electron deficient nitrogen
(c)	CHCl_2 $t\text{BuO}^- \text{K}^+$ (A) $\text{C} \equiv \text{C}$ (B) AlCl_3 (C)	(r)	Carbene will formed in this reaction
(d)	 KOH	(s)	N_2 will evolve.

6. Match the column I and II:

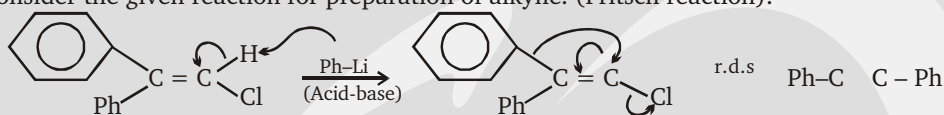
Column (I)		Column (II)	
Reaction		Intermediate	
(a)	$\text{CHCl}_3 \text{ KOH}$	(p)	Carbocation
(b)	 Ph Li	(q)	Carbanion
(c)	 Na	(r)	Free radical
(d)	 H	(s)	Carbene

7. Matrix :

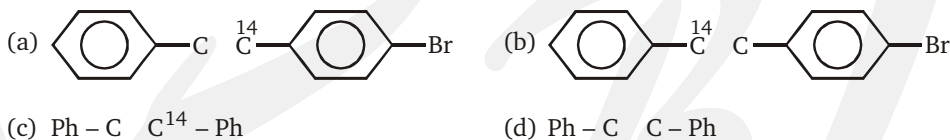
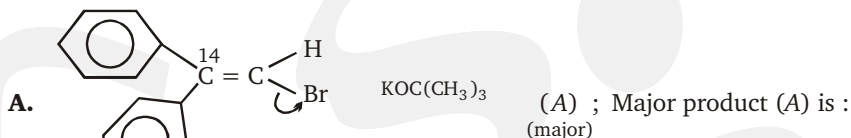
Column (I)		Column (II)	
Reaction		Product	
(a)	 CHCl_3 KOH,	(p)	
(b)	 CHClBr KOH,	(q)	
(c)	 CHCl_2Br KOH,	(r)	
(d)	 CHBr_2Cl KOH,	(s)	

8. Comprehension

1. Consider the given reaction for preparation of alkyne. (Fritsch reaction).

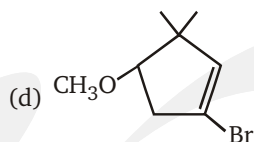
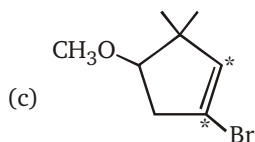


Anti group will migrate because of less steric hindrance.



- C.** Rate of reaction when the halide ion:



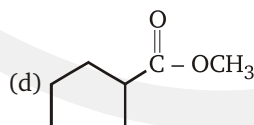
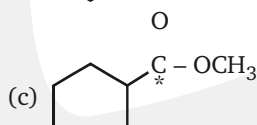
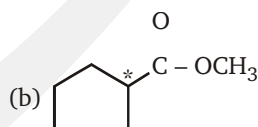
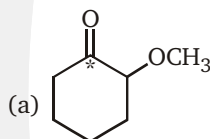
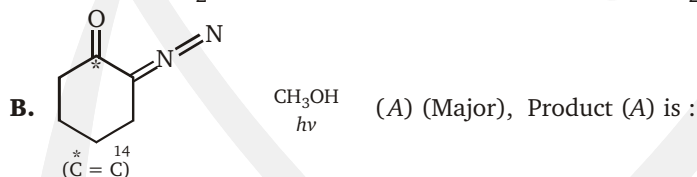
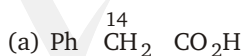
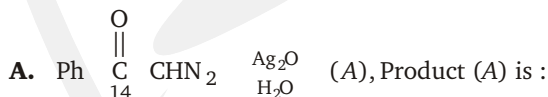
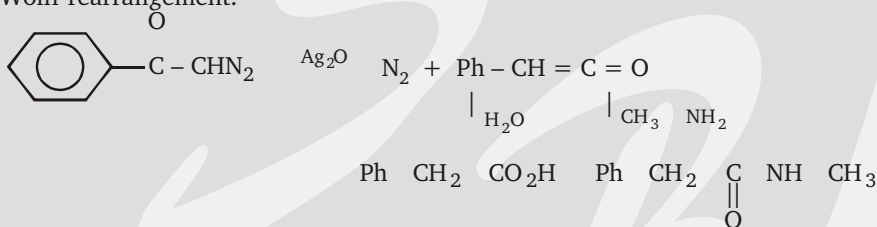


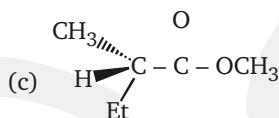
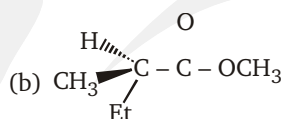
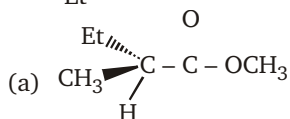
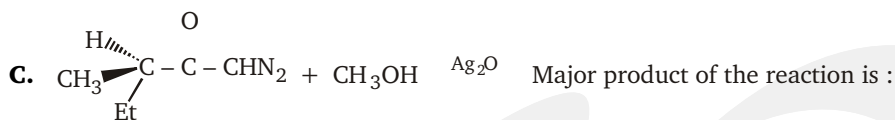
9. Comprehension

Wolff rearrangement

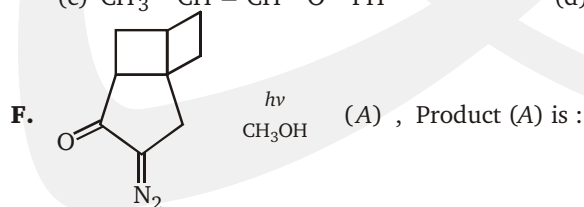
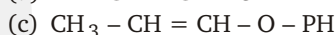
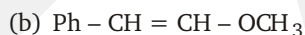
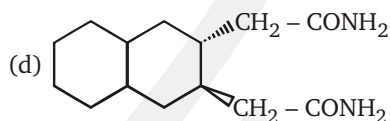
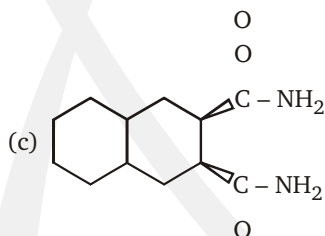
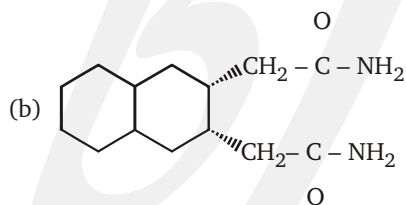
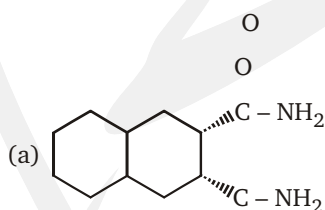
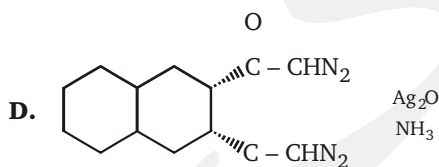
When α -Diazoketones are photo-irradiated or heated at high temperature or reacted with silver oxide or silver salts at room temperature, they lose nitrogen and rearrange to form ketene.

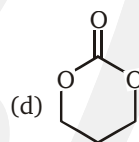
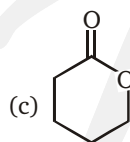
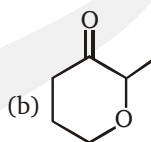
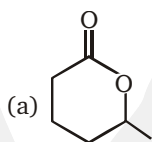
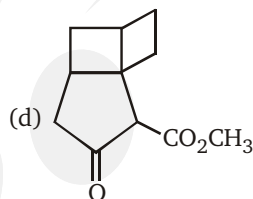
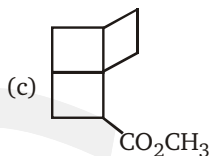
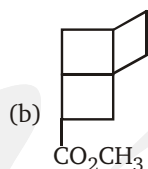
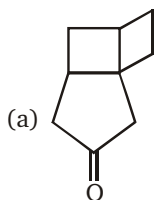
The ketenes react rapidly with water, alcohol and amines. Therefore, the reactions called Wolff-rearrangement.





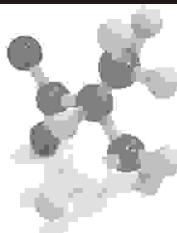
(d) None of these



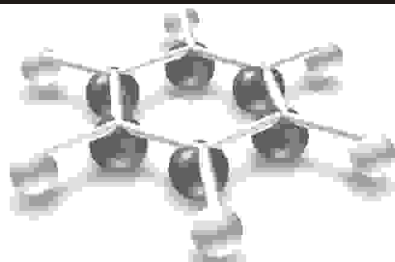


ANSWERS — LEVEL 2

1. A - d ; B - a ; C - d ; D - a,
2. A - b ; B - b ; C - d ; D - b
3. a - p, r, s ; b - q, r, s ; c - q, r, s ; d - p, r, s
4. a - q, s ; b - p, s ; c - s ; d - r, s
5. a - p, q, s ; b - p, r, s ; c - p, r ; d - p, q
6. a - q, s ; b - q, s ; c - q, s ; d - p
7. a - q ; b - p ; c - q ; d - q
8. A - a ; B - c ; C - b ; D - b
9. A - b ; B - c ; C - d ; D - b ; E - b ; F - b ; G - c



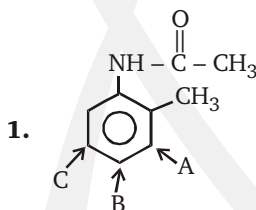
12



AROMATIC COMPOUNDS

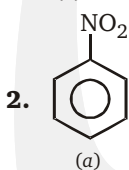


Level - 1

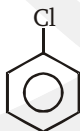


Identify the position where electrophilic aromatic substitution (EAS) is most favourable.

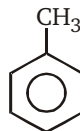
- (a) A
(b) B
(c) C
(d) A and C



(a)



(b)



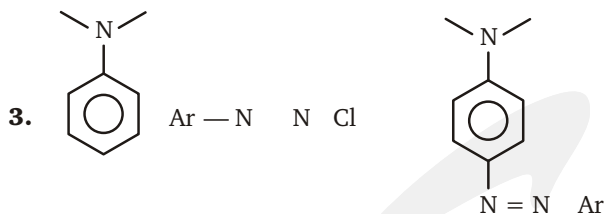
(c)



(d)

Correct order of rate of EAS (electrophilic aromatic substitution) is :

- (a) $c > b > a > d$
(b) $c > d > a > b$
(c) $a > b > c > d$
(d) $c > d > b > a$



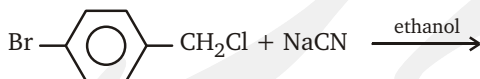
Above (C—N) coupling reaction take place at :

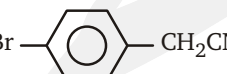
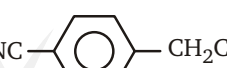
- (a) low pH (b) Intermediate pH
(c) high pH (d) any pH

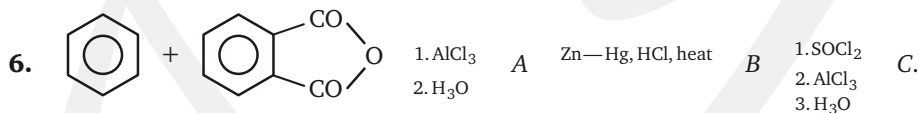
4. Which of the following has the lowest heat of combustion ?



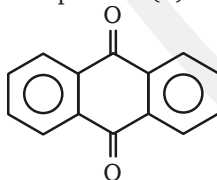
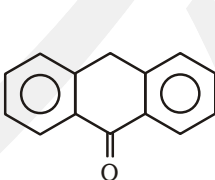
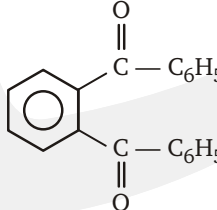
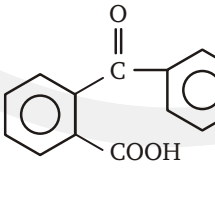
5. The product obtained from the reaction is :



- (a)  (b) 
(c)  (d) 



The end product (C) is :

- (a)  (b) 
(c)  (d) 

7. How many benzylic hydrogens are present in the hydrocarbon shown below ?



(a) 3

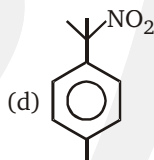
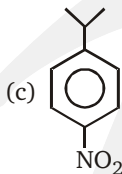
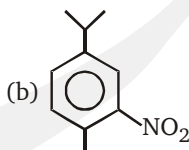
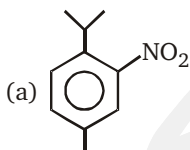
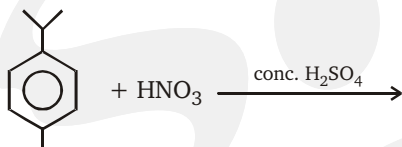
(b) 4

(c) 5

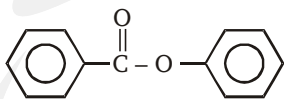
(d) 6

(e) 8

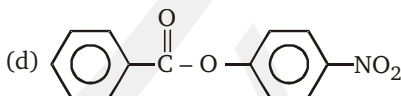
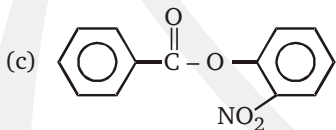
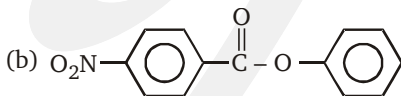
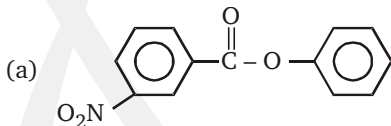
8. The major product formed in the reaction is :



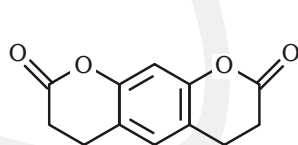
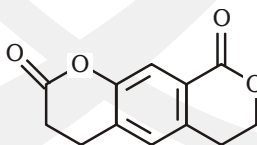
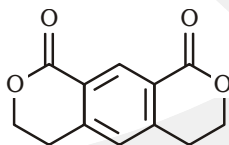
9. The major product formed in the reaction is :



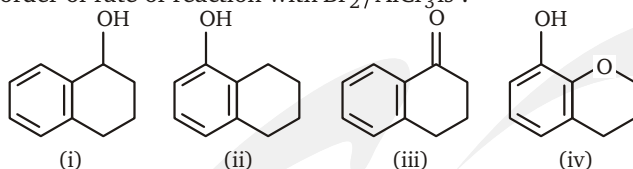
Conc. HNO_3 / conc. H_2SO_4
(mononitration)



10. Increasing order of rate of reaction with $\text{HNO}_3/\text{H}_2\text{SO}_4$ is :

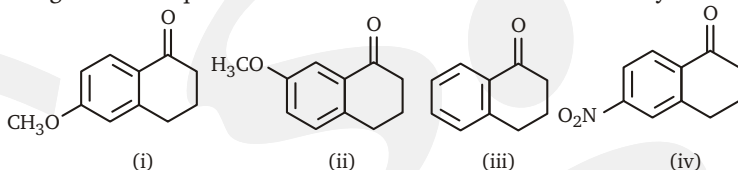
(a) $\text{iii} < \text{ii} < \text{i}$ (b) $\text{ii} < \text{iii} < \text{i}$ (c) $\text{i} < \text{iii} < \text{ii}$ (d) $\text{i} < \text{ii} < \text{iii}$

11. Increasing order of rate of reaction with $\text{Br}_2/\text{AlCl}_3$ is :



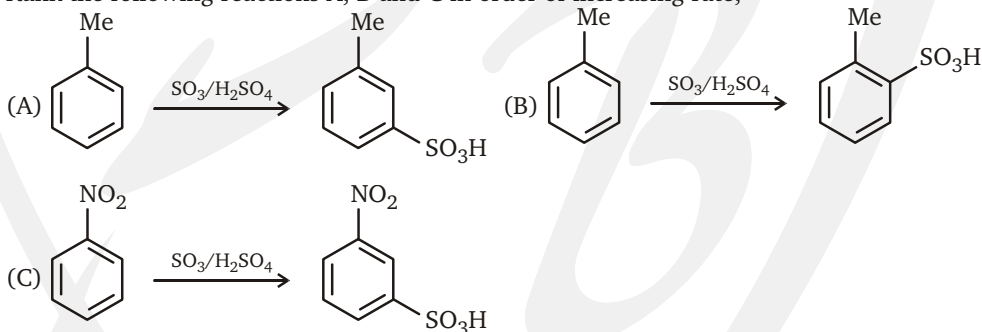
- (a) $\text{iii} < \text{i} < \text{ii} < \text{iv}$ (b) $\text{iv} < \text{ii} < \text{i} < \text{iii}$ (c) $\text{ii} < \text{iv} < \text{iii} < \text{i}$ (d) $\text{iv} < \text{ii} < \text{iii} < \text{i}$

12. Increasing order of equilibrium constant for the formation of a hydrate is :



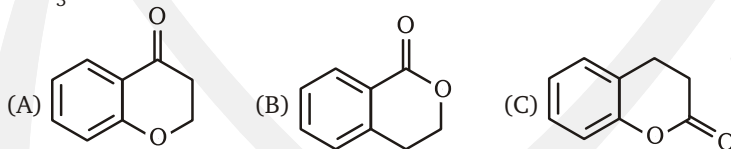
- (a) $\text{i} < \text{ii} < \text{iii} < \text{iv}$ (b) $\text{iv} < \text{ii} < \text{i} < \text{iii}$ (c) $\text{ii} < \text{iv} < \text{iii} < \text{i}$ (d) $\text{iv} < \text{ii} < \text{iii} < \text{i}$

13. Rank the following reactions A, B and C in order of increasing rate,

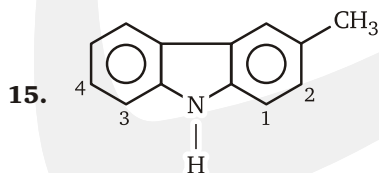


- (a) $\text{B} > \text{A} > \text{C}$ (b) $\text{B} > \text{C} > \text{A}$ (c) $\text{A} > \text{B} > \text{C}$ (d) $\text{A} > \text{C} > \text{B}$

14. Rank in order of increasing rate of reaction towards EAS with bromine in the presence of FeBr_3 .

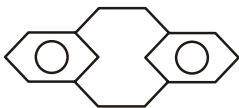


- (a) $\text{B} < \text{A} < \text{C}$ (b) $\text{B} < \text{C} < \text{A}$
(c) $\text{A} < \text{B} < \text{C}$ (d) $\text{A} < \text{C} < \text{B}$

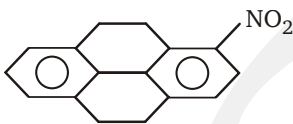
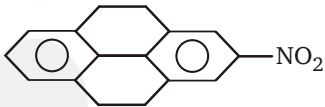
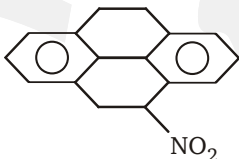


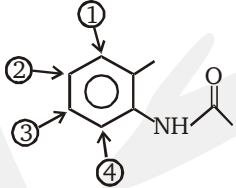
Identify the position where E.A.S. can take place.

- (a) 1 (b) 2 (c) 3 (d) 4

16.  H (A) HNO_3 (B).

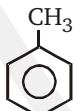
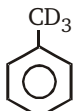
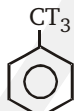
Product (B) in the above reactions is:

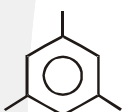
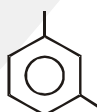
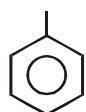
- (a)  (b) 
 (c)  (d) 

17.  Sulphonation is most favourable at the carbon number... .

- (a) 1 (b) 2 (c) 3 (d) 4

18. Arrange the following in decreasing order of reactivity towards EAS (electrophilic aromatic substitution)

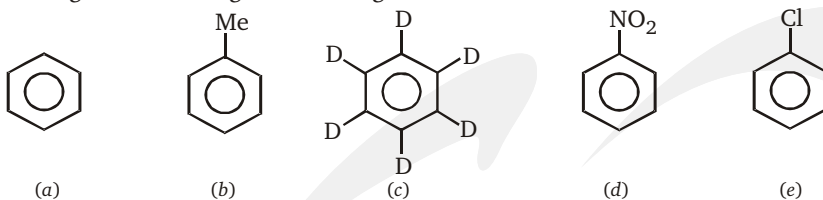
- (a)  (b)  (c) 
 (a) $a > b > c$ (b) $c > b > a$ (c) $a > c > b$ (d) $c > a > b$

19. (a)  (b)  (c)  (d) 

Decreasing order of rate of electrophilic aromatic substitution is :

- (a) $a > b > c > d$ (b) $a > c > b > d$
 (c) $b > a > c > d$ (d) $b > c > a > d$

20. Arrange the following in increasing order of rate of Nitration:



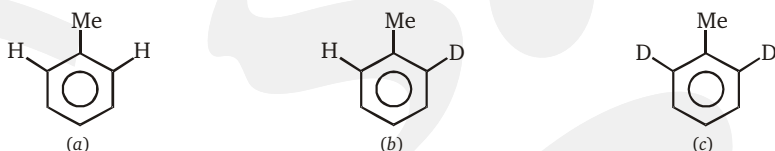
(a) $b < c < a < d < e$

(b) $d < e < a = c < b$

(c) $d < a = c < e < b$

(d) $a < c < b < e < d$

21.



The rate of nitration will be:

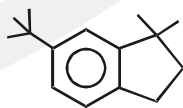
(a) $a > b > c$

(b) $a > c > b$

(c) $a = b = c$

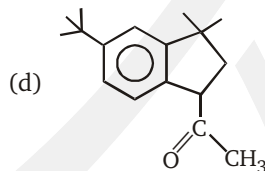
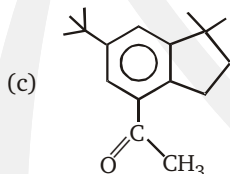
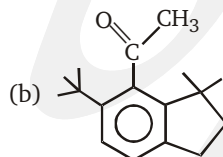
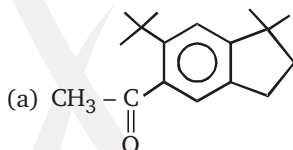
(d) $c > a > b$

22. The major product of the reaction is

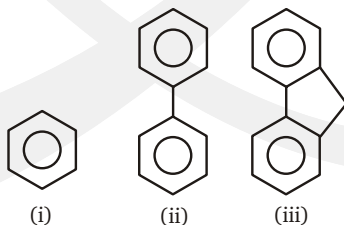


1. $\text{CH}_3\text{COCl}/\text{AlCl}_3$
2. H_2O

Product.



23. Arrange in their decreasing order of rate of electrophilic aromatic substitution :



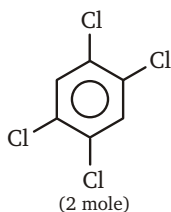
(a) $i > ii > iii$

(b) $iii > ii > i$

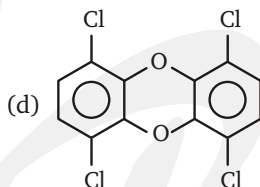
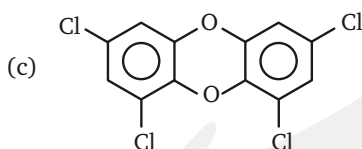
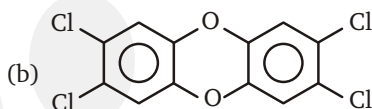
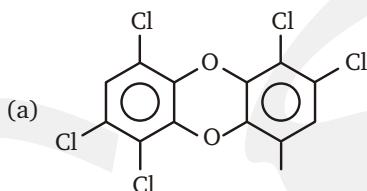
(c) $iii > i > ii$

(d) $i > iii > ii$

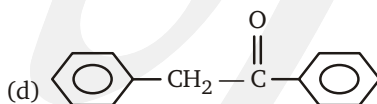
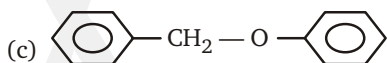
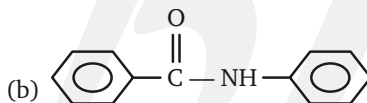
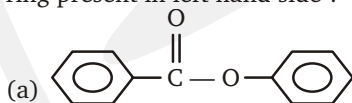
24.

HO
(4 mole)

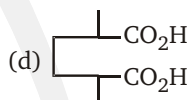
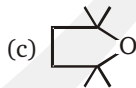
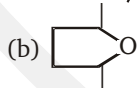
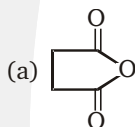
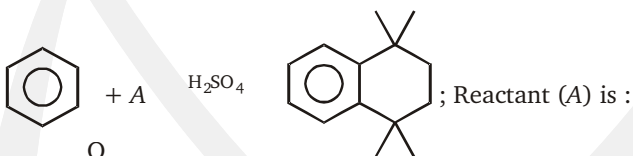
(A) ; Product (A) of the given reaction is :



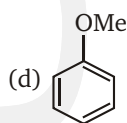
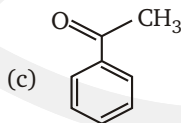
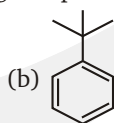
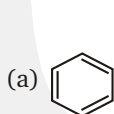
25. In which of the following compound electrophilic aromatic substitution take place in phenyl ring present in left hand side ?

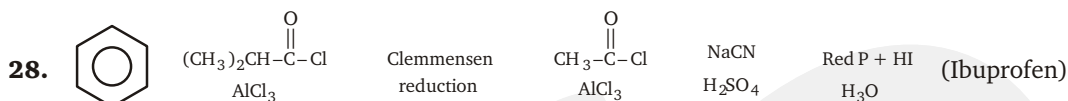


26.

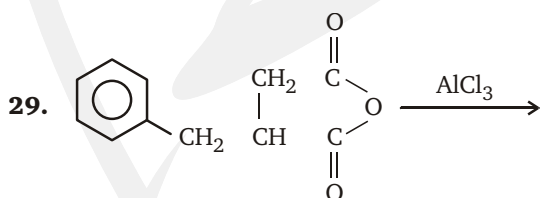
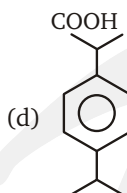
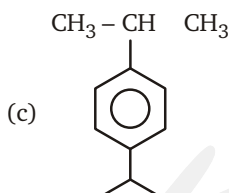
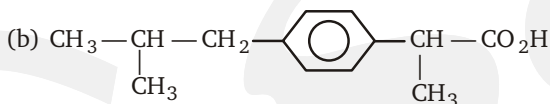
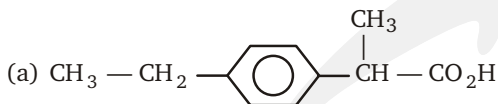


27. Which of the following compounds is the slowest to react with nitrosonium ion (NO^+) ?

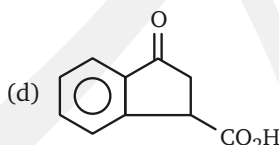
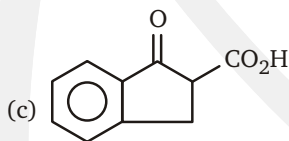
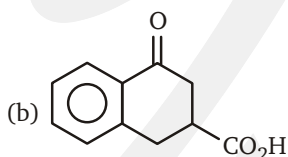
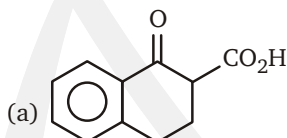




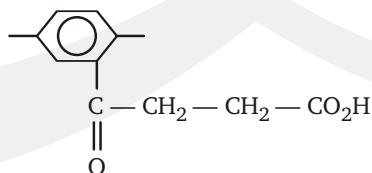
Ibuprofen is :

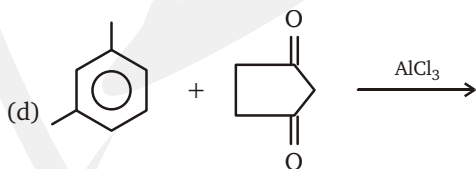
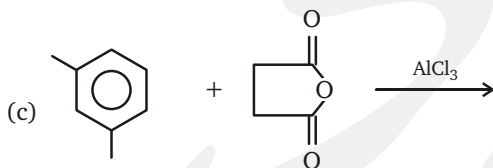
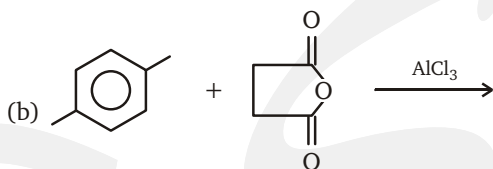
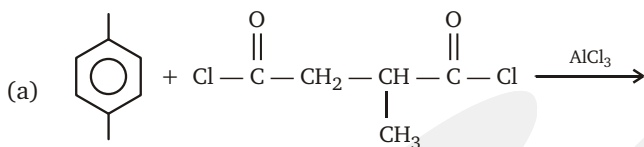


What is the major product of above Friedel-Craft reaction ?

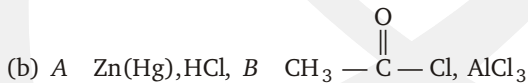
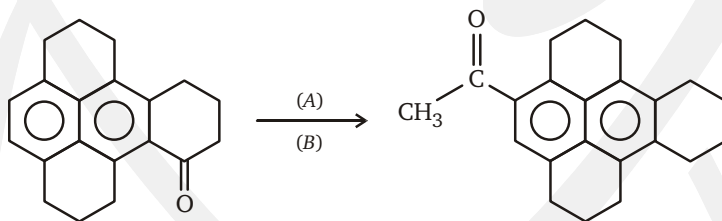


30. What combination of acid chloride or anhydride and arene would you choose to prepare given compound ?

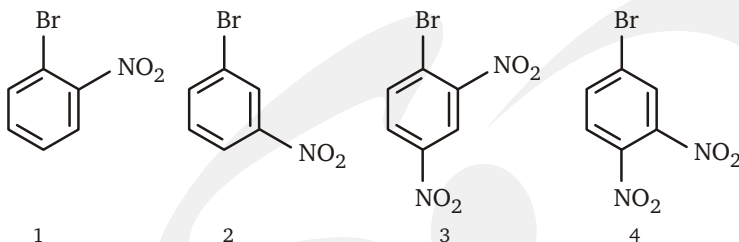




31. In the given conversion best yield will obtained with :

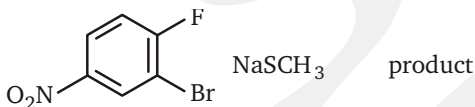


32. Rank the following in order of decreasing rate of reaction with alkoxide ion ($\text{CH}_3\text{CH}_2\text{O}^-$) in a nucleophilic aromatic substitution reaction :



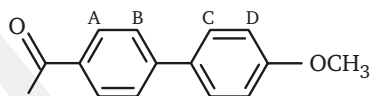
- (a) $3 > 4 > 1 > 2$ (b) $3 > 4 > 2 > 1$ (c) $2 > 1 > 4 > 3$ (d) $4 > 3 > 2 > 1$

33. Identify the principal organic product of the following reaction.



- (a)
- (b)
- (c)
- (d)

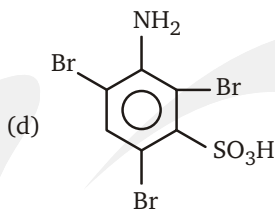
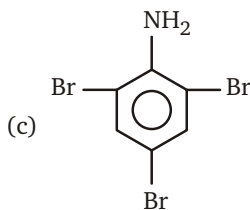
34. Which position will be attacked most rapidly by the nitronium ion ($-\text{NO}_2^+$) when the compound undergoes nitration with $\text{HNO}_3/\text{H}_2\text{SO}_4$:



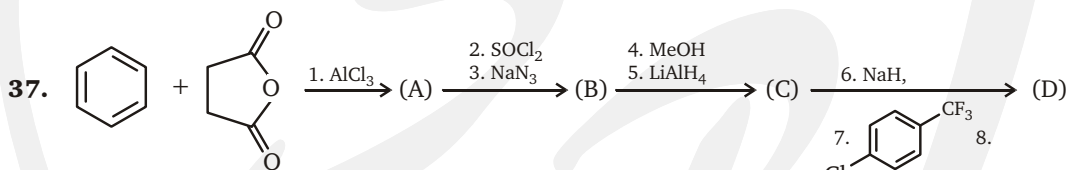
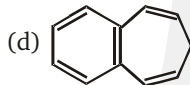
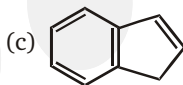
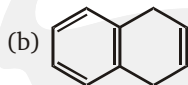
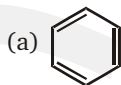
- (a) A (b) B (c) C (d) D

35. $\xrightarrow{\text{Conc. H}_2\text{SO}_4}$ (X) $\xrightarrow[\text{excess}]{\text{Br}_2/\text{H}_2\text{O}}$ (Y): Product (Y) of this reaction is :

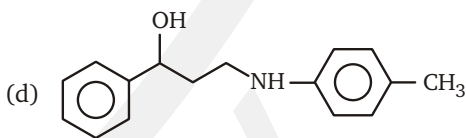
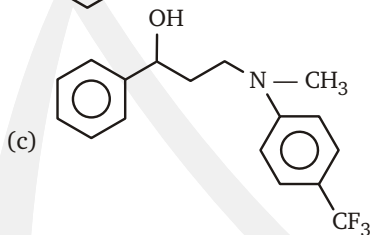
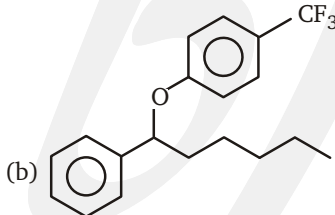
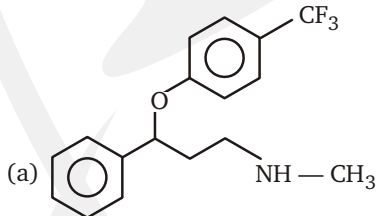
- (a)
- (b)



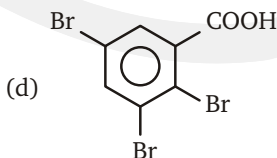
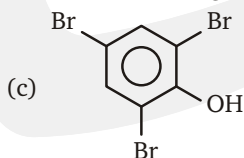
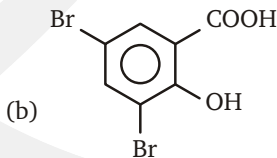
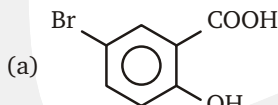
36. All the hydrocarbons shown are very weak acids. One, however, is far more acidic than the others. Which one is the strongest acid ?



Product (D) in above sequence is :



38. The action of bromine water (excess) on salicylic acid results in the formation of :

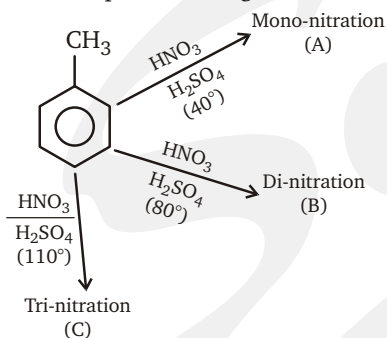


39. What is the correct order of *o/p* ratio when E attacks the following system ?



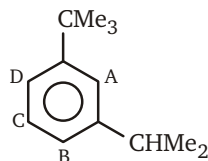
- (a) $A < B < C < D$ (b) $A = B = C = D$ (c) $D < C < B < A$ (d) $D < B < A < C$

40. How many products are capable of being formed from toluene in each of following reaction ?



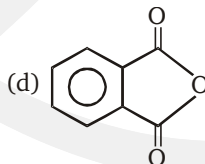
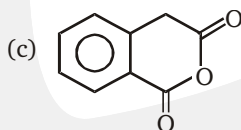
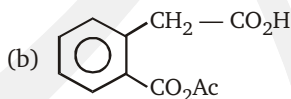
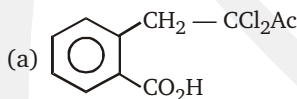
- (a) A 3, B 6, C 8 (b) A 3, B 6, C 6
(c) A 3, B 6, C 10 (d) A 3, B 4, C 6

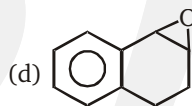
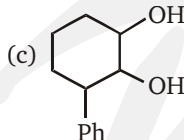
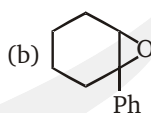
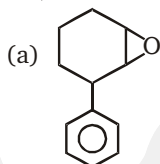
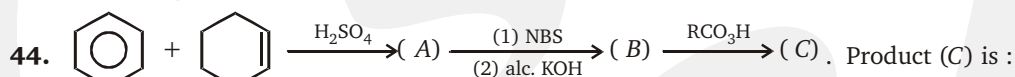
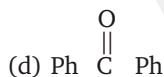
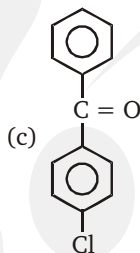
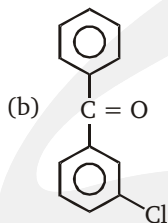
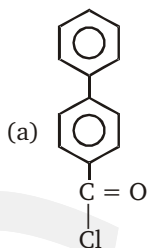
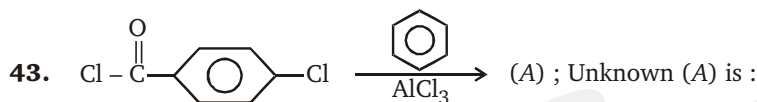
41. Nitration takes place at the which position of the given compound ?



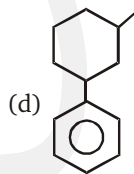
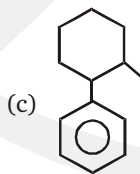
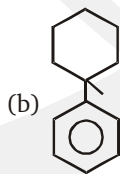
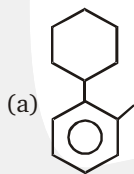
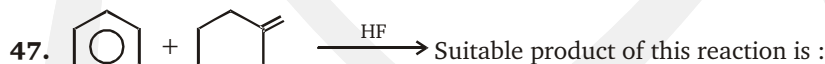
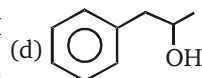
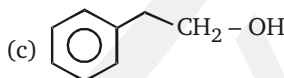
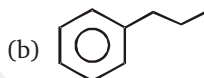
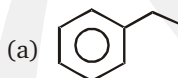
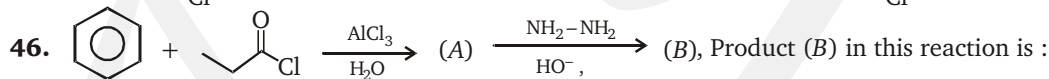
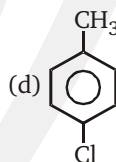
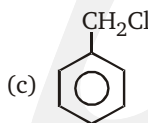
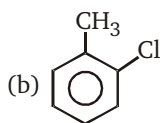
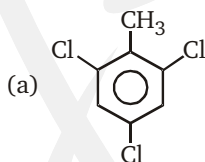
- (a) A (b) B
(c) C (d) D

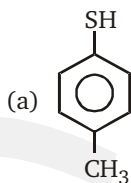
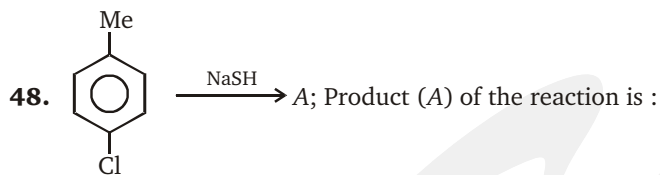
42. , Identify the product.



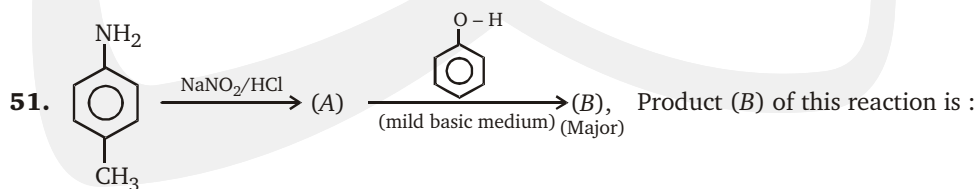
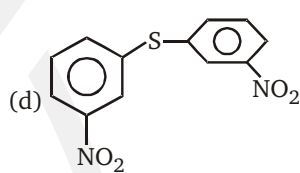
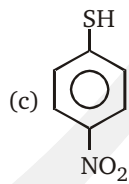
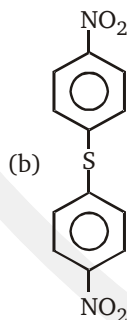
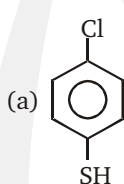
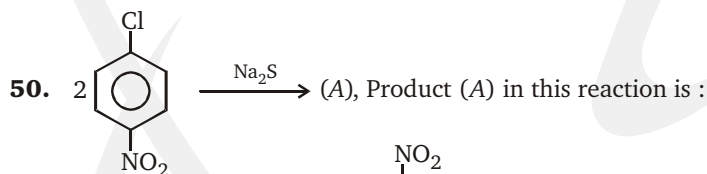
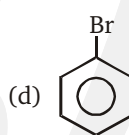
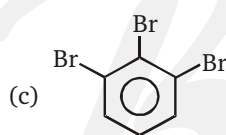
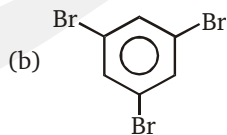
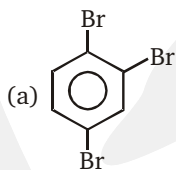
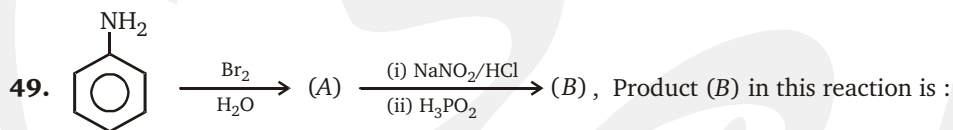
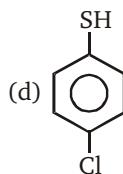
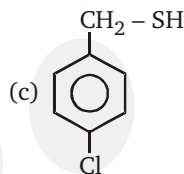


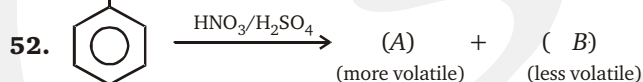
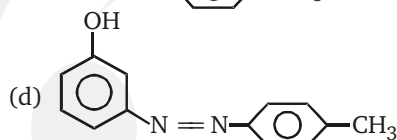
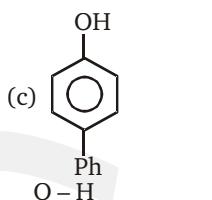
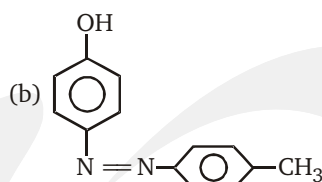
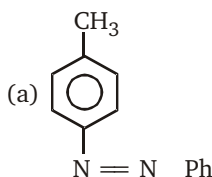
45. The reaction of toluene with chlorine in the presence of light gives :



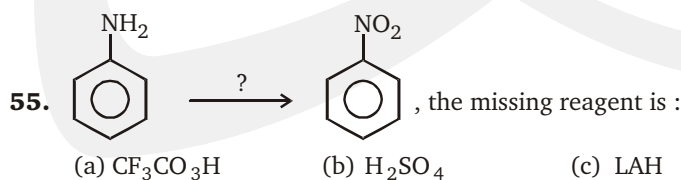
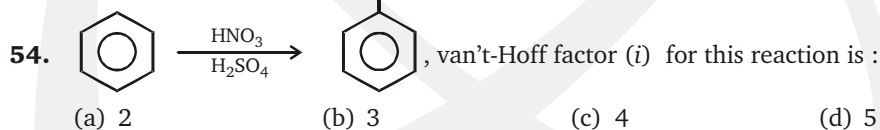
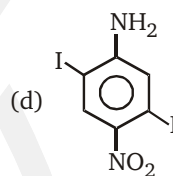
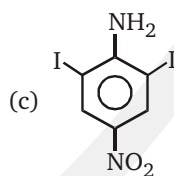
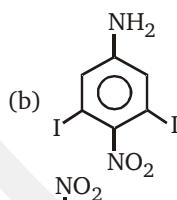
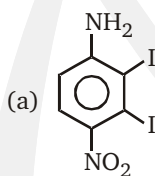
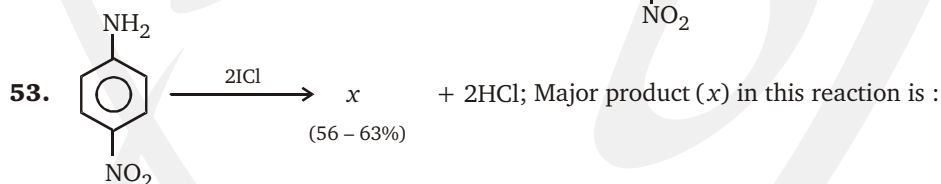
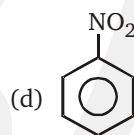
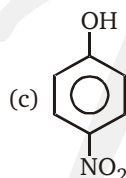
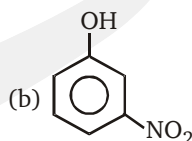
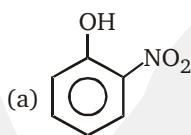


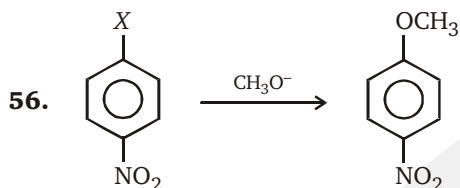
(b) no reaction





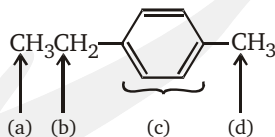
Product (A) of the above reaction is :





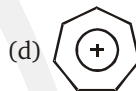
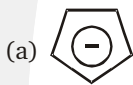
above reaction is an example of Nucleophilic aromatic substitution. Which of the following halide (X) is most readily replaced.

- (a) - F (b) - Cl (c) - Br (d) - I
57. When comparing the hydrogenation of benzene with that of a hypothetical 1, 3, 5-cyclohexatriene, benzene _____ than the cyclohexatriene.
- (a) absorbs 152 kJ/mol more heat (b) gives off 152 kJ/mol more heat
(c) absorbs 152 kJ/mol less heat (d) gives off 152 kJ/mol less heat
58. Which of the following hydrogens is most easily abstracted on reaction with bromine free radicals, $\text{Br}\cdot$?

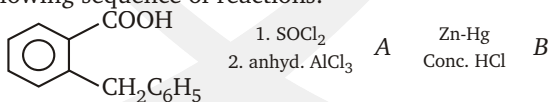


- (a) a (b) b (c) c (d) d
59. The electrophilic aromatic substitution proceeds through a :
- (a) free radical (b) sigma complex (c) benzyne (d) carbene
60. Which of the following substitution of benzene is ortho-para in electrophilic substitution and ortho-para in nucleophilic substitution ?
- (a) -NO₂ (b) -NO
(c) -SO₃H (d) -SO₂Me
61. The number of possible isomers of dichloronitrobenzene is :
- (a) 3 (b) 4 (c) 6 (d) 8

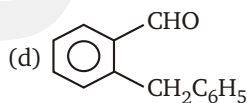
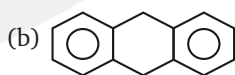
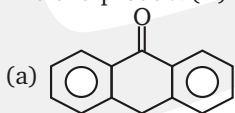
62. Which of the following is not an aromatic compound ?



63. Consider the following sequence of reactions.



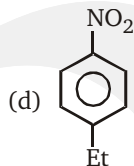
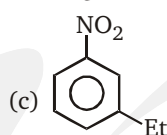
The end product (B) is:



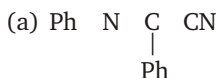
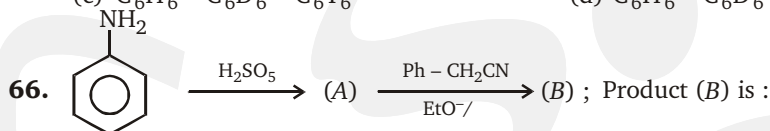
64. $\text{Ph} \text{ NO}_2 \text{ Et Cl} \xrightarrow{\text{AlCl}_3}$ (A), Product (A) of the given reaction is :

(a) $\text{Ph} - \text{NH} - \text{Et}$

(b) no-reaction



65. In nitration of benzene by mixed acid the rate of reaction will be :

(a) C_6H_6 C_6D_6 C_6T_6 (b) C_6H_6 C_6D_6 C_6T_6 (c) C_6H_6 C_6D_6 C_6T_6 (d) C_6H_6 C_6D_6 C_6T_6 

67. Which of the following ring compounds obeys Huckel's rule ?

(a) $\text{C}_4\text{H}_4^{1-}$ (b) $\text{C}_4\text{H}_4^{1+}$ (c) $\text{C}_4\text{H}_4^{2-}$ (d) C_4H_4

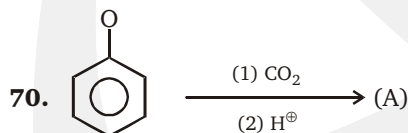
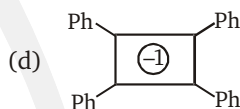
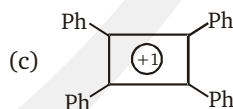
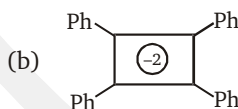
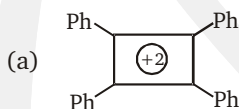
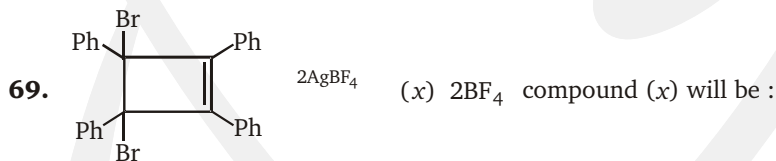
68. Nitration of which of the following reactant gives maximum % of meta product (using $\text{HNO}_3/\text{H}_2\text{SO}_4$) ?

(a) Toluene

(b) Aniline

(c) Benzene

(d) Isopropyl benzene



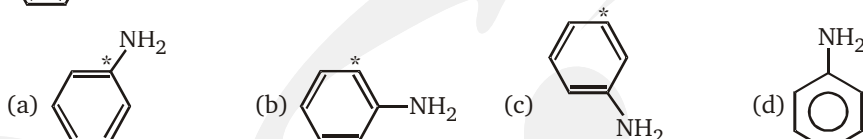
Which of the following is true statement about the reaction ?

(a) Ortho isomer is major if PhONa is used(b) Para isomer is major if PhOK is used

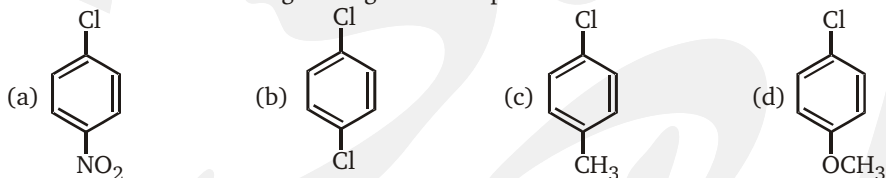
(c) Product formed is further used for preparation of drug aspirin

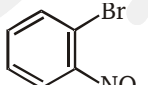
(d) All of these

71. Two benzyne intermediates are likely to be formed equally. Reaction with amide ion can occur in two different directions with each benzyne, giving three possible products. They are formed in a 1 : 2 : 1 ratio. Asterisk (*) refers to ^{14}C .



72. Which one of the following undergoes nucleophilic aromatic substitution at the fastest rate ?



73. For the reaction ; ?  ; the best combination of reactants is :



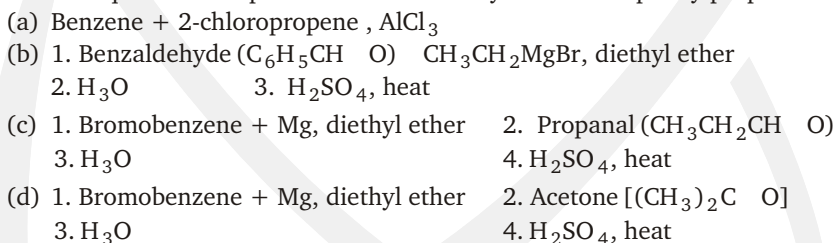
74. The action of AlCl_3 in Friedel Craft's reaction is:



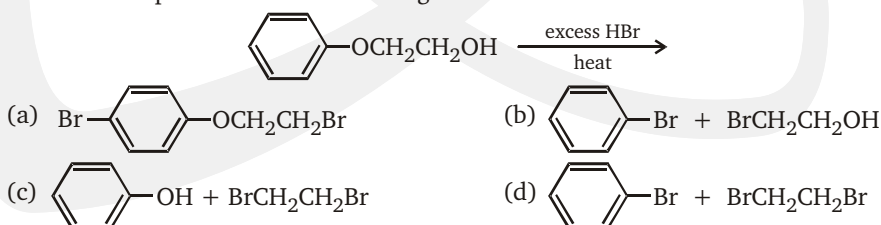
75. *n*-Butylbenzene on oxidation with hot alkaline KMnO_4 gives :



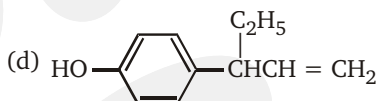
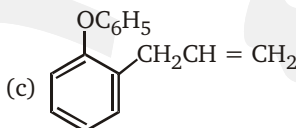
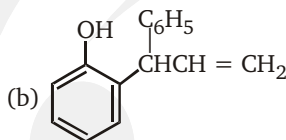
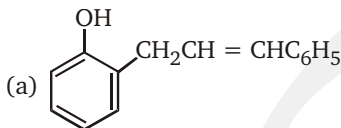
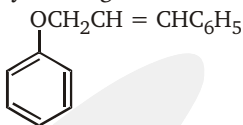
76. Which sequence of steps describes the best synthesis of 2-phenylpropene ?



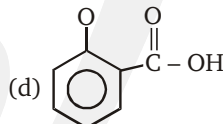
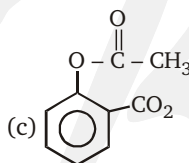
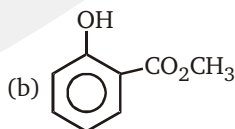
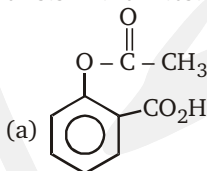
77. What are the products of the following reaction ?



78. What is the product obtained by heating the following allylic ether of phenol ?



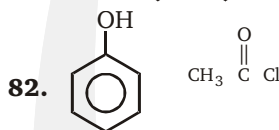
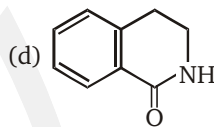
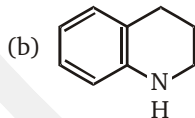
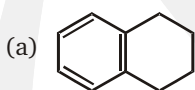
79. When you ingest aspirin, it passes through your stomach, which has an acidic pH, before traveling through the basic environment of your intestine. Provide the structure form as it exists in the intestine.



80. Which of the following sets of reagents, used in the order shown, would be enable for the preparation of *p*-chlorophenol from *p*-chloronitrobenzene ?

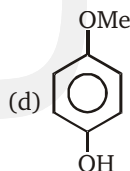
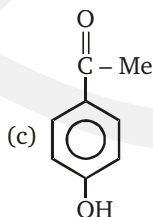
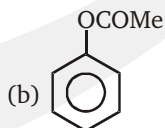
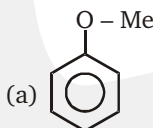
- (a) 1. Fe, HCl ; 2. NaOH ; 3. NaNO₂, H₂SO₄ ; 4. H₃PO₂
 (b) 1. Fe, HCl ; 2. NaOH ; 3. NaNO₂, H₂SO₄ ; 4. H₂O, heat
 (c) 1. Fe, HCl ; 2. NaOH ; 3. NaNO₂, H₂SO₄ ; 4. ethanol
 (d) 1. NaOH, heat ; 2. HCl

81. Which one of the following compounds undergoes bromination of its aromatic ring (electrophilic aromatic substitution) at the fastest rate ?

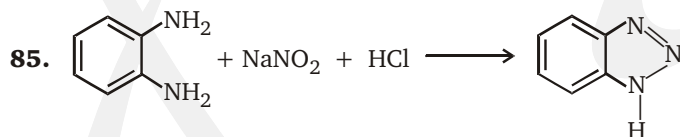
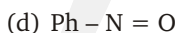
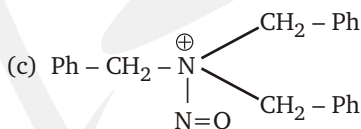
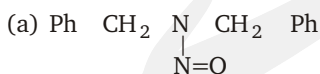
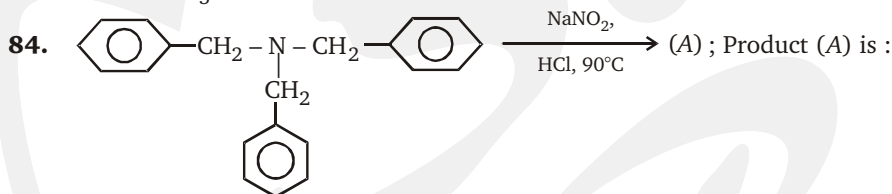
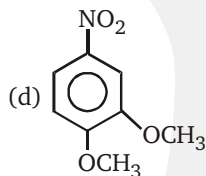
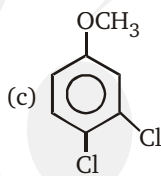
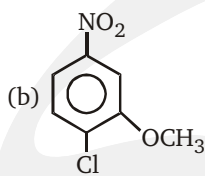
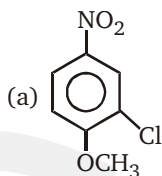
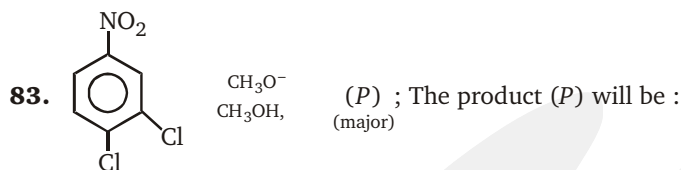


(P) Anhy. AlCl₃

(Q). Product (Q) in this reaction is :



Aromatic Compounds



This reaction is example of :

(a) Intermolecular C – N coupling

(b) Intramolecular C – N coupling

(c) Intermolecular N – N coupling

(d) Intramolecular N – N coupling

86. The total number of isomeric trimethylbenzene is :

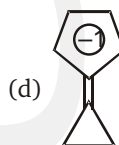
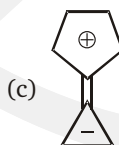
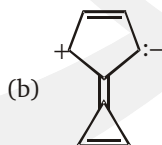
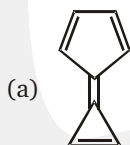
(a) 2

(b) 3

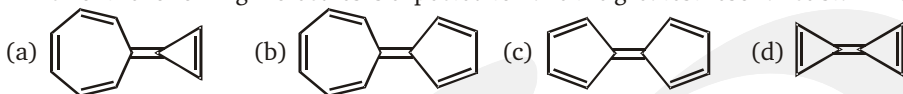
(c) 4

(d) 6

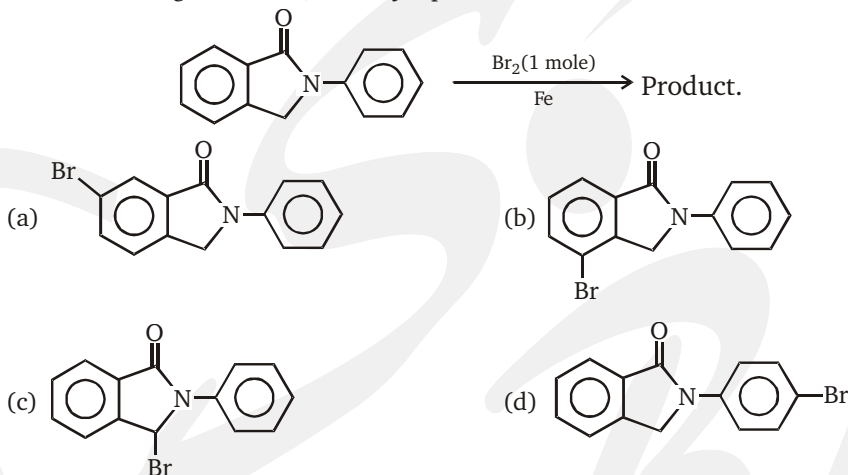
87. Caliene, C_7H_6 , is expected to be a fairly polar aromatic molecule. Which of the following resonance forms contributes to the greatest extent towards the real structure (resonance hybrid) of the molecule ?



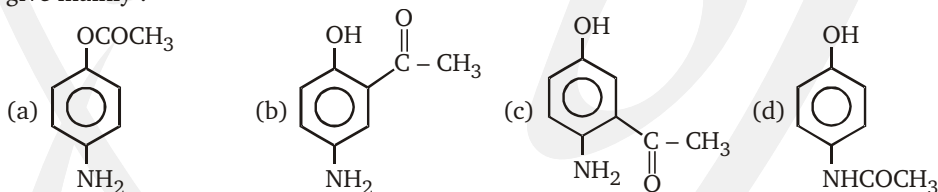
88. Which of the following molecules is expected to have the greatest resonance stabilization ?



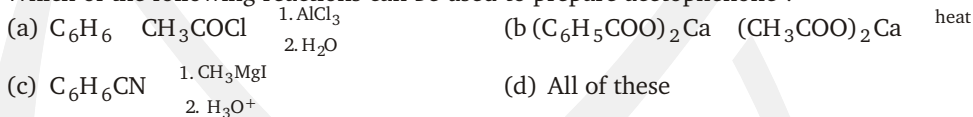
89. In the reaction given below, the major product formed is :



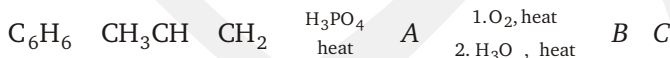
90. *p*-aminophenol reacts with one equivalent of acetyl chloride in the presence of pyridine to give mainly :



91. Which of the following reactions can be used to prepare acetophenone ?



92. Consider the following sequence of reactions.

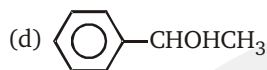
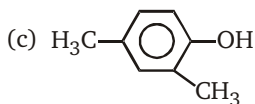


The products (B) and (C) are :

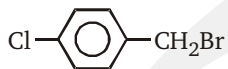
- (a) benzaldehyde and acetaldehyde (b) benzoic acid and acetic acid
(c) phenol and propionaldehyde (d) phenol and acetone

93. An organic compound having the molecular formula $\text{C}_8\text{H}_{10}\text{O}$ on being heated with I_2 and dilute NaOH gives a yellow precipitate. The expected compound is :





94. The product (B) of the reaction sequence is :

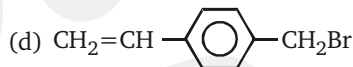
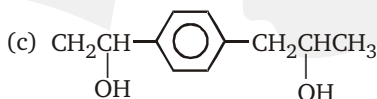
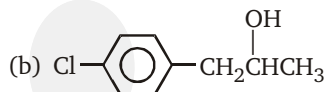
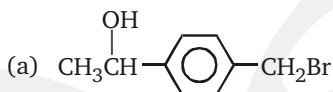


Mg, Et₂O

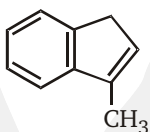
A

1. CH₃CHO
2. H₃O

B.



95. Consider the following sequence of reactions.

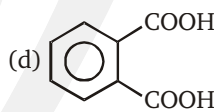
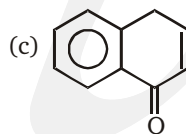
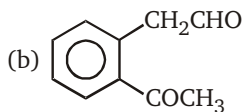
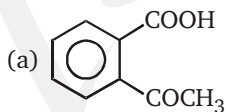


OsO₄
NaHSO₃

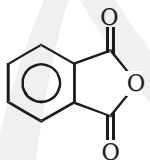
A

1. HIO₄
2. NaOH, heat

B. The final product (B) is :

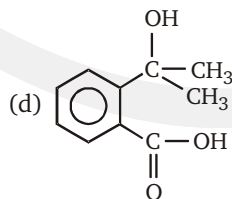
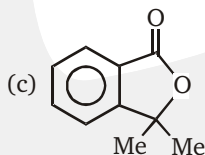
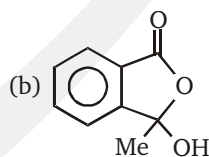
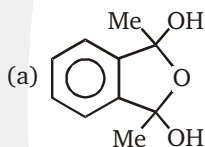


96. For the reaction, the product expected is :

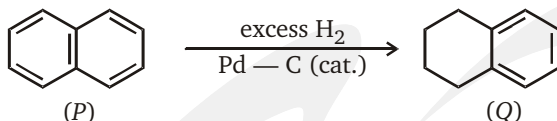


1. MeMgBr (2 mole)
2. H₃O

product,



97. Hydrogenation of naphthalene (*P*) with excess hydrogen gas stops cleanly at 1, 2, 3, 4-tetrahydronaphthalene (*Q*). What conclusion can be drawn from this experiment ?

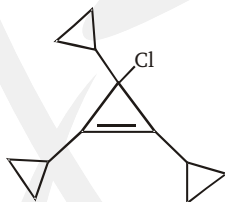


- (a) the hydrogenation of *P* is exothermic
 (b) one aromatic ring of *P* is more reactive than the aromatic ring of *Q*
 (c) one aromatic ring of *P* is less reactive than the other ring of *Q*
 (d) reduction of the first C — C of *P* is faster than reduction of the second or third C — C
98. Suggest the best reaction conditions for the synthesis shown below.



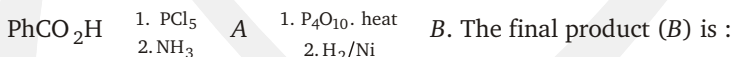
- (a) (1) HNO_3 , H_2SO_2 ; then (2) Br_2
 (b) (1) Br_2 ; then (2) HNO_3 , H_2SO_2
 (c) (1) CH_3Br , AlBr_3 ; then (2) HNO_3 , H_2SO_3
 (d) HNO_3 , H_2SO_2 , then (2) Br_2 , FeBr_3

99.



In the above compound Cl will be liberated easily in the form of :

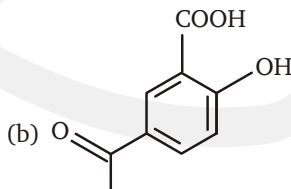
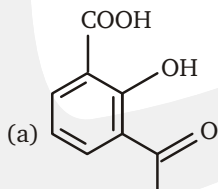
- (a) Cl (b) Cl (c) Cl (d) Cl^2
100. Consider the following sequence of reactions:

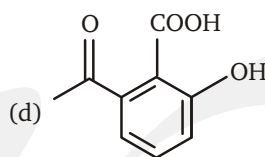
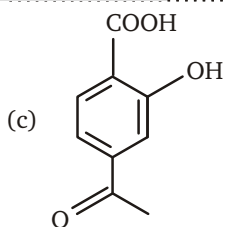


B. The final product (B) is :

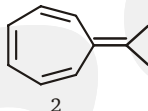
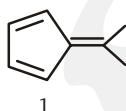
- (a) benzonitrile (b) benzylamine (c) aniline (d) benzamide

101. The major product of the acetylation of salicylic acid with $\text{Ac}_2\text{O}/\text{H}^-$ followed by heating with anhydrous AlCl_3 is :



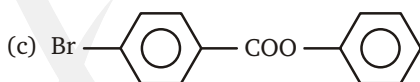
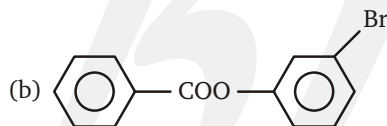
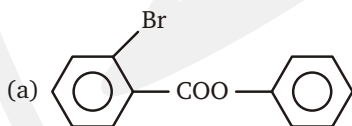


102. Which one of the following statements is **True**:

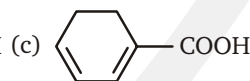
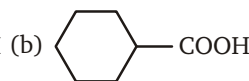
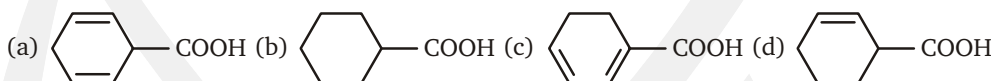


- (a) PhLi adds to both compounds with equal ease
 (b) PhLi does not add to either of the compounds
 (c) PhLi reacts readily with 1 but does not add to 2
 (d) PhLi reacts readily with 2 but does not add to 1

103. The major product expected from the mono-bromination of phenyl benzoate is :



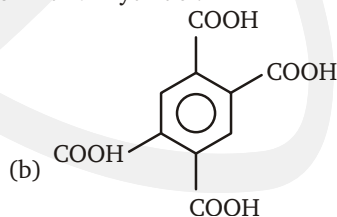
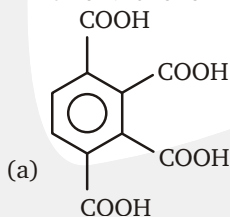
104. The Birch reduction of benzoic acid gives :



105. The decreasing order of reactivity of meta-nitrobromobenzene (I), 2,4,6-trinitrobromobenzene (II), para-nitrobromobenzene (III), and 2,4-dinitrobromobenzene (IV) towards HO^- ions is :

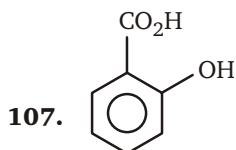
- (a) $\text{I} > \text{II} > \text{III} > \text{IV}$ (b) $\text{II} > \text{IV} > \text{III} > \text{I}$ (c) $\text{IV} > \text{II} > \text{III} > \text{I}$ (d) $\text{II} > \text{IV} > \text{I} > \text{III}$

106. Which of the following tetracarboxylic acid form di-anhydride :

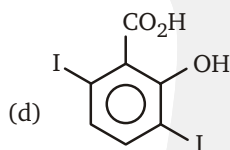
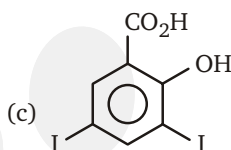
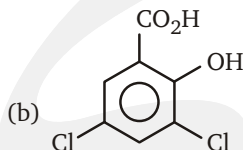
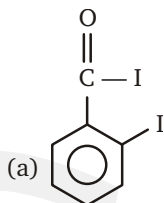


(c) neither (a) nor (b)

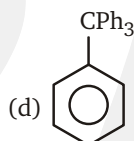
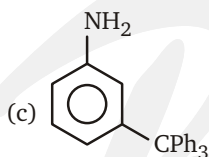
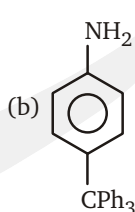
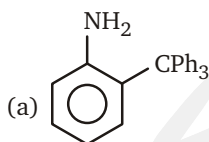
(d) both (a) and (b)



21 Cl (A); unknown (A) is :

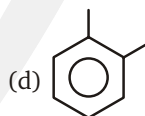
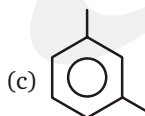
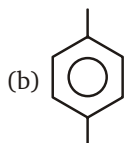
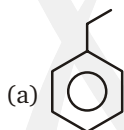


108. $\text{Ph}-\text{NH}_2 + \text{Ph}_3\text{COH} \xrightarrow{\text{H}^+/\text{D}^+} \text{P (major)}$ (not a N-derivative), Product (P) is :

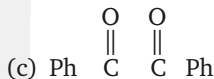
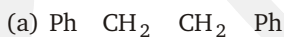


109. Deduce structure of (A).

(A) $\text{C}_8\text{H}_{10} \xrightarrow{\text{KMnO}_4}$ (B) $\text{C}_8\text{H}_6\text{O}_4 \xrightarrow[\text{Fe}]{\text{Br}_2}$ $\text{C}_8\text{H}_5\text{BrO}_4$ (C) (one-product only) :



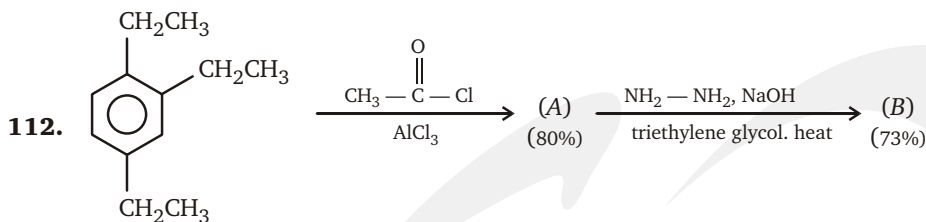
110. The deamination of $\text{Ph}_2\text{C}(\text{OH})\text{CH}_2\text{NH}_2$ with $\text{NaNO}_2 \cdot \text{HCl}$ gives a product (P), which on oxidation gives benzoic acid only. Identify the product (P).



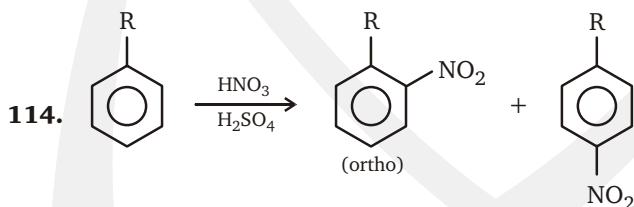
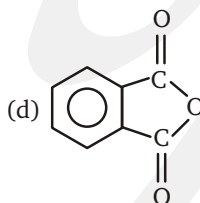
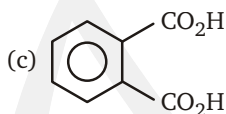
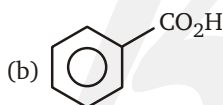
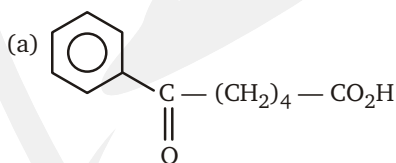
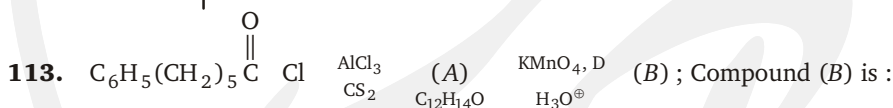
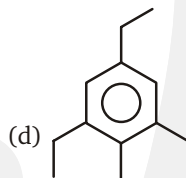
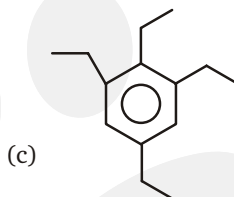
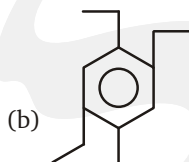
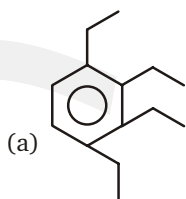
111. $\text{Ph}-\text{CO}_2\text{H} \xrightarrow{\text{SOCl}_2} \text{A} \xrightarrow{\text{Me}_2\text{NH}} \text{B} \xrightarrow{\text{(C)}} \text{Ph}-\text{CHO}$, unknown reagent (C) is :



Aromatic Compounds

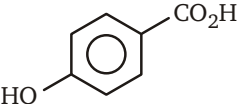


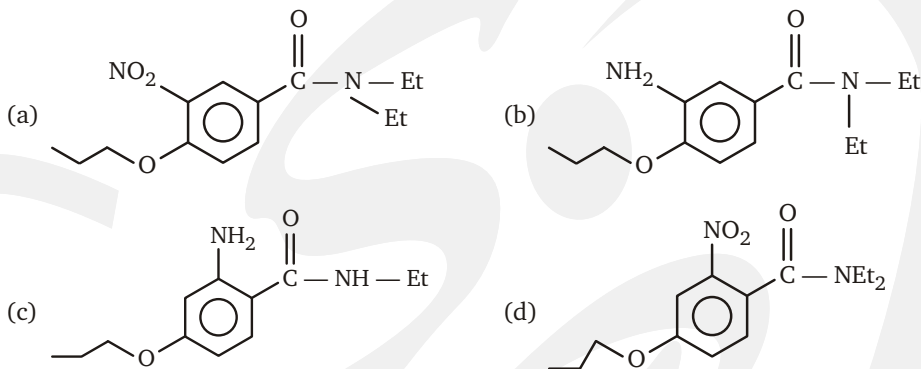
Product (B) is :

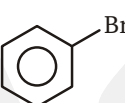


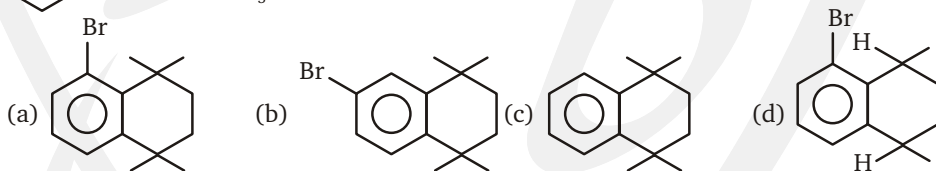
In the above reaction o/p ratio will be highest when :



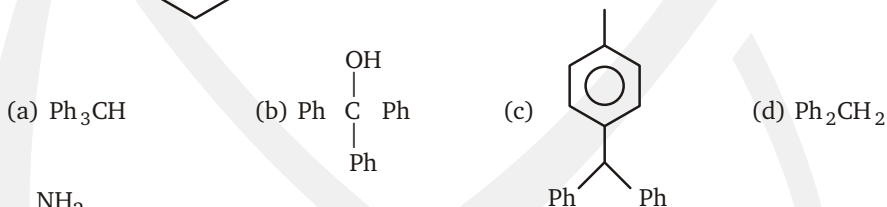
115.  $\xrightarrow{\text{HNO}_3}$ (1) $\xrightarrow[\text{PrCl}]{\text{Base}}$ (2) $\xrightarrow{\text{SOCl}_2}$ (3) $\xrightarrow[\text{(2) H}_2/\text{Pd}]{\text{(1) Et}_2\text{NH}}$ (4)
product (4) in the above reaction is :

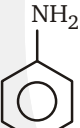


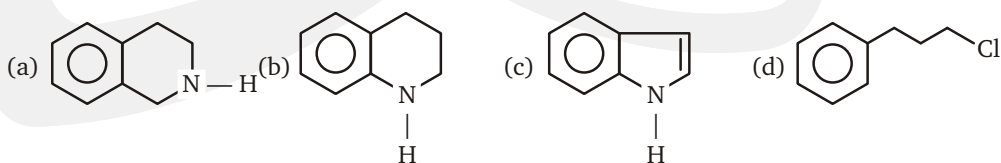
116.  $\xrightarrow[\text{AlCl}_3]{\text{1,1-dichloro-3,3-dimethylbutane}}$ (A); Product (A) of the reaction is :



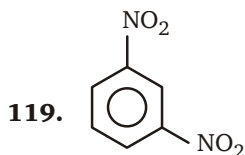
117. Ph CHO + 2  $\xrightarrow{\text{H}_2\text{SO}_4}$ major product of this reaction is :



118.  $\xrightarrow{\text{1-bromopropane}}$ (A); Product (A) of this reaction is :



Aromatic Compounds

 HNO_3 H_2SO_4 five days
to complete

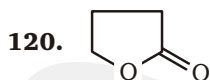
(X); Compound (X) is :

(a) 1,2,4-Trinitrobenzene

(c) 1,2,3-Trinitrobenzene

(b) 1,3,5-Trinitrobenzene

(d) Tri-nitro toluene (TNT)

(i) HCl
(ii) SOCl_2

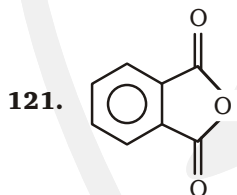
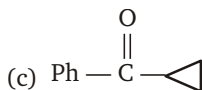
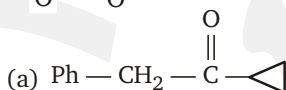
(A)

 Ph H
 AlCl_3

(B)

 KOH/MeOH

(C); Compound (C) is :

 C_6H_6
 AlCl_3

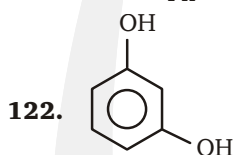
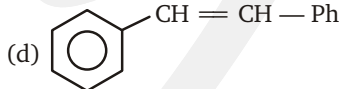
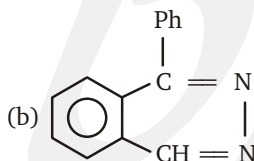
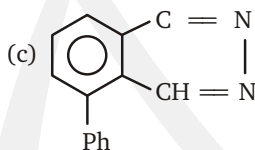
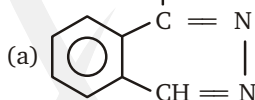
(A)

(i) PCl_5
(ii) H_2/Pd BaSO_4

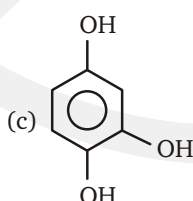
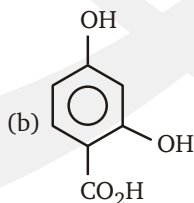
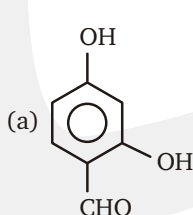
(B)

 NH_2 NH_2

(C); Compound (C) is :

(i) aq NaHCO_3 /boil
 H_2O

(A) , Product (A) in this reaction is :



(d) No reaction

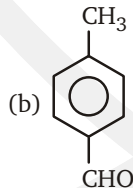
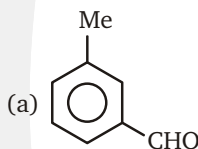
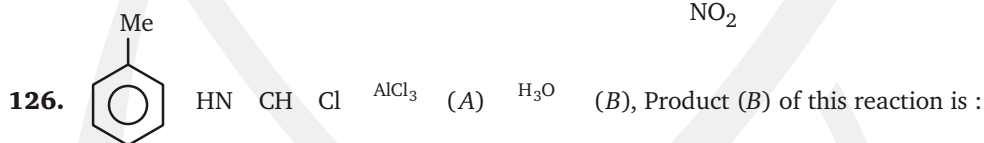
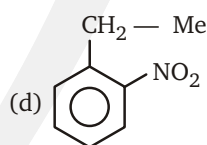
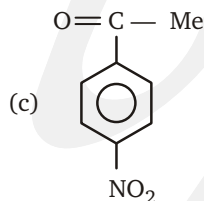
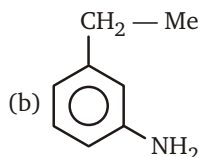
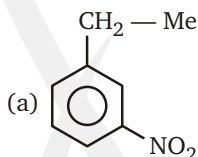
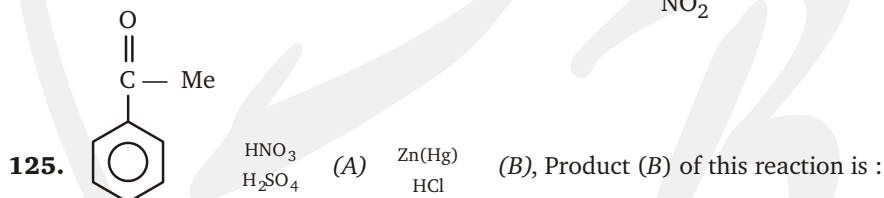
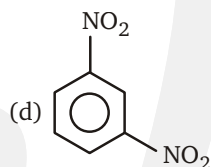
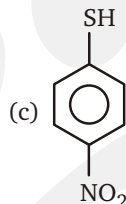
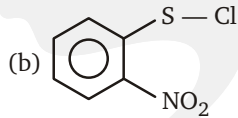
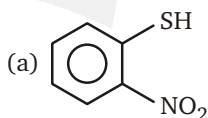
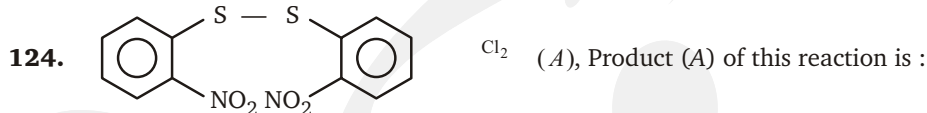


(a) i

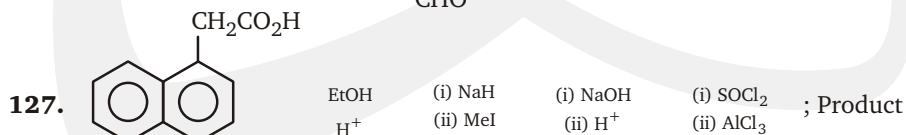
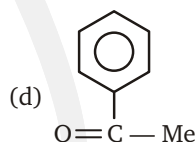
(b) ii

(c) iii

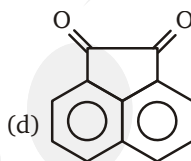
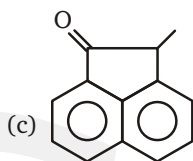
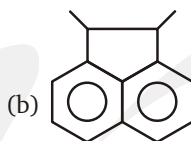
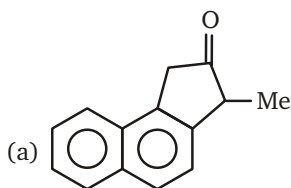
(d) iv



(c) No-reaction

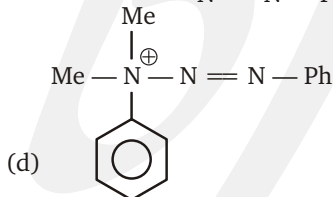
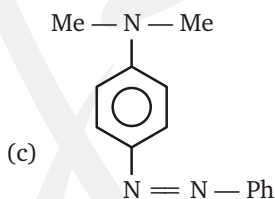
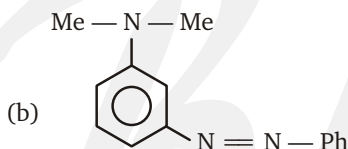
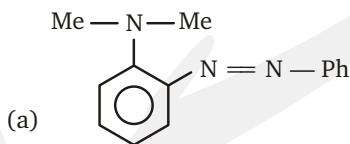


End product of the above reaction is :



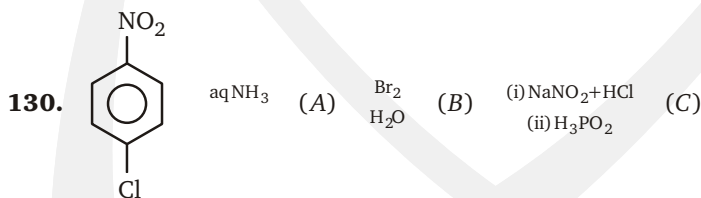
128. $\text{Ph-NH}_2 \xrightarrow{\text{CH}_3\text{Cl (2mole)}} \text{(A) Ph-N}_2^+\text{Cl}^- \text{ (B) (major) Butter yellow}$

Product of the above reaction is :

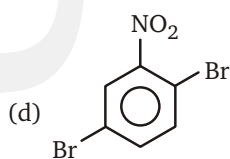
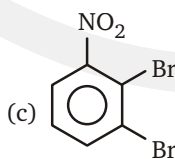
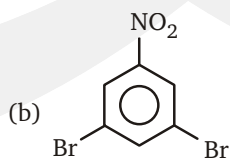
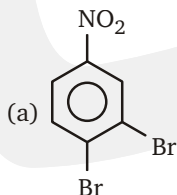


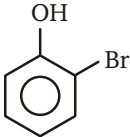
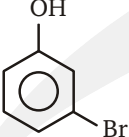
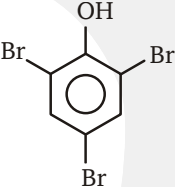
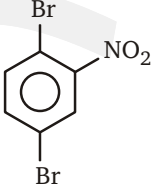
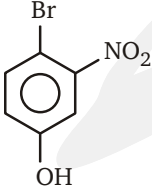
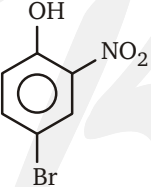
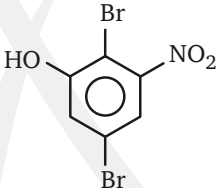
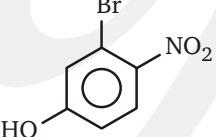
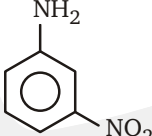
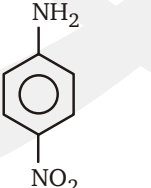
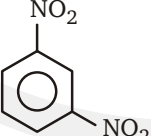
129. *p*-Toluidine reacts with benzene diazonium chloride to form compound, which on boiling with aq. H_2SO_4 give products :

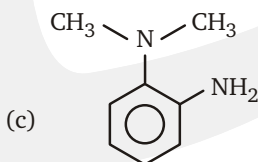
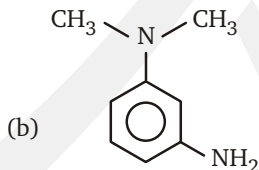
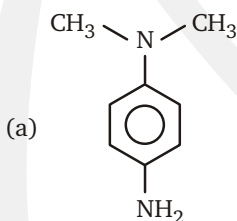
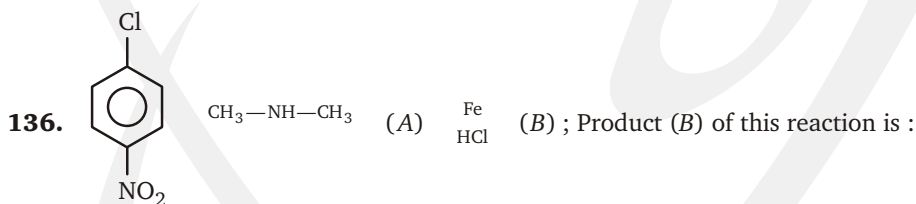
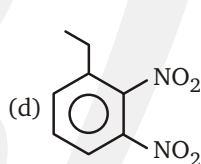
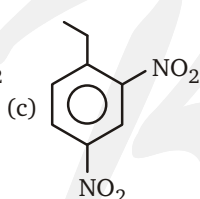
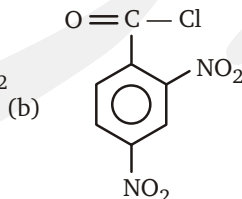
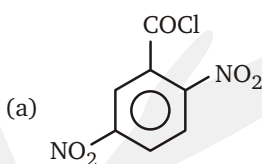
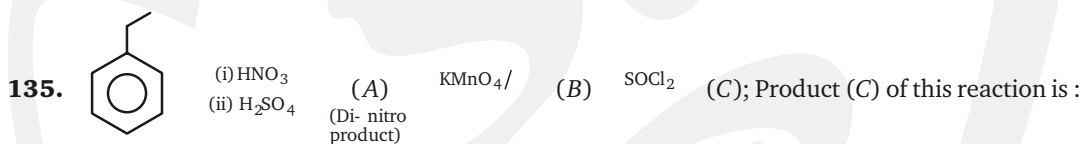
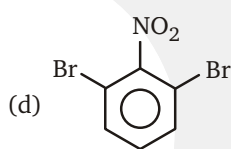
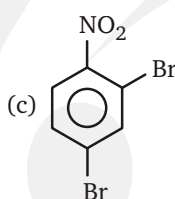
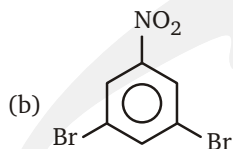
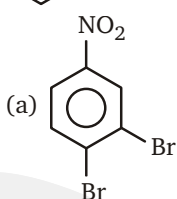
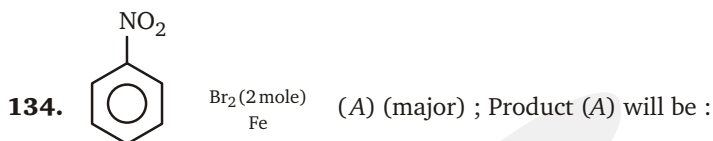
- (a) 3 (b) 2 (c) 4 (d) 5



Product (C) of the above reaction is :

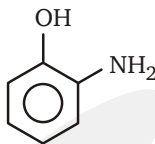


131.  $\xrightarrow[\text{(Fenton's reagent)}]{\text{Fe}^{+2}, \text{H}_2\text{O}_2}$ (A) $\xrightarrow[\text{H}_2\text{O}]{\text{Br}_2}$ (B) ; Major product (B) of this reaction is :
- (a)  (b)  (c)  (d) 
132.  $\xrightarrow{\text{HO}^-}$ (A) ; Product of the given reaction is :
- (a)  (b) 
- (c)  (d) 
133.  $\xrightarrow{(\text{CH}_3\text{CO})_2\text{O}}$ (A) $\xrightarrow[\text{H}_2\text{SO}_4]{\text{HNO}_3}$ (B) $\xrightarrow[\text{H}_2\text{O}]{\text{H}^+}$ (C), Product (C) of this reaction is :
- (a)  (b)  (c)  (d) 

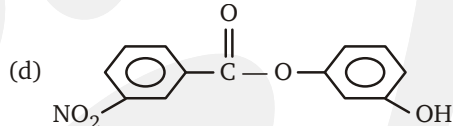
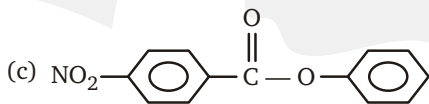
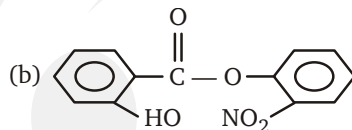
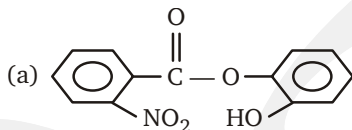


(d) None of these

137. (A) (i) SOCl_2 (B) $\text{Br}_2 + \text{KOH}$
(ii) NH_3

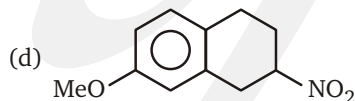
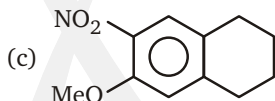
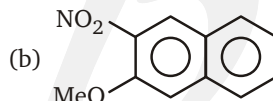
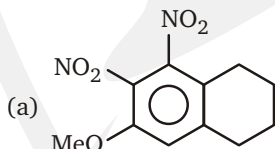


Which of the following compound on hydrolysis gives reactant (A) :

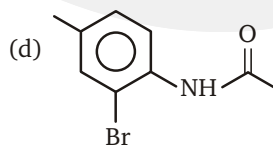
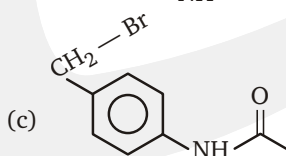
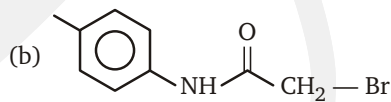
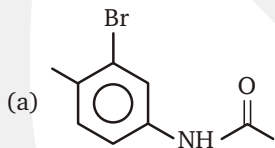


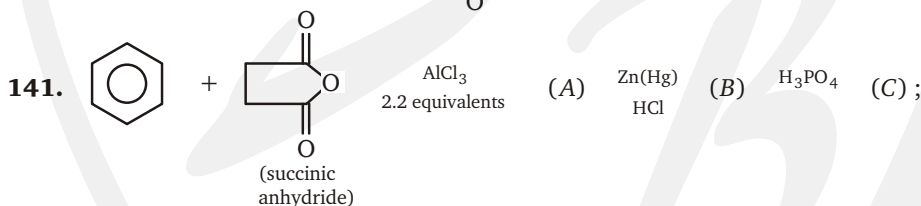
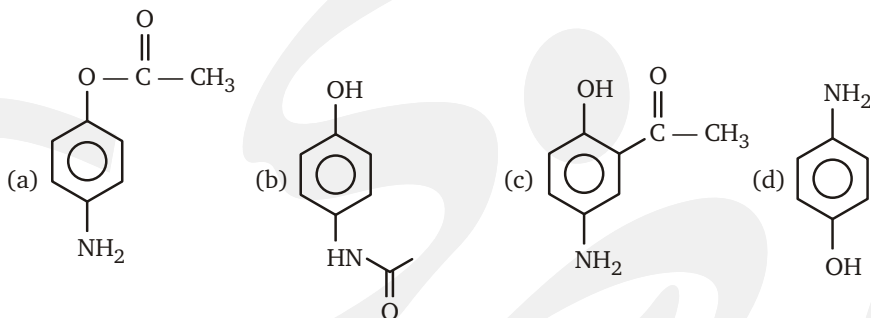
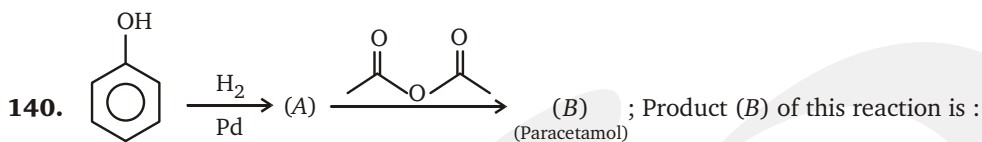
138. $\text{HNO}_3/\text{H}_2\text{SO}_4$ (A);

Product (A) of the above reaction is :

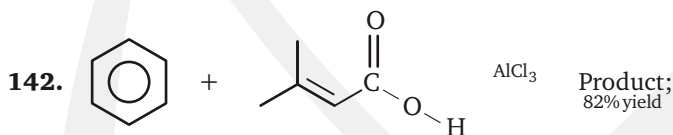
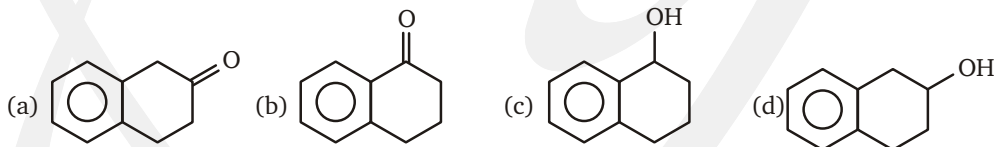


139. Br_2 ; Product of the reaction is :
 CS_2

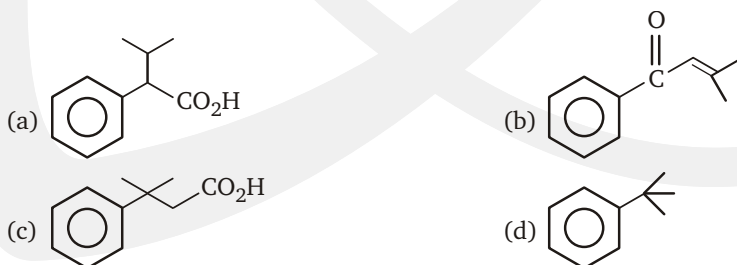




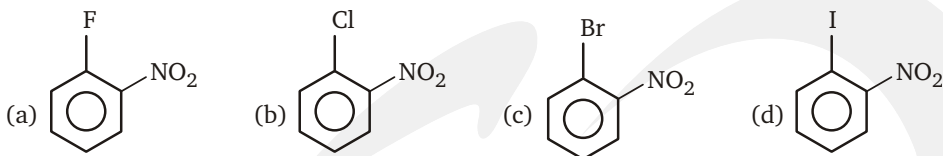
Product (C) of the above reaction is :



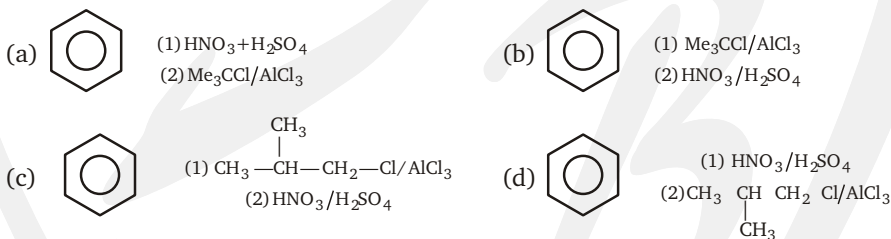
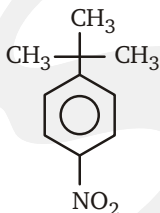
Product of the above Friedel-Craft reaction is :



143. Which of the following 2-halo nitrobenzene is most reactive towards nucleophilic aromatic substitution ?



144. Choose the best method to prepare given compound :

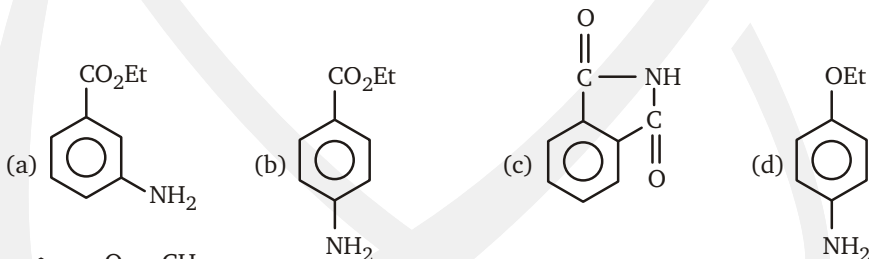


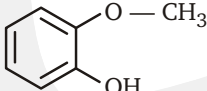
145.  $\xrightarrow[\text{H}_2\text{SO}_4]{\text{HNO}_3}$ (A) $\xrightarrow[\text{Pd, C}]{\text{H}_2}$ (B) $\xrightarrow{\text{KMnO}_4}$ (C) $\xrightarrow[\text{H}^+]{\text{EtOH}}$ (D)

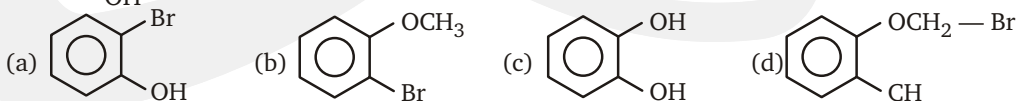
(Para isomer)

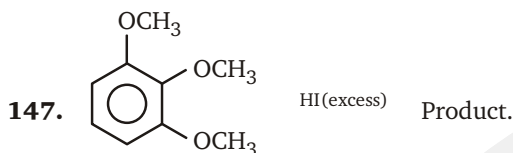
Benzocaine

Benzocaine has been used as a component of appetite suppressants, burn and sunburn remedies. Benzocaine is :

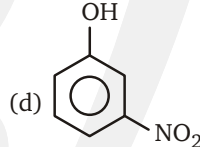
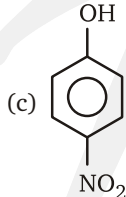
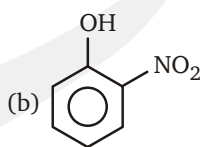
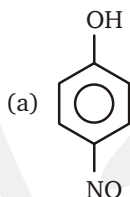
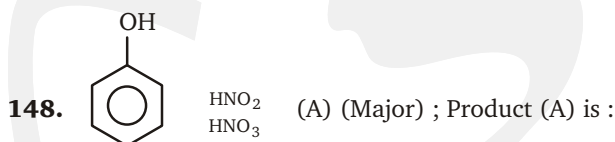
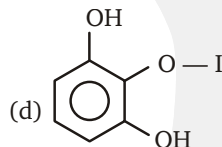
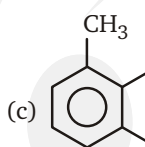
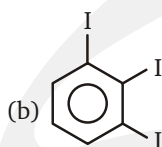
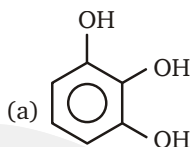


146.  $\xrightarrow[\text{86\%}]{\text{HBr}}$ (A) ; Product (A) of this reaction is :

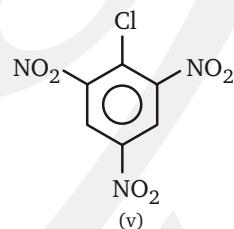
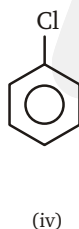
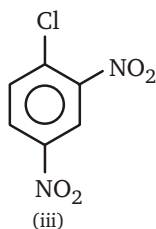
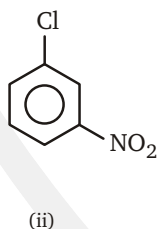




Predict major product of the above reaction is :



149. Arrange in their decreasing order of rate in $\text{S}_\text{N}\text{Ar}$.



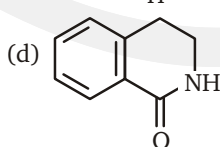
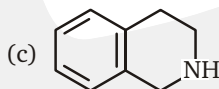
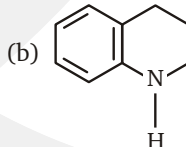
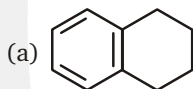
(a) $\text{i} > \text{ii} > \text{iv} > \text{iii} > \text{v}$

(c) $\text{v} > \text{iii} > \text{i} > \text{ii} > \text{iv}$

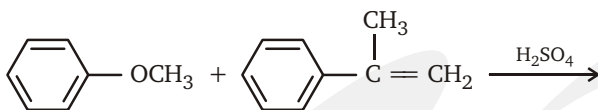
(b) $\text{ii} > \text{i} > \text{iii} > \text{v} > \text{iv}$

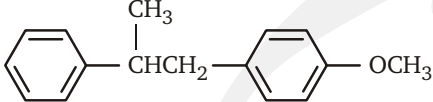
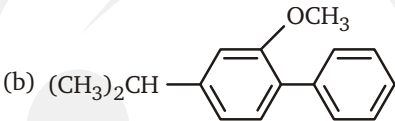
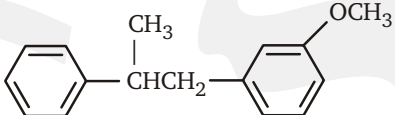
(d) $\text{v} > \text{iii} > \text{ii} > \text{i} > \text{iv}$

150. Which one of the following compounds undergoes bromination of its aromatic ring (electrophilic aromatic substitution) at the fastest rate ?

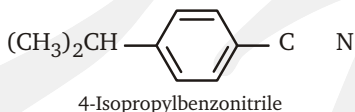


151. What is the product of the following reaction ?



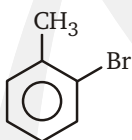
- (a)  (b) 
- (c)  (d) 

152. Which sequence represents the best synthesis of 4-isopropylbenzonitrile ?



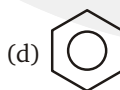
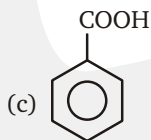
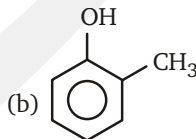
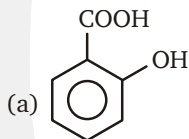
- | | | |
|---|--|---------------------------------|
| (a) 1. Benzene + $(\text{CH}_3)_2\text{CHCl}$, AlCl_3 ; | 2. Br_2 , FeBr_3 ; | 3. KCN |
| (b) 1. Benzene + $(\text{CH}_3)_2\text{CHCl}$, AlCl_3 ; | 2. HNO_3 , H_2SO_4 ; | 3. Fe , HCl ; |
| 4. NaOH | 5. NaNO_2 , HCl , H_2O | |
| (c) 1. Benzene + $(\text{CH}_3)_2\text{CHCl}$, AlCl_3 ; | 2. HNO_3 , H_2SO_4 ; | 3. Fe , HCl ; |
| 4. NaNO_2/HCl | 5. KCN | |
| (d) 1. Benzene + HNO_3 , H_2SO_4 ; | 2. $(\text{CH}_3)_2\text{CHCl}$, AlCl_3 ; | 3. Fe , HCl ; |
| 4. NaNO_2 , HCl , H_2O ; | 5. CuCN | |

153.

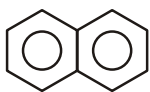


1. Mg/Ether
2. H_3O^+
3. KMnO_4/OH
4. H

A, Product A is :

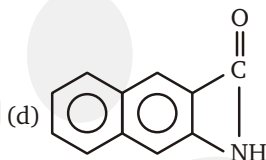
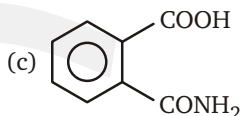
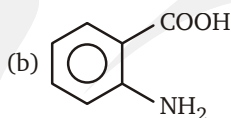
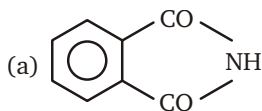


154.

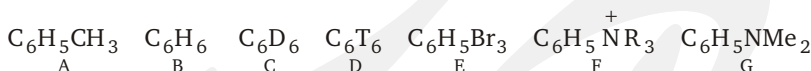


1. Air, V_2O_5 ,
2. NH_3 ,
3. $KOBr$,

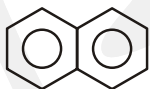
A, Product A is :



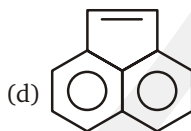
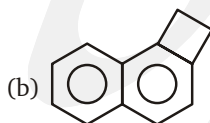
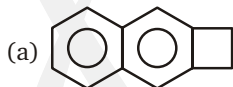
155. What is correct order of rate of nitration of the following compounds ?

(a) $G > A > B > C > D > E > F$ (b) $G > B > C > D > A > F$ (c) $G > A > B = C = D > E > F$ (d) $G > A > B > C = D > E > F$

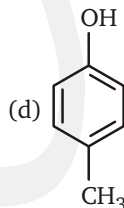
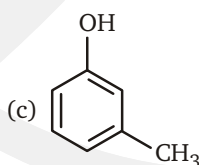
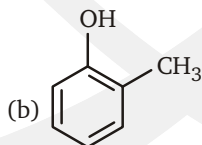
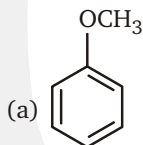
156.

 $AlCl_3$ [X] $Na-Hg, HCl$
excess

[Y] ; Product Y is :



157. Compound A (C_7H_8O) is insoluble in water, dilute HCl & aqueous $NaHCO_3$, but it dissolves in dilute $NaOH$. When A is treated with Br_2 water it is converted into a compound $C_7H_5OBr_3$ rapidly. The structure of A is :

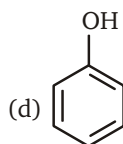
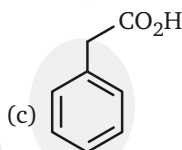
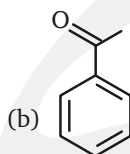
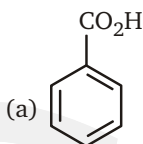


158. Give the product of the following reaction sequence :



1. $\text{HNO}_3/\text{H}_2\text{SO}_4$
2. $\text{H}_2/\text{Pd/C}$
3. NaNO_2/HCl
4. CuCN
5. MeMgBr
6. H_3O^+

Product

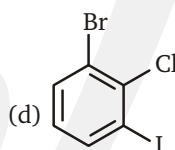
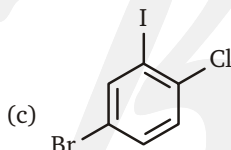
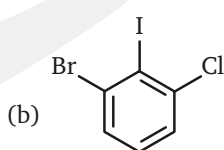
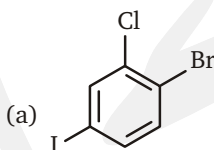


159. Give the product of the following reaction sequence:



1. $\text{HNO}_3/\text{H}_2\text{SO}_4$
2. $\text{Br}_2/\text{FeBr}_3$
3. $\text{H}_2/\text{Pd/C}$
4. $\text{Cl}_2/\text{FeBr}_3$
5. NaNO_2/HCl
6. KI

Product



160. Which represents an intermediate formed in the reaction of toluene and chlorine at elevated temperature in sunlight ?



161. The decreasing order of reactivity of *m*-nitrobromobenzene (I), 2, 4, 6- trinitrobromo-benzene (II), *p*-nitrobromobenzene (III), and 2,4-dinitrobromobenzene (IV), towards OH^- ions is :

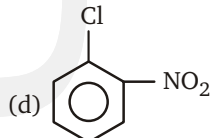
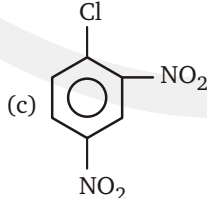
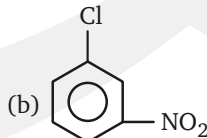
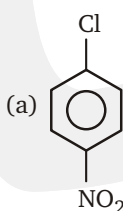
(a) $\text{I} > \text{II} > \text{III} > \text{IV}$

(b) $\text{II} > \text{IV} > \text{III} > \text{I}$

(c) $\text{IV} > \text{II} > \text{III} > \text{I}$

(d) $\text{II} > \text{IV} > \text{I} > \text{III}$

162. Which one of the following compounds is most reactive for $\text{ArS}_{\text{N}}2$ reaction ?



163. Which one amongst the following carbocations is most stable ?

- (a) $\text{C}_6\text{H}_5 - \text{CH} - \text{C}_6\text{H}_5$
- (b)
- (c) $\text{CH}_3 - \underset{\text{CH}_3}{\text{C}} - \text{CH}_3$
- (d)

164. Cyclopentadiene is much more acidic than cyclopentane. The reason is that :

- (a) cyclopentadiene has conjugated double bonds
- (b) cyclopentadiene has both sp^2 and sp^3 hybridized carbon atoms
- (c) cyclopentadiene is a strain-free cyclic system
- (d) cyclopentadienide ion, the conjugate base of cyclopentadiene, is an aromatic species and hence has higher stability

- 165.
- (I) (II) (III) (IV)

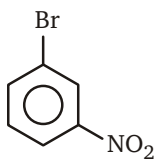
Friedel-Crafts acylation reaction can be used to obtain the compounds

- (a) II, III and IV
- (b) I, III and IV
- (c) I and II
- (d) II and III

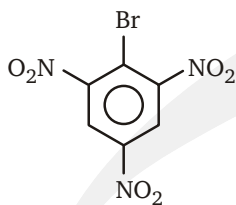
166. The major product of the reaction is :

-
- (a)
- (b)
- (c)
- (d)

167. The decreasing order of reactivity of given compound towards nucleophilic substitution with aqueous NaOH is :



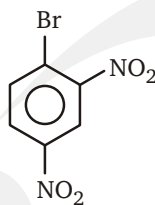
(I)



(II)



(III)



(IV)

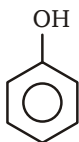
(a) I > II > III > IV

(b) II > IV > III > I

(c) IV > II > III > I

(d) II > IV > I > III

168. Identify the end product (B) of the following sequence of reactions.

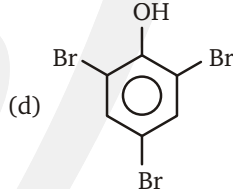
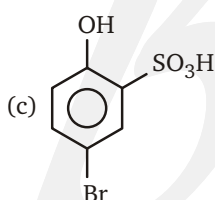
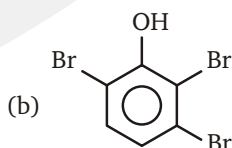
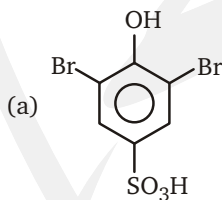


conc. H_2SO_4
100 $^\circ\text{C}$

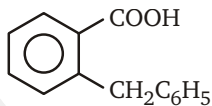
A

$\text{Br}_2/\text{H}_2\text{O}$
(excess)

B



169. Consider the following sequence of reactions :

 SOCl_2

A

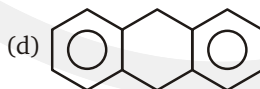
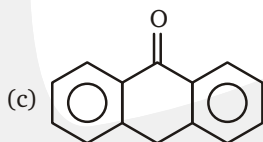
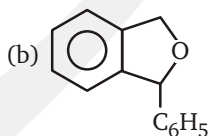
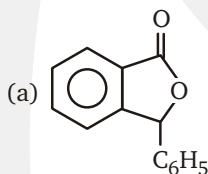
1. AlCl_3
2. H_3O^+

B

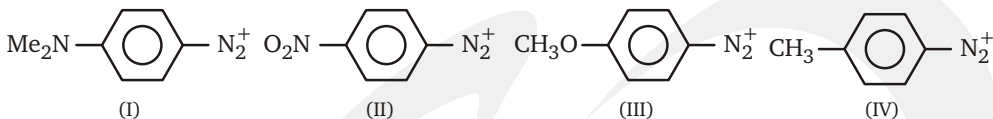
Zn-Hg
conc. HCl , heat

C

The end product (C) is :

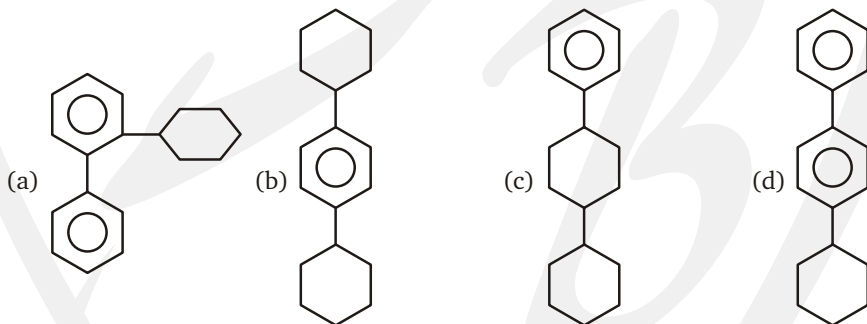
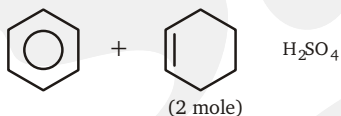


170. For the diazonium ions the order of reactivity towards diazo-coupling with phenol in the presence of dilute NaOH is :



- (a) $\text{I} < \text{IV} < \text{II} < \text{III}$ (b) $\text{I} < \text{III} < \text{IV} < \text{II}$
 (c) $\text{III} < \text{I} < \text{II} < \text{IV}$ (d) $\text{III} < \text{I} < \text{IV} < \text{II}$

171. Major product obtained in given reaction is :



172.  (A) $\xrightarrow[\text{or AlCl}_3]{\text{H}}$ (B) ; (A) & (B) are isomers. Product (B) is :
 Dewar's Benzene

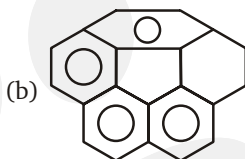
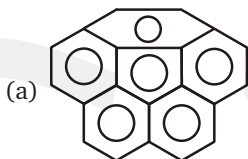


173. The step shown below is a recent synthesis of corannulene.

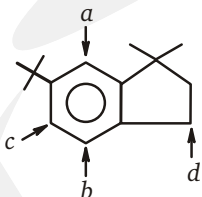


Pd/C
() (A) $C_{20}H_{10}$
Corannulene

Product (A) is :



174.



Identify the position where E.A.S. will take place :

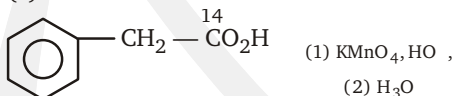
(a) a

(b) b

(c) c

(d) all the position are identical

175.



The labelled carbon goes with :

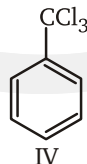
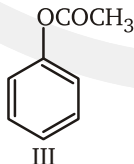
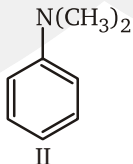
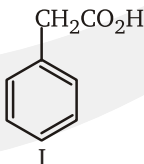
(a) $Ph - CO_2H$

(b) CO_2

(c) $Ph - CH_2 - CO_2H$

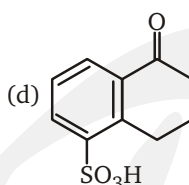
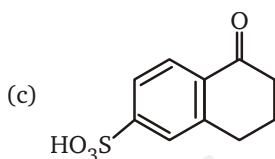
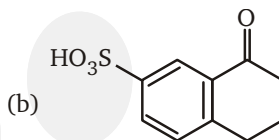
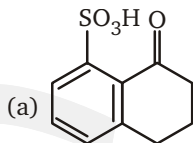
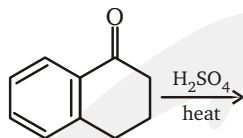
(d) CH_4

176. What is the expected order of reactivity of the following compounds in electrophilic chlorination (Cl_2 / $FeCl_3$) ?
(more reactive > less reactive)

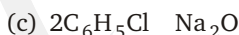
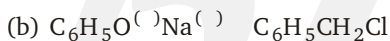


- (a) I > II > III > IV (b) IV > III > II > I (c) III > I > IV > II (d) II > III > I > IV

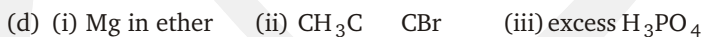
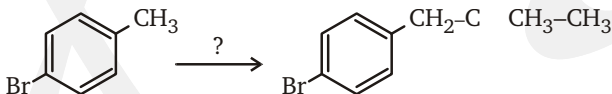
177. Which of the following is the major product from sulfonation of α -tetralone ?



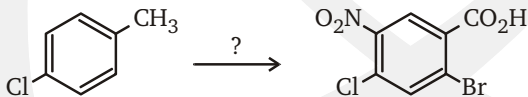
178. Which of the following procedures would be best for the preparation of phenyl benzyl ether ?



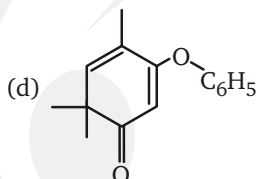
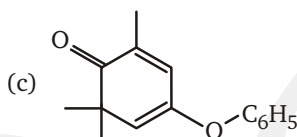
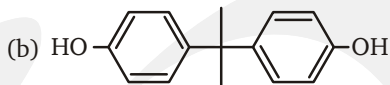
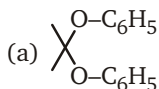
179. Which of the following procedures would be best for achieving the following reaction ?



180. Which of the following procedures would be best for achieving the following reaction ?



181. Phenol reacts with acetone in the presence of conc. sulphuric acid to form a $C_{15}H_{16}O_2$ product. Which of the following compounds is this product ?



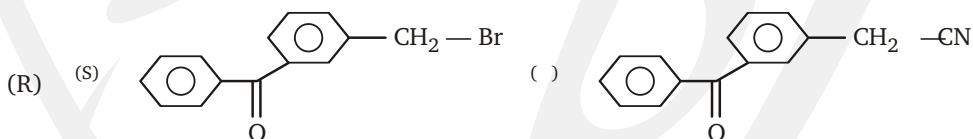
182. Heating benzene in a large excess of 80% D_2SO_4 in D_2O results in what product ?

- (a) $C_6H_5SO_3D$ (b) C_6H_5OD (c) C_6H_5D (d) C_6D_6

183. A solution of cyclohexene in benzene is stirred at $0^\circ C$ while concentrated sulphuric acid is added. After washing away the acid and removing the excess benzene, what product is isolated?

- (a) cyclohexylbenzene (b) 1-cyclohexylcyclohexene
(c) *trans*-1,2-diphenylcyclohexane (d) 1,1-diphenylcyclohexane

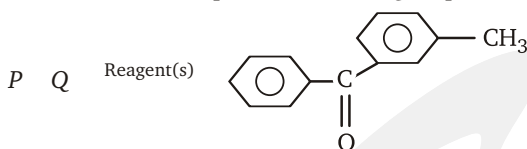
184. Identify the reagents S and in the scheme below in which R is converted to the nitrite V via the benzylic halide T.



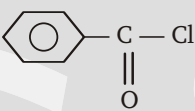
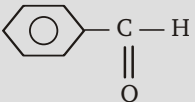
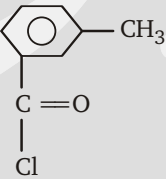
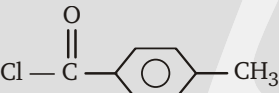
R, S and respectively are :

R	S	
(a)		($AlCl_3$) HCN
(b)		($AlCl_3$) HCN
(c)		($AlCl_3$) KCN
(d)		($AlCl_3$) KCN

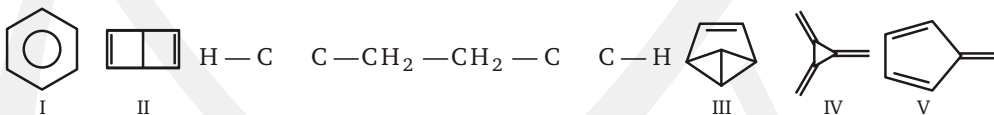
185. Two aromatic compounds *P* and *Q* give product *R*.



Reactant *P*, *Q* and reagent used in above reaction are :

P	Q	Reagent
(a) 		AlCl_3
(b) 		AlCl_3
(c) 		AlCl_3
(d) 		ZnCl_2

186. Which of the following C_6H_6 compounds has a single set of structurally equivalent hydrogen atoms ?



- (a) I and II (b) I and IV (c) I and V (d) I, II and III

187. Which of the following compounds would not be considered aromatic in its behaviour?



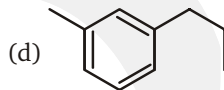
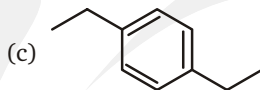
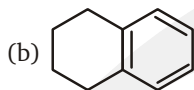
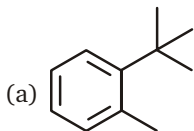
188. A C_8H_{10} hydrocarbon is nitrated by HNO_3 and sulphuric acid. Two, and only two, $\text{C}_8\text{H}_9\text{NO}_2$ isomers are obtained. Which of the following fits this evidence ?

- (a) ethyl benzene (b) *ortho*-xylene (c) *meta*-xylene (d) *para*-xylene

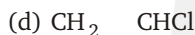
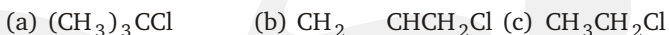
189. Which of the following benzene ring substituents is deactivating but *ortho-para* directing ?

- (a) $-\text{N}^+\text{O}_2^-$ (b) $-\text{OCH}_3$ (c) $-\text{COCH}_3$ (d) $-\text{NO}_2$

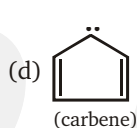
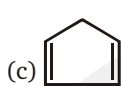
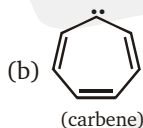
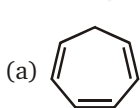
190. Which of the following compounds forms *ortho*-benzenedicarboxylic acid when oxidized by hot aqueous potassium permanganate ?



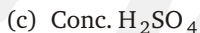
191. Which of the following organic chlorides will not give a Friedel-Craft alkylation product when heated with benzene and AlCl_3 ?



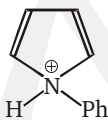
192. Which of the following is aromatic ?



193. Which of the following substance will increase the acidity of phenol ?



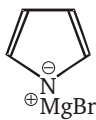
194. + PhMgBr $E + F$



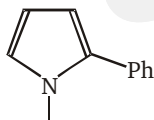
1



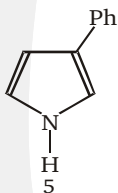
2



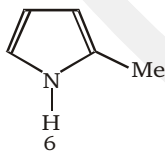
3



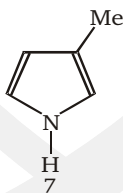
4



5

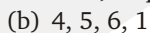


6

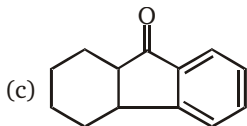
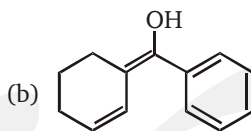
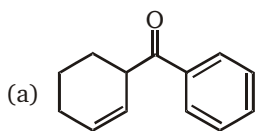


7

The structure of products $E - H$, respectively are

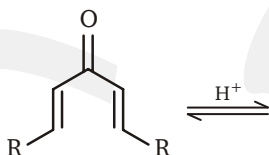


195.
(A); Product A is :

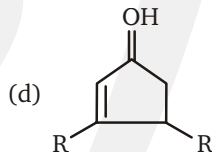
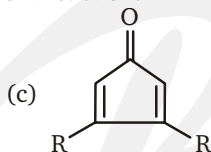
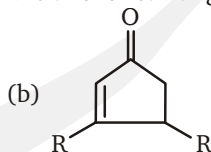
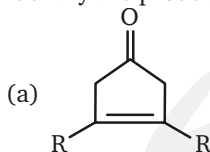


(d) none of these

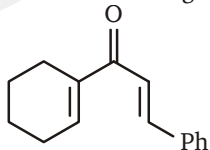
196.



Identify the product of the above rearrangement reaction.

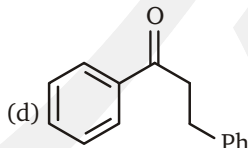
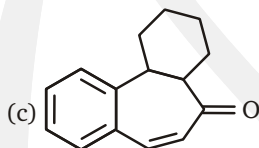
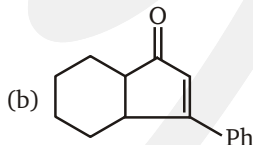
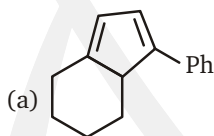


197. Product obtained in the following transformation is :

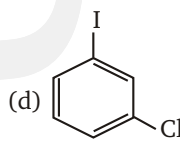
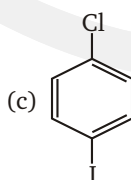
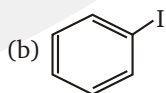
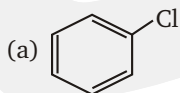
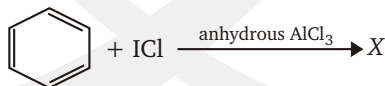


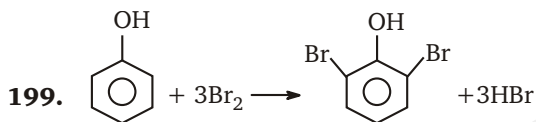
PPA

PPA = polyphosphoric acid

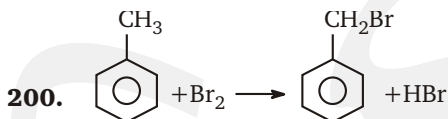


198. The compound X in the reaction.

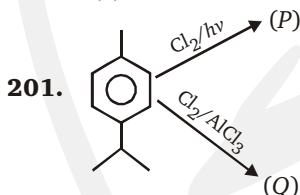




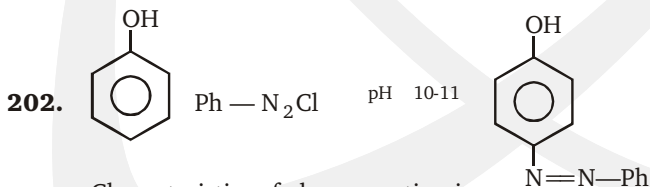
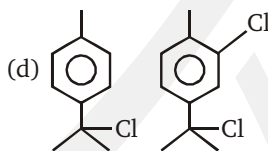
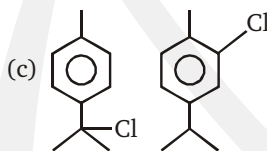
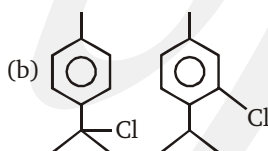
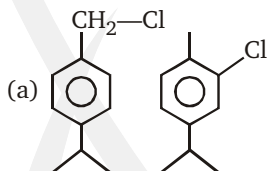
- (a) Nucleophilic addition (b) Nucleophilic substitution
(c) Electrophilic addition (d) Electrophilic substitution
(e) Free radical substitution



- (a) Nucleophilic addition (b) Nucleophilic substitution
(c) Electrophilic addition (d) Electrophilic substitution
(e) Free radical substitution



Identify major product of both respectively.



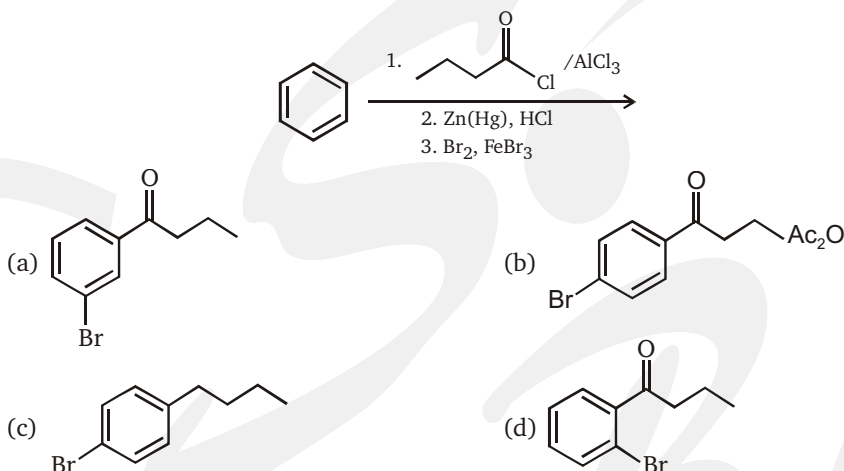
Characteristics of above reaction is :

- (a) C — N coupling reaction ; Carbocation is intermediate
(b) N — N coupling reaction ; Carbocation is intermediate
(c) C — N coupling reaction ; Carbanion is intermediate
(d) N — N coupling reaction ; Carbanion is intermediate

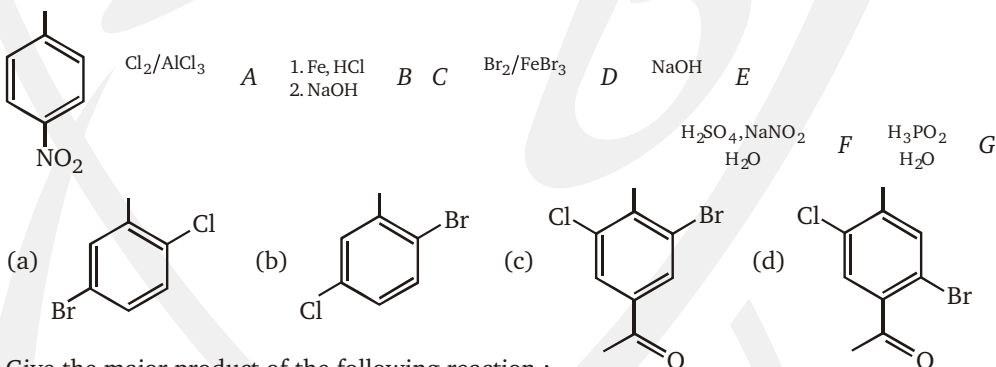
203. The compound formed on heating chlorobenzene with chloral in the presence of concentrated sulphuric acid, is :

- (a) Freon (b) DDT (c) Gammexene (d) Hexachloroethane

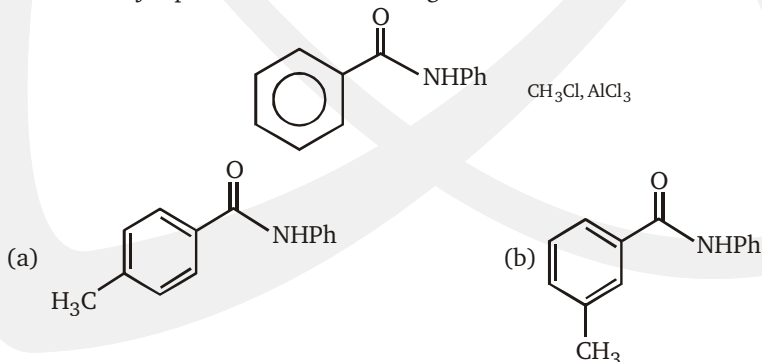
204. Predict the product of the following reaction.

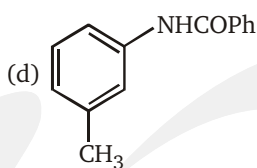
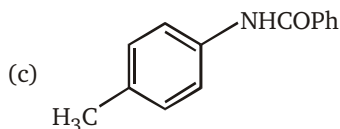


205. Predict the major product of the following reaction sequence.

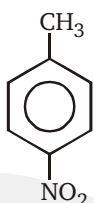


206. Give the major product of the following reaction :





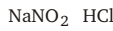
207.



(A)



(B)



(C)

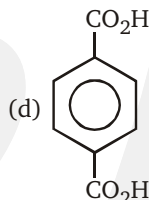
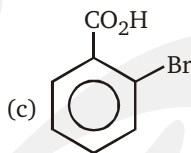
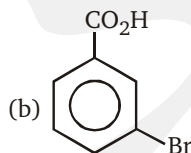
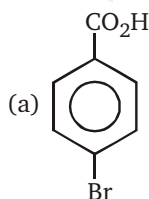


(D)

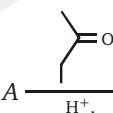
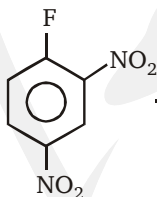


(E)

Product (E) is :



208.

A $\rightarrow$ B ; Products;

Incorrect statements regarding above reaction is

- (a) Product A is 2, 4-DNP
- (b) A to B dehydration reaction
- (c) A to B, geometrical isomersm will obtained as a product
- (d) B is known as oxime

209.

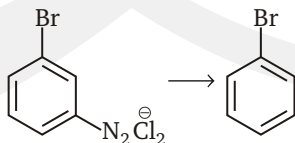
- (i) chlorobenzene is mono-nitrated to M
- (ii) nitrobenzene is mono-chlorinated to N
- (iii) anisole is mono-nitrated to P
- (iv) 2-nitrochlorobenzene is mono-nitrated to Q.

Out of M, N, P and Q the compound that undergoes reaction with aq. NaOH fastest is

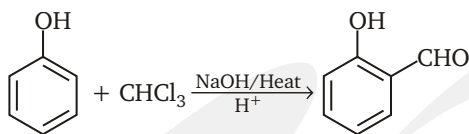
- (a) M
- (b) N
- (c) P
- (d) Q

210.

For the transformation the reagent used is

(a) LiAlH_4 (b) H_3PO_2 (c) H_3O^+ (d) H_2 / Pt

211. The reaction



is known as

- (a) Perkin reaction (b) Sandmeyer reaction
(c) Reimer-Tiemann reaction (d) Cannizzaro reaction

212. A compound X formed after heating coke with lime react with water to give Y which on passing over redhot iron at 873 produces Z. The compound Z is

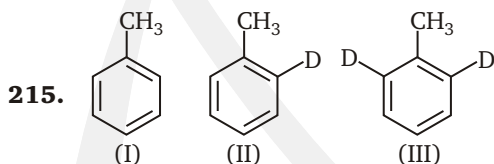
- (a) (b) (c) (d)

213. The reaction of 50% aq KOH on an equimolar mixture of 4-methylbenzaldehyde and formaldehyde followed by acidification gives :

- (a) (b) + HCOOH
(c) + CH₃OH (d)

214. Which isomer of xylene can give three different monochloroderivatives ?

- (a) o-xylene (b) m-xylene (c) p-xylene
(d) xylene cannot give a monochloro derivative



The rate of o-nitration of the above compounds, (I) toluene, (II) 2-D-toluene and (III) 2, 6-D₂-toluene is in the following order

- (a) I > II > III (b) II > I > III (c) III > I > II
(d) The rate is the same for all the three compounds

216. Cyclooctatetraene is expected to have :

- (a) a planar structure (b) a tub-shaped structure
(c) open chain isomeric structure (d) tautomeric bicyclic structure



Relation between C and D.

- (a) Identical (b) enantiomer (c) diastereomer (d) position

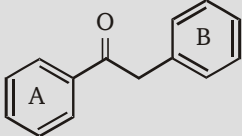
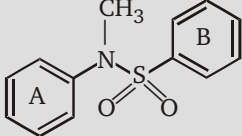
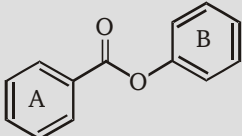
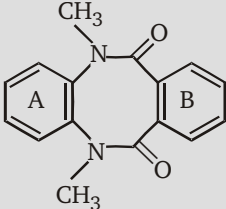
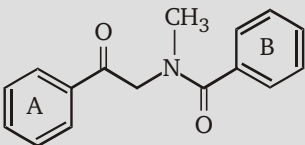
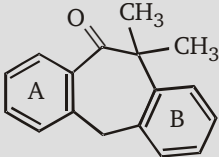
ANSWERS — LEVEL 1

1.	(b)	2.	(d)	3.	(b)	4.	(c)	5.	(a)	6.	(b)	7.	(c)	8.	(b)
9.	(d)	10.	(d)	11.	(a)	12.	(a)	13.	(a)	14.	(a)	15.	(a)	16.	(b)
17.	(b)	18.	(a)	19.	(b)	20.	(b)	21.	(c)	22.	(c)	23.	(b)	24.	(b)
25.	(d)	26.	(c)	27.	(c)	28.	(b)	29.	(b)	30.	(b)	31.	(b)	32.	(a)
33.	(b)	34.	(d)	35.	(c)	36.	(c)	37.	(a)	38.	(c)	39.	(a)	40.	(b)
41.	(b)	42.	(c)	43.	(c)	44.	(b)	45.	(c)	46.	(b)	47.	(b)	48.	(b)
49.	(b)	50.	(b)	51.	(b)	52.	(a)	53.	(c)	54.	(c)	55.	(a)	56.	(a)
57.	(d)	58.	(b)	59.	(b)	60.	(b)	61.	(c)	62.	(b)	63.	(b)	64.	(b)
65.	(a)	66.	(a)	67.	(c)	68.	(b)	69.	(a)	70.	(d)	71.	(b)	72.	(a)
73.	(a)	74.	(c)	75.	(a)	76.	(d)	77.	(c)	78.	(b)	79.	(c)	80.	(b)
81.	(b)	82.	(c)	83.	(a)	84.	(c)	85.	(d)	86.	(b)	87.	(d)	88.	(b)
89.	(d)	90.	(d)	91.	(d)	92.	(d)	93.	(d)	94.	(b)	95.	(c)	96.	(d)
97.	(b)	98.	(d)	99.	(b)	100.	(b)	101.	(b)	102.	(c)	103.	(d)	104.	(a)
105.	(b)	106.	(d)	107.	(c)	108.	(c)	109.	(b)	110.	(b)	111.	(c)	112.	(b)
113.	(c)	114.	(a)	115.	(b)	116.	(b)	117.	(a)	118.	(b)	119.	(b)	120.	(c)
121.	(b)	122.	(b)	123.	(c)	124.	(b)	125.	(b)	126.	(b)	127.	(c)	128.	(c)
129.	(c)	130.	(b)	131.	(d)	132.	(b)	133.	(b)	134.	(a)	135.	(b)	136.	(a)
137.	(b)	138.	(c)	139.	(d)	140.	(b)	141.	(b)	142.	(c)	143.	(a)	144.	(b)
145.	(b)	146.	(c)	147.	(a)	148.	(c)	149.	(c)	150.	(b)	151.	(d)	152.	(c)
153.	(c)	154.	(b)	155.	(c)	156.	(c)	157.	(c)	158.	(b)	159.	(c)	160.	(c)
161.	(b)	162.	(c)	163.	(d)	164.	(d)	165.	(c)	166.	(c)	167.	(b)	168.	(d)
169.	(d)	170.	(b)	171.	(b)	172.	(a)	173.	(a)	174.	(b)	175.	(b)	176.	(d)
177.	(b)	178.	(b)	179.	(c)	180.	(a)	181.	(b)	182.	(d)	183.	(a)	184.	(c)
185.	(c)	186.	(b)	187.	(b)	188.	(b)	189.	(a)	190.	(b)	191.	(d)	192.	(b)
193.	(c)	194.	(a)	195.	(c)	196.	(b)	197.	(b)	198.	(b)	199.	(d)	200.	(e)
201.	(c)	202.	(c)	203.	(b)	204.	(c)	205.	(a)	206.	(c)	207.	(c)	208.	(d)
209.	(d)	210.	(a)	211.	(c)	212.	(a)	213.	(b)	214.	(b)	215.	(d)	216.	(b)
217.	(b)														

Level - 2

1. Each of the six compounds shown at the bottom of the page has two aromatic (benzene) rings. In each case the two rings are different and are labeled A & B. **If an electrophilic substitution, such as nitration or bromination, is carried out on each compound, then identify which ring (A or B) will be preferentially attacked, and indicate the orientation of the substitution (ortho/para, meta or all sites).**

Compound	Reactivity	Substitution	Compound	Reactivity	Substitution
1.	A	ortho/para	2.	A	ortho/para
	B	meta		B	meta
		all sites			all sites
3.	A	ortho/para	4.	A	ortho/para
	B	meta		B	meta
		all sites			all sites
5.	A	ortho/para	6.	A	ortho/para
	B	meta		B	meta
		all sites			all sites

Compound		Compound	
1.		2.	
3.		4.	
5.		6.	

2. When given substituents on a benzene ring, as activating or de-activating and as ortho-para or meta directing for electrophilic aromatic substitution fill the following by appropriate (✓) right or (X) wrong.

	Substituent	Activating	De-activating	Ortho/para	Meta
1.	$-\text{OCH}_3$				
2.	$\begin{array}{c} \text{O} \\ \\ -\text{C}-\text{O}-\text{CH}_3 \end{array}$				
3.	$\begin{array}{c} -\text{O}-\text{C}-\text{CH}_3 \\ \\ \text{O} \end{array}$				
4.	$-\text{CH}_3$				
5.	$-\text{F}$				
6.	$-\text{Ph}$				
7.	$\begin{array}{c} \text{O} \\ \\ -\text{NH}-\text{C}-\text{CH}_3 \end{array}$				
8.	$\begin{array}{c} \text{O} \\ \\ -\text{C}-\text{NH}-\text{CH}_3 \end{array}$				
9.	$-\text{Br}$				
10.	$-\text{CN}$				
11.	$-\text{CF}_3$				
12.	$\begin{array}{c} \text{O} \\ \\ -\text{C}-\text{NH}_2 \end{array}$				
13.	$\begin{array}{c} \text{O} \\ \\ -\text{C}-\text{OH} \end{array}$				
14.	$-\text{CH}=\text{CH}_2$				

15.	$\begin{array}{c} \text{O} \\ \\ \text{CH} = \text{CH} - \text{C} - \text{OH} \end{array}$				
16.	$\begin{array}{c} \text{O} \\ \\ \text{CH} = \text{CH} - \text{C} - \text{H} \end{array}$				
17.	$-\text{S} - \text{Et}$				
18.	$\begin{array}{c} \text{S} \quad \text{Et} \\ \\ \text{O} \end{array}$				
19.	$\begin{array}{c} \text{O} \\ \\ \text{S} \quad \text{Et} \\ \\ \text{O} \end{array}$				
20.	$-\text{N} = \text{O}$				
21.	CH_2X				
22.	CHX_2				

3. Devise a series of reactions to convert benzene into *meta*-chlorobromobenzene.

Select reagents and conditions from the following table, listing them in the order of use.

Compound		Compound		Compound	
1.	sulphuric acid (conc.) heat	5.	Mg in ether	9.	Cu_2Br_2 HBr
2.	Cl_2 FeCl_3 and heat	6.	PBr_3	10.	$(\text{CH}_3\text{CO})_2\text{O}$ + Pyridine
3.	NaNO_2 H_3O^+ 0°C	7.	H_3PO_2		
4.	H_2 Pt catalyst	8.	HNO_3 (conc.) + H_2SO_4 (conc.) and heat		

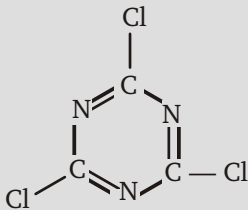
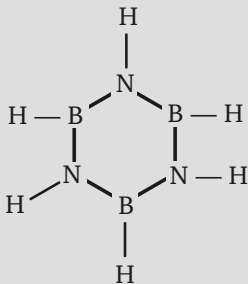
(a) 1 then 2 then 6

(b) 2 then 8 then 4 then 3 then 9

(c) 8 then 4 then 10 then 2 then 3 then 9

(d) 8 then 2 then 4 then 3 then 9

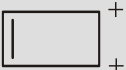
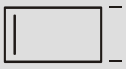
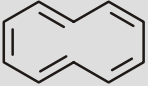
4. Match the Column (I) and Column (II). (Matrix)

Column (I)		Column (II)	
(a)		(p)	Aromatic
(b)		(q)	$(4n - 2)$ electron in a single ring
(c)	$\text{Fe}(\text{C}_5\text{H}_5)_2$	(r)	$4n$ electron in a single ring
(d)	$\text{Cr}(\text{C}_6\text{H}_6)_2$	(s)	Effective atomic number of metal = 36

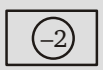
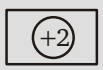
5. Match the Column (I) and Column (II).

Column (I)		Column (II)	
Compound (Monocyclic)		Number of π -electron	
(a)	$\text{C}_4\text{H}_4^{2+}$	(p)	2 e^-
(b)	$\text{C}_4\text{H}_4^{2-}$	(q)	6 e^-
(c)	$\text{C}_9\text{H}_9^{1+}$	(r)	8 e^-
(d)	$\text{C}_9\text{H}_9^{1-}$	(s)	10 e^-

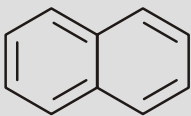
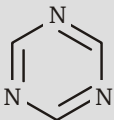
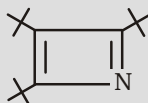
6. Match the Column (I), Column (II) and Column (III). (Matrix)

Column I		Column II		Column III	
(a)		(p)	Aromatic	(w)	$(4n + 2)$ electron. $n = 0, 1, 2, 3$
(b)		(q)	Non-aromatic	(x)	$4n$ electron $n = 1, 2, 3$
(c)		(r)	Anti-aromatic	(y)	Non-planar compound
(d)		(s)	Planar compound	(z)	Readily reacts with active metal

7. Match the Column (I), Column (II) and Column (III). (Matrix)

Column I		Column II		Column III	
(a)		(p)	Readily react with active metal	(w)	Aromatic
(b)		(q)	Readily undergo Dimerization at room temperature	(x)	Anti-aromatic
(c)		(r)	$(4n + 2)$ electron $n = 0, 1, 2, 3$	(y)	Non-aromatic
(d)		(s)	$4n$ electron	(z)	High dipole

8. Among the following compound.

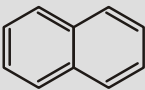
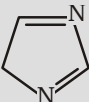
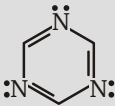
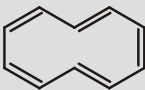
Compound		Compound		Compound	
(a)		(b)		(c)	
(d)		(e)		(f)	
(g)	$C_8H_8^{2+}$	(h)	C_3H_3	(i)	
(j)		(k)		(l)	

- (a) Number of compounds which are aromatic = P
 (b) Number of compounds which are anti-aromatic = Q
 (c) Number of compounds which are non-aromatic = R
 (d) Number of compounds which readily undergo dimerization at room temperature = S
 (e) Number of compound which reacts with active metal = T
 Sum of $P \quad Q \quad R \quad S \quad T$

9. Of the following compounds which will react with Br_2 at room temperature in dark.

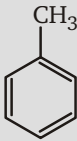
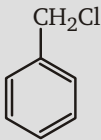
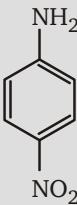
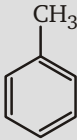
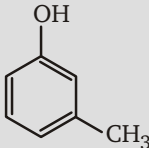
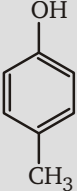
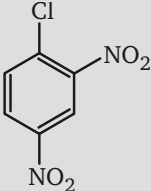
(a)	Benzene (C_6H_6)
(b)	Cyclohexene (C_6H_{10})
(c)	Cyclohexane (C_6H_{12})
(d)	Propanoic Acid ($C_2H_5CO_2H$)
(e)	Phenol (C_6H_5OH)
(f)	Nitrobenzene ($C_6H_5NO_2$)
(g)	Hexyne (C_6H_{10})
(h)	2,2-dichloropropane ($C_3H_6Cl_2$)

10. Among the following compound.

Compound		Compound		Compound	
(a)		(b)	$C_8H_8^2$	(c)	
(d)		(e)		(f)	
(g)		(h)		(i)	$C_3H_3^1$
(j)		(k)		(l)	

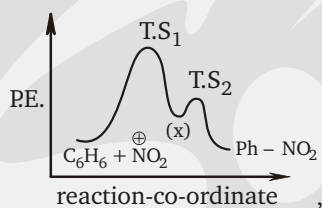
- (a) Number of compounds which are aromatic w
 (b) Number of compounds which are non-aromatic x
 (c) Number of compounds which are anti-aromatic y
 (d) Number of compounds which readily undergo Dimerization at room temperature z
 Sum of $w \ x \ y \ z \dots$

11. Complete the following table.

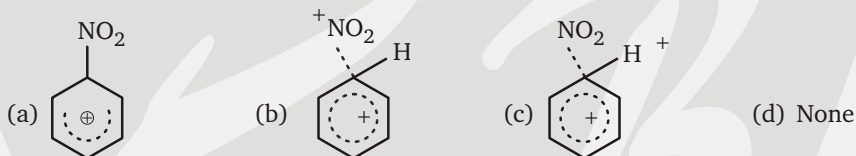
	Reactant	Reagents(s)/Conditions	Major Organic Products
(a)		(A)	
(b)		1. NaNO_2 in dilute H_2SO_4 / 0 – 5 °C 2. heat or boiling	(B)
(c)		SO_3 / conc. H_2SO_4	(C)
(d)	(D)	1. NaOH heated at 330°C 2. dilute H_3O	 and 
(e)		1. aqueous NaOH heated at 60°C 2. dilute H_3O	(E)

12. Comprehension

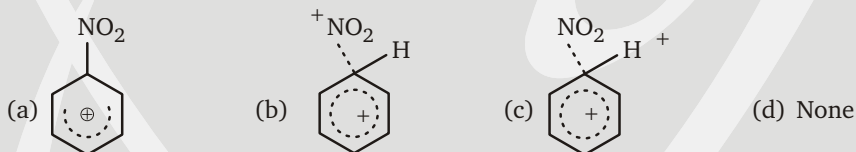
Given is the energy profile diagram of nitration of benzene using mixed acid. (HNO_3 H_2SO_4).



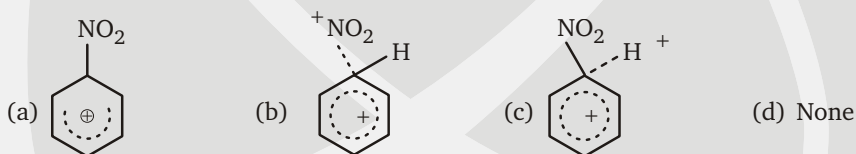
A. Identify (x) in above reaction :



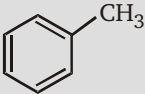
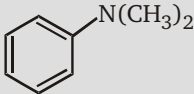
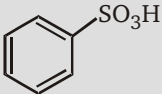
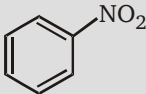
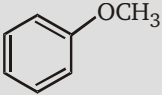
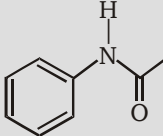
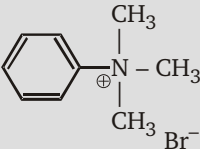
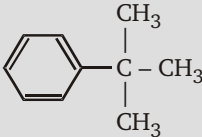
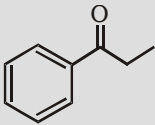
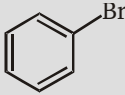
B. Identify T.S₁ in the above reaction.



C. Identify T.S₂ in the above reaction :

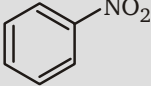
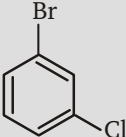
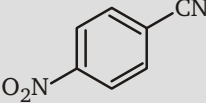
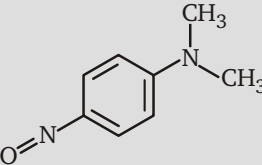
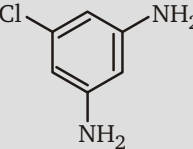
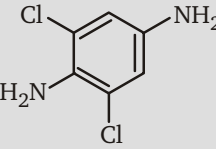


13. Examine the ten structural formulas shown below and select those that satisfy each of the following conditions. Enter one or more letters (a through j) in each answer box, reflecting your choice for each.

Compound		Compound	
a.		b.	
c.		d.	
e.		f.	
g.		h.	
i.		j.	

A.	Which compounds undergo electrophilic nitration more rapidly than benzene ?	
B.	Which compounds give meta substitution under electrophilic bromination conditions ?	

14. Nitrobenzene is a versatile compound that may be converted into a wide variety of substituted benzenes. Five such synthesis are shown below. In each reaction box above an arrow write letters designating the reagents and conditions, selected from the list at the bottom of the page, that would effect the transformation. The reagents must be written in the answer box in the correct order of their use. You may assume appropriate heating or cooling takes place, and more than one equivalent of the reagent may be used if needed.

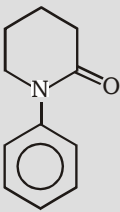
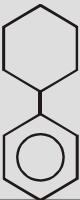
Reactant	Reagent		Product
 nitrobenzene	a.	v.	
	b.	w.	
	c.	x.	
	d.	y.	
	e.	z.	

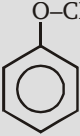
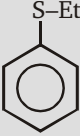
Reagents		Reagents	
A.	H ₂ , Ni catalyst	F.	Cl ₂ & FeCl ₃
B.	KBr & Cu ₂ Br ₂	G.	NaOH 10% solution
C.	KCN & Cu ₂ (CN) ₂	H.	(CH ₃ CO) ₂ O, pyridine
D.	HNO ₂ 0 °C	I.	HNO ₃ /H ₂ SO ₄
E.	CH ₃ I & pyridine		

15. Match the column I and II.

Column (I)		Column (II)	
Group		Effect on phenyl ring	
(a)	$\begin{array}{c} \text{CH} \quad \text{CH} \quad \text{CO}_2\text{H} \end{array}$	(p)	<i>o/p</i> -directors
(b)	$\begin{array}{c} \text{O} \\ \\ \text{O} \quad \text{S} \quad \text{CH}_3 \end{array}$	(q)	meta-directors
(c)	$\begin{array}{c} \text{O} \\ \\ \text{NH} \quad \text{C} \quad \text{CH}_3 \end{array}$	(r)	Activating group
(d)	$\begin{array}{c} \text{S} \quad \text{CH}_3 \\ \\ \text{O} \end{array}$	(s)	De-activating group

16. Match the column I and II.

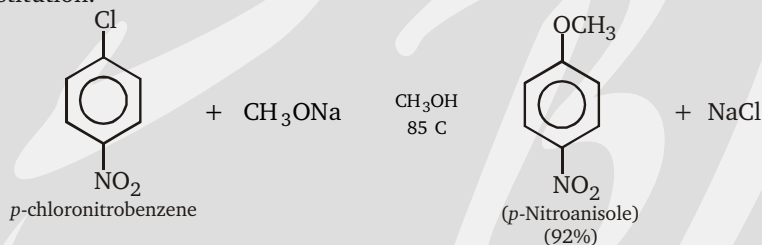
Column (I)		Column (II)	
Group		Effect on phenyl group	
(a)		(p)	Activating group
(b)		(q)	De-activating group

(c)		(r)	<i>o/p</i> -director
(d)		(s)	meta-director

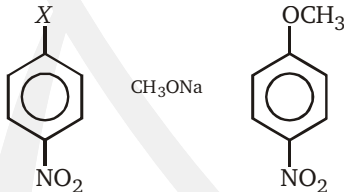
17. Comprehension

Nucleophilic Aromatic substitution ($\text{S}_{\text{N}}\text{Ar}$) :

A substituted benzene derivative containing- NO_2 and Cl group at *p*-position is subjected to Nu-substitution.

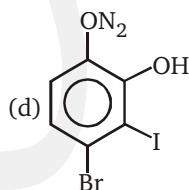
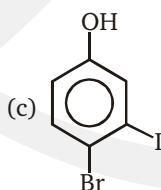
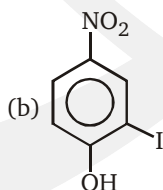
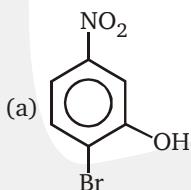
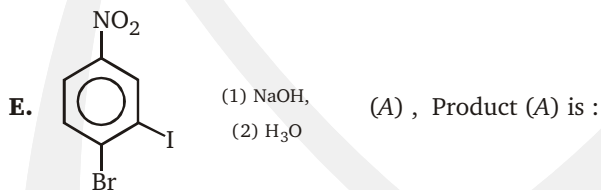
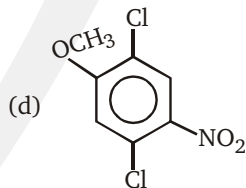
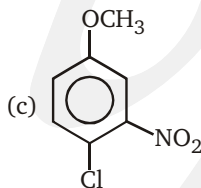
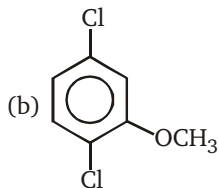
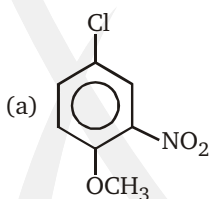
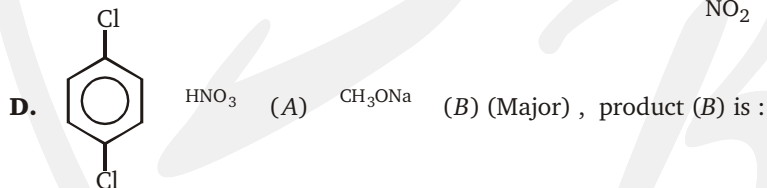
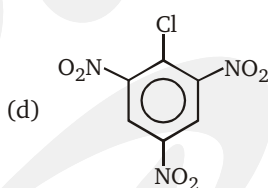
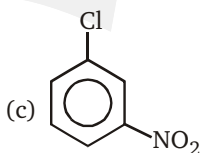
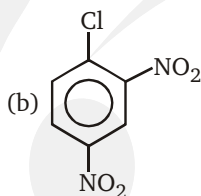
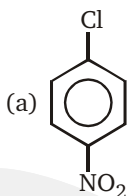


A. Match the column I and II :

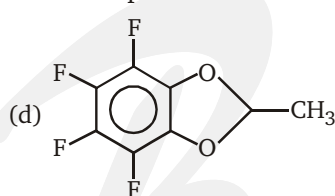
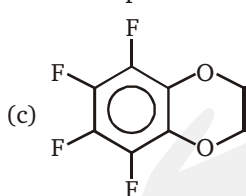
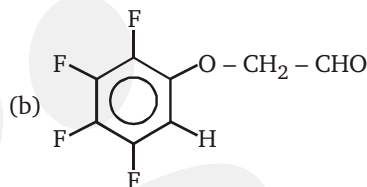
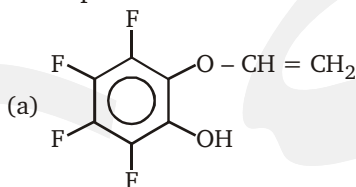
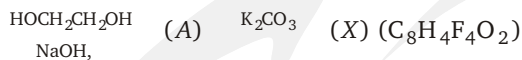
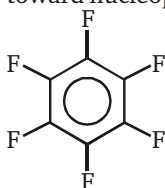


Column (I)		Column (II)	
$\text{X} = \text{halogen}$		relative reactivity toward ($\text{S}_{\text{N}}\text{Ar}$).	
(a)	- F	(p)	312
(b)	- Cl	(q)	1
(c)	- Br	(r)	0.8
(d)	- I	(s)	0.6

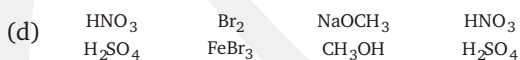
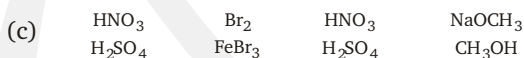
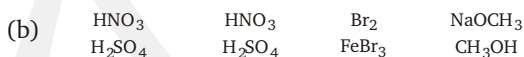
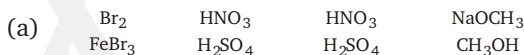
- B.** If step-2 were rate determining step, which halogen of aryl halide is most reactive toward SN_{Ar} .
- (a) Fluoride (b) Chloride (c) Bromide (D) Iodide
- C.** Which of the following is most reactive toward SN_{Ar} .



- F. The cumulative effect of their fluorine activate the rings of penta and hexa fluorobenzene toward nucleophilic aromatic substitution. What is compound X in the following synthesis ?



- G. Which is the best route for the synthesis of starting from benzene of ?

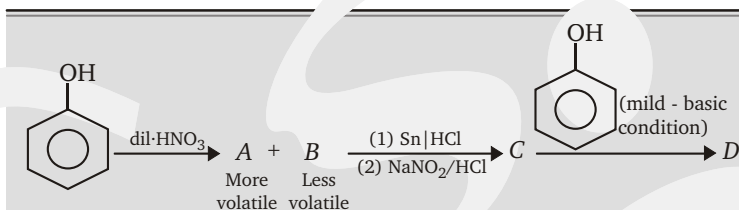


18. Identify product (A) and write its structure.



Subjective Problems

1.



Double bond equivalent of D is :

2. How many isomers 'x' of C_8H_{10} when reacts with hot alkaline KMnO_4 give only aromatic dicarboxylic acid ? How many isomers 'y' of C_4H_8 when reacts with hot alkaline KMnO_4 give carbondioxide ?
Sum of $x+y=?$
3. How many groups are *o/p* director in the electrophilic aromatic substitution ?

(i) $-\text{NH}_2$	(ii) $-\text{COH}$	(iii) $-\text{N} \quad \text{O}$	(iv) $-\text{COOH}$
(v) $-\text{OMe}$	(vi) $-\text{O} - \overset{\text{O}}{\parallel} \text{C} - \text{Me}$	(vii) $-\text{Et}$	(viii) $-\overset{\text{O}}{\parallel} \text{C} - \text{NH} - \text{Me}$
(ix) $-\text{N} \quad \text{NH}_2$	(x) $-\text{SO}_3\text{H}$		

ANSWERS — LEVEL 2

1.

Compound	Reactivity	Substitution
1	B	ortho/para
2	A	ortho/para
3	B	ortho/para
4	A	all position
5	B	meta
6	B	ortho/para

2.

	Substituent	Activating	De-activating	Ortho/para	Meta
1.	$-\text{OCH}_3$	✓	✗	✓	✗
2.	$\begin{array}{c} \text{O} \\ \parallel \\ -\text{C}-\text{O}-\text{CH}_3 \end{array}$	✗	✓	✗	✓
3.	$\begin{array}{c} \text{O} \\ \parallel \\ -\text{O}-\text{C}-\text{CH}_3 \end{array}$	✓	✗	✓	✗
4.	$-\text{CH}_3$	✓	✗	✓	✗
5.	$-\text{F}$	✗	✓	✓	✗
6.	$-\text{Ph}$	✓	✗	✓	✗
7.	$\begin{array}{c} \text{O} \\ \parallel \\ -\text{NH}-\text{C}-\text{CH}_3 \end{array}$	✓	✗	✓	✗
8.	$\begin{array}{c} \text{O} \\ \parallel \\ -\text{C}-\text{NH}-\text{CH}_3 \end{array}$	✗	✓	✗	✓
9.	$-\text{Br}$	✗	✓	✓	✗
10.	$-\text{CN}$	✗	✓	✗	✓

11.	CF_3	X	✓	X	✓
12.	$\begin{array}{c} \text{O} \\ \\ \text{C} \text{ NH}_2 \end{array}$	X	✓	X	✓
13.	$\begin{array}{c} \text{O} \\ \\ \text{C} \text{ OH} \end{array}$	X	✓	X	✓
14.	$\text{CH}=\text{CH}_2$	✓	X	✓	X
15.	$\begin{array}{c} \text{O} \\ \\ \text{CH}=\text{CH} \text{ C} \text{ OH} \end{array}$	X	✓	✓	X
16.	$\begin{array}{c} \text{O} \\ \\ \text{CH}=\text{CH} \text{ C} \text{ H} \end{array}$	X	✓	✓	X
17.	$-\text{S}-\text{Et}$	✓	X	✓	X
18.	$\begin{array}{c} \text{S} \text{ Et} \\ \\ \text{O} \end{array}$	X	✓	✓	X
19.	$\begin{array}{c} \text{O} \\ \\ \text{S} \text{ Et} \\ \\ \text{O} \end{array}$	X	✓	X	✓
20.	$-\text{N}=\text{O}$	X	✓	✓	X
21.	CH_2X	X	✓	X	✓
22.	CHX_2	X	✓	X	✓

3. d

4. a - p, q; b - p, q; c - p, q, s; d - p, q, s

5. a - q; b - p; c - r; d - s

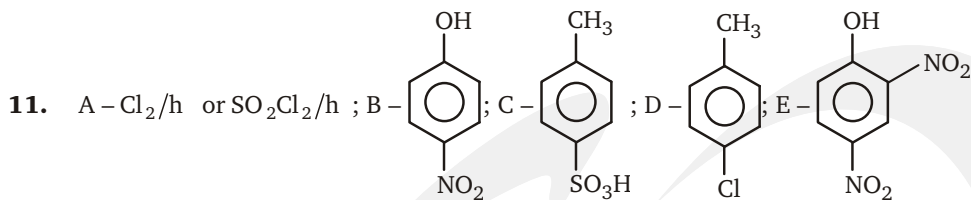
6. a - p, s - w; b - p, s - w; c - q - x, y, z; d - q - w - y

7. a - p, q, s - x; b - p - s - y; c - r - w, z; d - r - w, z

8. P Q R S T 18

9. b, e

10. w x y z 14



12. A - a; B - b; C - c

13. A - a, b, e, f, h; B - c, d, g, i (Note : yet $\text{C}_6\text{H}_5\text{Br}$ is less reactive than C_6H_6 but *o/p* directing)

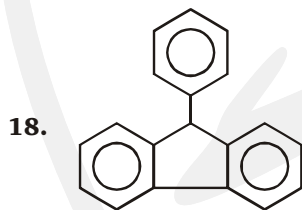
14. v - F, A, D, B; w - A, H, I, G, D, C; x - A, E, D; y - F, I, A or I, F, A;

z - A, H, I, F, G, A or A, H, I, F, A, G

15. a - p, s; b - p, r; c - p, r; d - p, s

16. a - p, r; b - p, r; c - p, r; d - p, r

17. A - a - p, b - q, c - r, d - s; B - d; C - d; D - a; E - b; F - c; G - a

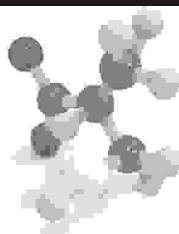


Subjective Problems

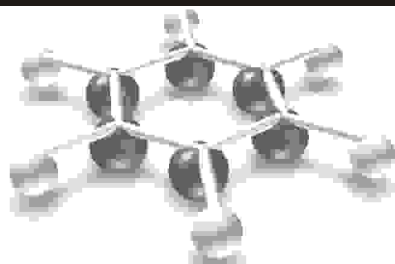
1. 9

2. 5

3. 6



13



PRACTICAL ORGANIC CHEMISTRY



Level - 1

1. and ; Compounds (X) and (Y) can be differentiated by :

(X) (Y)

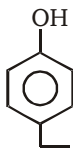
(a) H_3O^+ , NaOI (b) H_3O^+ , then Fehling test
(c) H_3O^+ , then Na (d) Both (b) and (c)
2. Compound $\text{CH}_3 - \text{CH} \begin{matrix} \text{OEt} \\ \text{OEt} \end{matrix}$ and $\text{CH}_3 - \text{CH}_2 - \text{O} - \text{CH}_2 - \text{CH}_3$ can be differentiated by :
(P) (Acetal) (Q)

(a) H_3O^+ , Na (b) H_3O^+ , Tollens' test
(c) H_3O^+ , Fehling test (d) All of these
3. and can be differentiated by :

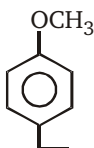
(aniline) (cyclohexyl amine)

(a) Hinsberg test (b) Iso-cyanide test
(c) NaNO_2 , HCl, then β -Naphthol (d) NaOH

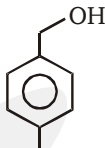
4.



(p-ethyl phenol)



(p-methyl anisole)

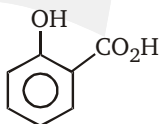


(p-ethyl benzyl alcohol)

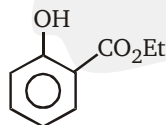
Above compounds can be differentiated by using the reagent:

- (a) NaOH, Tollen's reagent, FeCl_3 (b) CrO_3 , Tollen's reagent, FeCl_3
 (c) Tollen's reagent, CrO_3 , FeCl_3 (d) Na, Tollen's reagent, FeCl_3

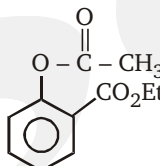
5.



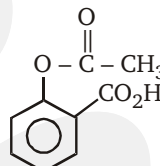
Salicyclic acid



Ethyl salicylate



Ethyl acetyl

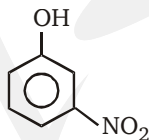


acetyl salicyclic acid

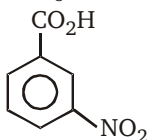
Above compounds can be differentiated by the salicylate. Which of the following chemical test? (used in decreasing order)

- (a) NaOH, FeCl_3 , NaHCO_3 (b) aq. NaHCO_3 , FeCl_3 , NaOH
 (c) NaOI, NaOH, NaHCO_3 (d) NaOH, Na, NaHCO_3

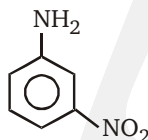
6.



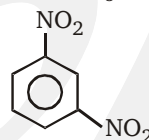
(m-nitrophenol)



(m-nitro benzoic acid)



(m-nitro aniline)

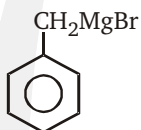


(m-dinitro benzene)

Above compounds can be differentiated by which of the following chemical test? (used in decreasing order)

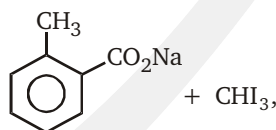
- (a) NaOH, NaHCO_3 , HCl (b) HCl, NaOH, NaHCO_3
 (c) NaHCO_3 , NaOH, HCl (d) NaOH, HCl, NaHCO_3

7.

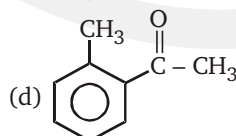
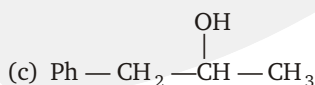
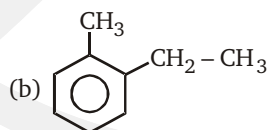
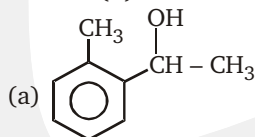


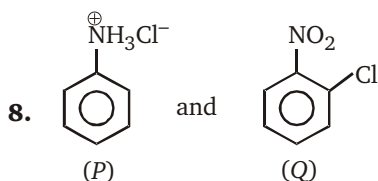
- (i) CH_3CHO
 (ii) NH_4Cl

(A)



Product (A) in the above reaction is :

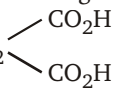
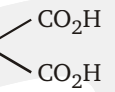




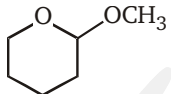
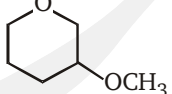
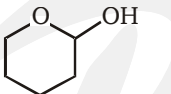
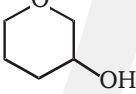
Above compounds (P) & (Q) can be differentiated by :

- (a) amm. AgNO_3 (b) NaOH
 (c) FeCl_3 (d) Both (a) & (b)

9. Which of following acid give positive Tollen's reagent test.

- (a) $(\text{CH}_2)_2$  (b) CH_2  (c)  (d) HCO_2H

10. Which of following compounds give positive Tollen's test?

- (a)  (b)  (c)  (d) 

11. Give a simple test to differentiate cyclohexane and cyclohexene

- (a) $\text{Br}_2/\text{H}_2\text{O}$ (b) Bayer's reagent
 (c) Tollen's reagent (d) Both (a) and (b)

12. Give test to differentiate (Bromobenzene) $\text{Ph}-\text{Br}$ and benzyl bromide (PhCH_2Br).

- (a) (i) aq. KOH (ii) Na (b) AgNO_3
 (c) KMnO_4 (d) All these

13. Give test to differentiate 1,1-dichloroethane and 1,2-dichloroethane :

- (a) 2,4 -DNP then aq. KOH (b) aq. KOH then 2, 4-DNP
 (c) NaHSO_3 (d) Lucas reagent

14. Test to differentiate between (CH_3OH) and $(\text{Ph}-\text{OH})$ is/are :
 (methanol) (Phenol)

- (a) Litmus test (b) FeCl_3
 (c) $\text{Br}_2/\text{H}_2\text{O}$ (d) All of these

15. Acetaldehyde and benzaldehyde can be differentiated by :

- (a) Fehling test (b) Iodoform test
 (c) Tollen's reagent (d) both (a) and (b)

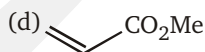
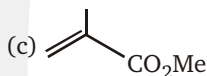
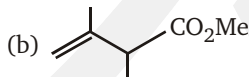
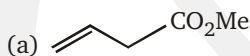
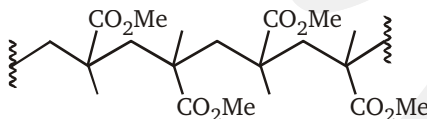
16. Ethylamine and diethylamine cannot be differentiated by :

- (a) Hinsberg test (b) carbylamine test
 (c) Iodoform test (d) both (a) and (b)

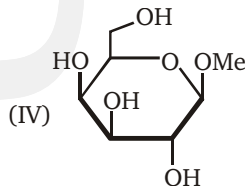
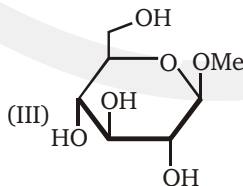
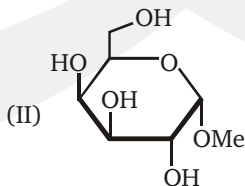
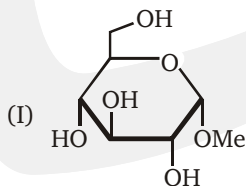
17. Lassaigne's test for the detection of nitrogen will fail in the case of :

- (a) NH_2CONH_2 (b) $\text{NH}_2\text{CONHNH}_2 \cdot \text{HCl}$
 (c) $\text{NH}_2\text{NH}_2 \cdot \text{HCl}$ (d) $\text{C}_6\text{H}_5\text{NHNH}_2 \cdot 2\text{HCl}$

18. Sodium nitroprusside when added to an alkaline solution of sulphide ions produces a colouration which is :
 (a) red (b) blue (c) brown (d) purple
19. In Kjeldahl's method, nitrogen present is estimated as :
 (a) N_2 (b) NH_3 (c) NO_2 (d) none of these
20. In Kjeldahl's method of estimation of nitrogen, K_2SO_4 acts as :
 (a) an oxidising agent (b) catalytic agent
 (c) hydrolysing agent (d) boiling point elevator
21. The prussian blue colour obtained during the test of nitrogen by Lassaigne's test is due to the formation of :
 (a) $Fe[Fe(CN)_6]_3$ (b) $Na_3[Fe(CN)_6]$
 (c) $Fe(CN)_3$ (d) $Na_4[Fe(CN)_5NOS]$
22. A compound which does not give a positive test in Lassaigne's test for nitrogen is:
 (a) urea (b) hydrazine (c) azobenzene (d) phenyl hydrazine
23. *p*-nitrophenol and *o*-nitrophenol are separated by :
 (a) distillation (b) steam distillation
 (c) crystallization (d) fractional crystallization
24. Which of the following reagent is used for the separation of acetaldehyde from acetophenone ?
 (a) NH_2OH (b) $NaOI$ (c) Tollen's reagent (d) $C_6H_5NHNH_2$
25. The formula of gas is $[CO]_x$. If its vapour density is 70, the value of x will be :
 (a) 2.5 (b) 3.0 (c) 5.0 (d) 6.0
26. The structure of the monomer that would give the following polymer by an addition mechanism is :

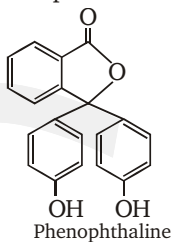


27. Identify the correct set of stereochemical relationships amongst the following monosaccharides I – IV

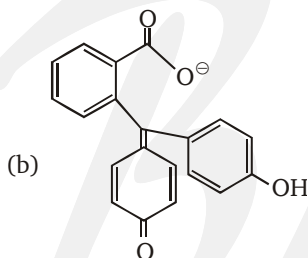
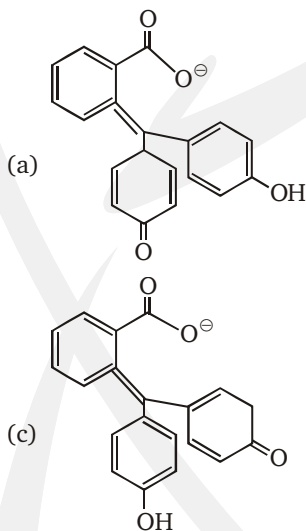


- (a) I and II are anomers ; III and IV are epimers
 (b) I and II are epimers ; III and IV are anomers
 (c) I and III are anomers ; I and II are epimers
 (d) I and III are epimers ; II and IV are anomers

28. A dye, phenolphthalein is prepared by reacting phenol with phthalic anhydride in acidic medium. It give pink colour in alkaline medium due to extended conjugation in a new complex formed (phthalein - dye test) identify the complex A :


 $\ominus \text{OH}$

(A) Product ; The structure of (A) is:



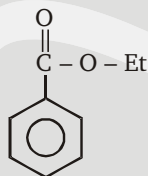
(d) None

ANSWERS — LEVEL 1

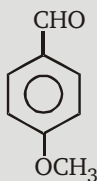
1.	(d)	2.	(d)	3.	(c)	4.	(b)	5.	(b)	6.	(c)	7.	(a)	8.	(d)
9.	(d)	10.	(c)	11.	(d)	12.	(d)	13.	(b)	14.	(d)	15.	(d)	16.	(c)
17.	(c)	18.	(b)	19.	(b)	20.	(d)	21.	(d)	22.	(b)	23.	(a)	24.	(c)
25.	(c)	26.	(c)	27.	(c)	28.	(b)								

Level - 2

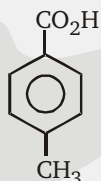
1. Comprehension



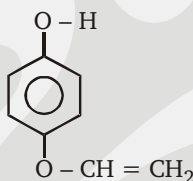
(a)



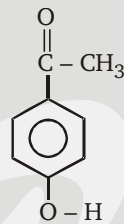
(b)



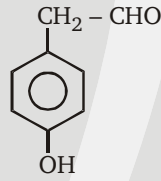
(c)



(d)



(e)

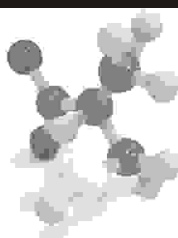


(f)

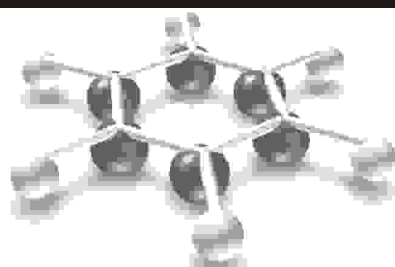
- A.** Which isomer gives positive iodoform test ?
 (a) a (b) b
 (c) d (d) e
- B.** Which isomer gives +ive Tollen's test, also reacts with FeCl_3 ?
 (a) b (b) f
 (c) c (d) d
- C.** Which isomer reacts with NaHCO_3 ?
 (a) c (b) d
 (c) e (d) f
- D.** Which isomer on hydrolysis gives 1, 4-di hydroxybenzene ?
 (a) a (b) d
 (c) e (d) f
- 2.** $\text{Ph} - \overset{\text{O}}{\parallel}{\text{C}} - \text{OH} \xrightarrow{\text{NaHCO}_3} \text{(A) gas} ; \text{Ph} - \text{OH} \xrightarrow{\text{Na}} \text{(B) gas}$
 Sum of molecular mass of gas ($A+B=?$)

ANSWERS — LEVEL 2

1. A - d; B - b; C - a; D - b
 2. 48



14



BIOMOLECULES

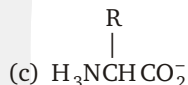
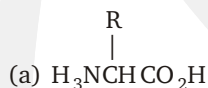
Level - 1

1. Which statement correctly completes the statement ?

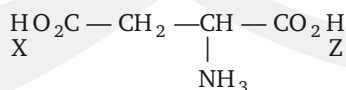
Except for glycine, which is achiral, all the amino acids present in proteins

- (a) are chiral, but racemic
- (b) have the L configuration at their carbon
- (c) have the R configuration at their carbon
- (d) have the S configuration at their carbon

2. Assume that a particular amino acid has an isoelectric point of 6.0. In a solution at pH 1.0, which of the following species will predominate ?



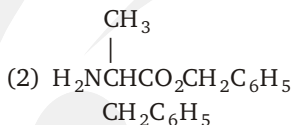
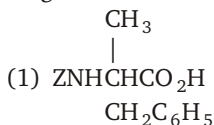
3. The pK_a values for the three ionizable groups X, Y and Z of glutamic acid are 4.3, 9.7 and 2.2 respectively



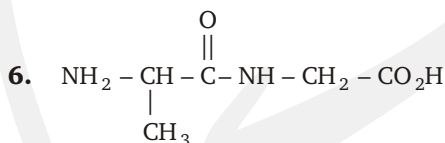
The isoelectric point for the amino acid is :

- (a) 7.00
- (b) 3.25
- (c) 4.95
- (d) 5.95

4. An amino acid may be represented by general formula $\text{H}_2\text{N}-\overset{\text{R}}{\underset{|}{\text{CH}}}-\text{COOH}$. If $\text{R} = \text{CH}_2\text{C}_6\text{H}_5$ then it is phenylalanine (Phe) and if $\text{R} = \text{CH}_3$ then it is alanine (Ala). Find the sequence of reagents from those given below to synthesize Phe - Ala.

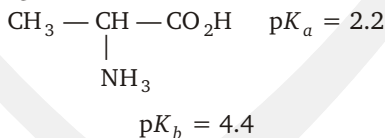


- (a) 1 and 2 (b) 1 and 4 (c) 2 and 3 (d) 3 and 4
5. Iso-electric point of alanine is (pH = 6). At which pH, maximum concentration of zwitter ion of alanine will be present ?
- (a) pH > 6 (b) pH < 6 (c) pH = 6 (d) pH = 7

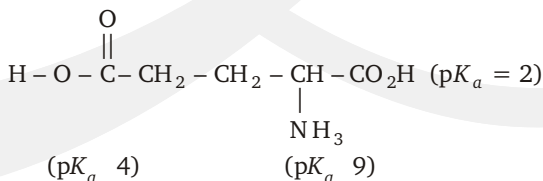


Identify the amino acid obtained by hydrolysis of the above compound:

- (a) Glycine (b) Alanine (c) Both (a) and (b) (d) None of these
7. At iso-electric point :
- (a) conc. of cation is equal to conc of anion
(b) Net charge is zero.
(c) Maximum conc. of di-polar ion (Zwitter ion) will be present
(d) All of the above
8. Which of following amino acid has lowest iso-electric point ?
- (a) Glycine (b) Alanine (c) Aspartic acid (d) Lysine
9. Find iso-electric point of given amino acid



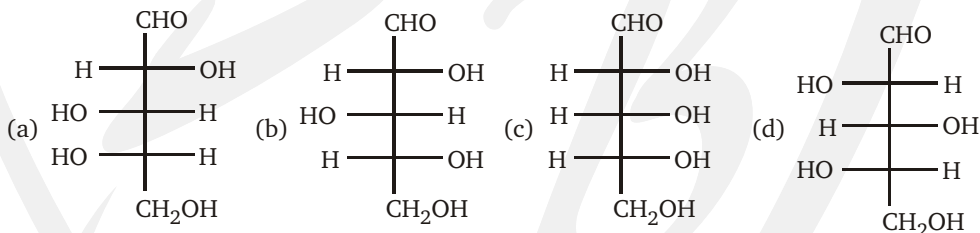
- (a) 3.3 (b) 5.9 (c) 9.6 (d) 11.8
10. Find iso-electric point of the given amino acid



- (a) 5.5 (b) 6.5 (c) 3 (d) 5

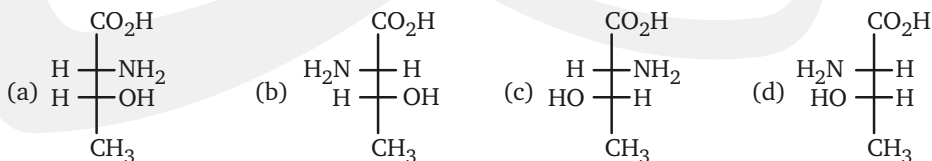
11. $\text{H}-\text{C}=\text{C}-\text{H}$ $\xrightarrow[\text{H}_2\text{SO}_4]{\text{HgSO}_4}$ (A) $\xrightarrow[\text{(2) H}_2\text{O}]{\text{(1) NH}_3, \text{HCN}}$ (B); Product (B) of given reaction is :
- (a) Glycine (b) Alanine
(c) valine (d) Leucine
12. Which amino acid does not contain chiral centre ?
(a) Valine (b) Leucine (c) Glycine (d) Iso-leucine
13. Which of the following is Sanger reagent ?
(a) 2,4-Di-nitro fluorebenzene (b) Phenyl isocyanate
(c) 2, 4-Di-nitro chlorobenzene (d) 2, 4-Di-nitro iodobenzene
14. A D-carbohydrate is :
(a) Always dextrorotatory
(b) Always laevorotatory
(c) Always the mirror of the corresponding L-carbohydrate
(d) None of these

15. Which L-sugar on oxidation gives an optically active dibasic acid (2 COOH groups) ?

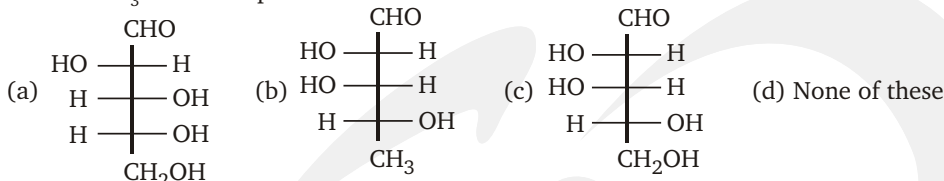


16. $\begin{array}{c} \text{CH} = \text{N} - \text{NH} - \text{Ph} \\ | \\ \text{C} = \text{N} - \text{NH} - \text{Ph} \\ | \\ \text{HO} - \text{C} - \text{H} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{CH}_2\text{OH} \end{array}$ The given osazone can be obtained by :

- (a) D-glucose (b) D-mannose (c) D-Idose (d) Both (a) & (b)
17. Which of the following pair gives same phenyl osazone ?
(a) D-Glucose and D-Allose (b) D-Glucose and D-Alfrose
(c) D-Glucose and D-Mannose (d) D-Glucose and D-Talose
18. Which of the following is the Fischer projection of L-threonine (also known as (2S, 3R)-2-amino-3-hydroxybutanoic acid) ?



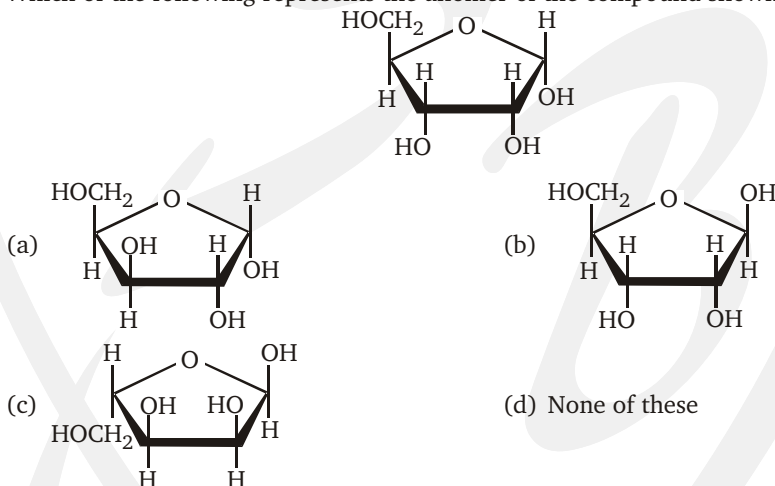
19. Among the three compounds shown below, two yield the same product on reaction with warm HNO_3 . The exception is :



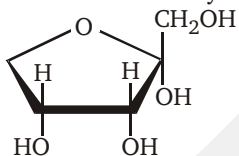
20. The optical rotation of the α -form of a pyranose is $+150.7^\circ$, that of the β -form is $+52.8^\circ$. In solution an equilibrium mixture of these anomers has an optical rotation of $+80.2^\circ$. The percentage of the α -form in equilibrium mixture is :

- (a) 28% (b) 32% (c) 68% (d) 72%

21. Which of the following represents the anomer of the compound shown ?



22. Which set of terms correctly identifies the carbohydrate shown ?



- (1) Pentose (2) Pentulose (3) Hexulose (4) Hexose
(5) Aldose (6) Ketose (7) Pyranose (8) Furanose
(a) 2, 6, 8 (b) 2, 6, 7
(c) 1, 5, 8 (d) A set of terms other than these

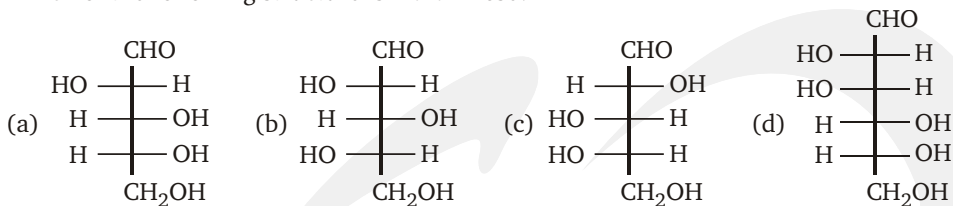
23. For the complex conversion of D-glucose into the corresponding osazone, the minimum number of equivalents of phenyl hydrazine required is :

- (a) two (b) three (c) four (d) five

24. Which one of the following compounds will form an osazone derivative ?

- (a) $\text{CH}_3\text{CH}_2\text{COCH}_2\text{OH}$ (b) $\text{CH}_3\text{COCH}_2\text{CH}_2\text{OH}$
(c) $\text{CH}_3\text{CH}_2\text{CHOHCH}_2\text{OH}$ (d) $\text{CH}_3\text{CH}_2\text{COCH}_2\text{OCH}_3$

25. Which of the following structure is L-arabinose?

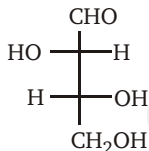


26. Which one of the statements concerning the equilibrium shown is true ?



- (a) The two structures are enantiomers of each other. They have equal but opposite optical rotations and racemize slowly at room temperature
- (b) The two structures are enantiomers of each other. They racemize too rapidly at room temperature for their optical rotations to be measured
- (c) The two structures are diastereomers of each other. Their interconversion is called mutarotation
- (d) The two structures are diastereomers of each other. Their interconversion does not require breaking and making bonds, only a change in conformation

27. The configurations of the chirality centres in D-threose (shown) are :

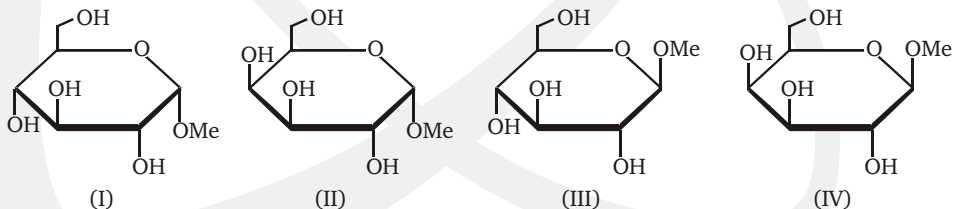


- (a) 2R, 3R (b) 2R, 3S (c) 2S, 3R (d) 2S, 3S

28. Rapid interconversion of α -D-glucose and β -D-glucose in solution is known as :

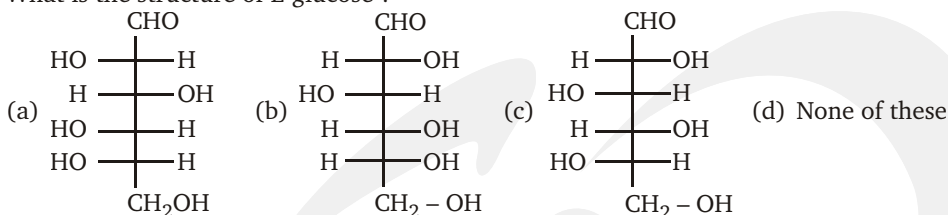
- (a) racemization (b) asymmetric induction
- (c) fluxional isomerization (d) mutarotation

29. Identify the correct set of stereochemical relationships amongst the following monosaccharides I-IV.

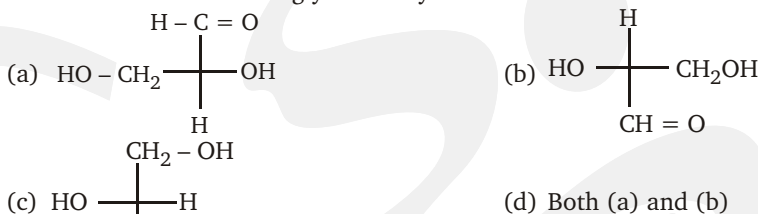


- (a) I and II are anomers ; III and IV are epimers
- (b) I and III are epimers ; II and IV are anomers
- (c) I and II are epimers ; III and IV are anomers
- (d) I and III are anomers ; I and II are epimers

30. What is the structure of L-glucose ?



31. What is the structure of L-glyceraldehyde?



32.
$$\begin{array}{c} \text{HC} - \text{OH} \\ || \\ \text{C} - \text{OH} \\ | \\ \text{HO} - \text{C} - \text{H} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{H} - \text{C} - \text{OH} \\ | \\ \text{CH}_2\text{OH} \end{array}$$
, the given is enol form of :

- (a) D-glucose (b) D-mannose (c) D-fructose (d) All of these

33. D-glucose $\xrightleftharpoons{\text{HO}}$ A + B; A and B are :

- (a) D-mannose & D-mannitol (b) D-mannose & D-Fructose
(c) D-Allose & D-Altrose (d) D-Glucose & D-Idose

34. Stereoisomers of aldohexose is (a) and stereoisomers of ketohexose is (b).

Ratio of a/b is :

- (a) $\frac{1}{2}$ (b) $\frac{2}{1}$ (c) $\frac{4}{1}$ (d) $\frac{1}{4}$

35. D-Glucose $\xrightarrow{\text{HNO}_3}$ (A); Product (A) is :

- (a) D-Gluconic acid (b) D-Glucitol (c) D-Fructose (d) D-Glucaric acid

36. D-glucose & D-fructose can be differentiated by :

- (a) Fehling solution (b) Tollens reagent (c) Benedict test (d) $\text{Br}_2/\text{H}_2\text{O}$

37. D-Glucose exist in x different forms. The value of x (stereoisomer) is :

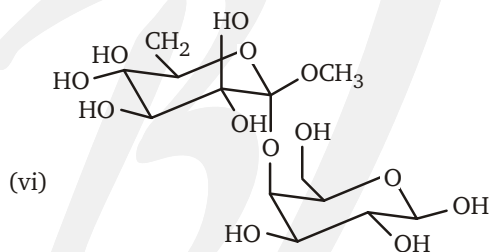
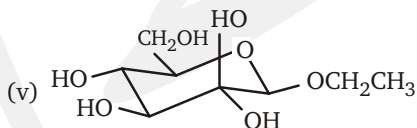
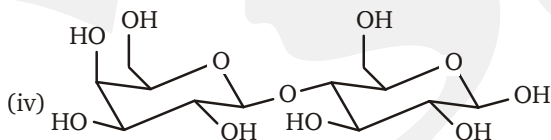
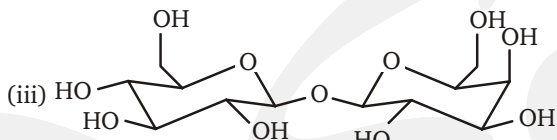
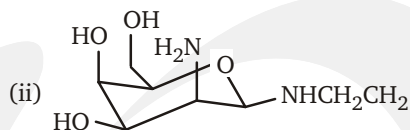
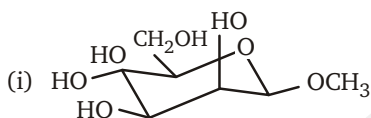
- (a) 2 (b) 3 (c) 4 (d) 5

38. D-Mannose $\xrightleftharpoons{\text{HO}}$ D-Glucose $\xrightleftharpoons{\text{HO}}$ (A) ;

Product (A) of above reaction is

- (a) D-glucose (b) D-fructose (c) D-talose (d) D-idose

39. Which of the molecules below will react with Ag ?



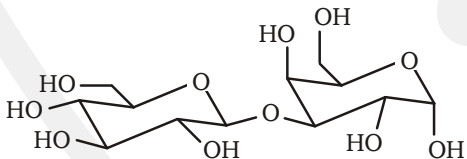
(a) (i), (iii) and (v)

(b) (ii) and (iv)

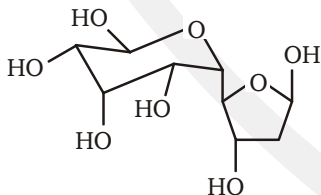
(c) (iv) and (vi)

(d) (i), (ii), (iii) and (vi)

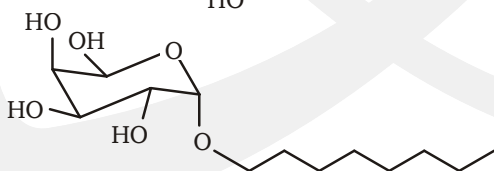
40. A.



B.



C.



D.

(a) Compound A (b) Compound B (c) Compound C (d) Compound D
(e) None of the above (they are all hemiacetals)

(a) $\begin{array}{c} \text{CH}_2\text{OH} \\ | \\ \text{OH}-\text{C}-\text{CHO} \\ | \\ \text{H} \end{array}$ (b) $\begin{array}{c} \text{CH}_2\text{OH} \\ | \\ \text{H}-\text{C}-\text{OH} \\ | \\ \text{CHO} \end{array}$ (c) $\begin{array}{c} \text{CHO} \\ | \\ \text{OH}-\text{C}-\text{CH}_2\text{OH} \\ | \\ \text{H} \end{array}$ (d) $\begin{array}{c} \text{CHO} \\ | \\ \text{H}-\text{C}-\text{CH}_2\text{OH} \\ | \\ \text{HO} \end{array}$

C(C1=CC=CC=C1)CCC(C2=CC=CC=C2)CCC(C3=CC=CC=C3)CCC(C4=CC=CC=C4)CCC

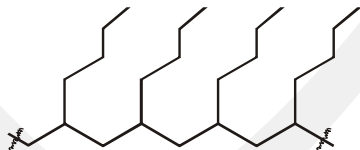
(a) 

(b) 

(c) 

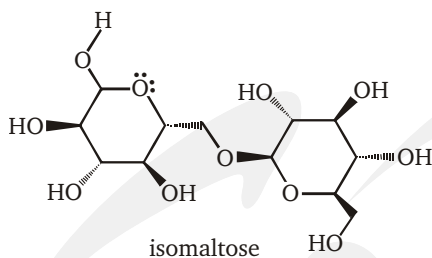
(d) 

43. The following structure represents a subunit of a hydrocarbon polymer that may be prepared by a radical polymerization method. Identify the monomer that has been polymerized to make this polymer chain.



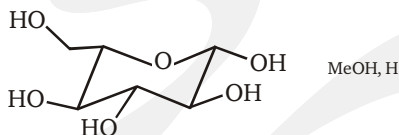
(a) (b) (c) (d)

44. Choose the answer that has correctly identified the number of acetals and hemiacetals in isomaltose.



Acetal	Hemiacetal	Acetal	Hemiacetal
(a) 0	0	(b) 1	0
(c) 0	1	(d) 1	1

45. Predict the product of the following reaction.



- (a)
- (b)
- (c)
- (d)

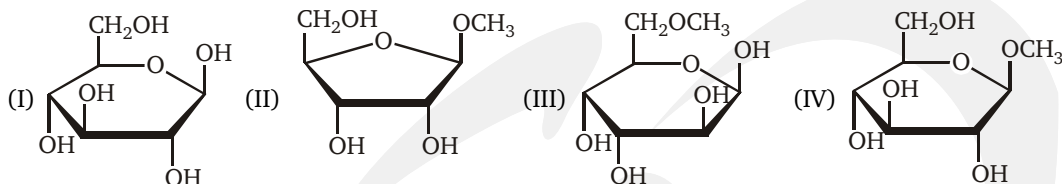
46. Which reagent/s can be used to distinguish glucose and fructose ?

- (I) Bromine water (II) Tollen's reagent (III) Schiff's reagent
 (a) (I), (II) and (III) (b) (II) and (III) (c) Only (I) (d) Only (III)

47. Choose the peptide that matches the abbreviation Phe-Val-Ala.

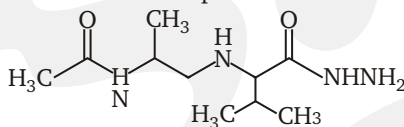
- (a)
- (b)
- (c)
- (d)

48. Which of the following carbohydrate(s) would not undergo mutarotation in aqueous solution?



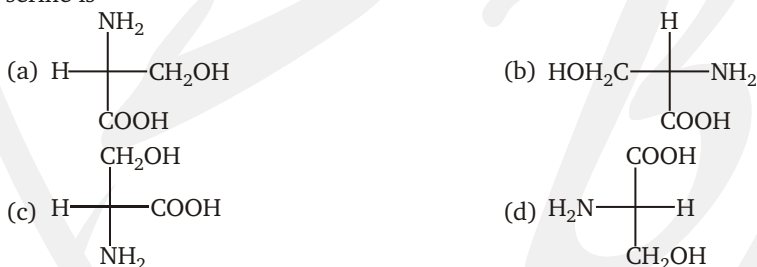
- (a) II only (b) I, III and IV only (c) II and IV only (d) I and III only

49. The number of peptide bonds in the compound.



- (a) 1 (b) 2 (c) 3 (d) 4

50. Serine ($\text{HOCH}_2\text{CH}(\text{NH}_2)\text{COOH}$) is an essential amino acid. The correct Fisher projection of serine is



ANSWERS — LEVEL 1

1.	(b)	2.	(a)	3.	(b)	4.	(b)	5.	(c)	6.	(c)	7.	(d)	8.	(c)
9.	(b)	10.	(c)	11.	(b)	12.	(c)	13.	(a)	14.	(d)	15.	(a)	16.	(d)
17.	(c)	18.	(b)	19.	(b)	20.	(a)	21.	(b)	22.	(a)	23.	(b)	24.	(a)
25.	(c)	26.	(c)	27.	(c)	28.	(d)	29.	(d)	30.	(a)	31.	(d)	32.	(d)
33.	(b)	34.	(b)	35.	(d)	36.	(d)	37.	(b)	38.	(b)	39.	(c)	40.	(e)
41.	(c)	42.	(b)	43.	(c)	44.	(d)	45.	(a)	46.	(c)	47.	(a)	48.	(c)
49.	(a)	50.	(d)												

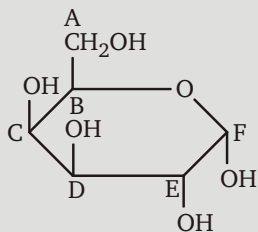


Level - 2

1. Match the Column (I) and Column (II). (Matrix)

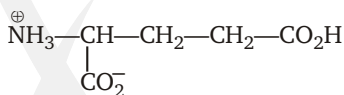
Column (I)		Column (II)	
	Molecule		Configuration
(a)	$ \begin{array}{c} \text{CHO} \\ \\ \text{H} - \text{C} - \text{OH} \\ \\ \text{CH}_2\text{OH} \end{array} $	(p)	R- (Rectus)
(b)	$ \begin{array}{c} \text{CHO} \\ \\ \text{HO} - \text{C} - \text{H} \\ \\ \text{CH}_2\text{OH} \end{array} $	(q)	S- (Sinister)
(c)	$ \begin{array}{c} \text{NH}_2 \\ \\ \text{H} - \text{C} - \text{CO}_2\text{H} \\ \\ \text{CH}_3 \end{array} $	(r)	D
(d)	$ \begin{array}{c} \text{NH}_2 \\ \\ \text{H} - \text{C} - \text{CH}_3 \\ \\ \text{CO}_2\text{H} \end{array} $	(s)	L

2. Comprehension



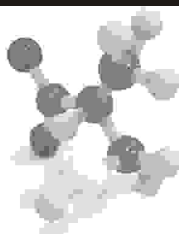
One cyclic acetal form of D-galactose is shown above.

- A.** Which atom is the anomeric carbon ?
 (a) Atom A (b) Atom B (c) Atom C (d) Atom D
 (e) Atom E (f) Atom F
- B.** Which name most completely describes this cyclic acetal form ?
 (a) -D-Galactofuranose (b) -P- Galactofuranose
 (c) -D- Galactopyranose (d) -D- Galactopyranose
- 3.** How many compound which is given below is isomer of *D*-Glucose ?
D-Mannose, *D*-Fructose, *D*-Gulose, *D*-Idose, *D*-Galactose, *D*-Arabinose, *D*-Ribose.
- 4.** How many acidic group is present in given amino acid ?

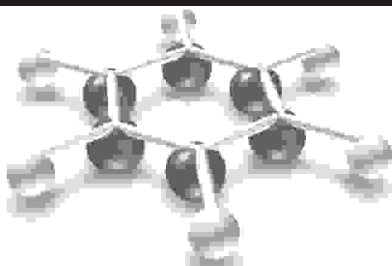


ANSWERS — LEVEL 2

1. a – p, r; b – q, s; c – q, s; d – p, r
2. A – f; B – c
3. 5
4. 2



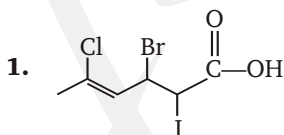
15



IUPAC NAME



Level - 1

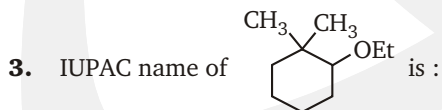


Total number of substituent present in the above compound :

- (a) 1 (b) 2
(c) 3 (d) 4

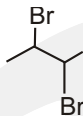


- (a) Hex-5-en-1-yne (b) Hex-1-en-5-yne
(c) Hex-6-en-1-yne (d) Hex-1-en-6-yne

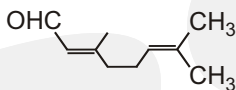


- (a) 1-Ethoxy-2, 2-dimethylcyclohexane (b) 2-ethoxy-1, 1-dimethyl cyclohexane
(c) 1, 1-Dimethyl-2-ethoxycyclohexane (d) 2-methyl-1, 1-ethoxy cyclohexane

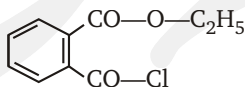
4. How many secondary carbon and hydrogen atoms are present in the molecule given below respectively ?



- (a) 2, 3 (b) 2, 2 (c) 3, 3 (d) 2, 0
5. Which IUPAC name is correct for the given compound ?

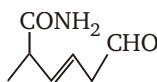


- (a) 3, 7-dimethylocta-2, 6-dienal (b) 2, 6-dimethyloct-2, 6-dienal-8
(c) 7-formyl-2, 6-dimethylhept-2, 6-diene (d) 7-aldo-2,6-dimethylhept-2, 6-diene
6. Write the IUPAC name of the following compound :



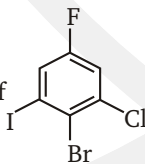
- (a) ethyl-2-(chlorocarbonyl) benzoate (b) ethyl-2-(chlorocarbonyl) hexanoate
(c) 2-(thoxycarbonyl) benzoyl chloride (d) None of these
7. The IUPAC name of the compound is :
- (a) *trans*-2-chloro-3-iodo-2-pentene (b) *cis*-2-chloro-3-iodo-2-pentene
(c) *trans*-3-iodo-4-chloro-3-pentene (d) *cis*-3-iodo-4-chloro-3-pentene

8. The IUPAC name of the compound is :



- (a) 2-methyl-6-oxohex-3-enamide (b) 6-keto-2-methyl hexanamide
(c) 2-carbamoylhexanal (d) 2-carbamoylhex-3-enal

9. The IUPAC name of is:



- (a) 1-Bromo-2-chloro-3-fluoro-6-iodo benzene
(b) 2-Bromo-1-chloro-5-fluoro-3-iodo benzene
(c) 4-Bromo-2-chloro-5-iodo-1-fluoro benzene
(d) 2-Bromo-3-chloro-1-iodo-5-fluoro benzene

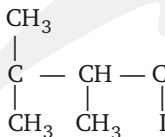
10. The IUPAC name of $\text{CH}_3-\text{CH}_2-\underset{\text{C}_2\text{H}_5}{\text{CH}}-\overset{\text{O}}{\underset{\parallel}{\text{C}}}-\text{OCH}_3$ is:
- (a) Methyl 2-ethylbutanoate (b) 1-methoxy-2-ethylbutanone
(c) 3-Methoxycarbonylpentane (d) 1-methoxy-2-ethylbutanal
11. The IUPAC name of $\text{CH}_3-\text{CH}=\underset{\text{O}}{\overset{\parallel}{\text{CH}}}-\text{C}-\text{OH}$ is:
- (a) But-1-en-4-oic acid (b) 1-hydroxybut-2-en-1-one
(c) But-2-en-1-oic acid (d) But-2-en-4-oic acid
12. The IUPAC name of $\text{CH}_3-\text{CH}=\overset{\text{OCH}_3}{\text{C}}-\text{CH}_3$ is:
- (a) 1-Methoxy-1-methylpropene (b) 2-Methoxybut-2-ene
(c) dimethylpropeneether (d) none of these
13. The IUPAC name of $\text{CH}_2=\overset{\text{Me}}{\text{C}}-\overset{\text{O}}{\underset{\parallel}{\text{C}}}-\text{OC}_2\text{H}_5$ is:
- (a) Ethyl 2-methylprop-2-enoate (b) Ethyl 2-methylprop-1-enoate
(c) 1-Ethoxy-2-methylprop-2-enone (d) 1-Ethoxy-2-methylprop-2-enal
14. The IUPAC name of $\text{CH}_3-\overset{\text{CH}_3}{\underset{|}{\text{CH}}}-\overset{\text{CH}_3}{\underset{|}{\text{CH}}}-\text{CH}_2\text{NH}_2$ is:
- (a) 2, 3-Dimethylbutan-4-amine (b) 2, 4-Dimethylbutan-1-amine
(c) 2,4-Dimethylbutan-4-amine (d) 2, 3-Dimethylbutan-1-amine
15. The IUPAC name of $\text{CH}_3-\overset{\text{O}}{\underset{\parallel}{\text{C}}}-\underset{\text{C}_2\text{H}_5}{\underset{|}{\text{CH}}}-\underset{\text{CH}_3}{\underset{|}{\text{CH}}}-\text{CH}_3$ is:
- (a) 3-(Methylethyl) pentan-2-one (b) 3-(Methylethyl)pentan-4-one
(c) 3-Ethyl-4-methylpentan-2-one (d) 3-Ethyl-2-methylpentan-4-one
16. The IUPAC name of $\text{CH}_3-\underset{\text{CH}_3}{\underset{|}{\text{CH}}}-\text{CH}_2-\overset{\text{O}}{\underset{\parallel}{\text{C}}}-\text{Br}$ is:
- (a) 2-Methylbutanoyl bromide (b) 2-Methylbutan-4-oyl bromide
(c) 1-Bromo-3-Methylbutanone (d) 3-Methylbutanoyl bromide

17. The IUPAC name of $\text{CH}_3 - \text{CH} - \text{CH}_2 - \text{OH}$ is:



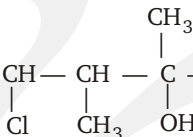
- (a) 2-Phenylpropan-1-ol (b) 2-Phenylpropan-3-ol
(c) 1-(2-Hydroxy-1-methylethyl) benzene (d) 1-((Hydroxymethyl)ethyl) benzene

18. The IUPAC name of $\text{CH}_3 - \text{C} - \text{CH} - \text{CH} - \text{CH}_2 - \text{CH}_3$ is:

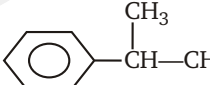


- (a) 3-Iodo-4,5,5-trimethylhexane (b) 4-Iodo-1, 1, 3-trimethylhexane
(c) 4-Iodo-2, 2-dimethylheptane (d) 4-Iodo-2, 2, 3-trimethylhexane

19. The IUPAC name of $\text{CH}_3 - \text{CH}_2 - \text{CH} - \text{CH} - \text{C} - \text{CH}_3$ is:



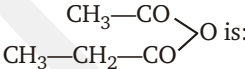
- (a) 4-Chloro-2, 3-dimethylhexane-2-ol (b) 4-Chloro-2-hydroxy-2, 3-dimethylhexane
(c) 4-Chloro-1, 1, 2-trimethylpentan-2-ol (d) 3-Chloro-2, 3-dimethylhexane-2-ol

20. The IUPAC name of  is:

- (a) 2-Phenylpropan-3-al (b) Formylethylbenzene
(c) 2-Phenylpropanal (d) Ethylformylbenzene

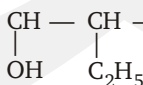
21. The IUPAC name of $\text{CH}_3 - \text{C}(=\text{O}) - \text{CH}(\text{CH}_3)_2$ is:

- (a) 2-Methylbutan-3-one (b) 3-Methylbutan-2-al
(c) 2-Methylbutan-3-al (d) 3-Methylbutan-2-one

22. The IUPAC name of  is:

- (a) Ethanoic propanoic anhydride (b) Propanoic ethanoic anhydride
(c) 1-Ethanoyloxypropanone (d) 3-Ethanoyloxypropan-3-one

23. The IUPAC name of $\text{CH}_3 - \text{CH} - \text{CH} - \text{CH}_2 - \text{OH}$ is:



- (a) 3-Ethylbutane-2, 4-diol (b) 2-Ethylbutane-1, 3-diol
(c) 3-Ethylbutane-1, 3-diol (d) 2-Ethyl-1-methylpropane-1, 3-diol

24. The IUPAC name of $\text{CH}_3 - \underset{\text{O}}{\underset{\parallel}{\text{C}}} - \underset{\text{O}}{\underset{\parallel}{\text{C}}} - \text{CH}_3$ is:
- (a) Butane-2, 3-dial (b) Butane-1, 3-dione
(c) Butane-2, 3-dione (d) 1, 2-dimethylethanedione
25. The IUPAC name of $\text{CH}_2 = \text{CH} - \text{CH} = \text{CH}_2$ is:
- (a) Butane (b) Buta-1, 3-diene (c) Butane-1, 3-diene (d) none of these
26. The IUPAC name of $\text{CH}_2(\text{COOH}) - \text{CH}_2 - \text{CH}_2(\text{COOH})$ is:
- (a) Pentane-1, 5-dioic acid (b) Pentane-1, 5-dicarboxylic acid
(c) Propane-1, 3-dioic acid (d) none of these
27. The IUPAC name of $\text{CH}_2(\text{CHO}) - \text{CH} = \text{CH} - \text{CHO}$ is:
- (a) propene-1, 3-dial (b) Propene-1, 3-dicarbaldehyde
(c) Pent-3-ene-1, 5-dial (d) Pent-2-ene-1, 5-dial
28. The IUPAC name of $\text{CH}_2(\text{CN}) - \text{CH}_2 - \text{CN}$ is:
- (a) Butane-1, 4-dicarbonitrile (b) Ethane-1, 2-dicarbonitrile
(c) Ethane-1, 2-dinitrile (d) Butane-1, 4-dinitrile
29. The IUPAC name of $\text{CH}_2(\text{NH}_2) - \text{CH}(\text{CH}_3) - \text{CH}_2 - \text{CH}_2(\text{NH}_2)$ is:
- (a) 2-Methylbutane-1, 4-diamine
(b) 3-Methylbutane-1, 4-diamine
(c) 3-(Aminomethyl)butanamine
(d) 2-(Aminomethyl)butan-4-amine
30. The IUPAC name of $\text{CH}_2(\text{Cl}) - \text{CH}(\text{CH}_2\text{Cl}) - \text{CH}_2(\text{Cl})$ is:
- (a) Tris(chloromethyl) methane
(b) 1, 3-Dichloro-2 (chloromethyl) propane
(c) 1-Chlorobis(chloromethyl) ethane
(d) none of these

31. The IUPAC name of $\text{CH}_3 - \underset{\text{OH}}{\text{CH}} - \underset{\text{CH}_3}{\text{CH}} - \underset{\text{OH}}{\text{CH}} - \overset{\text{CH}_3}{\underset{\text{CH}_3}{\text{C}}} - \text{CH}_3$ is:
- (a) 3, 5, 5-Trimethylhexane-2, 4-diol (b) 2, 2, 4-Trimethylhexane-3, 5-diol
(c) 1, 2, 4, 4-Tetramethylpentane-1, 3-diol (d) 2, 2, 4, 5-Tetramethylpentane-3, 5-diol
32. The IUPAC name of $\text{HOOC} - \text{CH} = \text{CH} - \text{COOH}$ is:
- (a) But-2-ene-1, 4-dicarboxylic acid (b) But-2-ene-1, 4-dioic acid
(c) Ethene dicarboxylic acid (d) Ethene dioic acid
33. The IUPAC name of $\text{CH}_3 - \overset{\text{OCH}_3}{\underset{|}{\text{CH}}} - \text{CHO}$ is:
- (a) 1-Formyl-1-methoxyethane (b) 2-Methoxypropan-3-one
(c) 2-Methoxypropanal (d) 2-Methoxypropan-3-al
34. The IUPAC name of $\text{CH}_2 = \overset{\text{CH}_3}{\underset{|}{\text{C}}} - \text{COOCH}_3$ is:
- (a) Methyl-2-methylprop-1-en-3-oate (b) 2-Methoxycarbonylpropene
(c) 2-Methoxycarbonylprop-2-ene (d) Methyl-2-methylprop-2-enoate
35. The IUPAC name of $\text{CH}_3 - \text{CH} = \text{CH} - \text{COOH}$ is:
- (a) But-2-ene-1-oic acid (b) But-1-ene-1-oic acid
(c) But-2-ene-1-carboxylic acid (d) Propene-1-carboxylic acid
36. The IUPAC name of $\text{CH}_3 - \overset{\text{OH}}{\underset{|}{\text{CH}}} - \text{COOH}$ is:
- (a) 2-Hydroxypropanoic acid (b) 1-Hydroxypropanoic acid
(c) 1-Hydroxyethane carboxylic acid (d) 1-Hydroxyethanoic acid
37. The IUPAC name of $\text{HO} - \overset{\text{OH}}{\underset{|}{\text{CH}}} - \text{COOH}$ is:
- (a) 2, 3-Dihydroxybutane-1, 4-carboxylic acid
(b) 2, 3-Dihydroxybutane-1, 4-dioic acid
(c) 1, 2-Dihydroxyethane dicarboxylic acid
(d) none of these

38. The IUPAC name of $\text{CH}_3 - \underset{\text{CH}_3}{\text{CH}} - \underset{\text{O}}{\text{C}} - \text{COOH}$ is:

- (a) 3-Methyl-2-oxobutanecarboxylic acid (b) 2-Methyl-3-oxobutan-4-oic acid
(c) 3-Methyl-2-oxobutanoic acid (d) 3-Methyl-1,2-dioxobutanoic acid

39. The IUPAC name of $\text{NC} - \text{CH}_2 - \text{CH}_2 - \text{COOH}$ is:

- (a) 3-Carboxy propanenitrile (b) 4-Cyanobutanoic acid
(c) 2-Cyanoethane Carboxylic acid (d) 3-Cyanopropanoic acid

40. The IUPAC name of $\begin{array}{c} \text{CH}_2\text{COOH} \\ | \\ \text{C} - \text{OH} \\ | \\ \text{COOH} \\ | \\ \text{CH}_2\text{COOH} \end{array}$ is:

- (a) 3-Carboxy-3-hydroxypentanedicarboxylic acid
(b) 2-Hydroxypropane-1, 2, 3-tricarboxylic acid
(c) 2-Hydroxypropane-1, 2, 3-trioic acid
(d) 3-Hydroxypropane-1, 2, 3-tricarboxylic acid

41. The IUPAC name of $\text{CH}_3 - \underset{\text{COOC}_2\text{H}_5}{\text{C}} - \text{CH} - \text{CH}_2 - \underset{\text{OH}}{\overset{\text{O}}{\text{C}}}$ is:

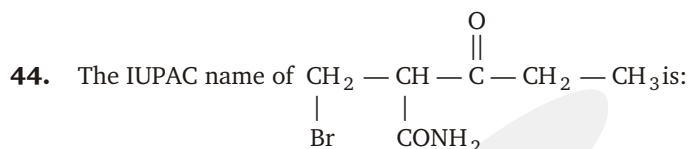
- (a) 4-ethoxycarbonylpent-3-enoic acid
(b) 4-ethanoyloxypent-3-enoic acid
(c) 3-ethoxycarbonylbut-2-enecarboxylic acid
(d) 3-ethoxycarbonylpent-3-enoic acid

42. The IUPAC name of $\text{CH}_3 - \underset{\text{CH}_3}{\text{CH}} - \underset{\text{O}}{\overset{\text{O}}{\text{C}}} - \underset{\text{O}}{\overset{\text{O}}{\text{C}}} - \underset{\text{Br}}{\text{NH}}$ is:

- (a) (N-Bromo)-2-keto-3-methylbutanamide
(b) (N-Bromo)-2-keto-4-methylbutanamide
(c) (N-Bromo)-1, 2-diketo-3-methylbutanamine carboxamide
(d) (N-Bromo)-1-keto-2-methylpropane

43. The IUPAC name of $\begin{array}{c} \text{CH}_2 - \text{C} - \text{CH} - \text{CH}_2\text{OH} \\ | \quad | \\ \text{Cl} \quad \text{CH}_3 \end{array}$ is:

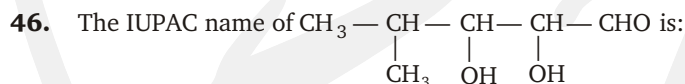
- (a) 4-Chloro-3-methylbut-2-en-1-ol (b) 1-Chloro-2-methylbut-2-en-4-ol
(c) 4-Chloro-1-hydroxy-3-methylbut-2-ene (d) 1-Chloro-4-hydroxy-2-methylbut-2-ene



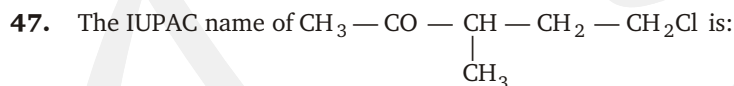
- (a) 2-(Bromomethyl)-3-oxopentane carboxamide
- (b) 1-Bromo-2-carbamoylpentan-3-one
- (c) 5-Bromo-4-carbamoylpentan-3-one
- (d) 2-(Bromomethyl)-3-oxopentanamide



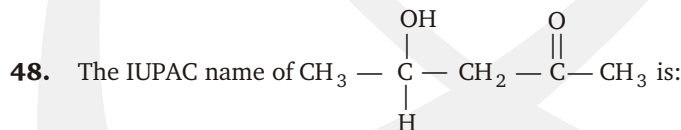
- (a) 2, 2-Dimethyl-4-chloro butane
- (b) 1-Chloro-3, 3-dimethylbutane
- (c) 4-Chloro-2, 2-dimethyl butane
- (d) none of these



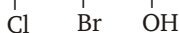
- (a) 2, 3-Dihydroxy-4-methylpentanal
- (b) 1-oxo-2, 3-Dihydroxy-4-methylpentane
- (c) 2,3-Dihydroxy-4-methylpentanone
- (d) 1, 2-Dihydroxy-3-methylbutanecarbaldehyde



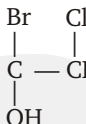
- (a) 1-Chloro-3-methylpentan-4-one
- (b) 1-Chloro-2-(oxoethyl)butane
- (c) 5-Chloro-3-methylpentan-2-one
- (d) 3-(2-Chloroethyl)butan-2-one



- (a) 2-Hydroxypentan-4-one
- (b) 4-Hydroxypentan-2-one
- (c) 4-oxopentan-2-ol
- (d) 2-oxopentan-4-ol



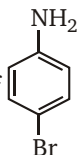
- 50.** The IUPAC name of $\text{CH}_3 - \text{CH}_2 - \overset{\text{Cl}}{\underset{\text{CH}_3}{\text{C}}} - \text{CH} - \text{CH}_2 - \text{Cl}$ is:



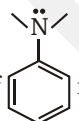
- 51.** The IUPAC name of $\text{CH}_3 - \text{CH} - \text{CH}_2 - \text{O} - \text{C}_2\text{H}_5$ is:



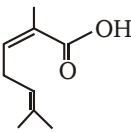
- 52.** The IUPAC name of  is:



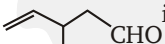
- 53.** The IUPAC name of  is:



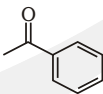
- (a) *N,N*-Dimethyl aminobenzene (b) *N,N*-Dimethyl benzenamine
(c) (a) and (b) both are correct (d) none of these

54. The IUPAC name of  is:

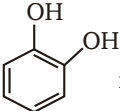
- (a) 2, 6-Dimethylhepta-2, 5-dienoic acid
- (b) 3, 7-Dimethylhepta-2, 5-dienoic acid
- (c) 1-Hydroxy-2, 6-dimethylhepta-2, 5-dienone
- (d) none of these

55. The IUPAC name of  is:

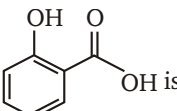
- (a) 3-Methylpent-1-en-4-al
- (b) 3-Methylpent-4-enal
- (c) 3-Methylpent-4-carbaldehyde
- (d) 3-Methyl-5-oxopent-1-ene

56. The IUPAC name of  is:

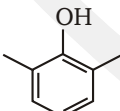
- (a) 2-Phenyl ethanone
- (b) 1-Phenyl ethanone
- (c) 1-(Oxoethyl)benzene
- (d) 1-(Ethylaloxo)-benzene

57. The IUPAC name of  is:

- (a) 2-Hydroxybenzenol
- (b) 1, 2-Dihydroxybenzene
- (c) Benzene-1, 2-diol
- (d) 2-Hydroxyphenol

58. The IUPAC name of  is:

- (a) 2-Carboxyphenol
- (b) 2-Hydroxybenzoic acid
- (c) 1-Carboxy-2-hydroxybenzene
- (d) 2-Carboxy-1-hydroxybenzene

59. The IUPAC name of  is:

- (a) 1,3-Dimethyl phenol
- (b) 1-Hydroxy-2-6-dimethyl benzene
- (c) 2, 6-Dimethyl benzenol
- (d) 2-Hydroxy-1-3-dimethylbenzene



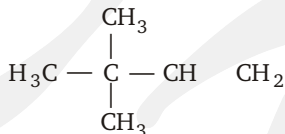
The IUPAC name is :

- (a) 3-phenyl prop-2-enoic acid (b) 3-phenol prop-1-enoic acid
(c) 3-carboxy-prop-1-ene benzene (d) but-2-enoic acid



- (a) Chloromethylbenzene (b) Chlorophenylmethane
(c) (a) and (b) both (d) none of these

62. The IUPAC name of the compound having the formula is :



- (a) 3, 3, 3-Trimethylprop-1-ene (b) 1, 1, 1-Trimethylprop-2-ene
(c) 3, 3-Dimethylbut-1-ene (d) 2, 2-Dimethylbut-3-ene

63. The IUPAC name of the compound $\text{CH}_2 = \text{CH} - \text{CH}(\text{CH}_3)_2$ is :

- (a) 1, 1-Dimethylprop-2-ene (b) 3-Methylbut-1-ene
(c) 2-Vinyl propane (d) none of these

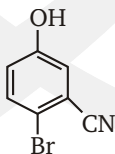
64. The number of sigma and pi-bonds in 1-butene 3-yne are :

- (a) 5 sigma and 5 pi (b) 7 sigma and 3 pi
(c) 8 sigma and 2 pi (d) 6 sigma and 4 pi

65. The IUPAC name of $\text{C}_6\text{H}_5\text{COCl}$ is :

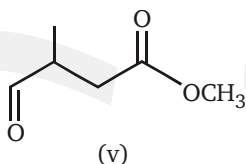
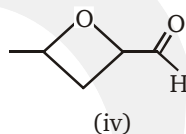
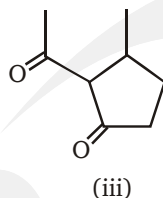
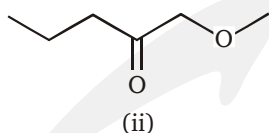
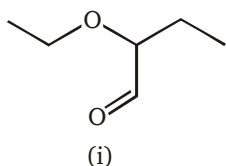
- (a) Benzoyl Chloride (b) Benzene chloro ketone
(c) Benzene carbonyl chloride (d) Chloro phenyl ketone

66. The IUPAC name of the following compound is :

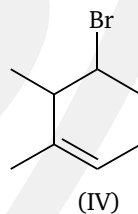
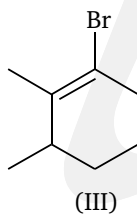
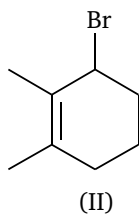
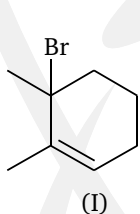


- (a) 4-Bromo-3-cyanophenol (b) 2-Bromo-5-hydroxybenzonitrile
(c) 2-Cyano-4-hydroxybromobenzene (d) 6-Bromo-3-hydroxybenzonitrile

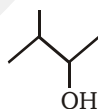
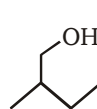
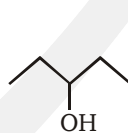
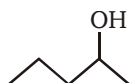
67. Many organic compounds contain more than one functional group. Which of the following is both an aldehyde and an ether ?



- (a) (i) only
(b) (i) and (iv)
(c) (ii) and (v)
(d) (iii) and (iv)
68. What is the sum of positions assigned to bromine while numbering the Parent Chain in the below compounds ?

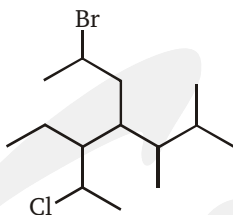


- (a) 13
(b) 14
(c) 15
(d) 16
69. How many number of compounds, which have same IUPAC name ?



- (a) 0
(b) 1
(c) 2
(d) 3

70. How many total number of substituents are present in the following compound ?



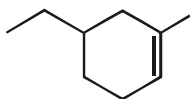
(a) 3

(b) 4

(c) 5

(d) 6

71.



Correct IUPAC name :

(a) 1-methyl-3-ethylcyclohexene

(b) 5-ethyl-1-methylcyclohexene

(c) 2-ethyl-4-methylcyclohexene

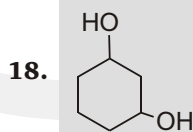
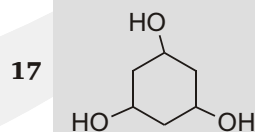
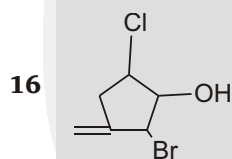
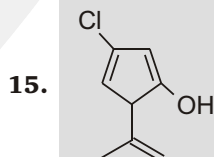
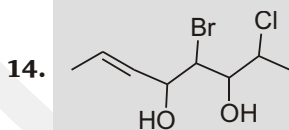
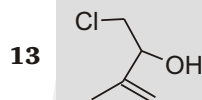
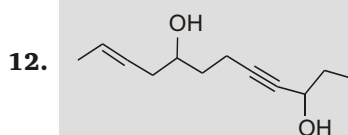
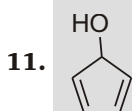
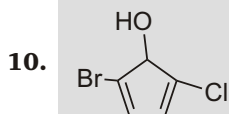
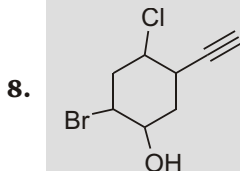
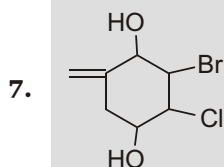
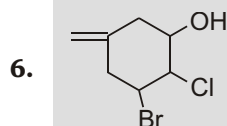
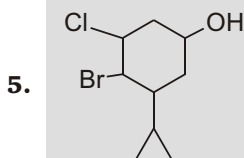
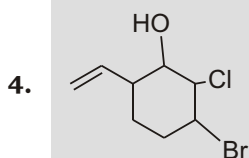
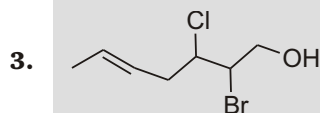
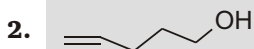
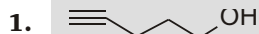
(d) 3-ethyl-1-methylcyclohexene

ANSWERS — LEVEL 1

1.	(c)	2.	(b)	3.	(b)	4.	(b)	5.	(a)	6.	(a)	7.	(a)	8.	(a)
9.	(b)	10.	(a)	11.	(c)	12.	(b)	13.	(a)	14.	(d)	15.	(c)	16.	(d)
17.	(a)	18.	(d)	19.	(a)	20.	(c)	21.	(d)	22.	(a)	23.	(b)	24.	(c)
25.	(b)	26.	(a)	27.	(d)	28.	(d)	29.	(a)	30.	(b)	31.	(a)	32.	(b)
33.	(c)	34.	(d)	35.	(a)	36.	(a)	37.	(b)	38.	(c)	39.	(d)	40.	(b)
41.	(a)	42.	(a)	43.	(a)	44.	(d)	45.	(b)	46.	(a)	47.	(c)	48.	(b)
49.	(a)	50.	(c)	51.	(a)	52.	(a)	53.	(b)	54.	(a)	55.	(b)	56.	(b)
57.	(c)	58.	(b)	59.	(c)	60.	(a)	61.	(b)	62.	(c)	63.	(b)	64.	(b)
65.	(c)	66.	(b)	67.	(b)	68.	(c)	69.	(a)	70.	(c)	71.	(b)		

Level - 2

Give the IUPAC name of the following compounds



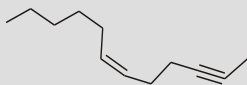
19.



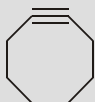
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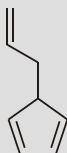
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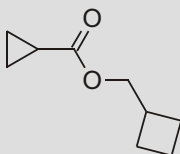
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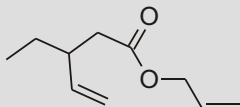
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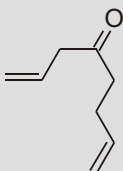
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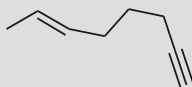
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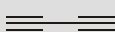
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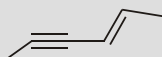
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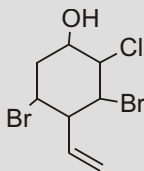
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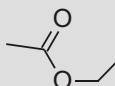
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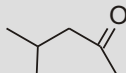
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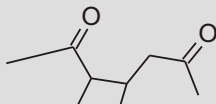
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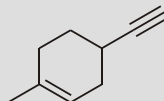
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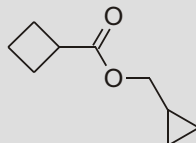
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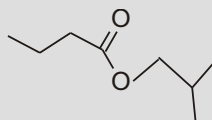
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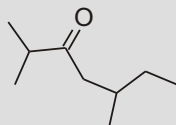
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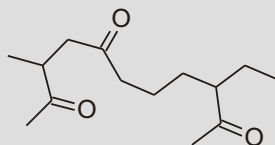
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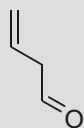
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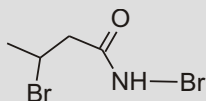
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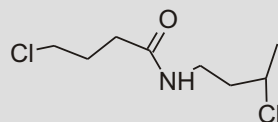
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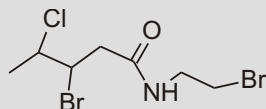
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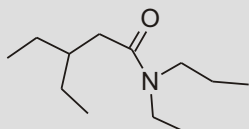
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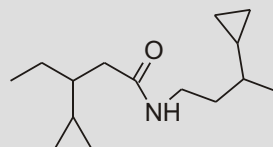
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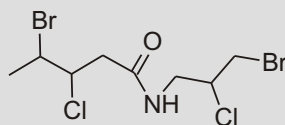
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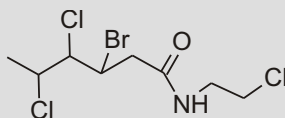
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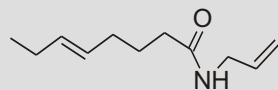
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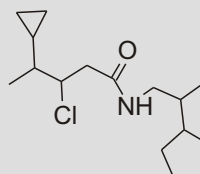
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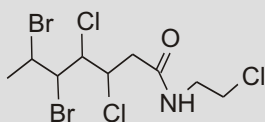
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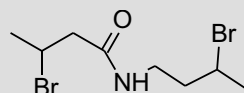
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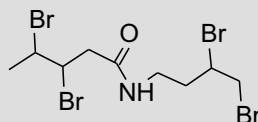
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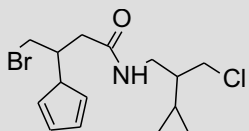
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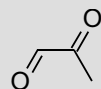
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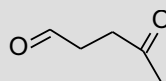
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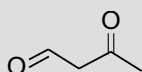
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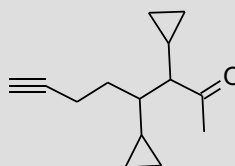
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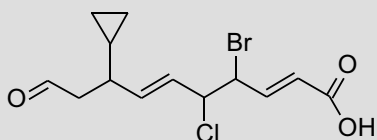


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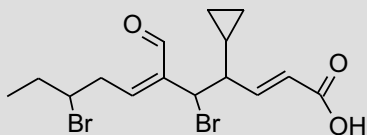


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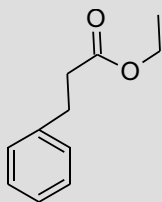




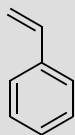
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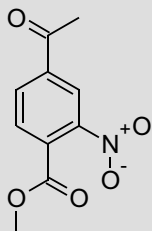
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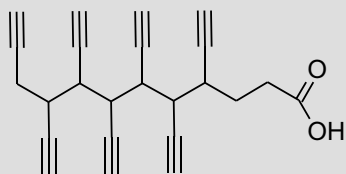
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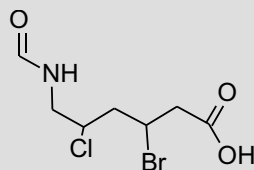
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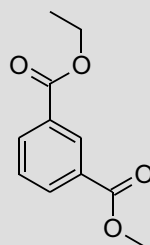
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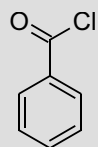
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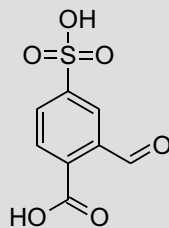
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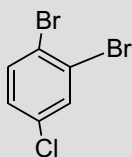
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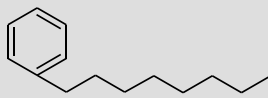
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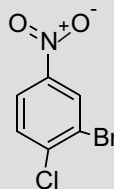
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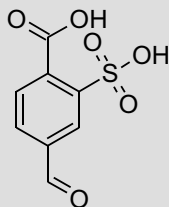
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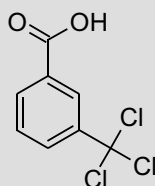
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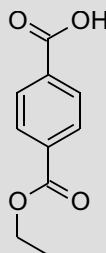
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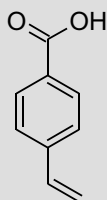
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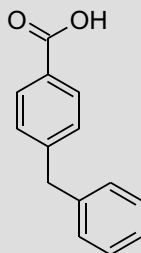
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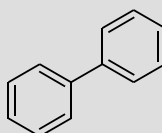
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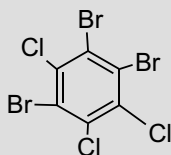
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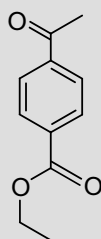
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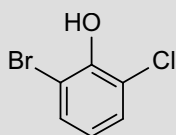
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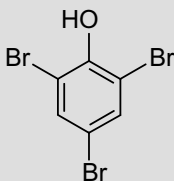
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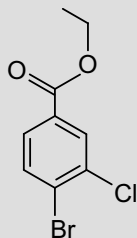
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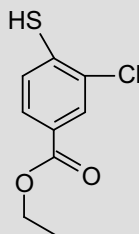
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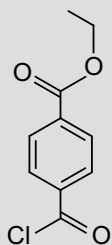
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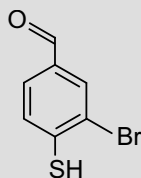
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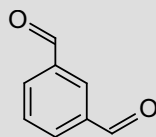
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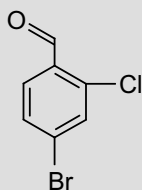
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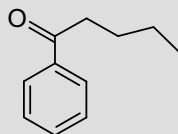
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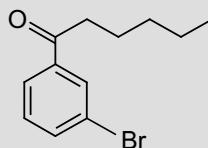
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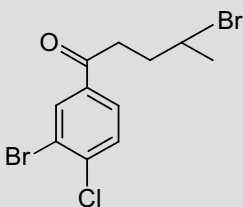
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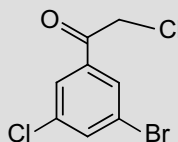
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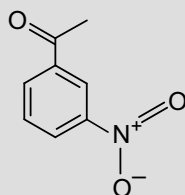
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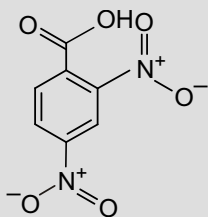
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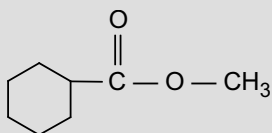
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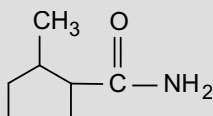
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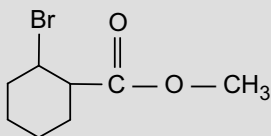
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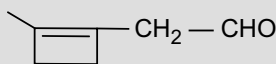
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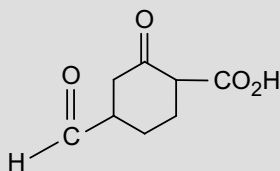
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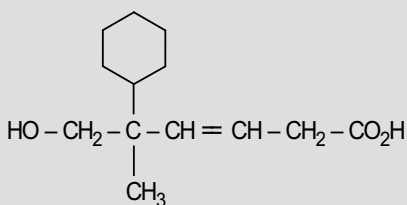
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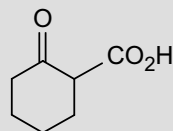
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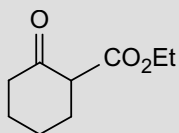


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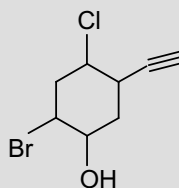


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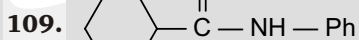
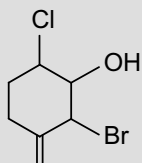
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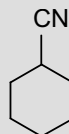
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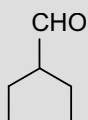
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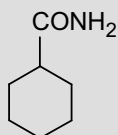
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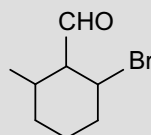
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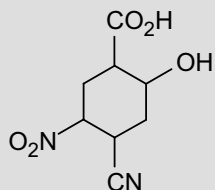
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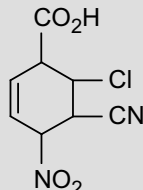
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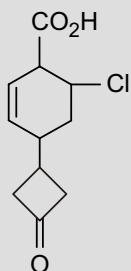
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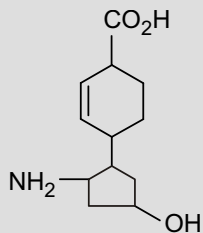
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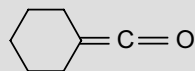
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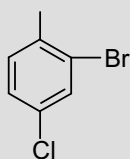
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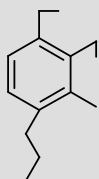
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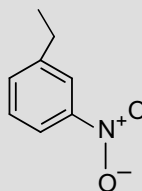
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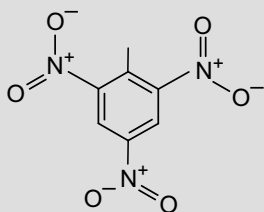
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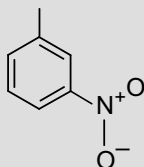
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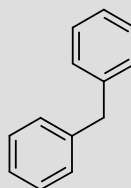
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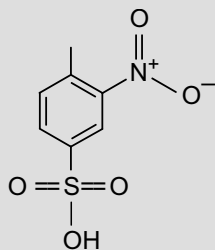
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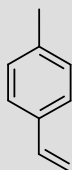
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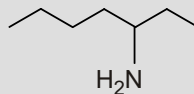
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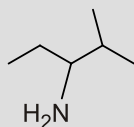
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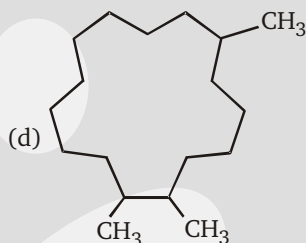
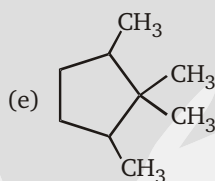
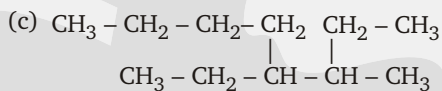
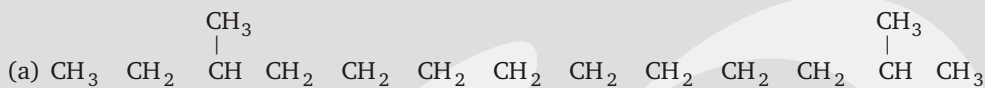
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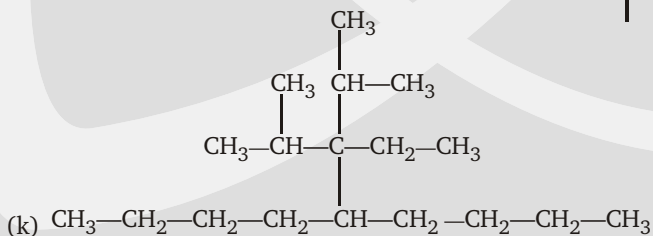
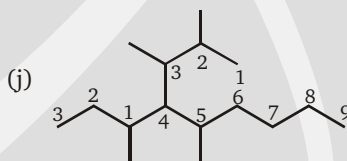
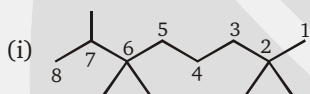
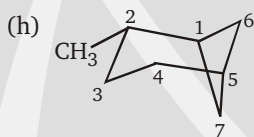
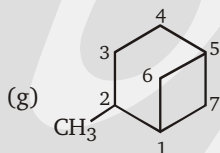
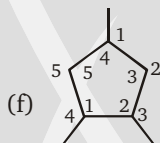
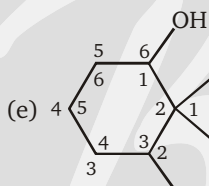
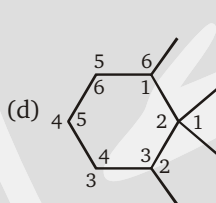
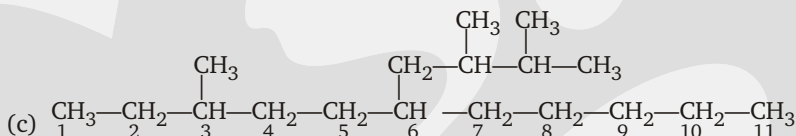
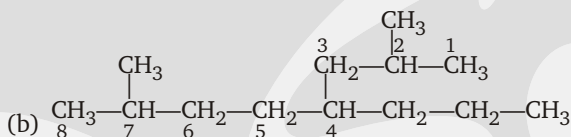
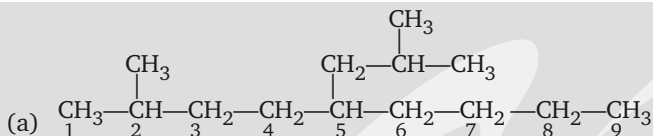
129.



130. Give the IUPAC name of the following compounds :



131. Give the IUPAC name of the following :



132. Match the Column :

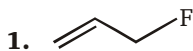
Column (I)		Column (II)	
Compound		IUPAC Name	
(a)		(p)	1, 3-dibromo-2-chlorocyclopropane
(b)		(q)	1, 2-dibromo-3-chlorocyclopropene
(c)		(r)	3-bromo-1-chlorocyclopropene
(d)		(s)	1-bromo-3-chlorocyclopropene

133. Match the Column :

Column (I)		Column (II)	
(a)		(p)	1-butyl-4-methyl cyclohexane
(b)		(q)	2-bromo-1,1-dichloro cyclohexane
(c)		(r)	1-Bromo-1,2-dichloro cyclohexane
(d)		(s)	Ethyl-2-formyl cyclohexane carboxylate



Subjective Problems



When IUPAC name of following compound is given, then double bond and substituent gets respectively

(x and y) number so the sum of (x + y) will be

ANSWERS – IUPAC Name

1.	pent-4-yn-1-ol	2.	pent-4-en-1-ol
3.	(5E)-2-bromo-3-chlorohept-5-en-1-ol	4.	3-bromo-2-chloro-6-ethenylcyclohexanol
5.	4-bromo-3-chloro-5-cyclopropylcyclohexanol		
6.	3-bromo-2-chloro-5-methylidenecyclohexanol		
7.	3-bromo-2-chloro-5-methylidenecyclohexane-1,4-diol		
8.	2-bromo-4-chloro-5-ethynylcyclohexanol	9.	cyclopenta-1,3-dien-1-ol
10.	2-bromo-5-chlorocyclopenta-2,4-dien-1-ol		
11.	cyclopenta-2,4-dien-1-ol	12.	dodec-10-en-4-yne-3,8-diol
13.	1-chloro-3-methylbut-3-en-2-ol	14.	4-bromo-2-chlorooct-6-ene-3,5-diol
15.	3-chloro-5-(1-methylethenyl)cyclopenta-1,3-dien-1-ol		
16.	2-bromo-5-chloro-3-methylidenecyclopentanol		
17.	cyclohexane-1,3,5-triol	18.	cyclohexane-1,3-diol
19.	cyclopenta-1,3-diene	20.	(6E)-oct-6-en-1-yne
21.	but-1-en-3-yne	22.	(3Z)-pent-3-en-1-yne
23.	buta-1,3-diyne	24.	(2E)-dec-2-en-5-yne
25.	(6Z)-dodec-6-en-2-yne	26.	pent-1-en-4-yne
27.	4-ethynyl-1-methylcyclohexene	28.	cyclooctyne
29.	(2E)-hex-2-en-4-yne	30.	5-ethynylcyclopenta-1,3-diene
31.	5-(prop-2-enyl) cyclopenta-1,3-diene		
32.	3,5-dibromo-2-chloro-4-ethenylcyclohexanol		
33.	cyclopropylmethyl cyclobutanecarboxylate		
34.	cyclobutylmethyl cyclopropanecarboxylate		
35.	ethyl ethanoate	36.	2-methylpropyl butanoate
37.	prop-2-enyl 3-ethylpent-4-enoate	38.	4-methylpentan-2-one
39.	2,5-dimethylheptan-3-one	40.	octa-1,7-dien-4-one
41.	3,4-dimethylheptane-2,6-dione	42.	9-ethyl-3-methylundecane-2,5,10-trione

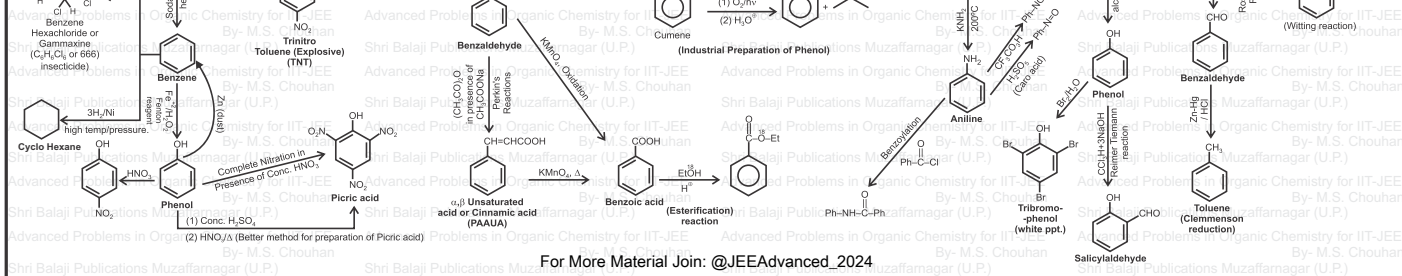
43.	5-methyloctanal	44.	but-3-enal
45.	3-ethylheptanal	46.	N,3-dibromobutanamide
47.	4-chloro-N-(3-chlorobutyl)butanamide		
48.	3-bromo-N-(2-bromoethyl)-4-chloropentanamide		
49.	N,3-diethyl-N-propylpentanamide		
50.	3-cyclopropyl-N-(3-cyclopropylbutyl)pentanamide		
51.	4-bromo-N-(3-bromo-2-chloropropyl)-3-chloropentanamide		
52.	3-bromo-4,5-dichloro-N-(2-chloroethyl)hexanamide		
53.	(5E)-N-(prop-2-en-yl)oct-5-enamide		
54.	3-chloro-N-(2-cyclopentylpropyl)-4-cyclopropylpentanamide		
55.	5,6-dibromo-3,4-dichloro-N-(2-chloroethyl)heptanamide		
56.	3-bromo-N-(3-bromobutyl)butanamide		
57.	3,4-dibromo-N-(3,4-dibromobutyl)pentanamide		
58.	4-bromo-N-(3-chloro-2-cyclopropylpropyl)-3-(cyclopenta-2,4-dien-yl)butanamide		
59.	2-oxopropanal	60.	4-oxopentanal
61.	3-oxobutanal	62.	3,4-dicyclopropyloct-7-yn-2-one
63.	(2E,6E)-4-bromo-5-chloro-8-cyclopropyl-10-oxodeca-2,6-dienoic acid		
64.	4,5,6,7,8,9-hexaethynyldec-11-ynoic acid		
65.	(2E,6E)-5,9-dibromo-4-cyclopropyl-6-formylundeca-2,6-dienoic acid		
66.	3-bromo-5-chloro-6-(formylamino)hexanoic acid		
67.	ethyl 3-phenylpropanoate	68.	ethyl methyl benzene-1,3-dicarboxylate
69.	ethenylbenzene	70.	benzoyl chloride
71.	methyl 4-acetyl-2-nitrobenzoate	72.	2-formyl-4-sulphobenzoic acid
73.	1,2-dibromo-4-chlorobenzene	74.	octylbenzene
75.	2-bromo-1-chloro-4-nitrobenzene	76.	4-formyl-2-sulphobenzoic acid
77.	3-(trichloromethyl)benzoic acid	78.	4-(ethoxycarbonyl)benzoic acid

79.	4-ethenylbenzoic acid	80.	4-benzylbenzoic acid
81.	biphenyl	82.	1,2,5-tribromo-3,4,6-trichlorobenzene
83.	ethyl 4-acetylbenzoate	84.	2-bromo-6-chlorophenol
85.	2, 4, 6-tribromophenol	86.	ethyl 4-bromo-3-chlorobenzoate
87.	ethyl 3-chloro-4-mercapto/sulpho benzoate	88.	ethyl 4-(chlorocarbonyl)benzoate
89.	3-bromo-4-sulphobenzaldehyde	90.	benzene-1, 3-dicarbaldehyde
91.	4-bromo-2-chlorobenzaldehyde	92.	1-phenylpentan-1-one
93.	1-(3-bromophenyl)hexan-1-one		
94.	4-bromo-1-(3-bromo-4-chlorophenyl)pentan-1-one		
95.	1-(3-bromo-5-chlorophenyl)-2-chloroethanone		
96.	1-(3-nitrophenyl)ethanone	97.	2,4-dinitrobenzoic acid
98.	methylcyclohexane carboxylate	99.	2-methylcyclopentane carboxamide
100.	methyl-2-bromocyclohexane carboxylate	101.	2-(2-methylcyclobut-1-enyl)ethanal
102.	4-formyl-2-oxocyclohexane-1-carboxylic acid		
103.	5-cyclohexyl-6-hydroxy-5-methylhex-3-en-1-oic acid		
104.	2-oxocyclohexane-1-carboxylic acid	105.	ethyl-2-oxocyclohexane-1-carboxylate
106.	<i>N</i> -methylmethanamide	107.	2-bromo-4-chloro-5-ethynylcyclohexanol
108.	2-bromo-6-chloro-3-methylidene-cyclohexanol		
109.	<i>N</i> -phenylcyclohexane carboxamide	110.	cyclohexane carbonitrile
111.	cyclopentanecarbaldehyde	112.	cyclohexanecarboxamide
113.	2-bromo-6-methyl-cyclohexanecarbaldehyde		
114.	cyclopropane carboxylic acid		
115.	4-cyno-2-hydroxy-5-nitrocyclohexane carboxylic acid		
116.	6-chloro-5-cyno-4-nitrocyclohex-2-ene carboxylic acid		
117.	6-chloro-4-(3-oxo cyclobutyl)cyclohex-2-ene carboxylic acid		
118.	4-(2-amino-4-hydroxycyclopentyl) cyclohex-2-enecarboxylic acid		

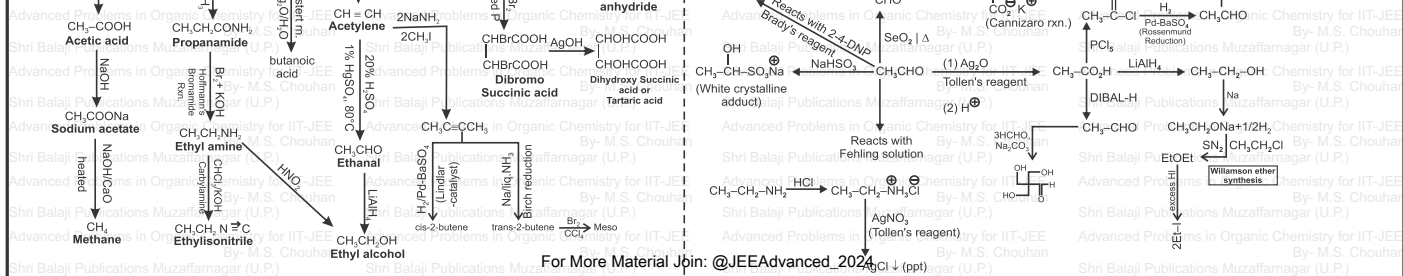
119.	cyclohexylidenemethanone	120.	2-bromo-4-chloro-1-methylbenzene
121.	1,2-diethyl-3-methyl-4-propylbenzene	122.	1-ethyl-3-nitrobenzene
123.	2-methyl-1, 3, 5-trinitrobenzene	124.	1-methyl-3-nitrobenzene
125.	Diphenylmethane	126.	4-Methyl-3-nitrobenzene sulphonic acid
127.	1-ethenyl-4-methylbenzene	128.	heptan-3-amine
129.	2-methylpentan-3-amine		
130.	(a) 2,11-dimethyltridecane		
	(b) 12-methyl-tridecan-3-ol		
	(c) 4-ethyl-3-methyloctane		
	(d) 1,2,7-trimethylcyclopentadecane		
	(e) 1,1,2,5-tetramethylcyclopentane		
131.	(a) 5-isobutyl-2-methylnonane, 5-2-methylnonane (2-methylpropyl)		
	(b) 2,7-dimethyl-4-propyloctane		
	(c) 6-(2,3-dimethylbutyl) 3-methyl undecane		
	(d) 1,1,2,6-tetramethylcyclohexane		
	(e) 2,2,3-trimethylcyclohexanol		
	(f) 1,2,4-trimethylcyclopentane		
	(g) 2-methylbicyclo[3.1.1] heptane		
	(h) 2-methylbicyclo[3.1.1] heptane		
	(i) 2,2,6,6,7-pentamethyloctane		
	(j) 4-(1methylpropyl)-2,3,5-trimethyl nonane Not 2-butyl		
	(k) 5-(1-ethyl-2-methyl-1-(1-methylethyl)propyl) nonane		
132.	a - r; b - s; c - q; d - p		
133.	a - q; b - p; c - s; d - r		

Subjective Problems

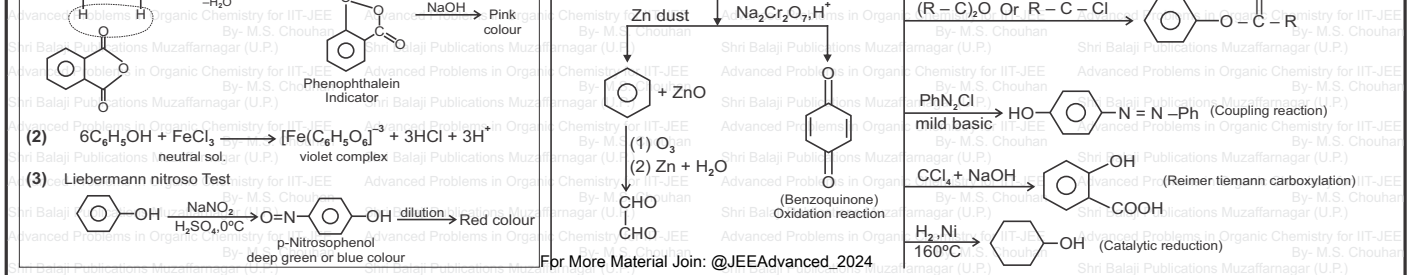
1. 4



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